# Hodge Theory

**— *EYNTKA* Series —**

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August 5, 2026

# *Contents*

# *Introduction*

A compact smooth manifold has de Rham cohomology, and a Riemannian metric on it picks out a distinguished representative in every class, the harmonic forms of *Riemannian Geometry* [9]. A smooth projective variety over $\mathbb{C}$ has coherent sheaf cohomology $H^q(X, \Omega_X^p)$, computed from the algebraic structure alone in *Algebraic Geometry* [4]. Nothing connects the two. The first is an invariant of the underlying smooth manifold and cannot detect which complex structure was placed on it; the second is defined only after that complex structure is fixed, and its individual pieces are not topological invariants at all. Hodge theory is the discovery that on a compact Kähler manifold each determines the other:

$$H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^q(X, \Omega_X^p)$$

Both sides of this isomorphism were already available to the reader; what is new is that they are the same.

The consequences run in both directions. Left to right, the topology bounds the algebra: the Hodge numbers $h^{p,q} = \dim_{\mathbb{C}} H^q(X, \Omega_X^p)$ sum along each anti-diagonal to a Betti number, so a purely topological count constrains a sheaf-cohomological one. Right to left, the algebra obstructs the geometry. Complex conjugation exchanges $H^{p,q}$ with $H^{q,p}$, forcing $h^{p,q} = h^{q,p}$, and therefore forcing every odd Betti number of a compact Kähler manifold to be even. The Hopf surface, a compact complex surface diffeomorphic to $S^1 \times S^3$, has $b_1 = 1$. It admits no Kähler metric, hence no projective embedding, and the proof is a count of the Betti numbers of a product of spheres.

The identity has a prehistory in the one-dimensional case. Riemann's bilinear relations of 1857 record symmetry and positivity constraints on the periods of holomorphic differentials on a compact Riemann surface, and they are the weight-one instance of everything in this book. Hodge's contribution, developed through the 1930s and published as *The Theory and Applications of Harmonic Integrals* in 1941 [19], was to read those constraints as coming from a decomposition of cohomology, and to produce the decomposition analytically by solving for harmonic representatives. The analysis was the difficult part, and the existence theorem it required was placed on secure foundations over the following years by Weyl [36] and Kodaira. Two further ingredients arrived from outside: Kähler's 1933 condition on a Hermitian metric, that its associated $(1,1)$-form be closed [23], which turned out to be exactly the hypothesis the decomposition needs, and Serre's GAGA [31], which identifies the analytic invariants with algebraic ones on a projective variety.

What kept the subject growing after the decomposition theorem was that the decomposition moves. In a family of compact Kähler manifolds the underlying topological cohomology is locally constant while the Hodge filtration varies holomorphically, and Griffiths, in the late 1960s, made that variation the object of study [17]: the period map, its derivative, and the transversality constraint the derivative satisfies. Deligne then removed the hypotheses. *Théorie de Hodge* II [13] and III [14] attach to the cohomology of an arbitrary complex algebraic variety, open or singular, a functorial mixed Hodge structure whose weight filtration measures the failure of purity. The subject's central open problem predates both: Hodge's 1950 conjecture, that on a smooth projective variety every class in $H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ is a rational combination of classes of codimension-$p$ subvarieties, is unproved and is one of the Millennium Prize problems.

A word on sources, since this book cites few. The EYNTKA series is written to be self-contained: a reader holding the prerequisites should never need an outside text to follow an argument, and where a result is taken rather than proved it is taken from another volume of the series. That is a constraint on the exposition, not a claim of originality. The treatment below is shaped throughout by Voisin's two volumes [33, 34], which set the order of the Hodge decomposition and of the theory of variations; by Huybrechts [21] for complex manifolds and the Kähler identities; by Griffiths and Harris [15] for the classical geometry; and, for the structural half, by Carlson, Müller-Stach and Peters [1] on period mappings and by Peters and Steenbrink [28] on mixed Hodge structures. A reader who wants a second account of any chapter should go to these first.

**Prerequisites and Place in the Series** This volume builds on *Complex Analysis* [25], *Algebraic Geometry* [4], *Riemannian Geometry* [9], *Differential Topology* [26], *Algebraic Topology* [24], *Homological Algebra* [6], *Lie Groups* [32], and *Algebraic Groups and Representations* [11]. The last two are not needed until Part II, and the algebraic-group material only from Chapter 8 onward, where Mumford-Tate groups are defined; Part I uses neither. It is naturally read alongside *Shimura Varieties* [10] and *p-adic Hodge Theory.* No later volume currently lists it as a prerequisite. The reader has access to every completed volume of the series but is not assumed to have read all of them; prerequisites are cited where invoked.

## Notation and Conventions

$X$ denotes a complex manifold, always connected, and $n$ its complex dimension $\dim_{\mathbb{C}} X$. Compactness is never silent: it is stated in the hypotheses of every result that uses it. Smooth means $C^\infty$, and the base field is $\mathbb{C}$ throughout.

The sheaf of holomorphic functions on $X$ is $\mathcal{O}_X$ and the sheaf of holomorphic $p$-forms is $\Omega_X^p$; $\mathcal{A}^k$ and $\mathcal{A}^{p,q}$ denote the sheaves of smooth complex-valued $k$-forms and $(p,q)$-forms, with $\mathcal{A}^k(X)$ and $\mathcal{A}^{p,q}(X)$ their global sections. Sheaf cohomology is $H^q(X,\mathcal{F})$, Dolbeault cohomology is $H^{p,q}_{\bar\partial}(X)$, and $H^k(X,\mathbb{C})$ is used for singular and de Rham cohomology with complex coefficients without distinguishing them, the de Rham theorem being available from *Differential Topology* [26]. Hodge and Betti numbers are $h^{p,q}$ and $b_k$. Complex projective space is $\mathbb{CP}^n$.

## Goal

Part I proves the Hodge decomposition, and is self-contained given the prerequisites above. Chapter 1 builds the objects: holomorphic functions of several variables, almost complex structures together with the Newlander-Nirenberg theorem identifying which of them arise from complex manifolds, the splitting of forms by type that turns the exterior derivative into $d = \partial + \bar{\partial}$, holomorphic vector bundles, and divisors and line bundles. Chapter 2 takes cohomology of the $\bar{\partial}$-complex alone. The $\bar{\partial}$-Poincaré lemma makes the sheaves $\mathcal{A}^{p,\bullet}$ a fine resolution of $\Omega_X^p$, which gives Dolbeault's theorem $H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X)$ and so identifies the two sides of the displayed isomorphism above as objects. The Frölicher spectral sequence then measures how far the bigraded data sits from de Rham cohomology, and the Iwasawa manifold shows the gap can be nonzero.

Chapter 3 supplies the hypothesis that closes it. A Kähler metric is a Hermitian metric whose fundamental form is closed; the Fubini-Study metric puts one on $\mathbb{CP}^n$ and hence on every projective manifold. The condition looks weak and is not: it forces a set of commutator identities among $\partial$, $\bar{\partial}$, their adjoints, and the Lefschetz operators, and from those identities follows the single equation $\Delta_d = 2\Delta_{\bar{\partial}}$ on which the whole decomposition rests. Chapter 4 cashes it. Harmonic theory is imported from *Riemannian Geometry* [9] and specialized to $\Delta_{\bar{\partial}}$, the decomposition follows, and with it Hodge symmetry, the parity of the odd Betti numbers, the Hodge index theorem on surfaces, and the Lefschetz $(1,1)$-theorem identifying integral $(1,1)$-classes as classes of divisors. Appendix A reduces the hard direction of the Newlander-Nirenberg theorem to a single analytic input and isolates exactly why that input is delicate.

The rest of the book abstracts the decomposition, then breaks its hypotheses. A Hodge structure is the bigrading together with the polarization the Kähler class supplies, taken as data on a rational vector space in its own right; those with fixed numerical invariants are parametrized by a period domain. Hard Lefschetz and the primitive decomposition sharpen the picture on a single manifold, while Lefschetz pencils supply the vanishing cycles and monodromy that make families tractable. In a family the Hodge structures vary: the underlying local system is flat, the filtration is holomorphic, and Griffiths transversality constrains its derivative, which makes the period map rigid enough to yield Torelli theorems, Noether-Lefschetz statements, and the algebraicity of Hodge loci. Dropping compactness or smoothness costs purity but not the theory, and Deligne's weight filtration produces a functorial mixed Hodge structure on any complex algebraic variety, built from the logarithmic de Rham complex in the smooth open case and from simplicial resolutions in general. The cycle class map and the Hodge conjecture close the mathematics, followed by a survey of what has been left undeveloped: Saito's mixed Hodge modules, the decomposition theorem, and nonabelian and $p$-adic Hodge theory.

A reader who finishes can prove the Hodge decomposition for a compact Kähler manifold, set up a variation of Hodge structure and apply Griffiths transversality, write down the mixed Hodge structure on the cohomology of a smooth open or singular variety, and read the current literature on period maps, Hodge loci, and algebraic cycles without further preparation.

# Part I

# Foundations of Compact Kähler Geometry

The Hodge decomposition is a statement about three structures carried by one space, and this Part introduces them in the order in which each becomes available. A complex manifold carries a splitting of its complexified tangent bundle into holomorphic and antiholomorphic parts. The splitting is linear algebra, it propagates to differential forms as the decomposition by type, and it breaks the exterior derivative into $d = \partial + \bar{\partial}$. Cohomology taken with respect to $\bar{\partial}$ alone is then available, and the relationship between the resulting Dolbeault groups and ordinary de Rham cohomology is the question the rest of the Part answers.

No metric has entered yet, and without one the answer is negative. On a general compact complex manifold the Dolbeault groups do not assemble into the de Rham groups: the Frölicher spectral sequence relating them need not degenerate, and on the Iwasawa manifold it does not. What repairs this is a Hermitian metric whose fundamental form is closed. That single closedness demand propagates into commutator identities among $\partial$, $\bar{\partial}$, their formal adjoints, and the Lefschetz operators, and the identities collapse the two natural Laplacians onto each other. Harmonic forms for one operator are then harmonic for the other, and since one Laplacian computes de Rham cohomology while the other computes Dolbeault cohomology, the two theories share a single space of representatives.

The last chapter of the Part imports the analytic input, the existence and uniqueness of harmonic representatives on a compact Riemannian manifold, from *Riemannian Geometry* [9], specializes it to $\Delta_{\bar{\partial}}$, and reads off the decomposition together with its first consequences.

# 1

# *Complex Manifolds and Holomorphic Bundles*

Requiring an atlas to have holomorphic rather than smooth transition maps is a small change to a definition, and the splitting of the complexified tangent bundle into holomorphic and antiholomorphic parts is its first consequence. This chapter constructs the objects that splitting acts on.

The first section develops the local theory of several complex variables and then gives the splitting its intrinsic form, as an endomorphism $J$ of the tangent bundle with $J^2 = -\mathrm{id}$. Not every such endomorphism comes from a holomorphic atlas, and the Newlander-Nirenberg theorem identifies the ones that do by the vanishing of a single tensor; the easy direction is proved here, the hard direction in appendix A. The remaining sections build the global objects on which Hodge theory operates: the Dolbeault complex, holomorphic vector bundles, and divisors and line bundles. Three examples enter here and recur throughout the book, complex projective space $\mathbb{CP}^n$, smooth hypersurfaces in $\mathbb{CP}^n$, and elliptic curves; each instantiates the later theory at a different weight, and together they span the range of Hodge structures that opens in chapter 5.

## 1.1 Holomorphic Functions and Almost Complex Structures

A complex manifold can be approached from two directions. The atlas-based definition (a smooth manifold whose transition functions are holomorphic) is intrinsic and well-suited to global constructions, but it conceals the linear-algebraic content that makes Hodge theory work. The second approach, via almost complex structures, isolates that content directly: one places a fiberwise endomorphism $J$ of the tangent bundle squaring to $-\mathrm{id}$ and asks when this comes from an atlas of holomorphic charts. The Newlander-Nirenberg theorem, which closes this section, says that the obstruction is encoded in a single tensor.

The bridge between the two approaches runs through the Cauchy-Riemann equations, which characterize holomorphy among smooth maps $U \subseteq \mathbb{C}^n \to \mathbb{C}$. Both the single-variable theory and the local several-variable theory are developed in [25], whose chapter on higher dimensional complex analysis carries the polydisk Cauchy formula and the power-series expansion in the form used below; they are recalled here in the notation this book will keep, rather than reproved. What that chapter does not take up, and what is characteristic of several variables, is the Hartogs extension phenomenon.

**Definition 1.1.1: Holomorphic Function (Several Variables)**

Let $U \subseteq \mathbb{C}^n$ be open. A continuously differentiable function $f\colon U \to \mathbb{C}$ is *holomorphic* if it is holomorphic in each variable separately, that is, for each $j \in \{1, \dots, n\}$ and each $z_0 \in U$, the function

$$\zeta \mapsto f(z_0^{(1)}, \dots, z_0^{(j-1)}, \zeta, z_0^{(j+1)}, \dots, z_0^{(n)})$$

is holomorphic in a neighborhood of $\zeta = z_0^{(j)}$ in $\mathbb{C}$.

The continuity hypothesis is theoretically removable (Hartogs proved that separate holomorphy in each variable already implies joint continuity) but the proof is technical and the additional generality plays no role in this book. Throughout this book, "holomorphic" on an open subset of $\mathbb{C}^n$ means $C^1$ and separately holomorphic; equivalently (corollary 1.1.5), it means the Cauchy-Riemann equations $\partial f/\partial \bar{z}_j = 0$ hold for $j = 1, \dots, n$.

The local theory of holomorphic functions in several variables rests on the Cauchy formula, which extends from disks to polydisks straightforwardly.

**Definition 1.1.2: Polydisk and Distinguished Boundary**

Let $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $r = (r_1, \dots, r_n) \in \mathbb{R}^n_{>0}$. The associated *polydisk* is

$$\Delta(a; r) = \{z \in \mathbb{C}^n : |z_j - a_j| < r_j \text{ for all } j\}$$

The *distinguished boundary* is the product of circular boundaries:

$$\partial^* \Delta(a; r) = \{z \in \mathbb{C}^n : |z_j - a_j| = r_j \text{ for all } j\}$$

The full topological boundary $\partial\Delta$ is strictly larger than $\partial^*\Delta$; the distinguished boundary is an $n$-dimensional torus, while $\partial\Delta$ has real dimension $2n-1$. The Cauchy formula extends from a single circle to the distinguished boundary, not to the full topological boundary.

**Theorem 1.1.3: Cauchy Integral Formula on Polydisks**

Let $f$ be holomorphic on an open neighborhood of the closed polydisk $\overline{\Delta(a; r)}$. Then for every $z \in \Delta(a; r)$,

$$f(z) = \frac{1}{(2\pi i)^n} \int_{\partial^* \Delta(a;r)} \frac{f(\zeta_1, \dots, \zeta_n)}{(\zeta_1 - z_1) \cdots (\zeta_n - z_n)} \, d\zeta_1 \cdots d\zeta_n$$

***Proof*:**
The argument is iterated application of the single-variable Cauchy formula [25]. Fix $z \in \overline{\Delta(a;r)}$. The function $\zeta_1 \mapsto f(\zeta_1, z_2, \ldots, z_n)$ is holomorphic on a neighborhood of the closed disk $\overline{\Delta(a_1;r_1)}$ by separate holomorphy of $f$, so

$$f(z_1, z_2, \ldots, z_n) = \frac{1}{2\pi i}\int_{|\zeta_1 - a_1| = r_1} \frac{f(\zeta_1, z_2, \ldots, z_n)}{\zeta_1 - z_1}\, d\zeta_1$$

Apply the same formula to $\zeta_2 \mapsto f(\zeta_1, \zeta_2, z_3, \ldots, z_n)$, holding $\zeta_1$ and the remaining $z_k$ fixed:

$$f(\zeta_1, z_2, z_3, \ldots, z_n) = \frac{1}{2\pi i}\int_{|\zeta_2 - a_2| = r_2} \frac{f(\zeta_1, \zeta_2, z_3, \ldots, z_n)}{\zeta_2 - z_2}\, d\zeta_2$$

Substituting into the first formula and continuing inductively through $\zeta_3, \ldots, \zeta_n$ produces a chain of iterated integrals. Joint continuity of the integrand permits the iterated integral to be combined into a single integral over the product torus $\partial^* \Delta(a;r)$ via Fubini's theorem, completing the proof.

The Cauchy formula has the same consequences in several variables as in one. Differentiating under the integral sign produces smoothness and real-analyticity; the integral representation also yields Cauchy estimates on derivatives in terms of supremum norms. The two consequences most important for the sequel are collected below.

**Corollary 1.1.4: Properties of Holomorphic Functions in Several Variables**

Let $f$ be holomorphic on a connected open set $U \subseteq \mathbb{C}^n$.

1. *Real-analyticity.* The function $f$ is real-analytic on $U$, with power series converging on every polydisk contained in $U$.
2. *Identity theorem.* If $f$ vanishes on a non-empty open subset of $U$, then $f \equiv 0$ on $U$.
3. *Maximum principle.* If $|f|$ attains a local maximum at a point of $U$, then $f$ is constant on $U$.
4. *Open mapping.* If $f$ is non-constant, then $f(U)$ is open in $\mathbb{C}$.

***Proof*:**
Each property reduces to its single-variable counterpart established in [25].

1. *Real-analyticity.* Expand the kernel

$$\frac{1}{(\zeta_j - a_j) - (z_j - a_j)} = \sum_{k_j \geq 0} \frac{(z_j - a_j)^{k_j}}{(\zeta_j - a_j)^{k_j+1}}$$

for $|z_j - a_j| < |\zeta_j - a_j| = r_j$, multiply across all $j$, substitute into theorem 1.1.3, and integrate term by term (uniform convergence on the distinguished boundary justifies the exchange). The result is the Taylor expansion

$$f(z) = \sum_{k\in\mathbb{N}^n} c_k(z-a)^k$$

with explicit integral expressions for the coefficients $c_k$.

2. *Identity theorem.* Suppose $f$ vanishes on a non-empty open $V \subseteq U$, and set

$$W := \{x \in U : \text{every partial derivative of } f \text{ vanishes at } x\}$$

where the derivatives of order 0 are included, so $f|_W = 0$. Then $W$ is closed in $U$, being an intersection of zero sets of continuous functions, and it contains $V$, so it is non-empty. It is also open: by item (1), $f$ agrees near any $x \in W$ with its Taylor series at $x$, every coefficient of which is a derivative of $f$ at $x$ and therefore vanishes; hence $f \equiv 0$ on a polydisk around $x$, and every derivative of $f$ vanishes throughout that polydisk. Connectedness of $U$ forces $W = U$, hence $f \equiv 0$ on $U$.

The set that is open here is $W$, not the zero set of $f$: a zero set is in general a hypersurface, as $f(z) = z_1$ on $\mathbb{C}^2$ shows.

3. *Maximum principle.* If $|f|$ has a local maximum at $p \in U$, restrict $f$ to the complex line $\zeta \mapsto p + \zeta v$ for an arbitrary direction $v \in \mathbb{C}^n$. The restriction has a local maximum of $|f|$ at $\zeta = 0$, so by the one-variable maximum principle it is constant on a small disk. Since $v$ was arbitrary, $f$ is constant on a polydisk neighborhood of $p$. The identity theorem then forces $f$ to be constant on $U$.

4. *Open mapping.* If $f$ is non-constant, choose a complex line through any $p \in U$ along which $f$ remains non-constant (such a line exists by item (2): otherwise $f$ would be constant on every line through $p$, hence locally constant, contradicting non-constancy). The one-variable open mapping theorem on the restriction of $f$ to that line places $f(p)$ in an open subset of $f(U)$. Since $p$ was arbitrary, $f(U)$ is open, completing the proof.

The Cauchy-Riemann equations make the algebraic content of holomorphy explicit. Decompose the complex coordinates as $z_j = x_j + iy_j$, so $\mathbb{C}^n$ is identified with $\mathbb{R}^{2n}$ via $(z_1, \ldots, z_n) \leftrightarrow (x_1, y_1, \ldots, x_n, y_n)$. Define the *Wirtinger operators*

$$\frac{\partial}{\partial z_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right), \qquad \frac{\partial}{\partial \bar{z}_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$$

acting on $C^1$ functions $U \subseteq \mathbb{C}^n \to \mathbb{C}$. These satisfy the duality relations $\partial z_k/\partial z_j = \delta_{jk}$, $\partial \bar{z}_k/\partial z_j = 0$, and conjugate counterparts (exercise 1.1.1 (1)).

**Corollary 1.1.5: Cauchy-Riemann Characterization (Several Variables)**

A function $f \in C^1(U, \mathbb{C})$ is holomorphic if and only if $\partial f/\partial \bar{z}_j = 0$ for $j = 1, \ldots, n$.

***Proof*:**
Holomorphy in the $j$-th variable separately is equivalent (by the single-variable Cauchy-Riemann equation [25]) to $\partial f/\partial \bar{z}_j = 0$. The joint $C^1$ regularity hypothesis combines the separate conditions into the single statement, completing the proof.

The contrast between several and single variables is sharpest in the Hartogs extension phenomenon, which has no one-variable analogue.

**1.1.6 Note: Hartogs Extension** When $n \geq 2$, holomorphic functions extend across compact "obstructions". If $K \subset U \subseteq \mathbb{C}^n$ is compact and $U \setminus K$ is connected, every holomorphic $f: U \setminus K \to \mathbb{C}$ extends uniquely to a holomorphic function on all of $U$. This fails dramatically in one variable: $1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$ and admits no extension. The phenomenon manifests in many ways (absence of isolated singularities, automatic compactness of zero sets) and is one reason several-variable complex geometry has a more rigid character than its one-variable counterpart. A full proof uses the $\bar{\partial}$-equation; see [20, Theorem 2.3.2] and [25].

With the local theory in place, the global object can be defined.

**Definition 1.1.7: Complex Manifold**

A *complex manifold* of complex dimension $n$ is a Hausdorff, second-countable topological space $X$ together with an atlas of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ where each $\varphi_\alpha: U_\alpha \to V_\alpha \subseteq \mathbb{C}^n$ is a homeomorphism onto an open subset, and the transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are holomorphic for every pair $\alpha, \beta$. A continuous map $f: X \to Y$ between complex manifolds is *holomorphic* if every local expression $\varphi_\beta \circ f \circ \varphi_\alpha^{-1}$ is holomorphic. A bijective holomorphic map with holomorphic inverse is a *biholomorphism*.

A complex manifold of complex dimension $n$ is a smooth manifold of real dimension $2n$ in a canonical way: each holomorphic chart, viewed under $\mathbb{C}^n \cong \mathbb{R}^{2n}$, is a smooth chart, and the transition maps are smooth (holomorphic transition maps are real-analytic by corollary 1.1.4). The orientation induced is canonical: the standard orientation on $\mathbb{C}^n$ is preserved by every biholomorphism, so a complex manifold is canonically oriented as a smooth manifold (exercise 1.1.1 (3)).

**Example 1.1.8: Complex Projective Space $\mathbb{CP}^n$**

Let $\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^\times$, with $[z_0 : \cdots : z_n]$ denoting the equivalence class of $(z_0, \ldots, z_n)$ under scalar multiplication by $\mathbb{C}^\times$. The standard atlas consists of $n+1$ charts

$$U_i = \{[z_0 : \cdots : z_n] \in \mathbb{CP}^n : z_i \neq 0\}, \quad \varphi_i([z]) = \left(\frac{z_0}{z_i}, \ldots, \widehat{\frac{z_i}{z_i}}, \ldots, \frac{z_n}{z_i}\right) \in \mathbb{C}^n$$

where the $i$-th coordinate is omitted from the image. Each $U_i$ is biholomorphic to $\mathbb{C}^n$, and the transition map $\varphi_j \circ \varphi_i^{-1}$ on the locus where its components are defined is the rational map sending $(w_0, \ldots, \widehat{w_i}, \ldots, w_n) \mapsto (w_0/w_j, \ldots, 1/w_j, \ldots, w_n/w_j)$ with $w_j$ taking the role of the formerly omitted coordinate. The transition maps are holomorphic, so $\mathbb{CP}^n$ is a compact complex manifold of dimension $n$.

**Example 1.1.9: Smooth Hypersurfaces in $\mathbb{CP}^n$**

Let $f \in \mathbb{C}[z_0, \ldots, z_n]$ be a homogeneous polynomial of degree $d \geq 1$ such that $\partial f/\partial z_0, \ldots, \partial f/\partial z_n$

have no common zero on $\mathbb{C}^{n+1} \setminus \{0\}$. The vanishing locus

$$V(f) = \{[z] \in \mathbb{CP}^n : f(z) = 0\}$$

is a compact complex manifold of dimension $n-1$. The complex structure is inherited from the implicit function theorem: in any affine chart $U_i$ on $\mathbb{CP}^n$, the dehomogenization of $f$ has a non-vanishing partial derivative on $V(f) \cap U_i$, and the holomorphic implicit function theorem ([25, the holomorphic implicit function theorem]) produces a holomorphic chart on the vanishing locus. Smooth hypersurfaces are the Hodge-theoretic engine of chapter 7, where their cohomology is computed via Jacobian rings.

**Example 1.1.10: Complex Tori**

Let $\Lambda \subset \mathbb{C}^n$ be a discrete subgroup of rank $2n$, that is, a lattice. The quotient $\mathbb{C}^n/\Lambda$ inherits a complex manifold structure from the translation action of $\Lambda$, since translations by lattice elements are biholomorphisms of $\mathbb{C}^n$. The resulting compact complex manifold, the *complex torus* associated to $\Lambda$, is diffeomorphic to the real torus $(\mathbb{R}/\mathbb{Z})^{2n}$. The case $n = 1$ recovers the elliptic curves of [5]; for $n \geq 2$, only a special class of complex tori (the *abelian varieties*) admits an embedding into projective space (a fact controlled by the polarization data of chapter 5).

The maximum principle takes a strong global form when the complex manifold is compact.

**Proposition 1.1.11: Maximum Principle on Compact Complex Manifolds**

Let $X$ be a connected, compact complex manifold. Every holomorphic function $f\colon X \to \mathbb{C}$ is constant.

***Proof*:**
Compactness ensures $|f|$ attains its supremum at some point $p \in X$. In a holomorphic chart around $p$, the function $f$ becomes a holomorphic function on an open subset of $\mathbb{C}^n$ with a local maximum of $|f|$ at the image of $p$; by corollary 1.1.4(3), $f$ is constant on the chart. The set on which $f$ equals its supremum value is therefore open, and it is closed by continuity. Connectedness of $X$ forces it to be all of $X$, completing the proof.

In particular, holomorphic functions on $\mathbb{CP}^n$ and on every compact complex submanifold of $\mathbb{C}^N$ are constant (exercise 1.1.1 (6)). Non-trivial holomorphic "data" on a compact complex manifold therefore lives not in functions but in sections of holomorphic vector bundles, the central object of section 1.3.

The atlas-based definition of a complex manifold conceals the linear-algebraic content that drives Hodge theory: the local model $\mathbb{C}^n \cong \mathbb{R}^{2n}$ carries a canonical $\mathbb{R}$-linear endomorphism $J$ of its tangent space (multiplication by $i$), and a complex manifold structure equips $TX$ fiberwise with this datum.

**Definition 1.1.12: Almost Complex Structure**

An *almost complex structure* on a smooth manifold $M$ is a smooth bundle endomorphism $J\colon TM \to TM$ satisfying $J^2 = -\mathrm{id}_{TM}$. The pair $(M, J)$ is called an *almost complex manifold.*

The condition $J^2 = -\mathrm{id}$ has immediate topological consequences. At each point $p \in M$, the operator

$J_p$ makes $T_pM$ into a complex vector space (via $i \cdot v := J_p v$); in particular, $T_pM$ has even real dimension, so $M$ has even real dimension. The complex structure on each tangent space induces a canonical orientation (exercise 1.1.1 (3)), so an almost complex manifold is canonically oriented. Higher-order topological constraints (i.e. vanishing of certain integral characteristic classes) are considered as the dimension grows, but the parity and orientation constraints suffice for the present discussion.

The decomposition that drives the rest of the book emerges after complexifying the tangent bundle.

**Proposition 1.1.13: The $T^{1,0} \oplus T^{0,1}$ Decomposition**

Let $(M, J)$ be an almost complex manifold. Extend $J$ $\mathbb{C}$-linearly to the complexified tangent bundle $TM_{\mathbb{C}} := TM \otimes_{\mathbb{R}} \mathbb{C}$. Then $J$ has eigenvalues $\pm i$, and $TM_{\mathbb{C}}$ splits as a direct sum of smooth subbundles

$$TM_{\mathbb{C}} = T^{1,0}M \oplus T^{0,1}M$$

where $T^{1,0}M$ is the $(+i)$-eigenbundle and $T^{0,1}M$ the $(-i)$-eigenbundle. The two summands are interchanged by complex conjugation $X \otimes \alpha \mapsto X \otimes \bar{\alpha}$ on $TM_{\mathbb{C}}$: $\overline{T^{1,0}M} = T^{0,1}M$. The projections $\pi^{1,0} \colon TM_{\mathbb{C}} \to T^{1,0}M$ and $\pi^{0,1} \colon TM_{\mathbb{C}} \to T^{0,1}M$ are given on every $Z \in TM_{\mathbb{C}}$, in particular on a real vector $X \in TM$, by

$$\pi^{1,0}(Z) = \frac{1}{2}(Z - iJZ), \qquad \pi^{0,1}(Z) = \frac{1}{2}(Z + iJZ)$$

***Proof*:**
Since $J^2 = -\mathrm{id}$ on $TM$ and the complexification is $\mathbb{C}$-linear, $J^2 = -\mathrm{id}$ on $TM_{\mathbb{C}}$. Hence $J$ satisfies the polynomial $T^2 + 1 = (T - i)(T + i)$, which has distinct roots, so $J$ is diagonalizable with eigenvalues $\pm i$. The projection formulas are verified directly: for $X \in TM$,

$$J\left(\frac{1}{2}(X - iJX)\right) = \frac{1}{2}(JX - iJ^2X) = \frac{1}{2}(JX + iX) = i \cdot \frac{1}{2}(X - iJX)$$

so $\pi^{1,0}(X) \in T^{1,0}M$, and similarly for $\pi^{0,1}$. Smoothness of the projections follows from smoothness of $J$. Complex conjugation maps the $+i$-eigenspace to the $-i$-eigenspace because $\bar{i} = -i$, completing the proof.

The model case is $\mathbb{C}^n$ with its standard almost complex structure.

**Example 1.1.14: The Standard Almost Complex Structure on $\mathbb{C}^n$**
On $\mathbb{C}^n$ with real coordinates $(x_1, y_1, \ldots, x_n, y_n)$ (so $z_j = x_j + iy_j$), the standard almost complex structure $J_0$ acts as

$$J_0\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \qquad J_0\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j}$$

The corresponding $\pm i$-eigenbundles are spanned by the Wirtinger operators:

$$T^{1,0}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\left\{\frac{\partial}{\partial z_j}\right\}_{j=1}^{n}, \qquad T^{0,1}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\left\{\frac{\partial}{\partial \bar{z}_j}\right\}_{j=1}^{n}$$

Direct calculation confirms $J_0(\partial/\partial z_j) = i\,\partial/\partial z_j$ and $J_0(\partial/\partial \bar{z}_j) = -i\,\partial/\partial \bar{z}_j$. The Cauchy-Riemann condition $\partial f/\partial \bar{z}_j = 0$ for the holomorphy of $f$ is the statement that the differential $df$, viewed in this basis, has no $T^{0,1}$ component, a coordinate-free reformulation that survives the passage to general complex manifolds.

The standard $J_0$ globalizes: every complex manifold inherits a canonical almost complex structure $J$ from its holomorphic charts, since holomorphic transition maps commute with $J_0$ (exercise 1.1.1 (2)). The reverse direction, when an almost complex structure comes from a complex manifold structure, is the content of the Newlander-Nirenberg theorem.

The obstruction is encoded in a single tensor:

**Definition 1.1.15: Nijenhuis Tensor**

Let $(M, J)$ be an almost complex manifold. The *Nijenhuis tensor* is the $\mathbb{R}$-bilinear map $N_J\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ defined on smooth real vector fields by

$$N_J(X, Y) = [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY]$$

where $[\cdot, \cdot]$ is the Lie bracket of vector fields.

A direct verification shows $N_J(fX, gY) = fg \cdot N_J(X, Y)$ for smooth functions $f, g$, so $N_J$ is in fact a tensor field of type $(1, 2)$ on $M$ (exercise 1.1.1 (4)). The tensor extends $\mathbb{C}$-linearly to $TM_{\mathbb{C}}$, and its vanishing has a clean geometric interpretation in terms of the $T^{1,0} \oplus T^{0,1}$ decomposition.

**Proposition 1.1.16: Integrability via Involutivity of $T^{1,0}$**

The Nijenhuis tensor $N_J$ vanishes if and only if $T^{1,0}M$ is closed under the Lie bracket, that is, $[T^{1,0}M, T^{1,0}M] \subseteq T^{1,0}M$. Equivalently, $N_J = 0$ if and only if $[T^{0,1}M, T^{0,1}M] \subseteq T^{0,1}M$.

***Proof*:**
Extend $J$, $N_J$, and the Lie bracket $\mathbb{C}$-linearly to $TM_{\mathbb{C}}$. The Nijenhuis formula

$$N_J(X, Y) = [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY]$$

holds for $X, Y \in TM_{\mathbb{C}}$ by $\mathbb{C}$-linear extension. Decompose any pair of vector fields according to the $T^{1,0} \oplus T^{0,1}$ splitting and check the three contributions separately.

*Step 1: cross terms automatically vanish.* If $X \in T^{1,0}M$ and $Y \in T^{0,1}M$, then $JX = iX$ and $JY = -iY$, so

$$\begin{aligned} N_J(X, Y) &= [X, Y] + J[iX, Y] + J[X, -iY] - [iX, -iY] \\ &= [X, Y] + iJ[X, Y] - iJ[X, Y] - (-i)(i)[X, Y] \\ &= [X, Y] - [X, Y] = 0 \end{aligned}$$

The Nijenhuis tensor therefore vanishes identically on mixed pairs from $T^{1,0}$ and $T^{0,1}$.

*Step 2: the $(1, 0)$-$(1, 0)$ contribution detects involutivity.* If $X, Y \in T^{1,0}M$, then $JX = iX$ and

$JY = iY$, so

$$\begin{aligned} N_J(X,Y) &= [X,Y] + J[iX,Y] + J[X,iY] - [iX,iY] \\ &= [X,Y] + iJ[X,Y] + iJ[X,Y] - i^2[X,Y] \\ &= 2[X,Y] + 2iJ[X,Y] \end{aligned}$$

The right side equals $4\,\pi^{0,1}([X,Y])$ by the projection formula $\pi^{0,1}(Z) = \frac{1}{2}(Z + iJZ)$. Hence $N_J(X,Y) = 0$ for all $X, Y \in T^{1,0}M$ exactly when $\pi^{0,1}([X,Y]) = 0$, that is, exactly when $[X,Y] \in T^{1,0}M$.

*Step 3: the* $(0,1)$-$(0,1)$ *case is conjugate.* A symmetric calculation shows that $N_J$ on pairs from $T^{0,1}M$ equals $4\,\pi^{1,0}([X,Y])$. Since $T^{0,1}M = \overline{T^{1,0}M}$ and the Lie bracket is real, the $(0,1)$-involutivity condition is equivalent to the $(1,0)$-involutivity condition.

The total Nijenhuis tensor is the sum of these three contributions, so $N_J = 0$ if and only if $T^{1,0}M$ (equivalently $T^{0,1}M$) is involutive, as we sought to show.

The involutivity formulation makes one direction of the Newlander-Nirenberg theorem an immediate calculation.

**Corollary 1.1.17: Complex Manifolds Have Vanishing Nijenhuis Tensor**

If $(M,J)$ is the underlying almost complex manifold of a complex manifold structure on $M$, then $N_J = 0$.

***Proof*:**
In any holomorphic chart with coordinates $z_1, \ldots, z_n$, the $+i$-eigenbundle $T^{1,0}M$ is spanned by $\partial/\partial z_1, \ldots, \partial/\partial z_n$ (example 1.1.14). The Lie bracket of any two such coordinate vector fields vanishes: $[\partial/\partial z_j, \partial/\partial z_k] = 0$ since partial derivatives commute. Hence $T^{1,0}M$ is closed under the Lie bracket on the chart, and proposition 1.1.16 gives $N_J = 0$ on the chart. The Nijenhuis tensor is a tensor field, so vanishing on a covering atlas implies vanishing globally, completing the proof.

The reverse implication is the substantive content of the theorem.

**Theorem 1.1.18: Newlander-Nirenberg**

Let $(M,J)$ be a smooth almost complex manifold. The almost complex structure $J$ comes from a complex manifold structure on $M$ (in which case the complex structure is unique) if and only if $N_J = 0$.

The forward implication is corollary 1.1.17; the converse, which produces a holomorphic atlas from the algebraic vanishing condition, requires elliptic-PDE machinery. Appendix A isolates the three obstacles precisely and cites [20] and [35] for the iteration at its heart. An almost complex structure with $N_J = 0$ is called *integrable*; the theorem identifies integrable almost complex structures with complex manifold structures.

The dimension-1 case is degenerate but worth noting: every almost complex structure on a smooth oriented surface is automatically integrable, since $T^{1,0}M$ has complex rank 1 and rank-1 distributions are trivially involutive (exercise 1.1.1 (5)). Almost complex surfaces and Riemann surfaces therefore coincide. In real dimension $\geq 4$, the integrability condition is non-trivial: there exist almost complex manifolds with non-vanishing Nijenhuis tensor, and the question of whether a smooth manifold admitting an almost complex structure also admits an integrable one is delicate. The six-sphere $S^6$ is the most famous open question. It carries an almost complex structure inherited from the octonions (whose Nijenhuis tensor does not vanish), but whether $S^6$ admits any complex structure at all remains unresolved ([21, §2.6]). The Hopf surface, a quotient of $\mathbb{C}^2 \setminus \{0\}$ by $\mathbb{Z}$ acting via dilation, will appear in chapter 3 as the canonical example of a complex manifold that admits no Kähler metric; it is a complex manifold whose underlying smooth manifold is $S^1 \times S^3$, and the failure to admit a Kähler metric is a phenomenon at the level of metric data, not at the level of integrability.

## Exercise 1.1.1

1. For the Wirtinger operators $\partial/\partial z_j$ and $\partial/\partial \bar{z}_j$ on $C^1$ functions $U \subseteq \mathbb{C}^n \to \mathbb{C}$, verify the duality relations
$$\frac{\partial z_k}{\partial z_j} = \delta_{jk}, \qquad \frac{\partial \bar{z}_k}{\partial z_j} = 0, \qquad \frac{\partial z_k}{\partial \bar{z}_j} = 0, \qquad \frac{\partial \bar{z}_k}{\partial \bar{z}_j} = \delta_{jk}$$
Conclude that $\{dz_j, d\bar{z}_j\}$ is dual to $\{\partial/\partial z_j, \partial/\partial \bar{z}_j\}$ as bases of the complexified cotangent and tangent spaces respectively.

2. Let $\varphi\colon U \to V$ be a holomorphic map between open subsets of $\mathbb{C}^n$. Show that the differential $d\varphi\colon T\mathbb{C}^n \to T\mathbb{C}^n$ commutes with the standard almost complex structure $J_0$:
$$d\varphi \circ J_0 = J_0 \circ d\varphi$$
Deduce that the standard almost complex structures on the local charts of a complex manifold $X$ patch consistently into a globally defined almost complex structure on $X$.

3. Let $V$ be a complex vector space of complex dimension $n$, viewed as a real vector space of dimension $2n$. Show that any $\mathbb{C}$-basis $v_1, \ldots, v_n$ of $V$ produces an $\mathbb{R}$-basis $v_1, Jv_1, \ldots, v_n, Jv_n$ whose orientation does not depend on the choice of $\mathbb{C}$-basis. Deduce that every almost complex manifold is canonically oriented.

4. Verify that the Nijenhuis tensor $N_J(X,Y) = [X,Y] + J[JX,Y] + J[X,JY] - [JX,JY]$ is $C^\infty(M)$-bilinear: $N_J(fX, gY) = fg \cdot N_J(X,Y)$ for smooth functions $f, g \in C^\infty(M)$. Conclude that $N_J$ is a $(1,2)$-tensor field on $M$. (Hint: $[fX, gY] = fg[X,Y] + fX(g)Y - gY(f)X$; track the four non-tensorial first-derivative terms produced by the four bracket pairs and verify they cancel.)

5. Let $(M, J)$ be a smooth almost complex manifold of complex dimension 1 (real dimension 2). Show that $N_J \equiv 0$ identically. Deduce that every almost complex structure on a smooth oriented surface is integrable, so that almost complex surfaces and Riemann surfaces coincide as categories.

6. Let $X \subseteq \mathbb{C}^N$ be a compact connected complex submanifold. Show that $X$ is a single point. (Hint: each coordinate function $z_i\colon X \to \mathbb{C}$ is holomorphic.)

## 1.2 The Dolbeault Complex

The complex structure on $TM$ from section 1.1 lifts to differential forms: every smooth complex-valued $k$-form on a complex manifold decomposes along the holomorphic-antiholomorphic bigrading inherited from the tangent bundle. The exterior derivative respects this bigrading and splits into two operators, $\partial$ (raising holomorphic degree) and $\bar{\partial}$ (raising antiholomorphic degree). The $\bar{\partial}$-complex is the Dolbeault complex; its cohomology, computed in chapter 2, is the analytic engine that drives the Hodge decomposition.

This section establishes the algebraic framework. The cotangent bundle decomposition mirrors the tangent decomposition from proposition 1.1.13; differential forms acquire a bidegree; the exterior derivative splits on a complex manifold; and the Dolbeault complex assembles from $\bar{\partial}$. Cohomological content waits for chapter 2; the present section sets up the operators.

**Proposition 1.2.1: Cotangent Decomposition**

Let $(M, J)$ be an almost complex manifold of complex dimension $n$. The complexified cotangent bundle decomposes as

$$T^*M_{\mathbb{C}} = T^{*1,0}M \oplus T^{*0,1}M$$

where $T^{*1,0}M$ is the dual of $T^{1,0}M$ and $T^{*0,1}M$ is the dual of $T^{0,1}M$ inside $T^*M_{\mathbb{C}}$. Complex conjugation on $T^*M_{\mathbb{C}}$ exchanges the two summands: $\overline{T^{*1,0}M} = T^{*0,1}M$.

***Proof*:**
Take complex-linear duals of the decomposition $TM_{\mathbb{C}} = T^{1,0}M \oplus T^{0,1}M$ from proposition 1.1.13. The annihilator of $T^{0,1}M$ in $T^*M_{\mathbb{C}}$ is dual to $T^{1,0}M$ and is by definition $T^{*1,0}M$; symmetrically for $T^{*0,1}M$. Smoothness of the projections follows from smoothness of $J$ on $TM$. The conjugation statement follows from $\overline{T^{1,0}M} = T^{0,1}M$ and the naturality of complex conjugation under duality, completing the proof.

On $\mathbb{C}^n$ with coordinates $z_j = x_j + iy_j$, the bases dual to the Wirtinger operators $\{\partial/\partial z_j, \partial/\partial \bar{z}_j\}$ of example 1.1.14 are precisely the differentials

$$dz_j = dx_j + i\,dy_j, \qquad d\bar{z}_j = dx_j - i\,dy_j$$

(exercise 1.1.1 (1)). Hence $T^{*1,0}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\{dz_j\}$ and $T^{*0,1}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\{d\bar{z}_j\}$, the cotangent analogues of the tangent decomposition. The wedge powers of this decomposition give the bigrading on differential forms.

**Definition 1.2.2: Forms of Type $(p, q)$**

Let $(M, J)$ be a complex manifold. The bundle of $(p, q)$-*forms* is

$$\Lambda^{p,q}T^*M := \Lambda^p T^{*1,0}M \otimes_{\mathbb{C}} \Lambda^q T^{*0,1}M$$

A smooth section of $\Lambda^{p,q}T^*M$ is a $(p, q)$-*form*, or a *form of type* $(p, q)$. The space of all such smooth sections is denoted $\mathcal{A}^{p,q}(M)$.

**Proposition 1.2.3: Bidegree Decomposition of $k$-Forms**

Let $(M, J)$ be a complex manifold of complex dimension $n$. The complexified bundle of $k$-forms decomposes as
$$\Lambda^k T^* M_{\mathbb{C}} = \bigoplus_{p+q=k} \Lambda^{p,q} T^* M$$
with summands non-zero precisely when $0 \leq p \leq n$ and $0 \leq q \leq n$. Consequently, the space of complex-valued $k$-forms splits as
$$\mathcal{A}^k(M; \mathbb{C}) = \bigoplus_{p+q=k} \mathcal{A}^{p,q}(M)$$

***Proof*:**
The wedge product is functorial: a direct-sum decomposition $V = V' \oplus V''$ of a finite-dimensional complex vector space induces
$$\Lambda^k V = \bigoplus_{p+q=k} \Lambda^p V' \otimes \Lambda^q V''$$
Apply this fiberwise to $T^* M_{\mathbb{C}} = T^{*1,0} M \oplus T^{*0,1} M$ from proposition 1.2.1, obtaining the bidegree decomposition of $\Lambda^k T^* M_{\mathbb{C}}$ at each point. The decomposition is smooth because the projections $\pi^{1,0}, \pi^{0,1}$ are smooth bundle maps. Taking smooth sections gives the corresponding decomposition of $\mathcal{A}^k(M; \mathbb{C})$, completing the proof.

In holomorphic local coordinates $z_1, \ldots, z_n$, a $(p, q)$-form admits the expression
$$\alpha = \sum_{|I|=p,\ |J|=q} f_{IJ}(z, \bar{z})\, dz_I \wedge d\bar{z}_J$$
where $I = (i_1 < \cdots < i_p)$ and $J = (j_1 < \cdots < j_q)$ range over strictly increasing multi-indices, $dz_I := dz_{i_1} \wedge \cdots \wedge dz_{i_p}$, $d\bar{z}_J := d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$, and $f_{IJ}$ is a smooth complex-valued function on the chart. The total degree is $|I| + |J| = p + q = k$.

**Example 1.2.4: $(p, q)$-Forms on $\mathbb{C}^n$**

A general 1-form on $\mathbb{C}^n$ is a smooth combination
$$\alpha = \sum_j f_j\, dz_j + \sum_j g_j\, d\bar{z}_j$$
with $\sum f_j\, dz_j \in \mathcal{A}^{1,0}(\mathbb{C}^n)$ and $\sum g_j\, d\bar{z}_j \in \mathcal{A}^{0,1}(\mathbb{C}^n)$. A general 2-form on $\mathbb{C}^2$ (complex dimension $n = 2$) splits into three bidegree components,
$$\begin{aligned}
\mathcal{A}^{2,0}(\mathbb{C}^2) : &\quad f\, dz_1 \wedge dz_2 \\
\mathcal{A}^{1,1}(\mathbb{C}^2) : &\quad f_{1\bar{1}} dz_1 \wedge d\bar{z}_1 + f_{1\bar{2}} dz_1 \wedge d\bar{z}_2 + f_{2\bar{1}} dz_2 \wedge d\bar{z}_1 + f_{2\bar{2}} dz_2 \wedge d\bar{z}_2 \\
\mathcal{A}^{0,2}(\mathbb{C}^2) : &\quad g\, d\bar{z}_1 \wedge d\bar{z}_2
\end{aligned}$$
with smooth functions $f, f_{j\bar{k}}, g$ on $\mathbb{C}^2$. The $(1, 1)$-component carries four independent coefficient functions; the $(2, 0)$ and $(0, 2)$ components carry one each. This pattern, $\dim_{\mathbb{C}} \Lambda^{p,q} = \binom{n}{p}\binom{n}{q}$, recovers $\dim_{\mathbb{C}} \Lambda^k = \binom{2n}{k}$ after summing over $p + q = k$ (exercise 1.2.1 (3)).

The exterior derivative $d$ acts on the complexified de Rham complex $\mathcal{A}^\bullet(M;\mathbb{C})$, and the bigrading refinement promotes this single operator to a pair $\partial, \bar\partial$. The construction starts on functions and extends to forms by the Leibniz rule.

For a smooth function $f\colon U \to \mathbb{C}$ on a holomorphic chart, the ordinary differential $df$ already lives in $\mathcal{A}^1(U;\mathbb{C}) = \mathcal{A}^{1,0}(U) \oplus \mathcal{A}^{0,1}(U)$. Splitting $df$ along the cotangent decomposition gives

$$df = \sum_j \frac{\partial f}{\partial z_j} dz_j + \sum_j \frac{\partial f}{\partial \bar z_j} d\bar z_j$$

where the Wirtinger operators on the right are those of example 1.1.14. The $(1,0)$-part is denoted $\partial f$ and the $(0,1)$-part $\bar\partial f$, so $df = \partial f + \bar\partial f$ on functions by definition. The notation extends to general $(p,q)$-forms through the antiderivation rule.

**Definition 1.2.5: The Operators $\partial$ and $\bar\partial$**

Let $M$ be a complex manifold and $\alpha = \sum_{|I|=p,|J|=q} f_{IJ}\, dz_I \wedge d\bar z_J$ a $(p,q)$-form expressed in holomorphic local coordinates. The *Dolbeault operators* $\partial$ and $\bar\partial$ act by differentiating coefficients separately:

$$\partial\alpha := \sum_{|I|=p,|J|=q} (\partial f_{IJ}) \wedge dz_I \wedge d\bar z_J$$
$$\bar\partial\alpha := \sum_{|I|=p,|J|=q} (\bar\partial f_{IJ}) \wedge dz_I \wedge d\bar z_J$$

Then $\partial : \mathcal{A}^{p,q}(M) \to \mathcal{A}^{p+1,q}(M)$ and $\bar\partial : \mathcal{A}^{p,q}(M) \to \mathcal{A}^{p,q+1}(M)$.

The chart-dependence is only apparent: the operators are intrinsic, as the next theorem shows by relating them to the global operator $d$.

**Theorem 1.2.6: Decomposition of the Exterior Derivative**

On a complex manifold $M$, the exterior derivative on complex-valued differential forms decomposes as

$$d = \partial + \bar\partial$$

on every $\mathcal{A}^{p,q}(M)$. In particular, the operators $\partial$ and $\bar\partial$ are independent of the choice of holomorphic chart and extend to globally defined first-order differential operators on $M$.

***Proof*:**
Verify the identity on a holomorphic chart with coordinates $z_1, \ldots, z_n$; intrinsic-ness then follows because $d$ is intrinsic and the decomposition is determined by the bigrading.

On a function $f$, the splitting $df = \partial f + \bar\partial f$ holds by definition of the Wirtinger operators. For a general $(p,q)$-form $\alpha = \sum f_{IJ}\, dz_I \wedge d\bar z_J$, the Leibniz rule and $d^2 = 0$ give

$$d(f_{IJ}\, dz_I \wedge d\bar z_J) = (df_{IJ}) \wedge dz_I \wedge d\bar z_J + f_{IJ}\, d(dz_I \wedge d\bar z_J)$$

The differentials $dz_i$ and $d\bar z_j$ are exact, so $d(dz_I \wedge d\bar z_J) = 0$ by the antiderivation rule applied iteratively. Hence

$$d\alpha = \sum (df_{IJ}) \wedge dz_I \wedge d\bar{z}_J = \sum (\partial f_{IJ} + \bar{\partial} f_{IJ}) \wedge dz_I \wedge d\bar{z}_J = \partial\alpha + \bar{\partial}\alpha$$

The first summand has type $(p+1, q)$ and the second has type $(p, q+1)$, matching the bidegrees promised in definition 1.2.5. The identity holds on every holomorphic chart, so it holds globally on $M$, completing the proof.

**1.2.7 Note: Non-Integrable Almost Complex Manifolds** On a general almost complex manifold $(M, J)$ where $J$ is not integrable, holomorphic coordinates do not exist (theorem 1.1.18), and the formula $d = \partial + \bar{\partial}$ fails. By bidegree, the exterior derivative on a $(p, q)$-form lands in four components,

$$d : \mathcal{A}^{p,q}(M) \to \mathcal{A}^{p+2,q-1}(M) \oplus \mathcal{A}^{p+1,q}(M) \oplus \mathcal{A}^{p,q+1}(M) \oplus \mathcal{A}^{p-1,q+2}(M)$$

The middle two components define $\partial$ and $\bar{\partial}$ in the general almost complex setting (as projections of $d$), and the outer two components, of bidegree $(2, -1)$ and $(-1, 2)$, are tensorial expressions in the Nijenhuis tensor $N_J$. They vanish identically if and only if $N_J = 0$, that is, if and only if $J$ is integrable (proposition 1.1.16). Throughout the rest of this book, $M$ is a complex manifold, so the two outer components are zero and $d = \partial + \bar{\partial}$. See [21, §2.6] for the full general formula.

The compatibility of $d^2 = 0$ with the bidegree decomposition produces three identities for $\partial$ and $\bar{\partial}$.

**Proposition 1.2.8: Dolbeault Identities**

On a complex manifold $M$,

1. $\partial^2 = 0$
2. $\bar{\partial}^2 = 0$
3. $\partial\bar{\partial} + \bar{\partial}\partial = 0$

***Proof*:**
By theorem 1.2.6, $d = \partial + \bar{\partial}$ on a complex manifold. Squaring and using $d^2 = 0$,

$$0 = d^2 = (\partial + \bar{\partial})^2 = \partial^2 + (\partial\bar{\partial} + \bar{\partial}\partial) + \bar{\partial}^2$$

The three summands have bidegrees $(2, 0)$, $(1, 1)$, and $(0, 2)$ respectively, mapping $\mathcal{A}^{p,q}$ into three distinct bidegree summands of $\mathcal{A}^{p+q+2}$. The total vanishes only when each component vanishes individually, yielding the three identities, completing the proof.

The identity $\bar{\partial}^2 = 0$ promotes the bigraded forms into a chain complex.

**Definition 1.2.9: Dolbeault Complex**

Let $M$ be a complex manifold of complex dimension $n$. For each fixed integer $p$ with $0 \le p \le n$, the *Dolbeault complex* in holomorphic degree $p$ is the cochain complex

$$0 \to \mathcal{A}^{p,0}(M) \xrightarrow{\bar\partial} \mathcal{A}^{p,1}(M) \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}^{p,n}(M) \to 0$$

with differential $\bar\partial$. The complex is well-defined because $\bar\partial^2 = 0$ (proposition 1.2.8).

**Definition 1.2.10: Dolbeault Cohomology**

The *Dolbeault cohomology* of $M$ in bidegree $(p, q)$ is

$$H^{p,q}_{\bar\partial}(M) := \frac{\ker\big(\bar\partial : \mathcal{A}^{p,q}(M) \to \mathcal{A}^{p,q+1}(M)\big)}{\operatorname{im}\big(\bar\partial : \mathcal{A}^{p,q-1}(M) \to \mathcal{A}^{p,q}(M)\big)}$$

with the convention that $\mathcal{A}^{p,-1}(M) = 0$.

The Dolbeault cohomology is the bidegree-refined invariant attached to a complex manifold; it sees both the holomorphic and the antiholomorphic structure. Chapter 2 computes $H^{p,q}_{\bar\partial}(M)$ via Dolbeault's theorem, identifying it with the sheaf cohomology of the holomorphic $p$-forms (defined below); on a compact Kähler manifold (chapter 4), the bigrading descends to the de Rham cohomology and produces the Hodge decomposition

$$H^k(M; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(M)$$

**Definition 1.2.11: Holomorphic Form**

A $(p, 0)$-form $\alpha \in \mathcal{A}^{p,0}(M)$ on a complex manifold $M$ is *holomorphic* if $\bar\partial\alpha = 0$. The space of holomorphic $p$-forms is denoted $\Omega^p(M)$, and the assignment $U \mapsto \Omega^p(U)$ for open $U \subseteq M$ defines a sheaf $\Omega^p_M$ of holomorphic $p$-forms on $M$.

In holomorphic coordinates, a $(p, 0)$-form $\alpha = \sum f_I\, dz_I$ is holomorphic precisely when each coefficient $f_I$ is a holomorphic function: by definition 1.2.5, $\bar\partial\alpha = \sum(\bar\partial f_I) \wedge dz_I$, and the linearly-independent forms $d\bar z_j \wedge dz_I$ all carry distinct multi-indices, so $\bar\partial\alpha = 0$ if and only if $\partial f_I/\partial \bar z_j = 0$ for all $I$ and $j$ (corollary 1.1.5). Holomorphic 0-forms are therefore holomorphic functions in the original sense, and holomorphic $p$-forms are the natural generalization.

**Example 1.2.12: $\bar z\, dz$ Is $(1, 0)$ but Not Holomorphic**

On $\mathbb{C}$, the form $\alpha = \bar z\, dz$ has type $(1, 0)$ since $dz \in T^{*1,0}\mathbb{C}$, yet it is not holomorphic:

$$\bar\partial\alpha = \bar\partial(\bar z\, dz) = (\bar\partial \bar z) \wedge dz = d\bar z \wedge dz \neq 0$$

Being $(1, 0)$ is a strictly weaker condition than being holomorphic: holomorphy is the $\bar\partial$-closed strengthening, and the holomorphic $(1, 0)$-forms on $\mathbb{C}$ are precisely $f(z)\, dz$ for $f$ holomorphic.

### Exercise 1.2.1

1. On $\mathbb{C}^n$ with holomorphic coordinates $z_1, \ldots, z_n$, verify directly that $df = \partial f + \bar{\partial} f$ for a smooth function $f\colon U \to \mathbb{C}$ by computing both sides in the real coordinates $(x_j, y_j)$ and the complex coordinates $(z_j, \bar{z}_j)$.

2. Show that $\partial$ and $\bar{\partial}$ are antiderivations on the bigraded algebra $\mathcal{A}^{\bullet,\bullet}(M)$: for $\alpha \in \mathcal{A}^{p,q}(M)$ and $\beta \in \mathcal{A}^{r,s}(M)$,
$$\partial(\alpha \wedge \beta) = (\partial\alpha) \wedge \beta + (-1)^{p+q}\, \alpha \wedge (\partial\beta)$$
and similarly for $\bar{\partial}$. (Hint: use $d = \partial + \bar{\partial}$ and the antiderivation property of $d$, then compare bidegree components.)

3. Show that $\dim_{\mathbb{C}} \Lambda^{p,q} T^*_x M = \binom{n}{p}\binom{n}{q}$ at every point $x \in M$, where $n = \dim_{\mathbb{C}} M$. Verify the consequence
$$\sum_{p+q=k} \binom{n}{p}\binom{n}{q} = \binom{2n}{k}$$
and identify this as the Vandermonde identity.

4. On a non-integrable almost complex manifold $(M, J)$, the exterior derivative $d : \mathcal{A}^{p,q} \to \mathcal{A}^{p+q+1}$ has four components in the bidegree decomposition. Identify the four components by writing $d\alpha$ for $\alpha \in \mathcal{A}^{0,1}(M)$ and using the projection $\pi^{r,s}\colon \mathcal{A}^{p+q+1} \to \mathcal{A}^{r,s}$. (Conceptual; do not compute the Nijenhuis-tensor expressions explicitly.)

5. On $\mathbb{C}$, define $\omega = (i/2\pi)\, \partial\bar{\partial} \log(1 + |z|^2)$. Compute $\omega$ explicitly as a $(1,1)$-form. Verify that $d\omega = 0$ and that $\omega$ is real-valued, that is, $\overline{\omega} = \omega$ in $\mathcal{A}^{1,1}(\mathbb{C})$.

6. Let $M$ be a complex manifold and $\alpha \in \mathcal{A}^{p,0}(M)$. Show that the following are equivalent:
   1. $\alpha$ is holomorphic, that is, $\bar{\partial}\alpha = 0$;
   2. In every holomorphic chart with coordinates $z_1, \ldots, z_n$, the coefficient functions of $\alpha = \sum f_I\, dz_I$ are holomorphic;
   3. $d\alpha = \partial\alpha$ (the antiholomorphic component of $d\alpha$ vanishes).

## 1.3 Holomorphic Vector Bundles

The Dolbeault formalism of section 1.2 acts on $\mathbb{C}$-valued differential forms, but the natural geometric objects on a complex manifold carry additional structure: vector bundles whose transition data is itself holomorphic. Sections of holomorphic vector bundles play the role that smooth functions play on a real manifold, with the qualitative difference that holomorphic sections survive the maximum principle of proposition 1.1.11 (which annihilated non-constant holomorphic functions on compact $X$). The tangent and cotangent bundles, the canonical bundle, and the line bundles $\mathcal{O}(d)$ on $\mathbb{CP}^n$ all live in this category and recur as running examples throughout the book.

This section develops the theory through the cocycle description, which makes operations such as tensor product and pullback transparent and provides the bridge to the sheaf-cohomology computation of the Picard group in chapter 2. The depth is calibrated to what Hodge theory uses; the K-theoretic perspective and the splitting principle are deferred to [27].

**Definition 1.3.1: Holomorphic Vector Bundle**

Let $X$ be a complex manifold. A *holomorphic vector bundle of rank $r$* on $X$ is a complex manifold $E$ together with a holomorphic surjection $\pi\colon E \to X$ such that:

1. Each fiber $E_x := \pi^{-1}(x)$ has the structure of a complex vector space of dimension $r$.
2. There exists an open cover $\{U_\alpha\}$ of $X$ and biholomorphisms $\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^r$ over $U_\alpha$ (that is, $\mathrm{pr}_1 \circ \Phi_\alpha = \pi$) whose restrictions $\Phi_\alpha|_{E_x}\colon E_x \to \{x\} \times \mathbb{C}^r$ are $\mathbb{C}$-linear isomorphisms for every $x \in U_\alpha$.

The maps $\Phi_\alpha$ are called *local trivializations*, and a rank-1 bundle is a *line bundle*. The smooth analogue of the definition is in [26]; the holomorphy of $\pi$ and the $\Phi_\alpha$ is the only departure from that setting.

The data of an atlas of trivializations is equivalent to the simpler combinatorial data of how the trivializations relate on overlaps.

**Proposition 1.3.2: Cocycle Description**

Let $X$ be a complex manifold with open cover $\{U_\alpha\}$.

1. Every holomorphic vector bundle $E \to X$ of rank $r$ trivialized over $\{U_\alpha\}$ determines a family of holomorphic functions $g_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{C})$ satisfying
$$g_{\alpha\beta}\, g_{\beta\gamma} = g_{\alpha\gamma} \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma, \qquad g_{\alpha\alpha} = I \quad \text{on } U_\alpha$$
2. Conversely, every family $\{g_{\alpha\beta}\}$ satisfying these conditions arises from a unique (up to canonical isomorphism) holomorphic vector bundle.
3. Two families $\{g_{\alpha\beta}\}$ and $\{g'_{\alpha\beta}\}$ determine isomorphic bundles if and only if there exist holomorphic functions $h_\alpha\colon U_\alpha \to \mathrm{GL}_r(\mathbb{C})$ with $g'_{\alpha\beta} = h_\alpha\, g_{\alpha\beta}\, h_\beta^{-1}$ on $U_\alpha \cap U_\beta$.

The family $\{g_{\alpha\beta}\}$ is the *cocycle of $E$ relative to* $\{U_\alpha\}$.

***Proof*:**
*(1).* Given the trivializations $\{\Phi_\alpha\}$, the composition $\Phi_\alpha \circ \Phi_\beta^{-1}$ on $(U_\alpha \cap U_\beta) \times \mathbb{C}^r$ has the form $(x, v) \mapsto (x, g_{\alpha\beta}(x)\, v)$ for some map $g_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{C})$, since both $\Phi_\alpha$ and $\Phi_\beta$ are linear on fibers. Holomorphy of $g_{\alpha\beta}$ follows from holomorphy of the $\Phi$ maps. The cocycle conditions are immediate from the associativity $\Phi_\alpha \circ \Phi_\beta^{-1} \circ \Phi_\beta \circ \Phi_\gamma^{-1} = \Phi_\alpha \circ \Phi_\gamma^{-1}$ and from $\Phi_\alpha \circ \Phi_\alpha^{-1} = \mathrm{id}$.

*(2).* Given a cocycle, set $E := \bigsqcup_\alpha (U_\alpha \times \mathbb{C}^r)/\sim$, where $(x, v) \in U_\beta \times \mathbb{C}^r$ is identified with $(x, g_{\alpha\beta}(x)v) \in U_\alpha \times \mathbb{C}^r$ for $x \in U_\alpha \cap U_\beta$, matching the direction fixed in part (1): $g_{\alpha\beta}$ carries $\beta$-coordinates to $\alpha$-coordinates. The cocycle condition makes $\sim$ an equivalence relation, the quotient inherits a complex manifold structure (transition maps are holomorphic by hypothesis), and the natural projection to $X$ is holomorphic with $\mathbb{C}^r$ fibers and tautological trivializations.

*(3).* An isomorphism $\varphi\colon E \to E'$ over $X$ in trivializations $\{\Phi_\alpha\}, \{\Phi'_\alpha\}$ corresponds to a family

of holomorphic maps $h_\alpha \colon U_\alpha \to \mathrm{GL}_r(\mathbb{C})$ via $\Phi'_\alpha \circ \varphi \circ \Phi_\alpha^{-1}(x, v) = (x, h_\alpha(x)\, v)$. Compatibility with the transitions on overlaps gives the coboundary relation $g'_{\alpha\beta} = h_\alpha\, g_{\alpha\beta}\, h_\beta^{-1}$, completing the proof.

The cocycle perspective is the practical one: most constructions below are described by transparent operations on cocycles, and most verifications reduce to checking the cocycle conditions.

**Definition 1.3.3: Holomorphic Section**

A *holomorphic section* of $E \to X$ over an open $U \subseteq X$ is a holomorphic map $s \colon U \to E$ satisfying $\pi \circ s = \mathrm{id}_U$. Equivalently (in trivializations), a family of holomorphic functions $s_\alpha \colon U_\alpha \cap U \to \mathbb{C}^r$ with $s_\alpha = g_{\alpha\beta}\, s_\beta$ on overlaps.
The space of holomorphic sections of $E$ over $U$ is denoted $H^0(U, E)$ or $\Gamma(U, E)$. The presheaf $U \mapsto H^0(U, E)$ is a sheaf, denoted $\mathcal{O}_X(E)$ (sheaf machinery developed in chapter 2, citing [4]).

The cocycle equivalence makes the standard linear-algebra operations on vector spaces extend automatically to holomorphic vector bundles.

**Definition 1.3.4: Operations on Holomorphic Vector Bundles**

Let $E$ and $F$ be holomorphic vector bundles on $X$ with cocycles $\{g^E_{\alpha\beta}\}$ and $\{g^F_{\alpha\beta}\}$ relative to a common trivializing cover $\{U_\alpha\}$. The following constructions produce holomorphic vector bundles whose cocycles are obtained from those of $E$ and $F$ by the corresponding linear-algebraic operation:

1. *Direct sum* $E \oplus F$: cocycle $\begin{pmatrix} g^E_{\alpha\beta} & 0 \\ 0 & g^F_{\alpha\beta} \end{pmatrix}$.
2. *Tensor product* $E \otimes F$: cocycle $g^E_{\alpha\beta} \otimes g^F_{\alpha\beta}$ (Kronecker product).
3. *Dual* $E^*$: cocycle $((g^E_{\alpha\beta})^{-1})^T$.
4. *Hom* bundle: $\mathrm{Hom}(E, F) := E^* \otimes F$.
5. *Exterior power* $\Lambda^k E$: cocycle $\Lambda^k g^E_{\alpha\beta}$, with $\det E := \Lambda^{\mathrm{rk}\, E} E$ the *determinant line bundle*.

**Definition 1.3.5: Pullback Bundle**

Let $f \colon X \to Y$ be a holomorphic map between complex manifolds and $E \to Y$ a holomorphic vector bundle with cocycle $\{g_{\alpha\beta}\}$ relative to $\{V_\alpha\}$. The *pullback bundle* $f^*E \to X$ is the holomorphic vector bundle with cocycle $\{g_{\alpha\beta} \circ f\}$ relative to the cover $\{f^{-1}(V_\alpha)\}$. Equivalently, the fiber satisfies $(f^*E)_x = E_{f(x)}$.

The cocycle data of bundle operations is mechanical, but the resulting bundles encode genuine geometric content. The first instance is the holomorphic tangent and cotangent bundles, which package the $T^{1,0}/T^{0,1}$ decomposition of proposition 1.1.13 into holomorphic vector bundles.

**Example 1.3.6: Holomorphic Tangent and Cotangent Bundles**

The bundle $T^{1,0}X$ from proposition 1.1.13 is a holomorphic vector bundle of rank $n = \dim_{\mathbb{C}} X$. In a holomorphic chart $(U_\alpha, \varphi_\alpha)$ with coordinates $z^\alpha = (z_1^\alpha, \ldots, z_n^\alpha)$, the local trivialization sends $\partial/\partial z_j^\alpha \mapsto e_j$. On $U_\alpha \cap U_\beta$, the transition is the holomorphic Jacobian

$$g_{\alpha\beta} = \left(\frac{\partial z_i^\alpha}{\partial z_j^\beta}\right)_{i,j=1}^{n} \in \mathrm{GL}_n(\mathbb{C})$$

holomorphic by the chain rule applied to the holomorphic transition maps of $X$. The dual bundle $T^{*1,0}X$ inherits the inverse-transpose cocycle, and its $p$-th exterior power $\Lambda^p T^{*1,0}X$ is the holomorphic vector bundle whose holomorphic sections are precisely the holomorphic $p$-forms of definition 1.2.11; that is, $\mathcal{O}_X(\Lambda^p T^{*1,0}X) = \Omega_X^p$ as sheaves.

The rank-1 case of the bundle operations defines a group structure on isomorphism classes.

**Proposition 1.3.7: The Picard Group**

Let $X$ be a complex manifold. The set of isomorphism classes of holomorphic line bundles on $X$, with operation $\otimes$, forms an abelian group: the identity is the trivial bundle $\mathcal{O}_X := X \times \mathbb{C}$, and the inverse of a line bundle $L$ is its dual $L^*$. This group is the *Picard group* of $X$, denoted $\mathrm{Pic}(X)$.

***Proof*:**

Tensor product is associative and commutative on the level of bundles (with canonical isomorphisms). The trivial bundle has cocycle $g_{\alpha\beta} \equiv 1$, and $L \otimes \mathcal{O}_X$ has the same cocycle as $L$. For $L$ with cocycle $\{g_{\alpha\beta}\}$, the dual $L^*$ has cocycle $\{g_{\alpha\beta}^{-1}\}$, and $L \otimes L^*$ has cocycle $\{g_{\alpha\beta} \cdot g_{\alpha\beta}^{-1}\} = \{1\}$, that is, the trivial bundle. The group axioms are verified, completing the proof.

The cocycle description identifies the Picard group with a sheaf-cohomology group. This is the entry point to the divisor-theoretic computation of chapter 2, and it is used repeatedly in section 4.4, so it is recorded as a statement rather than left as a remark.

**Proposition 1.3.8: The Picard Group as Sheaf Cohomology**

Let $X$ be a complex manifold and $\mathcal{O}_X^*$ the sheaf of nowhere-zero holomorphic functions on $X$. There is a canonical group isomorphism

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^*)$$

In particular every class in $H^1(X, \mathcal{O}_X^*)$ is the class of a holomorphic line bundle.

***Proof*:**

Fix an open cover $\{U_\alpha\}$ of $X$. By proposition 1.3.2(1) a line bundle trivialized over that cover determines holomorphic functions $g_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^*$ satisfying $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$, which is exactly the condition for $\{g_{\alpha\beta}\}$ to be a Čech 1-cocycle for the sheaf $\mathcal{O}_X^*$; part (2) says every such cocycle arises this way. By part (3), two cocycles give isomorphic bundles precisely when they differ by $h_\alpha h_\beta^{-1}$ for nowhere-zero holomorphic $h_\alpha$ on $U_\alpha$, which is exactly a Čech coboundary. Hence isomorphism classes of line bundles trivialized over $\{U_\alpha\}$ correspond bijectively to $\check{H}^1(\{U_\alpha\}, \mathcal{O}_X^*)$, and the correspondence carries $\otimes$ to the group operation, since the cocycle of a tensor product is the product of the cocycles (definition 1.3.4).

Every line bundle is trivial over some cover, and refining a cover is compatible with both constructions, so passing to the colimit over refinements gives $\operatorname{Pic}(X) \cong \check{H}^1(X, \mathcal{O}_X^*)$. Čech and derived-functor cohomology agree in degrees 0 and 1 for an arbitrary sheaf of abelian groups on an arbitrary topological space ([4, the degree-one comparison]), and degree 1 is all that is needed here, so $\check{H}^1(X, \mathcal{O}_X^*) = H^1(X, \mathcal{O}_X^*)$, completing the proof. No acyclicity of $\mathcal{O}_X^*$ is required, and none is available: $\mathcal{O}_X^*$ is not a fine sheaf.

The present section keeps the remaining discussion at the level of cocycles.

The basic non-trivial example, and one of the most important objects on $\mathbb{CP}^n$, is the tautological line bundle.

**Example 1.3.9: The Tautological Line Bundle on $\mathbb{CP}^n$**

The total space
$$\mathcal{O}_{\mathbb{CP}^n}(-1) := \{([v], w) \in \mathbb{CP}^n \times \mathbb{C}^{n+1} : w \in \mathbb{C} \cdot v\}$$
is a holomorphic line bundle over $\mathbb{CP}^n$, with projection to the first factor and fiber over $[v]$ equal to the line $\mathbb{C} \cdot v \subseteq \mathbb{C}^{n+1}$. In the standard chart $U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\}$ on $\mathbb{CP}^n$, the trivialization
$$\Phi_i \colon \mathcal{O}(-1)|_{U_i} \to U_i \times \mathbb{C}, \qquad ([v], w) \mapsto ([v], w_i)$$
sends a fiber element to its $i$-th coordinate as a vector in $\mathbb{C}^{n+1}$. To compute the transition on $U_i \cap U_j$: starting with $([v], \alpha) \in U_i \cap U_j \times \mathbb{C}$, the inverse $\Phi_i^{-1}$ produces the unique $w \in \mathbb{C} \cdot v$ with $w_i = \alpha$, namely $w = (\alpha/v_i)\, v$. Then $\Phi_j([v], w) = ([v], w_j) = ([v], \alpha\, v_j/v_i)$. The transition is therefore multiplication by the holomorphic function
$$g_{ji}([v]) = \frac{v_j}{v_i} = \frac{z_j}{z_i}$$
on $U_i \cap U_j$, where the right-hand side uses the homogeneous coordinates restricted to the overlap. The cocycle condition $g_{ki} = g_{kj}\, g_{ji}$ holds tautologically: $z_k/z_i = (z_k/z_j)(z_j/z_i)$.

**Definition 1.3.10: The Bundles $\mathcal{O}(d)$**

On $\mathbb{CP}^n$, the *hyperplane bundle* is $\mathcal{O}(1) := \mathcal{O}(-1)^*$, a holomorphic line bundle with cocycle $g_{ij}([z]) = z_j/z_i$ in the standard atlas (the inverse of the $\mathcal{O}(-1)$ cocycle from example 1.3.9). For each integer $d \in \mathbb{Z}$, define
$$\mathcal{O}(d) := \mathcal{O}(1)^{\otimes d}$$
(with the convention $\mathcal{O}(d) := \mathcal{O}(-1)^{\otimes |d|}$ for $d < 0$ and $\mathcal{O}(0) := \mathcal{O}_{\mathbb{CP}^n}$). The cocycle is $g_{ij}([z]) = (z_j/z_i)^d$.

The global sections of $\mathcal{O}(d)$ are the building blocks for projective embeddings, and they admit an explicit description.

**Theorem 1.3.11: Global Sections of $\mathcal{O}(d)$**

For each $d \in \mathbb{Z}$,

$$H^0(\mathbb{CP}^n, \mathcal{O}(d)) = \begin{cases} \mathbb{C}[z_0, \dots, z_n]_d & d \geq 0 \\ 0 & d < 0 \end{cases}$$

where $\mathbb{C}[z_0, \dots, z_n]_d$ denotes the vector space of homogeneous polynomials of degree $d$ in $n+1$ variables.

***Proof*:**
A holomorphic section of $\mathcal{O}(d)$ on $\mathbb{CP}^n$ corresponds, via the cocycle $g_{ij} = (z_j/z_i)^d$ from definition 1.3.10, to a family of holomorphic functions $\{s_i \colon U_i \to \mathbb{C}\}$ satisfying $s_i = (z_j/z_i)^d\, s_j$ on $U_i \cap U_j$. Multiplying both sides of this relation by $z_i^d$ and simplifying $z_i^d (z_j/z_i)^d = z_j^d$ gives

$$z_i^d\, s_i([z]) = z_j^d\, s_j([z]) \quad \text{on } U_i \cap U_j$$

Define $F \colon \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{C}$ by the formula $F(z) := z_i^d\, s_i([z])$ for any index $i$ with $z_i \neq 0$. The cocycle identity above shows the choice of $i$ does not matter, so $F$ is a well-defined holomorphic function on $\mathbb{C}^{n+1} \setminus \{0\}$.

Hartogs extension (box 1.1.6) and the codimension $n + 1 \geq 2$ of $\{0\}$ in $\mathbb{C}^{n+1}$ extend $F$ uniquely to a holomorphic function on all of $\mathbb{C}^{n+1}$. The function $F$ is homogeneous of degree $d$:

$$F(\lambda z) = (\lambda z_i)^d\, s_i([\lambda z]) = \lambda^d\, z_i^d\, s_i([z]) = \lambda^d\, F(z)$$

for $\lambda \in \mathbb{C}^\times$ (using $[\lambda z] = [z]$ in $\mathbb{CP}^n$ and the linearity of multiplication by $\lambda^d$). A holomorphic function on $\mathbb{C}^{n+1}$ has a Taylor expansion $F(z) = \sum_I a_I\, z^I$ at the origin, and homogeneity $F(\lambda z) = \lambda^d F(z)$ forces every non-vanishing coefficient $a_I$ to satisfy $|I| = d$. For $d < 0$ no such multi-index exists, so $F \equiv 0$, and consequently every $s_i \equiv 0$. For $d \geq 0$, $F$ is a homogeneous polynomial of degree $d$.

Conversely, given a homogeneous polynomial $F$ of degree $d \geq 0$, the formula $s_i([z]) := F(z)/z_i^d$ defines a holomorphic function on $U_i$ (since $z_i \neq 0$ there). On $U_i \cap U_j$, the identity $F(z) = z_i^d\, s_i = z_j^d\, s_j$ rearranges to $s_i = (z_j/z_i)^d\, s_j$, the cocycle relation. The two constructions are mutually inverse, completing the proof.

The bundles $\mathcal{O}(d)$ are the prototype of how line bundles encode polynomial degree; they reappear as the natural ample classes on $\mathbb{CP}^n$ and underlie the projective embeddings of chapter 7.

The last definition collects the Hodge-theoretically central line bundle.

**Definition 1.3.12: Canonical Bundle**

Let $X$ be a complex manifold of complex dimension $n$. The *canonical bundle* of $X$ is

$$K_X := \det(T^{*1,0}X) = \Lambda^n T^{*1,0}X$$

a holomorphic line bundle whose holomorphic sections are the holomorphic $n$-forms; in the sheaf notation of definition 1.2.11 its sheaf of sections is $\Omega^n_X$, and the two symbols are used interchangeably. The letter $\omega$ is reserved throughout for the fundamental form of a Hermitian metric (chapter 3) and is never used for this bundle.

The canonical bundle of $\mathbb{C}\mathrm{P}^n$ admits an explicit identification: $K_{\mathbb{C}\mathrm{P}^n} = \mathcal{O}(-n-1)$. The proof uses the Euler sequence, the central worked example of the section, and is given in corollary 1.3.17. Setting up that sequence requires categorical machinery (subbundles, quotient bundles, short exact sequences) which is the next thread.

A holomorphic morphism between vector bundles $E$ and $F$ on $X$ is a holomorphic map $\varphi\colon E \to F$ commuting with the projections to $X$ and acting fiberwise as a $\mathbb{C}$-linear map $\varphi_x\colon E_x \to F_x$. In trivializations relative to a common cover, $\varphi$ is described by holomorphic matrix-valued functions $A_\alpha\colon U_\alpha \to \mathrm{Mat}_{s\times r}(\mathbb{C})$ (where $r, s$ are the ranks of $E, F$) satisfying the compatibility $g^F_{\beta\alpha}\, A_\alpha = A_\beta\, g^E_{\beta\alpha}$ on overlaps. Bundle morphisms compose, and one calls $\varphi$ injective (resp. surjective) if every $\varphi_x$ is.

**Definition 1.3.13: Subbundle and Quotient Bundle**

A holomorphic vector bundle $F$ on $X$ is a *subbundle* of a holomorphic vector bundle $E$ on $X$ if there is a holomorphic morphism $\iota\colon F \hookrightarrow E$ over $X$ that is injective on every fiber. Equivalently, after refining the trivializing cover, the cocycle $g^E_{\alpha\beta}$ takes the upper-triangular block form

$$g^E_{\alpha\beta} = \begin{pmatrix} g^F_{\alpha\beta} & * \\ 0 & g^{E/F}_{\alpha\beta} \end{pmatrix}$$

The *quotient bundle* $E/F$ is the holomorphic vector bundle with fiber $(E/F)_x := E_x/F_x$ and cocycle equal to the lower diagonal block $g^{E/F}_{\alpha\beta}$.

**Definition 1.3.14: Short Exact Sequence**

A sequence

$$0 \to F \xrightarrow{\iota} E \xrightarrow{q} G \to 0$$

of holomorphic vector bundles and morphisms on $X$ is a *short exact sequence* if $\iota$ is the inclusion of $F$ as a subbundle of $E$ and $q$ is the quotient morphism $E \to E/F = G$. Equivalently, $\iota_x\colon F_x \to E_x$ is injective, $q_x\colon E_x \to G_x$ is surjective, and $\ker q_x = \operatorname{im} \iota_x$ at every $x \in X$.

The first canonical short exact sequence on $\mathbb{C}\mathrm{P}^n$ is built from the tautological line bundle.

**Example 1.3.15: Tautological Sequence on $\mathbb{C}\mathrm{P}^n$**

The tautological line bundle $\mathcal{O}(-1)$ from example 1.3.9 is a subbundle of the trivial bundle $\mathcal{O}^{\oplus(n+1)} := \mathbb{C}\mathrm{P}^n \times \mathbb{C}^{n+1}$ via the inclusion $([v], w) \mapsto ([v], w)$ that views the line $\mathbb{C}\cdot v$ as a subspace of $\mathbb{C}^{n+1}$. The quotient is the *tautological quotient bundle* $Q := \mathcal{O}^{\oplus(n+1)}/\mathcal{O}(-1)$, of rank $n$, with fiber $\mathbb{C}^{n+1}/\mathbb{C} \cdot v$ at $[v]$. The associated short exact sequence

$$0 \to \mathcal{O}(-1) \to \mathcal{O}^{\oplus(n+1)} \to Q \to 0$$

is the *tautological sequence* on $\mathbb{C}\mathrm{P}^n$.

The Hodge-theoretically central exact sequence is the result of tensoring the tautological sequence by $\mathcal{O}(1)$ and identifying the resulting quotient with the holomorphic tangent bundle.

**Theorem 1.3.16: The Euler Sequence**

On $\mathbb{C}\mathrm{P}^n$, there is a canonical short exact sequence of holomorphic vector bundles

$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)} \to T^{1,0}\mathbb{C}\mathrm{P}^n \to 0$$

called the *Euler sequence*. The injection sends the constant section 1 of $\mathcal{O}$ to the global section $(z_0, z_1, \dots, z_n)$ of $\mathcal{O}(1)^{\oplus(n+1)}$ (each $z_k$ regarded as a global section of $\mathcal{O}(1)$ per theorem 1.3.11); the surjection is induced by the differential of the projection $p\colon \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{C}\mathrm{P}^n$.

***Proof*:**
*Step 1: Tensor the tautological sequence.* Apply $- \otimes \mathcal{O}(1)$ to the tautological sequence (example 1.3.15). Tensor product preserves short exactness in the holomorphic vector bundle category because in cocycle trivializations the upper-triangular block structure of the middle cocycle is preserved when each block is multiplied by the scalar cocycle of the line bundle. Using $\mathcal{O}(-1) \otimes \mathcal{O}(1) = \mathcal{O}$ and $\mathcal{O}^{\oplus(n+1)} \otimes \mathcal{O}(1) = \mathcal{O}(1)^{\oplus(n+1)}$,

$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)} \to Q \otimes \mathcal{O}(1) \to 0$$

The task reduces to identifying $Q \otimes \mathcal{O}(1) \cong T^{1,0}\mathbb{C}\mathrm{P}^n$ as holomorphic vector bundles.

*Step 2: Pointwise differential.* The projection $p\colon \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{C}\mathrm{P}^n$ is a holomorphic submersion with $\mathbb{C}^\times$-orbits as fibers. At a point $z \in \mathbb{C}^{n+1} \setminus \{0\}$, the differential $dp_z\colon T_z(\mathbb{C}^{n+1} \setminus \{0\}) = \mathbb{C}^{n+1} \to T_{[z]}\mathbb{C}\mathrm{P}^n$ has kernel equal to the radial line $\mathbb{C} \cdot z = \mathcal{O}(-1)_{[z]}$ (the tangent direction along the $\mathbb{C}^\times$-orbit through $z$). The differential descends to a $\mathbb{C}$-linear isomorphism

$$\overline{dp_z} : \mathbb{C}^{n+1}/\mathbb{C} \cdot z = Q_{[z]} \xrightarrow{\sim} T_{[z]}\mathbb{C}\mathrm{P}^n$$

at every point. The isomorphism is pointwise; whether it globalizes to a bundle isomorphism depends on its scaling behavior under the $\mathbb{C}^\times$-action.

*Step 3: Globalize via the $\mathcal{O}(1)$ twist.* Under $z \mapsto \lambda z$ with $\lambda \in \mathbb{C}^\times$, computation in the chart $U_i$ gives

$$dp_{\lambda z}(\zeta) = \frac{1}{\lambda}\, dp_z(\zeta)$$

The differential scales inversely under the $\mathbb{C}^\times$-action on the total space. The compensating factor $\lambda^{-1}$ matches the scaling of a generator of $\mathcal{O}(1)_{[z]} = (\mathbb{C} \cdot z)^*$, since $\mathcal{O}(-1)$ generators scale by $\lambda$ and the dual scales by $\lambda^{-1}$. Tensoring $Q$ with $\mathcal{O}(1)$ therefore corrects the scaling, and $\overline{dp}$ promotes to a globally defined holomorphic isomorphism

$$Q \otimes \mathcal{O}(1) \xrightarrow{\sim} T^{1,0}\mathbb{C}\mathrm{P}^n$$

Composition with the twisted tautological sequence of Step 1 produces the Euler sequence as stated, completing the proof.

The Euler sequence has an immediate determinant-bundle consequence, yielding the canonical bundle of $\mathbb{C}\mathrm{P}^n$.

**Corollary 1.3.17: Canonical Bundle of $\mathbb{C}\mathrm{P}^n$**

$\det T^{1,0}\mathbb{C}\mathrm{P}^n = \mathcal{O}(n+1)$ and $K_{\mathbb{C}\mathrm{P}^n} = \mathcal{O}(-n-1)$.

***Proof*:**
The determinant of a short exact sequence of vector bundles $0 \to F \to E \to G \to 0$ satisfies the multiplicativity relation $\det E = \det F \otimes \det G$, immediate from the upper-triangular block form of the cocycle of $E$ and $\det\left(\begin{smallmatrix} A & * \\ 0 & B \end{smallmatrix}\right) = \det A \cdot \det B$. Applied to the Euler sequence (theorem 1.3.16),

$$\det \mathcal{O}(1)^{\oplus(n+1)} = \det \mathcal{O} \otimes \det T^{1,0}\mathbb{C}\mathrm{P}^n$$

The trivial bundle $\mathcal{O}$ has trivial determinant, and the determinant of a direct sum of line bundles is the tensor product: $\det \mathcal{O}(1)^{\oplus(n+1)} = \mathcal{O}(1)^{\otimes(n+1)} = \mathcal{O}(n+1)$. Hence $\det T^{1,0}\mathbb{C}\mathrm{P}^n = \mathcal{O}(n+1)$, and dualizing gives $K_{\mathbb{C}\mathrm{P}^n} = \det T^{*1,0}\mathbb{C}\mathrm{P}^n = \mathcal{O}(n+1)^* = \mathcal{O}(-n-1)$, completing the proof.

## Exercise 1.3.1

1. Let $f\colon X \to Y$ be a holomorphic map between complex manifolds and $E, F$ holomorphic vector bundles on $Y$. Show that there is a natural isomorphism
$$f^*(E \otimes F) \cong f^*E \otimes f^*F$$
of holomorphic vector bundles on $X$, and verify that it is compatible with composition of pullbacks.

2. Let $L$ be a holomorphic line bundle on $X$ with cocycle $\{g_{ij}\}$ relative to $\{U_i\}$. Show that $L$ is isomorphic to the trivial bundle $\mathcal{O}_X = X \times \mathbb{C}$ if and only if there exist non-vanishing holomorphic functions $h_i\colon U_i \to \mathbb{C}^\times$ with $g_{ij} = h_i\, h_j^{-1}$ on every overlap.

3. Show that $\mathcal{O}(d) \otimes \mathcal{O}(e) \cong \mathcal{O}(d+e)$ as holomorphic line bundles on $\mathbb{C}\mathrm{P}^n$ for all $d, e \in \mathbb{Z}$. Deduce that the assignment $d \mapsto \mathcal{O}(d)$ defines a group homomorphism $\mathbb{Z} \to \mathrm{Pic}(\mathbb{C}\mathrm{P}^n)$. (The image is in fact the entire Picard group of $\mathbb{C}\mathrm{P}^n$, but the surjectivity uses sheaf cohomology and is deferred to chapter 2.)

4. Compute the transition functions of $T^{*1,0}\mathbb{C}\mathrm{P}^1$ relative to the standard atlas $\{U_0, U_1\}$ with affine coordinates $w = z_1/z_0$ on $U_0$ and $w' = z_0/z_1 = 1/w$ on $U_1$. Identify the resulting line bundle as a power of $\mathcal{O}(1)$.

5. Show that for a rank-$r$ holomorphic vector bundle $E$ with cocycle $\{g_{\alpha\beta}\}$, the determinant line bundle $\det E := \Lambda^r E$ has cocycle $\{\det g_{\alpha\beta}\}$. Use this together with exercise 1.3.1 (4) to verify $\det T^{1,0}\mathbb{C}\mathrm{P}^1 = \mathcal{O}(2)$ directly from the standard atlas, in agreement with corollary 1.3.17 for $n = 1$.

6. Take the Euler sequence (theorem 1.3.16) and pass to global sections. Use theorem 1.3.11 to identify $H^0(\mathbb{C}\mathrm{P}^n, \mathcal{O}(1)^{\oplus(n+1)})$ with the space of $\mathbb{C}$-linear maps $\mathbb{C}^{n+1} \to \mathbb{C}^{n+1}$, and identify

the global section $(z_0, \ldots, z_n)$ of theorem 1.3.16 with the identity map. Conclude that the inclusion $\mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)}$ is non-trivial on global sections; specifically, the global section $1 \in H^0(\mathbb{C}\mathrm{P}^n, \mathcal{O})$ maps to a non-zero element. (No global sections of $T^{1,0}\mathbb{C}\mathrm{P}^n$ are produced by this argument; the exact sequence does not split, and $H^0(\mathbb{C}\mathrm{P}^n, T^{1,0}\mathbb{C}\mathrm{P}^n)$ requires sheaf cohomology to compute, deferred to chapter 2.)

## 1.4 Divisors, Line Bundles, and the Blow-Up

Line bundles on a complex manifold come in two complementary presentations. The first, through cocycles, was developed in section 1.3. The second, through divisors (formal $\mathbb{Z}$-linear combinations of codimension-1 subvarieties), exposes the geometric content directly: a section of $\mathcal{O}(D)$ is a meromorphic function with poles bounded by $D$, and the line bundle $\mathcal{O}(D)$ records the divisor's vanishing data globally. The blow-up is a complementary local surgery: it modifies a complex manifold at a single point by replacing the point with a copy of $\mathbb{C}\mathrm{P}^{n-1}$, the projective space of tangent directions through it.

The narrative role of this section runs through four later chapters. The Lefschetz $(1,1)$ theorem (chapter 4) identifies integral classes in $H^{1,1}$ with classes of divisors; the cycle class map (**??**) is the higher-codimension generalization. Lefschetz pencils (chapter 6) are constructed as families of hyperplane sections after blowing up the base locus. Mixed Hodge structures on smooth open varieties (chapter 9) are built from logarithmic-de-Rham complexes on smooth compactifications $\overline{X} \supset X$ where $\overline{X} \setminus X$ is a normal-crossings divisor, which is produced by iterated blow-ups. Without divisors and blow-ups as language, none of these constructions has firm footing.

The material below is also developed (in the algebraic-variety setting) in [4, divisors as the geometric representation of line bundles]; the present section is for readers who have not yet internalized that material, and it foregrounds the complex-analytic perspective (meromorphic functions over holomorphic, analytic hypersurfaces over algebraic subvarieties) that recurs in the rest of this book.

**Definition 1.4.1: Weil Divisor**

Let $X$ be a connected complex manifold. An *irreducible analytic hypersurface* in $X$ is an irreducible closed analytic subset of complex codimension one. A *Weil divisor* on $X$ is a formal $\mathbb{Z}$-linear combination

$$D = \sum_i n_i Y_i$$

with the $Y_i$ pairwise distinct irreducible analytic hypersurfaces and $n_i \in \mathbb{Z}$, the sum locally finite (every point has a neighborhood meeting only finitely many $Y_i$ with $n_i \neq 0$). Weil divisors form an abelian group $\mathrm{Div}(X)$ under formal addition. A divisor is *effective*, written $D \geq 0$, if all $n_i \geq 0$.

The Weil description records the geometry of a divisor as a sum of hypersurfaces. The Cartier description records its local equations, which is the data needed to build a line bundle.

**Definition 1.4.2: Cartier Divisor**

A *Cartier divisor* on a complex manifold $X$ is given by local data $\{(U_\alpha, f_\alpha)\}$ consisting of an open cover $\{U_\alpha\}$ of $X$ and non-zero meromorphic functions $f_\alpha$ on each $U_\alpha$ such that the ratios

$$f_\alpha/f_\beta \in \mathcal{O}^*(U_\alpha \cap U_\beta)$$

are holomorphic and non-vanishing on every overlap, where $\mathcal{O}^*(U)$ denotes the multiplicative group of nowhere-zero holomorphic functions on $U$. Two such families $\{(U_\alpha, f_\alpha)\}$ and $\{(U'_\beta, h_\beta)\}$ are *equivalent* if $f_\alpha/h_\beta \in \mathcal{O}^*(U_\alpha \cap U'_\beta)$ for every $\alpha$ and $\beta$, that is, if the two families record the same local vanishing data. The equivalence classes form an abelian group under the operation $\{(U_\alpha, f_\alpha)\} + \{(U_\alpha, h_\alpha)\} = \{(U_\alpha, f_\alpha\, h_\alpha)\}$.

On a smooth complex manifold, Weil and Cartier divisors are canonically identified: every irreducible analytic hypersurface has a local holomorphic equation $f$ at every point (a consequence of the local structure of analytic sets, see [21, §2.3] or [4, Cartier divisors, line bundles, and the class map]), and conversely the zero locus of a Cartier divisor's local equations assembles into a Weil divisor with multiplicities equal to the orders of vanishing. Throughout the rest of the book, $X$ is smooth and "divisor" refers to either notion.

The order of vanishing along a hypersurface gives the principal divisors and the linear-equivalence relation.

**Definition 1.4.3: Principal Divisor and Linear Equivalence**

Let $f$ be a non-zero global meromorphic function on a connected complex manifold $X$. The *principal divisor* of $f$ is

$$(f) := \sum_Y \mathrm{ord}_Y(f) \cdot Y$$

where the sum runs over irreducible analytic hypersurfaces $Y \subset X$ and $\mathrm{ord}_Y(f) \in \mathbb{Z}$ is the order of vanishing of $f$ along $Y$ (positive for zeros, negative for poles, zero if $Y$ is contained in neither $\{f = 0\}$ nor $\{f = \infty\}$). The sum is locally finite by the analytic-set structure of the zero and pole loci.
Two divisors $D$ and $D'$ are *linearly equivalent*, written $D \sim D'$, if $D - D' = (f)$ for some non-zero global meromorphic function $f$ on $X$. Linear equivalence is an equivalence relation, and the quotient

$$\mathrm{Cl}(X) := \mathrm{Div}(X)/\!\sim$$

is the *divisor class group* of $X$.

The divisor class group plays the geometric role that the Picard group played in section 1.3: an invariant attached to $X$ that records its low-codimension structure. The two are connected by a canonical homomorphism, built from local equations.

**Definition 1.4.4: Line Bundle of a Divisor**

Let $D$ be a Cartier divisor on a complex manifold $X$ with local equations $\{(U_\alpha, f_\alpha)\}$. The *line bundle of $D$* is the holomorphic line bundle $\mathcal{O}(D) \to X$ with cocycle

$$g_{\alpha\beta} := \frac{f_\alpha}{f_\beta} \in \mathcal{O}^*(U_\alpha \cap U_\beta)$$

relative to the trivializing cover $\{U_\alpha\}$.

The cocycle conditions $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ and $g_{\alpha\alpha} = 1$ hold automatically: the ratios telescope. The construction of $\mathcal{O}(D)$ is functorial in the divisor, in a sense made precise by the next theorem.

**Proposition 1.4.5: Properties of $\mathcal{O}(D)$**

Let $D, D'$ be Cartier divisors on a complex manifold $X$ and $f$ a non-zero global meromorphic function on $X$. Then:

1. $\mathcal{O}(D + D') \cong \mathcal{O}(D) \otimes \mathcal{O}(D')$ as holomorphic line bundles.
2. $\mathcal{O}(-D) \cong \mathcal{O}(D)^*$.
3. $\mathcal{O}(0) = \mathcal{O}_X$, the trivial line bundle.
4. $\mathcal{O}(D + (f)) \cong \mathcal{O}(D)$. In particular, linearly equivalent divisors have isomorphic line bundles.

***Proof*:**

1. Take a common refinement of trivializing covers; the cocycle of $D + D'$ is $f_\alpha h_\alpha/(f_\beta h_\beta) = (f_\alpha/f_\beta)(h_\alpha/h_\beta) = g^D_{\alpha\beta}\, g^{D'}_{\alpha\beta}$, the cocycle of $\mathcal{O}(D) \otimes \mathcal{O}(D')$.
2. The cocycle of $-D$ is $f_\alpha^{-1}/f_\beta^{-1} = f_\beta/f_\alpha = (g^D_{\alpha\beta})^{-1}$, the cocycle of $\mathcal{O}(D)^*$ for line bundles.
3. The zero divisor is given by the local data $\{(U_\alpha, 1)\}$, with cocycle $1/1 = 1$.
4. The principal divisor $(f)$ has local equations $(U_\alpha, f|_{U_\alpha})$ on a fine enough cover, so $D + (f)$ has local equations $(U_\alpha, f_\alpha\, f|_{U_\alpha})$. The cocycle is $(f_\alpha\, f)/(f_\beta\, f) = f_\alpha/f_\beta = g^D_{\alpha\beta}$, the same cocycle as $\mathcal{O}(D)$. Hence $\mathcal{O}(D + (f))$ and $\mathcal{O}(D)$ have identical cocycles and are isomorphic, completing the proof.

The fourth property says the line bundle depends only on the linear equivalence class. The assignment therefore descends to a group homomorphism on the class group.

**Theorem 1.4.6: Divisor Class to Picard Homomorphism**

The assignment $D \mapsto [\mathcal{O}(D)]$ defines a group homomorphism

$$\mathrm{Cl}(X) \to \mathrm{Pic}(X)$$

from the divisor class group to the Picard group of $X$. The homomorphism is injective. When $X$ is smooth and projective, it is also surjective (proof in chapter 2 via sheaf cohomology).

***Proof*:**
The map is well-defined by proposition 1.4.5(4) and a group homomorphism by part (1). For injectivity: if $\mathcal{O}(D) \cong \mathcal{O}(D')$, then by the cocycle equivalence (proposition 1.3.2(3)) there exist holomorphic non-vanishing functions $h_\alpha$ on $U_\alpha$ with $g^D_{\alpha\beta} = h_\alpha\, g^{D'}_{\alpha\beta}\, h_\beta^{-1}$, that is, $f_\alpha/f_\beta = (h_\alpha\, f'_\alpha)/(h_\beta\, f'_\beta)$. Equivalently, the local meromorphic functions $f_\alpha/(h_\alpha\, f'_\alpha)$ agree on overlaps and patch into a global meromorphic function $f$ with $D - D' = (f)$. Hence $D \sim D'$ in $\mathrm{Cl}(X)$.

Surjectivity for smooth projective $X$ uses the Picard exact sequence and the GAGA principle, both deferred to chapter 2, completing the present proof.

The homomorphism is the bridge that translates between the geometric language of divisors and the topological language of line bundles. In the projective case both sides are equal as groups, but the identification is more useful as a mediator: divisors give explicit representatives for line-bundle classes (allowing concrete computations), and line bundles give a topological/cohomological framework for their analysis.

The sections of $\mathcal{O}(D)$ acquire a parallel divisor-theoretic interpretation.

**Proposition 1.4.7: Sections of $\mathcal{O}(D)$ via Meromorphic Functions**

Let $D$ be a Cartier divisor on a complex manifold $X$. There is a canonical $\mathbb{C}$-vector-space isomorphism

$$H^0(X, \mathcal{O}(D)) \cong \{f \in \mathcal{M}^*(X) : (f) + D \geq 0\} \cup \{0\}$$

where $\mathcal{M}^*(X)$ denotes the global non-zero meromorphic functions. The zero locus of a non-zero section is a divisor in the linear-equivalence class of $D$.

***Proof*:**
Suppose $D$ is given by local data $\{(U_\alpha, f_\alpha)\}$. A holomorphic section of $\mathcal{O}(D)$ is, in trivializations, a family of holomorphic functions $\{s_\alpha \colon U_\alpha \to \mathbb{C}\}$ with $s_\alpha = (f_\alpha/f_\beta)\, s_\beta$ on overlaps, rearranging to $s_\alpha\, f_\beta = s_\beta\, f_\alpha$, that is, $s_\alpha/f_\alpha = s_\beta/f_\beta$. The local meromorphic functions $s_\alpha/f_\alpha$ patch into a global meromorphic function $f$ with the property that $f \cdot f_\alpha = s_\alpha$ is holomorphic on $U_\alpha$. The divisor of $f$ satisfies $\mathrm{ord}_Y(f) + \mathrm{ord}_Y(f_\alpha) \geq 0$ on each $U_\alpha$, that is, $(f) + D \geq 0$ globally.

Conversely, a global meromorphic $f$ with $(f) + D \geq 0$ determines holomorphic functions $s_\alpha := f \cdot f_\alpha$ on $U_\alpha$ that satisfy the cocycle relation; the assignment $f \mapsto \{s_\alpha\}$ is the inverse map.

The zero locus of a non-zero section corresponds, on each chart, to the locus where $f \cdot f_\alpha = 0$. Globally, this is the divisor $(f) + D$, which is effective and linearly equivalent to $D$ via $f$, completing the proof.

The first running example is the hyperplane class on $\mathbb{CP}^n$, which identifies the cocycle-defined $\mathcal{O}(1)$ from section 1.3 with a divisor.

**Example 1.4.8: Hyperplane Class on $\mathbb{CP}^n$**

Let $H := V(z_0) = \{[z_0 : \cdots : z_n] : z_0 = 0\} \subset \mathbb{CP}^n$, a smooth analytic hypersurface. In the standard chart $U_i = \{z_i \neq 0\}$, the hypersurface $H \cap U_i$ is cut out by the holomorphic function $f_i := z_0/z_i$ (well-defined and holomorphic on $U_i$). The Cartier-divisor cocycle is

$$g_{ij} = \frac{f_i}{f_j} = \frac{z_0/z_i}{z_0/z_j} = \frac{z_j}{z_i}$$

matching the cocycle of $\mathcal{O}(1)$ from definition 1.3.10. Hence $\mathcal{O}(H) \cong \mathcal{O}(1)$ as holomorphic line bundles. More generally, $\mathcal{O}(dH) \cong \mathcal{O}(d)$ for every $d \in \mathbb{Z}$, and the section-divisor correspondence (proposition 1.4.7) recovers $H^0(\mathbb{CP}^n, \mathcal{O}(d)) = \mathbb{C}[z_0, \ldots, z_n]_d$ from theorem 1.3.11: a homogeneous polynomial $F$ of degree $d$ defines the global meromorphic function $F/z_0^d$ on $\mathbb{CP}^n$, whose divisor is $V(F) - dH$, satisfying $(F/z_0^d) + dH = V(F) \geq 0$.

The second example is the divisor of a smooth hypersurface, the geometric object that drives the Hodge-theoretic content of chapter 7.

**Example 1.4.9: Smooth Hypersurfaces as Divisors**

Let $F \in \mathbb{C}[z_0, \ldots, z_n]_d$ be a homogeneous polynomial of degree $d \geq 1$ with $\partial F/\partial z_0, \ldots, \partial F/\partial z_n$ having no common zero on $\mathbb{C}^{n+1} \setminus \{0\}$. The smooth hypersurface $V(F) = \{[z] \in \mathbb{CP}^n : F(z) = 0\}$ from example 1.1.9 is a Weil divisor on $\mathbb{CP}^n$. In the standard chart $U_i$, the local equation is the dehomogenization $F/z_i^d$, holomorphic and non-vanishing outside $V(F) \cap U_i$. The cocycle of the associated line bundle is

$$g_{ij} = \frac{F/z_i^d}{F/z_j^d} = \left(\frac{z_j}{z_i}\right)^d$$

matching $\mathcal{O}(d)$. Hence $\mathcal{O}(V(F)) \cong \mathcal{O}(d)$, and $V(F)$ is realized as the zero locus of the global section $F \in H^0(\mathbb{CP}^n, \mathcal{O}(d))$ from theorem 1.3.11.

The blow-up of a point completes the geometric apparatus. The construction begins on $\mathbb{C}^n$ at the origin and globalizes via charts.

**Definition 1.4.10: Blow-Up of $\mathbb{C}^n$ at the Origin**

The *blow-up of $\mathbb{C}^n$ at the origin* is the closed analytic subset

$$\widetilde{\mathbb{C}^n} := \{(z, [\ell]) \in \mathbb{C}^n \times \mathbb{CP}^{n-1} : z \in \mathbb{C} \cdot \ell\}$$

together with the natural projection $\sigma \colon \widetilde{\mathbb{C}^n} \to \mathbb{C}^n$ to the first factor, $(z, [\ell]) \mapsto z$. The condition $z \in \mathbb{C} \cdot \ell$ requires $z$ to lie on the line spanned by $\ell$, equivalently $z_i \ell_j = z_j \ell_i$ for all $i, j$. Outside the origin, the projection is a biholomorphism $\widetilde{\mathbb{C}^n} \setminus \sigma^{-1}(0) \to \mathbb{C}^n \setminus \{0\}$; the fiber $E := \sigma^{-1}(0)$ over the origin, called the *exceptional divisor*, is canonically isomorphic to $\mathbb{CP}^{n-1}$.

**Proposition 1.4.11: Smoothness and Atlas of $\widetilde{\mathbb{C}^n}$**

The blow-up $\widetilde{\mathbb{C}^n}$ is a smooth complex manifold of complex dimension $n$. The standard atlas $\{V_k\}_{k=1}^n$ on $\widetilde{\mathbb{C}^n}$ is given by

$$V_k := \widetilde{\mathbb{C}^n} \cap (\mathbb{C}^n \times \{[\ell] : \ell_k \neq 0\})$$

with chart coordinates $(z_k, w_1, \ldots, \widehat{w_k}, \ldots, w_n)$ where $w_j := \ell_j/\ell_k$ for $j \neq k$. The embedding into $\mathbb{C}^n \times \mathbb{C}\mathrm{P}^{n-1}$ in chart $V_k$ sends $(z_k, w) \mapsto ((z_k w_1, \ldots, z_k w_{k-1}, z_k, z_k w_{k+1}, \ldots, z_k w_n), [w_1 : \cdots : 1 : \cdots : w_n])$ where the 1 is in position $k$.

***Proof*:**
On the open set where $\ell_k \neq 0$, the condition $z_i \ell_j = z_j \ell_i$ for all $i, j$ reduces to $z_j = (\ell_j/\ell_k)\, z_k = w_j\, z_k$ for $j \neq k$. The equations express $z_j$ as a holomorphic function of $z_k$ and the $w_j$, so $V_k$ is the graph of these functions over $\mathbb{C}^n$ (with coordinates $(z_k, (w_j)_{j\neq k})$), making $V_k$ a smooth complex submanifold of $\mathbb{C}^n \times \{\ell_k \neq 0\}$ of complex dimension $n$, where $\{\ell_k \neq 0\} \subset \mathbb{C}\mathrm{P}^{n-1}$ is the $k$-th standard chart. The charts $\{V_k\}$ cover $\widetilde{\mathbb{C}^n}$ because every $[\ell] \in \mathbb{C}\mathrm{P}^{n-1}$ has at least one non-zero coordinate. Transitions on overlaps $V_k \cap V_l$ are given by the standard $\mathbb{C}\mathrm{P}^{n-1}$ transitions on the projective factor, hence holomorphic, completing the proof.

The blow-up at a point on a general complex manifold is built from the local model.

**Definition 1.4.12: Blow-Up of a Complex Manifold at a Point**

Let $X$ be a complex manifold of complex dimension $n$ and $p \in X$. Choose a holomorphic chart $\varphi\colon U \to V \subseteq \mathbb{C}^n$ with $\varphi(p) = 0$, perform the blow-up of $V$ at the origin to obtain $\widetilde{V}$, and glue $\widetilde{V}$ to $X \setminus \{p\}$ along the biholomorphism $\widetilde{V} \setminus E \cong V \setminus \{0\} \cong U \setminus \{p\}$. The resulting complex manifold is the *blow-up of $X$ at $p$*, denoted $\widetilde{X} = \mathrm{Bl}_p X$, and the natural projection $\sigma\colon \widetilde{X} \to X$ is biholomorphic on $\widetilde{X} \setminus E$ and contracts $E := \sigma^{-1}(p) \cong \mathbb{C}\mathrm{P}^{n-1}$ to $p$. The construction is independent of the choice of chart up to canonical isomorphism.

The chart-independence is verified in exercise 1.4.1 (3); intuitively, the exceptional divisor remembers the tangent space $T_pX$ rather than the chart used to identify a neighborhood.

A worked computation makes the local geometry transparent in the two-dimensional case.

**Example 1.4.13: Blow-Up of $\mathbb{C}^2$ at the Origin**

The blow-up of $\mathbb{C}^2$ at the origin is

$$\widetilde{\mathbb{C}^2} = \{((x, y), [u : v]) \in \mathbb{C}^2 \times \mathbb{C}\mathrm{P}^1 : xv = yu\}$$

with two charts:

- In $V_u := \{u \neq 0\}$, set $t := v/u$. The condition $xv = yu$ becomes $y = xt$, parametrizing $V_u$ by $(x, t) \in \mathbb{C}^2$ via $(x, t) \mapsto ((x, xt), [1 : t])$.
- In $V_v := \{v \neq 0\}$, set $s := u/v$. The condition $xv = yu$ becomes $x = sy$, parametrizing $V_v$ by $(s, y) \in \mathbb{C}^2$ via $(s, y) \mapsto ((sy, y), [s : 1])$.

On the overlap $V_u \cap V_v$ (where both $u, v \neq 0$, so $t \neq 0$), the transition $(x,t) \mapsto (s,y)$ is given by $s = 1/t$ and $y = xt$, holomorphic where $t \neq 0$ and inverse to $(s,y) \mapsto (x,t) = (sy, 1/s)$ where $s \neq 0$.

The exceptional divisor $E$ is given by $x = 0$ in $V_u$ (parametrized by the coordinate $t$) and by $y = 0$ in $V_v$ (parametrized by $s$). The transition $s = 1/t$ on $E$ is the standard $\mathbb{C}\mathrm{P}^1$ chart change, confirming $E \cong \mathbb{C}\mathrm{P}^1$.

The line bundle $\mathcal{O}(E)$ has local equation $x$ on $V_u$ and $y$ on $V_v$. The cocycle on $V_u \cap V_v$ is

$$g = \frac{x}{y} = \frac{x}{xt} = \frac{1}{t} = s$$

(using $y = xt$ on the overlap). Restricting to $E$ where $x = y = 0$, the cocycle is $1/t = s$, viewed as a function on $E \cap (V_u \cap V_v) \cong \mathbb{C}^*$ in either coordinate. This is precisely the cocycle of $\mathcal{O}_{\mathbb{C}\mathrm{P}^1}(-1)$ from example 1.3.9, so $\mathcal{O}(E)|_E \cong \mathcal{O}_{\mathbb{C}\mathrm{P}^1}(-1)$.

The identification $\mathcal{O}(E)|_E \cong \mathcal{O}(-1)$ is not an accident: the line bundle of the exceptional divisor restricts to the tautological line bundle on the exceptional divisor itself. This appears in every blow-up computation, and the dimension-$n$ analogue $\mathcal{O}(E)|_E \cong \mathcal{O}_{\mathbb{C}\mathrm{P}^{n-1}}(-1)$ holds for the blow-up of any point on a complex manifold of dimension $n$ (exercise 1.4.1 (5)).

The last definition tracks how subvarieties transform under blow-up.

**Definition 1.4.14: Strict Transform**

Let $\sigma\colon \widetilde{X} \to X$ be the blow-up of a complex manifold $X$ at a point $p$, and $Y \subset X$ a closed analytic subspace. The *strict transform* of $Y$ is the closed analytic subspace

$$\widetilde{Y} := \overline{\sigma^{-1}(Y \setminus \{p\})}$$

the closure (in $\widetilde{X}$) of the preimage of the non-blown-up part of $Y$.

If $p \notin Y$, the strict transform $\widetilde{Y}$ is just $\sigma^{-1}(Y)$ and the blow-up is harmless. If $p \in Y$, the strict transform separates $Y$ from the exceptional divisor: the intersection $\widetilde{Y} \cap E$ records the tangent directions of $Y$ at $p$. For a smooth curve $C$ through the origin in $\mathbb{C}^2$ with tangent direction $[a : b] \in \mathbb{C}\mathrm{P}^1$, the strict transform $\widetilde{C}$ meets $E$ at the single point $[a : b] \in \mathbb{C}\mathrm{P}^1 \cong E$ (exercise 1.4.1 (4)). The blow-up has thus "resolved the tangent directions" of curves through $p$ into distinct points on $E$.

## Exercise 1.4.1

1. Show that any two distinct hyperplanes $H, H' \subset \mathbb{C}\mathrm{P}^n$ are linearly equivalent. Exhibit explicitly a non-zero global meromorphic function $f$ on $\mathbb{C}\mathrm{P}^n$ with $(f) = H - H'$.

2. For a homogeneous polynomial $F \in \mathbb{C}[z_0, \ldots, z_n]_d$ with $d \geq 1$ (not necessarily smooth), describe the divisor of the corresponding section of $\mathcal{O}(d)$. Conclude that the projective space of degree-$d$ hypersurfaces in $\mathbb{C}\mathrm{P}^n$ has dimension $\binom{n+d}{d} - 1$, and identify the dimension of the projective space of conics in $\mathbb{C}\mathrm{P}^2$ ($n = 2$, $d = 2$).

3. Show that the blow-up $\mathrm{Bl}_p X$ of definition 1.4.12 is well-defined up to canonical isomorphism: if $\varphi_1, \varphi_2\colon U \to V_1, V_2 \subseteq \mathbb{C}^n$ are two holomorphic charts around $p \in X$ with $\varphi_i(p) = 0$, the resulting blow-ups $\widetilde{V_1}$ and $\widetilde{V_2}$ glue to $X \setminus \{p\}$ in canonically isomorphic ways. (Hint: the gluing is determined by the differential $d(\varphi_2 \circ \varphi_1^{-1})_0 \in \mathrm{GL}_n(\mathbb{C})$.)

4. Let $C \subset \mathbb{C}^2$ be a smooth holomorphic curve through the origin with tangent direction $[a : b] \in \mathbb{CP}^1$ at the origin. Show that the strict transform $\widetilde{C} \subset \widetilde{\mathbb{C}^2}$ is a smooth curve meeting the exceptional divisor $E \cong \mathbb{CP}^1$ at the single point $[a : b]$. (Hint: parametrize $C$ near the origin and lift to $\widetilde{\mathbb{C}^2}$ in the appropriate chart.)

5. Generalize example 1.4.13: for the blow-up $\mathrm{Bl}_0\mathbb{C}^n$ with exceptional divisor $E \cong \mathbb{CP}^{n-1}$, show that $\mathcal{O}(E)|_E \cong \mathcal{O}_{\mathbb{CP}^{n-1}}(-1)$ by computing the cocycle of $\mathcal{O}(E)$ in the standard atlas of proposition 1.4.11 and restricting to $E$.

6. Let $L \to X$ be a holomorphic line bundle on a connected complex manifold $X$, and $s \in H^0(X, L)$ a non-zero global holomorphic section. Show that the zero locus $\{s = 0\}$ defines an effective Cartier divisor $D_s$ on $X$ with $\mathcal{O}(D_s) \cong L$. (Conclusion: every line bundle with a non-zero global section is the line bundle of an effective divisor. The converse to surjectivity in theorem 1.4.6 requires the existence of meromorphic sections, which is automatic on smooth projective $X$.)

# 2

# Sheaf and Dolbeault Cohomology

Chapter 1 attached to every complex manifold $X$ the Dolbeault complex and its cohomology $H^{p,q}_{\bar{\partial}}(X)$, a bidegree-refined invariant assembled from smooth $(p,q)$-forms and the operator $\bar{\partial}$. That construction is analytic: $H^{p,q}_{\bar{\partial}}(X)$ measures the failure of $\bar{\partial}$ to be locally solvable in the large. Left in this form the groups are hard to compute and their functoriality is opaque. This chapter identifies them with a holomorphic invariant, the sheaf cohomology $H^q(X, \Omega^p_X)$ of the holomorphic $p$-forms, and thereby makes Dolbeault cohomology as tractable as any derived-functor cohomology.

The identification is Dolbeault's theorem, and its mechanism is a soft resolution. Sheaf cohomology on a paracompact manifold can be computed from any acyclic resolution; the sheaves $\mathcal{A}^{p,q}$ of smooth forms are fine, hence acyclic, and the $\bar{\partial}$-Poincaré lemma shows that they resolve $\Omega^p_X$. Assembling these facts yields $H^q(X, \Omega^p_X) \cong H^{p,q}_{\bar{\partial}}(X)$. The chapter closes by organizing the individual bidegrees into one object, the Frölicher spectral sequence relating Dolbeault to de Rham cohomology, and by recording the comparison theorem (GAGA) that ties this analytic picture to the algebraic geometry of [4].

## 2.1 Sheaf Cohomology and Fine Resolutions

The Dolbeault cohomology $H^{p,q}_{\bar{\partial}}(X)$ built in chapter 1 is an analytic invariant: it is the cohomology of a complex of infinite-dimensional spaces of smooth forms, and nothing so far says it is finite-dimensional, computable, or tied to the holomorphic geometry of $X$. The bridge that makes it all three is sheaf cohomology. This chapter identifies $H^{p,q}_{\bar{\partial}}(X)$ with the sheaf cohomology $H^q(X, \Omega^p_X)$ of the holomorphic $p$-forms, and the identification runs through a single mechanism: the smooth-form sheaves resolve $\Omega^p_X$, and they do so by *fine* sheaves, which carry no higher cohomology on a paracompact space. This section supplies that mechanism.

The sheaf-cohomology machinery itself is developed in full in [4], and it is not redeveloped here. What this section owns is the piece that book does not: the fact that the sheaves of smooth forms on a complex manifold are fine, together with the abstract de Rham principle in the form the comparison theorem will use. The payoff, that a fine resolution computes sheaf cohomology, is proved here from partitions of unity, because it is exactly the tool the rest of the chapter runs on.

Recall from [4] the setup. To a sheaf $\mathcal{F}$ of abelian groups on a topological space $X$ one associates its cohomology groups $H^q(X, \mathcal{F})$ for $q \geq 0$. They are defined as the right derived functors of the global-sections functor $\mathcal{F} \mapsto \mathcal{F}(X) = \Gamma(X, \mathcal{F})$ on the abelian category of sheaves. Concretely, one chooses an injective resolution $0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$, applies global sections, and takes cohomology of the resulting complex $\Gamma(X, \mathcal{I}^\bullet)$; the groups $H^q(X, \mathcal{F})$ do not depend on the choice of resolution. Three properties are all this section needs, and each is proved in [4]:

1. $H^0(X, \mathcal{F}) = \Gamma(X, \mathcal{F})$, the global sections.
2. A short exact sequence of sheaves $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ induces a long exact cohomology sequence
$$\cdots \to H^q(X, \mathcal{F}') \to H^q(X, \mathcal{F}) \to H^q(X, \mathcal{F}'') \to H^{q+1}(X, \mathcal{F}') \to \cdots$$
3. A sheaf $\mathcal{F}$ with $H^q(X, \mathcal{F}) = 0$ for all $q \geq 1$ is called *acyclic*. Injective sheaves, and more generally flasque sheaves, are acyclic.

The long exact sequence is the engine: it converts geometric input (an exact sequence of sheaves) into relations among cohomology groups. The definition through injective resolutions is theoretically clean but computationally useless, since injective sheaves are enormous. The remedy, developed below, is that any resolution by acyclic sheaves serves just as well, and acyclic sheaves adapted to smooth geometry are abundant.

The abundance comes from partitions of unity. A sheaf built so that its sections can be cut apart and reassembled along a locally finite cover has no room for the cohomological obstructions that the derived functor measures. Two nested notions capture this. The stronger one, fineness, asks that the sheaf admit multiplication by smooth bump functions; the weaker one, softness, asks only that sections extend off closed sets. Both are engineered to kill higher cohomology on well-behaved spaces.

**Definition 2.1.1: Fine and Soft Sheaves**

Let $X$ be a paracompact Hausdorff space.
A sheaf $\mathcal{F}$ of abelian groups on $X$ is *soft* if every section over a closed set extends to a global section: for every closed $A \subseteq X$, the restriction map $\Gamma(X, \mathcal{F}) \to \Gamma(A, \mathcal{F}|_A)$ is surjective, where $\Gamma(A, \mathcal{F}|_A)$ denotes sections of the restricted sheaf on the subspace $A$.
Let now $\mathcal{R}$ be a sheaf of commutative rings on $X$ and $\mathcal{F}$ a sheaf of $\mathcal{R}$-modules. Then $\mathcal{F}$ is *fine over* $\mathcal{R}$ if for every locally finite open cover $\{U_i\}_{i\in I}$ of $X$ there is a family of endomorphisms $\eta_i \colon \mathcal{F} \to \mathcal{F}$ (a *partition of unity subordinate to the cover*, acting by $\mathcal{R}$-module maps) such that

1. $\eta_i$ is supported in $U_i$, meaning there is a closed set $S_i \subseteq U_i$ with $\eta_i$ acting as zero on stalks outside $S_i$;
2. the supports $\{S_i\}$ form a locally finite family;
3. $\sum_{i\in I} \eta_i = \mathrm{id}_{\mathcal{F}}$, a sum that is locally finite and hence well-defined.

Fineness is a property of the pair $(\mathcal{F}, \mathcal{R})$ rather than of $\mathcal{F}$ alone, since the $\eta_i$ are required to be $\mathcal{R}$-linear. Two choices of $\mathcal{R}$ occur below: the smooth-function sheaf $\mathcal{A}^0$, which is the ring for every sheaf shown to be fine in this chapter, and $\mathcal{F}$ itself, used when testing $\underline{\mathbb{C}}$ and $\mathcal{O}_X$ in the exercises. Where no ring is named, $\mathcal{R} = \mathcal{A}^0$ is meant.

The definition of fineness is stated for $\mathcal{R}$-modules because that is where it will be applied: the smooth-function sheaf multiplies the smooth-form sheaves, and a smooth partition of unity supplies exactly the endomorphisms $\eta_i$. Softness is the coarser property, phrased purely in terms of extending sections, and it is the one directly responsible for vanishing cohomology; fineness is the convenient sufficient condition that geometry hands over for free. On a paracompact space every fine sheaf is soft, since multiplication by a partition-of-unity element supported near a closed set produces the required extension, and the vanishing theorem below can be stated for soft sheaves. The proof is given for fine sheaves because that is the case the chapter uses and because the partition-of-unity endomorphisms make the argument transparent.

**2.1.2 Note: fine versus soft** On a paracompact space fine implies soft, and both imply acyclicity. The reverse implications fail: a soft sheaf need not be a module over any ring of functions, so need not be fine. For the purposes of this book the distinction is immaterial. Every sheaf that arises below is a module over the sheaf $\mathcal{A}^0$ of smooth functions and is fine by proposition 2.1.4, so fineness is the property invoked throughout.

The prototype to keep in mind is the sheaf $\mathcal{A}^0$ of smooth complex-valued functions on a smooth manifold. A smooth partition of unity $\{\rho_i\}$ subordinate to a locally finite cover $\{U_i\}$ gives endomorphisms $\eta_i(f) = \rho_i f$ of $\mathcal{A}^0$; they are supported in $U_i$, their supports are locally finite, and $\sum_i \rho_i = 1$ gives $\sum_i \eta_i = \mathrm{id}$. So $\mathcal{A}^0$ is fine over itself. By contrast, the constant sheaf $\underline{\mathbb{C}}$ and the structure sheaf $\mathcal{O}_X$ of holomorphic functions are *not* fine, because there are no holomorphic (or locally constant) bump functions to build the $\eta_i$; a nonzero holomorphic function supported in a small ball does not exist. The failure of fineness for $\underline{\mathbb{C}}$ and $\mathcal{O}_X$ is precisely why they carry interesting cohomology, and the fineness of the smooth-form sheaves is what lets those smooth sheaves compute it.

The payoff of fineness is that it forces higher cohomology to vanish. This is the technical heart of the section, and the proof is a direct partition-of-unity argument on Čech cochains. The strategy is standard for soft or fine sheaves: given a cohomology class of positive degree, one uses the partition of unity to split any cocycle into pieces small enough to be individually trivialized, and reassembles the trivializations into a global primitive.

**Proposition 2.1.3: Fine Sheaves Are Acyclic**

Let $X$ be a paracompact Hausdorff space and $\mathcal{F}$ a fine sheaf on $X$. Then $H^q(X, \mathcal{F}) = 0$ for all $q \geq 1$.

***Proof*:**

Two routes are available and they agree. The direct one is the Čech computation below, which uses the partition of unity explicitly and is worth seeing because it is the mechanism, not the conclusion; it shows $\check{H}^q(\mathcal{U}, \mathcal{F}) = 0$ for every locally finite cover, and Čech cohomology agrees with derived-functor cohomology in every degree on a paracompact space ([4, Čech cohomology on a paracompact space]). The structural route is shorter: a fine sheaf is soft, and soft sheaves are acyclic ([4, fine and flasque sheaves are soft] and [4, soft sheaves are acyclic]), which gives the result with no Čech computation at all.

*Step 1: Reduction to a locally finite cover and its partition of unity.* Fix $q \geq 1$ and a Čech cohomology class represented by a cocycle. By definition the Čech cohomology $\check{H}^q(X, \mathcal{F})$ is the colimit of $\check{H}^q(\mathcal{U}, \mathcal{F})$ over open covers $\mathcal{U}$ ordered by refinement, and every open cover of a paracompact space admits a locally finite refinement. So the class is represented by a Čech $q$-cocycle $c = (c_{i_0 \cdots i_q})$ on a locally finite open cover $\mathcal{U} = \{U_i\}_{i \in I}$, where $c_{i_0 \cdots i_q} \in \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_q})$ is alternating in its indices and satisfies the cocycle condition $\delta c = 0$, that is

$$\sum_{k=0}^{q+1} (-1)^k \, c_{i_0 \cdots \widehat{i_k} \cdots i_{q+1}} = 0$$

on each $(q+2)$-fold intersection $U_{i_0} \cap \cdots \cap U_{i_{q+1}}$. Because $\mathcal{F}$ is fine and $\mathcal{U}$ is locally finite, fix a partition of unity $\{\eta_i\}_{i \in I}$ subordinate to $\mathcal{U}$: endomorphisms with $\eta_i$ supported in a closed $S_i \subseteq U_i$, the family $\{S_i\}$ locally finite, and $\sum_i \eta_i = \mathrm{id}_{\mathcal{F}}$.

*Step 2: Construction of a candidate primitive.* The goal is a Čech $(q-1)$-cochain $b = (b_{i_0 \cdots i_{q-1}})$ with $\delta b = c$. For each multi-index $i_0, \ldots, i_{q-1}$ define

$$b_{i_0 \cdots i_{q-1}} = \sum_{j \in I} \eta_j \big( c_{j \, i_0 \cdots i_{q-1}} \big)$$

where the section $c_{j \, i_0 \cdots i_{q-1}} \in \mathcal{F}(U_j \cap U_{i_0} \cap \cdots \cap U_{i_{q-1}})$ is first cut off by $\eta_j$. Here the key point is that $\eta_j(c_{j \, i_0 \cdots i_{q-1}})$, though initially defined only on $U_j \cap U_{i_0} \cap \cdots \cap U_{i_{q-1}}$, extends by zero to a section over all of $U_{i_0} \cap \cdots \cap U_{i_{q-1}}$: the endomorphism $\eta_j$ acts as zero on stalks outside $S_j$, so $\eta_j(c_{j \, i_0 \cdots i_{q-1}})$ has support in the closed set $S_j$ and vanishes near the boundary of $U_j$ inside $U_{i_0} \cap \cdots \cap U_{i_{q-1}}$; extending it by 0 over the part of $U_{i_0} \cap \cdots \cap U_{i_{q-1}}$ outside $U_j$ produces a well-defined section on $U_{i_0} \cap \cdots \cap U_{i_{q-1}}$, because the two definitions agree on the overlap (both being $\eta_j$ applied to the same section, which vanishes there). The sum over $j$ is locally finite by local finiteness of the supports $\{S_j\}$, so $b_{i_0 \cdots i_{q-1}}$ is a well-defined section of $\mathcal{F}$ over $U_{i_0} \cap \cdots \cap U_{i_{q-1}}$, and $b$ is a genuine Čech $(q-1)$-cochain.

*Step 3: Verification that $\delta b = c$.* Compute the coboundary on a $q$-fold intersection $U_{i_0} \cap \cdots \cap U_{i_q}$.

By definition of $\delta$,

$$(\delta b)_{i_0\cdots i_q} = \sum_{k=0}^{q}(-1)^k\, b_{i_0\cdots\widehat{i_k}\cdots i_q} = \sum_{k=0}^{q}(-1)^k \sum_{j\in I}\eta_j\big(c_{j\, i_0\cdots\widehat{i_k}\cdots i_q}\big)$$

Interchange the two finite (locally finite) sums and pull out the endomorphism $\eta_j$, which commutes with the alternating sum because each $\eta_j$ is additive and compatible with restriction:

$$(\delta b)_{i_0\cdots i_q} = \sum_{j\in I}\eta_j\left(\sum_{k=0}^{q}(-1)^k\, c_{j\, i_0\cdots\widehat{i_k}\cdots i_q}\right)$$

Now examine the inner sum. Apply the cocycle condition $\delta c = 0$ to the $(q+1)$-tuple of indices $(j, i_0, \ldots, i_q)$: writing out $\sum_{k=0}^{q+1}(-1)^k c_{\cdots\widehat{k}\cdots} = 0$ for this tuple, the $k=0$ term drops the leading index $j$ and equals $c_{i_0\cdots i_q}$, while the terms $k = 1, \ldots, q+1$ drop one of $i_0, \ldots, i_q$ and contribute, with a shifted sign, exactly the inner sum above. Concretely, the cocycle identity reads

$$c_{i_0\cdots i_q} - \sum_{k=0}^{q}(-1)^k\, c_{j\, i_0\cdots\widehat{i_k}\cdots i_q} = 0$$

so the inner sum equals $c_{i_0\cdots i_q}$ (which does not involve $j$). Substituting,

$$(\delta b)_{i_0\cdots i_q} = \sum_{j\in I}\eta_j\big(c_{i_0\cdots i_q}\big) = \left(\sum_{j\in I}\eta_j\right)\big(c_{i_0\cdots i_q}\big) = c_{i_0\cdots i_q}$$

using $\sum_j \eta_j = \mathrm{id}_{\mathcal{F}}$. Thus $\delta b = c$, so the cocycle $c$ is a coboundary and its class in $\check{H}^q(\mathcal{U}, \mathcal{F})$ vanishes. Since every class over an arbitrary cover refines to such a locally finite cover, every class in $\check{H}^q(X, \mathcal{F}) = H^q(X, \mathcal{F})$ is zero for $q \geq 1$, as we sought to show.

Fineness lets a global cocycle be dismantled index by index and reassembled into a primitive, and the cocycle condition is exactly what guarantees the pieces fit. No local vanishing lemma is needed, only the ability to multiply sections by bump functions. This is why the smooth category is so much softer than the holomorphic or topological one. In the holomorphic category the analogue fails, because $\mathcal{O}_X$ is not fine, and recovering primitives requires a local solvability theorem instead, the $\bar\partial$-Poincaré lemma of section 2.2. The division of labor for the whole chapter is already visible: fineness handles the global gluing, and a local lemma handles the local exactness.

With the vanishing theorem in hand, the task is to certify that the specific sheaves of interest are fine. The sheaves $\mathcal{A}^{p,q}$ of smooth $(p,q)$-forms are modules over the sheaf $\mathcal{A}^0$ of smooth functions: a smooth function multiplies a smooth form to give a smooth form. Since $\mathcal{A}^0$ is fine, its module structure is inherited, and multiplication by a smooth partition of unity supplies the required endomorphisms.

**Proposition 2.1.4: Smooth Form Sheaves Are Fine**

Let $X$ be a complex manifold and $\mathcal{A}^0$ the sheaf of smooth complex-valued functions on $X$. For every $p, q \geq 0$ the sheaf $\mathcal{A}^{p,q}$ of smooth $(p,q)$-forms is a sheaf of $\mathcal{A}^0$-modules, and it is fine. In particular each $\mathcal{A}^{p,q}$ is acyclic: $H^k(X, \mathcal{A}^{p,q}) = 0$ for all $k \geq 1$. The same holds for the sheaf $\mathcal{A}^k$ of all smooth $k$-forms.

***Proof*:**
The underlying smooth manifold of a complex manifold is paracompact and Hausdorff (a complex manifold is a manifold, and manifolds are paracompact), so the ambient hypothesis of proposition 2.1.3 and definition 2.1.1 is met.

*Step 1: The module structure.* A section of $\mathcal{A}^{p,q}$ over an open set $U$ is a smooth $(p,q)$-form on $U$ (definition 1.2.2). If $f \in \mathcal{A}^0(U)$ is a smooth function and $\alpha \in \mathcal{A}^{p,q}(U)$, then $f\alpha$ is again a smooth $(p,q)$-form: multiplication by a function preserves the type $(p,q)$, since it multiplies each coefficient of the local expression $\alpha = \sum a_{IJ}\, dz_I \wedge d\bar{z}_J$ by $f$ without changing the differentials $dz_I \wedge d\bar{z}_J$. This makes $\mathcal{A}^{p,q}$ a sheaf of $\mathcal{A}^0$-modules; the case $\mathcal{A}^k = \bigoplus_{p+q=k} \mathcal{A}^{p,q}$ is a direct sum of such modules and is likewise an $\mathcal{A}^0$-module.

*Step 2: Construction of the partition of unity.* Let $\{U_i\}_{i\in I}$ be a locally finite open cover of $X$. Because $X$ is a paracompact smooth manifold, there is a smooth partition of unity $\{\rho_i\}_{i\in I}$ subordinate to $\{U_i\}$: each $\rho_i \in \mathcal{A}^0(X)$ is a smooth function with $\mathrm{supp}(\rho_i) \subseteq U_i$ a closed set, the family of supports is locally finite, and $\sum_i \rho_i \equiv 1$. Existence of such a partition is the defining feature of paracompact smooth manifolds and is established in the smooth-manifold foundations ([26]). Define $\eta_i \colon \mathcal{A}^{p,q} \to \mathcal{A}^{p,q}$ by multiplication, $\eta_i(\alpha) = \rho_i \alpha$. Each $\eta_i$ is an $\mathcal{A}^0$-module endomorphism, since multiplication by the function $\rho_i$ commutes with multiplication by any $f \in \mathcal{A}^0$.

*Step 3: Verification of the fineness axioms.* Check the three conditions of definition 2.1.1 against the closed sets $S_i = \mathrm{supp}(\rho_i) \subseteq U_i$:

1. $\eta_i$ is supported in $U_i$: if $\alpha$ is a germ at a point outside $S_i$, then $\rho_i$ vanishes on a neighborhood of that point, so $\eta_i(\alpha) = \rho_i\alpha = 0$ on stalks outside $S_i$.
2. The supports $\{S_i\}$ form a locally finite family, which is exactly the local finiteness of the partition of unity.
3. $\sum_i \eta_i = \mathrm{id}$: for any local section $\alpha$, the sum $\sum_i \rho_i\alpha = (\sum_i \rho_i)\alpha = \alpha$ is locally finite (only finitely many $\rho_i$ are nonzero near any point) and equals $\alpha$ because $\sum_i \rho_i \equiv 1$.

So $\{\eta_i\}$ is a partition of unity subordinate to $\{U_i\}$ in the sense of definition 2.1.1, and $\mathcal{A}^{p,q}$ is fine. By proposition 2.1.3, $H^k(X, \mathcal{A}^{p,q}) = 0$ for all $k \geq 1$; the identical argument applies to $\mathcal{A}^k$, completing the proof.

This proposition is the crux on which section 2.3 turns. The smooth-form sheaves are, by themselves, cohomologically inert; they contribute nothing to $H^k(X, -)$ in positive degrees. Whatever cohomology a complex manifold carries must therefore live in how these inert sheaves fit together into a complex, not in any single one of them. That is precisely the content of a resolution: a long exact sequence of acyclic sheaves whose cohomology is squeezed entirely into the object being resolved. The next principle makes this quantitative.

The definition of $H^q(X, \mathcal{F})$ through injective resolutions is replaced, for computation, by the observation that any acyclic resolution does the job. This is the abstract de Rham theorem. Its proof is a diagram chase through the long exact sequences of a filtered resolution and is carried out in full in [4]; per the cross-book convention the statement is owned here (it is the exact form the chapter applies) while the proof is a citation, not a sketch.

**Theorem 2.1.5: Abstract de Rham Theorem**

Let $X$ be a paracompact Hausdorff space and $\mathcal{F}$ a sheaf of abelian groups on $X$. Suppose

$$0 \to \mathcal{F} \xrightarrow{\varepsilon} \mathcal{L}^0 \xrightarrow{d^0} \mathcal{L}^1 \xrightarrow{d^1} \mathcal{L}^2 \to \cdots$$

is a resolution of $\mathcal{F}$ by *acyclic* sheaves $\mathcal{L}^q$, meaning the sequence is exact as a sequence of sheaves and $H^k(X, \mathcal{L}^q) = 0$ for all $q$ and all $k \geq 1$. Then for every $q \geq 0$ there is a canonical isomorphism

$$H^q(X, \mathcal{F}) \cong \frac{\ker\big(d^q \colon \Gamma(X, \mathcal{L}^q) \to \Gamma(X, \mathcal{L}^{q+1})\big)}{\operatorname{im}\big(d^{q-1} \colon \Gamma(X, \mathcal{L}^{q-1}) \to \Gamma(X, \mathcal{L}^q)\big)}$$

that is, the sheaf cohomology of $\mathcal{F}$ is computed as the cohomology of the complex of global sections $\Gamma(X, \mathcal{L}^\bullet)$ obtained by applying the global-sections functor to the resolution (dropping $\mathcal{F}$ itself). The isomorphism is natural in $\mathcal{F}$ and independent of the choice of acyclic resolution.

***Proof Key ideas*:**
The full proof is given in [6, derived functors and acyclic functors], where it is proved for an arbitrary right-derived functor; the sheaf-cohomology case is the specialization to global sections. It is a DAG citation to the upstream homological development, not a gap in the present argument.

$$0 \to \mathcal{Z}^q \to \mathcal{L}^q \to \mathcal{Z}^{q+1} \to 0, \qquad \mathcal{Z}^0 = \mathcal{F}, \quad \mathcal{Z}^{q+1} = \operatorname{im}(d^q) = \ker(d^{q+1})$$

where $\mathcal{Z}^q$ is the sheaf of $q$-cocycles. Each short exact sequence yields a long exact cohomology sequence; the acyclicity $H^k(X, \mathcal{L}^q) = 0$ for $k \geq 1$ collapses the connecting maps into isomorphisms $H^k(X, \mathcal{Z}^{q+1}) \cong H^{k+1}(X, \mathcal{Z}^q)$ for $k \geq 1$, and identifies $H^1(X, \mathcal{Z}^q)$ with the cokernel of $\Gamma(X, \mathcal{L}^q) \to \Gamma(X, \mathcal{Z}^{q+1})$. Iterating the dimension shift $H^q(X, \mathcal{F}) \cong H^1(X, \mathcal{Z}^{q-1}) \cong \cdots$ and unwinding the global-sections identifications produces the stated formula.

The abstract de Rham theorem is the license to compute. It says the injective (or flasque) resolution used to define $H^q(X, \mathcal{F})$ may be traded for any acyclic one, and the answer is unchanged. In practice the resolution one chooses is dictated by geometry: for the constant sheaf $\underline{\mathbb{C}}$ on a smooth manifold, the smooth de Rham complex; for the holomorphic $p$-forms $\Omega^p_X$ on a complex manifold, the Dolbeault complex of smooth $(p, q)$-forms. Both are acyclic resolutions by proposition 2.1.4, so both compute the relevant sheaf cohomology as the cohomology of a complex of global smooth forms. The first case is worked out immediately below; the second is the subject of section 2.3.

The clean illustration is the smooth de Rham complex. It resolves the constant sheaf, its sheaves are fine, and the abstract de Rham theorem then identifies sheaf cohomology of $\underline{\mathbb{C}}$ with de Rham cohomology. This is the de Rham theorem proper, and it is the exact template the Dolbeault comparison follows in the holomorphic setting, with $\underline{\mathbb{C}}$ replaced by $\Omega^p_X$ and $d$ by $\bar{\partial}$.

**Example 2.1.6: The Smooth de Rham Complex as a Fine Resolution**
Let $X$ be a complex manifold of complex dimension $n$, viewed as a smooth manifold of real dimension $2n$, and let $\mathcal{A}^k$ denote the sheaf of smooth complex-valued $k$-forms, with exterior derivative $d \colon \mathcal{A}^k \to \mathcal{A}^{k+1}$. Consider the complex of sheaves

$$0 \to \underline{\mathbb{C}} \xrightarrow{\iota} \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \mathcal{A}^2 \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{A}^{2n} \to 0 \tag{2.1}$$

where $\iota$ includes the locally constant functions into the smooth functions.

*Exactness.* The sequence (2.1) is exact as a sequence of sheaves. At $\mathcal{A}^0$: a smooth function $f$ with $df = 0$ is locally constant, so $\ker(d\colon \mathcal{A}^0 \to \mathcal{A}^1) = \underline{\mathbb{C}}$ and $\iota$ is injective onto this kernel. At $\mathcal{A}^k$ for $k \geq 1$: exactness is a stalk-level statement, and on stalks it is the smooth Poincaré lemma, that every closed $k$-form on a ball (equivalently, on a contractible neighborhood) is exact ([26]). Thus (2.1) is a resolution of $\underline{\mathbb{C}}$.

*Acyclicity.* Each $\mathcal{A}^k$ is fine by proposition 2.1.4, hence acyclic: $H^j(X, \mathcal{A}^k) = 0$ for $j \geq 1$.

*Conclusion.* Apply theorem 2.1.5 to the acyclic resolution (2.1). The complex of global sections is the smooth de Rham complex $\big(\Gamma(X, \mathcal{A}^\bullet), d\big) = \big(\mathcal{A}^\bullet(X), d\big)$, whose cohomology is by definition the de Rham cohomology $H^k_{\mathrm{dR}}(X, \mathbb{C})$. Therefore

$$H^k(X, \underline{\mathbb{C}}) \cong H^k_{\mathrm{dR}}(X, \mathbb{C})$$

for all $k$. Sheaf cohomology of the constant sheaf $\underline{\mathbb{C}}$, an object built from the topology of $X$, is computed by integrating differential forms. As a sanity check on $X = \mathbb{C}\mathrm{P}^1$ (example 1.1.8), which is diffeomorphic to the two-sphere $S^2$, one has $H^0_{\mathrm{dR}} = \mathbb{C}$ (connected), $H^1_{\mathrm{dR}} = 0$, and $H^2_{\mathrm{dR}} = \mathbb{C}$ (generated by the volume form / Fubini-Study class), matching $H^*(\mathbb{C}\mathrm{P}^1, \underline{\mathbb{C}})$.

The template is now explicit. To compute the sheaf cohomology of a sheaf $\mathcal{F}$, find a fine (hence acyclic) resolution of $\mathcal{F}$ by smooth sheaves and take cohomology of the global sections. For $\underline{\mathbb{C}}$ the resolution is the de Rham complex and the answer is de Rham cohomology. For the holomorphic $p$-forms $\Omega^p_X$, the resolution will be the Dolbeault complex $0 \to \Omega^p_X \to \mathcal{A}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1} \to \cdots$, and the answer will be Dolbeault cohomology $H^{p,q}_{\bar{\partial}}(X)$ (definition 1.2.10). The only missing ingredient is the exactness of that holomorphic resolution at the sheaf level, which is local solvability of $\bar{\partial}$.

### Exercise 2.1.1

1. Show that the constant sheaf $\underline{\mathbb{C}}$ on a connected complex manifold $X$ of positive dimension is not fine over itself, and that the structure sheaf $\mathcal{O}_X$ of holomorphic functions is not fine over itself either. Deduce that neither can be resolved by taking $\mathcal{L}^0 = \mathcal{F}$ alone (that is, that the identity resolution $0 \to \mathcal{F} \to \mathcal{F} \to 0$ does not exhibit $\mathcal{F}$ as acyclic in general). Explain in one sentence why this is consistent with $H^q(X, \underline{\mathbb{C}})$ and $H^q(X, \mathcal{O}_X)$ being nonzero for some $q \geq 1$.
2. Let $\mathcal{F}$ be a fine sheaf on a paracompact space with a partition of unity $\{\eta_i\}$ subordinate to a locally finite cover $\{U_i\}$. Verify the two facts used silently in the proof of proposition 2.1.3: that for a section $s$ defined on an intersection containing $U_i \cap V$, the section $\eta_i(s)$ extends by zero to a section on $V$; and that an $\mathcal{F}$-endomorphism $\eta_i$ commutes with the Čech coboundary $\delta$ applied index-wise.

## 2.2 The $\bar{\partial}$-Poincaré Lemma

The sheaf-cohomology engine of section 2.1 computes the cohomology of a sheaf from any acyclic resolution. To turn the Dolbeault complex into such a resolution, one input is still missing: local exactness. A resolution of a sheaf $\mathcal{F}$ is a complex of sheaves whose only cohomology sheaf is $\mathcal{F}$ itself in degree zero, which is to say the complex is exact as a sequence of sheaves. Exactness of a sequence of sheaves is a purely local condition, checked on stalks, so the question is whether the equation

$\bar{\partial}u = f$ can always be solved locally once the obvious necessary condition $\bar{\partial}f = 0$ holds. This section proves that it can. The result is the holomorphic analogue of the ordinary Poincaré lemma, which says that a closed differential form is locally exact for $d$; here the operator is $\bar{\partial}$ and the local model is a polydisc rather than a ball.

The strategy is the one that governs so much of complex geometry: reduce to one variable, solve there by an integral formula built from the Cauchy kernel, then bootstrap to several variables by induction. The one-variable step is an inhomogeneous Cauchy-Riemann equation, and its solution is the Cauchy transform. Everything downstream in this chapter rests on the local exactness established here, so the argument is given in full, including the full induction with no step abbreviated.

Fix coordinates and notation. Write $z = x + iy$ for a complex coordinate, so that

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\,\frac{\partial}{\partial y}\right)$$

is the Cauchy-Riemann operator, and a smooth function $g$ is holomorphic precisely when $\partial g/\partial \bar{z} = 0$ (this is definition 1.2.5 specialized to one variable). The area element in the coordinate $\zeta = \xi + i\eta$ is

$$\frac{i}{2}\,d\zeta \wedge d\bar{\zeta} = \frac{i}{2}(d\xi + i\,d\eta) \wedge (d\xi - i\,d\eta) = d\xi \wedge d\eta$$

so integration against $\frac{i}{2}\,d\zeta \wedge d\bar{\zeta}$ is integration against Lebesgue area measure. The one-variable engine is the *Cauchy-Pompeiu formula*, which upgrades the Cauchy integral formula to smooth functions that need not be holomorphic: for a smooth function $g$ on a neighborhood of the closed disc $\overline{D}$ of radius $r$ centered at the origin, and any $z \in D$,

$$g(z) = \frac{1}{2\pi i}\int_{\partial D} \frac{g(\zeta)}{\zeta - z}\,d\zeta + \frac{1}{2\pi i}\int_{D} \frac{\partial g/\partial \bar{\zeta}}{\zeta - z}\,d\zeta \wedge d\bar{\zeta}$$

The first term is the holomorphic Cauchy integral; the second measures the failure of $g$ to be holomorphic. Lemma 2.2.1 proves it. The $\bar{\partial}$-Poincaré lemma is what one gets by reading it backwards: the area integral, taken on its own, *produces* a function whose $\bar{\partial}$-derivative is a prescribed $f$.

**Lemma 2.2.1: The Cauchy-Pompeiu Formula**

Let $g$ be smooth on a neighborhood of the closed disc $\overline{D} \subseteq \mathbb{C}$ of radius $r$ about the origin. Then for every $z \in D$

$$g(z) = \frac{1}{2\pi i}\int_{\partial D} \frac{g(\zeta)}{\zeta - z}\,d\zeta + \frac{1}{2\pi i}\int_{D} \frac{\partial g/\partial \bar{\zeta}}{\zeta - z}\,d\zeta \wedge d\bar{\zeta}$$

and the area integral converges absolutely.

***Proof*:**

*Convergence.* In polar coordinates $\zeta = z + \rho e^{i\theta}$ the area element is $\rho\,d\rho\,d\theta$ while the integrand carries $\rho^{-1}$, so the integrand is $O(1)$ near $\zeta = z$ and the integral converges absolutely; this is the one place the two-dimensionality of the domain is used, and it is why no principal value is needed.

*Stokes on the punctured disc.* Fix $z \in D$ and $\varepsilon > 0$ small enough that $\overline{D_\varepsilon(z)} \subseteq D$, and put

$D_\varepsilon = D \setminus \overline{D_\varepsilon(z)}$. On $D_\varepsilon$ the function $\zeta \mapsto g(\zeta)/(\zeta - z)$ is smooth, and since $1/(\zeta - z)$ is holomorphic there,

$$d\left(\frac{g(\zeta)}{\zeta - z}\,d\zeta\right) = \frac{\partial g/\partial\bar{\zeta}}{\zeta - z}\,d\bar{\zeta} \wedge d\zeta = -\frac{\partial g/\partial\bar{\zeta}}{\zeta - z}\,d\zeta \wedge d\bar{\zeta}$$

The boundary of $D_\varepsilon$ is $\partial D$ traversed positively together with $\partial D_\varepsilon(z)$ traversed negatively, so Stokes gives

$$\int_{\partial D} \frac{g(\zeta)}{\zeta - z}\,d\zeta - \int_{\partial D_\varepsilon(z)} \frac{g(\zeta)}{\zeta - z}\,d\zeta = -\int_{D_\varepsilon} \frac{\partial g/\partial\bar{\zeta}}{\zeta - z}\,d\zeta \wedge d\bar{\zeta}$$

*The limit.* On $\partial D_\varepsilon(z)$ write $\zeta = z + \varepsilon e^{i\theta}$, so $d\zeta = i\varepsilon e^{i\theta} d\theta$ and $d\zeta/(\zeta - z) = i\,d\theta$. Hence

$$\int_{\partial D_\varepsilon(z)} \frac{g(\zeta)}{\zeta - z}\,d\zeta = i\int_0^{2\pi} g\big(z + \varepsilon e^{i\theta}\big)\,d\theta \xrightarrow[\varepsilon\to 0]{} 2\pi i\,g(z)$$

by continuity of $g$. The right-hand side converges to the integral over all of $D$ by absolute convergence. Rearranging and dividing by $2\pi i$ gives the formula.

The heart of the matter is the one-variable inhomogeneous equation. Given a smooth function $f$ on a disc, one seeks a smooth $u$ with $\partial u/\partial\bar{z} = f$. There is no obstruction to solving this in one variable: unlike the several-variable case, where $\bar{\partial} f = 0$ is forced, a single equation $\partial u/\partial\bar{z} = f$ imposes nothing on $f$ beyond smoothness. The solution is the Cauchy transform of $f$, and the proof that it works is a differentiation under the integral sign after the pole has been tamed by a change of variables.

**Lemma 2.2.2: The One-Variable Inhomogeneous $\bar{\partial}$-Equation**

Let $f$ be a smooth complex-valued function on a neighborhood of the closed disc $\overline{D} \subset \mathbb{C}$ of radius $r$ centered at the origin. Define the *Cauchy transform*

$$u(z) = \frac{1}{2\pi i}\int_D \frac{f(\zeta)}{\zeta - z}\,d\zeta \wedge d\bar{\zeta} = -\frac{1}{\pi}\int_D \frac{f(\zeta)}{\zeta - z}\,d\xi\,d\eta$$

for $z \in D$, where $\zeta = \xi + i\eta$. Then $u$ is smooth on $D$ and satisfies

$$\frac{\partial u}{\partial\bar{z}} = f$$

throughout $D$.

***Proof*:**

The two expressions for $u$ agree because $d\zeta \wedge d\bar{\zeta} = (d\xi + i\,d\eta) \wedge (d\xi - i\,d\eta) = -2i\,d\xi \wedge d\eta$, so that $\frac{1}{2\pi i}\,d\zeta \wedge d\bar{\zeta} = -\frac{1}{\pi}\,d\xi\,d\eta$. The proof proceeds in three stages: make the singular integral manifestly convergent by a change of variables, differentiate under the integral, and identify the derivative using the Cauchy-Pompeiu formula.

*Step 1: A translation-invariant form of the integral.* Substitute $w = \zeta - z$, so $\zeta = z + w$ and $d\xi\,d\eta = d\sigma\,d\tau$ where $w = \sigma + i\tau$. As $\zeta$ ranges over $D$, the new variable $w$ ranges over the translated disc $D - z$. Thus

$$u(z) = -\frac{1}{\pi} \int_{D-z} \frac{f(z+w)}{w} \, d\sigma \, d\tau$$

The integrand is singular at $w = 0$ because of the factor $1/w$, but this singularity is integrable: in polar coordinates $w = \rho e^{i\theta}$ one has $d\sigma \, d\tau = \rho \, d\rho \, d\theta$ and $1/w = e^{-i\theta}/\rho$, so $\frac{1}{w} \, d\sigma \, d\tau = e^{-i\theta} \, d\rho \, d\theta$ has no pole. To differentiate cleanly, fix a slightly larger radius $r' > r$ and extend $f$ to a smooth compactly supported function $\tilde{f}$ on all of $\mathbb{C}$ that agrees with $f$ on $\overline{D}$; this is possible by multiplying $f$ by a smooth cutoff equal to 1 on $\overline{D}$ and supported in the disc of radius $r'$. To avoid tracking the moving boundary $D - z$ in the differentiation, introduce the auxiliary function

$$v(z) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{\tilde{f}(z+w)}{w} \, d\sigma \, d\tau$$

whose domain of integration no longer depends on $z$. The integral converges absolutely because $\tilde{f}$ has compact support and $1/w$ is locally integrable. This $v$ is not equal to $u$ on $D$; the two differ by the integral over the exterior region $\mathbb{C} \setminus (D - z)$, which Step 3 shows is holomorphic in $z$ and so has vanishing $\bar{z}$-derivative. Thus $u$ and $v$ share the same $\bar{z}$-derivative on $D$, and it suffices to compute $\partial v/\partial \bar{z}$.

*Step 2: Differentiation under the integral.* Because the domain $\mathbb{C}$ is fixed and $\tilde{f}$ is smooth with compact support, the partial derivatives of $v$ may be computed by differentiating under the integral sign; the local integrability of $1/w$ and the compact support of the derivatives of $\tilde{f}$ justify the interchange. Since $\partial/\partial \bar{z}$ acts only through $\tilde{f}(z+w)$ and $\partial \tilde{f}(z+w)/\partial \bar{z} = (\partial \tilde{f}/\partial \bar{\zeta})(z+w)$, this gives

$$\frac{\partial v}{\partial \bar{z}}(z) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{(\partial \tilde{f}/\partial \bar{\zeta})(z+w)}{w} \, d\sigma \, d\tau$$

Now undo the translation: put $\zeta = z + w$ again, so $w = \zeta - z$ and $d\sigma \, d\tau = d\xi \, d\eta$, yielding

$$\frac{\partial v}{\partial \bar{z}}(z) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{(\partial \tilde{f}/\partial \bar{\zeta})(\zeta)}{\zeta - z} \, d\xi \, d\eta = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{\partial \tilde{f}/\partial \bar{\zeta}}{\zeta - z} \, d\zeta \wedge d\bar{\zeta}$$

where the last equality restores the wedge form of the area element.

*Step 3: Identification via Cauchy-Pompeiu.* Apply the Cauchy-Pompeiu formula to $\tilde{f}$ on any disc $D_R$ of radius $R$ large enough that $\tilde{f}$ vanishes on and outside $\partial D_R$. The boundary integral is then zero, because $\tilde{f} = 0$ on $\partial D_R$, and the formula reads

$$\tilde{f}(z) = \frac{1}{2\pi i} \int_{D_R} \frac{\partial \tilde{f}/\partial \bar{\zeta}}{\zeta - z} \, d\zeta \wedge d\bar{\zeta}$$

Since $\partial \tilde{f}/\partial \bar{\zeta}$ vanishes outside $D_R$, the domain of integration may be enlarged to all of $\mathbb{C}$ without changing the value, and the right-hand side is exactly the expression for $\partial v/\partial \bar{z}$ computed in Step 2. Hence

$$\frac{\partial v}{\partial \bar{z}}(z) = \tilde{f}(z)$$

for every $z \in \mathbb{C}$. On the disc $D$ one has $\tilde{f} = f$. It remains to justify the claim from Step 1 that $u$ and $v$ have the same $\bar{z}$-derivative on $D$. Their difference is

$$v(z) - u(z) = -\frac{1}{\pi} \int_{\mathbb{C} \setminus (D-z)} \frac{\tilde{f}(z+w)}{w} \, d\sigma \, d\tau$$

The region $\mathbb{C} \setminus (D - z)$ corresponds under $\zeta = z + w$ to $\mathbb{C} \setminus D$, where the integrand is $\tilde{f}(\zeta)/(\zeta - z)$ with $\zeta$ never equal to $z$ (since $z \in D$). On that region $1/(\zeta - z)$ is holomorphic in $z$, and $\tilde{f}(\zeta)$ does not depend on $z$, so the integrand is holomorphic in $z$; differentiation under the integral, again justified by the compact support of $\tilde{f}$, gives $\partial(v - u)/\partial\bar{z} = 0$ on $D$. Therefore

$$\frac{\partial u}{\partial \bar{z}}(z) = \frac{\partial v}{\partial \bar{z}}(z) = \tilde{f}(z) = f(z)$$

for all $z \in D$. Smoothness of $u$ on $D$ follows from the decomposition $u = v - (v - u)$ used above: $v$ is smooth because it is the convolution of the compactly supported $\tilde{f}$ with the locally integrable $1/w$ over the fixed domain $\mathbb{C}$, and $v - u$ is smooth because its integrand is holomorphic in $z$ over the fixed region $\mathbb{C} \setminus D$, so both may be differentiated under the integral to all orders. This gives $\partial u/\partial\bar{z} = f$ on $D$, as we sought to show.

The lemma is the one used throughout. Its content is that in one complex variable the operator $\partial/\partial\bar{z}$ is surjective onto smooth functions on a disc, with the Cauchy transform providing an explicit right inverse. The change of variables that moves the singularity of $1/(\zeta - z)$ to the origin and reads it in polar coordinates is what makes the integral behave; without it, the pole would obstruct differentiation. The one-variable solution is defined on all of $D$; the locality of the several-variable statement comes not from this step but from the induction, where each one-variable solve needs its coefficients smooth up to the closure of the disc it integrates over. One caution worth recording: the Cauchy transform does not preserve compact support even when $f$ has it, so the solution $u$ need not vanish near the boundary. This will matter when the construction is iterated across variables, where each step must not disturb the holomorphy already arranged in earlier variables.

With one variable solved, the several-variable statement follows by induction. The setting is a polydisc, the natural local model for a complex manifold. Recall that a form of type $(p, q)$ was defined in definition 1.2.2, the operator $\bar{\partial}\colon \mathcal{A}^{p,q} \to \mathcal{A}^{p,q+1}$ in definition 1.2.5, and the relation $\bar{\partial}^2 = 0$ in proposition 1.2.8; a form $\alpha$ is *$\bar{\partial}$-closed* if $\bar{\partial}\alpha = 0$ and *$\bar{\partial}$-exact* if $\alpha = \bar{\partial}\beta$ for some $(p, q-1)$-form $\beta$. The lemma asserts that on a polydisc these two conditions coincide once $q \geq 1$.

**Lemma 2.2.3: The $\bar{\partial}$-Poincaré Lemma**

Let $\Delta = \Delta_1 \times \cdots \times \Delta_n \subseteq \mathbb{C}^n$ be a polydisc, with coordinates $z_1, \ldots, z_n$, and let $q \geq 1$. If $\alpha$ is a smooth $\bar{\partial}$-closed form of type $(p, q)$ on $\Delta$, that is $\bar{\partial}\alpha = 0$, then there is a smooth $(p, q-1)$-form $\beta$ on any relatively compact subpolydisc $\Delta' \Subset \Delta$ with

$$\alpha = \bar{\partial}\beta$$

on $\Delta'$. Every $\bar{\partial}$-closed $(p, q)$-form with $q \geq 1$ is therefore locally $\bar{\partial}$-exact.

The motivation for the exact statement, with its shrinking to $\Delta' \Subset \Delta$, is that each one-variable solve integrates the Cauchy kernel over a disc across whose closure the integrand must be smooth; passing to a relatively compact subpolydisc $\Delta' \Subset \Delta$ guarantees this at every step of the induction. The proof is an induction on the number of coordinates $\bar{z}_1, \ldots, \bar{z}_n$ whose differentials can appear in $\alpha$. At each stage a single one-variable solve removes the highest such differential.

***Proof*:**
The index $p$ plays no role in what follows: $\bar{\partial}$ acts only on the antiholomorphic part of a form and leaves the holomorphic differentials $dz_{i_1} \wedge \cdots \wedge dz_{i_p}$ untouched, so the entire argument may be run coefficient block by coefficient block with these holomorphic factors carried along as inert spectators. Fix $p$ and suppress it in the notation, writing every $(p,q)$-form as a sum of terms $\varphi\, dz_I \wedge d\bar{z}_J$ with $|I| = p$, $|J| = q$, where $dz_I$ is held fixed throughout. It therefore suffices to treat the case $p = 0$; the general case is identical with $dz_I$ appended to every term.

For $0 \le k \le n$, say that a $(0,q)$-form *involves only* $\bar{z}_1, \dots, \bar{z}_k$ if, when written in the basis $d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$, every basis form that occurs has all its indices $j_1, \dots, j_q$ in $\{1, \dots, k\}$; equivalently, no $d\bar{z}_j$ with $j > k$ appears in a differential factor. Let $P(k)$ be the statement:

> Every smooth $\bar{\partial}$-closed $(0,q)$-form with $q \ge 1$ on a polydisc $\Delta$ that involves only $\bar{z}_1, \dots, \bar{z}_k$ is $\bar{\partial}$-exact on every relatively compact subpolydisc $\Delta' \Subset \Delta$.

The lemma is $P(n)$, since on $\Delta$ every $(0,q)$-form involves only $\bar{z}_1, \dots, \bar{z}_n$. The proof is by induction on $k$. Before the induction, record one consequence of closedness that every stage relies on. If a $(0,q)$-form $\alpha = \sum_J a_J\, d\bar{z}_J$ (sum over multi-indices $J \subseteq \{1, \dots, k\}$ of size $q$) is $\bar{\partial}$-closed and involves only $\bar{z}_1, \dots, \bar{z}_k$, then each coefficient $a_J$ is holomorphic in the later variables $z_{k+1}, \dots, z_n$. Indeed

$$\bar{\partial}\alpha = \sum_J \sum_{j=1}^{n} \frac{\partial a_J}{\partial \bar{z}_j}\, d\bar{z}_j \wedge d\bar{z}_J$$

and for a fixed $j > k$ and multi-index $J \subseteq \{1, \dots, k\}$, the basis form $d\bar{z}_j \wedge d\bar{z}_J$ receives a contribution only from the term $(\partial a_J / \partial \bar{z}_j)\, d\bar{z}_j \wedge d\bar{z}_J$: no other $J'$ can produce it, because every other differential factor available to $\alpha$ carries indices $\le k$, and $j > k$ appears nowhere else. Setting the coefficient of $d\bar{z}_j \wedge d\bar{z}_J$ to zero gives $\partial a_J / \partial \bar{z}_j = 0$ for all $j > k$, which is the asserted holomorphy in $z_{k+1}, \dots, z_n$.

*Base case $k = 0$.* A $(0,q)$-form with $q \ge 1$ that involves only the empty set of variables $\bar{z}_1, \dots, \bar{z}_0$ has no $d\bar{z}_j$ available to build a basis form of positive degree $q$, so it is the zero form. The zero form is $\bar{\partial}0$, hence $\bar{\partial}$-exact, and $P(0)$ holds vacuously.

*Inductive step.* Assume $P(k-1)$ for some $k$ with $1 \le k \le n$, and let $\alpha$ be a smooth $\bar{\partial}$-closed $(0,q)$-form, $q \ge 1$, on $\Delta$ that involves only $\bar{z}_1, \dots, \bar{z}_k$. Separate the terms of $\alpha$ according to whether $d\bar{z}_k$ appears:

$$\alpha = d\bar{z}_k \wedge \gamma + \rho$$

where $\gamma$ is a $(0, q-1)$-form and $\rho$ is a $(0,q)$-form, and both $\gamma$ and $\rho$ involve only $\bar{z}_1, \dots, \bar{z}_{k-1}$ (that is, neither contains a factor $d\bar{z}_k$). This decomposition is unique: collect every basis form of $\alpha$ that contains $d\bar{z}_k$, factor that $d\bar{z}_k$ out to the left, and call the remainder $\gamma$; the terms with no $d\bar{z}_k$ make up $\rho$.

*Recording the holomorphy of the coefficients.* Write $\gamma = \sum_L g_L\, d\bar{z}_L$ with each multi-index $L \subseteq \{1, \dots, k-1\}$ of size $q-1$, and $\rho = \sum_M h_M\, d\bar{z}_M$ with each $M \subseteq \{1, \dots, k-1\}$ of size $q$. The coefficients $g_L$ and $h_M$ are smooth functions of all $n$ variables. By the holomorphy consequence recorded before the induction, applied to $\alpha$ (which is $\bar{\partial}$-closed and involves only $\bar{z}_1, \dots, \bar{z}_k$), every coefficient of $\alpha$ is holomorphic in $z_{k+1}, \dots, z_n$. The coefficients of $\alpha$ are exactly the $g_L$ (coefficients of the $d\bar{z}_k$-terms) and the $h_M$ (coefficients of the $\rho$-terms), so

$$\frac{\partial g_L}{\partial \bar{z}_j} = 0 \quad \text{and} \quad \frac{\partial h_M}{\partial \bar{z}_j} = 0 \quad \text{for every } j > k$$

This holomorphy in the later variables is what keeps the construction below inside the variable range $\bar z_1, \dots, \bar z_k$, and it is the only use of $\bar\partial$-closedness in the inductive step until the residual form is handed to $P(k-1)$.

*Solving away $d\bar z_k$.* For each coefficient $g_L$ of $\gamma$, solve the one-variable inhomogeneous equation

$$\frac{\partial G_L}{\partial \bar z_k} = g_L$$

in the single variable $z_k$ by lemma 2.2.2, treating the remaining coordinates $z_1, \dots, z_{k-1}, z_{k+1}, \dots, z_n$ as parameters. Fix once and for all a nested chain of subpolydiscs $\Delta' \Subset \Delta'' \Subset \Delta$, so that in each variable the closed disc of $\Delta''$ lies strictly inside $\Delta$ and strictly contains that of $\Delta'$. The one-variable solve at stage $k$ is carried out over the $k$-th factor of $\Delta''$, so $\beta_1$ is defined on $\Delta$ in every slot except the $k$-th, where it is defined on $\Delta''_k$. Each variable is shrunk at its own stage and at no other, so the single intermediate polydisc $\Delta''$ accommodates all $n$ stages and the final primitive is valid on $\Delta'$. Set

$$G_L(z_1, \dots, z_n) = \frac{1}{2\pi i} \int_{\Delta''_k} \frac{g_L(z_1, \dots, \zeta, \dots, z_n)}{\zeta - z_k}\, d\zeta \wedge d\bar\zeta$$

where $\zeta$ occupies the $k$-th slot and $\Delta''_k$ is the $k$-th factor of the intermediate polydisc $\Delta''$, whose closure contains the $k$-th factor of $\Delta'$ and lies inside $\Delta_k$. Because $g_L$ is smooth and its dependence on the parameters is smooth, the resulting $G_L$ is smooth in all $n$ variables: differentiation in the parameter directions passes under the integral sign by the same compact-support and local-integrability estimates used in lemma 2.2.2, and lemma 2.2.2 itself gives $\partial G_L/\partial \bar z_k = g_L$. Assemble

$$\beta_1 = \sum_L G_L\, d\bar z_L$$

a $(0, q-1)$-form involving only $\bar z_1, \dots, \bar z_{k-1}$ in its differential factors, though its coefficients $G_L$ depend on all of $z_1, \dots, z_n$ including $\bar z_k$.

The primitive $G_L$ inherits the holomorphy of $g_L$ in the later variables. For $j > k$, differentiating the integral defining $G_L$ under the integral sign in the parameter direction $\bar z_j$ gives $\partial G_L/\partial \bar z_j$ as the Cauchy transform of $\partial g_L/\partial \bar z_j$, which is zero by the holomorphy recorded above; hence $\partial G_L/\partial \bar z_j = 0$ for all $j > k$, so each $G_L$ is holomorphic in $z_{k+1}, \dots, z_n$.

Now compute $\bar\partial \beta_1$ and compare with $\alpha$. Expanding and using $\partial G_L/\partial \bar z_j = 0$ for $j > k$,

$$\bar\partial \beta_1 = \sum_L \sum_{j=1}^{n} \frac{\partial G_L}{\partial \bar z_j}\, d\bar z_j \wedge d\bar z_L = \sum_L \frac{\partial G_L}{\partial \bar z_k}\, d\bar z_k \wedge d\bar z_L + \sum_L \sum_{j=1}^{k-1} \frac{\partial G_L}{\partial \bar z_j}\, d\bar z_j \wedge d\bar z_L$$

the middle term dropping out for $j > k$. Using $\partial G_L/\partial \bar z_k = g_L$ in the first sum,

$$\bar\partial \beta_1 = d\bar z_k \wedge \left( \sum_L g_L\, d\bar z_L \right) + \sum_L \sum_{j=1}^{k-1} \frac{\partial G_L}{\partial \bar z_j}\, d\bar z_j \wedge d\bar z_L = d\bar z_k \wedge \gamma + \eta$$

where

$$\eta = \sum_L \sum_{j=1}^{k-1} \frac{\partial G_L}{\partial \bar z_j}\, d\bar z_j \wedge d\bar z_L$$

is a $(0, q)$-form involving only $\bar{z}_1, \ldots, \bar{z}_{k-1}$, since every differential factor appearing in it is $d\bar{z}_j \wedge d\bar{z}_L$ with $j \leq k-1$ and $L \subseteq \{1, \ldots, k-1\}$. The first term $d\bar{z}_k \wedge \gamma$ matches the $d\bar{z}_k$-part of $\alpha$ exactly. Therefore

$$\alpha - \bar{\partial}\beta_1 = (d\bar{z}_k \wedge \gamma + \rho) - (d\bar{z}_k \wedge \gamma + \eta) = \rho - \eta$$

and this form $\rho - \eta$ has no $d\bar{z}_k$ factor at all: both $\rho$ and $\eta$ involve only $\bar{z}_1, \ldots, \bar{z}_{k-1}$.

*Applying the inductive hypothesis.* The residual $\rho - \eta$ is a $(0, q)$-form of the same degree $q \geq 1$ as $\alpha$; passing from $\alpha$ to $\rho - \eta$ lowers the number of antiholomorphic variables from $k$ to $k-1$ but does not change the form degree, since $\beta_1$ carries the drop in degree, not the residual. The residual involves only $\bar{z}_1, \ldots, \bar{z}_{k-1}$, as recorded above. It is $\bar{\partial}$-closed: because $\bar{\partial}\alpha = 0$ and $\bar{\partial}(\bar{\partial}\beta_1) = 0$ by $\bar{\partial}^2 = 0$ (proposition 1.2.8), one has

$$\bar{\partial}(\rho - \eta) = \bar{\partial}(\alpha - \bar{\partial}\beta_1) = \bar{\partial}\alpha - \bar{\partial}^2\beta_1 = 0$$

Thus $\rho - \eta$ is a $\bar{\partial}$-closed $(0, q)$-form with $q \geq 1$ involving only $\bar{z}_1, \ldots, \bar{z}_{k-1}$, so the inductive hypothesis $P(k-1)$ supplies a $(0, q-1)$-form $\beta_2$ on any relatively compact subpolydisc with $\rho - \eta = \bar{\partial}\beta_2$. This is the one place $P(k-1)$ is invoked, and it applies uniformly for every $q \geq 1$: when $q = 1$ the residual $\rho - \eta$ is a $\bar{\partial}$-closed $(0,1)$-form to which $P(k-1)$ still speaks, with the recursion bottoming out at the base case $P(0)$ where a positive-degree form in no variables is forced to be zero. Setting $\beta = \beta_1 + \beta_2$ then gives

$$\bar{\partial}\beta = \bar{\partial}\beta_1 + \bar{\partial}\beta_2 = \bar{\partial}\beta_1 + (\rho - \eta) = (d\bar{z}_k \wedge \gamma + \eta) + (\rho - \eta) = d\bar{z}_k \wedge \gamma + \rho = \alpha$$

on $\Delta'$, establishing $P(k)$ and completing the induction.

By induction $P(k)$ holds for all $0 \leq k \leq n$; in particular $P(n)$ is the assertion of the lemma. Restoring the suppressed holomorphic factor $dz_I$ to every term extends the conclusion from $p = 0$ to arbitrary $p$ without change, since $dz_I$ is inert under $\bar{\partial}$. Every $\bar{\partial}$-closed $(p, q)$-form with $q \geq 1$ is thus $\bar{\partial}$-exact on a relatively compact subpolydisc, completing the proof.

The lemma is the local vanishing theorem that powers the whole chapter. Read at the level of stalks it says that the Dolbeault complex has no local cohomology in positive $\bar{\partial}$-degree: at each point, a $\bar{\partial}$-closed germ of positive type is a $\bar{\partial}$-coboundary. That is exactly the exactness needed to make the sequence $0 \to \Omega^p_X \to \mathcal{A}^{p,0} \to \mathcal{A}^{p,1} \to \cdots$ a resolution, which is the subject of section 2.3. The single point where the sequence is *not* exact is $q = 0$: a $\bar{\partial}$-closed $(p, 0)$-form is one whose coefficients are holomorphic, and there is no lower-degree form to write it as a coboundary of. That surviving kernel is precisely $\Omega^p_X$ (definition 1.2.11), which is why the holomorphic $p$-forms sit at the left end of the resolution rather than being killed off with the rest.

The hypothesis $q \geq 1$ is not a technicality one could hope to remove; it marks the boundary between the holomorphic and the smooth. The contrast with the ordinary Poincaré lemma for $d$ is instructive: there the domain need only be star-shaped or contractible, whereas for $\bar{\partial}$ the natural local model is the polydisc, and the statement is local on polydiscs. The Cauchy transform, unlike the homotopy operator for $d$, does not extend to an arbitrary contractible domain because it is tied to the one-variable integral kernel. This is the first place where the complex structure constrains the geometry of the local model.

To see the mechanism at work with no induction to track, here is the base of the induction made concrete: a single $(0, 1)$-form on a bidisc, solved by one application of the Cauchy transform.

**Example 2.2.4: Solving $\bar{\partial}u = f$ on a Bidisc**

Take $n = 2$ with coordinates $z_1, z_2$ on a polydisc $\Delta = \Delta_1 \times \Delta_2 \subseteq \mathbb{C}^2$, and consider the $(0,1)$-form

$$\alpha = \bar{z}_2\, d\bar{z}_1 + \bar{z}_1\, d\bar{z}_2$$

First check that $\alpha$ is $\bar{\partial}$-closed, as the lemma requires. Compute

$$\bar{\partial}\alpha = \left(\frac{\partial \bar{z}_2}{\partial \bar{z}_2}\right) d\bar{z}_2 \wedge d\bar{z}_1 + \left(\frac{\partial \bar{z}_1}{\partial \bar{z}_1}\right) d\bar{z}_1 \wedge d\bar{z}_2 = d\bar{z}_2 \wedge d\bar{z}_1 + d\bar{z}_1 \wedge d\bar{z}_2 = 0$$

using $d\bar{z}_2 \wedge d\bar{z}_1 = -\, d\bar{z}_1 \wedge d\bar{z}_2$, so $\alpha$ is closed and a solution $u$ with $\bar{\partial}u = \alpha$ should exist.

Run the inductive scheme with $k = 2$. The decomposition $\alpha = d\bar{z}_2 \wedge \gamma + \rho$ has $\gamma = \bar{z}_1$ (a function, the coefficient of $d\bar{z}_2$) and $\rho = \bar{z}_2\, d\bar{z}_1$. Solve $\partial G/\partial \bar{z}_2 = \bar{z}_1$ in the variable $z_2$ with $z_1$ a parameter; a solution is $G = \bar{z}_1\bar{z}_2$, since $\partial(\bar{z}_1\bar{z}_2)/\partial\bar{z}_2 = \bar{z}_1$. This $G$ is not obtained here from the Cauchy integral but by inspection, which is legitimate: the integral formula produces *a* solution, and any solution differing from it by a $z_2$-holomorphic function serves equally well; $\bar{z}_1\bar{z}_2$ is the cleanest representative. Then

$$\bar{\partial}G = \frac{\partial(\bar{z}_1\bar{z}_2)}{\partial \bar{z}_1}\, d\bar{z}_1 + \frac{\partial(\bar{z}_1\bar{z}_2)}{\partial \bar{z}_2}\, d\bar{z}_2 = \bar{z}_2\, d\bar{z}_1 + \bar{z}_1\, d\bar{z}_2 = \alpha$$

so in fact $u = G = \bar{z}_1\bar{z}_2$ already solves $\bar{\partial}u = \alpha$ on all of $\Delta$, and the second inductive stage (removing $d\bar{z}_1$) contributes nothing because $\alpha - \bar{\partial}G = 0$ already. The final answer is

$$u(z_1, z_2) = \bar{z}_1\, \bar{z}_2$$

The example shows two features at once: the single one-variable solve suffices when the residual form vanishes, and the freedom to add a holomorphic function lets one replace the Cauchy-transform output by any convenient antiderivative.

The example also illustrates why the solution is local rather than global. The function $u = \bar{z}_1\bar{z}_2$ is defined on all of $\mathbb{C}^2$, but that is an accident of how simple $\alpha$ is; for a general smooth $(0,1)$-form the Cauchy transform in each variable is defined only after shrinking to a relatively compact subpolydisc, and the resulting $u$ generically does not extend. The passage from local to global solvability, when it is possible at all, is measured precisely by the Dolbeault cohomology $H^{p,q}_{\bar{\partial}}(X)$ of definition 1.2.10, whose computation is the goal of section 2.3.

## Exercise 2.2.1

1. Let $f$ be the smooth function $f(z) = \bar{z}$ on the closed unit disc $\overline{D} \subseteq \mathbb{C}$. Using lemma 2.2.2, exhibit an explicit smooth $u$ on $D$ with $\partial u/\partial \bar{z} = \bar{z}$, and verify directly that your $u$ solves the equation. Then describe the full set of smooth solutions on $D$.
2. On the bidisc $\Delta \subseteq \mathbb{C}^2$ consider the $(0,1)$-form $\alpha = z_2\, d\bar{z}_1$, where $z_2$ is the holomorphic coordinate (not $\bar{z}_2$). Show that $\alpha$ is $\bar{\partial}$-closed, and find a smooth $u$ with $\bar{\partial}u = \alpha$ on $\Delta$.
3. Show that the $(0,1)$-form $\alpha = \bar{z}_1\, d\bar{z}_2 - \bar{z}_2\, d\bar{z}_1$ on $\mathbb{C}^2$ is *not* $\bar{\partial}$-closed, and conclude that no smooth $u$ can satisfy $\bar{\partial}u = \alpha$ on any open set. Explain how this is consistent with the $\bar{\partial}$-Poincaré lemma.

4. The proof of lemma 2.2.3 records, before the induction, that a $\bar{\partial}$-closed $(0, q)$-form involving only $\bar{z}_1, \ldots, \bar{z}_k$ has every coefficient holomorphic in the later variables $z_{k+1}, \ldots, z_n$. Explain in one paragraph where this holomorphy is used in the inductive step, and what would go wrong in the construction of the primitive $\beta_1$ if it failed.

5. The ordinary Poincaré lemma for the operator $d$ holds on any star-shaped open set. Give an example of a smooth $\bar{\partial}$-closed $(0, 1)$-form on the punctured bidisc $\Delta \setminus \{0\}$, or explain why the local statement of lemma 2.2.3 does not immediately give a solution there, contrasting the roles of the domain in the two Poincaré lemmas.

6. Let $f(z) = 1/z$ on the annulus $1/2 < |z| < 2$, which is smooth there. Explain why the Cauchy transform of lemma 2.2.2 does not directly produce a solution of $\partial u/\partial \bar{z} = f$ on the annulus, and state on which domain the lemma *does* apply to a restriction of $f$.

## 2.3 Dolbeault's Theorem

The two previous sections each contribute one half of a machine. The first supplied an abstract principle: the sheaf cohomology $H^q(X, \mathcal{F})$ of any sheaf can be read off from a resolution of $\mathcal{F}$ by acyclic sheaves (theorem 2.1.5), and the smooth form sheaves $\mathcal{A}^{p,q}$ are acyclic because they are fine (proposition 2.1.4). The second supplied local exactness: away from the bottom, the $\bar{\partial}$-complex of $(p, \bullet)$-forms has no cohomology, since every $\bar{\partial}$-closed form is locally $\bar{\partial}$-exact (lemma 2.2.3). This section fits the two halves together. The output is Dolbeault's theorem, which identifies the analytic Dolbeault cohomology $H^{p,q}_{\bar{\partial}}(X)$ from definition 1.2.10 with the sheaf cohomology $H^q(X, \Omega^p_X)$ of the holomorphic $p$-forms. This is the theorem that makes Dolbeault cohomology computable: the left side is defined analytically, by solving $\bar{\partial}$-equations, while the right side is a sheaf-cohomology group amenable to the tools of homological algebra, Serre duality, and the machinery of coherent sheaves.

The identification rests on a single sequence of sheaves. Fix a holomorphic degree $p$ and consider the smooth $(p, q)$-forms as $q$ ranges over $0, 1, 2, \ldots, n$, connected by the $\bar{\partial}$-operator. Prepending the holomorphic $p$-forms $\Omega^p_X$ produces a complex of sheaves, and the claim is that this complex is a resolution of $\Omega^p_X$ by acyclic sheaves. Once that is established, theorem 2.1.5 does the rest.

The first task is to identify the kernel of $\bar{\partial}$ in the bottom degree. A smooth $(p, 0)$-form is a section of $\mathcal{A}^{p,0}$; applying $\bar{\partial}$ lands it in $\mathcal{A}^{p,1}$. The forms killed by this operator are precisely the holomorphic ones.

**2.3.1 The kernel of $\bar\partial$ on $(p,0)$-forms** A $(p,0)$-form written in local holomorphic coordinates as

$$\alpha = \sum_{|I|=p} f_I \, dz^I$$

with $f_I$ smooth and $dz^I = dz^{i_1} \wedge \cdots \wedge dz^{i_p}$ satisfies $\bar\partial\alpha = 0$ if and only if every coefficient $f_I$ is holomorphic. Indeed, since the $dz^{i_j}$ have no antiholomorphic part,

$$\bar\partial\alpha = \sum_{|I|=p} \sum_{j=1}^{n} \frac{\partial f_I}{\partial \bar z^j} \, d\bar z^j \wedge dz^I$$

and the forms $d\bar z^j \wedge dz^I$ are linearly independent, so $\bar\partial\alpha = 0$ forces $\partial f_I / \partial \bar z^j = 0$ for all $I$ and $j$. These are the Cauchy-Riemann equations, whose solutions are exactly the holomorphic functions.

This is the sheaf-theoretic content of definition 1.2.11: a holomorphic $p$-form is a $(p,0)$-form with holomorphic coefficients, and box 2.3.1 says this is the same as being $\bar\partial$-closed. The inclusion $\Omega_X^p \hookrightarrow \mathcal{A}^{p,0}$ therefore identifies $\Omega_X^p$ with the kernel of the first $\bar\partial$-arrow. Everything now assembles into a resolution.

Before stating the resolution formally, note what has to be checked. A resolution is an exact sequence of sheaves, and exactness of a sequence of sheaves is a purely local condition: it can be checked on stalks, hence in an arbitrarily small polydisc around each point. At the bottom, exactness at $\Omega_X^p$ and at $\mathcal{A}^{p,0}$ records that $\Omega_X^p$ is the kernel of $\bar\partial$, which is box 2.3.1. In every higher degree $q \geq 1$, exactness at $\mathcal{A}^{p,q}$ says that a $\bar\partial$-closed $(p,q)$-form is locally $\bar\partial$-exact, which is precisely the $\bar\partial$-Poincaré lemma. The fineness that makes the resolution acyclic is inherited from $\mathcal{A}^0$.

**Proposition 2.3.2: The Dolbeault Resolution of $\Omega_X^p$**

Let $X$ be a complex manifold and fix $0 \leq p \leq n$. The sequence of sheaves

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}^{p,0} \xrightarrow{\bar\partial} \mathcal{A}^{p,1} \xrightarrow{\bar\partial} \mathcal{A}^{p,2} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}^{p,n} \longrightarrow 0$$

where the first map is the inclusion of holomorphic $p$-forms as $\bar\partial$-closed smooth $(p,0)$-forms, is a resolution of $\Omega_X^p$ by fine sheaves. In particular it is an acyclic resolution.

***Proof*:**
The maps $\bar\partial \colon \mathcal{A}^{p,q} \to \mathcal{A}^{p,q+1}$ compose to zero, since $\bar\partial^2 = 0$ (definition 1.2.5, definition 1.2.9), so the displayed diagram is a complex of sheaves. Three claims establish that it is an acyclic resolution: fineness of the terms, exactness at the bottom, and exactness in positive degrees.

*Step 1: The terms $\mathcal{A}^{p,q}$ are fine.* This is proposition 2.1.4: each $\mathcal{A}^{p,q}$ is a sheaf of modules over the sheaf $\mathcal{A}^0$ of smooth functions, and a sheaf of $\mathcal{A}^0$-modules is fine because $\mathcal{A}^0$ admits smooth partitions of unity subordinate to any open cover. By proposition 2.1.3, a fine sheaf on the paracompact manifold $X$ is acyclic, that is, $H^j(X, \mathcal{A}^{p,q}) = 0$ for all $j \geq 1$ and all $q$. Acyclicity of the resolving terms is what theorem 2.1.5 requires.

*Step 2: Exactness at $\Omega_X^p$ and at $\mathcal{A}^{p,0}$.* Injectivity of the inclusion $\Omega_X^p \hookrightarrow \mathcal{A}^{p,0}$ is immediate. Exactness at $\mathcal{A}^{p,0}$ asserts that the image of the inclusion equals the kernel of $\bar\partial \colon \mathcal{A}^{p,0} \to \mathcal{A}^{p,1}$.

By box 2.3.1, a $(p,0)$-form is $\bar\partial$-closed if and only if its coefficients are holomorphic, and this is exactly the condition that it be a holomorphic $p$-form (definition 1.2.11). The image and the kernel therefore coincide.

*Step 3: Exactness at $\mathcal{A}^{p,q}$ for $q \geq 1$.* Exactness of a sequence of sheaves is checked on stalks, so fix a point $x \in X$ and a germ at $x$ represented by a $\bar\partial$-closed $(p,q)$-form $\alpha$ defined on a neighborhood of $x$. Shrinking the neighborhood, one may take it to be a polydisc centered at $x$. The $\bar\partial$-Poincaré lemma (lemma 2.2.3) produces, on that polydisc, a $(p,q-1)$-form $\beta$ with $\bar\partial\beta = \alpha$. Thus every $\bar\partial$-closed germ in degree $q \geq 1$ is $\bar\partial$-exact, so the kernel of $\bar\partial$ at $\mathcal{A}^{p,q}$ equals the image of $\bar\partial$ from $\mathcal{A}^{p,q-1}$ on stalks, which is exactness at $\mathcal{A}^{p,q}$.

Steps 2 and 3 together give exactness of the whole sequence, so it is a resolution of $\Omega_X^p$; Step 1 makes it a resolution by fine, hence acyclic, sheaves, completing the proof.

The resolution in proposition 2.3.2 is the holomorphic analogue of the smooth de Rham resolution of the constant sheaf $\underline{\mathbb{C}}$ used in the worked example of section 2.1. There, the constant sheaf was resolved by the sheaves $\mathcal{A}^k$ of smooth $k$-forms via the exterior derivative $d$, with local exactness supplied by the ordinary Poincaré lemma; the resulting computation recovered de Rham cohomology as sheaf cohomology of $\underline{\mathbb{C}}$. Here $\Omega_X^p$ replaces $\underline{\mathbb{C}}$, the type-$(p,q)$ forms replace the $k$-forms, and $\bar\partial$ replaces $d$, with the $\bar\partial$-Poincaré lemma supplying local exactness. The two resolutions are structurally identical, and each computes a cohomology theory of interest as the sheaf cohomology of the sheaf sitting at the bottom.

With a fine resolution of $\Omega_X^p$ in hand, the abstract de Rham principle delivers the theorem at once. Recall the mechanism: to compute $H^q(X,\mathcal{F})$ from an acyclic resolution $0 \to \mathcal{F} \to \mathcal{L}^\bullet$, take global sections to obtain the complex $\mathcal{L}^\bullet(X)$ and read off its cohomology (theorem 2.1.5). Applying this to the Dolbeault resolution, the complex of global sections is exactly the Dolbeault complex of definition 1.2.9, and its cohomology is $H_{\bar\partial}^{p,q}(X)$ by definition.

**Theorem 2.3.3: Dolbeault's Theorem**

Let $X$ be a complex manifold. For all $0 \leq p,q \leq n$ there is a canonical isomorphism

$$H^q(X,\Omega_X^p) \;\cong\; H_{\bar\partial}^{p,q}(X)$$

between the sheaf cohomology of the holomorphic $p$-forms and the Dolbeault cohomology of $X$.

***Proof*:**
By proposition 2.3.2, the sequence

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}^{p,0} \xrightarrow{\bar\partial} \mathcal{A}^{p,1} \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}^{p,n} \longrightarrow 0$$

is a resolution of $\Omega_X^p$ by fine sheaves, and fine sheaves on the paracompact manifold $X$ are acyclic (proposition 2.1.3). The abstract de Rham theorem (theorem 2.1.5, proved in full in [4]) states that the sheaf cohomology $H^q(X,\Omega_X^p)$ is computed by the cohomology of the complex of global sections of any acyclic resolution. Taking global sections of the resolving terms gives the spaces $\mathcal{A}^{p,q}(X)$ of global smooth $(p,q)$-forms, with the boundary maps induced by $\bar\partial$; this is precisely the Dolbeault complex

$$\mathcal{A}^{p,0}(X) \xrightarrow{\bar\partial} \mathcal{A}^{p,1}(X) \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \mathcal{A}^{p,n}(X)$$

of definition 1.2.9. Its cohomology in degree $q$ is

$$\frac{\ker\big(\bar\partial\colon \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p,q+1}(X)\big)}{\operatorname{im}\big(\bar\partial\colon \mathcal{A}^{p,q-1}(X) \to \mathcal{A}^{p,q}(X)\big)}$$

which is the Dolbeault cohomology group $H^{p,q}_{\bar\partial}(X)$ of definition 1.2.10. Equating the two computations of the same cohomology gives the isomorphism $H^q(X, \Omega^p_X) \cong H^{p,q}_{\bar\partial}(X)$, as we sought to show.

The isomorphism is canonical in a concrete sense worth spelling out. A class in $H^q(X, \Omega^p_X)$, presented for instance by a Čech cocycle for a cover, corresponds under the theorem to the class of a global $\bar\partial$-closed $(p, q)$-form, obtained by patching local $\bar\partial$-primitives with a partition of unity. No choice enters the identification beyond the resolution itself, which is canonical, so the isomorphism commutes with the maps induced by holomorphic morphisms. This is the precise sense in which $H^q(X, \Omega^p_X)$ and $H^{p,q}_{\bar\partial}(X)$ are two names for one object.

The value of Dolbeault's theorem lies in what each side of the isomorphism contributes. The Dolbeault side $H^{p,q}_{\bar\partial}(X)$ is analytic and hands-on: a class is a smooth form, and computing the group means solving $\bar\partial$-equations. The sheaf-cohomology side $H^q(X, \Omega^p_X)$ is algebraic and structural, and it is through this side that the full weight of coherent-sheaf theory bears on Hodge theory. Three consequences illustrate the point.

The first is functoriality. Sheaf cohomology is a functor: a holomorphic map $f\colon X \to Y$ induces a pullback $f^*\colon H^q(Y, \Omega^p_Y) \to H^q(X, \Omega^p_X)$, compatible with composition. Under Dolbeault's theorem this pullback corresponds to pullback of $(p, q)$-forms, so the two functorialities agree. This is what lets Dolbeault cohomology be treated as a genuine cohomology theory on the category of complex manifolds, rather than a collection of unrelated vector spaces.

The second is finite-dimensionality on compact manifolds. On a compact complex manifold $X$, the sheaves $\Omega^p_X$ are coherent, and the sheaf cohomology of a coherent sheaf on a compact complex manifold is finite-dimensional over $\mathbb{C}$ by the Cartan-Serre finiteness theorem. Through Dolbeault's theorem, this transfers to the analytic side, where it is otherwise not at all evident: the Dolbeault cohomology group $H^{p,q}_{\bar\partial}(X)$, a priori the cohomology of an infinite-dimensional complex of smooth forms, is finite-dimensional. Compactness cannot be dropped here; on a noncompact manifold such as $\mathbb{C}^n$ the groups may vanish or behave differently, and finiteness can fail.

**Definition 2.3.4: Hodge Numbers**

Let $X$ be a compact complex manifold. The *Hodge numbers* of $X$ are the dimensions

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^{p,q}_{\bar\partial}(X) = \dim_{\mathbb{C}} H^q(X, \Omega^p_X)$$

for $0 \le p, q \le n$, where the two expressions agree by Dolbeault's theorem (theorem 2.3.3).

The third consequence, then, is the very existence of the Hodge numbers as finite invariants. The equality of the two dimensions in definition 2.3.4 is the whole content of Dolbeault's theorem read at the level of dimensions. It says that the analytically defined count of independent $\bar\partial$-cohomology classes coincides with an algebraically defined count of sheaf-cohomology classes. Arranging the $h^{p,q}$ in a diamond, the *Hodge diamond*, gives a table of numerical invariants that will organize much of the rest of this book; on a compact Kähler manifold these numbers will satisfy the symmetries

$h^{p,q} = h^{q,p}$ and $h^{p,q} = h^{n-p,n-q}$ and will refine the Betti numbers through the Hodge decomposition of chapter 4. For now the point is only that they are well-defined finite numbers, and that Dolbeault's theorem is what guarantees it.

The theory is best absorbed on the smallest nontrivial complete example, the Riemann sphere $\mathbb{CP}^1$. Here $n = 1$, so the only bidegrees are $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$, and all four Dolbeault groups can be computed directly from the sheaf-cohomology side using facts about line bundles on $\mathbb{CP}^1$.

**Example 2.3.5: Dolbeault Cohomology of the Riemann Sphere**

Let $X = \mathbb{CP}^1$, the complex projective line (example 1.1.8), with $n = \dim_{\mathbb{C}} X = 1$. The relevant holomorphic-form sheaves are

$$\Omega^0_X = \mathcal{O}_X, \qquad \Omega^1_X = K_X = \mathcal{O}_X(-2)$$

the first because $\Omega^0_X$ is by definition the structure sheaf, the second because the canonical bundle $K_X = \Omega^1_X$ (definition 1.3.12) of $\mathbb{CP}^1$ is $\mathcal{O}(-2)$ by corollary 1.3.17 at $n = 1$. By Dolbeault's theorem (theorem 2.3.3),

$$H^{p,q}_{\bar\partial}(\mathbb{CP}^1) \cong H^q\big(\mathbb{CP}^1, \Omega^p_{\mathbb{CP}^1}\big)$$

so it suffices to compute the sheaf cohomology of $\mathcal{O}$ and of $\mathcal{O}(-2)$ in degrees $q = 0, 1$. The needed input is the standard computation of line-bundle cohomology on $\mathbb{CP}^1$: for $d \geq 0$,

$$\dim H^0\big(\mathbb{CP}^1, \mathcal{O}(d)\big) = d + 1, \qquad H^1\big(\mathbb{CP}^1, \mathcal{O}(d)\big) = 0$$

while for $d \leq -2$, $H^0(\mathbb{CP}^1, \mathcal{O}(d)) = 0$ and, by Serre duality ([4, twisted sheaf cohomology on projective space] and [4, Serre duality for a projective scheme]) $H^1(\mathbb{CP}^1, \mathcal{O}(d)) \cong H^0(\mathbb{CP}^1, \mathcal{O}(-2-d))^\vee$, one gets $\dim H^1(\mathbb{CP}^1, \mathcal{O}(d)) = -d - 1$. The two cases needed are $d = 0$ and $d = -2$.

1. *Bidegree* $(0,0)$. $H^{0,0}_{\bar\partial}(\mathbb{CP}^1) = H^0(\mathbb{CP}^1, \mathcal{O}) = \mathbb{C}$, the global holomorphic functions, which on the compact connected manifold $\mathbb{CP}^1$ are the constants. Thus $h^{0,0} = 1$.
2. *Bidegree* $(0,1)$. $H^{0,1}_{\bar\partial}(\mathbb{CP}^1) = H^1(\mathbb{CP}^1, \mathcal{O}) = 0$, since $\mathcal{O} = \mathcal{O}(0)$ has $d = 0 \geq 0$. Thus $h^{0,1} = 0$.
3. *Bidegree* $(1,0)$. $H^{1,0}_{\bar\partial}(\mathbb{CP}^1) = H^0(\mathbb{CP}^1, \mathcal{O}(-2)) = 0$, since a nonzero global section of $\mathcal{O}(-2)$ would have an effective divisor of degree $\deg \mathcal{O}(-2) = -2$, and no effective divisor has negative degree. (A global holomorphic section may of course vanish; what it cannot do is vanish to a total order that is negative.) Thus $h^{1,0} = 0$: the sphere carries no nonzero global holomorphic 1-form.
4. *Bidegree* $(1,1)$. $H^{1,1}_{\bar\partial}(\mathbb{CP}^1) = H^1(\mathbb{CP}^1, \mathcal{O}(-2))$. By Serre duality this is dual to $H^0(\mathbb{CP}^1, \mathcal{O}(-2-(-2))) = H^0(\mathbb{CP}^1, \mathcal{O}) = \mathbb{C}$, so $H^{1,1}_{\bar\partial}(\mathbb{CP}^1) \cong \mathbb{C}$ and $h^{1,1} = 1$.

The Hodge numbers assemble into the diamond

$$\begin{array}{ccc} & h^{0,0} & \\ h^{1,0} & & h^{0,1} \\ & h^{1,1} & \end{array} \quad = \quad \begin{array}{ccc} & 1 & \\ 0 & & 0 \\ & 1 & \end{array}$$

Two independent checks confirm the answer. First, the sum $\sum_{p+q=k} h^{p,q}$ over each row reproduces the Betti numbers of $\mathbb{CP}^1 = S^2$: the diagonals give $h^{0,0} = 1 = b_0$, $h^{1,0} + h^{0,1} = 0 = b_1$, and $h^{1,1} = 1 = b_2$, matching $b_0 = b_2 = 1$, $b_1 = 0$. Second, the diamond exhibits the Hodge symmetries $h^{p,q} = h^{q,p}$ (here $h^{1,0} = h^{0,1} = 0$) and $h^{p,q} = h^{n-p,n-q}$ (here $h^{0,0} = h^{1,1} = 1$), which every compact Kähler manifold satisfies and which $\mathbb{CP}^1$, being projective hence Kähler, must obey.

The computation on $\mathbb{C}\mathrm{P}^1$ is the seed of the general projective-space calculation. On $\mathbb{C}\mathrm{P}^n$ the Hodge numbers are $h^{p,q} = 1$ when $p = q$ and 0 otherwise, so the Hodge diamond is concentrated on its vertical axis; the $\mathbb{C}\mathrm{P}^1$ diamond above is the $n = 1$ instance. This pattern, and the check that its diagonal sums reproduce the Betti numbers of $\mathbb{C}\mathrm{P}^n$, is taken up in section 2.4, where the Frölicher spectral sequence explains why the diagonal-sum check works and when it is an equality rather than an inequality.

### Exercise 2.3.1

1. Let $X$ be a complex manifold and $\alpha$ a smooth $(p,0)$-form on an open set $U \subseteq X$. Show directly from the local coordinate expression that $\bar\partial\alpha = 0$ on $U$ if and only if $\alpha$ has holomorphic coefficients, and explain why this identifies $\ker(\bar\partial\colon \mathcal{A}^{p,0} \to \mathcal{A}^{p,1})$ with $\Omega^p_X$ as subsheaves of $\mathcal{A}^{p,0}$.
2. Show that $H^{p,0}_{\bar\partial}(X) \cong H^0(X, \Omega^p_X)$, the space of global holomorphic $p$-forms, and deduce that $H^{p,0}_{\bar\partial}(X)$ is the kernel of $\bar\partial$ on global $(p,0)$-forms with no quotient. Conclude that on a compact connected complex manifold $h^{0,0} = 1$.
3. Let $X$ be a compact connected complex manifold of dimension $n$. Assuming Serre duality in the form $H^q(X, \Omega^p_X) \cong H^{n-q}(X, \Omega^{n-p}_X)^\vee$, show that $h^{p,q} = h^{n-p,n-q}$. Then compute $h^{n,n}(X)$ and interpret the resulting class in $H^{n,n}_{\bar\partial}(X)$.
4. Explain why the reasoning behind Dolbeault's theorem does not require $X$ to be compact, yet the finite-dimensionality of the $H^{p,q}_{\bar\partial}(X)$ does. Illustrate by describing $H^{0,0}_{\bar\partial}(\mathbb{C})$ and arguing that $H^{0,1}_{\bar\partial}(\mathbb{C}) = 0$, so that $\bar\partial$ is surjective onto smooth functions on $\mathbb{C}$.
5. Let $f\colon X \to Y$ be a holomorphic map. Describe the induced pullback $f^*\colon H^{p,q}_{\bar\partial}(Y) \to H^{p,q}_{\bar\partial}(X)$ on the Dolbeault side, and explain how Dolbeault's theorem matches it with the pullback $f^*\colon H^q(Y, \Omega^p_Y) \to H^q(X, \Omega^p_X)$ on sheaf cohomology. What does $f^*$ do on $H^{0,0}$ when $X$ and $Y$ are compact and connected?
6. Using the fact that $h^{p,q}(\mathbb{C}\mathrm{P}^n) = 1$ for $p = q$ and 0 otherwise, write out the Hodge diamond of $\mathbb{C}\mathrm{P}^2$, verify that the diagonal sums reproduce the Betti numbers $b_0 = b_2 = b_4 = 1$, $b_1 = b_3 = 0$ of $\mathbb{C}\mathrm{P}^2$, and confirm the two Hodge symmetries $h^{p,q} = h^{q,p}$ and $h^{p,q} = h^{n-p,n-q}$.

## 2.4 The Frölicher Spectral Sequence and GAGA

Dolbeault's theorem (theorem 2.3.3) computes each individual group $H^{p,q}_{\bar\partial}(X)$ as the sheaf cohomology $H^q(X, \Omega^p_X)$. These groups form a bigraded refinement of the complex de Rham cohomology $H^k_{\mathrm{dR}}(X, \mathbb{C})$: the de Rham complex is built from the operator $d = \partial + \bar\partial$ (theorem 1.2.6), whose two pieces separately assemble the Dolbeault groups. The question this section answers is how the bigraded data $H^{p,q}_{\bar\partial}(X)$ reassembles into the single-graded data $H^k_{\mathrm{dR}}(X, \mathbb{C})$. On an arbitrary complex manifold the two are not simply related: the bidegree decomposition of forms does not descend to a decomposition of cohomology, because $d$ mixes the pieces. What organizes the discrepancy is a spectral sequence, the Frölicher (or Hodge-de Rham) spectral sequence, whose first page is the Dolbeault cohomology and whose limit is the de Rham cohomology. Its degeneration is exactly the statement that the bigrading survives to cohomology, and this degeneration, proved for compact Kähler manifolds in chapter 4, is one form of the Hodge decomposition. The section closes with GAGA, which identifies the analytic groups $H^q(X, \Omega_X^p)$ with their algebraic counterparts when $X$ is projective, so that transcendental Hodge theory and algebraic geometry compute the same numbers.

The de Rham complex $(\mathcal{A}^\bullet(X), d)$ of global smooth forms carries a decreasing filtration by holomorphic degree. Recall from proposition 1.2.3 that a smooth $k$-form decomposes into its $(p,q)$-components with $p+q=k$. Collecting all components whose holomorphic degree is at least $p$ produces a subspace of $\mathcal{A}^k(X)$, and letting $k$ vary gives a graded subspace of the whole complex.

**Definition 2.4.1: The Hodge Filtration on the de Rham Complex**

Let $X$ be a complex manifold. For each integer $p \geq 0$, let

$$F^p\mathcal{A}^k(X) = \bigoplus_{\substack{r+s=k \\ r\geq p}} \mathcal{A}^{r,s}(X)$$

be the space of smooth $k$-forms all of whose bidegree components have holomorphic degree at least $p$. The collection $F^\bullet$ is the *Hodge filtration* (or *filtration bête* by holomorphic degree) on the de Rham complex $(\mathcal{A}^\bullet(X), d)$.

For this to be a filtration of the complex, the differential must respect it: $d$ must carry $F^p\mathcal{A}^k(X)$ into $F^p\mathcal{A}^{k+1}(X)$. This is where the decomposition $d = \partial + \bar\partial$ does the work. On a component of bidegree $(r,s)$ with $r \geq p$, the operator $\partial$ raises the bidegree to $(r+1,s)$ and $\bar\partial$ raises it to $(r,s+1)$; both outputs still have holomorphic degree at least $p$. So $d\big(F^p\mathcal{A}^k(X)\big) \subseteq F^p\mathcal{A}^{k+1}(X)$, and the filtration is compatible with $d$. It is decreasing, $F^{p+1}\mathcal{A}^k(X) \subseteq F^p\mathcal{A}^k(X)$, exhaustive ($F^0\mathcal{A}^k(X) = \mathcal{A}^k(X)$), and bounded (it terminates, since $F^p\mathcal{A}^k(X) = 0$ once $p > k$).

A filtered complex has an associated spectral sequence. This is the machinery developed in *Homological Algebra* [6], "Convergence for a Decreasing Filtration of a Cochain Complex": to a complex with a bounded, exhaustive, decreasing filtration $F^\bullet$ one associates a sequence of bigraded pages $(E_r^{p,q}, d_r)$, each the cohomology of the previous with respect to a differential $d_r$ of bidegree $(r, 1-r)$, converging to the associated graded of the cohomology of the total complex with respect to the induced filtration. The construction and its convergence are proved there in full; the content of this section is the identification of the pages in the present geometric situation. The convention places the filtration index $p$ as the first (column) coordinate and the complementary index $q = k-p$ as the second (row) coordinate, so that the total degree $p+q=k$ is the de Rham degree.

The zeroth page reads off directly. By definition of the associated graded of a filtration, $E_0^{p,q} = F^p\mathcal{A}^{p+q}(X)/F^{p+1}\mathcal{A}^{p+q}(X)$, which is the single bidegree slot $\mathcal{A}^{p,q}(X)$: passing to the quotient discards every component of holomorphic degree $> p$ and every component of holomorphic degree $< p$ was already excluded by $F^p$. The induced differential $d_0$ is the part of $d$ that preserves the filtration level $p$ while raising total degree, namely the part that raises the antiholomorphic degree. That part is $\bar\partial$. So the zeroth page is the Dolbeault complex, column by column, and its cohomology is the Dolbeault cohomology. This is the geometric heart of the whole construction, and it gives the first page.

**Theorem 2.4.2: The Frölicher Spectral Sequence**

Let $X$ be a complex manifold. The Hodge filtration (definition 2.4.1) on the de Rham complex $(\mathcal{A}^\bullet(X), d)$ gives rise to a spectral sequence, the *Frölicher spectral sequence*, whose first page is the Dolbeault cohomology,

$$E_1^{p,q} = H_{\bar\partial}^{p,q}(X)$$

with first differential $d_1 \colon E_1^{p,q} \to E_1^{p+1,q}$ induced by $\partial$, and which converges to the complex de Rham cohomology:

$$E_1^{p,q} = H_{\bar\partial}^{p,q}(X) \implies H_{\mathrm{dR}}^{p+q}(X, \mathbb{C})$$

The abutment is filtered by the Hodge filtration induced on $H_{\mathrm{dR}}^k(X, \mathbb{C})$, and $E_\infty^{p,q} \cong \mathrm{gr}_F^p H_{\mathrm{dR}}^{p+q}(X, \mathbb{C})$.

***Proof*:**
The general construction of the spectral sequence of a bounded filtered complex, together with its convergence to the associated graded of the total cohomology, is proved in *Homological Algebra* [6], "Convergence for a Decreasing Filtration of a Cochain Complex"; that machinery applies verbatim to the bounded exhaustive decreasing filtration $F^\bullet$ of definition 2.4.1. What remains is to identify the first page and its differential in the present situation.

*Step 1: The zeroth page is the Dolbeault complex.* By the definition of the zeroth page of the spectral sequence of a filtered complex (*Homological Algebra* [6], "The Exact Couple of a Filtered Complex"), $E_0^{p,q}$ is the associated graded piece $F^p\mathcal{A}^{p+q}(X)/F^{p+1}\mathcal{A}^{p+q}(X)$. By definition 2.4.1 the numerator is the sum of all bidegree slots of total degree $p+q$ with holomorphic degree at least $p$, and the denominator is the sum of those with holomorphic degree at least $p+1$; the quotient is the single slot of bidegree $(p, q)$, so

$$E_0^{p,q} = \mathcal{A}^{p,q}(X)$$

The differential $d_0$ is the map induced by $d$ on the associated graded; it has bidegree $(0, 1)$ in the $(p, q)$ indexing, so it raises total degree by one while preserving the filtration level $p$. Write $d = \partial + \bar\partial$ (theorem 1.2.6). On a form of bidegree $(p, q)$ the summand $\partial$ lands in bidegree $(p+1, q)$, hence in $F^{p+1}$, and so vanishes in the quotient defining $E_0^{p,q+1}$; the summand $\bar\partial$ lands in bidegree $(p, q+1)$ and survives. Therefore $d_0 = \bar\partial$, and the $p$-th column of the zeroth page is the Dolbeault complex

$$\mathcal{A}^{p,0}(X) \xrightarrow{\bar\partial} \mathcal{A}^{p,1}(X) \xrightarrow{\bar\partial} \mathcal{A}^{p,2}(X) \xrightarrow{\bar\partial} \cdots$$

of definition 1.2.9.

*Step 2: The first page is Dolbeault cohomology.* The first page is the cohomology of $(E_0, d_0)$. By Step 1 this is the cohomology of the Dolbeault complex in bidegree $(p, q)$, which is $H_{\bar\partial}^{p,q}(X)$ by definition 1.2.10. Hence

$$E_1^{p,q} = H_{\bar\partial}^{p,q}(X)$$

*Step 3: The first differential is induced by $\partial$.* The differential $d_1 \colon E_1^{p,q} \to E_1^{p+1,q}$ has bidegree $(1, 0)$ and is computed as follows. Represent a class in $E_1^{p,q}$ by a $\bar\partial$-closed form $\alpha \in \mathcal{A}^{p,q}(X)$. Then $d\alpha = \partial\alpha + \bar\partial\alpha = \partial\alpha$ since $\alpha$ is $\bar\partial$-closed, and $\partial\alpha$ lies in $\mathcal{A}^{p+1,q}(X) \subseteq F^{p+1}$. The recipe for $d_1$ in the spectral sequence of a filtered complex, which is that on representatives the differential is induced by $d$ (*Homological Algebra* [6], "Explicit Description of the Pages", carried to cochain complexes by "The Cochain Form of the Section"), sends the class of $\alpha$ to the class of $d\alpha = \partial\alpha$ in $E_1^{p+1,q} = H_{\bar\partial}^{p+1,q}(X)$. This is well defined precisely because $\partial\alpha$ is $\bar\partial$-closed: $\bar\partial\partial\alpha = -\partial\bar\partial\alpha = 0$ using the identity $\partial\bar\partial = -\bar\partial\partial$ from proposition 1.2.8. Thus $d_1$ is the map $[\alpha]_{\bar\partial} \mapsto [\partial\alpha]_{\bar\partial}$ induced by $\partial$ on Dolbeault cohomology.

*Step 4: Convergence.* The filtration $F^\bullet$ is exhaustive and bounded, so by the convergence theorem of *Homological Algebra* [6], "Convergence for a Decreasing Filtration of a Cochain Complex", the spectral sequence converges: on each bidegree the pages stabilize to a limit term $E_\infty^{p,q}$, and this limit is the associated graded of the total cohomology $H_{\mathrm{dR}}^{p+q}(X, \mathbb{C})$ for the filtration induced by $F^\bullet$. Explicitly, the filtration induced on $H_{\mathrm{dR}}^k(X, \mathbb{C})$ is $F^p H_{\mathrm{dR}}^k(X, \mathbb{C}) = \operatorname{im}\big(H^k(F^p\mathcal{A}^\bullet(X)) \to H_{\mathrm{dR}}^k(X, \mathbb{C})\big)$, and $E_\infty^{p,q} \cong \operatorname{gr}_F^p H_{\mathrm{dR}}^{p+q}(X, \mathbb{C})$, completing the identification and the proof.

The theorem is a bookkeeping device with sharp content. It says that de Rham cohomology is assembled from Dolbeault cohomology by a succession of correction steps: start from $\bigoplus_{p+q=k} H_{\bar\partial}^{p,q}(X)$, take the cohomology of the differential $d_1$ induced by $\partial$ to get the second page, then the cohomology of $d_2$, and so on, until the pages stabilize; the surviving graded pieces reassemble $H_{\mathrm{dR}}^k(X, \mathbb{C})$. Each differential $d_r$ measures a way in which a Dolbeault class fails to extend to a genuine de Rham class, or in which two Dolbeault classes become identified in de Rham cohomology. On a general complex manifold these higher differentials can be nonzero, and then the Dolbeault numbers over-count the de Rham cohomology. The extreme opposite case, when every $d_r$ with $r \geq 1$ vanishes, is called degeneration at $E_1$; there the first page already equals the limit page, and the bigrading passes intact to cohomology.

The distinction between degeneration and its failure is not academic. The first differential $d_1$ is the map $\partial\colon H_{\bar\partial}^{p,q}(X) \to H_{\bar\partial}^{p+1,q}(X)$, and there exist compact complex manifolds where it is nonzero, so the Frölicher spectral sequence does not degenerate for them. The Iwasawa manifold, a compact quotient of a complex Heisenberg group of complex dimension 3, is the standard example: a direct computation of its Dolbeault cohomology exhibits a nonvanishing $d_1$, so its first Betti number is strictly smaller than $h^{1,0} + h^{0,1}$ (see [15] for the full calculation on the Iwasawa manifold). Such manifolds are necessarily non-Kähler, and the reason is exactly the degeneration statement below.

Once the abutment is filtered with $E_\infty^{p,q}$ as graded pieces, counting dimensions across a fixed total degree gives an inequality between Betti numbers and Hodge numbers. This is the first hard consequence of the spectral sequence and the numerical form of the entire theory.

**Corollary 2.4.3: The Frölicher Inequality**

Let $X$ be a compact complex manifold, and write $b_k = \dim_{\mathbb{C}} H_{\mathrm{dR}}^k(X, \mathbb{C})$ for its $k$-th Betti number and $h^{p,q} = \dim_{\mathbb{C}} H_{\bar\partial}^{p,q}(X)$ for its Hodge numbers. Then

$$b_k \leq \sum_{p+q=k} h^{p,q}$$

for every $k$, with equality for all $k$ if and only if the Frölicher spectral sequence of $X$ degenerates at $E_1$.

***Proof*:**
Fix $k$. On a compact complex manifold each Dolbeault group $H^{p,q}_{\bar\partial}(X)$ is finite dimensional (this is Dolbeault's theorem theorem 2.3.3 together with the finiteness of the sheaf cohomology of a coherent sheaf on a compact complex manifold; the Betti numbers are likewise finite), so every dimension below is a well-defined nonnegative integer.

*Step 1: Each page cannot grow.* For $r \geq 1$ the page $E_{r+1}$ is the cohomology of $(E_r, d_r)$, so in each bidegree $E^{p,q}_{r+1}$ is a subquotient of $E^{p,q}_r$, giving $\dim_{\mathbb{C}} E^{p,q}_{r+1} \leq \dim_{\mathbb{C}} E^{p,q}_r$. Passing to the limit, $\dim_{\mathbb{C}} E^{p,q}_\infty \leq \dim_{\mathbb{C}} E^{p,q}_1 = h^{p,q}$.

*Step 2: The limit page computes the Betti number.* By theorem 2.4.2 the limit terms are the graded pieces of a finite filtration of $H^k_{\mathrm{dR}}(X, \mathbb{C})$, so

$$b_k = \dim_{\mathbb{C}} H^k_{\mathrm{dR}}(X, \mathbb{C}) = \sum_{p+q=k} \dim_{\mathbb{C}} E^{p,q}_\infty$$

Combining with Step 1,

$$b_k = \sum_{p+q=k} \dim_{\mathbb{C}} E^{p,q}_\infty \leq \sum_{p+q=k} \dim_{\mathbb{C}} E^{p,q}_1 = \sum_{p+q=k} h^{p,q}$$

which is the asserted inequality.

*Step 3: The equality criterion.* Equality holds in the displayed line, for a fixed $k$, if and only if $\dim_{\mathbb{C}} E^{p,q}_\infty = \dim_{\mathbb{C}} E^{p,q}_1$ for every $(p,q)$ with $p+q=k$ (a sum of nonnegative terms each bounded by the corresponding term of another sum is equal to it precisely when the terms match). By Step 1, the chain of inequalities $\dim_{\mathbb{C}} E^{p,q}_1 \geq \dim_{\mathbb{C}} E^{p,q}_2 \geq \cdots \geq \dim_{\mathbb{C}} E^{p,q}_\infty$ collapses to equalities throughout exactly when each map does not drop dimension, that is when every $d_r$ $(r \geq 1)$ is zero at $E^{p,q}_r$ and at the source $E^{p-r,q+r-1}_r$ of the incoming differential. Requiring equality for all $k$ forces every $d_r$ on the entire page to vanish for all $r \geq 1$, which is the definition of degeneration at $E_1$; conversely if the sequence degenerates at $E_1$ then $E_\infty = E_1$ and equality holds in every total degree. This is the claimed equivalence, completing the proof.

The Frölicher inequality is the first quantitative link between the analytic bigrading and the topology of $X$. Read from left to right it bounds a topological invariant, the Betti number, by a sum of analytic invariants, the Hodge numbers; the gap $\sum_{p+q=k} h^{p,q} - b_k$ measures the total drop caused by the higher differentials. Read as an equality it becomes a strong constraint on the manifold: equality in every degree is the assertion that the entire spectral sequence collapses immediately, so that the Hodge numbers add up to the Betti numbers with no cancellation. The parity consequence is immediate and worth recording: if the sequence degenerates and moreover Hodge symmetry $h^{p,q} = h^{q,p}$ holds, then $b_{2k+1} = \sum_{p+q=2k+1} h^{p,q}$ is a sum of terms that pair off, so every odd Betti number is even. Both hypotheses, degeneration and Hodge symmetry, are supplied by the Kähler condition, and this is the route by which chapter 4 deduces that odd Betti numbers of compact Kähler manifolds are even.

The equality case is not exotic. It holds for the class of manifolds this book is built around.

**2.4.4 Foreshadowing: Degeneration on Compact Kähler Manifolds** If $X$ is a compact Kähler manifold, then the Frölicher spectral sequence degenerates at $E_1$, and hence $b_k = \sum_{p+q=k} h^{p,q}$ for every $k$ (equality in the Frölicher inequality, corollary 2.4.3). The proof rests on the Hodge decomposition, which produces an actual direct-sum splitting $H^k_{\mathrm{dR}}(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ of the abutment realizing the Hodge filtration, forcing every higher differential to vanish. This is established in full in chapter 4 via the Kähler identity $\Delta_d = 2\Delta_{\bar\partial}$; it is stated here to complete the interpretation of the inequality but is not proved at this stage, since the harmonic-theory input is developed only in chapter 3 and chapter 4.

Degeneration is genuine information about $X$, not a formal fact. Its failure on the Iwasawa manifold shows that it can fail, and the dimension there is not incidental: non-degeneration cannot occur on a compact complex *surface*, whose Frölicher spectral sequence always degenerates at $E_1$ (chapter 4); its success on compact Kähler manifolds is a theorem requiring the metric input of the Kähler identities. Deferring the proof is the correct organization: the spectral sequence and the inequality are elementary consequences of the filtration and belong here, while the vanishing of the higher differentials is a consequence of harmonic theory and belongs where that theory is built. What can be extracted now, with no metric hypothesis, is the numerology on projective space, which turns out already to satisfy the equality.

**Example 2.4.5: Hodge Numbers and Degeneration for $\mathbb{C}\mathrm{P}^n$**

Let $X = \mathbb{C}\mathrm{P}^n$ (example 1.1.8), of complex dimension $n$. Its complex cohomology is well known from [24]: $H^{2j}_{\mathrm{dR}}(\mathbb{C}\mathrm{P}^n,\mathbb{C}) = \mathbb{C}$ for $0 \le j \le n$, generated by the $j$-th power of the hyperplane class, and $H^{\mathrm{odd}}_{\mathrm{dR}}(\mathbb{C}\mathrm{P}^n,\mathbb{C}) = 0$. Thus the Betti numbers are

$$b_k = \begin{cases} 1 & k \text{ even},\ 0 \le k \le 2n \\ 0 & k \text{ odd} \end{cases}$$

The Dolbeault numbers $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}_{\bar\partial}(\mathbb{C}\mathrm{P}^n) = \dim_{\mathbb{C}} H^q(\mathbb{C}\mathrm{P}^n, \Omega^p_{\mathbb{C}\mathrm{P}^n})$ (via Dolbeault's theorem theorem 2.3.3) were computed for $\mathbb{C}\mathrm{P}^1$ in example 2.3.5; the pattern for all $n$ is

$$h^{p,q} = \begin{cases} 1 & 0 \le p = q \le n \\ 0 & p \ne q \end{cases}$$

so the Dolbeault cohomology is concentrated on the diagonal, one copy of $\mathbb{C}$ in each bidegree $(p,p)$ for $0 \le p \le n$ and zero off the diagonal. This is the algebraic-geometry computation $H^q(\mathbb{C}\mathrm{P}^n, \Omega^p) = \mathbb{C}$ for $p = q$ and 0 otherwise, carried out in [4] through the Bott vanishing / Euler-sequence method, and it agrees with the analytic Dolbeault computation by Dolbeault's theorem.

Now compare the two sides of the Frölicher inequality. Fix $k$. The right-hand sum is

$$\sum_{p+q=k} h^{p,q} = \begin{cases} h^{k/2,k/2} = 1 & k \text{ even},\ 0 \le k \le 2n \\ 0 & k \text{ odd} \end{cases}$$

because the only nonzero Hodge number with $p+q=k$ occurs when $k$ is even and $p = q = k/2$. This is exactly $b_k$ in every degree, so

$$b_k = \sum_{p+q=k} h^{p,q} \qquad \text{for all } k$$

Equality holds throughout, and by corollary 2.4.3 the Frölicher spectral sequence of $\mathbb{C}\mathrm{P}^n$ degenerates at $E_1$. This is consistent with box 2.4.4, since $\mathbb{C}\mathrm{P}^n$ carries the Fubini-Study Kähler metric (constructed in chapter 3); but here the degeneration is verified directly from the numbers, with no appeal to harmonic theory. The reason there is no room for a higher differential is structural: with the cohomology supported on the diagonal $p = q$, the differential $d_1 \colon E_1^{p,q} \to E_1^{p+1,q}$ maps a nonzero group $H^{p,p}$ into $H^{p+1,p}$, which is zero, and likewise every $d_r$ $(r \geq 1)$ has zero target, so all higher differentials vanish for want of a place to land.

The example is the cleanest instance of degeneration: the spectral sequence collapses not because of a metric argument but because the Dolbeault cohomology is so sparse that no nonzero differential can exist. The same diagonal concentration recurs for every smooth projective toric variety and, more generally, wherever the Hodge numbers off the diagonal vanish. It also anchors the numerology used throughout the book: the Hodge diamond of $\mathbb{C}\mathrm{P}^n$ is a single string of 1's down the diagonal, and every later Hodge diamond is measured against this baseline.

The Dolbeault numbers of $\mathbb{C}\mathrm{P}^n$ were quoted from two sources that agree, the analytic Dolbeault computation and the algebraic sheaf-cohomology computation $H^q(\mathbb{C}\mathrm{P}^n, \Omega^p_{\mathbb{C}\mathrm{P}^n})$. That they agree is a theorem, GAGA, which asserts the agreement for every smooth projective variety. This is the bridge that lets Hodge theory, an analytic subject, compute invariants of algebraic varieties, and lets algebraic geometry supply the Hodge numbers that feed the Frölicher inequality.

The setup requires a word on the two categories being compared. A smooth projective variety $X$ over $\mathbb{C}$ carries two structure sheaves and two cohomology theories. Algebraically, $X$ is a scheme of finite type over $\mathbb{C}$ with its Zariski topology and its sheaf $\mathcal{O}_X^{\mathrm{alg}}$ of regular functions, and one has the coherent sheaf cohomology $H^q(X^{\mathrm{alg}}, \Omega^p_{X^{\mathrm{alg}}})$ of algebraic Kähler differentials. Analytically, the same variety has an underlying complex manifold $X^{\mathrm{an}}$ (the analytification), with its classical topology and holomorphic structure sheaf $\mathcal{O}_{X^{\mathrm{an}}}$, and the sheaf cohomology $H^q(X^{\mathrm{an}}, \Omega^p_{X^{\mathrm{an}}})$ of holomorphic $p$-forms (definition 1.2.11) is the analytic object identified with Dolbeault cohomology in theorem 2.3.3. There is always a comparison map from the algebraic groups to the analytic groups, induced by the map of ringed spaces $X^{\mathrm{an}} \to X^{\mathrm{alg}}$. GAGA says this map is an isomorphism.

**Theorem 2.4.6: GAGA for Coherent Cohomology**

Let $X$ be a smooth projective variety over $\mathbb{C}$, with analytification $X^{\mathrm{an}}$ the associated compact complex manifold. Then for all $p, q \geq 0$ the comparison map

$$H^q(X^{\mathrm{alg}}, \Omega^p_{X^{\mathrm{alg}}}) \xrightarrow{\sim} H^q(X^{\mathrm{an}}, \Omega^p_{X^{\mathrm{an}}})$$

from algebraic to analytic coherent cohomology is an isomorphism. In particular the algebraic and analytic Hodge numbers agree, so the Dolbeault numbers $h^{p,q}$ of $X^{\mathrm{an}}$ may be computed by algebraic geometry.

***Proof*:**
This is Serre's GAGA theorem in the form comparing coherent cohomology of a projective variety with that of its analytification. It is proved in full in [4]: for a projective scheme $X$ over $\mathbb{C}$ and any coherent algebraic sheaf $\mathcal{F}$, analytification $\mathcal{F} \mapsto \mathcal{F}^{\mathrm{an}}$ is an equivalence between coherent algebraic and coherent analytic sheaves, and the natural maps $H^q(X^{\mathrm{alg}}, \mathcal{F}) \to H^q(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}})$ are isomorphisms for all $q$. The differential sheaves $\Omega^p_{X^{\mathrm{alg}}}$ are coherent, and their analytifications are the holomorphic differential sheaves $\Omega^p_{X^{\mathrm{an}}}$, so the general statement specializes to the one

above. The proof rests on the finiteness and comparison theorems for projective morphisms, which are developed in [4] rather than in this book.

GAGA is the reason the transcendental theory of this book is a theory about algebraic varieties and not only about abstract complex manifolds. It has two payoffs used repeatedly. First, computability: the analytic Hodge numbers of a projective variety, hard to reach through partial differential equations, become algebraic quantities computable by sheaf cohomology, Serre duality, and exact sequences, exactly as the $h^{p,q}$ of $\mathbb{C}\mathrm{P}^n$ were read off above. Second, rigidity: a holomorphic object on a projective variety, such as a holomorphic form or a coherent subsheaf, is automatically algebraic, so the analytic and algebraic worlds have the same coherent geometry. The Lefschetz $(1,1)$-theorem and the Hodge conjecture, which ask whether certain analytic cohomology classes are algebraic, live in the space GAGA opens up: GAGA settles the coherent-sheaf half of the question, and Hodge theory addresses the topological half.

One boundary is worth marking. GAGA compares coherent cohomology, the groups $H^q(X, \Omega_X^p)$; it does not by itself supply the Hodge decomposition of the topological cohomology $H^k(X, \mathbb{C})$. That decomposition is the content of chapter 4 and requires the Kähler metric. What GAGA and the Frölicher spectral sequence contribute is the input and the framework: the Dolbeault numbers (computed algebraically, via GAGA) and their relationship to the Betti numbers (organized by the spectral sequence). Degeneration, proved by harmonic theory in chapter 4, is the missing statement that turns the Frölicher inequality into the Hodge decomposition.

## Exercise 2.4.1

1. Verify directly that the Hodge filtration $F^\bullet$ of definition 2.4.1 is a filtration of the de Rham complex, that is, that $d(F^p\mathcal{A}^k(X)) \subseteq F^p\mathcal{A}^{k+1}(X)$ for all $p, k$. Then show it is bounded on each $\mathcal{A}^k(X)$ by exhibiting the smallest and largest $p$ for which $F^p\mathcal{A}^k(X)$ is nonzero.
2. Let $X$ be a compact Riemann surface (complex dimension 1) of genus $g$. Its Betti numbers are $b_0 = b_2 = 1$ and $b_1 = 2g$, and its Hodge numbers are $h^{0,0} = h^{1,1} = 1$, $h^{1,0} = h^{0,1} = g$. Verify the Frölicher inequality in every degree and conclude that the spectral sequence degenerates at $E_1$ for every compact Riemann surface, with no appeal to the Kähler condition.
3. Show that on a compact complex manifold with $h^{p,q} = 0$ whenever $p \neq q$, the Frölicher spectral sequence degenerates at $E_1$. (This is the structural reason $\mathbb{C}\mathrm{P}^n$ degenerates in example 2.4.5; isolate the argument that no nonzero differential can exist.)
4. The Hodge-de Rham (or Hirzebruch) alternating sum is $\chi(X) = \sum_k (-1)^k b_k$ and the analytic Euler characteristic in each column is $\chi(X, \Omega^p) = \sum_q (-1)^q h^{p,q}$. Show that regardless of whether the Frölicher spectral sequence degenerates, one has $\sum_k (-1)^k b_k = \sum_{p,q} (-1)^{p+q} h^{p,q}$, so that the topological Euler characteristic equals the alternating sum of all Hodge numbers.
5. Use GAGA (theorem 2.4.6) together with Serre duality on a smooth projective variety $X$ of dimension $n$ to prove the duality $h^{p,q} = h^{n-p,n-q}$ among the Hodge numbers. State clearly which input is analytic and which is algebraic.
6. A Hopf surface is a compact complex surface diffeomorphic to $S^1 \times S^3$; it is not Kähler. From the diffeomorphism type its Betti numbers are $b_0 = b_4 = 1$, $b_1 = b_3 = 1$, $b_2 = 0$. Explain why the odd Betti number $b_1 = 1$ shows, via corollary 2.4.3 and box 2.4.4, that either the Frölicher spectral sequence of a Hopf surface fails to degenerate at $E_1$ or Hodge symmetry $h^{p,q} = h^{q,p}$ fails. (In fact it is Hodge symmetry that fails: $h^{1,0} = 0$ but $h^{0,1} = 1$. The spectral sequence

does degenerate, as it does for every compact complex surface.)

# 3

# Kähler Metrics and the Kähler Identities

Chapter 2 computed the Dolbeault cohomology of a complex manifold and organized its bidegrees, through the Frölicher spectral sequence, against the de Rham cohomology. On a general complex manifold that spectral sequence need not degenerate: the Hodge bigrading is an invariant of the complex structure alone, and nothing yet forces it to survive to the topology. What is missing is a compatible metric. This chapter supplies it. A Hermitian metric whose fundamental form is closed (a *Kähler* metric) binds the complex structure to the Riemannian one tightly enough that the two Laplacians $\Delta_d$ and $\Delta_{\bar\partial}$ become proportional.

That proportionality, $\Delta_d = 2\Delta_{\bar\partial}$, is the engine of the theory, and it follows from a short list of commutator relations among $\partial$, $\bar\partial$, their formal adjoints, and the Lefschetz operators $L$ and $\Lambda$: the Kähler identities. This chapter builds them from the ground up, beginning with Hermitian metrics and the Kähler condition, passing through the model Fubini-Study metric that makes every projective manifold Kähler, and arriving at the identities and the $\partial\bar\partial$-lemma. It closes with a surface that fails the Kähler condition, confirming that the Kähler hypothesis of the next chapter is a restriction and not an automatic property. The harmonic theory that turns $\Delta_d = 2\Delta_{\bar\partial}$ into the Hodge decomposition is the subject of chapter 4.

## 3.1 Hermitian Metrics and the Kähler Condition

The two previous chapters extracted everything the holomorphic structure alone can give: the operators $\partial, \bar\partial$, the type decomposition of forms, and the Dolbeault cohomology $H^{p,q}_{\bar\partial}(X)$ that refines de Rham cohomology. What they could not deliver is a comparison between the two bigradings the manifold carries: the topological grading by total degree $k$ and the holomorphic grading by type

$(p, q)$. The Frölicher spectral sequence of chapter 2 measured the gap between them and produced the inequality $b_k \leq \sum_{p+q=k} h^{p,q}$, but it could not force equality. Equality is a statement about harmonic representatives, and harmonic representatives require a metric. This section introduces the metric structure and singles out the one compatibility condition, the Kähler condition, that makes the holomorphic and metric structures cooperate. Everything in the remaining foundations of compact Kähler geometry rests on it.

A complex manifold carries two pieces of linear-algebraic data at each point: the complex structure $J$ on the real tangent space, and, once a metric is chosen, an inner product. A Hermitian metric is precisely a Riemannian metric that respects $J$. The associated fundamental form $\omega$ repackages this compatible pair as a single differential 2-form, and the Kähler condition $d\omega = 0$ asks that this form be closed. The payoff of closedness, made precise by proposition 3.1.6, is that a Kähler metric osculates the flat metric of $\mathbb{C}^n$ to second order in suitable holomorphic coordinates. This second-order flatness is what lets every first-order identity proved on flat $\mathbb{C}^n$ propagate to an arbitrary Kähler manifold, and it is the engine behind the Kähler identities of section 3.3.

Throughout, $X$ is a complex manifold with $\dim_{\mathbb{C}} X = n$. At a point $p$ the holomorphic tangent space $T_p^{1,0}X$ is the $+i$-eigenspace of $J$ acting on the complexified tangent space $T_pX \otimes \mathbb{C}$; in holomorphic coordinates $(z^1, \ldots, z^n)$ it has the basis $\partial/\partial z^1, \ldots, \partial/\partial z^n$ (proposition 1.1.13 and example 1.1.14). A metric on this complex vector space is a Hermitian inner product varying smoothly with $p$.

**Definition 3.1.1: Hermitian Metric**

A *Hermitian metric* on a complex manifold $X$ is a choice, for each $p \in X$, of a positive-definite Hermitian inner product

$$h_p \colon T_p^{1,0}X \times T_p^{1,0}X \to \mathbb{C}$$

depending smoothly on $p$, where $h_p$ is $\mathbb{C}$-linear in its first argument, conjugate-linear in its second, and satisfies $h_p(u, v) = \overline{h_p(v, u)}$ and $h_p(u, u) > 0$ for $u \neq 0$. In holomorphic coordinates the metric is recorded by the smooth functions

$$h_{j\bar{k}} = h\left(\frac{\partial}{\partial z^j}, \frac{\partial}{\partial z^k}\right)$$

which form, at each point, a positive-definite Hermitian matrix $(h_{j\bar{k}})$. A *Hermitian manifold* is a pair $(X, h)$.

The bar in $h_{j\bar{k}}$ is bookkeeping, not conjugation of a number: it records that the second slot of $h$ is the conjugate-linear one, so that under a change of holomorphic coordinates the index $\bar{k}$ transforms by the conjugate Jacobian while $j$ transforms by the Jacobian. Hermitian symmetry reads $h_{j\bar{k}} = \overline{h_{k\bar{j}}}$, and positive-definiteness is the statement that $\sum_{j,k} h_{j\bar{k}}\, \xi^j \overline{\xi^k} > 0$ for every nonzero $\xi \in \mathbb{C}^n$.

Every complex manifold admits a Hermitian metric. Cover $X$ by holomorphic charts, put the standard flat Hermitian form $\sum_j dz^j \otimes d\bar{z}^j$ on each chart, and glue with a partition of unity subordinate to the cover; a convex combination of positive-definite Hermitian forms is again positive-definite Hermitian, so the result is a global Hermitian metric. The interesting question is therefore never existence of a Hermitian metric but whether one can be chosen compatibly with the holomorphic structure in the stronger sense captured below.

A Hermitian metric $h$ carries two pieces of real data. Its real part $g = \operatorname{Re} h$ is a Riemannian metric on the underlying real manifold, compatible with $J$ in the sense that $g(Ju, Jv) = g(u, v)$; its imaginary part is an alternating form, and it is this alternating part, viewed as a differential 2-form, that organizes Kähler geometry.

**Definition 3.1.2: The Fundamental Form**

Let $(X, h)$ be a Hermitian manifold. The *fundamental form* of $h$ is the differential 2-form $\omega$ defined in holomorphic coordinates by

$$\omega = i \sum_{j,k} h_{j\bar{k}}\, dz^j \wedge d\bar{z}^k \tag{3.1}$$

where $dz^j = dx^j + i\, dy^j$ and $d\bar{z}^k = dx^k - i\, dy^k$ in the real coordinates $z^j = x^j + iy^j$.

The formula (3.1) looks coordinate-dependent, but it is not: the collection $(h_{j\bar{k}})$ transforms exactly so that $\sum_{j,k} h_{j\bar{k}}\, dz^j \wedge d\bar{z}^k$ is invariant, being the coordinate expression of the intrinsic tensor $-2\,\mathrm{Im}\, h$ transported to forms (proposition 3.1.3). The normalization is the Huybrechts one, and the factor $i$ is what makes $\omega$ real, as the next proposition records.

**Proposition 3.1.3: The Fundamental Form is Real and Positive**

The fundamental form $\omega$ of a Hermitian metric $h$ is a real $(1,1)$-form: $\bar{\omega} = \omega$. Moreover $\omega$ is positive, meaning that for every nonzero $(1,0)$-tangent vector $v$ at any point,

$$-i\,\omega(v, \bar{v}) > 0$$

and the underlying Riemannian metric is recovered by $g(u, v) = \frac{1}{2}\,\omega(u, Jv)$.

***Proof*:**
That $\omega$ is of type $(1,1)$ is immediate from (3.1), since every summand is $dz^j \wedge d\bar{z}^k$.

*Step 1: $\omega$ is real.* Conjugating (3.1) and using $\overline{dz^j} = d\bar{z}^j$ gives

$$\bar{\omega} = -i \sum_{j,k} \overline{h_{j\bar{k}}}\, d\bar{z}^j \wedge dz^k$$

where the sign flips because $\bar{i} = -i$. Swapping the summation names $j \leftrightarrow k$ and reordering the wedge, $d\bar{z}^k \wedge dz^j = -dz^j \wedge d\bar{z}^k$, we obtain

$$\bar{\omega} = -i \sum_{j,k} \overline{h_{k\bar{j}}}\, d\bar{z}^k \wedge dz^j = i \sum_{j,k} \overline{h_{k\bar{j}}}\, dz^j \wedge d\bar{z}^k$$

Hermitian symmetry $\overline{h_{k\bar{j}}} = h_{j\bar{k}}$ turns the right-hand side back into $\omega$, so $\bar{\omega} = \omega$.

*Step 2: positivity.* Fix a point and a nonzero $v = \sum_j \xi^j\, \partial/\partial z^j \in T^{1,0}X$. Then $\bar{v} = \sum_k \overline{\xi^k}\, \partial/\partial \bar{z}^k$, and pairing $\omega$ against $(v, \bar{v})$ picks out the coefficient of $dz^j \wedge d\bar{z}^k$ evaluated on $\partial/\partial z^j$ and $\partial/\partial \bar{z}^k$, which is 1. Thus

$$\omega(v, \bar{v}) = i \sum_{j,k} h_{j\bar{k}}\, \xi^j \overline{\xi^k}$$

so $-i\,\omega(v, \bar{v}) = \sum_{j,k} h_{j\bar{k}}\, \xi^j \overline{\xi^k}$, which is positive for $v \neq 0$ by positive-definiteness of $(h_{j\bar{k}})$.

*Step 3: recovering $g$.* On real tangent vectors the Hermitian metric decomposes as $h = g + i\, g(-, J-)$ into its symmetric part $g = \operatorname{Re} h$ and its alternating part $\operatorname{Im} h = g(-, J-)$; that the second summand is alternating and $J$-invariant is exactly the compatibility $g(Ju, Jv) = g(u, v)$. Unwinding (3.1) on real tangent vectors identifies $\omega$ with $-2 \operatorname{Im} h$ under the normalization (3.1) (the factor 2 is the same one that produces $\omega_0 = 2 \sum_j dx^j \wedge dy^j$ from the flat metric in example 3.1.4), giving

$$\omega(u, v) = -2 \operatorname{Im} h(u, v) = -2\, g(u, Jv)$$

Replacing $v$ by $Jv$ and using $J^2 = -1$ yields $\omega(u, Jv) = -2\, g(u, J^2 v) = 2\, g(u, v)$, that is $g(u, v) = \frac{1}{2}\, \omega(u, Jv)$, as we sought to show.

The proposition says that a Hermitian metric and its fundamental form are two encodings of the same object. The metric $g$ is the symmetric datum a geometer measures lengths with; the form $\omega$ is the alternating datum an algebraic geometer integrates against and takes cohomology of. The relation $g(u, v) = \frac{1}{2}\, \omega(u, Jv)$ lets one pass between them freely, and it is why a single form $\omega$ suffices to encode all the metric geometry on a complex manifold. Positivity of $\omega$ is the condition that distinguishes fundamental forms among all real $(1, 1)$-forms: an arbitrary real $(1, 1)$-form need not come from a metric, but a positive one does.

The natural first example is the model on which every local computation will be tested.

**Example 3.1.4: The Flat Metric on $\mathbb{C}^n$**

On $\mathbb{C}^n$ take the standard Hermitian metric $h_{j\bar{k}} = \delta_{jk}$, the constant identity matrix. Its fundamental form is

$$\omega_0 = i \sum_{j=1}^{n} dz^j \wedge d\bar{z}^j$$

Writing $z^j = x^j + iy^j$, each summand expands as $dz^j \wedge d\bar{z}^j = (dx^j + i\, dy^j) \wedge (dx^j - i\, dy^j) = -2i\, dx^j \wedge dy^j$, so

$$\omega_0 = 2 \sum_{j=1}^{n} dx^j \wedge dy^j$$

a constant-coefficient form, and $\omega_0^n = 2^n\, n!\, dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n$ is a positive multiple of the Euclidean volume form. The underlying Riemannian metric $g = \operatorname{Re} h$ is $\sum_j \big((dx^j)^2 + (dy^j)^2\big)$, the Euclidean metric on $\mathbb{R}^{2n}$.

The flat form $\omega_0$ has constant coefficients in the real coordinates $x^j, y^j$, so it is closed: $d\omega_0 = 0$. This closedness is the property the Kähler condition abstracts. It is the one condition under which the metric structure and the complex structure are compatible enough to interact with cohomology, and it is far from automatic once the coefficients $h_{j\bar{k}}$ are allowed to vary.

**Definition 3.1.5: Kähler Metric and Kähler Manifold**

A Hermitian metric $h$ on a complex manifold $X$ is a *Kähler metric* if its fundamental form is closed:

$$d\omega = 0$$

In this case $\omega$ is a *Kähler form*, and the pair $(X, \omega)$ is a *Kähler manifold.*

Since $\omega$ is a real $(1, 1)$-form, its exterior derivative splits by type as $d\omega = \partial\omega + \bar{\partial}\omega$, where $\partial\omega$ has type $(2,1)$ and $\bar{\partial}\omega$ has type $(1,2)$ (theorem 1.2.6). Because $\omega$ is real, $\overline{\partial\omega} = \bar{\partial}\bar{\omega} = \bar{\partial}\omega$, so the two pieces are complex conjugates of each other. The single real equation $d\omega = 0$ is therefore equivalent to either one of $\partial\omega = 0$ or $\bar{\partial}\omega = 0$: closedness, $\partial$-closedness, and $\bar{\partial}$-closedness of $\omega$ all coincide. This trichotomy is used constantly and is worth keeping in view.

The Kähler condition is a system of first-order partial differential equations on the metric coefficients. To see what it demands, differentiate (3.1). Applying $\partial$ and collecting terms,

$$\partial\omega = i \sum_{j,k,l} \frac{\partial h_{j\bar{k}}}{\partial z^l}\, dz^l \wedge dz^j \wedge d\bar{z}^k \tag{3.2}$$

and antisymmetry of $dz^l \wedge dz^j$ in $l, j$ means $\partial\omega = 0$ is the requirement

$$\frac{\partial h_{j\bar{k}}}{\partial z^l} = \frac{\partial h_{l\bar{k}}}{\partial z^j} \qquad \text{for all } j,k,l \tag{3.3}$$

Conjugating gives the companion symmetry $\partial h_{j\bar{k}}/\partial\bar{z}^l = \partial h_{j\bar{l}}/\partial\bar{z}^k$, which is $\bar{\partial}\omega = 0$. So the Kähler condition is exactly the symmetry (3.3) of the first holomorphic derivatives of the metric. It says nothing about the values $h_{j\bar{k}}(p)$ themselves, only about how they vary to first order. This is the precise sense in which a Kähler metric is one that is flat to first order, and it is what the characterization proved next makes geometric.

The characterization upgrades (3.3) into a geometric statement: a metric is Kähler at $p$ precisely when holomorphic coordinates can be chosen in which it agrees with the flat metric to second order. This is the central result of the chapter. Every Kähler identity in section 3.3 is first-order in the metric, so it holds on a general Kähler manifold as soon as it holds on flat $\mathbb{C}^n$; the following proposition is what licenses that reduction.

**Proposition 3.1.6: Existence of Kähler Normal Coordinates**

Let $(X, h)$ be a Hermitian manifold and $p \in X$. The following are equivalent.

1. $d\omega = 0$ at the point $p$.
2. There exist holomorphic coordinates $(z^1, \ldots, z^n)$ centered at $p$ in which the metric osculates the flat metric to second order:

$$h_{j\bar{k}}(z) = \delta_{jk} + O(|z|^2)$$

equivalently $h_{j\bar{k}}(p) = \delta_{jk}$ and all first derivatives $\partial h_{j\bar{k}}/\partial z^l$ and $\partial h_{j\bar{k}}/\partial \bar{z}^l$ vanish at $p$.

If $d\omega = 0$ on all of $X$, such coordinates exist at every point; they are called *Kähler normal coordinates* at $p$.

***Proof*:**
*Step 1:* (2) $\Rightarrow$ (1). Suppose coordinates exist with $h_{j\bar{k}}(z) = \delta_{jk} + O(|z|^2)$. Then every first derivative $\partial h_{j\bar{k}}/\partial z^l$ vanishes at $p$, so (3.2) shows $\partial\omega|_p = 0$, and by conjugation $\bar{\partial}\omega|_p = 0$; hence $d\omega|_p = 0$.

*Step 2: normalizing the value at $p$.* Now assume $d\omega|_p = 0$; we produce the coordinates. Start from any holomorphic coordinates $(w^1, \ldots, w^n)$ centered at $p$. The matrix $H = (h_{j\bar{k}}(p))$ is positive-definite Hermitian, so by simultaneous diagonalization there is a constant invertible complex matrix $B = (B^r_j)$ with

$$\sum_{r,s} \overline{B^s_k}\, h_{r\bar{s}}(p)\, B^r_j = \delta_{jk}$$

The linear holomorphic change $u^j = \sum_r (B^{-1})^j_r\, w^r$, that is $w^r = \sum_j B^r_j u^j$, has constant Jacobian $\partial w^r/\partial u^j = B^r_j$. Since the metric coefficients transform covariantly in the first index and by the conjugate in the second, the coefficients in the $u$-coordinates are

$$h'_{j\bar{k}}(u) = \sum_{r,s} \frac{\partial w^r}{\partial u^j}\, \overline{\frac{\partial w^s}{\partial u^k}}\, h_{r\bar{s}}(w(u)) = \sum_{r,s} B^r_j\, \overline{B^s_k}\, h_{r\bar{s}}(w(u))$$

Evaluating at $u = 0$ gives $h'_{j\bar{k}}(0) = \sum_{r,s} \overline{B^s_k}\, h_{r\bar{s}}(p)\, B^r_j = \delta_{jk}$ by the choice of $B$. So after this linear change $h'_{j\bar{k}}(0) = \delta_{jk}$. Rename the $u$-coordinates $w$; there is now no loss in assuming $h_{j\bar{k}}(p) = \delta_{jk}$, and it remains to kill the first derivatives.

*Step 3: a quadratic holomorphic change kills the first derivatives.* Set $c^j_{lm} = -\,\partial h_{m\bar{j}}/\partial w^l\big|_p$ and seek new coordinates

$$z^j = w^j - \frac{1}{2}\sum_{l,m} c^j_{lm}\, w^l w^m \tag{3.4}$$

The Kähler equation (3.3) at $p$ says $\partial h_{m\bar{j}}/\partial w^l = \partial h_{l\bar{j}}/\partial w^m$, so $c^j_{lm} = c^j_{ml}$ is symmetric in its lower indices and (3.4) is a well-defined quadratic change. It fixes $p$ with Jacobian $\partial z^j/\partial w^r|_p = \delta_{jr}$, so it preserves $h_{j\bar{k}}(p) = \delta_{jk}$; inverting, $w^j = z^j + \frac{1}{2}\sum_{l,m} c^j_{lm}\, z^l z^m + O(|z|^3)$ with holomorphic Jacobian

$$\frac{\partial w^r}{\partial z^j} = \delta_{rj} + \sum_m c^r_{jm}\, z^m + O(|z|^2)$$

Writing $\tilde{h}$ for the coefficients in the $z$-coordinates, the tensor transformation law reads

$$\tilde{h}_{j\bar{k}}(z) = \sum_{r,s} \frac{\partial w^r}{\partial z^j}\, \overline{\frac{\partial w^s}{\partial z^k}}\, h_{r\bar{s}}(w(z))$$

Differentiating in the holomorphic direction $z^l$ and evaluating at $p$, the antiholomorphic Jacobian factor is constant to first order and contributes nothing, so only two terms survive:

$$\left.\frac{\partial \tilde{h}_{j\bar{k}}}{\partial z^l}\right|_p = \left.\frac{\partial h_{j\bar{k}}}{\partial w^l}\right|_p + c^k_{jl}$$

the first term from differentiating $h_{r\bar{s}}(w(z))$ through the leading Jacobian $\delta_{rj}\delta_{sk}$ and the chain rule, the second from differentiating the holomorphic Jacobian factor $\partial w^r/\partial z^j$ against the leading metric value $h_{r\bar{k}}(p) = \delta_{rk}$. By the definition of $c$ and (3.3), $c^k_{jl} = -\,\partial h_{l\bar{k}}/\partial w^j|_p = -\,\partial h_{j\bar{k}}/\partial w^l|_p$, so the two terms cancel and $\partial\tilde{h}_{j\bar{k}}/\partial z^l|_p = 0$. Conjugating, the antiholomorphic derivatives $\partial\tilde{h}_{j\bar{k}}/\partial\bar{z}^l|_p$ are the complex conjugates of $\partial\tilde{h}_{k\bar{j}}/\partial z^l|_p$, which have just been arranged to vanish, so they vanish too.

*Step 4: reading off the osculation.* After Steps 2 and 3, $\tilde{h}_{j\bar{k}}(p) = \delta_{jk}$ and every first partial derivative of $\tilde{h}_{j\bar{k}}$ vanishes at $p$. Taylor expansion in the real coordinates then gives $\tilde{h}_{j\bar{k}}(z) = \delta_{jk} + O(|z|^2)$, which is statement (2), completing the proof.

The content of the proposition is that closedness of $\omega$ is exactly the obstruction to matching the flat metric to second order, and when the obstruction vanishes the match can be made. The value $h_{j\bar{k}}(p) = \delta_{jk}$ can always be arranged by a linear change on any complex manifold, Kähler or not; it is the vanishing of the first derivatives that is special, and the symmetry (3.3) required for the quadratic change of Step 3 to succeed is exactly the Kähler condition. A general Hermitian metric can be made to agree with the flat one only to first order, not second: the first derivatives obstruct, and (3.3) is the statement that half of them can be gauged away.

The name is chosen by analogy with Riemannian normal coordinates, in which a Riemannian metric agrees with the Euclidean one to second order at a point. The subtlety in the complex setting is that the coordinate change is required to be holomorphic; one cannot use an arbitrary diffeomorphism. That is why not every Hermitian metric admits normal coordinates, and why the Kähler condition is a genuine restriction rather than a normalization always available. A Hermitian metric that fails to be Kähler cannot be made to osculate the flat metric to second order by any holomorphic change of coordinates.

This second-order flatness is the reason Kähler geometry is so much more rigid than Hermitian geometry. Any identity between differential operators that involves the metric only through its value and first derivatives at a point holds on a Kähler manifold exactly when it holds on $\mathbb{C}^n$ with the flat metric, because Kähler normal coordinates make the two settings indistinguishable to that order. The Kähler identities of section 3.3 are of precisely this first-order character, and their proof will consist of verifying them once on flat $\mathbb{C}^n$ and invoking proposition 3.1.6 to conclude the general case.

Two families of examples make the definition concrete. The first is the flat metric itself, the second a class of quotients that will recur throughout the book.

**Example 3.1.7: Flat $\mathbb{C}^n$ is Kähler**

The flat metric of example 3.1.4 is Kähler. Its fundamental form $\omega_0 = 2\sum_j dx^j \wedge dy^j$ has constant coefficients, so $d\omega_0 = 0$ immediately. Equivalently, the metric coefficients $h_{j\bar{k}} = \delta_{jk}$ are constant, so all their derivatives vanish and the symmetry (3.3) holds trivially. The standard coordinates on $\mathbb{C}^n$ are already Kähler normal coordinates at every point, with no error term at all: $h_{j\bar{k}}(z) = \delta_{jk}$ exactly. Flat $\mathbb{C}^n$ is thus the model Kähler manifold, and proposition 3.1.6 says every Kähler manifold looks like it to second order.

**Example 3.1.8: Complex Tori with a Constant Metric**

A complex torus $X = \mathbb{C}^n/\Lambda$, where $\Lambda \subset \mathbb{C}^n$ is a lattice of rank $2n$ (example 1.1.10), inherits from $\mathbb{C}^n$ its holomorphic coordinates: the projection $\pi\colon \mathbb{C}^n \to X$ is a local biholomorphism, and the standard coordinates $z^j$ on $\mathbb{C}^n$ give local holomorphic coordinates on $X$ (the functions $z^j$ are not single-valued on the quotient, but the differentials $dz^j$ descend globally). Fix any positive-definite constant Hermitian matrix $(h_{j\bar{k}})$ with $h_{j\bar{k}} \in \mathbb{C}$ independent of the point, and let $h$ be the Hermitian metric on $\mathbb{C}^n$ with these coefficients. Because the coefficients are constant, $h$ is invariant under all translations, in particular under translation by the lattice $\Lambda$, so it descends to a Hermitian metric on the quotient $X$.

Its fundamental form is $\omega = i\sum_{j,k} h_{j\bar{k}}\, dz^j \wedge d\bar{z}^k$ with constant coefficients, hence $d\omega = 0$: the metric is Kähler. The simplest choice $h_{j\bar{k}} = \delta_{jk}$ gives the flat torus, whose fundamental form is the descent of $\omega_0$; a general constant $(h_{j\bar{k}})$ gives a Kähler metric of "constant type" whose Kähler class $[\omega] \in H^2(X,\mathbb{R})$ can be computed by integrating against the 2-cycles of the torus. Every complex torus therefore admits a Kähler metric, so tori supply an inexhaustible testing ground for Kähler theory; the case $n = 1$ recovers the elliptic curve with its flat metric.

The torus example isolates the mechanism. The obstruction to the Kähler condition lives in the first derivatives of the metric coefficients, and a translation-invariant metric has coefficients that do not vary at all, so the obstruction is absent for the most elementary reason available. Contrast this with the Fubini-Study metric of section 3.2, whose coefficients vary from point to point; there the Kähler condition holds not because the coefficients are constant but because they are the second derivatives of a single potential function, and $d(\partial\bar{\partial}\varphi) = 0$ automatically. Both mechanisms, constancy and the $\partial\bar{\partial}$-potential, will be exploited repeatedly.

One structural consequence of the Kähler condition deserves mention now, since it explains why the Kähler class is a topological invariant of the metric. On a compact Kähler manifold $X$ of dimension $n$, the form $\omega^n/n!$ is a positive multiple of the volume form (as example 3.1.4 illustrates for the flat model), so its integral is the total volume and in particular is nonzero. Since $\omega$ is closed, it defines a de Rham class $[\omega] \in H^2(X, \mathbb{R})$, and $\int_X \omega^n = \int_X \omega \wedge \cdots \wedge \omega$ depends only on this class by Stokes' theorem. A nonzero integral forces $[\omega]^n \neq 0$ in $H^{2n}(X, \mathbb{R})$, so $[\omega] \neq 0$, and more strongly $[\omega]^k \neq 0$ for every $k \leq n$. This is the first topological constraint the Kähler condition imposes: a compact Kähler manifold of complex dimension $n$ has $H^{2k}(X, \mathbb{R}) \neq 0$ for all $0 \leq k \leq n$. The Hopf surface of section 3.4 will fail this constraint and thereby fail to be Kähler.

### Exercise 3.1.1

1. Let $\omega = i \sum_{j,k} h_{j\bar{k}}\, dz^j \wedge d\bar{z}^k$ be the fundamental form of a Hermitian metric. Verify directly, without invoking proposition 3.1.3, that the two-form $\eta = i\, dz^1 \wedge d\bar{z}^1$ on $\mathbb{C}^1$ satisfies $\bar{\eta} = \eta$, and compute $\eta$ in the real coordinates $z = x + iy$. Then show that $i\, dz^1 \wedge dz^2$ is *not* real, so that the factor of $i$ alone does not make a form real; the type $(1,1)$ and Hermitian symmetry are needed.

2. Let $\omega$ be a Kähler form on a complex manifold $X$ and let $f\colon X \to \mathbb{R}_{>0}$ be a smooth positive function. Show that the conformally rescaled metric with fundamental form $f\omega$ is Kähler if and only if $df \wedge \omega = 0$. On a compact connected Kähler manifold of complex dimension $n \geq 2$, deduce that $f\omega$ is Kähler if and only if $f$ is constant, and explain why the deduction fails for $n = 1$. (Hint: the map $\alpha \mapsto \alpha \wedge \omega^{n-1}$ on 1-forms is a pointwise isomorphism.)

3. Let $(X, \omega_X)$ and $(Y, \omega_Y)$ be Kähler manifolds with projections $p\colon X \times Y \to X$ and $q\colon X \times Y \to Y$. Show that $\omega = p^*\omega_X + q^*\omega_Y$ is the fundamental form of a Kähler metric on $X \times Y$. Conclude that a product of complex tori, and more generally a finite product of Kähler manifolds, is Kähler.

4. On the complex torus $X = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\operatorname{Im}\tau > 0$, equip $X$ with the flat metric $h_{1\bar{1}} = 1$. Compute the total area $\int_X \omega$ of the fundamental form in terms of $\operatorname{Im}\tau$, and confirm it is positive. (This is the $n = 1$ case of the volume computation, and $[\omega] \neq 0$ in $H^2(X, \mathbb{R})$.)

## 3.2 The Fubini-Study Metric

The Kähler condition of section 3.1 is a closedness requirement on a positive $(1,1)$-form, and closedness alone can be hard to verify by hand: it asks that the exterior derivative of $\omega$ vanish identically, a condition on first derivatives of the metric coefficients. The cleanest way to produce a closed positive form is to write it as $i\partial\bar{\partial}\varphi$ for a real function $\varphi$, for then closedness is automatic. Recall from proposition 1.2.8 that $\partial^2 = \bar{\partial}^2 = 0$ and $\partial\bar{\partial} = -\bar{\partial}\partial$, and from theorem 1.2.6 that $d = \partial + \bar{\partial}$; a

$(1,1)$-form of the shape $i\partial\bar\partial\varphi$ therefore satisfies $d(i\partial\bar\partial\varphi) = i(\partial+\bar\partial)\partial\bar\partial\varphi = 0$, since $\partial^2\bar\partial\varphi = 0$ and $\bar\partial\partial\bar\partial\varphi = -\partial\bar\partial\bar\partial\varphi = 0$. A function $\varphi$ whose $i\partial\bar\partial\varphi$ equals a given Kähler form is called a *Kähler potential*. This section carries out the construction for the one example that makes the whole theory apply to algebraic geometry: complex projective space $\mathbb{C}\mathrm{P}^n$ (example 1.1.8), equipped with the Fubini-Study metric. Because the metric is built from a local potential, it is Kähler almost for free; and because the Kähler condition passes to complex submanifolds, this single example proves that every projective manifold, hence every smooth projective algebraic variety, is Kähler.

The subtlety is that $\mathbb{C}\mathrm{P}^n$ has no global holomorphic coordinate, so no single $\varphi$ can serve as a potential on all of it. What saves the construction is that the natural candidate, $\log\|Z\|^2$ built from a homogeneous representative $Z = (z_0 : \cdots : z_n)$, is not globally defined but changes only by the addition of $\log|f|^2$ for a nowhere-zero holomorphic $f$ when the representative is rescaled, and such a term is killed by $\partial\bar\partial$. The forms $i\partial\bar\partial\varphi$ on the various charts therefore agree on overlaps and glue to a global closed positive $(1,1)$-form.

The identity that makes the gluing work comes first.

**Lemma 3.2.1: Vanishing of $\partial\bar\partial\log|f|^2$**

Let $f$ be a holomorphic function on an open set $U \subseteq \mathbb{C}^m$ with $f$ nowhere zero on $U$. Then $\log|f|^2$ is smooth on $U$ and

$$\partial\bar\partial\log|f|^2 = 0$$

***Proof*:**
Since $f$ is holomorphic and nowhere zero, $\log f$ admits a smooth local branch near every point (a holomorphic logarithm exists on any simply connected neighborhood on which $f$ does not vanish, and the two local branches differ by a locally constant $2\pi i$-multiple, which does not affect the derivatives below). Write $\log|f|^2 = \log(f\bar f) = \log f + \overline{\log f}$ locally. Now $\log f$ is holomorphic, so $\bar\partial(\log f) = 0$, whence $\partial\bar\partial(\log f) = 0$; and $\overline{\log f}$ is antiholomorphic, so $\partial(\overline{\log f}) = 0$, whence $\partial\bar\partial(\overline{\log f}) = -\bar\partial\partial(\overline{\log f}) = 0$ by proposition 1.2.8. Adding the two contributions gives $\partial\bar\partial\log|f|^2 = 0$, as we sought to show.

The content of the lemma is that a holomorphic factor is invisible to the operator $\partial\bar\partial$: only the size of $f$ enters $\log|f|^2$, but the operator sees only the second-order mixed variation, which a holomorphic function does not contribute. This is exactly what is needed to make the transition between two homogeneous representatives on $\mathbb{C}\mathrm{P}^n$ harmless, since those two representatives differ by a nowhere-zero holomorphic factor on the overlap.

Recall the standard charts on $\mathbb{C}\mathrm{P}^n$ from example 1.1.8: writing a point as $[Z] = [z_0 : \cdots : z_n]$ in homogeneous coordinates, the chart $U_j = \{[Z] : z_j \neq 0\}$ carries the affine coordinates $w_k^{(j)} = z_k/z_j$ for $k \neq j$, and $U_j \cong \mathbb{C}^n$ via these. On $U_j$ the function $[Z] \mapsto \log(\sum_{k=0}^n |z_k/z_j|^2)$ is a well-defined smooth function, since the ratios $z_k/z_j$ are coordinates there. This is the local potential from which the Fubini-Study form is built.

**Definition 3.2.2: The Fubini-Study Metric**

The *Fubini-Study form* on $\mathbb{C}\mathrm{P}^n$ is the real $(1,1)$-form $\omega_{\mathrm{FS}}$ defined on the chart $U_j = \{z_j \neq 0\}$ by

$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\,\partial\bar\partial\,\log\Bigl(\sum_{k=0}^{n}\lvert z_k/z_j\rvert^2\Bigr) = \frac{i}{2\pi}\,\partial\bar\partial\,\log\lVert Z\rVert^2$$

where $\lVert Z\rVert^2 = \sum_{k=0}^{n}\lvert z_k\rvert^2$ is computed from any homogeneous representative $Z$. The associated Hermitian metric on $T^{1,0}\mathbb{C}\mathrm{P}^n$ (definition 3.1.1), whose fundamental form (definition 3.1.2) is $\omega_{\mathrm{FS}}$, is the *Fubini-Study metric*.

Two points in the definition need justification before it can be trusted: that the local formula is independent of the homogeneous representative $Z$ (so the forms on different charts agree where they overlap and glue to one global form), and that $\omega_{\mathrm{FS}}$ is positive (so it is the fundamental form of a Hermitian metric, not a closed real $(1,1)$-form). Both are settled in the theorem below. The normalizing constant $\frac{1}{2\pi}$ is the Huybrechts convention; it is chosen so that the cohomology class $[\omega_{\mathrm{FS}}]$ is the integral generator of $H^2(\mathbb{C}\mathrm{P}^n,\mathbb{Z})$, a point taken up at the end of the section.

**Theorem 3.2.3: $\mathbb{C}\mathrm{P}^n$ Is Kähler**

The Fubini-Study form $\omega_{\mathrm{FS}}$ of definition 3.2.2 is a well-defined global real $(1,1)$-form on $\mathbb{C}\mathrm{P}^n$; it is positive; and it is closed. Hence $(\mathbb{C}\mathrm{P}^n,\omega_{\mathrm{FS}})$ is a compact Kähler manifold.

***Proof*:**
The proof is in four steps: the local formula is independent of the chosen representative and chart, the resulting form is real, it is positive, and it is closed.

*Step 1: Independence of the homogeneous representative.* Fix the chart $U_j$ and let $Z, Z'$ be two homogeneous representatives of the same points, so $Z' = fZ$ for a nowhere-zero holomorphic function $f$ on $U_j$ (concretely, on $U_j$ one may take $Z = (z_0,\dots,z_n)$ and the normalized representative with $z_j = 1$, related by $f = 1/z_j$, which is holomorphic and nonvanishing on $U_j$). Then

$$\log\lVert Z'\rVert^2 = \log\bigl(\lvert f\rvert^2\lVert Z\rVert^2\bigr) = \log\lvert f\rvert^2 + \log\lVert Z\rVert^2$$

Applying $\partial\bar\partial$ and using lemma 3.2.1 to kill the first term, $\partial\bar\partial\log\lVert Z'\rVert^2 = \partial\bar\partial\log\lVert Z\rVert^2$. The local formula for $\omega_{\mathrm{FS}}$ therefore does not depend on which representative is used.

Now consider two charts $U_j$ and $U_l$ and a point $[Z]\in U_j\cap U_l$. On the overlap both charts compute $\omega_{\mathrm{FS}}$ as $\frac{i}{2\pi}\partial\bar\partial\log\lVert Z\rVert^2$ for a homogeneous representative, and by the previous paragraph the answer is insensitive to the choice of representative. The transition from the $U_j$-representative (normalized by $z_j = 1$) to the $U_l$-representative (normalized by $z_l = 1$) is multiplication by the nowhere-zero holomorphic function $z_j/z_l$ on the overlap, again killed by $\partial\bar\partial$. The two local definitions agree on $U_j\cap U_l$, so $\omega_{\mathrm{FS}}$ is a well-defined global $(1,1)$-form. It is smooth because on each chart it is $\partial\bar\partial$ of the smooth function $\log\lVert Z\rVert^2$.

*Step 2: The form is real.* A $(1,1)$-form $\alpha$ is real when $\bar\alpha = \alpha$. Since $\log\lVert Z\rVert^2$ is a real-valued function, complex conjugation exchanges $\partial$ and $\bar\partial$ on it, giving $\overline{\partial\bar\partial\log\lVert Z\rVert^2} = \bar\partial\partial\log\lVert Z\rVert^2 =$

$-\partial\bar{\partial}\log\|Z\|^2$ by proposition 1.2.8. Therefore

$$\overline{\omega_{\mathrm{FS}}} = \overline{\frac{i}{2\pi}\partial\bar{\partial}\log\|Z\|^2} = -\frac{i}{2\pi}\cdot\left(-\partial\bar{\partial}\log\|Z\|^2\right) = \omega_{\mathrm{FS}}$$

where the sign from $\bar{i} = -i$ cancels the sign produced by swapping $\partial$ and $\bar{\partial}$. Thus $\omega_{\mathrm{FS}}$ is real.

*Step 3: Positivity.* By homogeneity and the transitive action of the unitary group $U(n+1)$ on $\mathbb{C}\mathrm{P}^n$ (which acts by holomorphic isometries of the construction, since it preserves $\|Z\|^2$), it suffices to check positivity at a single point, say the point $p = [1\colon 0\colon \cdots : 0] \in U_0$. On $U_0$ use the affine coordinates $w_k = z_k/z_0$ for $k = 1, \ldots, n$, so that $p$ is the origin $w = 0$ and

$$\log\|Z\|^2 = \log\Big(1 + \sum_{k=1}^{n}|w_k|^2\Big)$$

Write $\rho = 1 + \sum_k|w_k|^2 = 1 + \sum_k w_k\bar{w}_k$. Then $\partial\rho = \sum_k \bar{w}_k\, dw_k$ and $\bar{\partial}\rho = \sum_k w_k\, d\bar{w}_k$, and a direct computation gives

$$\partial\bar{\partial}\log\rho = \partial\Big(\frac{\bar{\partial}\rho}{\rho}\Big) = \frac{\partial\bar{\partial}\rho}{\rho} - \frac{\partial\rho\wedge\bar{\partial}\rho}{\rho^2}$$

Here $\partial\bar{\partial}\rho = \sum_k dw_k\wedge d\bar{w}_k$ (since $\partial\bar{\partial}(w_k\bar{w}_k) = dw_k\wedge d\bar{w}_k$), so

$$\partial\bar{\partial}\log\rho = \frac{1}{\rho}\sum_k dw_k\wedge d\bar{w}_k - \frac{1}{\rho^2}\Big(\sum_k \bar{w}_k\, dw_k\Big)\wedge\Big(\sum_l w_l\, d\bar{w}_l\Big)$$

At the origin $w = 0$ we have $\rho = 1$ and the second term vanishes, so

$$\omega_{\mathrm{FS}}\big|_p = \frac{i}{2\pi}\sum_{k=1}^{n} dw_k\wedge d\bar{w}_k$$

Comparing with the Huybrechts normalization $\omega = i\sum h_{k\bar{l}}\, dw_k\wedge d\bar{w}_l$ of definition 3.1.2, the Hermitian matrix at $p$ is $h_{k\bar{l}} = \frac{1}{2\pi}\delta_{kl}$, which is positive definite. By $U(n+1)$-homogeneity the same holds at every point, so $\omega_{\mathrm{FS}}$ is positive.

*Step 4: Closedness.* On each chart $\omega_{\mathrm{FS}} = \frac{i}{2\pi}\partial\bar{\partial}\varphi$ with $\varphi = \log\|Z\|^2$, and

$$d(\partial\bar{\partial}\varphi) = (\partial + \bar{\partial})\partial\bar{\partial}\varphi = \partial^2\bar{\partial}\varphi + \bar{\partial}\partial\bar{\partial}\varphi = 0 - \partial\bar{\partial}\bar{\partial}\varphi = 0$$

using $\partial^2 = 0$, $\bar{\partial}\partial = -\partial\bar{\partial}$, and $\bar{\partial}^2 = 0$ from proposition 1.2.8. Thus $d\omega_{\mathrm{FS}} = 0$ on each chart, hence globally. Combined with Steps 1 through 3, $\omega_{\mathrm{FS}}$ is a closed positive real $(1,1)$-form, so it is the fundamental form of a Kähler metric (definition 3.1.5). Since $\mathbb{C}\mathrm{P}^n$ is compact (example 1.1.8), $(\mathbb{C}\mathrm{P}^n, \omega_{\mathrm{FS}})$ is a compact Kähler manifold, completing the proof.

The theorem is what makes the book's methods apply so widely. Every smooth projective variety embeds in some $\mathbb{C}\mathrm{P}^N$, and the next corollary shows that the Kähler condition is inherited by complex submanifolds. Together they place the whole of smooth projective algebraic geometry inside the reach of Kähler theory, and hence, after chapter 4, of the Hodge decomposition. The local-potential mechanism is what makes the argument short: the hard-looking condition $d\omega = 0$ was verified by a one-line type computation once the form was recognized as $\partial\bar{\partial}$ of a function.

Before turning to submanifolds, the potential deserves a concrete look on the smallest case, where the round sphere makes an appearance.

**Example 3.2.4: The Round Metric on $\mathbb{C}\mathrm{P}^1$**

Take $n = 1$ and the chart $U_0 = \{z_0 \neq 0\}$ with affine coordinate $w = z_1/z_0$, identifying $U_0 \cong \mathbb{C}$. By the Step 3 computation with a single coordinate,

$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\,\partial\bar{\partial}\log(1+|w|^2) = \frac{i}{2\pi}\cdot\frac{dw\wedge d\bar{w}}{(1+|w|^2)^2}$$

where the second equality follows because with one coordinate $\rho = 1+w\bar{w}$ gives $\partial\bar{\partial}\log\rho = \frac{dw\wedge d\bar{w}}{\rho} - \frac{\bar{w}\,dw\wedge w\,d\bar{w}}{\rho^2} = \frac{(1+|w|^2)-|w|^2}{\rho^2}\,dw\wedge d\bar{w} = \frac{dw\wedge d\bar{w}}{\rho^2}$. Writing $w = x+iy$, one has $dw\wedge d\bar{w} = -2i\,dx\wedge dy$, so

$$\omega_{\mathrm{FS}} = \frac{1}{\pi}\cdot\frac{dx\wedge dy}{(1+x^2+y^2)^2}$$

which is a positive multiple of the standard area form on the plane, blown down toward infinity by the factor $(1+|w|^2)^{-2}$. This is exactly the pushforward, under stereographic projection, of the area form of a round metric on $S^2$. One caution about the arithmetic: the quantity $\int_{\mathbb{C}\mathrm{P}^1}\omega_{\mathrm{FS}}$ is the integral of the *fundamental* 2*-form*, which under the Huybrechts normalization is twice the Riemannian area of the metric $g = \operatorname{Re} h$ (the flat model $\omega = i\,dw\wedge d\bar{w} = 2\,dx\wedge dy$ against area form $dx\wedge dy$ exhibits the factor 2). With this understood: the unnormalized potential $i\partial\bar{\partial}\log(1+|w|^2)$ gives the round sphere of radius $\frac{1}{2}$, whose Riemannian area is $\pi$ and whose fundamental form therefore integrates to $2\pi$; the cohomological factor $\frac{1}{2\pi}$ rescales that form to total integral 1. So $\mathbb{C}\mathrm{P}^1$ with the Fubini-Study metric is a round sphere up to constant scale: since $\int_{\mathbb{C}\mathrm{P}^1}\omega_{\mathrm{FS}} = 1$ is a form integral, the Riemannian area is $\frac{1}{2}$, so the sphere has radius $\frac{1}{2\sqrt{2\pi}}$, the radius-$\frac{1}{2}$ sphere shrunk by the normalization. The total integral is indeed finite,

$$\int_{\mathbb{C}\mathrm{P}^1}\omega_{\mathrm{FS}} = \frac{1}{\pi}\int_{\mathbb{C}}\frac{dx\,dy}{(1+x^2+y^2)^2} = \frac{1}{\pi}\int_0^{2\pi}\!\!\int_0^{\infty}\frac{r\,dr\,d\theta}{(1+r^2)^2} = \frac{1}{\pi}\cdot 2\pi\cdot\frac{1}{2} = 1$$

where the omitted point at infinity has measure zero and does not affect the integral. The value 1 is the normalization promised after definition 3.2.2: $\int_{\mathbb{C}\mathrm{P}^1}\omega_{\mathrm{FS}} = 1$ says that $[\omega_{\mathrm{FS}}]$ is the generator of $H^2(\mathbb{C}\mathrm{P}^1,\mathbb{Z}) \cong \mathbb{Z}$ dual to the fundamental class.

The example makes the geometry vivid: the Fubini-Study metric on $\mathbb{C}\mathrm{P}^1$ is nothing more exotic than the round sphere, and the abstract potential $\log(1+|w|^2)$ recovers a metric the reader has met since calculus. The factor $(1+|w|^2)^{-2}$ is the stereographic conformal factor, and the finiteness of the total area reflects the compactness of $\mathbb{C}\mathrm{P}^1$. This is the $n=1$ instance of the general principle that $\omega_{\mathrm{FS}}$ integrates to 1 over any line $\mathbb{C}\mathrm{P}^1 \subseteq \mathbb{C}\mathrm{P}^n$, which pins down the cohomology class.

The inheritance property that makes theorem 3.2.3 do its work across all of projective geometry comes next.

**Corollary 3.2.5: Projective Manifolds Are Kähler**

Let $(X, \omega)$ be a Kähler manifold and let $\iota\colon Y \hookrightarrow X$ be a complex submanifold. Then $(Y, \iota^*\omega)$ is a Kähler manifold. In particular, every complex submanifold of $\mathbb{C}\mathrm{P}^n$ is Kähler; hence every smooth projective algebraic variety over $\mathbb{C}$, being a complex submanifold of some $\mathbb{C}\mathrm{P}^N$, is a compact Kähler manifold.

***Proof*:**
The claim is that $\iota^*\omega$ is a closed positive real $(1,1)$-form on $Y$. Each property is checked by pulling back along the holomorphic inclusion $\iota$.

*Type and reality.* The inclusion $\iota$ is holomorphic, so its differential $d\iota$ maps $T^{1,0}Y$ into $T^{1,0}X$ and $T^{0,1}Y$ into $T^{0,1}X$; pullback of forms therefore preserves the $(p,q)$-bigrading, and $\iota^*\omega$ is again of type $(1,1)$. Pullback commutes with complex conjugation, so $\overline{\iota^*\omega} = \iota^*\bar{\omega} = \iota^*\omega$ since $\omega$ is real; thus $\iota^*\omega$ is real.

*Closedness.* Pullback commutes with the exterior derivative, so $d(\iota^*\omega) = \iota^*(d\omega) = \iota^*0 = 0$, using $d\omega = 0$ from the hypothesis that $(X, \omega)$ is Kähler.

*Positivity.* Fix $y \in Y$ and a nonzero tangent vector $v \in T^{1,0}_y Y$. The Hermitian form associated to $\iota^*\omega$ evaluates on $v$ as the Hermitian form of $\omega$ evaluated on $d\iota(v) \in T^{1,0}_{\iota(y)}X$. Since $\iota$ is an immersion, $d\iota$ is injective, so $d\iota(v) \neq 0$; and $\omega$ is positive, so its Hermitian form is positive on the nonzero vector $d\iota(v)$. Therefore the Hermitian form of $\iota^*\omega$ is positive on every nonzero $v$, and $\iota^*\omega$ is positive.

Thus $\iota^*\omega$ is a closed positive real $(1,1)$-form, hence the fundamental form of a Kähler metric on $Y$ (definition 3.1.5). Applying this with $(X, \omega) = (\mathbb{C}\mathrm{P}^n, \omega_{\mathrm{FS}})$ from theorem 3.2.3, every complex submanifold of $\mathbb{C}\mathrm{P}^n$ is Kähler. A smooth projective variety over $\mathbb{C}$ is by definition a closed complex submanifold of some $\mathbb{C}\mathrm{P}^N$, and it is compact because $\mathbb{C}\mathrm{P}^N$ is; so it is a compact Kähler manifold, as we sought to show.

The corollary is the reason Hodge theory is a tool of algebraic geometry and not only of complex differential geometry. The restriction $\iota^*\omega_{\mathrm{FS}}$ of the Fubini-Study form is the metric a projective variety carries by virtue of its embedding, and no further work is required to know it is Kähler. Every theorem this book proves for compact Kähler manifolds, the Hodge decomposition (chapter 4), Hodge symmetry, the Lefschetz theorems, applies at once to every smooth projective variety. The converse fails: there exist compact complex manifolds carrying no Kähler metric at all, a phenomenon isolated in section 3.4 through the Hopf surface, and there even exist compact Kähler manifolds that are not projective (the general complex torus of dimension at least 2). The Kähler world properly contains the projective one.

There is one loose end worth tying off: the precise meaning of the normalizing constant $\frac{1}{2\pi}$. The claim, foreshadowed after definition 3.2.2 and confirmed for $n = 1$ in example 3.2.4, is that $[\omega_{\mathrm{FS}}]$ is an integral class, indeed the positive generator of $H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z})$. The clean statement identifies it with a Chern class.

**Proposition 3.2.6: The Fubini-Study Class and $c_1(\mathcal{O}(1))$**

In $H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z})$ one has

$$[\omega_{\mathrm{FS}}] = c_1(\mathcal{O}(1))$$

the first Chern class of the hyperplane bundle $\mathcal{O}(1)$, dual to the tautological bundle $\mathcal{O}(-1)$ of example 1.3.9. In particular $[\omega_{\mathrm{FS}}]$ is the positive generator of $H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z}) \cong \mathbb{Z}$, and $\int_{\mathbb{C}\mathrm{P}^1} \omega_{\mathrm{FS}} = 1$ for any line $\mathbb{C}\mathrm{P}^1 \subseteq \mathbb{C}\mathrm{P}^n$.

The same curvature computation is carried out in [9, the Chern connection of a Hermitian holomorphic line bundle], and the proof below duplicates it. It is kept here because the normalizations differ and the difference matters downstream. That book writes $\omega_{FS} = i\,\partial\bar\partial \log(1+|w|^2)$ and concludes $c_1(\mathcal{O}(1)) = \frac{1}{2\pi}\omega_{FS}$, whereas $\omega_{\mathrm{FS}}$ here carries the factor $\frac{1}{2\pi}$ in its definition, so that

$$\omega_{FS}^{\mathrm{RG}} = 2\pi\,\omega_{\mathrm{FS}}$$

writing $\omega_{FS}^{\mathrm{RG}}$ for that book's form, and the two statements agree. The convention used throughout this book is the one that makes $[\omega_{\mathrm{FS}}]$ an *integral* class, which is what section 4.4 and the projectivity criteria of chapter 5 require; importing the upstream normalization unchanged would put a $2\pi$ into every integrality statement.

***Proof*:**
Recall from example 1.3.9 the tautological line bundle $\mathcal{O}(-1) \subseteq \mathbb{C}\mathrm{P}^n \times \mathbb{C}^{n+1}$, whose fiber over $[Z]$ is the line $\mathbb{C}\cdot Z \subseteq \mathbb{C}^{n+1}$, and its dual $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$. Equip $\mathcal{O}(-1)$ with the Hermitian metric inherited from the standard Hermitian inner product on $\mathbb{C}^{n+1}$: a section of $\mathcal{O}(-1)$ over $U_j$ is represented by the holomorphic frame $s_j\colon [Z] \mapsto Z/z_j$ (the point of the line normalized to have $j$th coordinate 1), whose squared length is $h(s_j, s_j) = \|Z/z_j\|^2 = \|Z\|^2/|z_j|^2$.

The first Chern class of a Hermitian holomorphic line bundle $(L, h)$ is represented by the first Chern form $\frac{i}{2\pi}\Theta$, where $\Theta$ is the curvature of the Chern connection of $(L, h)$; the general statement that this closed 2-form represents the integral class $c_1(L)$, and that on a line bundle with local holomorphic frame $s$ of squared length $|h| = h(s,s)$ the curvature is $\Theta = -\partial\bar\partial\log|h|$, is the Chern-Weil description of the first Chern class, developed for Hermitian vector bundles in [9] (Part II, Chern-Weil theory); this book uses that formula rather than re-deriving it. For $\mathcal{O}(-1)$ with the frame $s_j$ above,

$$\frac{i}{2\pi}\Theta_{\mathcal{O}(-1)} = -\frac{i}{2\pi}\partial\bar\partial\log\frac{\|Z\|^2}{|z_j|^2} = -\frac{i}{2\pi}\partial\bar\partial\log\|Z\|^2 + \frac{i}{2\pi}\partial\bar\partial\log|z_j|^2 = -\omega_{\mathrm{FS}}$$

where the term $\frac{i}{2\pi}\partial\bar\partial\log|z_j|^2$ vanishes by lemma 3.2.1 ($z_j$ is nowhere zero and holomorphic on $U_j$). Thus $c_1(\mathcal{O}(-1)) = [-\omega_{\mathrm{FS}}]$. Since $c_1(L^\vee) = -c_1(L)$ for any line bundle, $c_1(\mathcal{O}(1)) = -c_1(\mathcal{O}(-1)) = [\omega_{\mathrm{FS}}]$.

The class $c_1(\mathcal{O}(1))$ is the positive generator of $H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z}) \cong \mathbb{Z}$: the hyperplane bundle $\mathcal{O}(1)$ has a section vanishing simply along a hyperplane $H \cong \mathbb{C}\mathrm{P}^{n-1}$, and $c_1(\mathcal{O}(1))$ is the Poincaré dual of $[H]$, which generates $H^2$. Pairing with a line $\mathbb{C}\mathrm{P}^1 \subseteq \mathbb{C}\mathrm{P}^n$ (which meets a general hyperplane in one point) gives $\int_{\mathbb{C}\mathrm{P}^1}\omega_{\mathrm{FS}} = \langle c_1(\mathcal{O}(1)), [\mathbb{C}\mathrm{P}^1]\rangle = 1$, matching the direct computation of example 3.2.4, completing the proof.

The proposition settles the choice of constant. Stripping the $\frac{1}{2\pi}$ would rescale $\omega_{\mathrm{FS}}$ and its class off the integral lattice; with it, $[\omega_{\mathrm{FS}}]$ is not just a real Kähler class but the integral generator, and the metric is the curvature form of a positive line bundle. This is the analytic face of a definition that recurs throughout the book: a Kähler class that is integral, meaning it lies in the image of $H^2(X,\mathbb{Z}) \to H^2(X,\mathbb{R})$, is called a *polarization*, and a compact Kähler manifold admitting one is projective by the Kodaira embedding theorem. The Fubini-Study class is the prototype, the polarization every projective variety inherits from its ambient $\mathbb{CP}^N$. The identification $[\omega_{\mathrm{FS}}] = c_1(\mathcal{O}(1))$ also foreshadows the Lefschetz (1,1)-theorem of chapter 4, where cup product with a Kähler class becomes the Lefschetz operator $L$ driving Hard Lefschetz.

**Exercise 3.2.1**

1. Let $\varphi, \psi$ be smooth real functions on an open set $U \subseteq \mathbb{C}^n$ with $i\partial\bar{\partial}\varphi = i\partial\bar{\partial}\psi$. Show that $\varphi - \psi$ is *pluriharmonic*, meaning $\partial\bar{\partial}(\varphi - \psi) = 0$, and deduce that two Kähler potentials for the same form differ by the real part of a holomorphic function. (This is the local ambiguity in a Kähler potential, and the mechanism behind the gluing in theorem 3.2.3.)
2. Verify directly that the group $U(n+1)$ acts on $\mathbb{CP}^n$ by biholomorphisms preserving $\omega_{\mathrm{FS}}$. (Hint: a matrix $A \in U(n+1)$ preserves $\|Z\|^2$, so it preserves the local potential; conclude that it preserves $\frac{i}{2\pi}\partial\bar{\partial}\log\|Z\|^2$.) Explain why this reduces the positivity check in theorem 3.2.3 to a single point.
3. Compute the total volume $\int_{\mathbb{CP}^n} \frac{\omega_{\mathrm{FS}}^n}{n!}$. (Hint: use that $\int_{\mathbb{CP}^n} \omega_{\mathrm{FS}}^n = \langle [\omega_{\mathrm{FS}}]^n, [\mathbb{CP}^n] \rangle$ and that $[\omega_{\mathrm{FS}}]$ is the generator of $H^2(\mathbb{CP}^n,\mathbb{Z})$ with $\int_{\mathbb{CP}^n} [\omega_{\mathrm{FS}}]^n = 1$, from proposition 3.2.6 and the ring structure of $H^*(\mathbb{CP}^n,\mathbb{Z})$.)
4. Let $C \subseteq \mathbb{CP}^2$ be a smooth plane curve of degree $d$, defined by a homogeneous polynomial $F$ of degree $d$. Using corollary 3.2.5, explain why $C$ is a compact Kähler manifold, and compute $\int_C \iota^*\omega_{\mathrm{FS}}$ where $\iota\colon C \hookrightarrow \mathbb{CP}^2$ is the inclusion. (Hint: a line $\mathbb{CP}^1$ meets $C$ in $d$ points by Bézout, so $[C] = d[\mathbb{CP}^1]$ in $H_2(\mathbb{CP}^2,\mathbb{Z})$; use proposition 3.2.6.)
5. Show that the closed real $(1,1)$-form $\eta = \frac{i}{2\pi}\, dw \wedge d\bar{w}$ on $\mathbb{C}$ (with $w = x + iy$) is positive but does *not* extend to a smooth positive $(1,1)$-form on $\mathbb{CP}^1$. (Hint: compute $\int_{\mathbb{C}} \eta$ and compare with the finiteness forced by compactness in example 3.2.4.) Contrast this with $\omega_{\mathrm{FS}}$, whose $(1+|w|^2)^{-2}$ factor is exactly what allows the extension.

## 3.3 The Kähler Identities

The Kähler condition $d\omega = 0$ is a single first-order equation on the metric, and by itself it looks modest. Its force comes from what it does to the operators of the Dolbeault complex. On a general Hermitian manifold the operators $\partial$, $\bar{\partial}$, and their formal adjoints $\partial^*$, $\bar{\partial}^*$ are related only loosely; the three Laplacians $\Delta_d$, $\Delta_\partial$, $\Delta_{\bar{\partial}}$ are different second-order operators, and there is no reason for the de Rham cohomology of $X$ to see the $(p,q)$-decomposition at all. On a Kähler manifold this changes completely. The Kähler condition $d\omega = 0$ forces a set of commutation relations among $\partial$, $\bar{\partial}$, and the Lefschetz operator $L = \omega \wedge (-)$ and its adjoint $\Lambda$, and these relations collapse the three Laplacians into one: $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_\partial$. Once the harmonic forms of chapter 4 are in hand, this one identity is what splits $H^k(X,\mathbb{C})$ along the bigrading and produces the Hodge decomposition.

The strategy of proof is worth stating at the outset, because it is what makes the computation manageable. The Kähler identities are commutator relations between first-order differential operators whose coefficients are built from the metric and its first derivatives. The Kähler condition says exactly that, in suitable coordinates, the metric osculates the flat metric to second order (proposition 3.1.6); the first derivatives of the metric therefore vanish at the chosen point. So both sides of each identity, evaluated at that point, agree with the corresponding expression for the flat metric on $\mathbb{C}^n$. It then suffices to verify the identities once and for all on flat $\mathbb{C}^n$, where every operator has constant coefficients and the whole thing reduces to combinatorics of wedge and contraction. Both halves of this argument are carried out in full below.

The operators that enter the identities come first. The metric supplies the Hodge star $\star$, and through it the formal adjoints $\partial^*$ and $\bar{\partial}^*$; the fundamental form supplies the Lefschetz operator $L$ and its pointwise adjoint $\Lambda$.

**Definition 3.3.1: The Lefschetz Operators**

Let $(X, \omega)$ be a Hermitian manifold of complex dimension $n$. The *Lefschetz operator* is

$$L\colon \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p+1,q+1}(X), \qquad L\alpha = \omega \wedge \alpha$$

Its pointwise adjoint with respect to the Hermitian inner product on forms induced by the metric is the *adjoint Lefschetz operator*

$$\Lambda = L^*\colon \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p-1,q-1}(X)$$

characterized by $\langle L\alpha, \beta\rangle = \langle \alpha, \Lambda\beta\rangle$ at every point of $X$.

Because $\omega$ is a real $(1,1)$-form, $L$ raises the bidegree by $(1,1)$ and is a real operator; $\Lambda$ lowers it by $(1,1)$. Both are algebraic (zeroth-order) operators: they act fibrewise on forms, involving no derivatives, and their coefficients are the components of $\omega$. In terms of the Hodge star one has $\Lambda = \star^{-1} L \star$ up to a sign fixed by the bidegree, but the pointwise-adjoint characterization is the one used here. The pair $(L, \Lambda)$ together with the counting operator that measures degree generates a copy of $\mathfrak{sl}_2$ acting on the exterior algebra at each point; that representation-theoretic structure is what drives the Hard Lefschetz theorem in chapter 6, but here only the two operators themselves are needed.

The formal adjoints of $\partial$ and $\bar{\partial}$ are built from the Hodge star exactly as the adjoint $d^*$ of the exterior derivative is on a Riemannian manifold. The general framework, that on a compact oriented manifold the operator $d^* = -\star d \star$ (up to a degree-dependent sign that vanishes in even real dimension) is the formal adjoint of $d$ for the $L^2$ inner product $\langle\!\langle \alpha, \beta\rangle\!\rangle = \int_X \langle \alpha, \beta\rangle \,\mathrm{vol}$, is developed for Riemannian manifolds in [9]. What is specific to the complex setting is the splitting of $d^*$ along the bigrading, which is recorded next.

**Definition 3.3.2: The Formal Adjoints $\partial^*$ and $\bar\partial^*$**

Let $(X,\omega)$ be a compact Hermitian manifold of complex dimension $n$, so the underlying real dimension $2n$ is even. Define

$$\partial^* = -\star\,\bar\partial\star, \qquad \bar\partial^* = -\star\,\partial\star$$

where $\star\colon \mathcal{A}^{p,q}(X)\to\mathcal{A}^{n-q,\,n-p}(X)$ is the $\mathbb{C}$-linear Hodge star of the metric. Then $\partial^*$ is the formal adjoint of $\partial$ and $\bar\partial^*$ the formal adjoint of $\bar\partial$ with respect to $\langle\!\langle -,-\rangle\!\rangle$; that is, $\langle\!\langle \partial\alpha,\beta\rangle\!\rangle = \langle\!\langle \alpha,\partial^*\beta\rangle\!\rangle$ and likewise for $\bar\partial$.

The star sends $(p,q)$-forms to $(n-q,n-p)$-forms, so it exchanges the two factors of the bigrading. This is why $\partial^* = -\star\bar\partial\star$ carries a $\bar\partial$ in its middle: conjugating $\bar\partial$ by the type-exchanging star produces an operator that lowers the holomorphic degree, matching the adjoint of $\partial$. The adjointness itself is the Riemannian statement $d^* = -\star d\star$ (in real dimension $2n$ the sign $(-1)^{2n(k+1)+1} = -1$ is uniform) split into its $(-1,0)$ and $(0,-1)$ parts using $d = \partial+\bar\partial$ from theorem 1.2.6: $d^* = \partial^*+\bar\partial^*$, and matching bidegrees identifies $\partial^* = -\star\bar\partial\star$ and $\bar\partial^* = -\star\partial\star$. These lower the holomorphic and antiholomorphic degrees by one respectively, so $\partial^*\colon\mathcal{A}^{p,q}\to\mathcal{A}^{p-1,q}$ and $\bar\partial^*\colon\mathcal{A}^{p,q}\to\mathcal{A}^{p,q-1}$. Complex conjugation exchanges $\partial\leftrightarrow\bar\partial$ and hence $\partial^*\leftrightarrow\bar\partial^*$, a symmetry that will halve the work below.

Before stating the identities, it helps to see the flat model that the whole proof reduces to, since the identities are ultimately a statement about constant-coefficient operators on $\mathbb{C}^n$.

**Example 3.3.3: The Operators on Flat $\mathbb{C}^n$**

On $\mathbb{C}^n$ with coordinates $z^j = x^j+iy^j$ and the flat metric for which $\{dz^j, d\bar z^k\}$ are the standard covectors, the fundamental form is

$$\omega = i\sum_{j=1}^{n} dz^j\wedge d\bar z^j$$

Introduce, for each index $j$, the wedge operators $e_j = dz^j\wedge(-)$ and $\bar e_j = d\bar z^j\wedge(-)$, and their pointwise adjoints, the contraction (interior-product) operators $\iota_j = e_j^*$ and $\bar\iota_j = \bar e_j^*$. With the normalization that $\{dz^j\}$ and $\{d\bar z^j\}$ are orthonormal up to the scaling built into the Hermitian inner product, these satisfy the anticommutation relations

$$\{e_j,\iota_k\} = \delta_{jk}, \quad \{\bar e_j,\bar\iota_k\} = \delta_{jk}, \quad \{e_j,e_k\} = \{e_j,\bar e_k\} = \{e_j,\bar\iota_k\} = 0$$

and all brackets of like-type wedges or of unlike-type wedge and contraction vanish. In these terms

$$L = i\sum_j e_j\bar e_j, \qquad \Lambda = L^* = -i\sum_j \bar\iota_j\iota_j$$

The operator $\Lambda$ is minus $i$ times a sum of double contractions because $L$ is $i$ times a sum of double wedges and taking adjoints reverses the order and turns each wedge into its contraction; the sign of $L$ flips under adjunction because $i$ is imaginary. These are the constant-coefficient models to which the general identities are reduced.

The anticommutation relations recorded here are the Clifford-algebra relations of the exterior algebra; they are what make the flat computation a finite piece of bookkeeping rather than an analytic problem. Everything below is assembled from them.

Now the central result. The Kähler identities are a small system of commutator relations. Writing $[A, B] = AB - BA$ for the commutator of operators (with no sign twist, since $\Lambda$ has even total degree), they read as follows.

**Theorem 3.3.4: The Kähler Identities**

Let $(X, \omega)$ be a Kähler manifold. Then the following commutator identities hold on $\mathcal{A}^{\bullet,\bullet}(X)$:

$$[\Lambda, \bar{\partial}] = -i\,\partial^*, \qquad [\Lambda, \partial] = i\,\bar{\partial}^*$$

and, by adjunction,

$$[L, \partial^*] = i\,\bar{\partial}, \qquad [L, \bar{\partial}^*] = -i\,\partial$$

The two lines are equivalent: the second is obtained from the first by taking adjoints, which turns $\Lambda$ into $L$ and $\bar{\partial}$ into $\bar{\partial}^*$. Conjugation alone will not do it, since $\Lambda$ is real and conjugation therefore never converts a commutator with $\Lambda$ into one with $L$; what conjugation relates is the two identities *within* each line. Only the first pair needs to be proved, and within that pair the second identity is the complex conjugate of the first, since conjugation fixes $\Lambda$ (it is a real operator, being adjoint to the real operator $L$), exchanges $\partial \leftrightarrow \bar{\partial}$, and exchanges $\partial^* \leftrightarrow \bar{\partial}^*$: conjugating $[\Lambda, \bar{\partial}] = -i\partial^*$ gives $[\Lambda, \partial] = +i\bar{\partial}^*$ (the sign flips because $\overline{-i} = i$). So the entire theorem reduces to the single identity $[\Lambda, \bar{\partial}] = -i\partial^*$.

***Proof*:**
By the reduction just described it suffices to prove

$$[\Lambda, \bar{\partial}] = -i\,\partial^* \tag{3.5}$$

the remaining three identities following by complex conjugation and adjunction. The proof has two stages: first the reduction of the general Kähler case to the flat model at a point, then the direct verification of (3.5) on flat $\mathbb{C}^n$.

*Step 1: Reduction to the flat model at a point.* Fix a point $p \in X$. Both sides of (3.5) are differential operators of order one: $\bar{\partial}$ and $\partial^*$ are first order, and $\Lambda$ is a zeroth-order (algebraic) operator, so the commutator $[\Lambda, \bar{\partial}]$ is again first order. A first-order operator, evaluated on a form at $p$, depends only on the value and the first derivatives at $p$ of the coefficients of the operator, that is, on the components of $\omega$ and their first derivatives at $p$. By proposition 3.1.6, the Kähler condition provides holomorphic coordinates $(z^1, \ldots, z^n)$ centered at $p$ in which

$$h_{j\bar{k}}(z) = \delta_{jk} + O(|z|^2)$$

so that the metric coefficients equal the flat coefficients $\delta_{jk}$ at $p$ and their first derivatives vanish at $p$. Consequently every operator appearing in (3.5) has, at $p$, exactly the same value and the same first derivatives as the corresponding flat operator built from $\omega_{\text{flat}} = i \sum dz^j \wedge d\bar{z}^j$. Since a first-order operator at $p$ is determined by that data, both sides of (3.5) at $p$ coincide with their flat counterparts. Here is the point of the Kähler hypothesis: the second-order osculation kills the first derivatives of the metric, which is precisely the data a first-order operator could otherwise see. If the metric were only Hermitian, the first derivatives of $\omega$ would enter and the two sides would differ by a zeroth-order torsion term built from $d\omega$; the Kähler condition $d\omega = 0$ is what makes that term disappear. Because $p$ was arbitrary, it now suffices to prove (3.5) for the flat metric on $\mathbb{C}^n$, which is the content of Step 2.

*Step 2: Verification on flat $\mathbb{C}^n$.* Work with the constant-coefficient operators of example 3.3.3. In these terms

$$\bar\partial = \sum_k \bar e_k\,\partial_{\bar z^k}, \qquad \Lambda = -i\sum_j \bar\iota_j\iota_j$$

where $\partial_{\bar z^k} = \partial/\partial\bar z^k$ denotes the (scalar, coefficient-wise) differentiation operator, which commutes with all the algebraic wedge and contraction operators because the flat coframe is constant. The formal adjoint $\partial^*$ likewise has constant coefficients; its expression will emerge from the computation rather than being assumed. We compute the commutator $[\Lambda,\bar\partial]$ directly:

$$[\Lambda,\bar\partial] = \Big[-i\sum_j \bar\iota_j\iota_j,\ \sum_k \bar e_k\,\partial_{\bar z^k}\Big] = -i\sum_{j,k}\big[\bar\iota_j\iota_j,\ \bar e_k\big]\,\partial_{\bar z^k}$$

the derivative $\partial_{\bar z^k}$ passing through the algebraic operators freely. It remains to evaluate the algebraic commutator $[\bar\iota_j\iota_j,\ \bar e_k]$. Since $\iota_j$ (a $dz$-contraction) anticommutes with $\bar e_k$ (a $d\bar z$-wedge), and both $\bar\iota_j$ and $\bar e_k$ are $d\bar z$-type, we compute using the Clifford relations of example 3.3.3. First,

$$\iota_j\,\bar e_k = -\bar e_k\,\iota_j$$

because a $dz$-contraction and a $d\bar z$-wedge anticommute (their bracket $\{e_j,\bar e_k\}$-type relation vanishes, and $\iota_j = e_j^*$, $\bar e_k$ are of unlike type). Therefore

$$\bar\iota_j\iota_j\,\bar e_k = -\bar\iota_j\,\bar e_k\,\iota_j$$

Now use the anticommutator $\{\bar\iota_j,\bar e_k\} = \delta_{jk}$, that is $\bar\iota_j\bar e_k = \delta_{jk} - \bar e_k\bar\iota_j$, to move $\bar\iota_j$ past $\bar e_k$:

$$-\bar\iota_j\,\bar e_k\,\iota_j = -\big(\delta_{jk} - \bar e_k\bar\iota_j\big)\iota_j = -\delta_{jk}\,\iota_j + \bar e_k\bar\iota_j\iota_j$$

On the other hand $\bar e_k\,\bar\iota_j\iota_j$ is the product in the opposite order, so the commutator is

$$[\bar\iota_j\iota_j,\ \bar e_k] = \bar\iota_j\iota_j\bar e_k - \bar e_k\bar\iota_j\iota_j = \big(-\delta_{jk}\iota_j + \bar e_k\bar\iota_j\iota_j\big) - \bar e_k\bar\iota_j\iota_j = -\delta_{jk}\,\iota_j$$

Substituting this into the sum,

$$[\Lambda,\bar\partial] = -i\sum_{j,k}(-\delta_{jk}\,\iota_j)\,\partial_{\bar z^k} = i\sum_j \iota_j\,\partial_{\bar z^j}$$

It remains to identify the right-hand side with $-i\partial^*$. On flat $\mathbb{C}^n$ the formal adjoint of $\partial = \sum_j e_j\,\partial_{z^j}$ is computed by integration by parts against the Hermitian $L^2$ inner product. The adjoint of the wedge $e_j$ is the contraction $\iota_j = e_j^*$; the adjoint of the differentiation $\partial_{z^j}$ is $-\partial_{\bar z^j}$, since $\int_{\mathbb{C}^n}(\partial_{z^j} f)\,\bar g\,dV = -\int_{\mathbb{C}^n} f\,\overline{\partial_{\bar z^j} g}\,dV$ with no boundary term for compactly supported forms. Applying $(AB)^* = B^*A^*$,

$$\partial^* = \sum_j(\partial_{z^j})^*\,e_j^* = \sum_j(-\partial_{\bar z^j})\,\iota_j = -\sum_j \iota_j\,\partial_{\bar z^j}$$

the derivative and the algebraic contraction commuting because the flat coframe is constant. The same expression follows from $\partial^* = -\star\bar\partial\star$: conjugating $\bar\partial = \sum_j \bar e_j\partial_{\bar z^j}$ by the type-exchanging star turns each $d\bar z$-wedge $\bar e_j$ into the $dz$-contraction $\iota_j$ while leaving $\partial_{\bar z^j}$ untouched, recovering $\partial^* = -\sum_j \iota_j\partial_{\bar z^j}$. Since $\partial^* = -\sum_j \iota_j\partial_{\bar z^j}$, the sum $\sum_j \iota_j\partial_{\bar z^j}$ equals $-\partial^*$, and comparing with the computed commutator gives

$$[\Lambda,\bar\partial] = i\sum_j \iota_j\,\partial_{\bar z^j} = i\,(-\partial^*) = -i\,\partial^*$$

This is (3.5) on flat $\mathbb{C}^n$. By Step 1 it holds at every point of any Kähler manifold, and the remaining three identities follow by adjunction and conjugation, completing the proof.

The Kähler identities package the interaction of the metric, the complex structure, and the exterior calculus into four relations, but their content is a single computation on $\mathbb{C}^n$ propagated to the general case by osculation. The reduction step is where the Kähler hypothesis is spent: it is exactly the vanishing of the first derivatives of $\omega$ that lets the flat answer stand in for the curved one, and it is exactly what fails on a non-Kähler Hermitian manifold, where the identities pick up a torsion correction. Everything downstream, the collapse of the Laplacians and ultimately the Hodge decomposition, flows from these four relations.

The immediate payoff is the master identity relating the three Laplacians. On a general Hermitian manifold $\Delta_d$, $\Delta_\partial$, and $\Delta_{\bar\partial}$ are unrelated; the Kähler identities force them to be proportional, with $\Delta_\partial$ and $\Delta_{\bar\partial}$ equal and $\Delta_d$ their sum.

**Theorem 3.3.5: The Master Identity**

On a Kähler manifold $(X, \omega)$ the three Laplacians coincide up to a factor of two:

$$\Delta_d = 2\,\Delta_{\bar\partial} = 2\,\Delta_\partial$$

Moreover $\Delta_{\bar\partial}$ commutes with the Lefschetz operators $L$ and $\Lambda$, and it preserves the bidegree, so it acts on each space $\mathcal{A}^{p,q}(X)$.

***Proof*:**
The proof is a sequence of algebraic manipulations of the Kähler identities of theorem 3.3.4, with no further analysis.

*Step 1:* $\Delta_\partial = \Delta_{\bar\partial}$. Expand $\Delta_\partial = \partial\partial^* + \partial^*\partial$ using the Kähler identity $\partial^* = i[\Lambda, \bar\partial] = i(\Lambda\bar\partial - \bar\partial\Lambda)$ from (3.5):

$$\begin{aligned}\Delta_\partial &= \partial\big(i(\Lambda\bar\partial - \bar\partial\Lambda)\big) + \big(i(\Lambda\bar\partial - \bar\partial\Lambda)\big)\partial \\ &= i\big(\partial\Lambda\bar\partial - \partial\bar\partial\Lambda + \Lambda\bar\partial\partial - \bar\partial\Lambda\partial\big) \\ &= i\big(\partial\Lambda\bar\partial - \bar\partial\Lambda\partial\big) + i\big(\Lambda\bar\partial\partial - \partial\bar\partial\Lambda\big)\end{aligned}$$

For $\Delta_{\bar\partial}$, expand with the conjugate Kähler identity $\bar\partial^* = -i[\Lambda, \partial] = -i(\Lambda\partial - \partial\Lambda)$:

$$\begin{aligned}\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial &= \bar\partial\big(-i(\Lambda\partial - \partial\Lambda)\big) + \big(-i(\Lambda\partial - \partial\Lambda)\big)\bar\partial \\ &= -i\big(\bar\partial\Lambda\partial - \bar\partial\partial\Lambda + \Lambda\partial\bar\partial - \partial\Lambda\bar\partial\big) \\ &= -i\big(\bar\partial\Lambda\partial - \partial\Lambda\bar\partial\big) - i\big(\Lambda\partial\bar\partial - \bar\partial\partial\Lambda\big)\end{aligned}$$

Compare the two expressions grouping by grouping, using $\partial\bar\partial = -\bar\partial\partial$ from proposition 1.2.8 throughout. The first grouping of $\Delta_\partial$ is $i(\partial\Lambda\bar\partial - \bar\partial\Lambda\partial)$, and the first grouping of $\Delta_{\bar\partial}$ is $-i(\bar\partial\Lambda\partial - \partial\Lambda\bar\partial) = i(\partial\Lambda\bar\partial - \bar\partial\Lambda\partial)$; these are equal. For the second groupings, $\Lambda\bar\partial\partial = -\Lambda\partial\bar\partial$ and $\partial\bar\partial\Lambda = -\bar\partial\partial\Lambda$, so the second grouping of $\Delta_\partial$ is

$$i(\Lambda\bar\partial\partial - \partial\bar\partial\Lambda) = i(-\Lambda\partial\bar\partial + \bar\partial\partial\Lambda) = -i(\Lambda\partial\bar\partial - \bar\partial\partial\Lambda)$$

which is exactly the second grouping of $\Delta_{\bar\partial}$. Both groupings agree, so

$$\Delta_\partial = \Delta_{\bar\partial}$$

*Step 2:* $\Delta_d = \Delta_\partial + \Delta_{\bar\partial}$. Expand $\Delta_d$ using $d = \partial + \bar\partial$ from theorem 1.2.6 and $d^* = \partial^* + \bar\partial^*$:

$$\Delta_d = dd^* + d^*d = (\partial + \bar\partial)(\partial^* + \bar\partial^*) + (\partial^* + \bar\partial^*)(\partial + \bar\partial)$$

Multiplying out gives $\Delta_\partial + \Delta_{\bar\partial}$ (the pure terms) plus the four cross terms

$$(\partial\bar\partial^* + \bar\partial^*\partial) + (\bar\partial\partial^* + \partial^*\bar\partial)$$

The claim is that both parenthesized cross terms vanish. Consider $\partial\bar\partial^* + \bar\partial^*\partial$. Substitute $\bar\partial^* = -i(\Lambda\partial - \partial\Lambda)$:

$$\partial\bar\partial^* + \bar\partial^*\partial = -i\big(\partial\Lambda\partial - \partial\partial\Lambda + \Lambda\partial\partial - \partial\Lambda\partial\big)$$

Now $\partial\partial = 0$ (this is $\partial^2 = 0$, the holomorphic half of proposition 1.2.8), so the two middle terms vanish, and the first and last terms cancel:

$$\partial\bar\partial^* + \bar\partial^*\partial = -i\big(\partial\Lambda\partial - 0 + 0 - \partial\Lambda\partial\big) = 0$$

The other cross term $\bar\partial\partial^* + \partial^*\bar\partial$ vanishes by the analogous computation, using $\partial^* = i(\Lambda\bar\partial - \bar\partial\Lambda)$ and $\bar\partial^2 = 0$ (exercise 3.3.1 (4)). Hence

$$\Delta_d = \Delta_\partial + \Delta_{\bar\partial} = 2\,\Delta_{\bar\partial}$$

the last equality by Step 1. Combining Steps 1 and 2 gives $\Delta_d = 2\Delta_{\bar\partial} = 2\Delta_\partial$.

*Step 3:* $\Delta_{\bar\partial}$ *commutes with* $L$ *and* $\Lambda$*, and preserves bidegree.* Since $\Delta_{\bar\partial} = \frac{1}{2}\Delta_d$ and $\Delta_d = dd^* + d^*d$ preserves total degree $k = p + q$, and since $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ preserves the antiholomorphic degree $q$ (both $\bar\partial\bar\partial^*$ and $\bar\partial^*\bar\partial$ map $\mathcal{A}^{p,q}$ to itself, as $\bar\partial$ raises $q$ by one and $\bar\partial^*$ lowers it by one), the operator $\Delta_{\bar\partial}$ preserves both $k$ and $q$, hence preserves $(p, q)$. For the commutation with $L$: the Kähler identities give $[L, \partial^*] = i\bar\partial$ and $[L, \bar\partial^*] = -i\partial$, while $L$ commutes with $\partial$ and $\bar\partial$ because $L\alpha = \omega \wedge \alpha$ with $\omega$ closed, so that $d(\omega \wedge \alpha) = \omega \wedge d\alpha$ and separating types gives $\partial L = L\partial$, $\bar\partial L = L\bar\partial$. Then

$$[L, \Delta_{\bar\partial}] = [L, \bar\partial\bar\partial^* + \bar\partial^*\bar\partial] = \bar\partial[L, \bar\partial^*] + [L, \bar\partial^*]\bar\partial = \bar\partial(-i\partial) + (-i\partial)\bar\partial = -i(\bar\partial\partial + \partial\bar\partial) = 0$$

using $[L, \bar\partial] = 0$ to pull $L$ through $\bar\partial$ and $\bar\partial\partial + \partial\bar\partial = 0$ from proposition 1.2.8 at the end. For the commutation with $\Lambda$, take formal $L^2$ adjoints of $[L, \Delta_{\bar\partial}] = 0$. Since $L^* = \Lambda$ and $\Delta_{\bar\partial}$ is self-adjoint (it is $\bar\partial\bar\partial^* + \bar\partial^*\bar\partial$, a sum of an operator with its own adjoint), the adjoint of the vanishing commutator is $[\Delta_{\bar\partial}, \Lambda] = -[\Lambda, \Delta_{\bar\partial}] = 0$, so $[\Lambda, \Delta_{\bar\partial}] = 0$ as well. This completes the proof.

The factor of two in $\Delta_d = 2\Delta_{\bar\partial}$ says that harmonicity for the de Rham complex and harmonicity for the Dolbeault complex are the same condition on a Kähler manifold: a form is $\Delta_d$-harmonic if and only if it is $\Delta_{\bar\partial}$-harmonic. Since $\Delta_{\bar\partial}$ preserves the bidegree, the space of harmonic $k$-forms splits as a direct sum of harmonic $(p, q)$-forms with $p + q = k$. Feeding this into the harmonic representation of cohomology, once that theory is available in chapter 4, is exactly what yields the Hodge decomposition $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$. Everything analytic in the preceding chapters exists to make this identity available, and the bigraded splitting it produces drives the structural theory that follows.

That $\Delta_{\bar\partial}$ commutes with $L$ and $\Lambda$ is the seed of the Hard Lefschetz theorem. It means the $\mathfrak{sl}_2$-action generated by $(L, \Lambda)$ descends from forms to harmonic forms, hence to cohomology; the Lefschetz decomposition of cohomology into primitive pieces is then a matter of representation theory of $\mathfrak{sl}_2$. That development belongs to chapter 4 and beyond, but the commutation established here is what makes it possible.

To see the identities in action on the ground, the first Kähler identity is verified by hand in the smallest case below.

**Example 3.3.6: Verifying $[\Lambda, \bar\partial] = -i\partial^*$ on $\mathbb{C}^1$**

On $\mathbb{C} = \mathbb{C}^1$ with the flat metric, the fundamental form is $\omega = i\,dz \wedge d\bar z$, so $L\alpha = i\,dz \wedge d\bar z \wedge \alpha$ and, dropping the index since $n = 1$,

$$L = i\,e\bar e, \qquad \Lambda = -i\,\bar\iota\,\iota$$

The full form algebra is four-dimensional, spanned by 1, $dz$, $d\bar z$, $dz \wedge d\bar z$, with coefficients smooth functions. The instructive case, where both sides are nonzero, is a $(1,0)$-form $\alpha = g\,dz$ with $g$ a smooth function. Compute the left side. First $\Lambda\alpha = -i\,\bar\iota\iota(g\,dz) = -ig\,\bar\iota(1) = 0$, since $\iota$ removes the $dz$ factor to leave the function $g$ and $\bar\iota$ then annihilates it (no $d\bar z$ factor remains), so $\bar\partial\Lambda\alpha = 0$. Next $\bar\partial\alpha = \bar\partial(g\,dz) = \partial_{\bar z}g\;d\bar z \wedge dz = -\partial_{\bar z}g\;dz \wedge d\bar z$, and applying $\Lambda$ to the top form, $\Lambda(dz \wedge d\bar z) = -i\,\bar\iota\iota(dz \wedge d\bar z) = -i\,\bar\iota(d\bar z) = -i$, gives

$$\Lambda\bar\partial\alpha = -\partial_{\bar z}g \cdot (-i) = i\,\partial_{\bar z}g$$

Therefore $[\Lambda, \bar\partial]\alpha = \Lambda\bar\partial\alpha - \bar\partial\Lambda\alpha = i\,\partial_{\bar z}g$. Now the right side. On $\mathbb{C}^1$, $\partial^* = -\iota\,\partial_{\bar z}$ contracts $dz$ after differentiating; applied to $\alpha = g\,dz$,

$$\partial^*\alpha = -\iota(\partial_{\bar z}g\;dz) = -\partial_{\bar z}g \cdot \iota(dz) = -\partial_{\bar z}g$$

so $-i\,\partial^*\alpha = i\,\partial_{\bar z}g$, matching the left side exactly. The identity holds on $\alpha = g\,dz$. The three remaining pure types, a function, a $(0,1)$-form, and the top form (each with a smooth coefficient), are checked the same way in exercise set 3.3.1; the first two give zero on both sides, and the top form reproduces the same nonvanishing pattern seen here.

This hand computation exposes the mechanism behind the general proof: the identity is enforced degree by degree, and on each fixed form it reduces to matching a contraction-after-derivative on one side against a commutator of wedge and contraction on the other. The Clifford relations of example 3.3.3 do all the work; the analysis contributes only the transport from the flat point back to the curved manifold, which proposition 3.1.6 supplies.

## Exercise 3.3.1

1. Complete example 3.3.6 by verifying the Kähler identity $[\Lambda, \bar\partial] = -i\partial^*$ on the three remaining basis directions of $\mathbb{C}^1$: a function $\beta = f$, a $(0,1)$-form $\beta = g\,d\bar z$, and the top form $\beta = h\,dz \wedge d\bar z$, with $f, g, h$ smooth functions. For each, compute $\Lambda\bar\partial\beta$, $\bar\partial\Lambda\beta$, and $-i\partial^*\beta$ separately, and confirm that the two sides agree (both vanish in the first two cases; both equal $i\,\partial_{\bar z}h\;d\bar z$ in the last).

2. Using the Clifford relations of example 3.3.3, show that on flat $\mathbb{C}^n$ the adjoint Lefschetz operator satisfies $\Lambda = -i\sum_j \bar\iota_j\iota_j$ by computing the pointwise adjoint of $L = i\sum_j e_j\bar e_j$ directly. (Recall that $(AB)^* = B^*A^*$, that $e_j^* = \iota_j$ and $\bar e_j^* = \bar\iota_j$, and that $(iA)^* = -iA^*$.)

3. On a general Hermitian (non-Kähler) manifold the Kähler identity acquires a correction. Without computing the correction explicitly, explain precisely where the proof of theorem 3.3.4

breaks down when $d\omega \neq 0$, identifying which step uses the Kähler hypothesis and what data of the metric the two sides of the identity would then fail to share at the point $p$.

4. In the proof of theorem 3.3.5, Step 2, the cross term $\partial\bar{\partial}^* + \bar{\partial}^*\partial$ was shown to vanish using $\partial^2 = 0$. Carry out the analogous computation for the other cross term $\bar{\partial}\partial^* + \partial^*\bar{\partial}$ in full, substituting $\partial^* = i(\Lambda\bar{\partial} - \bar{\partial}\Lambda)$ and using $\bar{\partial}^2 = 0$, and confirm it too vanishes.

5. Deduce from theorem 3.3.5 that on a compact Kähler manifold a $d$-harmonic $k$-form $\alpha$ (that is, $\Delta_d\alpha = 0$) decomposes as $\alpha = \sum_{p+q=k} \alpha^{p,q}$ with each component $\alpha^{p,q}$ itself $\Delta_d$-harmonic. State clearly which property of $\Delta_{\bar{\partial}}$ from the theorem you use, and why $\Delta_d$-harmonicity of $\alpha$ passes to each $\alpha^{p,q}$.

6. Show directly, from the Kähler identities and the closedness of $\omega$, that the Lefschetz operator $L$ commutes with $\Delta_\partial$ as well (not only with $\Delta_{\bar{\partial}}$). Explain why this is consistent with theorem 3.3.5.

## 3.4 The $\partial\bar{\partial}$-Lemma and a Non-Kähler Surface

The master identity $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_\partial$ of theorem 3.3.5 is the analytic heart of Kähler geometry, but its cohomological force is not yet visible: a statement about Laplacians only becomes a statement about forms once harmonic representatives enter, and the harmonic theory belongs to chapter 4. This section extracts the one purely form-level consequence that the equality of Laplacians will produce, the $\partial\bar{\partial}$-lemma, and puts it to work in two directions. In the positive direction it tightens the relation between the various notions of exactness on a compact Kähler manifold, collapsing $d$-exactness, $\partial$-exactness, $\bar{\partial}$-exactness, and $\partial\bar{\partial}$-exactness into a single condition for forms of pure type. In the negative direction it becomes a test: a compact complex manifold on which the lemma fails cannot carry any Kähler metric. The Hopf surface supplies the first example in the book of a compact complex manifold that is not Kähler, and it does so through the topological consequence of the lemma, the constraint that odd Betti numbers be even.

The projective manifolds of section 3.2 are the source of positive examples: every one of them is Kähler, by corollary 3.2.5, so every one of them obeys the $\partial\bar{\partial}$-lemma and has even odd Betti numbers. The point of this section is that these constraints are not automatic for compact complex manifolds. Complex structures are abundant, Kähler complex structures are not, and the Hopf surface is the standard first example of the gap.

The statement of the lemma requires no new machinery beyond the operators $\partial$ and $\bar{\partial}$ of definition 1.2.5 and the decomposition $d = \partial + \bar{\partial}$ of theorem 1.2.6. A form $\alpha$ of type $(p, q)$ is *of pure type*; a general $k$-form decomposes into its pure-type components, and the lemma is about the pure-type pieces.

**Theorem 3.4.1: The $\partial\bar{\partial}$-Lemma**

Let $X$ be a compact Kähler manifold and let $\alpha$ be a form of pure type $(p, q)$ on $X$. Suppose $\alpha$ is $d$-closed. Then the following conditions are equivalent:

1. $\alpha$ is $d$-exact;
2. $\alpha$ is $\partial$-exact;
3. $\alpha$ is $\bar{\partial}$-exact;
4. $\alpha = \partial\bar{\partial}\gamma$ for some form $\gamma$ of type $(p-1, q-1)$.

In particular, a $d$-exact form of pure type is $\partial\bar{\partial}$-exact.

The proof rests on the Hodge decomposition of chapter 4, which produces for each form a unique harmonic representative and orthogonal decompositions adapted to each of $d$, $\partial$, and $\bar{\partial}$. The identity $\Delta_d = 2\Delta_{\bar{\partial}}$ forces these three decompositions to be compatible, and that compatibility is exactly what makes the four exactness conditions coincide. The harmonic input is developed only in the next chapter, so the proof is deferred there.

***Proof Deferred*:**
The equivalence is proved in chapter 4, once the Hodge decomposition and the identity $\Delta_d = 2\Delta_{\bar{\partial}}$ (theorem 3.3.5) are available: the equality of Laplacians makes the $d$-, $\partial$-, and $\bar{\partial}$-harmonic forms coincide, and the resulting compatibility of the three orthogonal decompositions collapses the four conditions above onto one another. The argument uses harmonic representatives, which are not yet at hand, so the equivalence is deferred to that chapter rather than sketched with unavailable tools; the deferral is discharged in chapter 4.

The lemma is a statement about the difference between two a priori distinct facts. That $\alpha = \partial\bar{\partial}\gamma$ implies $\alpha$ is $d$-exact is immediate, since $\partial\bar{\partial} = \frac{1}{2}d(\bar{\partial} - \partial)$ on a pure-type form (using $d = \partial + \bar{\partial}$ and $\bar{\partial}\partial = -\partial\bar{\partial}$ from proposition 1.2.8, whence $d(\bar{\partial} - \partial) = (\partial + \bar{\partial})(\bar{\partial} - \partial) = \partial\bar{\partial} - \bar{\partial}\partial = 2\partial\bar{\partial}$, using $\partial^2 = \bar{\partial}^2 = 0$); the content is the reverse implication, that a form which is exact for the total differential is exact for the finer operator $\partial\bar{\partial}$. On a general compact complex manifold this reverse implication is false, and its failure is obstructed at the level of Bott-Chern and Aeppli cohomology, which measure precisely the gap between the finer and coarser notions of exactness. On a compact Kähler manifold it holds, and this is what allows Hodge theory to read off holomorphic data from topological data.

A first consequence records what the lemma says about closed forms of type $(1, 1)$, the type of a Kähler form. Two cohomologous real closed $(1, 1)$-forms differ by a $d$-exact real $(1, 1)$-form, and the lemma turns that difference into $\partial\bar{\partial}$ of a single global function. This is the analytic origin of the notion of a Kähler potential.

**Corollary 3.4.2: Cohomologous Closed $(1,1)$-Forms Differ by a Potential**

Let $X$ be a compact Kähler manifold and let $\omega, \omega'$ be two real closed $(1,1)$-forms on $X$ with $[\omega] = [\omega']$ in $H^2(X, \mathbb{R})$. Then there is a real-valued function $\varphi \in \mathcal{C}^\infty(X, \mathbb{R})$ with

$$\omega' - \omega = i\,\partial\bar{\partial}\varphi$$

***Proof*:**
Since $[\omega] = [\omega']$ the difference $\beta := \omega' - \omega$ is a real $d$-exact form of type $(1,1)$, hence $d$-closed of pure type. By theorem 3.4.1 there is a form $\gamma$ of type $(0,0)$, that is a function $\gamma \in \mathcal{C}^\infty(X, \mathbb{C})$, with $\beta = \partial\bar{\partial}\gamma$. It remains to arrange that the potential is real and that the constant $i$ appears. Because $\beta$ is real, $\beta = \overline{\beta}$; taking conjugates and using $\overline{\partial\bar{\partial}\gamma} = \bar{\partial}\partial\overline{\gamma} = -\partial\bar{\partial}\overline{\gamma}$ (by proposition 1.2.8) gives $\beta = -\partial\bar{\partial}\overline{\gamma}$. Averaging the two expressions for $\beta$,

$$\beta = \tfrac{1}{2}\big(\partial\bar{\partial}\gamma - \partial\bar{\partial}\overline{\gamma}\big) = \partial\bar{\partial}\Big(\tfrac{\gamma - \overline{\gamma}}{2}\Big)$$

The function $\frac{1}{2}(\gamma - \overline{\gamma})$ is purely imaginary, so it equals $i\varphi$ for a real-valued $\varphi := \frac{1}{2i}(\gamma - \overline{\gamma}) \in \mathcal{C}^\infty(X, \mathbb{R})$. Substituting, $\beta = \partial\bar{\partial}(i\varphi) = i\,\partial\bar{\partial}\varphi$, which is the claimed identity, as we sought to show.

The corollary says that on a compact Kähler manifold the Kähler class $[\omega] \in H^2(X, \mathbb{R})$ determines the Kähler form up to the addition of $i\,\partial\bar{\partial}\varphi$; the space of Kähler forms in a fixed class is, in this precise sense, a space of potentials. This is the starting point for the study of canonical metrics: fixing the class and searching within it for a form satisfying a curvature condition becomes a scalar partial differential equation for $\varphi$, which is how the Calabi conjecture and the existence of Kähler-Einstein metrics are formulated. That translation from a form-level problem to a scalar equation is only possible because the lemma reduces the freedom in a Kähler class to a single function.

A second consequence works in complementary direction, extracting holomorphic rather than metric information. A holomorphic $p$-form is a $(p,0)$-form annihilated by $\bar{\partial}$; on an arbitrary compact complex manifold such a form need not be closed for the full differential $d$, since $\partial$ of it may be nonzero. On a compact Kähler manifold it is always closed, and the proof again routes through the harmonic theory that the Kähler condition tames.

**Corollary 3.4.3: Holomorphic Forms Are Closed**

Let $X$ be a compact Kähler manifold and let $\alpha$ be a holomorphic $p$-form, that is a global section of $\Omega^p_X$ regarded as a $(p,0)$-form with $\bar{\partial}\alpha = 0$ (definition 1.2.11). Then $d\alpha = 0$; that is, $\partial\alpha = 0$ as well.

***Proof*:**
Write $d\alpha = \partial\alpha + \bar{\partial}\alpha$. By hypothesis $\bar{\partial}\alpha = 0$, so $d\alpha = \partial\alpha$, a form of type $(p+1, 0)$. The claim is that $\partial\alpha = 0$.

*Step 1: $\alpha$ is $\bar{\partial}$-harmonic.* By theorem 3.3.5, $\Delta_{\bar{\partial}} = \frac{1}{2}\Delta_d$. A holomorphic $p$-form satisfies $\bar{\partial}\alpha = 0$; it also satisfies $\bar{\partial}^*\alpha = 0$ for type reasons, since $\bar{\partial}^*$ lowers antiholomorphic degree and $\alpha$ has antiholomorphic degree 0, so $\bar{\partial}^*\alpha$ would be a form of antiholomorphic degree $-1$, necessarily zero. Hence $\Delta_{\bar{\partial}}\alpha = (\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial})\alpha = 0$, so $\alpha$ is $\bar{\partial}$-harmonic.

*Step 2: $\alpha$ is $d$-harmonic, hence $d$-closed.* By the master identity, $\Delta_d\alpha = 2\Delta_{\bar{\partial}}\alpha = 0$, so $\alpha$ is harmonic for $\Delta_d$. On a compact Riemannian manifold a form is $\Delta_d$-harmonic if and only if it is both $d$-closed and $d^*$-closed ([9, harmonic equals closed and coclosed]): the identity $\langle\!\langle \Delta_d\alpha, \alpha\rangle\!\rangle = \|d\alpha\|^2 + \|d^*\alpha\|^2$ (integration by parts using the formal adjoint) forces $d\alpha = 0$ and $d^*\alpha = 0$ when the left side vanishes. In particular $d\alpha = 0$, and since $d\alpha = \partial\alpha$ this gives $\partial\alpha = 0$, completing the proof.

The corollary is the mechanism behind one half of the Hodge decomposition: it says the holomorphic $p$-forms, defined by the purely holomorphic condition $\bar{\partial}\alpha = 0$, automatically represent de Rham cohomology classes, so there is a map $H^0(X, \Omega^p_X) \to H^p(X, \mathbb{C})$. On a compact Kähler manifold this map is injective and lands in the $(p, 0)$-part of the Hodge decomposition, identifying $H^{p,0}$ with the space of global holomorphic $p$-forms. The Kähler hypothesis is doing real work: on a general compact complex manifold a global holomorphic form can fail to be closed, and the identification breaks. The Iwasawa manifold, the quotient of the complex Heisenberg group of upper-triangular unipotent $3 \times 3$ matrices by its lattice of Gaussian-integer entries, is the standard witness: it carries a holomorphic 1-form $\omega_3$ with $d\omega_3 = -\omega_1 \wedge \omega_2 \neq 0$, so $H^{1,0}$ is strictly larger than the space of closed holomorphic 1-forms and cannot inject into $H^1$. The Hopf surface below fails to be Kähler for a different, purely topological reason, and its own holomorphic forms are too scarce to display this particular breakdown.

Beyond these form-level consequences, the lemma also has a topological consequence that requires none of its analytic content to state, only the Hodge decomposition it feeds into. On a compact Kähler manifold the decomposition $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ together with the symmetry $H^{p,q} = \overline{H^{q,p}}$ forces every odd Betti number to be even; for $k$ odd the summands pair off into complex-conjugate pairs $H^{p,q} \oplus H^{q,p}$ with $p \neq q$, and $b_k = \sum_{p+q=k} h^{p,q}$ is a sum of equal pairs. This is proved in full in chapter 4; what matters here is its contrapositive as an obstruction.

**Proposition 3.4.4: Odd Betti Obstruction**

If $X$ is a compact complex manifold admitting some Kähler metric, then every odd Betti number $b_{2k+1}(X)$ is even. Equivalently, a compact complex manifold with an odd Betti number that is odd carries no Kähler metric.

***Proof*:**

Suppose $X$ carries a Kähler metric. By the Hodge decomposition (chapter 4), $H^{2k+1}(X, \mathbb{C}) = \bigoplus_{p+q=2k+1} H^{p,q}(X)$ with $H^{p,q} = \overline{H^{q,p}}$, so $h^{p,q} = h^{q,p}$. Since $p + q = 2k + 1$ is odd, no summand has $p = q$; the indices pair off into distinct pairs $\{(p, q), (q, p)\}$, and

$$b_{2k+1}(X) = \sum_{p+q=2k+1} h^{p,q} = 2 \sum_{\substack{p+q=2k+1 \\ p<q}} h^{p,q}$$

which is even. The contrapositive is the stated obstruction, completing the proof.

The Frölicher inequality of corollary 2.4.3 already gives a form of this: it bounds $b_k \leq \sum_{p+q=k} h^{p,q}$, with equality precisely when the Frölicher spectral sequence degenerates at $E_1$. The Kähler condition guarantees degeneration (this is one of the payoffs of chapter 4), so on a Kähler manifold the inequality is an equality and the conjugation symmetry then forces the parity constraint. For a general compact complex manifold neither the degeneration nor the symmetry is available, and $b_k$ can carry the "wrong" parity. The Hopf surface exhibits exactly this.

The Hopf surface is built as a quotient of $\mathbb{C}^2 \setminus \{0\}$ by a single contracting biholomorphism. It is the standard example of a compact complex surface with no Kähler metric, and the reason is visible at the level of its underlying topology: it is diffeomorphic to $S^1 \times S^3$, whose first Betti number is 1, which is odd.

**Example 3.4.5: The Hopf Surface**

Fix the biholomorphism $f\colon \mathbb{C}^2 \setminus \{0\} \to \mathbb{C}^2 \setminus \{0\}$ given by $f(z) = 2z$, and let $\Gamma = \langle f \rangle \cong \mathbb{Z}$ be the infinite cyclic group it generates. The *Hopf surface* is the quotient

$$S = (\mathbb{C}^2 \setminus \{0\})/\Gamma$$

Its structure is unwound in three steps.

*Step 1: $S$ is a compact complex manifold of dimension* 2. The map $f$ is a biholomorphism of $\mathbb{C}^2 \setminus \{0\}$, and the group $\Gamma \cong \mathbb{Z}$ acts holomorphically. A free, properly discontinuous action of a discrete group by biholomorphisms has a quotient that is again a complex manifold, the charts descending from local sections of the projection ([26, the quotient manifold theorem] for the smooth structure; holomorphy of the transition maps is inherited because the deck transformations are biholomorphic). Both hypotheses hold here. The action is free: $f^m(z) = 2^m z = z$ with $z \neq 0$ forces $2^m = 1$, hence $m = 0$. It is properly discontinuous: any $z \neq 0$ has a neighborhood, for instance the open annulus $U = \{w : \frac{1}{\sqrt{2}}|z| < |w| < \sqrt{2}\,|z|\}$, meeting only finitely many of its $\Gamma$-translates. Indeed $f^m(U) = \{w : 2^m \frac{1}{\sqrt{2}}|z| < |w| < 2^m\sqrt{2}\,|z|\}$ is the shell of radii $(2^{m-1/2}|z|,\, 2^{m+1/2}|z|)$, whose outer radius $2^{m+1/2}|z|$ equals the inner radius of the next shell $f^{m+1}(U)$; the open shells therefore meet at most along excluded boundary spheres and are pairwise disjoint, so $U$ meets only $f^0(U) = U$ itself. A free, properly discontinuous holomorphic action of a discrete group yields a complex manifold quotient with the quotient holomorphic structure, so $S$ is a complex manifold of dimension $\dim_{\mathbb{C}} S = 2$. Compactness is settled in the next step by exhibiting a diffeomorphism to a compact manifold.

*Step 2: $S$ is diffeomorphic to $S^1 \times S^3$.* Introduce the radial-logarithmic map

$$\Phi\colon \mathbb{C}^2 \setminus \{0\} \to S^3 \times \mathbb{R}, \qquad \Phi(z) = \left(\tfrac{z}{|z|},\ \log_2 |z|\right)$$

where $S^3 = \{z \in \mathbb{C}^2 : |z| = 1\}$ is the unit sphere and $|z| = (|z_1|^2 + |z_2|^2)^{1/2}$. This $\Phi$ is a diffeomorphism, with inverse $(u,t) \mapsto 2^t u$. Under $\Phi$ the generator $f(z) = 2z$ becomes $(u,t) \mapsto (u, t+1)$, since $|2z| = 2|z|$ gives $\log_2 |2z| = \log_2 |z| + 1$ and $\frac{2z}{|2z|} = \frac{z}{|z|}$. Thus $\Phi$ conjugates the $\Gamma$-action to the action of $\mathbb{Z}$ on $S^3 \times \mathbb{R}$ by translation in the second factor, and

$$S = (\mathbb{C}^2 \setminus \{0\})/\Gamma \ \cong\ (S^3 \times \mathbb{R})/\mathbb{Z} \ \cong\ S^3 \times (\mathbb{R}/\mathbb{Z}) = S^3 \times S^1$$

where $\mathbb{Z}$ acts only on the $\mathbb{R}$ factor. Hence $S$ is diffeomorphic to $S^1 \times S^3$, which is compact.

*Step 3: $b_1(S) = 1$, so $S$ admits no Kähler metric.* The Künneth theorem ([24, the Künneth formula]) gives $H^1(S^1 \times S^3, \mathbb{R}) \cong H^1(S^1, \mathbb{R}) \oplus H^1(S^3, \mathbb{R})$. Since $H^1(S^1, \mathbb{R}) \cong \mathbb{R}$ and $H^1(S^3, \mathbb{R}) = 0$ (the sphere $S^3$ is 2-connected), $b_1(S) = 1$. This odd Betti number is odd, so by proposition 3.4.4 the surface $S$ carries no Kähler metric. Equivalently, via corollary 2.4.3: any Kähler structure would force $b_1 = h^{1,0} + h^{0,1}$ with $h^{1,0} = h^{0,1}$, hence $b_1$ even, contradicting $b_1 = 1$.

The Hopf surface answers the question left open in section 3.2. There the Fubini-Study construction and corollary 3.2.5 produced Kähler metrics on every projective manifold, and theorem 3.2.3 showed $\mathbb{CP}^n$ itself is Kähler; the abundance of projective examples might suggest that the Kähler condition is nearly automatic. The Hopf surface shows it is not. It is a compact complex manifold built from elementary data, a scaling of $\mathbb{C}^2$, yet no metric on it can be Kähler, because its topology already forbids the Hodge symmetry that a Kähler metric would impose. The obstruction is not analytic subtlety but a parity: $b_1(S) = 1$.

The contrast sharpens the meaning of the projective examples. A smooth projective variety inherits its Kähler form by restriction from the ambient $\mathbb{CP}^N$ (corollary 3.2.5), and this reflects the fact that projective varieties satisfy every Kähler constraint, including even odd Betti numbers and the $\partial\bar{\partial}$-lemma. The Hopf surface, by contrast, is never projective, because a projective manifold is Kähler and $S$ is not. It is the model non-algebraic compact complex manifold, and it marks the boundary of the theory this book develops: the Hodge decomposition of chapter 4 applies to Kähler manifolds, and the Hopf surface is the reminder that outside that class the decomposition, and everything built on it, can fail.

## Exercise 3.4.1

1. Let $X$ be any complex manifold (Kähler or not) and let $\gamma$ be a form of type $(p-1, q-1)$. Show directly, using only $d = \partial + \bar{\partial}$ (theorem 1.2.6) and the relations of proposition 1.2.8, that $\alpha := \partial\bar{\partial}\gamma$ is $d$-closed. This is the "easy direction" of the $\partial\bar{\partial}$-lemma and needs no Kähler hypothesis.

2. On a compact Kähler manifold $X$, corollary 3.4.2 produces a potential $\varphi$ with $\omega' - \omega = i\,\partial\bar{\partial}\varphi$. Determine the extent to which $\varphi$ is unique: describe all real functions $\psi$ with $i\,\partial\bar{\partial}\psi = 0$, and conclude that $\varphi$ is unique up to an additive real constant. (Use that a global function annihilated by $\partial\bar{\partial}$ is pluriharmonic, hence harmonic for the underlying Riemannian metric, and that a harmonic function on a compact connected manifold is constant.)

3. On the Hopf surface $S = (\mathbb{C}^2 \setminus \{0\})/\Gamma$, consider the holomorphic 2-form $\eta = dz_1 \wedge dz_2$ on $\mathbb{C}^2 \setminus \{0\}$. Show that $\eta$ is *not* $\Gamma$-invariant, and find the scalar by which $f^*\eta$ differs from $\eta$. Explain why this shows $S$ has no nonzero global holomorphic 2-form, so that $h^{2,0}(S) = 0$. Note that corollary 3.4.3 has nothing to say here, but for the structural reason that $S$ is not Kähler, not because $H^0(S, \Omega^2_S) = 0$.

4. The complex torus $\mathbb{C}^2/\Lambda$ of example 1.1.10 (with $\Lambda$ a lattice of rank 4) is diffeomorphic to $(S^1)^4$ and carries the flat Kähler metric of section 3.1. Compute $b_1$ and $b_3$ and verify they are even, as proposition 3.4.4 requires. Contrast the outcome with the Hopf surface: both are quotients of an open subset of $\mathbb{C}^2$, yet only one is Kähler; identify which topological feature distinguishes them.

5. Assume, as a direct Dolbeault computation, that the Hopf surface has Hodge numbers $h^{1,0} = 0$ and $h^{0,1} = 1$. Using $b_1 = 1$, show that the Frölicher inequality corollary 2.4.3 holds here as an equality $b_1 = h^{1,0} + h^{0,1}$ even though $S$ is not Kähler. Explain why this does not contradict proposition 3.4.4: which of the two ingredients (Frölicher degeneration, Hodge symmetry $h^{p,q} = h^{q,p}$) fails on $S$?

6. Deduce from proposition 3.4.4 that the Hopf surface $S$ is not projective, and more generally that no compact complex manifold with an odd first Betti number can be embedded as a complex submanifold of any $\mathbb{CP}^N$. Which result of section 3.2 does this argument combine

with proposition 3.4.4 to reach the conclusion?

# 4

# Hodge Decomposition

Chapter 3 produced the master identity $\Delta_d = 2\Delta_{\bar{\partial}}$: on a compact Kähler manifold the de Rham and Dolbeault Laplacians agree up to a factor. This chapter turns that identity into a statement about cohomology. The bridge is harmonic theory: on a compact manifold every cohomology class has a unique harmonic representative, so the equality of Laplacians descends to an equality of harmonic spaces, and the $(p, q)$-grading of forms passes to cohomology.

The result is the Hodge decomposition, $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ with $H^{p,q} = \overline{H^{q,p}}$: cohomology splits along the holomorphic bigrading, so a compact Kähler manifold's topology records its complex structure. The elliptic analysis behind harmonic representatives is imported from [9]; this chapter supplies the complex refinement, proves the decomposition, and collects its consequences: the parity of odd Betti numbers, the Hodge index theorem, and the Lefschetz $(1, 1)$-theorem identifying which classes come from divisors.

## 4.1 The $\bar{\partial}$-Hodge Theorem

The Kähler identities of chapter 3 tied the three Laplacians together but left them as operators on forms; nothing yet says that a Dolbeault class contains a distinguished representative, or even that the Dolbeault groups are finite dimensional. Both facts come from one source: $\Delta_{\bar{\partial}}$ is an elliptic operator, and the elliptic package on a compact manifold produces a finite-dimensional space of harmonic forms with a unique harmonic representative in every cohomology class. That analytic package is developed in full elsewhere in the series; this section imports it and applies it to the complex setting, where it becomes the $\bar{\partial}$-analogue of the classical Hodge theorem for the de Rham complex.

The real case is the model to keep in view. On a compact oriented Riemannian manifold $M$ the Laplace-Beltrami operator $\Delta_d = dd^* + d^*d$ is a formally self-adjoint elliptic operator of order two, and the Hodge-de Rham theorem identifies its kernel with cohomology: every de Rham class has a unique harmonic representative, so the space $\mathcal{H}^k(M)$ of $\Delta_d$-harmonic $k$-forms satisfies $\mathcal{H}^k(M) \cong H^k_{\mathrm{dR}}(M)$, and one has the orthogonal decomposition $\Omega^k(M) = \mathcal{H}^k(M) \oplus d\,\Omega^{k-1}(M) \oplus d^*\,\Omega^{k+1}(M)$. This is proved in *Riemannian Geometry* [9], where the underlying elliptic machinery (the Gårding-Sobolev a-priori estimate, elliptic regularity, and the Rellich compactness that forces the kernel to be finite dimensional and the range closed) is established once for *any* formally self-adjoint elliptic operator of order two on a closed manifold, not just for $\Delta_d$. The package is stated at a level of generality that applies verbatim to a second self-adjoint elliptic operator, and $\Delta_{\bar\partial}$ is exactly such an operator. This book owns none of that analysis; it owns the complex refinement, and to it we now turn.

The refinement replaces $d$ by $\bar\partial$ and real forms by $(p,q)$-forms. The relevant Laplacian is the $\bar\partial$-Laplacian $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ built from $\bar\partial$ and its formal adjoint $\bar\partial^*$ of definition 3.3.2, acting on the smooth $(p,q)$-forms $\mathcal{A}^{p,q}(X)$ of a compact Hermitian manifold $X$. Its kernel is what plays the role of the harmonic forms.

**Definition 4.1.1: $\bar\partial$-Harmonic Forms**

Let $(X,\omega)$ be a compact Hermitian manifold of complex dimension $n$ and let $0 \le p,q \le n$. A smooth $(p,q)$-form $\alpha \in \mathcal{A}^{p,q}(X)$ is *$\bar\partial$-harmonic* if $\Delta_{\bar\partial}\alpha = 0$. The space of $\bar\partial$-harmonic $(p,q)$-forms is denoted $\mathcal{H}^{p,q}_{\bar\partial}(X)$.

The condition $\Delta_{\bar\partial}\alpha = 0$ has the same two-sided reformulation as its real counterpart. Because $\Delta_{\bar\partial}$ is self-adjoint and nonnegative with respect to the Hermitian inner product $\langle\!\langle -,- \rangle\!\rangle$ on forms, a form is in its kernel exactly when it is annihilated by both $\bar\partial$ and $\bar\partial^*$.

**Proposition 4.1.2: Two-Sided Characterization of $\bar\partial$-Harmonicity**

On a compact Hermitian manifold $X$, a form $\alpha \in \mathcal{A}^{p,q}(X)$ is $\bar\partial$-harmonic if and only if $\bar\partial\alpha = 0$ and $\bar\partial^*\alpha = 0$.

***Proof*:**
If $\bar\partial\alpha = 0$ and $\bar\partial^*\alpha = 0$ then $\Delta_{\bar\partial}\alpha = \bar\partial\bar\partial^*\alpha + \bar\partial^*\bar\partial\alpha = 0$, so $\alpha$ is $\bar\partial$-harmonic. Conversely, suppose $\Delta_{\bar\partial}\alpha = 0$. Pairing with $\alpha$ and using that $\bar\partial^*$ is the formal adjoint of $\bar\partial$ (definition 3.3.2),

$$0 = \langle\!\langle \Delta_{\bar\partial}\alpha, \alpha \rangle\!\rangle = \langle\!\langle \bar\partial\bar\partial^*\alpha, \alpha \rangle\!\rangle + \langle\!\langle \bar\partial^*\bar\partial\alpha, \alpha \rangle\!\rangle = \langle\!\langle \bar\partial^*\alpha, \bar\partial^*\alpha \rangle\!\rangle + \langle\!\langle \bar\partial\alpha, \bar\partial\alpha \rangle\!\rangle = \|\bar\partial^*\alpha\|^2 + \|\bar\partial\alpha\|^2$$

A sum of two nonnegative real numbers is zero only when both vanish, so $\bar\partial\alpha = 0$ and $\bar\partial^*\alpha = 0$, as we sought to show.

The proposition is the reason harmonic forms compute cohomology at all. A $\bar\partial$-harmonic form is in particular $\bar\partial$-closed, so it defines a Dolbeault class; and being also $\bar\partial^*$-closed makes it orthogonal to the exact forms, which is what will pin it down as the unique harmonic representative once the decomposition is in hand. The argument used compactness only to make the integration-by-parts of definition 3.3.2 valid with no boundary term, and it is identical in form to the real Hodge-theory computation for $\Delta_d$; only the operators have changed.

To apply the imported package, $\Delta_{\bar\partial}$ must be a formally self-adjoint elliptic operator of order two. Self-adjointness is built into its definition: $\Delta_{\bar\partial}^* = (\bar\partial\bar\partial^* + \bar\partial^*\bar\partial)^* = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial = \Delta_{\bar\partial}$, using $(\bar\partial^*)^* = \bar\partial$. Ellipticity is a statement about the principal symbol.

**Proposition 4.1.3: $\Delta_{\bar\partial}$ is Self-Adjoint Elliptic of Order Two**

On a compact Hermitian manifold $X$, the operator $\Delta_{\bar\partial}\colon \mathcal{A}^{p,q}(X) \to \mathcal{A}^{p,q}(X)$ is a formally self-adjoint differential operator of order two whose principal symbol at a nonzero covector $\xi$ is $\sigma_{\Delta_{\bar\partial}}(\xi) = \frac{1}{2}|\xi|^2\,\mathrm{id}$. In particular $\Delta_{\bar\partial}$ is elliptic.

***Proof*:**
Both $\bar\partial$ and $\bar\partial^*$ are first-order differential operators, so $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ is a differential operator of order at most two, and formal self-adjointness was checked above. It remains to compute the principal symbol. The principal symbol is multiplicative and additive on the leading parts, so $\sigma_{\Delta_{\bar\partial}}(\xi) = \sigma_{\bar\partial}(\xi)\sigma_{\bar\partial^*}(\xi) + \sigma_{\bar\partial^*}(\xi)\sigma_{\bar\partial}(\xi)$. The symbol of $\bar\partial$ at $\xi$ is exterior multiplication by the $(0,1)$-part $\xi^{0,1}$ of $\xi$, and the symbol of its adjoint $\bar\partial^*$ is the corresponding interior product (contraction) $\iota_{\xi^{0,1}}$, up to the sign and factor recorded in definition 3.3.2. The Clifford-type anticommutation of exterior and interior multiplication then gives

$$\sigma_{\Delta_{\bar\partial}}(\xi) = \xi^{0,1} \wedge \iota_{\xi^{0,1}} + \iota_{\xi^{0,1}}\, \xi^{0,1} \wedge (-) = |\xi^{0,1}|^2\,\mathrm{id} = \tfrac{1}{2}|\xi|^2\,\mathrm{id}$$

where the last equality is the pointwise identity $|\xi^{0,1}|^2 = \frac{1}{2}|\xi|^2$ for a real covector split into its $(1,0)$- and $(0,1)$-parts. This is a positive multiple of the identity for every $\xi \neq 0$, so $\sigma_{\Delta_{\bar\partial}}(\xi)$ is invertible and $\Delta_{\bar\partial}$ is elliptic, completing the proof.

The symbol came out to $\frac{1}{2}|\xi|^2\,\mathrm{id}$, exactly half the symbol $|\xi|^2\,\mathrm{id}$ of the real Laplacian $\Delta_d$. On a Kähler manifold the master identity $\Delta_d = 2\Delta_{\bar\partial}$ of theorem 3.3.5 makes the two operators equal up to the factor of two, so their symbols must agree up to that factor. On a general Hermitian manifold the master identity fails, yet the symbol computation still returns $\frac{1}{2}|\xi|^2\,\mathrm{id}$, because the symbol sees only the leading Clifford term and is blind to the lower-order torsion by which $\Delta_d$ and $2\Delta_{\bar\partial}$ differ. Ellipticity of $\Delta_{\bar\partial}$ therefore holds on any compact Hermitian manifold, with no Kähler hypothesis, and that is all the imported package requires.

Applying the elliptic package to the self-adjoint elliptic operator $\Delta_{\bar\partial}$ yields the $\bar\partial$-Hodge theorem.

**Theorem 4.1.4: The $\bar\partial$-Hodge Theorem**

Let $X$ be a compact Hermitian manifold of complex dimension $n$, and fix $0 \le p, q \le n$. Then the space $\mathcal{H}^{p,q}_{\bar\partial}(X)$ of $\bar\partial$-harmonic $(p,q)$-forms is finite dimensional, and there is an orthogonal direct-sum decomposition

$$\mathcal{A}^{p,q}(X) = \mathcal{H}^{p,q}_{\bar\partial}(X) \oplus \bar\partial\,\mathcal{A}^{p,q-1}(X) \oplus \bar\partial^*\,\mathcal{A}^{p,q+1}(X)$$

The projection onto the first summand induces a canonical isomorphism

$$\mathcal{H}^{p,q}_{\bar\partial}(X) \xrightarrow{\cong} H^{p,q}_{\bar\partial}(X)$$

so every Dolbeault class contains a unique $\bar\partial$-harmonic representative. Combining with Dolbeault's theorem (theorem 2.3.3),

$$\mathcal{H}^{p,q}_{\bar\partial}(X) \cong H^{p,q}_{\bar\partial}(X) \cong H^q(X, \Omega^p_X)$$

and in particular the sheaf cohomology $H^q(X, \Omega^p_X)$ is finite dimensional.

***Proof*:**
By proposition 4.1.3, $\Delta_{\bar\partial}$ is a formally self-adjoint elliptic operator of order two on the compact manifold $X$, acting on the smooth sections of the vector bundle $\bigwedge^{p,q} T^*X$ whose sections are the $(p,q)$-forms. The elliptic package of [9], stated there for any such operator, provides two facts: the kernel $\ker \Delta_{\bar\partial} = \mathcal{H}^{p,q}_{\bar\partial}(X)$ is finite dimensional, and there is an $L^2$-orthogonal decomposition

$$\mathcal{A}^{p,q}(X) = \mathcal{H}^{p,q}_{\bar\partial}(X) \oplus \Delta_{\bar\partial}\,\mathcal{A}^{p,q}(X) \tag{4.1}$$

of the smooth forms into the harmonic forms and the image of the Laplacian, orthogonal because $\Delta_{\bar\partial}$ is self-adjoint. The splitting, together with finite dimensionality and smoothness of the harmonic space, is the abstract Hodge decomposition for a formally self-adjoint elliptic operator on a closed manifold ([8, the abstract Hodge decomposition]); [9, the Hodge-de Rham theorem] is the same statement for $\Delta_d$, and is the model this specializes. That is the only input taken from outside; the rest is bookkeeping with $\bar\partial$ and $\bar\partial^*$.

*Step 1: Refining the splitting.* Expand the image term in (4.1). Since $\Delta_{\bar\partial}\beta = \bar\partial(\bar\partial^*\beta) + \bar\partial^*(\bar\partial\beta)$, every element of $\Delta_{\bar\partial}\,\mathcal{A}^{p,q}(X)$ lies in $\bar\partial\,\mathcal{A}^{p,q-1}(X) + \bar\partial^*\,\mathcal{A}^{p,q+1}(X)$. These last two subspaces are mutually orthogonal: for $\gamma \in \mathcal{A}^{p,q-1}$ and $\delta \in \mathcal{A}^{p,q+1}$,

$$\langle\!\langle \bar\partial\gamma, \bar\partial^*\delta \rangle\!\rangle = \langle\!\langle \bar\partial\bar\partial\gamma, \delta \rangle\!\rangle = 0$$

because $\bar\partial^2 = 0$. Each is orthogonal to the harmonics as well: if $\alpha$ is $\bar\partial$-harmonic then $\bar\partial^*\alpha = 0$ by proposition 4.1.2, so $\langle\!\langle \alpha, \bar\partial\gamma \rangle\!\rangle = \langle\!\langle \bar\partial^*\alpha, \gamma \rangle\!\rangle = 0$, and symmetrically $\langle\!\langle \alpha, \bar\partial^*\delta \rangle\!\rangle = \langle\!\langle \bar\partial\alpha, \delta \rangle\!\rangle = 0$. The three subspaces $\mathcal{H}^{p,q}_{\bar\partial}(X)$, $\bar\partial\,\mathcal{A}^{p,q-1}(X)$, $\bar\partial^*\,\mathcal{A}^{p,q+1}(X)$ are therefore pairwise orthogonal, and by (4.1) they span $\mathcal{A}^{p,q}(X)$. This gives the claimed orthogonal decomposition.

*Step 2: Harmonic representatives.* A class in $H^{p,q}_{\bar\partial}(X)$ is represented by a $\bar\partial$-closed form $\alpha$, taken modulo $\bar\partial\,\mathcal{A}^{p,q-1}(X)$. Decompose $\alpha = \alpha_{\mathcal{H}} + \bar\partial\gamma + \bar\partial^*\delta$ by the orthogonal splitting. Applying $\bar\partial$ and using $\bar\partial\alpha = 0$, $\bar\partial\alpha_{\mathcal{H}} = 0$, and $\bar\partial\bar\partial\gamma = 0$ leaves $\bar\partial\bar\partial^*\delta = 0$; pairing this with $\delta$ gives $\|\bar\partial^*\delta\|^2 = \langle\!\langle \bar\partial\bar\partial^*\delta, \delta \rangle\!\rangle = 0$, so the term $\bar\partial^*\delta$ vanishes. Hence $\alpha = \alpha_{\mathcal{H}} + \bar\partial\gamma$ differs from its harmonic part by a $\bar\partial$-exact form, so $[\alpha] = [\alpha_{\mathcal{H}}]$: every class has a harmonic representative. It is unique, since a $\bar\partial$-harmonic form that is also $\bar\partial$-exact, say $\alpha_{\mathcal{H}} = \bar\partial\eta$, satisfies $\|\alpha_{\mathcal{H}}\|^2 = \langle\!\langle \bar\partial\eta, \alpha_{\mathcal{H}}\rangle\!\rangle = \langle\!\langle \eta, \bar\partial^*\alpha_{\mathcal{H}}\rangle\!\rangle = 0$ by proposition 4.1.2, so $\alpha_{\mathcal{H}} = 0$. The harmonic projection $\mathcal{H}^{p,q}_{\bar\partial}(X) \to H^{p,q}_{\bar\partial}(X)$ is thus a bijection, and it is linear, hence the stated isomorphism.

*Step 3: The sheaf-theoretic identification.* Dolbeault's theorem (theorem 2.3.3) supplies the canonical isomorphism $H^{p,q}_{\bar\partial}(X) \cong H^q(X, \Omega^p_X)$. Composing with the harmonic isomorphism just constructed gives the chain $\mathcal{H}^{p,q}_{\bar\partial}(X) \cong H^{p,q}_{\bar\partial}(X) \cong H^q(X, \Omega^p_X)$. Finite dimensionality of the left-hand space, from the elliptic package, transports across the isomorphisms to the sheaf cohomology, completing the proof.

The theorem does for the $\bar\partial$-complex what the classical Hodge theorem does for the de Rham complex: it selects, in each Dolbeault class, one preferred form, the one killed by both $\bar\partial$ and $\bar\partial^*$. Three consequences are worth separating. First, the Dolbeault groups $H^{p,q}_{\bar\partial}(X)$, and with them the sheaf cohomology $H^q(X, \Omega^p_X)$, are finite dimensional on any compact complex manifold, so the Hodge numbers $h^{p,q}$ of definition 2.3.4 are well-defined finite invariants; this discharges the finiteness assumed in chapter 2. Second, harmonic representatives are canonical, depending only on the Hermitian metric, so a metric-independent cohomology class acquires a metric-dependent but canonical incarnation as a form, and analytic operations on forms (wedging, contraction, the Lefschetz operators) can be performed on cohomology by performing them on harmonic representatives. Third, the whole statement holds on a compact Hermitian manifold, before any Kähler hypothesis. The Kähler condition enters only in section 4.2, where the master identity $\Delta_d = 2\Delta_{\bar\partial}$ makes the $\bar\partial$-harmonic and $d$-harmonic forms coincide and thereby ties this bigraded decomposition to the single-graded de Rham cohomology.

One caution on what the theorem does *not* yet give. On a general compact Hermitian manifold the harmonic projection depends on which operator one harmonizes against: a $\Delta_{\bar\partial}$-harmonic form need not be $\Delta_d$-harmonic, and the direct sum $\bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar\partial}(X)$ need not equal the de Rham harmonic space $\mathcal{H}^k(X)$. The three Laplacians are distinct off the Kähler locus, and forcing them to agree is exactly the work chapter 3 did through the Kähler identities and the work section 4.2 will complete.

The cleanest place to see harmonic representatives explicitly is a complex torus, where the flat metric has constant coefficients and the harmonic condition becomes a condition of constant coefficients on the form.

**Example 4.1.5: Harmonic Forms on a Complex Torus**

Let $X = \mathbb{C}^n/\Lambda$ be a complex torus (example 1.1.10) with the flat Hermitian metric descending from the standard metric $\sum dz^j \otimes d\bar z^j$ on $\mathbb{C}^n$, whose coefficients are constant and hence $\Lambda$-invariant. Write $z^1, \dots, z^n$ for the standard coordinates on $\mathbb{C}^n$; the differentials $dz^j$ and $d\bar z^j$ are $\Lambda$-invariant and descend to global 1-forms on $X$. For a multi-index $I = (i_1 < \cdots < i_p)$ and $J = (j_1 < \cdots < j_q)$ set $dz^I = dz^{i_1} \wedge \cdots \wedge dz^{i_p}$ and $d\bar z^J = d\bar z^{j_1} \wedge \cdots \wedge d\bar z^{j_q}$. A general $(p,q)$-form is $\alpha = \sum_{I,J} f_{IJ}\, dz^I \wedge d\bar z^J$ with smooth coefficients $f_{IJ}$ on $X$.

*The constant-coefficient forms are harmonic.* If every $f_{IJ}$ is constant, then $\bar\partial\alpha = 0$ since $\bar\partial dz^I = \bar\partial d\bar z^J = 0$ and the coefficients contribute no derivative. For the flat metric the Hodge star sends a constant-coefficient form to a constant-coefficient form, so $\bar\partial^*\alpha = -\star\,\partial\star\alpha = 0$ by the same computation applied after $\star$. By proposition 4.1.2, $\alpha$ is $\bar\partial$-harmonic. The $\binom{n}{p}\binom{n}{q}$ forms $dz^I \wedge d\bar z^J$ are linearly independent, so they span a space of $\bar\partial$-harmonic forms of dimension $\binom{n}{p}\binom{n}{q}$.

*These are all of them.* On the flat torus the $\bar\partial$-Laplacian acts on a form $\alpha = \sum_{I,J} f_{IJ}\, dz^I \wedge d\bar z^J$ coefficientwise as the ordinary scalar Laplacian: $\Delta_{\bar\partial}\alpha = \sum_{I,J} (\Delta_0 f_{IJ})\, dz^I \wedge d\bar z^J$, where $\Delta_0$ is the flat Laplacian on functions, because the frame forms are parallel and the metric coefficients are constant. Thus $\alpha$ is harmonic if and only if each $f_{IJ}$ is a harmonic function on the compact manifold $X$, and a harmonic function on a compact connected manifold is constant (pairing $\Delta_0 f_{IJ} = 0$ with $f_{IJ}$ and integrating by parts gives $\|\nabla f_{IJ}\|^2 = 0$, so $f_{IJ}$ is constant). Hence

$$\mathcal{H}^{p,q}_{\bar\partial}(X) = \left\{ \textstyle\sum_{I,J} c_{IJ}\, dz^I \wedge d\bar z^J : c_{IJ} \in \mathbb{C} \right\}$$

is exactly the space of constant-coefficient $(p,q)$-forms.

By theorem 4.1.4 the Hodge numbers are $h^{p,q} = \dim_{\mathbb{C}} \mathcal{H}^{p,q}_{\bar\partial}(X) = \binom{n}{p}\binom{n}{q}$, which agrees with the direct sheaf computation $H^q(X, \Omega^p_X) \cong \bigwedge^p \mathbb{C}^n \otimes \bigwedge^q \overline{\mathbb{C}^n}$: the cotangent bundle of a torus is trivial, so $\Omega^p_X \cong \mathcal{O}_X^{\oplus\binom{n}{p}}$ and $H^q(X, \Omega^p_X) \cong H^q(X, \mathcal{O}_X)^{\oplus\binom{n}{p}}$, with $\dim H^q(X, \mathcal{O}_X) = \binom{n}{q}$. Every Dolbeault class on a torus is thus represented by a unique translation-invariant form.

The torus makes the abstract selection of a harmonic representative completely concrete: harmonizing a $(p,q)$-form there means averaging away its non-constant part, and the survivor is the unique translation-invariant form in the class. For $n = 1$ the count $h^{p,q} = \binom{1}{p}\binom{1}{q}$ reproduces the Hodge diamond of an elliptic curve, $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1} = 1$; the classes $dz$ and $d\bar z$ are the two harmonic 1-forms, and their span is $H^1(X, \mathbb{C})$. This is the weight-one Hodge structure that reappears in chapter 5.

## Exercise 4.1.1

1. Show that on a compact Hermitian manifold every $\bar\partial$-harmonic $(p,q)$-form is $\bar\partial$-closed and $\bar\partial^*$-closed, and deduce that the space $\mathcal{H}^{p,q}_{\bar\partial}(X)$ injects into $H^{p,q}_{\bar\partial}(X)$ by the class map $\alpha \mapsto [\alpha]$, without invoking the full $\bar\partial$-Hodge theorem. Which single ingredient of theorem 4.1.4 is still needed to conclude that the map is surjective?

2. Verify the pointwise identity $|\xi^{0,1}|^2 = \frac{1}{2}|\xi|^2$ used in the symbol computation of proposition 4.1.3, for a real covector $\xi$ decomposed as $\xi = \xi^{1,0} + \xi^{0,1}$ with $\xi^{1,0} = \overline{\xi^{0,1}}$. Explain how this factor of $\frac{1}{2}$ is consistent with the master identity $\Delta_d = 2\Delta_{\bar\partial}$ of theorem 3.3.5 at the level of principal symbols on a Kähler manifold.

3. For the elliptic curve $X = \mathbb{C}/\Lambda$ with the flat metric, write down explicit harmonic representatives for a basis of each of $H^{0,0}_{\bar\partial}(X)$, $H^{1,0}_{\bar\partial}(X)$, $H^{0,1}_{\bar\partial}(X)$, and $H^{1,1}_{\bar\partial}(X)$, and confirm that the resulting Hodge numbers are $h^{p,q} = \binom{1}{p}\binom{1}{q}$. Which of these four groups computes the holomorphic 1-forms, and why does its dimension equal the genus?

4. The $\bar\partial$-Hodge theorem is stated for a compact Hermitian manifold, with no Kähler hypothesis, yet the Hodge decomposition of section 4.2 requires Kähler. Identify precisely which statement of theorem 4.1.4 would fail on a general compact Hermitian manifold if one tried to replace $\Delta_{\bar\partial}$ throughout by $\Delta_d$, and name the property of the three Laplacians that is available only in the Kähler case.

5. Show that complex conjugation carries $\bar\partial$-harmonic $(p,q)$-forms to $\partial$-harmonic $(q,p)$-forms, where $\Delta_\partial = \partial\partial^* + \partial^*\partial$ and a form is $\partial$-harmonic when $\Delta_\partial$ annihilates it. On a compact *Kähler* manifold, deduce using theorem 3.3.5 that $\overline{\mathcal{H}^{p,q}_{\bar\partial}(X)} = \mathcal{H}^{q,p}_{\bar\partial}(X)$, and interpret this as a symmetry of the Hodge numbers.

## 4.2 The Hodge Decomposition

The previous section produced, on a compact Hermitian manifold, a unique $\bar{\partial}$-harmonic representative for each Dolbeault class, so that $\mathcal{H}^{p,q}_{\bar{\partial}}(X) \cong H^q(X, \Omega^p_X)$. That statement uses no relation between the $\bar{\partial}$-theory and the underlying real de Rham theory; the two live on separate footings, one bigraded by type and one graded by total degree. On a compact Kähler manifold the master identity $\Delta_d = 2\Delta_{\bar{\partial}}$ of theorem 3.3.5 fuses them. A form is $d$-harmonic exactly when it is $\bar{\partial}$-harmonic, and $\Delta_{\bar{\partial}}$ preserves bidegree, so the real harmonic space acquires a bigrading it had no reason to carry. Transported through the two Hodge isomorphisms, that bigrading becomes a splitting of complex cohomology by type. This section proves that splitting, records the symmetry it enjoys under conjugation, and uses it to settle two facts earlier chapters left open here: the degeneration of the Frölicher spectral sequence and the $\partial\bar{\partial}$-lemma.

Throughout, $X$ is a compact Kähler manifold of complex dimension $n$, with a fixed Kähler metric. We write $\mathcal{H}^k(X)$ for the space of real ($\Delta_d$-)harmonic $k$-forms with complex coefficients and $\mathcal{H}^{p,q}_{\bar{\partial}}(X)$ for the $\Delta_{\bar{\partial}}$-harmonic $(p,q)$-forms of definition 4.1.1.

The first observation is that the two Laplacians, agreeing up to the scalar 2, have the same kernel. This is what lets the bidegree decomposition of forms descend to harmonic forms.

**Theorem 4.2.1: Hodge Decomposition**

Let $X$ be a compact Kähler manifold of complex dimension $n$. Then the space of complex-valued harmonic $k$-forms decomposes by bidegree,

$$\mathcal{H}^k(X) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar{\partial}}(X)$$

and this induces a direct sum decomposition of complex cohomology,

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X) \qquad \text{where} \qquad H^{p,q}(X) \cong \mathcal{H}^{p,q}_{\bar{\partial}}(X) \cong H^q(X, \Omega^p_X)$$

The summand $H^{p,q}(X)$ is the image in $H^k(X,\mathbb{C})$ of the $\bar{\partial}$-harmonic $(p,q)$-forms. Moreover the decomposition satisfies the *Hodge symmetry*

$$H^{p,q}(X) = \overline{H^{q,p}(X)}$$

where the bar denotes complex conjugation on $H^k(X,\mathbb{C}) = H^k(X,\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$.

***Proof*:**
*Step 1: The two Laplacians share a kernel.* By theorem 3.3.5 the identity $\Delta_d = 2\Delta_{\bar{\partial}}$ holds on every form on $X$. A form $\alpha$ satisfies $\Delta_d \alpha = 0$ if and only if $\Delta_{\bar{\partial}}\alpha = 0$, so the $d$-harmonic forms and the $\bar{\partial}$-harmonic forms coincide as subspaces of the smooth complex-valued forms. The real Hodge-de Rham theorem, applied to the operator $\Delta_d$ on the closed manifold $X$ and cited to [9], identifies $\mathcal{H}^k(X)$ with $H^k(X,\mathbb{C})$: every de Rham class has a unique harmonic representative.

*Step 2: Harmonic forms split by type.* A smooth complex $k$-form decomposes uniquely into its components of each bidegree, $\alpha = \sum_{p+q=k} \alpha^{p,q}$ with $\alpha^{p,q} \in \mathcal{A}^{p,q}(X)$. The operator $\Delta_{\bar{\partial}} =$

$\bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ preserves bidegree, since $\bar\partial$ raises $q$ by one and $\bar\partial^*$ lowers it by one and neither changes $p$. Hence

$$\Delta_{\bar\partial}\alpha = \sum_{p+q=k} \Delta_{\bar\partial}(\alpha^{p,q})$$

and, the summands lying in distinct bidegrees, $\Delta_{\bar\partial}\alpha = 0$ holds if and only if $\Delta_{\bar\partial}\alpha^{p,q} = 0$ for every $(p,q)$. So a form is $\bar\partial$-harmonic exactly when each of its bihomogeneous components is $\bar\partial$-harmonic. Combined with Step 1,

$$\mathcal{H}^k(X) = \bigoplus_{p+q=k} \mathcal{H}^{p,q}_{\bar\partial}(X)$$

the sum being direct because the bidegree components of a form are unique.

*Step 3: Passing to cohomology.* Define $H^{p,q}(X) \subseteq H^k(X,\mathbb{C})$ as the image, under the isomorphism $\mathcal{H}^k(X) \cong H^k(X,\mathbb{C})$ of Step 1, of the summand $\mathcal{H}^{p,q}_{\bar\partial}(X)$. Applying that isomorphism to the direct sum of Step 2 gives

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$$

By construction $H^{p,q}(X) \cong \mathcal{H}^{p,q}_{\bar\partial}(X)$, and the $\bar\partial$-Hodge theorem theorem 4.1.4 together with Dolbeault's theorem theorem 2.3.3 identifies $\mathcal{H}^{p,q}_{\bar\partial}(X) \cong H^{p,q}_{\bar\partial}(X) \cong H^q(X,\Omega^p_X)$. This proves the first two displays.

*Step 4: Hodge symmetry.* Complex conjugation acts on complex-valued forms, sending $\mathcal{A}^{p,q}(X)$ to $\mathcal{A}^{q,p}(X)$: if $\alpha = f\,dz_{i_1} \wedge \cdots \wedge d\bar z_{j_q}$ has type $(p,q)$, then $\bar\alpha$ has type $(q,p)$. Conjugation interchanges the two Dolbeault differentials, $\overline{\bar\partial\alpha} = \partial\bar\alpha$, hence interchanges their formal adjoints and the associated Laplacians, $\overline{\Delta_{\bar\partial}\alpha} = \Delta_\partial\bar\alpha$. By theorem 3.3.5 the Kähler condition forces $\Delta_\partial = \Delta_{\bar\partial}$, so

$$\overline{\Delta_{\bar\partial}\alpha} = \Delta_{\bar\partial}\bar\alpha$$

Thus $\alpha$ is $\bar\partial$-harmonic if and only if $\bar\alpha$ is, and conjugation restricts to a conjugate-linear isomorphism $\mathcal{H}^{p,q}_{\bar\partial}(X) \to \mathcal{H}^{q,p}_{\bar\partial}(X)$. On cohomology this is exactly the map induced by conjugation on $H^k(X,\mathbb{C})$, which carries $H^k(X,\mathbb{R})$ to itself. Therefore $H^{p,q}(X) = \overline{H^{q,p}(X)}$, completing the proof.

The decomposition draws together the foundational chapters: the holomorphic structure of complex manifolds, the Dolbeault theory of chapter 2, and the Kähler identities of chapter 3 combine so that a purely topological invariant, the complex cohomology of $X$, remembers the complex geometry. A class in $H^k(X,\mathbb{C})$ is no longer an unstructured object; it carries a canonical expression $\sum_{p+q=k}\alpha^{p,q}$ into pieces of definite type, and each piece $H^{p,q}(X)$ is a sheaf-cohomology group $H^q(X,\Omega^p_X)$ computable by algebraic means. The two isomorphisms in the theorem run in opposite directions: $H^{p,q} \cong \mathcal{H}^{p,q}_{\bar\partial}$ is analytic, produced by harmonic theory, while $\mathcal{H}^{p,q}_{\bar\partial} \cong H^q(X,\Omega^p_X)$ is the algebraic reading through Dolbeault. That the same group admits both descriptions is the reason Hodge theory transfers information between the transcendental and algebraic sides of complex geometry.

Two features stand out before an example. First, the splitting depends on the Kähler metric only through the identity $\Delta_d = 2\Delta_{\bar\partial}$; the resulting subspaces $H^{p,q}(X) \subseteq H^k(X,\mathbb{C})$ are independent of the chosen Kähler metric. Abstract isomorphism with the metric-free group $H^q(X,\Omega^p_X)$ is not what shows this, since it fixes only the dimension of the subspace and not its position inside $H^k(X,\mathbb{C})$; what does is the metric-free characterization $H^{p,q}(X) = \{[\alpha] : \alpha \text{ a } d\text{-closed form of type } (p,q)\}$, equivalently

$H^{p,q} = F^p \cap \overline{F^q}$ for the Hodge filtration of definition 2.4.1. Second, the Hodge symmetry $H^{p,q} = \overline{H^{q,p}}$ is what obstructs an arbitrary complex manifold from being Kähler: it forces constraints on the Hodge numbers, taken up in section 4.3.

The complex torus illustrates the decomposition concretely, since every cohomology class has a translation-invariant harmonic representative and the decomposition can be written down by hand.

**Example 4.2.2: The Hodge Diamond of a Complex Torus**

Let $X = \mathbb{C}^n/\Lambda$ be a complex torus of dimension $n$ (example 1.1.10), carrying the flat Kähler metric descended from the standard Hermitian metric on $\mathbb{C}^n$. Write $z_1, \ldots, z_n$ for the linear complex coordinates on $\mathbb{C}^n$; the forms $dz_1, \ldots, dz_n, d\bar{z}_1, \ldots, d\bar{z}_n$ are translation-invariant and descend to global forms on $X$.

For the flat metric the Laplacian $\Delta_{\bar\partial}$ has constant coefficients, and a $(p,q)$-form with constant coefficients has $\bar\partial$ and $\bar\partial^*$ both zero, hence is harmonic. Conversely a harmonic form on the compact torus is translation-invariant, so

$$\mathcal{H}^{p,q}_{\bar\partial}(X) = \left\{ \sum a_{IJ}\, dz_I \wedge d\bar{z}_J : a_{IJ} \in \mathbb{C},\ |I| = p,\ |J| = q \right\}$$

where $I, J$ range over increasing multi-indices. The dimension is $h^{p,q} = \binom{n}{p}\binom{n}{q}$, and the Hodge decomposition reads

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X), \qquad \dim H^{p,q}(X) = \binom{n}{p}\binom{n}{q}$$

For $n = 1$ this is an elliptic curve $E = \mathbb{C}/\Lambda$: the diamond has $h^{0,0} = h^{1,1} = 1$ at top and bottom and $h^{1,0} = h^{0,1} = 1$ in the middle row, recovering $b_1 = 2$ with $H^1(E, \mathbb{C}) = H^{1,0} \oplus H^{0,1} = \mathbb{C}\, dz \oplus \mathbb{C}\, d\bar{z}$. For $n = 2$ the diamond is

$$\begin{array}{ccccc} & & h^{0,0} & & \\ & h^{1,0} & & h^{0,1} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{2,1} & & h^{1,2} & \\ & & h^{2,2} & & \end{array} \quad = \quad \begin{array}{ccccc} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{array}$$

so $b_0 = b_4 = 1$, $b_1 = b_3 = 4$, $b_2 = 6$. The total dimension across the diamond is $\sum_{p,q} \binom{2}{p}\binom{2}{q} = (1+1)^2(1+1)^2 = 16 = 2^4$, matching $H^*(X, \mathbb{C}) = \Lambda^*(\mathbb{C}^4)$ for the real four-torus.

Two checks confirm the theorem. The Hodge symmetry $h^{p,q} = h^{q,p}$ is visible in $\binom{n}{p}\binom{n}{q} = \binom{n}{q}\binom{n}{p}$, the diamond being symmetric across its vertical axis. And each row sums to the correct Betti number: $\sum_{p+q=k} \binom{n}{p}\binom{n}{q} = \binom{2n}{k}$ by Vandermonde's identity, which is exactly $b_k = \dim \Lambda^k(\mathbb{C}^{2n})$ for the underlying $2n$-real-dimensional torus.

The torus makes the mechanism transparent because harmonicity reduces to constancy of coefficients, so the abstract $\mathcal{H}^{p,q}_{\bar\partial}$ is a space one can write with a basis. For a general Kähler manifold the harmonic representatives are inaccessible by hand, but the theorem guarantees the same bookkeeping holds: the Betti numbers are recovered as row sums of the Hodge numbers, and the diamond is symmetric under $(p,q) \mapsto (q,p)$.

The decomposition of theorem 4.2.1 settles a question raised in chapter 2. There the Frölicher spectral sequence, abutting to $H^k(X, \mathbb{C})$ from the $E_1$-page $E_1^{p,q} = H^q(X, \Omega^p_X)$, produced only the inequality

$b_k \le \sum_{p+q=k} h^{p,q}$ of corollary 2.4.3, with equality holding if and only if the sequence degenerates at $E_1$. On a compact Kähler manifold the Hodge decomposition forces equality, hence degeneration.

**Corollary 4.2.3: Frölicher Degeneration**

On a compact Kähler manifold $X$, the Frölicher spectral sequence degenerates at $E_1$: all differentials $d_r$ for $r \ge 1$ vanish, and $E_1^{p,q} = E_\infty^{p,q}$. Equivalently, equality holds in corollary 2.4.3,

$$b_k = \sum_{p+q=k} h^{p,q}$$

for every $k$.

***Proof*:**
The spectral sequence converges to $H^k(X, \mathbb{C})$ with $E_1^{p,q} = H^q(X, \Omega_X^p)$, so for each $k$

$$\dim H^k(X, \mathbb{C}) = \sum_{p+q=k} \dim E_\infty^{p,q} \le \sum_{p+q=k} \dim E_1^{p,q} = \sum_{p+q=k} h^{p,q}$$

the inequality of corollary 2.4.3, arising because each later page is a subquotient of the previous. By theorem 4.2.1 the Kähler manifold satisfies

$$\dim H^k(X, \mathbb{C}) = \sum_{p+q=k} \dim H^{p,q}(X) = \sum_{p+q=k} \dim H^q(X, \Omega_X^p) = \sum_{p+q=k} h^{p,q}$$

So the inequality is an equality for every $k$. Equality of the total dimensions forces $\dim E_\infty^{p,q} = \dim E_1^{p,q}$ in each bidegree, since $E_\infty^{p,q}$ is a subquotient of $E_1^{p,q}$ and a proper subquotient would strictly drop the dimension in that slot. A subquotient of equal dimension is the whole space, so $E_1^{p,q} = E_\infty^{p,q}$ and every intermediate differential $d_r$ $(r \ge 1)$ vanishes, as we sought to show.

Degeneration at $E_1$ restates the Hodge decomposition from the spectral-sequence viewpoint: the associated graded of $H^k(X, \mathbb{C})$ under the filtration by holomorphic type is already visible on the first page, with no higher differentials to compute. It also gives a metric-free reason the Hodge numbers are cohomological invariants of the Kähler manifold, since $E_1^{p,q} = H^q(X, \Omega_X^p)$ is defined without reference to a metric. For a compact complex manifold that is not Kähler the sequence may fail to degenerate; the Iwasawa manifold (complex dimension 3) is the standard witness, its Frölicher spectral sequence having a nonzero $d_1$. (Non-degeneration cannot occur on a surface: the Frölicher spectral sequence of every compact complex surface degenerates at $E_1$, Kähler or not.)

The second deferred result is the $\partial\bar{\partial}$-lemma of theorem 3.4.1, whose proof was postponed to here. It states that on a compact Kähler manifold a $(p,q)$-form $\alpha$ that is both $\partial$-closed and $\bar{\partial}$-closed (equivalently, $d$-closed of pure type) and that is $d$-exact, $\partial$-exact, or $\bar{\partial}$-exact admits a $(p-1, q-1)$-form $\beta$ with $\alpha = \partial\bar{\partial}\beta$. The proof is the three-operator refinement of the harmonic decomposition: the Hodge decomposition holds separately for each of the three Laplacians, and $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_\partial$ makes their harmonic spaces and orthogonal complements coincide.

***Proof of theorem 3.4.1, deferred from section 3.4*:**
Fix the Kähler metric on $X$ and use the associated $L^2$ inner product on forms. By theorem 3.3.5 the three Laplacians coincide up to scalar, $\Delta_d = 2\Delta_{\bar\partial} = 2\Delta_\partial$, so they share a kernel: on each bidegree $(p,q)$ there is a single harmonic space $\mathcal{H}^{p,q}(X)$, and a single Green operator $G$ inverting the common Laplacian on the orthogonal complement of the harmonics. Because $\partial^*, \bar\partial^*, \partial, \bar\partial$ each commute with the common Laplacian on a Kähler manifold, they commute with $G$. These commutation facts are the only input beyond the harmonic decomposition of theorem 4.1.4.

*Step 1: Reduce to a $\bar\partial$-primitive.* Suppose first that $\alpha$ is $d$-exact. The harmonic projection $\mathcal{H}(\alpha)$ vanishes: a harmonic form is $L^2$-orthogonal to every exact form, since for harmonic $\eta$ and $\alpha = d\rho$ one has $\langle \eta, d\rho\rangle = \langle d^*\eta, \rho\rangle = 0$. The $\bar\partial$-Hodge decomposition of theorem 4.1.4 then gives, using $G$,

$$\alpha = \mathcal{H}(\alpha) + \bar\partial\bar\partial^* G\alpha + \bar\partial^*\bar\partial G\alpha = \bar\partial\bar\partial^* G\alpha + \bar\partial^*\bar\partial G\alpha$$

The two terms are the components of $\alpha$ in the orthogonal summands $\mathrm{im}\bar\partial$ and $\mathrm{im}\bar\partial^*$. Now $\bar\partial\alpha = 0$, so $\alpha \in \ker\bar\partial = \mathcal{H}^{p,q}(X) \oplus \mathrm{im}\bar\partial$, which is orthogonal to $\mathrm{im}\bar\partial^*$; hence the $\mathrm{im}\bar\partial^*$-component vanishes, $\bar\partial^*\bar\partial G\alpha = 0$. Therefore

$$\alpha = \bar\partial\gamma, \qquad \gamma := \bar\partial^* G\alpha \in \mathcal{A}^{p,q-1}(X)$$

*Step 2: $\gamma$ is $\partial$-closed, and decomposing it gives a potential.* First, $\gamma$ is $\partial$-closed. The vanishing of the cross term $\partial\bar\partial^* + \bar\partial^*\partial$, established in Step 2 of the proof of theorem 3.3.5, together with $\bar\partial^*$ commuting with $G$ and $\partial\alpha = 0$, gives

$$\partial\gamma = \partial\bar\partial^* G\alpha = -\bar\partial^*\partial G\alpha = -\bar\partial^* G\,\partial\alpha = 0$$

Now decompose $\gamma$ by the $\partial$-Hodge theorem, which is available because $\Delta_\partial = \Delta_{\bar\partial}$ shares the same harmonic space and Green operator $G$:

$$\gamma = \mathcal{H}(\gamma) + \partial\partial^* G\gamma + \partial^*\partial G\gamma$$

The three terms are the components of $\gamma$ in $\mathcal{H}^{p,q-1}(X)$, $\mathrm{im}\partial$, and $\mathrm{im}\partial^*$. Since $\gamma$ is $\partial$-closed, $\gamma \in \ker\partial = \mathcal{H}^{p,q-1}(X) \oplus \mathrm{im}\partial$, which is orthogonal to $\mathrm{im}\partial^*$; so the last component vanishes, $\partial^*\partial G\gamma = 0$, and $\gamma = \mathcal{H}(\gamma) + \partial\partial^* G\gamma$. Apply $\bar\partial$: the harmonic term is annihilated by $\bar\partial$ (a form harmonic for the common Laplacian is killed by both $\partial$ and $\bar\partial$), so

$$\alpha = \bar\partial\gamma = \bar\partial\partial\partial^* G\gamma = -\partial\bar\partial\partial^* G\gamma$$

using $\bar\partial\partial = -\partial\bar\partial$. Set $\beta' := -\partial^* G\gamma$. Its bidegree is $(p-1, q-1)$, since $\gamma$ has bidegree $(p, q-1)$ and $\partial^*$ lowers $p$ by one while $G$ preserves bidegree. Then

$$\alpha = \partial\bar\partial\beta'$$

which is the required expression with $\beta = \beta'$.

*Step 3: The other exactness hypotheses.* If $\alpha$ is $\bar\partial$-exact rather than $d$-exact, then since $\alpha$ is also $\partial$-closed it is orthogonal to the harmonics (a $\bar\partial$-exact form is orthogonal to $\ker\Delta_{\bar\partial} = \mathcal{H}^{p,q}(X)$), so $\mathcal{H}(\alpha) = 0$ and Steps 1 and 2 apply verbatim. If $\alpha$ is $\partial$-exact, conjugate: $\bar\alpha$ is $\bar\partial$-exact of type $(q,p)$, the previous case yields $\bar\alpha = \partial\bar\partial\beta''$, and conjugating back gives $\alpha = \bar\partial\partial\overline{\beta''} = \partial\bar\partial\big(-\overline{\beta''}\big)$. In every case $\alpha$ is $\partial\bar\partial$-exact, completing the proof.

On a compact Kähler manifold the only obstruction to writing a pure-type exact form as $\partial\bar\partial$ of something is measured by harmonic forms, which vanish for exact classes. In chapter 5 the lemma underlies the fact that a variation of Hodge structure is controlled by the Hodge filtration alone. Its failure on a non-Kähler manifold accounts for the behaviour of the Frölicher spectral sequence there.

**4.2.4 Warning: the Kähler hypothesis** Every step of the $\partial\bar\partial$-lemma used $\Delta_\partial = \Delta_{\bar\partial}$, which is the Kähler condition (theorem 3.3.5). On a compact complex manifold that is not Kähler the three Laplacians differ, their harmonic spaces separate, and the conclusion can fail: the Hopf surface $S = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z\rangle$, diffeomorphic to $S^1 \times S^3$, has $b_1(S) = 1$. An odd first Betti number is incompatible with the Hodge symmetry $h^{1,0} = h^{0,1}$ (which would force $b_1 = 2h^{1,0}$), so $S$ is not Kähler, its Hodge decomposition fails, and with it the $\partial\bar\partial$-lemma. The lemma and the decomposition require the Kähler condition together.

### Exercise 4.2.1

1. A K3 surface is a compact Kähler surface ($n = 2$) with $h^{1,0} = 0$ and $h^{2,0} = 1$; its remaining independent Hodge number is $h^{1,1} = 20$. Using Hodge symmetry and corollary 4.2.3, fill in the full Hodge diamond and compute the Betti numbers $b_0, b_1, b_2, b_3, b_4$. Check that $b_2 = 22$ and that the Euler characteristic is 24.
2. Show directly from theorem 4.2.1 that if $X$ is a compact Kähler manifold, then $\dim H^{p,q}(X) = \dim H^{q,p}(X)$, and deduce that $b_{2k+1}(X)$ is even for every $k$. (This is the discharge of the forward reference; give the two-line argument.)
3. For the complex torus $X = \mathbb{C}^2/\Lambda$ of example 4.2.2, write down an explicit basis of $H^{1,1}(X)$ in terms of the invariant forms $dz_i \wedge d\bar z_j$, and confirm $\dim H^{1,1} = 4$. Express the class of the flat Kähler form $\omega$ in this basis.
4. The subspaces $H^{p,q}(X) \subseteq H^k(X, \mathbb{C})$ are defined in theorem 4.2.1 using a chosen Kähler metric (through the harmonic representatives). Explain why $H^{p,q}(X)$ is nonetheless independent of the choice of Kähler metric. (Hint: identify $H^{p,q}(X)$ with a group defined without a metric.)
5. Let $\alpha$ be a real $d$-exact $(1,1)$-form on a compact Kähler manifold $X$. Use the $\partial\bar\partial$-lemma to produce a real function $f$ with $\alpha = i\,\partial\bar\partial f$. (Hint: apply theorem 3.4.1 to get $\alpha = \partial\bar\partial\beta$, then use that $\alpha$ is real to arrange $\beta$ of the form $if$.)
6. Suppose a compact complex manifold $Y$ has $b_1(Y)$ odd. Explain why $Y$ cannot be Kähler, and why the Frölicher spectral sequence of $Y$ and its Hodge symmetry cannot both hold: at least one of "$E_1$-degeneration" and "$h^{1,0} = h^{0,1}$" must fail. (No computation is needed; combine corollary 4.2.3 with theorem 4.2.1. Note that $b_1$ odd does not by itself force non-degeneration: the Hopf surface has $b_1 = 1$ yet its Frölicher spectral sequence degenerates, so it is Hodge symmetry that fails there.)

## 4.3 First Consequences

The decomposition $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ of theorem 4.2.1 is a constraint on the cohomology of any compact Kähler manifold, and its first payoff is a list of relations among the Hodge numbers

$h^{p,q} = \dim_{\mathbb{C}} H^{p,q}(X)$ that hold for no reason on a general compact complex manifold. Two symmetries organize the Hodge diamond into its familiar shape, and the parity of the odd Betti numbers, already extracted in chapter 3, falls out again as a special case. Beyond the numerical symmetries lies a statement about the intersection pairing: on a compact Kähler surface the signature of the cup-product form on $H^2$ is pinned down by the Hodge decomposition alone. This is the Hodge index theorem, the low-degree instance of the polarization theory that chapter 5 develops in general.

Recall from theorem 4.2.1 that complex conjugation on $H^k(X,\mathbb{C}) = H^k(X,\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$ interchanges the summands $H^{p,q}$ and $H^{q,p}$: one has $H^{p,q} = \overline{H^{q,p}}$. Conjugation is a $\mathbb{C}$-antilinear isomorphism, so it carries a $\mathbb{C}$-basis of $H^{q,p}$ to a $\mathbb{C}$-basis of $H^{p,q}$, and the two spaces have the same complex dimension. That is the first symmetry. The second comes from Serre duality, which identifies $H^q(X,\Omega^p_X)$ with the dual of $H^{n-q}(X,\Omega^{n-p}_X)$ through the canonical bundle $\Omega^n_X = \det \Omega^1_X$ of definition 1.3.12. Together they fold the array $(h^{p,q})_{0\le p,q\le n}$ onto itself under two reflections.

**Corollary 4.3.1: Hodge Symmetries**

Let $X$ be a compact Kähler manifold of complex dimension $n$. The Hodge numbers $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}(X)$ satisfy

$$h^{p,q} = h^{q,p} \qquad \text{and} \qquad h^{p,q} = h^{n-p,n-q}$$

for all $0 \le p,q \le n$. Consequently each odd Betti number $b_{2k+1}(X) = \sum_{p+q=2k+1} h^{p,q}$ is even.

***Proof*:**
*Conjugation symmetry.* By theorem 4.2.1 the conjugate-linear map $\alpha \mapsto \bar\alpha$ on $H^k(X,\mathbb{C})$ restricts to a $\mathbb{C}$-antilinear isomorphism $H^{q,p}(X) \xrightarrow{\sim} H^{p,q}(X)$. An antilinear isomorphism preserves complex dimension, so $h^{p,q} = h^{q,p}$.
*Duality symmetry.* This one is internal, and deliberately so: Serre duality as stated upstream ([4, Serre duality for a projective scheme]) assumes projectivity, which a compact Kähler manifold need not have, so it cannot be used here without narrowing the hypothesis. The Hodge star supplies the symmetry directly. By definition 3.3.2 the star carries $(p,q)$-forms to $(n-q,n-p)$-forms, so the conjugate-linear map

$$\alpha \longmapsto \star\bar\alpha$$

carries $(p,q)$-forms to $(n-p,n-q)$-forms, conjugation exchanging the two indices first. It commutes with $\Delta_{\bar\partial}$, since $\star$ commutes with the Laplacian on a compact Hermitian manifold and conjugation exchanges $\Delta_\partial$ with $\Delta_{\bar\partial}$, which agree up to the factor of two of theorem 3.3.5 on a Kähler manifold. So it restricts to a conjugate-linear isomorphism $\mathcal{H}^{p,q}_{\bar\partial}(X) \to \mathcal{H}^{n-p,n-q}_{\bar\partial}(X)$, and it is injective because $\star$ is invertible. By theorem 4.1.4 these harmonic spaces compute $H^{p,q}(X)$, so $h^{p,q} = h^{n-p,n-q}$. On a projective $X$ this is the Hodge-theoretic form of Serre duality, but no projectivity was used.
*Odd Betti parity.* Fix an odd degree $k = 2m+1$. Pair each term $h^{p,q}$ with $p+q=k$ against $h^{q,p}$; since $p \ne q$ when $p+q$ is odd, the summands split into disjoint pairs $\{(p,q),(q,p)\}$, each contributing $2h^{p,q}$ by the conjugation symmetry. Hence $b_k(X) = \sum_{p+q=k} h^{p,q}$ is a sum of even numbers, completing the proof.

The two symmetries say the Hodge diamond is invariant under reflection across its vertical axis (conjugation) and under the central involution $(p,q) \mapsto (n-p,n-q)$ (Serre duality). For a surface

$n = 2$ the diamond is

$$\begin{array}{ccccc} & & h^{2,2} & & \\ & h^{2,1} & & h^{1,2} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{1,0} & & h^{0,1} & \\ & & h^{0,0} & & \end{array}$$

with $h^{2,2} = h^{0,0} = 1$ (the top and bottom are $H^0$ and $H^{2n}$ of a connected compact manifold), $h^{1,0} = h^{0,1} = h^{2,1} = h^{1,2}$, and $h^{2,0} = h^{0,2}$; only $h^{1,0}$, $h^{2,0}$, and $h^{1,1}$ are free. The odd Betti parity here reads $b_1 = 2h^{1,0}$ and $b_3 = 2h^{2,1} = 2h^{1,0}$, recovering exactly the obstruction of proposition 3.4.4: that proposition was proved in chapter 3 by citing this decomposition forward, and the forward reference is now discharged. The content is identical, so nothing new is proved; the diamond simply displays the parity as the pairing of a term with its conjugate.

Hodge symmetry is a Kähler phenomenon, not a feature of complex manifolds in general. On the Hopf surface of chapter 3 one has $b_1 = 1$, forcing $h^{1,0} \neq h^{0,1}$, so conjugation symmetry fails; this is one route to seeing that the Hopf surface carries no Kähler metric.

The symmetries above are statements about dimensions. The decomposition also controls the cup-product pairing, and on a surface it does so completely. On a compact oriented real 4-manifold the intersection form

$$Q(\alpha, \beta) = \int_X \alpha \wedge \beta$$

on $H^2(X, \mathbb{R})$ is a nondegenerate symmetric bilinear form (nondegenerate by Poincaré duality, [24, nondegeneracy of the cup pairing]), so it has a well-defined signature $(b^+, b^-)$: the number of positive and negative eigenvalues over $\mathbb{R}$. For a smooth oriented 4-manifold the signature is a diffeomorphism invariant, not determined by $b_2$ alone. The Hodge index theorem says that when $X$ is a compact Kähler surface, the Hodge decomposition forces the value: $b^+$ and $b^-$ are determined by $h^{2,0}$ and $h^{1,1}$ alone.

The mechanism is the sign of $\int_X \alpha \wedge \bar{\alpha}$ on the pieces of the decomposition. The general statement that positivity of the polarizing form alternates with the Hodge bigrading is the Hodge-Riemann bilinear relations, developed for arbitrary weight in chapter 5. Here the only input needed is the degree-2 case on a surface, which is elementary enough to verify by hand.

**Lemma 4.3.2: Signs of the Cup Form in Bidegree Two**

Let $X$ be a compact Kähler surface with Kähler form $\omega$, and let $Q(\alpha, \beta) = \int_X \alpha \wedge \beta$ be the cup-product form on $H^2(X, \mathbb{R})$, extended $\mathbb{C}$-bilinearly to $H^2(X, \mathbb{C})$. Then:

1. For $0 \neq \alpha \in H^{2,0}(X)$ one has $\int_X \alpha \wedge \bar{\alpha} > 0$; the same holds for $\alpha \in H^{0,2}(X)$.
2. The class $[\omega] \in H^{1,1}(X)$ satisfies $\int_X \omega \wedge \omega > 0$.
3. On the primitive part $P^{1,1}(X) = \{\, \beta \in H^{1,1}(X) \,\big|\, \beta \wedge \omega = 0 \,\}$ of the real $(1,1)$-classes, one has $\int_X \beta \wedge \beta < 0$ for $0 \neq \beta$ real.

***Proof*:**
Work with harmonic representatives throughout, so a cohomology class of type $(p, q)$ is represented by a harmonic $(p, q)$-form. Fix a point and choose a local unitary coframe $\varphi^1, \varphi^2$ of $(1, 0)$-forms in which the Kähler form is $\omega = i(\varphi^1 \wedge \bar{\varphi}^1 + \varphi^2 \wedge \bar{\varphi}^2)$. The metric volume form is $\mathrm{vol} = \frac{1}{2}\omega^2$,

and expanding,

$$\tfrac{1}{2}\omega^2 = i^2\,\varphi^1 \wedge \bar\varphi^1 \wedge \varphi^2 \wedge \bar\varphi^2 = (i\,\varphi^1 \wedge \bar\varphi^1) \wedge (i\,\varphi^2 \wedge \bar\varphi^2)$$

so vol is the product of the two positive real 2-forms $i\,\varphi^j \wedge \bar\varphi^j$, and $\int_X f\,\mathrm{vol} > 0$ for a positive function $f$.

*Part (1).* A nonzero $(2,0)$-form is $\alpha = a\,\varphi^1 \wedge \varphi^2$ at each point with $a$ a complex function, so

$$\alpha \wedge \bar\alpha = |a|^2\,\varphi^1 \wedge \varphi^2 \wedge \bar\varphi^1 \wedge \bar\varphi^2$$

Reordering $\varphi^1 \wedge \varphi^2 \wedge \bar\varphi^1 \wedge \bar\varphi^2$ to $\varphi^1 \wedge \bar\varphi^1 \wedge \varphi^2 \wedge \bar\varphi^2$ moves $\bar\varphi^1$ left past $\varphi^2$, one transposition, contributing a sign $-1$; and $\varphi^1 \wedge \bar\varphi^1 \wedge \varphi^2 \wedge \bar\varphi^2 = i^{-2}\,\mathrm{vol} = -\mathrm{vol}$. The two signs cancel, giving $\alpha \wedge \bar\alpha = |a|^2\,\mathrm{vol}$ pointwise. Integrating, $\int_X \alpha \wedge \bar\alpha > 0$ unless $\alpha \equiv 0$. The case $\alpha \in H^{0,2}$ is the conjugate statement.

*Part (2).* From $\omega^2 = 2\,\mathrm{vol}$ (the definition of the induced volume form on a Kähler surface) one gets $\int_X \omega \wedge \omega = 2\int_X \mathrm{vol} = 2\,\mathrm{Vol}(X) > 0$.

*Part (3).* Let $\beta$ be a real primitive $(1,1)$-form, $\beta \wedge \omega = 0$. The claim is that the Hodge star, defined in definition 3.3.2, acts on such forms by $\star\beta = -\beta$. Primitivity says $\Lambda\beta = 0$ (definition 3.3.1), which in the unitary coframe is the trace condition on the Hermitian matrix of $\beta$; the real primitive $(1,1)$-forms are spanned by

$$\gamma_1 = i(\varphi^1 \wedge \bar\varphi^1 - \varphi^2 \wedge \bar\varphi^2), \quad \gamma_2 = \varphi^1 \wedge \bar\varphi^2 - \varphi^2 \wedge \bar\varphi^1, \quad \gamma_3 = i(\varphi^1 \wedge \bar\varphi^2 + \varphi^2 \wedge \bar\varphi^1)$$

(each is real and has vanishing $\varphi^1 \wedge \bar\varphi^1, \varphi^2 \wedge \bar\varphi^2$ trace). On an oriented Riemannian 4-manifold $\star$ is an involution on 2-forms, giving the splitting $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$ into self-dual and anti-self-dual parts; a general 2-form is a sum of the two and is *not* an eigenform, so this alone does not place $\gamma_j$ in one summand. What does is a norm count. Write $\gamma_j = \gamma_j^+ + \gamma_j^-$; then $\gamma_j \wedge \gamma_j = (|\gamma_j^+|^2 - |\gamma_j^-|^2)\,\mathrm{vol}$ while $\|\gamma_j\|^2 = |\gamma_j^+|^2 + |\gamma_j^-|^2$. Direct computation in the unitary coframe gives $\gamma_j \wedge \gamma_j = -2\,\mathrm{vol}$ and $\|\gamma_j\|^2 = 2$, and the two force $|\gamma_j^+|^2 = 0$. So each $\gamma_j$ is anti-self-dual, $\star\gamma_j = -\gamma_j$. The sign is read off from $\gamma_j \wedge \star\gamma_j = \|\gamma_j\|^2\,\mathrm{vol}$, a positive multiple of vol: since $\gamma_1 \wedge \gamma_1 = -2\,\mathrm{vol} < 0$ (and likewise $\gamma_2 \wedge \gamma_2, \gamma_3 \wedge \gamma_3 < 0$), each eigenvalue is $-1$. Hence $\star\beta = -\beta$ on the whole primitive space. Since $\beta$ is real, $\bar\beta = \beta$, and

$$\int_X \beta \wedge \beta = -\int_X \beta \wedge \star\beta = -\int_X |\beta|^2\,\mathrm{vol} < 0$$

for $\beta \not\equiv 0$, completing the proof.

The three signs are the whole content: the cup form is positive on $H^{2,0} \oplus H^{0,2}$ (once one passes to a real basis), positive along the Kähler class inside $H^{1,1}$, and negative on the orthogonal complement of the Kähler class inside the real $(1,1)$-classes. Counting dimensions of the positive and negative eigenspaces gives the signature.

**Theorem 4.3.3: Hodge Index Theorem**

Let $X$ be a compact Kähler surface. The cup-product form $Q(\alpha,\beta) = \int_X \alpha \wedge \beta$ on $H^2(X,\mathbb{R})$ has signature

$$(b^+, b^-) = \big(2h^{2,0} + 1,\ h^{1,1} - 1\big)$$

In particular its restriction to the real $(1,1)$-classes $H^{1,1}(X) \cap H^2(X,\mathbb{R})$ has signature $(1, h^{1,1} - 1)$: one positive direction, spanned by the Kähler class, and $h^{1,1} - 1$ negative directions.

***Proof*:**
*Step 1: Orthogonality of the Hodge pieces.* Extend $Q$ $\mathbb{C}$-bilinearly to $H^2(X,\mathbb{C}) = H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$. A product $\alpha \wedge \beta$ of a $(p,q)$-form and a $(p',q')$-form is a $(p+p', q+q')$-form, and only a $(2,2)$-form has nonzero integral over the surface $X$. So $Q(\alpha,\beta) = 0$ unless the bidegrees are complementary: $Q$ pairs $H^{2,0}$ with $H^{0,2}$, pairs $H^{1,1}$ with itself, and kills every other combination. The decomposition is $Q$-orthogonal into the block $H^{2,0} \oplus H^{0,2}$ and the block $H^{1,1}$.
*Step 2: The block $H^{2,0} \oplus H^{0,2}$.* This complex subspace has complex dimension $2h^{2,0}$ and is defined over $\mathbb{R}$: complex conjugation swaps its two summands, so its real points form a real subspace $V \subseteq H^2(X,\mathbb{R})$ of real dimension $2h^{2,0}$, and every $w \in V$ is $w = \alpha + \bar{\alpha}$ for a unique $\alpha \in H^{2,0}$. On a surface $\alpha \wedge \alpha$ is a $(4,0)$-form, hence zero, and likewise $\bar{\alpha} \wedge \bar{\alpha} = 0$, so

$$Q(w,w) = \int_X (\alpha + \bar{\alpha}) \wedge (\alpha + \bar{\alpha}) = 2\int_X \alpha \wedge \bar{\alpha}$$

which is $> 0$ for $\alpha \neq 0$ by part (1) of lemma 4.3.2. Thus $Q|_V$ is positive definite, and $V$ contributes $(2h^{2,0}, 0)$ to the signature.
*Step 3: The block $H^{1,1} \cap H^2(X,\mathbb{R})$.* On the real $(1,1)$-classes $Q$ is real-valued and nondegenerate. The Kähler class $[\omega]$ satisfies $Q([\omega],[\omega]) > 0$ by part (2) of lemma 4.3.2, so $\mathbb{R}[\omega]$ is a positive line. The Lefschetz operator $L = \omega \wedge (-)$ of definition 3.3.1 gives, in degree two on a surface, an orthogonal decomposition of the real $(1,1)$-classes into $\mathbb{R}[\omega]$ and the primitive part $P^{1,1} = \ker(\Lambda\omega)$: any real $(1,1)$-class $\gamma$ writes as $\gamma = \frac{Q(\gamma,[\omega])}{Q([\omega],[\omega])}[\omega] + \gamma_0$ with $\gamma_0$ primitive, and the two summands are $Q$-orthogonal because $Q([\omega],\gamma_0) = \int_X \omega \wedge \gamma_0 = 0$ by primitivity. Part (3) of lemma 4.3.2 makes $Q$ negative definite on $P^{1,1}$, of dimension $h^{1,1} - 1$. So the real $(1,1)$-classes contribute $(1, h^{1,1} - 1)$.
*Step 4: Assemble.* By Step 1 the two blocks are $Q$-orthogonal, so their signatures add:

$$(b^+, b^-) = (2h^{2,0}, 0) + (1, h^{1,1} - 1) = (2h^{2,0} + 1,\ h^{1,1} - 1)$$

The restriction to the real $(1,1)$-classes is $(1, h^{1,1} - 1)$ from Step 3, as we sought to show.

The theorem earns its name from the negative-definite conclusion on the primitive $(1,1)$-classes: apart from the single positive direction of the Kähler class, the intersection form on divisor-type classes is negative definite. Since the Néron-Severi group $\mathrm{NS}(X)$ sits inside $H^{1,1}(X) \cap H^2(X,\mathbb{Z})$ (this is the Lefschetz $(1,1)$-theorem of the next section), the algebraic classes inherit signature $(1, \rho - 1)$ when $X$ is projective, where $\rho = \operatorname{rk} \mathrm{NS}(X)$ is the Picard number. Projectivity is needed and not automatic: the restriction of a form of signature $(1, h^{1,1} - 1)$ to a subspace has a positive direction only if that subspace contains a class of positive square, which an ample class supplies. On a non-projective Kähler surface $\mathrm{NS}(X)$ can be negative definite, or zero. For an algebraic surface with an ample class $H$, this recovers the classical Hodge index theorem: if a divisor class $D$ satisfies $D \cdot H = 0$, then

$D^2 \le 0$, with equality only for $D$ numerically trivial. The transcendental part $H^{2,0} \oplus H^{0,2}$ supplies the extra positive directions $2h^{2,0}$, which is why a K3 surface, with $h^{2,0} = 1$, has $b^+ = 3$ rather than 1.

The value $b^- = h^{1,1} - 1$ also expresses the topological signature $\sigma(X) = b^+ - b^- = 2 + 2h^{2,0} - h^{1,1}$ purely through Hodge numbers. With $\chi_{\mathrm{top}}(X) = 2 - 4h^{1,0} + 2h^{2,0} + h^{1,1}$ (from $b_1 = 2h^{1,0}$, $b_2 = 2h^{2,0} + h^{1,1}$), the two combine to

$$\tfrac{1}{4}\big(\sigma(X) + \chi_{\mathrm{top}}(X)\big) = 1 - h^{1,0} + h^{2,0} = \chi(\mathcal{O}_X)$$

the holomorphic Euler characteristic $\chi(\mathcal{O}_X) = \sum_q (-1)^q h^{0,q}$. This is the topological form of Noether's formula: the alternating sum $\sum_q (-1)^q h^{0,q}$ is pinned down by the signature and the Euler characteristic together.

**Example 4.3.4: The Intersection Form on a K3 and an Abelian Surface**

Two surfaces show the two regimes of the index theorem.

1. *K3 surface.* A K3 surface $X$ has $h^{1,0} = 0$ (so $b_1 = 0$) and $h^{2,0} = 1$, hence $b_2 = 2 \cdot 1 + h^{1,1} = 22$ forces $h^{1,1} = 20$. The Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

   By theorem 4.3.3 the intersection form on $H^2(X, \mathbb{Z}) \cong \mathbb{Z}^{22}$ has signature $(2 \cdot 1 + 1,\ 20 - 1) = (3, 19)$. The intersection form is unimodular ([24, integral unimodularity of the intersection pairing]) and even. Those two facts do not by themselves pin down the lattice; the classification of indefinite even unimodular lattices, which this book does not prove, identifies it with $H^2(X, \mathbb{Z}) \cong U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$, the K3 lattice, where $U$ is the hyperbolic plane and $E_8(-1)$ the negative $E_8$ lattice; the $(3, 19)$ signature is exactly $3 \cdot (1, 1) + 2 \cdot (0, 8)$. On the real $(1, 1)$-classes the signature is $(1, 19)$, and for a *projective* K3 the algebraic classes $\mathrm{NS}(X)$, of rank $\rho \le 20$, carry signature $(1, \rho - 1)$.

2. *Abelian surface.* An abelian surface $X = \mathbb{C}^2/\Lambda$ has, by example 4.2.2 with $n = 2$, Hodge numbers $h^{1,0} = 2$, $h^{2,0} = 1$, $h^{1,1} = 4$, so $b_2 = 2 \cdot 1 + 4 = 6$. The intersection form on $H^2(X, \mathbb{Z}) \cong \mathbb{Z}^6$ has signature $(2 \cdot 1 + 1,\ 4 - 1) = (3, 3)$. Indeed $H^2(X, \mathbb{Z}) \cong \bigwedge^2 H^1(X, \mathbb{Z})$ with $H^1 \cong \mathbb{Z}^4$, and the cup product $\bigwedge^2 \mathbb{Z}^4 \to \bigwedge^4 \mathbb{Z}^4 \cong \mathbb{Z}$ is the standard split form of signature $(3, 3)$. On the real $(1, 1)$-classes, of dimension 4, the signature is $(1, 3)$.

## Exercise 4.3.1

1. Let $X$ be a compact Kähler surface. Using $b_1 = 2h^{1,0}$, $b_2 = 2h^{2,0} + h^{1,1}$, and $b_3 = b_1$, express the topological Euler characteristic $\chi_{\mathrm{top}}(X) = \sum_k (-1)^k b_k$ in terms of $h^{1,0}, h^{2,0}, h^{1,1}$, and show $\chi_{\mathrm{top}}(X) \equiv b_2 \equiv h^{1,1} \pmod 2$. (Do not expect $\chi_{\mathrm{top}}$ itself to be even: $\mathbb{P}^2$ has $\chi_{\mathrm{top}} = 3$.)

2. Show that on any compact Kähler surface the topological signature $\sigma(X) = b^+ - b^-$ equals $2 + 2h^{2,0} - h^{1,1}$. Deduce that $\sigma(X) \equiv b_2 \pmod 2$.

3. The Hopf surface (a quotient of $\mathbb{C}^2 \setminus \{0\}$) is a compact complex surface with $b_1 = 1$. Show that it admits no Kähler metric, using only corollary 4.3.1.

4. Let $X$ be a compact Kähler surface and $H$ an ample class with $H^2 = d > 0$. For a real $(1,1)$-class $D$ set $D_0 = D - \frac{D\cdot H}{d}H$. Show $D_0$ is primitive and deduce the classical inequality: if $D \cdot H = 0$ and $D \neq 0$ then $D^2 < 0$.

5. On a compact Kähler surface, use corollary 4.3.1 to show $h^{2,0} = h^{0,2} = p_g$ (the geometric genus $\dim H^0(X, \Omega_X^2)$) and $h^{1,0} = h^{0,1} = q$ (the irregularity $\dim H^0(X, \Omega_X^1)$), identifying $H^{2,0}(X)$ with $H^0(X, \Omega_X^2)$ and $H^{1,0}(X)$ with $H^0(X, \Omega_X^1)$.

6. For the abelian surface of example 4.3.4, exhibit an explicit basis of $H^2(X, \mathbb{R})$ in which the cup-product form is diagonal with entries $(+1, +1, +1, -1, -1, -1)$, starting from a real basis $e_1, e_2, e_3, e_4$ of $H^1(X, \mathbb{R})$.

## 4.4 The Lefschetz $(1,1)$-Theorem

The Hodge decomposition sorts $H^2(X, \mathbb{C})$ into the summands $H^{2,0}$, $H^{1,1}$, $H^{0,2}$; the topology alone cannot see this splitting, but the complex structure does. A holomorphic line bundle produces a class in $H^2(X, \mathbb{Z})$, its first Chern class, and this class always lands in the $(1,1)$-part. Lefschetz's theorem is the converse: on a compact Kähler manifold every integral class of type $(1,1)$ comes from a holomorphic line bundle, hence from a divisor. This section proves that converse through the exponential sequence, which is the exact sequence linking the additive sheaf $\mathcal{O}_X$ to the multiplicative sheaf $\mathcal{O}_X^*$ of nowhere-vanishing holomorphic functions.

The mechanism is a single connecting homomorphism. The holomorphic line bundles on $X$ are classified by $H^1(X, \mathcal{O}_X^*)$, and the exponential sequence produces a map from this group to $H^2(X, \mathbb{Z})$ which is exactly $c_1$. The theorem describes the kernel and image of that map.

**Proposition 4.4.1: The Exponential Sequence**

Let $X$ be a complex manifold. The map $\exp\colon \mathcal{O}_X \to \mathcal{O}_X^*$, $f \mapsto \exp(2\pi i f)$, fits into a short exact sequence of sheaves of abelian groups

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}_X^* \to 0$$

where $\underline{\mathbb{Z}}$ is the constant sheaf and the multiplicative group structure is used on $\mathcal{O}_X^*$. The induced long exact sequence in cohomology contains

$$\cdots \to H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X) \to \cdots$$

and the connecting homomorphism $c_1\colon H^1(X, \mathcal{O}_X^*) \to H^2(X, \mathbb{Z})$ sends the class of a holomorphic line bundle $L$ to its first Chern class.

***Proof:***
*Step 1: Exactness of the sheaf sequence.* The constant sheaf $\underline{\mathbb{Z}}$ includes into $\mathcal{O}_X$ as the locally constant integer-valued functions, and $\exp(2\pi i n) = 1$ for $n \in \mathbb{Z}$, so the composite $\underline{\mathbb{Z}} \to \mathcal{O}_X^*$ is trivial and the sequence is a complex. Injectivity of $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is clear. For surjectivity of exp on

stalks, let $g$ be a nowhere-vanishing holomorphic function on a connected open $U$. Shrinking $U$ to a polydisc neighborhood of any point, $g$ admits a holomorphic logarithm: $g$ has no zeros, so the holomorphic 1-form $dg/g$ is closed, and on a simply connected domain it has a holomorphic primitive $h$ with $d(ge^{-h}) = 0$, whence $g = c\,e^h$ for a constant $c \neq 0$; absorbing $\log c$ into $h$ and dividing by $2\pi i$ gives a holomorphic $f$ with $\exp(2\pi i f) = g$. Thus exp is surjective as a map of sheaves. Exactness in the middle is the observation that $\exp(2\pi i f) = 1$ forces $f$ to be a locally constant integer, so the kernel of exp is exactly $\underline{\mathbb{Z}}$.

*Step 2: The connecting map is $c_1$.* A holomorphic line bundle $L$ is given by a cocycle $(g_{\alpha\beta})$ of transition functions $g_{\alpha\beta} \in \mathcal{O}_X^*(U_\alpha \cap U_\beta)$ satisfying the cocycle condition $g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = 1$; this is precisely a Čech 1-cocycle representing a class in $H^1(X, \mathcal{O}_X^*)$ (proposition 1.3.8). The connecting homomorphism of the long exact sequence computes as follows. Choose logarithms $f_{\alpha\beta} = \frac{1}{2\pi i} \log g_{\alpha\beta} \in \mathcal{O}_X(U_\alpha \cap U_\beta)$ on the overlaps, so $\exp(2\pi i f_{\alpha\beta}) = g_{\alpha\beta}$. The failure of $(f_{\alpha\beta})$ to be a cocycle,

$$n_{\alpha\beta\gamma} = f_{\beta\gamma} - f_{\alpha\gamma} + f_{\alpha\beta}$$

is a 2-cocycle valued in $\underline{\mathbb{Z}}$ (it exponentiates to $g_{\beta\gamma}g_{\alpha\gamma}^{-1}g_{\alpha\beta} = 1$, so it is integer-valued), and its class in $H^2(X, \mathbb{Z})$ is by definition the image of $[L]$ under the connecting map. That integer 2-cocycle is the standard Čech representative of the first Chern class of $L$, completing the proof.

The two boxed groups at the ends of the six-term stretch are the obstruction groups. Reading the sequence

$$H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \xrightarrow{\iota} H^2(X, \mathcal{O}_X)$$

as an exact sequence, a class $\gamma \in H^2(X, \mathbb{Z})$ lies in the image of $c_1$ if and only if it dies in $H^2(X, \mathcal{O}_X)$. The map $\iota$ is the composite of the change-of-coefficients map $H^2(X, \mathbb{Z}) \to H^2(X, \mathbb{C})$ with the projection of $H^2(X, \mathbb{C})$ onto its $H^{0,2}$-summand, since Dolbeault's theorem (theorem 2.3.3) identifies $H^2(X, \mathcal{O}_X) = H^{0,2}(X)$. The Hodge decomposition turns the vanishing of $\iota(\gamma)$ into a statement about Hodge type, and that is the theorem.

Before stating it, one point of bookkeeping. The image of an integral class $\gamma$ in $H^2(X, \mathbb{C})$ decomposes as $\gamma^{2,0} + \gamma^{1,1} + \gamma^{0,2}$. Because $\gamma$ is real, complex conjugation fixes it, and conjugation exchanges $H^{2,0}$ with $H^{0,2}$ (theorem 4.2.1), so $\gamma^{2,0} = \overline{\gamma^{0,2}}$. Hence $\gamma^{0,2} = 0$ if and only if $\gamma^{2,0} = 0$, and both vanish exactly when $\gamma$ is of pure type $(1,1)$.

**Theorem 4.4.2: The Lefschetz $(1,1)$-Theorem**

Let $X$ be a compact Kähler manifold and $\gamma \in H^2(X, \mathbb{Z})$. The following are equivalent.

1. $\gamma$ is the first Chern class $c_1(L)$ of a holomorphic line bundle $L$ on $X$.
2. The image of $\gamma$ in $H^2(X, \mathcal{O}_X)$ vanishes.
3. The image of $\gamma$ in $H^2(X, \mathbb{C})$ lies in the summand $H^{1,1}(X)$; that is, $\gamma$ is an integral class of type $(1,1)$.

***Proof*:**

*(1) ⇔ (2).* By proposition 4.4.1 the segment $H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \xrightarrow{\iota} H^2(X, \mathcal{O}_X)$ is exact, so $\gamma$ lies in the image of $c_1$ precisely when $\iota(\gamma) = 0$. Since every element of $H^1(X, \mathcal{O}_X^*)$ is the class of a holomorphic line bundle (proposition 1.3.8), the image of $c_1$ is exactly the set of Chern classes of line bundles, which is (1).

*(2) ⇔ (3).* Dolbeault's theorem (theorem 2.3.3) identifies $H^2(X, \mathcal{O}_X) = H^{0,2}(X)$, and under this identification $\iota$ is the composite

$$H^2(X, \mathbb{Z}) \to H^2(X, \mathbb{C}) = H^{2,0}(X) \oplus H^{1,1}(X) \oplus H^{0,2}(X) \to H^{0,2}(X)$$

of the coefficient map with projection onto the $(0,2)$-summand; the first equality is the Hodge decomposition (theorem 4.2.1), available because $X$ is compact Kähler. Thus $\iota(\gamma) = 0$ says $\gamma^{0,2} = 0$. As noted before the statement, the image of $\gamma$ is real, so $\gamma^{2,0} = \overline{\gamma^{0,2}}$, and $\gamma^{0,2} = 0$ forces $\gamma^{2,0} = 0$ as well. Hence $\gamma^{0,2} = 0$ if and only if $\gamma = \gamma^{1,1}$ lies in $H^{1,1}(X)$, which is (3), as we sought to show.

The equivalence of (1) and (3) is the usable form: it converts a transcendental condition (Hodge type, read off the complex structure) into an algebro-geometric one (representability by a line bundle). The Chern classes of line bundles are exactly the integral $(1,1)$-classes. Naming the group of such classes records this.

**Definition 4.4.3: The Néron-Severi Group**

Let $X$ be a compact Kähler manifold. The *Néron-Severi group* of $X$ is

$$\mathrm{NS}(X) = H^{1,1}(X) \cap H^2(X, \mathbb{Z})$$

the group of integral cohomology classes of type $(1,1)$, where the intersection is taken inside $H^2(X, \mathbb{C})$. Its rank is the *Picard number* $\rho(X)$.

By theorem 4.4.2 the Néron-Severi group is the image of $c_1 \colon H^1(X, \mathcal{O}_X^*) \to H^2(X, \mathbb{Z})$, so it is the group of first Chern classes of holomorphic line bundles. On a projective manifold every holomorphic line bundle arises from a divisor. Combining the theorem with the divisor-to-Picard correspondence of chapter 1 gives the geometric reading in that case.

An element $D \in \mathrm{Div}(X)$ is a formal $\mathbb{Z}$-combination of irreducible hypersurfaces; the class map $D \mapsto \mathcal{O}_X(D)$ sends it to a holomorphic line bundle (definition 1.4.4), and this descends to a homomorphism $\mathrm{Cl}(X) \to H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X)$ from divisor classes to line bundles (theorem 1.4.6). Composing with $c_1$, an integral $(1,1)$-class is the class of a divisor: on a projective manifold every holomorphic line bundle is $\mathcal{O}_X(D)$ for some divisor $D$, so $\mathrm{NS}(X)$ is exactly the group of divisor classes modulo those algebraically equivalent to zero. This is why the theorem is stated as "integral $(1,1)$-classes are divisor classes".

**4.4.4 Warning: Kähler versus projective** On a general compact Kähler manifold, $c_1(L)$ is an integral $(1,1)$-class for every line bundle $L$, and theorem 4.4.2 produces a line bundle from each such class. The further step "every line bundle comes from a divisor" needs a nonzero *meromorphic* section of $L$, equivalently that $L \otimes \mathcal{O}(mH)$ be globally generated for $m \gg 0$ with $H$ ample, which writes $L = \mathcal{O}(D_1 - mH)$. That is available only when $X$ is projective (equivalently, when $X$ carries a Kähler class in $H^2(X, \mathbb{Z})$). A compact Kähler manifold with $\mathrm{NS}(X) = 0$, such as a generic complex torus (example 1.1.10), carries no divisors at all, consistent with an empty Néron-Severi group.

The running example makes these groups concrete. A complex torus $X = \mathbb{C}^g/\Lambda$ (example 1.1.10) has $H^1(X, \mathcal{O}_X)$ and $H^2(X, \mathbb{Z})$ both computable from the lattice, so the Néron-Severi group can be described by hand. The case $g = 2$, an abelian surface, is the smallest one where the Picard number varies with the complex structure.

**Example 4.4.5: The Néron-Severi Group of an Abelian Surface**

Let $X = \mathbb{C}^2/\Lambda$ be a complex torus of dimension 2, with $\Lambda \cong \mathbb{Z}^4$ a lattice. Its integral cohomology is the exterior algebra on $H^1(X,\mathbb{Z}) = \operatorname{Hom}(\Lambda,\mathbb{Z}) \cong \mathbb{Z}^4$, so

$$H^2(X,\mathbb{Z}) = \bigwedge^2 \operatorname{Hom}(\Lambda,\mathbb{Z}) \cong \mathbb{Z}^6$$

The complex structure on $\mathbb{C}^2$ makes $\Lambda \otimes \mathbb{R} = \mathbb{C}^2$ a complex vector space, and the Hodge decomposition of $H^2(X,\mathbb{C})$ reads, in terms of the splitting of alternating $\mathbb{R}$-bilinear forms on $\mathbb{C}^2$ into their types,

$$H^{2,0} \oplus H^{1,1} \oplus H^{0,2}, \qquad h^{2,0} = 1, \quad h^{1,1} = 4, \quad h^{0,2} = 1$$

An integral class $\gamma \in H^2(X,\mathbb{Z})$ is an alternating form $E\colon \Lambda \times \Lambda \to \mathbb{Z}$. By theorem 4.4.2, $\gamma \in \operatorname{NS}(X)$ if and only if $\gamma$ is of type $(1,1)$, and for an alternating form on the complex vector space $\mathbb{C}^2$ the $(1,1)$-condition is exactly the Riemann bilinear relation

$$E(iu,iv) = E(u,v) \qquad \text{for all } u,v \in \mathbb{C}^2$$

(the vanishing of the $(2,0)$ and $(0,2)$ parts of $E$ is the invariance of $E$ under the complex structure $J = i$). Thus

$$\operatorname{NS}(X) = \{\, E \in \bigwedge^2 \operatorname{Hom}(\Lambda,\mathbb{Z}) : E(iu,iv) = E(u,v) \,\}$$

The rank of this group depends on $\Lambda$. For a very general lattice the only integral $J$-invariant alternating form is 0, so $\rho(X) = 0$ and $X$ has no divisors: it is a non-algebraic complex torus. Imposing the $J$-invariance on one form (a single nonzero solution together with a positivity condition) already forces $X$ to be an abelian variety; the maximal Picard number for a 2-torus is $\rho = 4$, attained when $X$ is isogenous to a product of elliptic curves with complex multiplication. The interpolation between these, and the Appell-Humbert description of line bundles by their Chern class $E$ together with a semicharacter, is developed in [3]; the point here is that the Lefschetz theorem reduces the whole computation to counting integral solutions of the linear condition $E(iu,iv) = E(u,v)$.

The example exhibits the two extremes the theorem allows. When $\operatorname{NS}(X) = 0$ the manifold is Kähler but carries no line bundle with nonzero Chern class, so no divisor; when $\rho$ is maximal the torus is an abelian variety with many divisors. Both are governed by the single linear condition that an integral class be of type $(1,1)$, and that condition is what theorem 4.4.2 extracts from the exponential sequence.

## Exercise 4.4.1

1. Show that the exponential sequence $0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$ does not split as a sequence of sheaves of abelian groups when $X = \mathbb{C}^*$ (the punctured plane, a noncompact Riemann surface). (Hint: consider whether the identity function $z \in \mathcal{O}_X^*(X)$ admits a global holomorphic logarithm.)

2. Let $X$ be a compact Kähler manifold. Using proposition 4.4.1, show that the kernel of $c_1\colon H^1(X,\mathcal{O}_X^*) \to H^2(X,\mathbb{Z})$ is the quotient $H^1(X,\mathcal{O}_X)/\operatorname{im}\big(H^1(X,\mathbb{Z})\big)$, the group $\operatorname{Pic}^0(X)$

of line bundles with vanishing first Chern class. Identify this quotient as a complex torus using the Hodge decomposition of $H^1$.

3. For a compact Kähler manifold $X$, show that the Picard number satisfies $\rho(X) \leq h^{1,1}(X)$. Give an example (a very general complex torus of dimension $\geq 2$ will do) where the inequality is strict.
4. Let $E = \mathbb{C}/\Lambda$ be an elliptic curve. Compute $H^2(E, \mathbb{Z})$ and show that $\mathrm{NS}(E) = H^2(E, \mathbb{Z}) \cong \mathbb{Z}$, so that every integral class of the top-degree cohomology of $E$ is the class of a divisor. Identify the generator with the class of a single point.
5. Let $X = E_1 \times E_2$ be a product of two elliptic curves. Show that $\mathrm{NS}(X) = \mathbb{Z}F_1 \oplus \mathbb{Z}F_2 \oplus \mathrm{Hom}(E_1, E_2)$, where $F_1, F_2$ are the two fiber classes, and deduce that $\rho(X) = 2 + \mathrm{rank}\mathrm{Hom}(E_1, E_2)$. Conclude that $\rho(X) = 2$ for $E_1, E_2$ generic (non-isogenous), $\rho(X) = 3$ when $E_1, E_2$ are isogenous without complex multiplication, and $\rho(X) = 4$ when they are isogenous with complex multiplication.
6. State the Hodge conjecture for $H^2$ and observe that on a compact Kähler manifold it is a theorem, namely theorem 4.4.2. Explain in one or two sentences why the analogous statement in degree $H^4$ (for $(2,2)$-classes) is not accessible by the exponential-sequence argument.

# Part II

# Hodge Structures and Their Variations

Part I ended with a theorem about a single manifold: the complex cohomology of a compact Kähler $X$ splits by type, and the splitting is an invariant of $X$. Read again, the conclusion barely mentions $X$. What it produces is a piece of linear algebra on a rational vector space, a conjugate-symmetric decomposition of its complexification together with a bilinear form supplied by the Kähler class, and every use of the geometry has been spent by the time that data appears.

This Part takes the linear algebra as the object. The gain is reach rather than economy. Structures of the same shape arise where no Kähler manifold is in sight, and the ones that do come from geometry satisfy constraints that a single manifold cannot exhibit: fix the numerical invariants and the polarized structures of that type are the points of a complex manifold, so a family of varieties traces a path in it and a degeneration is a limit. Questions that cannot even be posed about one $X$ become questions about points and paths. Whether two varieties with the same cohomology are isomorphic is a question of injectivity; which classes are algebraic is a question about the locus where the structure acquires extra pieces.

The chapters that follow build the structure and the space that parametrizes it, prove the Lefschetz theorems that constrain it on a single manifold, and then let it move: a family of varieties gives a holomorphically varying structure whose derivative is constrained, and that constraint is the source of most of what is known about how varieties fit together in moduli.

# 5

# *Hodge Structures and Polarizations*

The Hodge decomposition attaches to a compact Kähler manifold a bigrading of its complex cohomology. This chapter isolates that datum from the manifold that produced it. A *Hodge structure* is a conjugate-symmetric bigrading of the complexification of a rational vector space, and a *polarization* is the extra bilinear form that cup product against a Kähler class supplies. The polarization is what makes the notion rigid: unpolarized structures of a given type form a set with no useful geometry, while polarized ones form a complex manifold.

Two features surface once the manifold is gone, and both are invisible while it is present. The bigrading turns out to be equivalent to a decreasing filtration, and it is the filtration, not the bigrading, that behaves well in families. And the polarized structures of a fixed numerical type are themselves the points of a complex manifold, the period domain, whose geometry the last two sections begin.

## 5.1 Pure Hodge Structures and the Hodge Filtration

The definition is read off from theorem 4.2.1. That theorem produces, on a compact Kähler manifold $X$, a decomposition $H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ satisfying $H^{p,q} = \overline{H^{q,p}}$, where the conjugation is the one fixing the real cohomology $H^k(X,\mathbb{R})$. Both halves matter. The decomposition alone is a bigrading of a complex vector space and carries no information beyond a list of dimensions; the conjugation symmetry is what ties it to a real, and in fact rational, structure underneath.

**Definition 5.1.1: Pure Hodge Structure of Weight $k$**

Let $V_{\mathbb{Z}}$ be a finitely generated free abelian group, $V_{\mathbb{Q}} = V_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ and $V_{\mathbb{C}} = V_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C}$, and let $k \in \mathbb{Z}$. A *pure Hodge structure of weight $k$* on $V_{\mathbb{Z}}$ is a direct sum decomposition

$$V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$$

into complex subspaces, subject to the symmetry

$$\overline{V^{p,q}} = V^{q,p}$$

where the bar is the conjugation of $V_{\mathbb{C}} = V_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ fixing $V_{\mathbb{R}}$. The integers $h^{p,q} = \dim_{\mathbb{C}} V^{p,q}$ are the *Hodge numbers* of the structure. The same definition with $V_{\mathbb{Q}}$ in place of $V_{\mathbb{Z}}$ defines a *rational* Hodge structure; $V_{\mathbb{Z}}$ is called the *lattice* and is retained because the geometric examples carry one.

The weight is part of the data, not a consequence of it: the same bigraded vector space underlies a weight-$k$ structure and, after relabelling, structures of other weights. Only the sum $p+q$ is pinned, and pinning it is what makes the Hodge numbers of two structures comparable.

The symmetry deserves a second look, because it is the only place the rational structure enters. Conjugation on $V_{\mathbb{C}}$ is defined by $V_{\mathbb{R}}$, hence by $V_{\mathbb{Q}}$, hence by $V_{\mathbb{Z}}$; a bigrading of an abstract complex vector space has no conjugation to be symmetric under. So $\overline{V^{p,q}} = V^{q,p}$ is a compatibility between the bigrading and the lattice, and it immediately forces $h^{p,q} = h^{q,p}$, since conjugation is a conjugate-linear isomorphism and those preserve complex dimension.

**Example 5.1.2: The Cohomology of a Compact Kähler Manifold**

Let $X$ be a compact Kähler manifold and let $V_{\mathbb{Z}} = H^k(X,\mathbb{Z})/(\text{torsion})$, so that $V_{\mathbb{C}} = H^k(X,\mathbb{C})$. Setting $V^{p,q} = H^{p,q}(X)$, theorem 4.2.1 supplies both the decomposition and the symmetry $H^{p,q} = \overline{H^{q,p}}$, the conjugation there being the one fixing $H^k(X,\mathbb{R})$. So $H^k(X,\mathbb{Z})$ carries a pure Hodge structure of weight $k$, whose Hodge numbers are the $h^{p,q}(X)$ of definition 2.3.4.

This is the example the definition was extracted from, and for the rest of the book it is the source of every geometric instance. The weight equals the cohomological degree, which is the reason the abstract theory indexes by weight at all.

**Example 5.1.3: The Lowest Weights**

The first two weights are already instructive. Both descriptions below assume the structure is *effective*, meaning $V^{p,q} = 0$ unless $p \geq 0$ and $q \geq 0$. Effectivity holds for the cohomology of a compact Kähler manifold, where $p$ and $q$ are form degrees, but it is not part of definition 5.1.1 and it fails after a Tate twist (definition 5.1.10); without it, weight alone constrains only the sum $p+q$.

1. *Weight 0, effective.* The only pair with $p+q=0$ and $p,q \geq 0$ is $(0,0)$, so $V_{\mathbb{C}} = V^{0,0}$ and the structure carries no information beyond $V_{\mathbb{Z}}$ itself. Effective weight-0 Hodge structures are just lattices.

2. *Weight 1, effective.* The available pairs are $(1,0)$ and $(0,1)$, giving $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ with $\overline{V^{1,0}} = V^{0,1}$. The structure is therefore determined by the single subspace $V^{1,0}$, which must satisfy $V^{1,0} \oplus \overline{V^{1,0}} = V_{\mathbb{C}}$. Such a subspace is the same thing as a complex structure on the

real vector space $V_{\mathbb{R}}$: the projection $V_{\mathbb{R}} \hookrightarrow V_{\mathbb{C}} \to V^{1,0}$ along $V^{0,1}$ is an $\mathbb{R}$-linear isomorphism, and pulling back multiplication by $i$ gives an endomorphism $J$ of $V_{\mathbb{R}}$ with $J^2 = -\mathrm{id}$ (exercise 5.1.1 (1)). In particular $V_{\mathbb{R}}$ has even dimension and $h^{1,0} = h^{0,1} = \frac{1}{2}\dim_{\mathbb{R}} V_{\mathbb{R}}$.

The weight-1 case is the one that will reappear as the Hodge-theoretic description of complex tori in section 5.4, and the appearance of $J$ is expected: it is the linear-algebraic form of the almost complex structure of definition 1.1.12.

For a single vector space the bigrading is the natural presentation. In a family it is not, and the reason is worth stating now even though the payoff is several chapters away: as a manifold varies, its subspaces $H^{p,q}$ move smoothly and no better, while certain sums of them move holomorphically. Those sums are the terms of a filtration.

**Definition 5.1.4: The Hodge Filtration**

Let $V$ carry a pure Hodge structure of weight $k$. Its *Hodge filtration* is the decreasing chain of complex subspaces

$$F^pV_{\mathbb{C}} := \bigoplus_{r \geq p} V^{r,\,k-r}$$

so that $F^pV_{\mathbb{C}} \supseteq F^{p+1}V_{\mathbb{C}}$, with $F^pV_{\mathbb{C}} = V_{\mathbb{C}}$ for $p \leq \min\{r : V^{r,k-r} \neq 0\}$ and $F^pV_{\mathbb{C}} = 0$ for $p$ large.

Nothing has been lost in passing from the bigrading to the filtration, because the bigrading can be read back off. The recovery formula is the content of the next theorem, and it is the identity that makes the two presentations interchangeable.

**Theorem 5.1.5: Bigradings and Filtrations**

Let $V_{\mathbb{Z}}$ be a finitely generated free abelian group and $k \in \mathbb{Z}$. The following data are equivalent.

1. A pure Hodge structure of weight $k$ on $V_{\mathbb{Z}}$, that is, a decomposition $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$.

2. A decreasing filtration $\cdots \supseteq F^pV_{\mathbb{C}} \supseteq F^{p+1}V_{\mathbb{C}} \supseteq \cdots$ of $V_{\mathbb{C}}$ by complex subspaces, which is exhaustive and separated ($F^pV_{\mathbb{C}} = V_{\mathbb{C}}$ for $p \ll 0$ and $F^pV_{\mathbb{C}} = 0$ for $p \gg 0$), satisfying
$$F^pV_{\mathbb{C}} \oplus \overline{F^{k-p+1}V_{\mathbb{C}}} = V_{\mathbb{C}} \qquad \text{for every } p \tag{5.1}$$

The passage from (1) to (2) is $F^pV_{\mathbb{C}} = \bigoplus_{r \geq p} V^{r,k-r}$, and the passage from (2) to (1) is

$$V^{p,q} = F^pV_{\mathbb{C}} \cap \overline{F^qV_{\mathbb{C}}}, \qquad p+q=k \tag{5.2}$$

The two constructions are mutually inverse.

***Proof*:**
Write $F^p$ for $F^pV_{\mathbb{C}}$ throughout.

*Step 1: a Hodge structure gives a filtration satisfying* (5.1). Let $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$ be a Hodge structure and set $F^p = \bigoplus_{r \geq p} V^{r,k-r}$. Decreasingness is clear, and the sum being finite gives exhaustiveness and separatedness. For (5.1), apply conjugation to the definition of $F^{k-p+1}$ and use $\overline{V^{r,k-r}} = V^{k-r,r}$:

$$\overline{F^{k-p+1}} = \overline{\bigoplus_{r \geq k-p+1} V^{r,k-r}} = \bigoplus_{r \geq k-p+1} V^{k-r,\,r} = \bigoplus_{s \leq p-1} V^{s,\,k-s}$$

the last equality by the substitution $s = k-r$, which turns $r \geq k-p+1$ into $s \leq p-1$. So $F^p$ collects the summands with first index at least $p$ and $\overline{F^{k-p+1}}$ collects those with first index at most $p-1$. Every summand lies in exactly one of the two, so their sum is $V_{\mathbb{C}}$ and their intersection is 0.

*Step 2: a filtration satisfying* (5.1) *gives a Hodge structure.* Let $F^\bullet$ be as in (2) and define $V^{p,q} := F^p \cap \overline{F^q}$ for $p+q=k$. The symmetry is immediate from the definition: conjugating,

$$\overline{V^{p,q}} = \overline{F^p \cap \overline{F^q}} = \overline{F^p} \cap F^q = V^{q,p}$$

since conjugation is an involution exchanging a subspace with its conjugate. It remains to prove that the $V^{p,q}$ sum to $V_{\mathbb{C}}$ and do so directly.

Fix $p$. The claim is

$$F^p = V^{p,k-p} \oplus F^{p+1} \tag{5.3}$$

Granting (5.3), descending induction on $p$ from the top of the filtration gives $F^p = \bigoplus_{r \geq p} V^{r,k-r}$ for every $p$, and taking $p \ll 0$, where $F^p = V_{\mathbb{C}}$, gives the required decomposition of $V_{\mathbb{C}}$.

To prove (5.3), first note $V^{p,k-p} \cap F^{p+1} = 0$. An element of the intersection lies in $F^{p+1}$, and lies in $\overline{F^{k-p}}$ because $V^{p,k-p} = F^p \cap \overline{F^{k-p}}$. Applying (5.1) at $p+1$ gives $F^{p+1} \oplus \overline{F^{k-p}} = V_{\mathbb{C}}$, so the intersection is zero. For the sum, let $v \in F^p$. By (5.1) at $p+1$ write $v = a+b$ with $a \in F^{p+1}$ and $b \in \overline{F^{k-p}}$. Then $b = v-a$ lies in $F^p$, since $v \in F^p$ and $a \in F^{p+1} \subseteq F^p$; so $b \in F^p \cap \overline{F^{k-p}} = V^{p,k-p}$, and $v = b+a$ exhibits $v$ in $V^{p,k-p} + F^{p+1}$. This proves (5.3).

*Step 3: the constructions are mutually inverse.* Step 1 computed $\overline{F^{k-p+1}} = \bigoplus_{s \leq p-1} V^{s,k-s}$ for the filtration attached to a Hodge structure, and the same substitution gives $\overline{F^q} = \bigoplus_{s \leq p} V^{s,k-s}$ when $q = k-p$. Intersecting with $F^p = \bigoplus_{r \geq p} V^{r,k-r}$ leaves exactly the single summand $V^{p,k-p}$, which is (5.2): starting from a Hodge structure, building its filtration and applying the recovery formula returns the original bigrading. Conversely, Step 2 built a Hodge structure from $F^\bullet$ and established $F^p = \bigoplus_{r \geq p} V^{r,k-r}$ along the way, so the filtration attached to that Hodge structure is $F^\bullet$ again. The two constructions are therefore mutually inverse, completing the proof.

Condition (5.1) is the whole content, and it is worth isolating what it says. A decreasing filtration on $V_{\mathbb{C}}$ is cheap: any flag of subspaces will do. What (5.1) demands is that each $F^p$ be transverse to the conjugate of its complementary step, and transversality is an open condition. This is the first sign that the structures of a given numerical type form an open subset of a space of flags rather than a closed subvariety, a theme section 5.3 takes up.

Maps between Hodge structures are constrained by the bigrading, and the constraint has a consequence strong enough to be worth proving at once.

**Definition 5.1.6: Morphism of Hodge Structures**

Let $V$ and $W$ carry pure Hodge structures of weights $k$ and $l$. A *morphism of Hodge structures* is a homomorphism $\varphi\colon V_{\mathbb{Z}} \to W_{\mathbb{Z}}$ whose complexification satisfies

$$\varphi_{\mathbb{C}}\big(V^{p,q}\big) \subseteq W^{p,q} \qquad \text{for all } p, q$$

Equivalently, $\varphi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}}$ for all $p$ (exercise 5.1.1 (2)).

A morphism between structures of different weights is necessarily zero, since the source has $V^{p,q}$ concentrated in $p+q=k$ and the target in $p+q=l$ (exercise 5.1.1 (3)). Hodge structures of all weights therefore form a category that is the direct sum of its weight-$k$ pieces, and no information is lost by fixing the weight.

**Proposition 5.1.7: Morphisms Are Strict**

Let $\varphi\colon V \to W$ be a morphism of pure Hodge structures of weight $k$. Then $\varphi$ is *strict* for the Hodge filtrations:

$$\varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^pW_{\mathbb{C}} = \varphi_{\mathbb{C}}\big(F^pV_{\mathbb{C}}\big) \qquad \text{for every } p$$

***Proof*:**
The inclusion $\varphi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq \varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^pW_{\mathbb{C}}$ is immediate from $\varphi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}}$ (definition 5.1.6).

For the reverse, let $w \in \varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^pW_{\mathbb{C}}$ and choose $v \in V_{\mathbb{C}}$ with $\varphi_{\mathbb{C}}(v) = w$. Decompose $v = \sum_{r+s=k} v^{r,s}$ into its bihomogeneous components. Since $\varphi_{\mathbb{C}}(V^{r,s}) \subseteq W^{r,s}$, the decomposition of $w$ is

$$w = \sum_{r+s=k} \varphi_{\mathbb{C}}\big(v^{r,s}\big), \qquad \varphi_{\mathbb{C}}\big(v^{r,s}\big) \in W^{r,s}$$

and this is *the* bihomogeneous decomposition of $w$, the components of a vector being unique. That $w$ lies in $F^pW_{\mathbb{C}}$ says precisely that its components with $r < p$ vanish, so $\varphi_{\mathbb{C}}\big(v^{r,s}\big) = 0$ whenever $r < p$. Set $v' := \sum_{r \geq p} v^{r,k-r}$. Then $v' \in F^pV_{\mathbb{C}}$ by definition 5.1.4, and

$$\varphi_{\mathbb{C}}(v') = \sum_{r\geq p} \varphi_{\mathbb{C}}\big(v^{r,k-r}\big) = \sum_{r+s=k} \varphi_{\mathbb{C}}\big(v^{r,s}\big) = w$$

the middle equality because the discarded terms were shown to vanish. Hence $w \in \varphi_{\mathbb{C}}(F^pV_{\mathbb{C}})$, as we sought to show.

Strictness is not a formality. For an arbitrary map of filtered vector spaces the inclusion $\varphi(F^p) \subseteq \varphi(V) \cap F^p$ is typically proper: a vector can land deep in the target filtration without any preimage lying correspondingly deep in the source (exercise 5.1.1 (4)). What rules this out here is the bigrading, which lets a preimage be truncated componentwise. The consequence is that kernels, images and quotients of morphisms of Hodge structures again carry Hodge structures of the same weight, so the weight-$k$ structures form an abelian category; without strictness the induced filtration on an image and the filtration it inherits as a subobject would differ.

Hodge structures are more than a class of objects. They form a tensor category: they can be multiplied, dualized and twisted, and the resulting bigradings are forced. The constructions are recorded here because everything downstream uses them. The Tate twist defined next is the special case $V \otimes \mathbb{Z}(n)$.

**Definition 5.1.8: Tensor Constructions on Hodge Structures**

Let $A$ be $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$ and let $V$ and $W$ be $A$-Hodge structures of weights $k$ and $l$. The *tensor product* $V \otimes W$ is the Hodge structure of weight $k+l$ on $V_A \otimes_A W_A$ with

$$(V \otimes W)^{p,q} = \bigoplus_{\substack{p_1+p_2=p \\ q_1+q_2=q}} V^{p_1,q_1} \otimes W^{p_2,q_2}$$

and the *dual* $V^\vee$ is the Hodge structure of weight $-k$ on $\mathrm{Hom}_A(V_A, A)$ with $(V^\vee)^{p,q} = (V^{-p,-q})^\vee$, where $(V^{-p,-q})^\vee \subseteq (V_\mathbb{C})^\vee$ is the annihilator of the summands $V^{r,s}$ with $(r,s) \neq (-p,-q)$. For $a, b \geq 0$ and $m \in \mathbb{Z}$ write

$$T^{a,b,m}(V) = V^{\otimes a} \otimes (V^\vee)^{\otimes b} \otimes \mathbb{Q}(m)$$

an $A$-Hodge structure of weight $k(a-b) - 2m$. The rational case is the one used throughout; the integral case is what definition 5.1.10 needs.

That the bidegrees above define a Hodge structure is a check on the conjugation symmetry: $\overline{(V \otimes W)^{p,q}} = (V \otimes W)^{q,p}$ because conjugation acts factorwise and the index set is symmetric under swapping the two coordinates, and similarly for the dual.

The classes these constructions exist to isolate are the ones that sit in the middle of the bigrading and are rational. They appear informally in section 5.4 and in the Lefschetz $(1,1)$-theorem of section 4.4; here is the definition.

**Definition 5.1.9: Hodge Classes**

Let $W$ be a rational Hodge structure of weight $2p$. A *Hodge class* of $W$ is an element of

$$\mathrm{Hdg}(W) = W_\mathbb{Q} \cap W^{p,p}$$

the intersection being taken inside $W_\mathbb{C}$. If the weight of $W$ is odd, set $\mathrm{Hdg}(W) = 0$.

Hodge classes are what chapter 8 characterizes group-theoretically and what **??** asks to be algebraic.

One further construction is needed before the examples, because it is what allows structures of different weights to be compared at all.

**Definition 5.1.10: The Tate Structure and Tate Twists**

The *Tate structure* $\mathbb{Z}(1)$ is the weight $-2$ Hodge structure on the lattice $2\pi i\,\mathbb{Z} \subset \mathbb{C}$ with $V^{-1,-1} = \mathbb{C}$ and all other summands zero. For $n \in \mathbb{Z}$ write $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$, a weight $-2n$ structure on a rank-one lattice concentrated in bidegree $(-n,-n)$.
The *Tate twist* of a weight-$k$ Hodge structure $V$ is $V(n) := V \otimes_\mathbb{Z} \mathbb{Z}(n)$, a Hodge structure of weight $k - 2n$ with

$$\big(V(n)\big)^{p,q} = V^{p+n,\,q+n}$$

The lattice of $\mathbb{Z}(1)$ is $2\pi i\,\mathbb{Z}$ rather than $\mathbb{Z}$, and the factor is not decoration. It is the period of the exponential, and it is exactly the constant that appears when the exponential sequence is used to compute Chern classes, as in proposition 4.4.1. Choosing the lattice this way makes the maps that arise geometrically into morphisms of Hodge structures on the nose rather than up to a power of $2\pi i$.

**Example 5.1.11: Twisting the Cohomology of a Surface**

Let $X$ be a compact Kähler surface, so $H^2(X,\mathbb{Z})$ carries a weight-2 structure with $h^{2,0} = h^{0,2}$ and $h^{1,1}$ as in corollary 4.3.1. Its twist $H^2(X,\mathbb{Z})(1)$ has weight 0, and its bidegrees are shifted so that the former $(1,1)$-part becomes the $(0,0)$-part.

This is the twist under which the cycle class map becomes weight-preserving. A curve $C \subset X$ has a class in $H^2(X,\mathbb{Z})$ of type $(1,1)$ (theorem 4.4.2), and after twisting it lands in the $(0,0)$-part of a weight-0 structure, which is where the class of a point sits for the trivial structure $\mathbb{Z}$. The twist is the bookkeeping that lets the comparison be stated as a map of Hodge structures at all, and **??** takes it up in general.

### Exercise 5.1.1

1. Let $V_{\mathbb{R}}$ be a real vector space. Show that giving a subspace $V^{1,0} \subseteq V_{\mathbb{C}}$ with $V^{1,0} \oplus \overline{V^{1,0}} = V_{\mathbb{C}}$ is the same as giving an endomorphism $J$ of $V_{\mathbb{R}}$ with $J^2 = -\mathrm{id}$, the correspondence sending $J$ to its $(+i)$-eigenspace in $V_{\mathbb{C}}$. Deduce that $V_{\mathbb{R}}$ must have even dimension, and that a weight-1 Hodge structure on $V_{\mathbb{Z}}$ is the same as a complex structure on $V_{\mathbb{R}}$. (Compare proposition 1.1.13, which is this statement applied fibrewise to a tangent bundle.)

2. Let $V, W$ carry pure Hodge structures of weight $k$ and let $\varphi\colon V_{\mathbb{Z}} \to W_{\mathbb{Z}}$ be a homomorphism. Show that $\varphi_{\mathbb{C}}(V^{p,q}) \subseteq W^{p,q}$ for all $p, q$ if and only if $\varphi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}}$ for all $p$. (For the converse, use that $\varphi_{\mathbb{C}}$ commutes with conjugation, since $\varphi$ is defined over $\mathbb{Z}$, together with (5.2).)

3. Let $V$ and $W$ carry pure Hodge structures of weights $k \neq l$. Show that the only morphism of Hodge structures $V \to W$ is zero. Conclude that the category of Hodge structures of all weights is the direct sum of the categories of weight-$k$ structures.

4. Strictness (proposition 5.1.7) is a property of morphisms of Hodge structures, not of filtered maps in general. Give two filtered complex vector spaces $(V, F^\bullet)$ and $(W, F^\bullet)$ and a linear map $\varphi$ with $\varphi(F^pV) \subseteq F^pW$ for all $p$ for which $\varphi(V) \cap F^1W \neq \varphi(F^1V)$. (Two-dimensional spaces suffice.) Then identify which hypothesis of proposition 5.1.7 your example violates.

5. Let $V$ carry a pure Hodge structure of weight $k$ with Hodge numbers $h^{p,q}$. Verify that $V(n)$ is a Hodge structure of weight $k - 2n$, that its Hodge numbers are $h^{p,q}(V(n)) = h^{p+n,q+n}(V)$, and that the symmetry $\overline{V^{p,q}} = V^{q,p}$ is preserved. Then compute the Hodge numbers of $H^1(E,\mathbb{Z})(1)$ for $E$ an elliptic curve, using example 4.2.2 with $n = 1$.

## 5.2 Polarizations and the Hodge-Riemann Bilinear Relations

Section 5.1 attached to a lattice a bigrading of its complexification and stopped there. Nothing in definition 5.1.1 refers to a bilinear form, and without one the structures of a given numerical type carry no positivity: there is no notion of a bounded family of them, and no constraint separating the bigradings that arise from geometry from those that do not.

Where the missing form comes from is already visible in low degree. On a compact Kähler surface the cup product on $H^2$ takes a definite sign on each piece of the Hodge decomposition: positive on a class of type $(2,0)$ paired against its conjugate, positive along the Kähler class, and negative on the primitive real $(1,1)$-classes (lemma 4.3.2). This section states the pattern those signs follow in every weight, adopts it as an axiom on a form attached to an abstract Hodge structure, and then identifies the part of the cohomology of a manifold on which the axiom holds.

The signs in lemma 4.3.2 are not arbitrary, and the alternation is governed by $p-q$. The bookkeeping device that makes this uniform across weights is an operator that multiplies the $(p,q)$-piece by $i^{p-q}$, absorbing the alternation into the form and leaving a positive quantity behind.

**Definition 5.2.1: Polarization of a Hodge Structure**

Let $V$ carry a pure Hodge structure of weight $k$. The *Weil operator* of the structure is the endomorphism $C$ of $V_{\mathbb{C}}$ acting on $V^{p,q}$ as multiplication by $i^{p-q}$.
A *polarization* of $V$ is a bilinear form $Q\colon V_{\mathbb{Q}} \times V_{\mathbb{Q}} \to \mathbb{Q}$, extended $\mathbb{C}$-bilinearly to $V_{\mathbb{C}}$, satisfying

$$Q(v,w) = (-1)^k\, Q(w,v)$$

together with the two *Hodge-Riemann bilinear relations*:

1. $Q\big(V^{p,q}, V^{p',q'}\big) = 0$ whenever $(p',q') \neq (q,p)$;
2. $i^{p-q}\, Q(v,\bar{v}) > 0$ for every nonzero $v \in V^{p,q}$.

A Hodge structure equipped with a chosen polarization is called *polarized*, and *polarizable* if some polarization exists. When $Q$ takes integer values on $V_{\mathbb{Z}}$ the polarization is called *integral*.

The two relations play different roles. The first is a vanishing condition: $Q$ pairs $V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$ and with nothing else, so the form breaks into independent blocks indexed by the unordered pairs $\{p,q\}$ with $p+q=k$. The second is where the content sits, since it converts $Q$ into a positive quantity on each block once the factor $i^{p-q}$ is inserted. The expression $Q(v,\bar{v})$ is exactly the pairing the first relation leaves alive: for $v \in V^{p,q}$ the vector $\bar{v}$ lies in $V^{q,p}$, the unique summand $Q$ does not annihilate against $V^{p,q}$.

The sign $(-1)^k$ is forced by the geometry the definition abstracts. A form built from the cup product in degree $k$ satisfies $\alpha \wedge \beta = (-1)^{k^2} \beta \wedge \alpha = (-1)^k \beta \wedge \alpha$, so it is symmetric in even weight and alternating in odd weight, and the definition records this rather than imposing it.

The first relation also has a compact reading in the language of definition 5.1.10. Condition (1) says exactly that $Q$ is a morphism of Hodge structures $Q\colon V \otimes V \to \mathbb{Z}(-k)$: the target is concentrated in bidegree $(k,k)$ and has weight $2k$, matching the weight of $V \otimes V$, and the bigrading of a tensor product is the convolution of the bigradings, so the morphism condition says $Q$ kills $V^{p,q} \otimes V^{p',q'}$ unless $p+p'=k$ and $q+q'=k$, which with $p+q=p'+q'=k$ is precisely $(p',q')=(q,p)$. Nothing more than condition (1) is captured that way. The morphism condition does not see the interchange of the two factors, so it says nothing about the $(-1)^k$-symmetry, and it is insensitive to size: the zero form is a morphism $V \otimes V \to \mathbb{Z}(-k)$ for every $V$ and polarizes nothing. The positivity condition (2) is therefore extra data of a different kind, and it is what rules out most forms.

A curve makes both conditions concrete, and there they are classical.

**Example 5.2.2: The Polarization on the $H^1$ of a Curve**

Let $C$ be a compact Riemann surface, that is, a compact Kähler manifold of complex dimension 1, and set $g = h^{1,0}(C)$. By theorem 4.2.1 the group $V_{\mathbb{Z}} = H^1(C, \mathbb{Z})$ carries a pure Hodge structure of weight 1 with

$$H^1(C, \mathbb{C}) = H^{1,0}(C) \oplus H^{0,1}(C), \qquad H^{1,0}(C) \cong H^0(C, \Omega^1_C)$$

so $V_{\mathbb{C}}$ splits into the holomorphic 1-forms and their conjugates, and $h^{1,0} = h^{0,1} = g$ by corollary 4.3.1. Take

$$Q(\alpha, \beta) = \int_C \alpha \wedge \beta$$

the cup-product form, which is integral on $H^1(C, \mathbb{Z})$ and alternating, as $(-1)^k = -1$ demands.

*The first relation.* A class in $H^{1,0}$ is represented by a holomorphic 1-form: the isomorphism $H^{1,0} \cong H^0(C, \Omega^1_C)$ above identifies it with one, and a holomorphic 1-form $\alpha$ is closed, since $\bar{\partial}\alpha = 0$ by holomorphy while $\partial\alpha$ is a $(2,0)$-form on a curve and hence zero. For $\alpha, \beta \in H^{1,0}$ the product $\alpha \wedge \beta$ is again of type $(2,0)$, so it vanishes and $Q(H^{1,0}, H^{1,0}) = 0$; conjugating gives $Q(H^{0,1}, H^{0,1}) = 0$. The surviving pairing is $H^{1,0}$ against $H^{0,1} = \overline{H^{1,0}}$, which is condition (1).

*The second relation.* Let $0 \neq \alpha \in H^{1,0}$ and write $\alpha = f\,dz$ in a holomorphic chart with coordinate $z = x + iy$. Then

$$\alpha \wedge \bar{\alpha} = |f|^2\, dz \wedge d\bar{z} = -2i\,|f|^2\, dx \wedge dy$$

so $i\,\alpha \wedge \bar{\alpha} = 2|f|^2\, dx \wedge dy$. The left side is a globally defined real 2-form on $C$, and the computation shows it is nonnegative in every chart and vanishes at a point exactly where $\alpha$ does. Hence

$$i^{1-0}\, Q(\alpha, \bar{\alpha}) = i \int_C \alpha \wedge \bar{\alpha} > 0$$

for $\alpha \neq 0$, which is condition (2) for the summand $H^{1,0}$. For $H^{0,1}$ write $\alpha = \bar{\beta}$ with $\beta \in H^{1,0}$; then $i^{0-1}Q(\bar{\beta}, \beta) = -i\int_C \bar{\beta} \wedge \beta = i\int_C \beta \wedge \bar{\beta} > 0$, using that 1-forms anticommute.

So $H^1(C, \mathbb{Z})$ with the cup product is an integrally polarized Hodge structure of weight 1. Written out in a basis of $H_1(C, \mathbb{Z})$ these two conditions are the classical Riemann bilinear relations on the period matrix of $C$; the complex-torus form of that statement belongs to [3], and section 5.4 takes up the weight-one dictionary in general.

The curve exhibits both relations but hides their structure, because in weight 1 there is only one block and only one sign. To see what the relations say in general the right move is to fold the factor $i^{p-q}$ into the form once and for all, which is what the Weil operator was introduced to do. The result is a single Hermitian form whose positivity encodes the whole alternating pattern, and a filtration-level restatement of the first relation that will matter as soon as the structure is allowed to move.

**Theorem 5.2.3: The Hodge-Riemann Relations in Arbitrary Weight**

Let $V$ carry a pure Hodge structure of weight $k$, with Hodge filtration $F^\bullet$ (definition 5.1.4) and Weil operator $C$, and let $Q$ be a bilinear form on $V_{\mathbb{Q}}$ with $Q(v,w) = (-1)^k Q(w,v)$, extended $\mathbb{C}$-bilinearly to $V_{\mathbb{C}}$.

1. Condition (1) of definition 5.2.1 holds if and only if

$$Q\big(F^p V_{\mathbb{C}},\ F^{k-p+1} V_{\mathbb{C}}\big) = 0 \qquad \text{for every } p$$

2. If condition (1) holds, then

$$h(u,v) := Q(Cu, \bar{v})$$

is a Hermitian form on $V_{\mathbb{C}}$ for which the decomposition $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$ is orthogonal.

3. Assume condition (1). Then condition (2) of definition 5.2.1 holds if and only if $h$ is positive definite, if and only if the real bilinear form $S(u,v) := Q(Cu,v)$ on $V_{\mathbb{R}}$ is symmetric and positive definite.

***Proof*:**

Throughout, $C$ is defined over $\mathbb{R}$: for $v \in V^{p,q}$ one has $\bar{v} \in V^{q,p}$, so $\overline{C\bar{v}} = \overline{i^{q-p}\,\bar{v}} = i^{p-q} v = Cv$, and $C$ therefore commutes with conjugation and preserves $V_{\mathbb{R}}$. Since $Q$ is defined over $\mathbb{Q}$, it satisfies $\overline{Q(u,v)} = Q(\bar{u},\bar{v})$.

1. Suppose condition (1) holds. Fix $p$ and take $v \in V^{r,k-r}$ with $r \geq p$ and $w \in V^{s,k-s}$ with $s \geq k-p+1$, so that $v \in F^p$ and $w \in F^{k-p+1}$ range over spanning sets of the two filtration steps. Condition (1) forces $Q(v,w) = 0$ unless $s = k-r$; but $r \geq p$ gives $k - r \leq k - p < k - p + 1 \leq s$, so $s \neq k - r$ always. Hence $Q(F^p, F^{k-p+1}) = 0$.

   Conversely, suppose $Q(F^p, F^{k-p+1}) = 0$ for every $p$, and let $r + s \neq k$. If $r + s > k$ then $s \geq k - r + 1$, so $V^{s,k-s} \subseteq F^s \subseteq F^{k-r+1}$ while $V^{r,k-r} \subseteq F^r$, and the hypothesis at $p = r$ gives $Q(V^{r,k-r}, V^{s,k-s}) = 0$. If instead $r + s < k$, put $r' = k - r$ and $s' = k - s$, so that $r' + s' = 2k - (r+s) > k$ and the case just settled gives $Q(V^{r',k-r'}, V^{s',k-s'}) = 0$. Since $\overline{V^{r,k-r}} = V^{k-r,r} = V^{r',k-r'}$ and likewise for $s$, reality of $Q$ turns this into

$$Q\big(V^{r,k-r}, V^{s,k-s}\big) = \overline{Q\big(V^{r',k-r'}, V^{s',k-s'}\big)} = 0$$

   Both cases together are condition (1). The second case used reality of $Q$ and nothing else; the filtration condition on its own constrains only the pairs with $r + s > k$.

2. Sesquilinearity is immediate: $h$ is $\mathbb{C}$-linear in $u$ because $C$ and $Q$ are, and $h(u, \lambda v) = Q(Cu, \bar{\lambda}\bar{v}) = \bar{\lambda}\, h(u,v)$.

   For orthogonality, let $u \in V^{p,q}$ and $v \in V^{p',q'}$. Then $Cu \in V^{p,q}$ and $\bar{v} \in V^{q',p'}$, so condition (1) gives $h(u,v) = 0$ unless $(q',p') = (q,p)$, that is unless $(p',q') = (p,q)$. So distinct summands are $h$-orthogonal.

   For Hermitian symmetry it therefore suffices to treat $u, v$ in the same summand $V^{p,q}$. Using reality of $Q$ and then the symmetry rule,

$$\overline{h(u,v)} = \overline{i^{p-q} Q(u,\bar{v})} = i^{q-p}\, Q(\bar{u}, v) = (-1)^k\, i^{q-p}\, Q(v, \bar{u})$$

while $h(v,u) = i^{p-q}Q(v,\bar{u})$. The two agree precisely when $(-1)^k i^{q-p} = i^{p-q}$, that is when $(-1)^k = i^{2(p-q)} = (-1)^{p-q}$. Since $p+q=k$, the integers $p-q$ and $p+q$ have the same parity, so $(-1)^{p-q} = (-1)^k$ and the identity holds. Hence $\overline{h(u,v)} = h(v,u)$ and $h$ is Hermitian.

3. For $0 \neq v \in V^{p,q}$ one has $h(v,v) = i^{p-q}Q(v,\bar{v})$, so condition (2) says exactly that $h$ is positive on each nonzero vector of each summand. By part (2) the summands are $h$-orthogonal, so a general $v = \sum_{p+q=k} v^{p,q}$ has $h(v,v) = \sum_{p+q=k} h(v^{p,q}, v^{p,q})$, and positivity on the summands is equivalent to positivity of $h$ on $V_{\mathbb{C}}$.

   It remains to compare $h$ with $S$. For $u, v \in V_{\mathbb{R}}$ one has $\bar{v} = v$, so $S(u,v) = Q(Cu,v) = h(u,v)$; this is a real number, being the value of the real form $Q$ on the real vectors $Cu$ and $v$. Hermitian symmetry of $h$ then reads $S(u,v) = \overline{S(v,u)} = S(v,u)$, so $S$ is symmetric, and $h$ positive definite gives $S(u,u) = h(u,u) > 0$ for $u \neq 0$.

   Conversely suppose $S$ is symmetric and positive definite on $V_{\mathbb{R}}$. Any $w \in V_{\mathbb{C}}$ is $w = u + iv$ with $u, v \in V_{\mathbb{R}}$, and expanding by sesquilinearity,

   $$h(w,w) = h(u,u) + h(v,v) - i\,h(u,v) + i\,h(v,u) = S(u,u) + S(v,v)$$

   the two middle terms cancelling because $h(u,v) = S(u,v) = S(v,u) = h(v,u)$. This is positive unless $u = v = 0$, so $h$ is positive definite, completing the proof.

Part (1) is the statement that will be used whenever the structure is allowed to vary. The bigraded form of the first relation refers to the subspaces $V^{p,q}$, which do not move holomorphically in a family; the filtration form refers only to $F^\bullet$, which does, and it is a closed condition on a flag. Part (3) has an immediate consequence worth recording: a polarization is nondegenerate, since $Q(u,v) = S(C^{-1}u, v)$ and $S$ is a positive definite form on $V_{\mathbb{R}}$, so no nonzero vector is $Q$-orthogonal to everything.

The operator $C$ deserves a second look in weight 1, where $C$ acts as $i$ on $V^{1,0}$ and as $-i$ on $V^{0,1}$. That is exactly the complex structure $J$ on $V_{\mathbb{R}}$ produced in exercise 5.1.1 (1) from the subspace $V^{1,0}$, so a polarization of a weight-1 structure is an alternating form $Q$ on $V_{\mathbb{Q}}$ for which $J$ is an isometry and $Q(Ju,u) > 0$ for $u \neq 0$ (exercise 5.2.1 (1)). Those are the conditions a complex torus must satisfy to be an abelian variety, and section 5.4 makes the identification.

The signs the polarization takes on $V_{\mathbb{R}}$ are now determined by the Hodge numbers alone, which is the general form of the phenomenon theorem 4.3.3 exhibited on a surface.

**Corollary 5.2.4: Alternating Definiteness and the Signature**

Let $(V, Q)$ be a polarized Hodge structure of weight $k$, with Hodge numbers $h^{p,q}$.

1. Suppose $k$ is even, so $Q$ is symmetric on $V_{\mathbb{R}}$. For each pair $p+q=k$ the subspace $V^{p,q} \oplus V^{q,p}$ (read as $V^{p,p}$ when $p=q$) is stable under conjugation, and $Q$ restricted to its real points is definite of sign $(-1)^{(p-q)/2}$. Distinct unordered pairs $\{p, q\}$ give $Q$-orthogonal subspaces, so the signature of $Q$ on $V_{\mathbb{R}}$ is
$$(b^+, b^-) = \Bigg( \sum_{\substack{p+q=k \\ p-q\equiv 0\ (4)}} h^{p,q}, \quad \sum_{\substack{p+q=k \\ p-q\equiv 2\ (4)}} h^{p,q} \Bigg)$$
2. Suppose $k$ is odd. Then $Q$ is a nondegenerate alternating form on $V_{\mathbb{R}}$, so $\dim_{\mathbb{R}} V_{\mathbb{R}}$ is even, and $S(u, v) = Q(Cu, v)$ is a positive definite symmetric form.

***Proof*:**

1. Conjugation exchanges $V^{p,q}$ and $V^{q,p}$, so $W_{p,q} := V^{p,q} \oplus V^{q,p}$ is conjugation-stable and its real points $W_{p,q} \cap V_{\mathbb{R}}$ form a real subspace of dimension $\dim_{\mathbb{C}} W_{p,q}$. By condition (1) of definition 5.2.1 the form $Q$ pairs $V^{p,q}$ only with $V^{q,p}$, so $W_{p,q}$ and $W_{p',q'}$ are $Q$-orthogonal whenever $\{p, q\} \neq \{p', q'\}$, and $V_{\mathbb{R}}$ is the $Q$-orthogonal direct sum of the $W_{p,q} \cap V_{\mathbb{R}}$ as $\{p, q\}$ runs over the unordered pairs with $p + q = k$.

   Fix such a pair with $p > q$. A real vector of $W_{p,q}$ has the form $w = v + \bar{v}$ for a unique $v \in V^{p,q}$, and $v \mapsto v + \bar{v}$ is an $\mathbb{R}$-linear isomorphism from $V^{p,q}$ (viewed as a real space of dimension $2h^{p,q}$) onto $W_{p,q} \cap V_{\mathbb{R}}$. Since $p \neq q$, condition (1) gives $Q(v, v) = 0 = Q(\bar{v}, \bar{v})$, so
$$Q(w, w) = 2\, Q(v, \bar{v}) = 2\, i^{q-p}\, h(v, v)$$
   by the definition of $h$ in theorem 5.2.3. As $k$ is even, $p-q$ is even and $i^{q-p} = (-1)^{(q-p)/2} = (-1)^{(p-q)/2}$; and $h(v, v) > 0$ for $v \neq 0$. So $Q$ is definite on $W_{p,q} \cap V_{\mathbb{R}}$ of sign $(-1)^{(p-q)/2}$, contributing $2h^{p,q}$ to $b^+$ or to $b^-$ according as $p - q \equiv 0$ or 2 modulo 4.

   For the middle pair $p = q = k/2$, a real vector $w \in V^{p,p} \cap V_{\mathbb{R}}$ satisfies $\bar{w} = w$, so $Q(w, w) = Q(w, \bar{w}) = i^0 h(w, w) = h(w, w) > 0$; this contributes $h^{k/2,k/2}$ to $b^+$, consistent with $p - q = 0 \equiv 0$ modulo 4. Summing the contributions over the unordered pairs is the same as summing $h^{p,q}$ over all ordered $(p, q)$ with $p + q = k$, since $p - q$ and $q - p$ are congruent modulo 4 to the same residue among $\{0, 2\}$. This gives the stated signature.

2. For $k$ odd, $Q(v, w) = -Q(w, v)$, so $Q$ is alternating. Part (3) of theorem 5.2.3 makes $S(u, v) = Q(Cu, v)$ symmetric and positive definite, hence nondegenerate; since $C$ is invertible and $Q(u, v) = S(C^{-1}u, v)$, the form $Q$ is nondegenerate as well. A nondegenerate alternating form on a real vector space forces the dimension to be even, as we sought to show.

The corollary is the precise sense in which positivity alternates with the bigrading. Moving one step along the row of the Hodge diamond raises $p$ by one and $q$ by one less, so $p - q$ changes by 2 and $(p - q)/2$ changes by 1: the sign of $Q$ flips at every step. In even weight the polarization is therefore definite exactly when the nonzero Hodge numbers all sit in bidegrees with $p - q \equiv 0$ modulo 4, or all in bidegrees with $p - q \equiv 2$ modulo 4; as soon as both residues occur, both $b^+$ and $b^-$ are nonzero and the form is indefinite. Several bidegrees may be occupied and the form still definite, as the weight-2 case $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 0$ shows. Either way the signature is read off the Hodge numbers without any further information about $V$. In odd weight there is no signature to compute, and the content of the corollary is the pair consisting of a symplectic form $Q$ and a compatible operator $C$ making $S$ positive definite, the data that will reappear as a period matrix.

What remains is to say where such a form comes from on a manifold. The obvious candidate, the cup product on $H^k(X, \mathbb{R})$, is not itself a polarization: on a surface it fails condition (2) whichever overall sign is chosen, because the Kähler class and a class of type $(2, 0)$ demand opposite signs (exercise 5.2.1 (5)). The repair is to cut $H^k$ down to the part on which the Kähler class acts trivially in the appropriate sense.

**Definition 5.2.5: Primitive Cohomology**

Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$, and let $L = \omega \wedge (-)$ be the Lefschetz operator of definition 3.3.1, acting on cohomology by $L[\alpha] = [\omega \wedge \alpha]$. For $0 \le k \le n$ the *primitive cohomology* in degree $k$ is

$$H^k(X, \mathbb{R})_{\mathrm{prim}} := \ker\Big(L^{n-k+1} \colon H^k(X, \mathbb{R}) \to H^{2n-k+2}(X, \mathbb{R})\Big)$$

and $H^{p,q}(X)_{\mathrm{prim}} := H^{p,q}(X) \cap H^k(X, \mathbb{C})_{\mathrm{prim}}$ for $p + q = k$. When $[\omega]$ lies in the image of $H^2(X, \mathbb{Q})$ the same kernel taken on $H^k(X, \mathbb{Q})$ defines $H^k(X, \mathbb{Q})_{\mathrm{prim}}$.

That the primitive part inherits the bigrading is the first thing to check, and it is what makes the polarization below a polarization of a Hodge structure rather than of a bare vector space.

**Proposition 5.2.6: Primitive Cohomology Is a Sub-Hodge Structure**

Let $X$ be a compact Kähler manifold of dimension $n$ with Kähler form $\omega$, and let $0 \le k \le n$. Then

$$H^k(X, \mathbb{C})_{\mathrm{prim}} = \bigoplus_{p+q=k} H^{p,q}(X)_{\mathrm{prim}}, \qquad \overline{H^{p,q}(X)_{\mathrm{prim}}} = H^{q,p}(X)_{\mathrm{prim}}$$

so the primitive cohomology carries a weight-$k$ bigrading with the symmetry required by definition 5.1.1. It is a Hodge structure in the sense of that definition as soon as it is defined over $\mathbb{Q}$, and it is so defined whenever the Kähler class is rational.

***Proof*:**
The operator $L$ is well defined on cohomology because $\omega$ is closed: for closed $\alpha$ the form $\omega \wedge \alpha$ is closed, and for $\alpha = d\beta$ it is $\pm\, d(\omega \wedge \beta)$, hence exact. It raises the bidegree by $(1, 1)$: a class in $H^{p,q}(X)$ is represented by a $d$-closed form of type $(p, q)$ (the metric-free description recorded after theorem 4.2.1), and wedging with the $(1, 1)$-form $\omega$ leaves a $d$-closed form of type $(p + 1, q + 1)$.

Consequently $L^{n-k+1}$ carries $H^{p,q}(X)$ into $H^{p+n-k+1,\, q+n-k+1}(X)$, and distinct bidegrees in the source have distinct bidegrees in the target. A class annihilated by $L^{n-k+1}$ therefore has each of its bihomogeneous components annihilated separately, which is the asserted direct sum. Since $\omega$ is a real form, $L$ commutes with conjugation, so the primitive part is conjugation-stable and $\overline{H^{p,q}_{\text{prim}}} = H^{q,p}_{\text{prim}}$.

For the last statement, a rational Kähler class makes $L^{n-k+1}$ an operator defined over $\mathbb{Q}$, so its kernel is a rational subspace, completing the proof.

Rationality of $[\omega]$ is sufficient and not necessary. Rescaling $\omega$ by a positive irrational factor multiplies $L^{n-k+1}$ by a scalar and so leaves the kernel untouched while destroying rationality of the class, and in the degrees where $L^{n-k+1}$ vanishes identically no condition on $[\omega]$ is needed at all.

Rationality of the Kähler class is the point at which the theory stops being about Kähler manifolds and starts being about projective ones. A general Kähler class lies in $H^2(X, \mathbb{R})$ and nowhere better, and then nothing forces $H^k_{\text{prim}}$ to be rational: in the degrees where $L^{n-k+1}$ is not identically zero it is in general only a real subspace with a bigrading, carrying no lattice. If instead $X \subseteq \mathbb{C}\mathrm{P}^N$ is a projective manifold and $\omega$ is the restriction of the Fubini-Study form, then $[\omega]$ is the restriction of the integral generator of $H^2(\mathbb{C}\mathrm{P}^N, \mathbb{Z})$ (proposition 3.2.6, corollary 3.2.5), hence integral, and the primitive cohomology is a genuine rational Hodge structure.

Two degrees are worth recording at once. For $k = 1$ the operator is $L^n\colon H^1 \to H^{2n+1} = 0$, so $H^1(X, \mathbb{R})_{\text{prim}} = H^1(X, \mathbb{R})$: the whole first cohomology is primitive. For $k = n$ the operator is $L\colon H^n \to H^{n+2}$, and on a surface ($n = k = 2$) this is $\alpha \mapsto [\omega \wedge \alpha]$ with values in $H^4(X, \mathbb{R}) \cong \mathbb{R}$, so

$$H^2(X, \mathbb{R})_{\text{prim}} = \left\{\alpha \in H^2(X, \mathbb{R}) \;\middle|\; \int_X \alpha \wedge \omega = 0\right\}$$

the cup-product orthogonal complement of the Kähler class. This is the subspace that appeared in Step 3 of the proof of theorem 4.3.3, where the primitive $(1,1)$-classes carried the negative-definite part of the intersection form.

On the primitive part the cup product, suitably normalized, is a polarization, and only part of the verification is available at this point. For $k \leq n$ the relevant form is

$$Q(\alpha, \beta) = (-1)^{k(k-1)/2} \int_X \alpha \wedge \beta \wedge \omega^{n-k}$$

Its symmetry type is immediate: interchanging $\alpha$ and $\beta$ costs $(-1)^{k^2} = (-1)^k$, as definition 5.2.1 requires. The first relation is a bidegree count. For $\alpha \in H^{p,q}$ and $\beta \in H^{p',q'}$ with $p + q = p' + q' = k$, the integrand has type $(p + p' + n - k,\ q + q' + n - k)$, and only a form of type $(n, n)$ has nonzero integral over $X$; this forces $p + p' = k$ and $q + q' = k$, hence $(p', q') = (q, p)$. The second relation is another matter. It rests on an identity of Weil expressing the Hodge star of a primitive form $\alpha$ of type $(p, q)$ through the Lefschetz operator.

**Theorem 5.2.7: Weil's Identity**

Let $X$ be a compact Kähler manifold of dimension $n$ and let $\alpha$ be a primitive form of type $(p, q)$ with $p + q = k \leq n$. Then

$$\star\, \alpha = (-1)^{k(k+1)/2}\, i^{p-q}\, \frac{1}{(n-k)!}\, L^{n-k} \alpha$$

The identity is a statement about the exterior algebra at a single point, where $L$ and $\Lambda$ generate an $\mathfrak{sl}_2$-action whose representation theory splits $\Lambda^\bullet T^*_x X$ into primitive pieces. That machinery is developed in chapter 6 and the proof is given there; the identity is stated here because the polarization is what needs it and nothing else in this chapter does. Applied to $\bar{\alpha}$, which is primitive of type $(q,p)$, and inserted into $\alpha \wedge \star\bar{\alpha} = |\alpha|^2 \,\mathrm{vol}$, it gives

$$\alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)!\,(-1)^{k(k+1)/2}\, i^{p-q}\, |\alpha|^2 \,\mathrm{vol}$$

The coefficient there is not positive, and for odd $k$ not even real: for a real primitive $(1,1)$-class on a surface it is $-1$, which is part (3) of lemma 4.3.2, and for $k=1$ it is $i^{-1}(n-1)!$. Positivity appears only once the two factors surrounding the integral are restored. The normalizing sign $(-1)^{k(k-1)/2}$ carried by $Q$ meets the $(-1)^{k(k+1)/2}$ above and leaves $(-1)^{k^2}$; the factor $i^{p-q}$ that condition (2) prescribes meets the $i^{p-q}$ above and leaves $(-1)^{p-q} = (-1)^k$; and $(-1)^{k^2+k} = 1$. So for a harmonic representative of a nonzero class in $H^{p,q}(X)_{\mathrm{prim}}$,

$$i^{p-q}\, Q(\alpha, \bar{\alpha}) = (n-k)! \int_X |\alpha|^2 \,\mathrm{vol} > 0$$

which is what condition (2) asks. Collecting the three verifications gives a statement worth a label, since the rest of the chapter refers back to this normalization.

**Proposition 5.2.8: The Geometric Polarization on Primitive Cohomology**

Let $X$ be a compact Kähler manifold of dimension $n$ with Kähler form $\omega$, and let $0 \le k \le n$. If $H^k(X,\mathbb{Q})_{\mathrm{prim}}$ is defined over $\mathbb{Q}$, then

$$Q(\alpha,\beta) = (-1)^{k(k-1)/2} \int_X \alpha \wedge \beta \wedge \omega^{n-k}$$

is a polarization of the weight-$k$ Hodge structure on $H^k(X,\mathbb{Q})_{\mathrm{prim}}$ in the sense of definition 5.2.1. The sign $(-1)^{k(k-1)/2}$ is not optional: without it condition (2) fails, and in weight 2 the polarizing form is the *negative* of the cup product.

***Proof Key ideas*:**
The symmetry type, the first relation and the second relation are the three verifications just carried out; only the second uses anything beyond bookkeeping, and what it uses is theorem 5.2.7, whose proof belongs to chapter 6. That is why the proof is given as key ideas rather than in full here; the general positivity statement is proved there. The two cases the present chapter settles directly are weight 1 in any dimension (exercise 5.2.1 (2)) and weight 2 on a surface, which is the case Part I already computed, completing the proof.

**Example 5.2.9: The Polarized Weight-Two Structure of a Surface**
Let $X$ be a compact Kähler surface with Kähler form $\omega$, so $n = k = 2$ and $\omega^{n-k} = 1$. The normalizing sign is $(-1)^{k(k-1)/2} = (-1)^1$, so the candidate polarization on $H^2(X,\mathbb{R})_{\mathrm{prim}}$ is minus the cup form,

$$Q(\alpha,\beta) = -\int_X \alpha \wedge \beta$$

which is symmetric, as $(-1)^k = 1$ demands.

*The decomposition.* By $\int_X \omega \wedge \omega > 0$ (lemma 4.3.2, part (2)) the Kähler class spans a line on which the cup form is nonzero, so

$$H^2(X,\mathbb{R}) = \mathbb{R}[\omega] \oplus H^2(X,\mathbb{R})_{\mathrm{prim}}$$

the primitive part being the cup-orthogonal complement described above. A class of type $(2,0)$ is automatically primitive, since $\omega \wedge \alpha$ is then of type $(3,1)$ and vanishes on a surface, and the same holds in type $(0,2)$; the primitive $(1,1)$-classes are the space $P^{1,1}(X) = \{\beta \in H^{1,1} \big| \beta \wedge \omega = 0\}$ of lemma 4.3.2. On the real $(1,1)$-classes the map $\beta \mapsto \int_X \beta \wedge \omega$ is onto $\mathbb{R}$, because $[\omega]$ itself has positive image, so $\dim_{\mathbb{R}}\big(P^{1,1}(X) \cap H^2(X,\mathbb{R})\big) = h^{1,1} - 1$ and the primitive Hodge numbers are

$$h^{2,0}_{\mathrm{prim}} = h^{0,2}_{\mathrm{prim}} = h^{2,0}, \qquad h^{1,1}_{\mathrm{prim}} = h^{1,1} - 1$$

*The two relations.* The first is the bidegree count already given, and on a surface it says only that $H^{2,0}$ pairs with $H^{0,2}$ and $H^{1,1}$ with itself. For the second, take $0 \neq \alpha \in H^{2,0}$; then $p - q = 2$ and

$$i^2\, Q(\alpha, \bar\alpha) = (-1) \cdot \Big( - \int_X \alpha \wedge \bar\alpha \Big) = \int_X \alpha \wedge \bar\alpha > 0$$

by part (1) of lemma 4.3.2. For $\alpha \in H^{0,2}$ write $\alpha = \bar\beta$ with $\beta \in H^{2,0}$; then $p - q = -2$, $i^{-2} = -1$, and $\int_X \bar\beta \wedge \beta = \int_X \beta \wedge \bar\beta$ since 2-forms commute, so the same computation applies. For $0 \neq \beta \in P^{1,1}(X)$ one has $p - q = 0$. The space $P^{1,1}(X)$ is stable under conjugation, since $H^{1,1}$ is and $\omega$ is real, so $\beta = \beta_1 + i\beta_2$ with $\beta_1 = \frac{1}{2}(\beta + \bar\beta)$ and $\beta_2 = \frac{1}{2i}(\beta - \bar\beta)$ real primitive $(1,1)$-classes, and

$$\beta \wedge \bar\beta = \beta_1 \wedge \beta_1 + \beta_2 \wedge \beta_2$$

the cross terms cancelling because 2-forms commute. Part (3) of lemma 4.3.2 makes each summand integrate to a nonpositive number, strictly negative unless the corresponding $\beta_j$ vanishes, so the total is negative unless $\beta = 0$. Hence $Q(\beta, \bar\beta) = -\int_X \beta \wedge \bar\beta > 0$. So $Q$ polarizes $H^2(X,\mathbb{R})_{\mathrm{prim}}$, and integrally so whenever $[\omega]$ is an integral class.

*Recovering the index theorem.* With the polarization in hand, corollary 5.2.4 computes the signature of $Q$ on the primitive part without any further geometry. The residues are $p - q = \pm 2 \equiv 2$ for the types $(2,0)$ and $(0,2)$, and $p - q = 0$ for type $(1,1)$, so

$$(b^+, b^-)\big(Q\big|_{H^2_{\mathrm{prim}}}\big) = \big(h^{1,1} - 1,\ 2h^{2,0}\big)$$

Since $Q$ is minus the cup form, the latter has signature $(2h^{2,0}, h^{1,1} - 1)$ on the primitive part, and restoring the positive line $\mathbb{R}[\omega]$ gives signature $(2h^{2,0} + 1,\ h^{1,1} - 1)$ on all of $H^2(X,\mathbb{R})$. This is exactly theorem 4.3.3, now obtained as the weight-2 instance of a statement about polarized Hodge structures rather than as a computation with the Hodge star on a four-manifold.

The example shows what the abstraction gave. The signature in theorem 4.3.3 came from lemma 4.3.2, three signs computed in a unitary coframe on a four-manifold, and every step of that computation was tied to the dimension. Corollary 5.2.4 delivers the same statement from the axioms of definition 5.2.1 in every weight at once, with the Hodge numbers as the only input. What it does not deliver is the verification that the axioms hold; that is where the geometry sits, and on a surface it was lemma 4.3.2 that supplied it.

## Exercise 5.2.1

1. Let $V$ carry a pure Hodge structure of weight $k$ with Weil operator $C$.
   1. Show that $C^2 = (-1)^k \operatorname{id}$, and that $C$ preserves $V_{\mathbb{R}}$.
   2. For $k = 1$, show that the restriction of $C$ to $V_{\mathbb{R}}$ is the complex structure $J$ attached to $V^{1,0}$ in exercise 5.1.1 (1).
   3. Deduce that a polarization of a weight-1 Hodge structure is the same thing as an alternating form $Q$ on $V_{\mathbb{Q}}$ satisfying $Q(Ju, Jv) = Q(u, v)$ for all $u, v \in V_{\mathbb{R}}$ and $Q(Ju, u) > 0$ for all $u \neq 0$.
2. Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$. Show that $H^1(X, \mathbb{R})$ is entirely primitive, and that

$$Q(\alpha, \beta) = \int_X \alpha \wedge \beta \wedge \omega^{n-1}$$

   satisfies both Hodge-Riemann bilinear relations on it. (For the second relation, work at a point in a unitary coframe $\varphi^1, \ldots, \varphi^n$ of $(1,0)$-forms with $\omega = i \sum_j \varphi^j \wedge \bar{\varphi}^j$, as in the proof of lemma 4.3.2, write $\alpha = \sum_j a_j \varphi^j$, and expand $\omega^{n-1}$.)
3. Let $(V, Q)$ be a polarized Hodge structure of weight $k$, and let $W \subseteq V$ be a sub-Hodge structure, meaning a $\mathbb{Q}$-subspace $W_{\mathbb{Q}} \subseteq V_{\mathbb{Q}}$ with $W_{\mathbb{C}} = \bigoplus_{p+q=k} \big(W_{\mathbb{C}} \cap V^{p,q}\big)$. Show that $Q|_W$ is a polarization of $W$, that $W^{\perp} = \{v \in V_{\mathbb{Q}} \big| Q(v, W_{\mathbb{Q}}) = 0\}$ is again a sub-Hodge structure, and that $V = W \oplus W^{\perp}$ as Hodge structures.
4. Let $X$ be a K3 surface, so $h^{1,0} = 0$, $h^{2,0} = 1$ and $h^{1,1} = 20$ as in exercise 4.2.1 (1). Compute $\dim_{\mathbb{R}} H^2(X, \mathbb{R})_{\text{prim}}$ and its primitive Hodge numbers, then use corollary 5.2.4 to find the signature of the polarization $Q$ of example 5.2.9. Check that the answer reproduces the signature of the cup form recorded in example 4.3.4.
5. Let $X$ be a compact Kähler surface with $h^{2,0} \geq 1$. Show that neither the cup form $\alpha, \beta \mapsto \int_X \alpha \wedge \beta$ nor its negative is a polarization of the weight-2 Hodge structure $H^2(X, \mathbb{R})$, by exhibiting for each choice of sign a class violating condition (2) of definition 5.2.1. Explain what goes right on $H^2(X, \mathbb{R})_{\text{prim}}$ that fails on $H^2(X, \mathbb{R})$.
6. Let $E = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\operatorname{Im} \tau > 0$, and let $a, b$ be the cycles in $H_1(E, \mathbb{Z})$ traced by the segments from 0 to 1 and from 0 to $\tau$. Let $u, v \in H^1(E, \mathbb{Z})$ be the dual basis, normalized by the complex orientation so that $\int_E u \wedge v = 1$. Using that $H^{1,0}(E) = \mathbb{C}\,[dz]$ (example 4.2.2) and that $dz$ has periods $\int_a dz = 1$ and $\int_b dz = \tau$, express $[dz]$ in terms of $u$ and $v$ and verify the second Hodge-Riemann relation for the cup form directly, computing $i\,Q([dz], \overline{[dz]})$ in closed form.

## 5.3 The Period Domain and Its Compact Dual

Fix the discrete invariants of a polarized Hodge structure: the lattice, the weight, the polarizing form and the Hodge numbers. What is left to vary is the filtration, and a filtration of a fixed vector space with fixed jump dimensions is a point of a flag manifold. This section carries out that identification and reads off the geometry it produces. The two conditions of definition 5.2.1 sit differently inside the flag manifold. The first is a system of bilinear equations, hence a closed condition; the second is a positivity, hence an open one. Imposing the first alone cuts out a compact complex manifold, the compact dual; imposing the second as well cuts out an open subset of it, the period domain.

Both spaces are homogeneous, the compact dual under the complex automorphism group of the polarizing form and the domain under its real points, and the isotropy group of the real action is compact. The metric geometry this suggests, and the distribution along which a family of varieties is constrained to move, belong to chapter 7 and chapter 8; here the task is the construction, the homogeneity that supplies the manifold structure, and the two examples the rest of the chapter uses.

Throughout, fix a finitely generated free abelian group $V_{\mathbb{Z}}$ of rank $m$, an integer $k$, a nondegenerate bilinear form $Q$ on $V_{\mathbb{Q}}$ satisfying $Q(u,v) = (-1)^k Q(v,u)$ and extended $\mathbb{C}$-bilinearly to $V_{\mathbb{C}}$, and integers $h^{p,q} \geq 0$ for $p+q=k$ with

$$h^{p,q} = h^{q,p}, \qquad \sum_{p+q=k} h^{p,q} = m$$

These are the data of definition 5.2.1 stripped of the filtration. Write $F^p$ for $F^pV_{\mathbb{C}}$, and set

$$f^p = \sum_{r \geq p} h^{r,\,k-r}$$

for the dimension a Hodge filtration with these Hodge numbers must have at level $p$. Substituting $s = k-r$ and using $h^{r,k-r} = h^{k-r,r}$ turns $f^{k-p+1} = \sum_{r \geq k-p+1} h^{r,k-r}$ into $\sum_{s \leq p-1} h^{s,k-s}$, so the two complementary levels have complementary dimensions:

$$f^p + f^{k-p+1} = m \qquad \text{for every } p \tag{5.4}$$

Finally, assume throughout that at least one polarized Hodge structure with these invariants exists. This is automatic in the geometric case, where the data is read off a compact Kähler manifold through theorem 4.2.1 and theorem 5.2.3, and it is what keeps the spaces below from being empty.

The first of the two conditions in definition 5.2.1 is the orthogonality $Q(V^{p,q}, V^{r,s}) = 0$ except when $(r,s) = (q,p)$. It can be read off the filtration alone:

$$Q\big(F^p,\, F^{k-p+1}\big) = 0 \qquad \text{for every } p \tag{5.5}$$

On a Hodge structure the two versions agree, which is part (1) of theorem 5.2.3. The filtration form has the advantage of making sense for a filtration that is not yet known to come from a bigrading, and that is why it, rather than the bigraded orthogonality, is what cuts out the space below.

Let Fl denote the manifold of decreasing filtrations $F^\bullet$ of $V_{\mathbb{C}}$ by complex subspaces with $\dim_{\mathbb{C}} F^p = f^p$. It is a compact complex manifold on which $\mathrm{GL}(V_{\mathbb{C}})$ acts transitively and holomorphically, and the orbit map admits local holomorphic sections [32]. Cutting it with (5.5) is the first of the two steps.

**Definition 5.3.1: The Compact Dual**

The *compact dual* attached to the data $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$ is

$$\check{D} = \big\{F^\bullet \in \mathrm{Fl} \big| Q(F^p, F^{k-p+1}) = 0 \text{ for every } p\big\}$$

the set of filtrations of the prescribed type satisfying the first bilinear relation (5.5).

By (5.4) and the nondegeneracy of $Q$, the subspace $(F^p)^\perp$ has dimension $m - f^p = f^{k-p+1}$, so the condition $Q(F^p, F^{k-p+1}) = 0$, which says $F^{k-p+1} \subseteq (F^p)^\perp$, is an equality of subspaces:

$$\check{D} = \big\{F^\bullet \in \mathrm{Fl} \big| (F^p)^\perp = F^{k-p+1} \text{ for every } p\big\}$$

A point of $\check{D}$ is a self-dual flag. The defining condition is closed: in local frames for the tautological subspaces it is the vanishing of the matrix of pairings $Q(v_i(F), w_j(F))$, whose entries depend continuously on $F$. So $\check{D}$ is a closed subset of the compact Fl, hence compact, which is what the name records. It is also cut out by algebraic equations, so once it is known to be a complex submanifold it is a smooth projective variety by GAGA ([4]); smoothness comes from proposition 5.3.5 below.

Nothing in $\check{D}$ refers to the lattice. Conjugation on $V_{\mathbb{C}}$ has not been used, and a self-dual flag need not be transverse to its own conjugate, so a point of $\check{D}$ need not define a Hodge structure at all. The second condition repairs this and does so with a strict inequality.

**Definition 5.3.2: The Period Domain**

The *period domain* attached to the data $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$ is the subset

$$D = \left\{ F^\bullet \in \check{D} \;\middle|\; i^{\,p-q}\,Q(v,\bar{v}) > 0 \text{ for all } p+q=k \text{ and all } v \in \left(F^p \cap \overline{F^q}\right) \setminus \{0\} \right\}$$

of the compact dual.

The positivity is imposed on the subspaces $F^p \cap \overline{F^q}$, which are defined for any flag; no bigrading is presupposed. That a point of $D$ nevertheless *is* a polarized Hodge structure, with the prescribed Hodge numbers, is the first half of proposition 5.3.4. Granting it for the moment, $D$ is exactly the set of polarized Hodge structures on $V_{\mathbb{Z}}$ of weight $k$ with Hodge numbers $h^{p,q}$ and polarization $Q$, presented through their Hodge filtrations rather than their bigradings. This is the sense in which $D$ classifies: its points are the objects, not isomorphism classes of them.

One piece of notation packages the positivity condition. Let $F$ be any flag satisfying the transversality (5.1), so that by theorem 5.1.5 the subspaces $V^{p,q} = F^p \cap \overline{F^q}$ form a Hodge structure of weight $k$, and let $v \mapsto v^{p,q}$ denote the associated projections. Define

$$H_F(u,v) = \sum_{p+q=k} i^{\,p-q}\,Q\big(u^{p,q},\,\overline{v^{p,q}}\big) \tag{5.6}$$

This form is Hermitian. Conjugating a term and using that $Q$ is defined over $\mathbb{R}$, then that $Q(a,b) = (-1)^k Q(b,a)$,

$$\overline{i^{\,p-q}Q\big(u^{p,q},\overline{v^{p,q}}\big)} = i^{\,q-p}Q\big(\overline{u^{p,q}},v^{p,q}\big) = (-1)^k i^{\,q-p}Q\big(v^{p,q},\overline{u^{p,q}}\big)$$

and $(-1)^k i^{\,q-p} = i^{\,2k+q-p} = i^{\,p-q}$, the two exponents differing by $2k + 2(q-p) = 4q$ when $p+q=k$. So $\overline{H_F(u,v)} = H_F(v,u)$, and in particular $H_F(v,v)$ is real. Comparing (5.6) with definition 5.3.2, a flag $F \in \check{D}$ satisfying (5.1) lies in $D$ precisely when $H_F$ is positive definite: the sum in (5.6) evaluated at $u = v$ is a sum of the quantities $i^{p-q}Q(v^{p,q},\overline{v^{p,q}})$, each of which the positivity condition makes strictly positive unless the component vanishes.

**Example 5.3.3: The Siegel Upper Half Space**

Take $k=1$ and $h^{1,0} = h^{0,1} = g$, so $m = 2g$ and $Q$ is a nondegenerate alternating form on $V_{\mathbb{Q}}$. Here $f^1 = g$ and $f^0 = 2g$, so Fl is the Grassmannian of $g$-dimensional subspaces of $V_{\mathbb{C}}$ and a flag is a single subspace $L = F^1$. The relation (5.5) at $p=1$ reads $Q(L,L) = 0$, so

$$\check{D} = \mathrm{LG} = \{\, L \in \mathrm{Fl} \,|\, L \text{ is Lagrangian} \,\}$$

the Lagrangian Grassmannian, and the positivity condition at $p=1$ reads $i\,Q(v,\bar{v}) > 0$ for $0 \neq v \in L$ (the condition at $p=0$ is its conjugate and adds nothing).

Choose a symplectic basis $e_1, \dots, e_g, f_1, \dots, f_g$ of $V_{\mathbb{Q}}$, which exists because $Q$ is a nondegenerate alternating form on a $\mathbb{Q}$-vector space, so $Q(e_a, f_b) = \delta_{ab}$ and $Q(e_a, e_b) = Q(f_a, f_b) = 0$; it is in particular a $\mathbb{C}$-basis of $V_{\mathbb{C}}$, and in coordinates $Q(x, y) = x^T J_0\, y$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$. Present $L$ as the column span of a $2g \times g$ matrix $\left(\begin{smallmatrix} \Lambda \\ M \end{smallmatrix}\right)$ of rank $g$. Then $Q(v, \bar{v}) = c^T \big(\Lambda^T \bar{M} - M^T \bar{\Lambda}\big)\bar{c}$ for $v$ the column combination with coefficient vector $c$, and the Lagrangian condition is $\Lambda^T M = M^T \Lambda$.

The block $\Lambda$ is invertible on $D$. Rewriting the two terms as $(\Lambda c)^T \bar{M}\bar{c}$ and $(Mc)^T \overline{\Lambda c}$ shows that each carries a factor $\Lambda c$ or its conjugate, so $\Lambda c = 0$ forces $Q(v, \bar{v}) = 0$; and $v \neq 0$ for $c \neq 0$ because the columns are independent, contradicting positivity. Normalizing on the right by $\Lambda^{-1}$ replaces the presenting matrix by $\left(\begin{smallmatrix} I \\ Z \end{smallmatrix}\right)$ with $Z = M\Lambda^{-1}$, which changes no subspace. In this normalization the Lagrangian condition becomes $Z = Z^T$ and the positivity becomes

$$i\, Q(v, \bar{v}) = i\, c^T \big(\bar{Z} - Z\big)\bar{c} = 2\, c^T \big(\operatorname{Im} Z\big)\bar{c} > 0$$

for all $c \neq 0$. Writing $d = \bar{c}$ turns the last expression into $2\, d^* (\operatorname{Im} Z) d$, so the condition says that the real symmetric matrix $\operatorname{Im} Z$ is positive definite. Hence

$$D \cong \mathfrak{H}_g = \big\{ Z \in \mathbb{C}^{g \times g} \big| Z = Z^T,\ \operatorname{Im} Z > 0 \big\}$$

the Siegel upper half space, of complex dimension $g(g+1)/2$; for $g = 1$ it is the upper half plane (exercise 5.3.1 (2)). The polarized weight-one structure on $H^1$ of a compact Riemann surface (example 5.2.2) is a point of it, and its coordinates $Z$ are the classical period matrix. The identification of $\mathfrak{H}_g$ with the parameter space for principally polarized abelian varieties belongs to [3]; the Hodge-theoretic half of that dictionary is section 5.4.

The example already exhibits the shape of the general situation: a compact object, the Lagrangian Grassmannian, containing an open piece cut out by a positivity. The next proposition establishes that shape in general, and simultaneously discharges the claim that a point of $D$ is a Hodge structure.

**Proposition 5.3.4: The Period Domain Is Open in the Compact Dual**

With the data and notation above:

1. Every $F \in D$ satisfies the transversality condition (5.1). The bigrading $V^{p,q} = F^p \cap \overline{F^q}$ it therefore defines is a Hodge structure of weight $k$ with Hodge numbers $h^{p,q}$, polarized by $Q$; conversely the Hodge filtration of any such polarized structure is a point of $D$.
2. $D$ is open in $\check{D}$.

***Proof*:**

1. Let $F \in D$ and fix $p$. By (5.4) the dimensions of $F^p$ and $\overline{F^{k-p+1}}$ add to $m$, so (5.1) holds as soon as $F^p \cap \overline{F^{k-p+1}} = 0$. Let $v$ lie in that intersection. Since $F^{k-p+1} \subseteq F^{k-p}$, the vector $v$ lies in $F^p \cap \overline{F^{k-p}}$, so the positivity condition of definition 5.3.2 applies to it. On the other hand $v \in F^p$ and $\bar{v} \in F^{k-p+1}$, so $Q(v, \bar{v}) = 0$ by (5.5). The positivity condition then forces $v = 0$, which gives (5.1).

   By theorem 5.1.5 the subspaces $V^{p,q} = F^p \cap \overline{F^q}$ now form a Hodge structure of weight $k$ with $F^p = \bigoplus_{r \geq p} V^{r,k-r}$; comparing dimensions, $\dim_{\mathbb{C}} V^{p,k-p} = f^p - f^{p+1} = h^{p,k-p}$, so the Hodge numbers are the prescribed ones. The bigraded orthogonality required by definition 5.2.1 follows from (5.5) by the equivalence established before definition 5.3.1, and the positivity required there is the defining condition of $D$. So $Q$ polarizes the structure. Conversely, the Hodge filtration of a polarized structure with these invariants has $\dim F^p = f^p$, satisfies (5.5) by the same equivalence, and satisfies the positivity because $F^p \cap \overline{F^q} = V^{p,q}$ there; so it is a point of $D$.

2. Write $U \subseteq \mathrm{Fl}$ for the set of flags satisfying (5.1) and $U^+ \subseteq U$ for the set of flags in $U$ at which $H_F$ is positive definite. By part (1), $D = \check{D} \cap U \cap U^+$. It suffices to prove that $U$ is open in Fl and that $U^+$ is open in $U$.

   Fix $F_0 \in U$. Since $\mathrm{GL}(V_{\mathbb{C}})$ acts transitively and holomorphically on Fl with local holomorphic sections of the orbit map [32], there are a neighbourhood $W$ of $F_0$ and a holomorphic $\gamma\colon W \to \mathrm{GL}(V_{\mathbb{C}})$ with $\gamma(F_0) = \mathrm{id}$ and $\gamma(F)F_0^r = F^r$ for every $r$ and every $F \in W$. Fix $p$, a basis $a_1, \dots, a_{f^p}$ of $F_0^p$ and a basis $b_1, \dots, b_{f^{k-p+1}}$ of $F_0^{k-p+1}$, and let $M_p(F)$ be the $m \times m$ matrix whose columns are the vectors $\gamma(F)a_i$ followed by the vectors $\overline{\gamma(F)b_j}$. Its columns span $F^p$ and $\overline{F^{k-p+1}}$ respectively, and by (5.4) there are exactly $m$ of them, so (5.1) holds at $p$ if and only if $\det M_p(F) \neq 0$. The entries of $M_p$ are continuous in $F$, and $\det M_p(F_0) \neq 0$; only finitely many $p$ impose a nonvacuous condition, so all the determinants are nonzero on a neighbourhood of $F_0$. Hence $U$ is open.

   On $U$ let $A_p(F) = M_p(F)\, E_p\, M_p(F)^{-1}$, where $E_p$ is the diagonal idempotent selecting the first $f^p$ coordinates. Then $A_p(F)$ is the projection onto $F^p$ with kernel $\overline{F^{k-p+1}}$, and by Cramer's rule it depends continuously on $F$. The computation in Step 1 of theorem 5.1.5 identifies $F^p = \bigoplus_{r \geq p} V^{r,k-r}$ and $\overline{F^{k-p+1}} = \bigoplus_{r \leq p-1} V^{r,k-r}$, so $A_p(F)$ is the projection onto the summands with first index at least $p$, and therefore $A_p(F) - A_{p+1}(F)$ is the projection $v \mapsto v^{p,k-p}$. Substituting into (5.6) exhibits the matrix of $H_F$ as a continuous function of $F$ on $U$.

   Fix now $F_1 \in U^+$ and an auxiliary Hermitian inner product on $V_{\mathbb{C}}$ with unit sphere $S$, which is compact. The function $(F, v) \mapsto H_F(v, v)$ is continuous on $U \times S$ and strictly positive on $\{F_1\} \times S$, so it has a positive minimum $c$ there. For each $v \in S$ continuity supplies a product neighbourhood $W_v \times S_v$ of $(F_1, v)$ on which $H_F(w, w) > c/2$; finitely many $S_v$ cover $S$, and on the intersection of the corresponding $W_v$ the form $H_F$ is positive on all of $S$, hence positive definite. So $U^+$ is open in $U$, and $D = \check{D} \cap U \cap U^+$ is open in $\check{D}$, as we sought to show.

The two halves of definition 5.2.1 have now been separated by their topological character. The first relation is a closed condition and produces a compact object; the second is open and produces a domain inside it. The difference is visible already in example 5.3.3: the Lagrangian Grassmannian is compact, while $\mathfrak{H}_g$ is connected and carries the nonconstant holomorphic functions $Z \mapsto Z_{ab}$, which a connected compact complex manifold does not admit (proposition 1.1.11), so it is not compact. In general $D$, being open in $\check{D}$, is compact only when it is a union of connected components of $\check{D}$. What is still missing is that $\check{D}$ is a manifold at all, so that "open subset" upgrades to "open complex submanifold". That comes from a group action.

**Proposition 5.3.5: Homogeneity of the Domain and Its Dual**

Let $G_{\mathbb{C}} = \{g \in \mathrm{GL}(V_{\mathbb{C}}) \big| Q(gu, gv) = Q(u, v) \text{ for all } u, v\}$ and let $G_{\mathbb{R}} = G_{\mathbb{C}} \cap \mathrm{GL}(V_{\mathbb{R}})$ be its subgroup of real points. Fix $F_o \in D$ and let $P = \{g \in G_{\mathbb{C}} \big| gF_o^p = F_o^p \text{ for all } p\}$ be its stabilizer. Then $G_{\mathbb{C}}$ acts transitively on $\check{D}$ and the orbit map induces a biholomorphism $G_{\mathbb{C}}/P \to \check{D}$, so that $\check{D}$ is a compact complex manifold and $D$ is an open complex submanifold of it. The subgroup $G_{\mathbb{R}}$ acts transitively on $D$, so that $D \cong G_{\mathbb{R}}/(G_{\mathbb{R}} \cap P)$, and the isotropy group $G_{\mathbb{R}} \cap P$ is compact.

***Proof*:**
Write $\varepsilon = (-1)^k$, so $Q(u, v) = \varepsilon Q(v, u)$, and call a subspace $N \subseteq V_{\mathbb{C}}$ isotropic if $Q(N, N) = 0$.

*Step 1: isotropic complements.* Let $N \subseteq V_{\mathbb{C}}$ be isotropic of dimension $d$. There is an isotropic subspace $N'$ of dimension $d$ such that $Q$ restricts to a perfect pairing between $N$ and $N'$ and $V_{\mathbb{C}} = (N \oplus N') \oplus (N \oplus N')^{\perp}$.

The linear map $V_{\mathbb{C}} \to N^*$ sending $w$ to $Q(-, w)|_N$ has kernel $N^{\perp}$, of dimension $m - d$, so it is surjective. Choosing a basis $n_1, \ldots, n_d$ of $N$, pick $u_1, \ldots, u_d$ with $Q(n_a, u_b) = \delta_{ab}$ and set $B_{ab} = Q(u_a, u_b)$, so that $B_{ba} = \varepsilon B_{ab}$. Put

$$u'_b = u_b - \frac{\varepsilon}{2} \sum_l B_{lb}\, n_l$$

Then $Q(n_a, u'_b) = \delta_{ab}$ because $N$ is isotropic, and expanding $Q(u'_a, u'_b)$ into $Q(u_a, u_b)$ together with the two cross terms $Q(u_a, u'_b - u_b)$ and $Q(u'_a - u_a, u_b)$, each evaluated from $Q(n_a, u_b) = \delta_{ab}$,

$$Q(u'_a, u'_b) = B_{ab} + \varepsilon\Big(-\frac{\varepsilon}{2} B_{ab}\Big) + \Big(-\frac{\varepsilon}{2} B_{ba}\Big) = B_{ab} - \tfrac{1}{2} B_{ab} - \tfrac{1}{2} B_{ab} = 0$$

using $\varepsilon^2 = 1$ and $B_{ba} = \varepsilon B_{ab}$. Let $N' = \mathrm{span}(u'_1, \ldots, u'_d)$; it is isotropic by the display, and $Q(n_a, u'_b) = \delta_{ab}$ makes the pairing between $N$ and $N'$ perfect. The Gram matrix of $Q$ on $N \oplus N'$ in the bases $(n_a)$, $(u'_b)$ is $\left(\begin{smallmatrix} 0 & I \\ \varepsilon I & 0 \end{smallmatrix}\right)$, which is invertible; so the $2d$ vectors are independent and $Q$ is nondegenerate on their span, whence $V_{\mathbb{C}}$ is the direct sum of that span and its perpendicular.

*Step 2: adapted decompositions.* Every $F \in \check{D}$ admits a decomposition $V_{\mathbb{C}} = \bigoplus_p A^p$ with $F^p = \bigoplus_{r \geq p} A^r$ and $Q(A^p, A^r) = 0$ whenever $p + r \neq k$.

Induct on the number of levels at which the flag jumps, the statement being understood for a self-dual flag in any nondegenerate $\varepsilon$-symmetric space. The base case is the zero space, where the flag jumps at no level and the decomposition with every $A^p = 0$ serves. So assume the space is nonzero and let $p_0$ be the largest $p$ with $F^p \neq 0$. Self-duality at $p_0 + 1$ gives $F^{k-p_0} = (F^{p_0+1})^{\perp} = V_{\mathbb{C}}$. If $p_0 < k - p_0$ then $F^{k-p_0} = V_{\mathbb{C}}$ is a nonzero level above $p_0$, contradicting maximality; so $p_0 \geq k - p_0$. When $p_0 = k - p_0$ the flag jumps once, $F^{p_0} = V_{\mathbb{C}}$, and $A^{p_0} = V_{\mathbb{C}}$ works, the condition $p_0 + p_0 = k$ imposing nothing.

Suppose $p_0 > k - p_0$, so $p_0 \geq k - p_0 + 1$ and $N := F^{p_0} \subseteq F^{k-p_0+1}$ is isotropic by (5.5). Let $d = \dim N = f^{p_0}$, take $N'$ as in Step 1, and set $W = (N \oplus N')^{\perp}$, a nondegenerate space of dimension $m - 2d$. Put $F_1^p = F^p \cap W$. For $k - p_0 < p \leq p_0$ one has $F^p \subseteq F^{k-p_0+1} = N^{\perp} = N \oplus W$ and $F^p \supseteq N$, so $F^p = N \oplus F_1^p$: decomposing $v \in F^p$ along $N \oplus W$, the $N$-component already lies in $F^p$, hence so does the $W$-component. For $p \le k - p_0$ one has $F^p = V_{\mathbb{C}}$ and $F_1^p = W$, while $F_1^p = 0$ for $p \ge p_0$ because $F^{p_0} = N$ meets $W$ trivially. Counting, $\dim F_1^{k-p_0+1} = f^{k-p_0+1} - d = (m-d) - d = \dim W$, so $F_1$ equals $W$ for every $p \le k - p_0 + 1$: it jumps at strictly fewer levels than $F$. It is also self-dual in $W$. For $k - p_0 < p \le p_0$, intersecting $(F^p)^\perp = F^{k-p+1} = N \oplus F_1^{k-p+1}$ with $W$ gives $F_1^{k-p+1}$, since $N \cap W = 0$, while the same intersection computes as

$$(F^p)^\perp \cap W = \left(N \oplus F_1^p\right)^\perp \cap W = N^\perp \cap (F_1^p)^\perp \cap W = \{w \in W \,|\, Q(w, F_1^p) = 0\}$$

using $W \subseteq N^\perp$; outside that range of $p$ both sides are 0 or $W$ and the claim is immediate. By induction $W = \bigoplus_p A_1^p$ adapted to $F_1$; setting $A^{p_0} = N$, $A^{k-p_0} = N'$ and $A^p = A_1^p$ for $k - p_0 < p < p_0$ gives the required decomposition of $V_{\mathbb{C}}$, since $A_1^{p_0}$ and $A_1^{k-p_0}$ both vanish by the shape of $F_1$, and since $N \oplus N'$ is $Q$-orthogonal to $W$ with $Q(N,N) = Q(N',N') = 0$.

*Step 3: transitivity on the compact dual.* Let $F, F' \in \check{D}$ with adapted decompositions $\{A^p\}$, $\{A'^p\}$ from Step 2. Since $Q$ is nondegenerate and $A^p$ pairs to zero with every summand but $A^{k-p}$, the induced pairing $A^p \times A^{k-p} \to \mathbb{C}$ is perfect, and likewise for the primed decomposition; in particular $\dim A^p = \dim A^{k-p}$, and both equal $f^p - f^{p+1}$. For each $p$ with $2p > k$ choose any linear isomorphism $\varphi_p \colon A^p \to A'^p$ and let $\varphi_{k-p} \colon A^{k-p} \to A'^{k-p}$ be the unique map with $Q(\varphi_p a, \varphi_{k-p} b) = Q(a,b)$, which exists and is an isomorphism because both pairings are perfect. If $k$ is even, $\varepsilon = 1$ and $Q$ restricts to nondegenerate symmetric forms on $A^{k/2}$ and $A'^{k/2}$ of equal rank; over $\mathbb{C}$ any two such are isometric, so choose an isometry $\varphi_{k/2}$. Let $g = \bigoplus_p \varphi_p$. It carries $F$ to $F'$, and it preserves $Q$: on a pair of summands with $p + r \ne k$ both sides vanish, on a pair with $p + r = k$ and $p \ne r$ it holds by construction in one order and by $\varepsilon$-symmetry in the other, and on the middle summand it holds because $\varphi_{k/2}$ is an isometry. So $g \in G_{\mathbb{C}}$ and the action is transitive.

Consequently $\check{D}$ is a single orbit of the complex Lie group $G_{\mathbb{C}}$ acting holomorphically on Fl, and it is closed in Fl; a closed orbit is an embedded complex submanifold and the orbit map descends to a biholomorphism $G_{\mathbb{C}}/P \to \check{D}$ ([32]). Since Fl is compact, so is $\check{D}$, and proposition 5.3.4 makes $D$ an open complex submanifold of it.

*Step 4: transitivity on the domain.* Let $F, F' \in D$, with bigradings $V^{p,q}$, $V'^{p,q}$ and Hermitian forms $H = H_F$, $H' = H_{F'}$ from (5.6), both positive definite by proposition 5.3.4. The two structures have the same Hodge numbers, so for each $p > q$ there is a $\mathbb{C}$-linear isometry $\varphi_{p,q} \colon (V^{p,q}, H) \to (V'^{p,q}, H')$; define $\varphi_{q,p}$ on $V^{q,p} = \overline{V^{p,q}}$ by $\varphi_{q,p}(\bar{u}) = \overline{\varphi_{p,q}(u)}$. If $k$ is even, write $r = k/2$; then $V^{r,r}$ is stable under conjugation and its real points carry $Q$ positive definite, since $H(v,v) = Q(v,\bar{v}) = Q(v,v)$ for real $v$. The same holds for $V'^{r,r}$, so there is an isometry of the two real quadratic spaces, and $\varphi_{r,r}$ is its complexification.

Let $g = \bigoplus \varphi_{p,q}$. It commutes with conjugation, hence is defined over $\mathbb{R}$, and it carries $F$ to $F'$. It preserves $Q$: the form is recovered from $H$ and the bigrading by $Q(u,v) = i^{\,q-p} H(u,\bar{v})$ for $u \in V^{p,q}$ and $v \in V^{q,p}$, and $Q(u,v) = 0$ for all other pairs of summands, so

$$Q(gu, gv) = i^{\,q-p} H'\big(\varphi_{p,q} u, \varphi_{p,q} \bar{v}\big) = i^{\,q-p} H(u, \bar{v}) = Q(u,v)$$

on the summands that pair, using $\overline{gv} = g\bar{v}$ and $\bar{v} \in V^{p,q}$; on the middle summand $\varphi_{r,r}$ was chosen to preserve $Q$ outright. Hence $g \in G_{\mathbb{R}}$ and $G_{\mathbb{R}}$ acts transitively on $D$.

Finally, an element of $G_{\mathbb{R}} \cap P$ fixes each $F_o^p$ and, being real, fixes each $\overline{F_o^q}$, hence fixes each $V^{p,q}$; preserving $Q$ and commuting with conjugation, it preserves $H_{F_o}$, which is $i^{\,p-q} Q(u, \bar{v})$ on $V^{p,q}$

by (5.6). So $G_{\mathbb{R}} \cap P$ is a closed subgroup of the unitary group of the positive definite form $H_{F_o}$, which is compact, completing the proof.

Two consequences follow. First, $\dim_{\mathbb{C}} D = \dim_{\mathbb{C}} \check{D} = \dim_{\mathbb{C}} G_{\mathbb{C}} - \dim_{\mathbb{C}} P$, so the dimension of the period domain is a Lie-theoretic count in the numerical data alone; the weight-one case gives $g(g+1)/2$, as computed by hand in example 5.3.3. Second, compactness of the isotropy group means an inner product on the tangent space at $F_o$ can be averaged over $G_{\mathbb{R}} \cap P$, so $D$ carries a $G_{\mathbb{R}}$-invariant Riemannian metric. Which further invariant structures $D$ carries, and the curvature of the resulting geometry, depend on the Hodge numbers; chapter 8 takes this up, together with the subbundle of the tangent bundle along which a varying Hodge structure is constrained to move, which is Griffiths transversality in chapter 7. The terminology reflects the classical cases: when $D$ is a bounded symmetric domain, $\check{D}$ is its compact dual in the sense of symmetric spaces, and $\mathfrak{H}_g \subseteq \mathrm{LG}$ is the model.

**Example 5.3.6: The Period Domain of a Polarized K3 Surface**

Take $k = 2$ with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = n$, so $m = n+2$ and $Q$ is symmetric. Here $f^2 = 1$ and $f^1 = n+1$, and a flag is a pair $F^2 \subseteq F^1$ of those dimensions.

The compact dual collapses to a hypersurface in a projective space. Writing $F^2 = \mathbb{C}\sigma$, the condition $(F^2)^\perp = F^1$ determines $F^1 = \sigma^\perp$, whose dimension is $m-1 = n+1$ as required, and the inclusion $F^2 \subseteq F^1$ says exactly $Q(\sigma,\sigma) = 0$; conversely any isotropic line assembles into such a flag. So

$$\check{D} \cong \left\{ [\sigma] \in \mathbb{P}(V_{\mathbb{C}}) \middle| Q(\sigma,\sigma) = 0 \right\}$$

a smooth quadric hypersurface in $\mathbb{P}^{n+1}$, of dimension $n$. The positivity conditions at $p = 2$ and $p = 0$ are conjugate to one another and read $-Q(\sigma,\bar{\sigma}) > 0$, since $i^{2-0} = -1$; the condition at $p = 1$ reads $Q(v,\bar{v}) > 0$ on $F^1 \cap \overline{F^1}$, equivalently that $Q$ is positive definite on the real points of that conjugation-stable space.

Write $\sigma = x + iy$ with $x, y \in V_{\mathbb{R}}$ and expand:

$$Q(\sigma,\sigma) = Q(x,x) - Q(y,y) + 2i\,Q(x,y), \qquad Q(\sigma,\bar{\sigma}) = Q(x,x) + Q(y,y)$$

So $Q(\sigma,\sigma) = 0$ says $Q(x,x) = Q(y,y)$ and $Q(x,y) = 0$, and then $Q(\sigma,\bar{\sigma}) = 2Q(x,x) < 0$ says the real plane $\mathrm{span}_{\mathbb{R}}(x,y)$ is negative definite for $Q$. Its $Q$-orthogonal complement is the space of real points of $\sigma^\perp \cap \overline{\sigma^\perp}$, so the remaining condition says that complement is positive definite, and the two together force $Q$ to have signature $(n,2)$. Conversely, if $Q$ has signature $(n,2)$ then a negative definite plane has positive definite complement automatically, and

$$D \cong \left\{ [\sigma] \in \mathbb{P}(V_{\mathbb{C}}) \middle| Q(\sigma,\sigma) = 0,\ Q(\sigma,\bar{\sigma}) < 0 \right\}$$

an open subset of the quadric, of dimension $n$. Sending $[\sigma]$ to $\mathrm{span}_{\mathbb{R}}(x,y)$ with the orientation $(x,y)$ identifies $D$ with the set of oriented negative definite real planes in $(V_{\mathbb{R}}, Q)$: scaling $\sigma$ by $re^{i\theta}$ rotates the ordered basis and preserves both plane and orientation, and an oriented negative definite plane admits an oriented basis $x, y$ with $Q(x,x) = Q(y,y)$ and $Q(x,y) = 0$, unique up to that rotation and scaling. Conjugation $[\sigma] \mapsto [\bar{\sigma}]$ is a fixed-point-free involution of $D$ that reverses this orientation.

For $n = 19$ the data is that of a polarized K3 surface. Let $X$ be a projective K3 surface with polarization class $h$, an integral $(1,1)$-class of positive square, and let $V_{\mathbb{Z}}$ be the primitive part of $H^2(X,\mathbb{Z})$, the orthogonal complement of $h$ for the cup form. By example 4.3.4 the cup form on $H^2(X,\mathbb{Z}) \cong \mathbb{Z}^{22}$ has signature $(3,19)$ and Hodge numbers $h^{2,0}=1$, $h^{1,1}=20$; removing the positive direction spanned by $h$ leaves a lattice of rank 21 with $h^{1,1}=19$ on which the cup form has signature $(2,19)$. The polarizing form of proposition 5.2.8 is the negative of the cup form in weight two, so $Q$ has signature $(19,2)$, and the plane spanned by the real and imaginary parts of a holomorphic 2-form $\sigma$ is the one the description above singles out. Passing to the primitive part is not a convenience. On all of $H^2(X,\mathbb{R})$ the Hodge numbers give $n=20$, but there $Q$, still minus the cup form, has signature $(19,3)$ rather than the $(20,2)$ just forced: the plane spanned by the real and imaginary parts of $\sigma$ is negative definite as before, while its complement, the real points of $\sigma^\perp \cap \overline{\sigma^\perp}$, contains the Kähler class, on which $Q([\omega],[\omega]) = -\int_X \omega\wedge\omega < 0$ by part (2) of lemma 4.3.2. So the positivity at $p=1$ fails there and the unpolarized $H^2$ of a K3 is not a point of the period domain for $n=20$, which is exercise 5.2.1 (5) read inside $\check{D}$. The domain has dimension 19; identifying it as the target of the period map for polarized K3 surfaces, and the injectivity of that map, is the Torelli theorem taken up in chapter 8, with the lattice-theoretic input cited to [4].

## Exercise 5.3.1

1. Derive (5.4) from the symmetry $h^{p,q}=h^{q,p}$, and deduce that for a filtration $F^\bullet \in \mathrm{Fl}$ the first bilinear relation (5.5) holds if and only if $(F^p)^\perp = F^{k-p+1}$ for every $p$. Where is the nondegeneracy of $Q$ used?

2. Take $k=1$ and $g=1$ in example 5.3.3, so $m=2$ and $Q$ is a nondegenerate alternating form on $V_{\mathbb{C}} \cong \mathbb{C}^2$. Show that $\check{D} \cong \mathbb{P}^1$ and that $D$ is the upper half plane, computing $iQ(v,\bar{v})$ directly for $v = e+\tau f$ in a symplectic basis. Then exhibit a point of $\check{D}\setminus D$ and verify that both the transversality (5.1) and the positivity fail at it.

3. Let $F_o \in D$ and let $G_{\mathbb{R}} \cap P$ be its isotropy group as in proposition 5.3.5. Show that restriction to the summands with $p > k/2$, together with the middle summand when $k$ is even, gives an isomorphism
$$G_{\mathbb{R}} \cap P \;\cong\; \prod_{p>k/2} \mathrm{U}\big(h^{p,k-p}\big) \times \mathrm{O}\big(h^{k/2,k/2}\big)$$
where the last factor is present only for $k$ even. Check the answer against example 5.3.3 and example 5.3.6 by comparing dimensions with $\dim_{\mathbb{C}} D$ in each case.

4. In the coordinates of example 5.3.3, write an element of $G_{\mathbb{R}} = \mathrm{Sp}(V_{\mathbb{R}},Q)$ in block form $g = \left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)$ relative to the symplectic basis. Show that the action of $G_{\mathbb{R}}$ on $D \cong \mathfrak{H}_g$ furnished by proposition 5.3.5 is
$$Z \longmapsto (\gamma+\delta Z)(\alpha+\beta Z)^{-1}$$
and that $\alpha+\beta Z$ is invertible for every $Z\in\mathfrak{H}_g$ and every $g$. Specialize to $g=1$ and identify the action with the Möbius action of $\mathrm{SL}(2,\mathbb{R})$ on the upper half plane.

5. Take $k=3$ with $h^{3,0}=h^{0,3}=1$ and $h^{2,1}=h^{1,2}=n$, the numerical type of $H^3$ of a compact Kähler threefold with a one-dimensional space of holomorphic 3-forms. Show that a point of $\check{D}$ is the same as a pair consisting of a Lagrangian subspace $F^2 \subseteq V_{\mathbb{C}}$ and a line $F^3 \subseteq F^2$, with $F^1$ determined. Using the normalization of example 5.3.3 to compute the dimension of the Lagrangian Grassmannian, conclude
$$\dim_{\mathbb{C}} \check{D} = \frac{(n+1)(n+2)}{2} + n$$

## 5.4 First Examples: Weight One and Weight Two

Two weights make the apparatus of section 5.1 through section 5.3 completely explicit, and they are the two the geometry supplies first. In weight 1 a Hodge structure is a single subspace $V^{1,0}$ of half the total dimension, and the objects it classifies are complex tori; the polarization becomes the classical Riemann form, and its existence is exactly the condition for the torus to be projective. In weight 2 with $h^{2,0} = 1$, the case of a K3 surface, the structure collapses to a single *line* in $H^2(X, \mathbb{C})$, and the two Hodge-Riemann relations become one quadratic equation and one inequality.

Example 5.1.3 already identified the linear algebra of weight 1: an effective weight-1 structure on $V_{\mathbb{Z}}$ is the same thing as a complex structure $J$ on the real vector space $V_{\mathbb{R}}$, the correspondence sending $J$ to its $(+i)$-eigenspace $V^{1,0}$ in $V_{\mathbb{C}}$. A lattice together with a complex structure on the real space it spans is exactly the data of a complex torus (example 1.1.10), so the dictionary is forced. What has to be checked is that it is a dictionary at the level of morphisms as well, and what has to be computed is what a polarization becomes on the other side.

**Theorem 5.4.1: Weight-One Hodge Structures and Complex Tori**

Let $g \geq 1$.

1. Let $V$ be an effective pure Hodge structure of weight 1 on a free abelian group $V_{\mathbb{Z}}$ of rank $2g$, let $\pi\colon V_{\mathbb{C}} \to V^{1,0}$ be the projection with kernel $V^{0,1}$, and let $J$ be the complex structure on $V_{\mathbb{R}}$ of example 5.1.3. Then $\pi(V_{\mathbb{Z}})$ is a lattice in the $g$-dimensional complex vector space $V^{1,0}$, and

$$X_V := V_{\mathbb{C}} \big/ \big(\overline{F^1 V_{\mathbb{C}}} + V_{\mathbb{Z}}\big) = V^{1,0} / \pi(V_{\mathbb{Z}})$$

   is a complex torus of dimension $g$, namely $V_{\mathbb{R}}/V_{\mathbb{Z}}$ carrying the complex structure $J$.

2. The assignment $V \mapsto X_V$ is an equivalence from the category of effective weight-1 Hodge structures on lattices of rank $2g$, with morphisms of Hodge structures, to the category of complex tori of dimension $g$, with holomorphic group homomorphisms. A quasi-inverse sends $X = W/\Gamma$ to the weight-1 structure on $\Gamma$ determined by multiplication by $i$ on $\Gamma \otimes_{\mathbb{Z}} \mathbb{R} = W$. The construction $V \mapsto X_V$ is functorial without reference to a fixed rank: a morphism of weight-1 Hodge structures $V \to V'$, of ranks $2g$ and $2g'$ not necessarily equal, induces a holomorphic homomorphism $X_V \to X_{V'}$, and every such homomorphism arises this way.

3. A bilinear form $Q\colon V_{\mathbb{Z}} \times V_{\mathbb{Z}} \to \mathbb{Z}$ is a polarization of $V$ if and only if it is alternating and satisfies

$$Q(Ju, Jv) = Q(u, v) \qquad \text{for all } u, v \in V_{\mathbb{R}} \tag{5.7}$$

$$Q(Ju, u) > 0 \qquad \text{for all } u \in V_{\mathbb{R}} \setminus \{0\} \tag{5.8}$$

   Equivalently, $H(u, v) := Q(Ju, v) + i\, Q(u, v)$ is a positive definite Hermitian form on the complex vector space $(V_{\mathbb{R}}, J)$ whose imaginary part takes integer values on $V_{\mathbb{Z}}$.

Conditions (5.7) and (5.8) are the Riemann bilinear relations in their classical, complex-torus form; a form satisfying them is called a Riemann form of the torus, and it is the object from which [3] builds line bundles on $X_V$.

***Proof*:**
Write $\Gamma_V := \pi(V_{\mathbb{Z}})$, and begin with two facts used throughout. Conjugation is a conjugate-linear isomorphism $V^{1,0} \to V^{0,1}$ by the symmetry in definition 5.1.1, so $h^{1,0} = h^{0,1}$; since these sum to $\dim_{\mathbb{C}} V_{\mathbb{C}} = 2g$, the space $V^{1,0}$ has complex dimension $g$. And for $u \in V_{\mathbb{R}}$,

$$u = \tfrac{1}{2}\big(u - iJu\big) + \tfrac{1}{2}\big(u + iJu\big)$$

where the first summand lies in $V^{1,0}$, since $J(u - iJu) = Ju + iu = i(u - iJu)$ exhibits it as a $(+i)$-eigenvector of $J$, and the second is its conjugate and so lies in $V^{0,1}$. Hence

$$\pi(u) = \tfrac{1}{2}\big(u - iJu\big), \qquad \overline{\pi(u)} = \tfrac{1}{2}\big(u + iJu\big) \qquad (u \in V_{\mathbb{R}}) \tag{5.9}$$

1. By example 5.1.3 the restriction $\pi|_{V_{\mathbb{R}}} : V_{\mathbb{R}} \to V^{1,0}$ is an isomorphism of real vector spaces, and $J$ is by construction the pullback along it of multiplication by $i$. Now $V_{\mathbb{Z}}$ is a discrete subgroup of rank $2g$ spanning $V_{\mathbb{R}}$, so its image $\Gamma_V$ is a discrete subgroup of rank $2g$ spanning the real $2g$-dimensional space $V^{1,0}$, that is, a lattice. The quotient $X_V = V^{1,0}/\Gamma_V$ is therefore a complex torus of dimension $g$ (example 1.1.10), and $\pi$ identifies it with $V_{\mathbb{R}}/V_{\mathbb{Z}}$ carrying the complex structure $J$. Finally $\overline{F^1 V_{\mathbb{C}}} = \overline{V^{1,0}} = V^{0,1}$ by definition 5.1.4, so $V_{\mathbb{C}}/(V^{0,1} + V_{\mathbb{Z}}) = V^{1,0}/\Gamma_V$, which is the displayed description.

2. *Objects.* Let $X = W/\Gamma$ be a complex torus of dimension $g$, so $\Gamma$ is a lattice of rank $2g$ in $W$ and the inclusion extends to an isomorphism $\Gamma \otimes_{\mathbb{Z}} \mathbb{R} \cong W$. Multiplication by $i$ on $W$ is an endomorphism $J$ of $\Gamma_{\mathbb{R}}$ with $J^2 = -\mathrm{id}$, so by exercise 5.1.1 (1) it determines an effective weight-1 Hodge structure $V$ on $\Gamma$ whose $(1,0)$-part is the $(+i)$-eigenspace of $J$. The map $\pi|_{\Gamma_{\mathbb{R}}} : W \to V^{1,0}$ intertwines multiplication by $i$ on $W$ with multiplication by $i$ on $V^{1,0}$, hence is an isomorphism of complex vector spaces, and it carries $\Gamma$ onto $\Gamma_V$; passing to quotients gives $X \cong X_V$. In the other direction, start from $V$ and form $X_V = V^{1,0}/\Gamma_V$. Its lattice is $\Gamma_V$, identified with $V_{\mathbb{Z}}$ by $\pi$, and multiplication by $i$ on $V^{1,0}$ pulls back along $\pi$ to $J$ on $V_{\mathbb{R}}$; so the Hodge structure attached to $X_V$ by the previous sentence is the one determined by $J$, which is $V$.

   *Morphisms.* Let $V, V'$ be effective weight-1 structures with complex structures $J, J'$, and let $\varphi\colon V_{\mathbb{Z}} \to V'_{\mathbb{Z}}$ be a homomorphism. By definition 5.1.6, $\varphi$ is a morphism of Hodge structures exactly when $\varphi_{\mathbb{C}}(V^{1,0}) \subseteq (V')^{1,0}$. Since $\varphi$ is defined over $\mathbb{Z}$, its complexification commutes with conjugation, so this inclusion forces $\varphi_{\mathbb{C}}(V^{0,1}) \subseteq (V')^{0,1}$ as well; as $J$ acts by $+i$ on $V^{1,0}$ and by $-i$ on $V^{0,1}$, the two inclusions together say precisely that $\varphi_{\mathbb{R}} \circ J = J' \circ \varphi_{\mathbb{R}}$. Such a $\varphi$ transports along $\pi$ and $\pi'$ to a $\mathbb{C}$-linear map $V^{1,0} \to (V')^{1,0}$ carrying $\Gamma_V$ into $\Gamma_{V'}$, hence descends to a holomorphic homomorphism $X_V \to X_{V'}$.

   Conversely let $f\colon X_V \to X_{V'}$ be a holomorphic group homomorphism. The universal covers are the vector spaces $V^{1,0}$ and $(V')^{1,0}$, and since $V^{1,0}$ is simply connected, $f$ lifts to a holomorphic $F\colon V^{1,0} \to (V')^{1,0}$ with $F(0) = 0$. That $f$ is a homomorphism says $F(x+y) - F(x) - F(y) \in \Gamma_{V'}$ for all $x, y$; the left-hand side is continuous in $(x, y)$, vanishes at $(0,0)$, and takes values in a discrete group, so it vanishes identically and $F$ is additive. Differentiating $F(x+y) = F(x) + F(y)$ with respect to $y$ at $y = 0$ gives $dF_x = dF_0$ for every $x$, so $F$ equals the $\mathbb{C}$-linear map $dF_0$. It satisfies $F(\Gamma_V) \subseteq \Gamma_{V'}$, and transporting back along $\pi, \pi'$ produces a homomorphism $\varphi\colon V_{\mathbb{Z}} \to V'_{\mathbb{Z}}$ commuting with $J$ and $J'$, that is, a morphism of Hodge structures inducing $f$. The two assignments are inverse to one another, so $\mathrm{Hom}(V, V') \to \mathrm{Hom}(X_V, X_{V'})$ is a bijection. Together with the previous paragraph this makes $V \mapsto X_V$ essentially surjective, full and faithful, hence an equivalence.

3. Specialized to $k = 1$, definition 5.2.1 and theorem 5.2.3 ask of $Q$ that it be $(-1)^1$-symmetric, that is alternating, and that

$$Q\big(F^1 V_{\mathbb{C}},\, F^1 V_{\mathbb{C}}\big) = 0, \qquad i\, Q(v, \bar{v}) > 0 \ \text{ for } 0 \neq v \in V^{1,0}$$

there being no primitivity restriction in weight 1, where every class is primitive. Extend $Q$ first $\mathbb{R}$-bilinearly to $V_{\mathbb{R}}$ and then $\mathbb{C}$-bilinearly to $V_{\mathbb{C}}$, and recall $F^1 V_{\mathbb{C}} = V^{1,0} = \pi(V_{\mathbb{R}})$.

For the first relation, (5.9) gives, for $u, v \in V_{\mathbb{R}}$,

$$4\, Q(\pi u, \pi v) = Q\big(u - iJu,\ v - iJv\big) = \big(Q(u,v) - Q(Ju,Jv)\big) - i\big(Q(u,Jv) + Q(Ju,v)\big)$$

Both bracketed quantities are real, so $Q(V^{1,0}, V^{1,0}) = 0$ holds if and only if $Q(Ju, Jv) = Q(u, v)$ and $Q(u, Jv) = -Q(Ju, v)$ for all real $u, v$. Each of these two identities implies the other: replacing $v$ by $Jv$ in the first gives $Q(Ju, J^2 v) = Q(u, Jv)$, which is the second, and replacing $v$ by $Jv$ in the second gives $Q(u, J^2 v) = -Q(Ju, Jv)$, which is the first. So the first Hodge-Riemann relation is exactly (5.7).

For the second, (5.9) and the vanishing of $Q(u, u)$ and $Q(Ju, Ju)$ give

$$4\, Q\big(\pi u, \overline{\pi u}\big) = Q\big(u - iJu,\ u + iJu\big) = i\, Q(u, Ju) - i\, Q(Ju, u) = 2i\, Q(u, Ju)$$

whence $i\, Q(\pi u, \overline{\pi u}) = \frac{i^2}{2} Q(u, Ju) = \frac{1}{2} Q(Ju, u)$. Since $\pi$ maps $V_{\mathbb{R}}$ onto $V^{1,0}$, the positivity condition is exactly (5.8). Nondegeneracy of $Q$ is a consequence rather than an extra hypothesis: if $Q(u, -)$ vanished identically then $Q(Ju, u) = -Q(u, Ju) = 0$, forcing $u = 0$.

It remains to identify $H$. Using (5.7) in the form $Q(u, Jv) = -Q(Ju, v)$,

$$\begin{aligned} H(Ju, v) &= Q(J^2 u, v) + i\, Q(Ju, v) = -Q(u, v) + i\, Q(Ju, v) = i\, H(u, v) \\ H(v, u) &= Q(Jv, u) + i\, Q(v, u) = -Q(u, Jv) - i\, Q(u, v) = Q(Ju, v) - i\, Q(u, v) = \overline{H(u, v)} \end{aligned}$$

so $H$ is $\mathbb{C}$-linear in its first argument and Hermitian, and $H(u, u) = Q(Ju, u)$ is positive for $u \neq 0$ by (5.8); its imaginary part is $Q$, which is $\mathbb{Z}$-valued on $V_{\mathbb{Z}}$. Conversely, a positive definite Hermitian $H$ on $(V_{\mathbb{R}}, J)$ with $\mathrm{Im}\, H$ integral on $V_{\mathbb{Z}}$ yields $Q := \mathrm{Im}\, H$, which is alternating, satisfies (5.7) because $H(Ju, Jv) = i\,\bar{i}\, H(u, v) = H(u, v)$, and satisfies (5.8) because $Q(Ju, u) = \mathrm{Im}\big(i\, H(u, u)\big) = H(u, u) > 0$. The two descriptions carry the same information, completing the proof.

The equivalence says that at weight 1 the abstract theory and classical complex geometry describe the same objects. A complex torus admits no nonconstant holomorphic functions (proposition 1.1.11) and no obvious algebraic data, yet the pair consisting of its lattice and its complex structure is a complete invariant, and that pair is a Hodge structure. Part (3) adds that the polarization, introduced in section 5.2 as linear algebra, is on the torus side a concrete and classical object: a positive definite Hermitian form with integral imaginary part.

The two relations have different characters, and the difference is what governs projectivity. Relation (5.7) is linear in $Q$ and is a nontrivial condition as soon as $g \geq 2$: it requires the alternating form to be invariant under the complex structure. Relation (5.8) is an open positivity condition, and it selects a cone inside the solutions of the first. The next statement reads this off, and it is the criterion promised in example 1.1.10.

**Corollary 5.4.2: Projectivity Criterion for a Complex Torus**

Let $X = W/\Gamma$ be a complex torus of dimension $g$ and let $V$ be the weight-1 Hodge structure on $\Gamma$ of theorem 5.4.1. The following are equivalent.

1. $X$ admits a closed holomorphic embedding into some $\mathbb{CP}^N$.
2. $V$ admits a polarization.
3. There is an alternating form $E\colon \Gamma \times \Gamma \to \mathbb{Z}$ with $E(iu, iv) = E(u, v)$ for all $u, v \in W$ and $E(iu, u) > 0$ for all $u \in W \setminus \{0\}$.

A complex torus satisfying these conditions is an *abelian variety*. Every complex torus of dimension 1 is an abelian variety, while for $g \geq 2$ a very general complex torus is not.

***Proof*:**
*(2) ⇔ (3).* The complex structure $J$ of theorem 5.4.1 is multiplication by $i$ on $W = \Gamma_{\mathbb{R}}$, so the two conditions in (3) are (5.7) and (5.8) verbatim, and part (3) of that theorem identifies the forms satisfying them with the polarizations of $V$.

*(1) ⇔ (3).* This is the classical criterion for a complex torus to be projective. Given $E$ as in (3), the Appell-Humbert construction attaches to it a holomorphic line bundle whose theta functions embed $X$; conversely an embedding produces such an $E$ from the hyperplane class. Both directions are proved in [3], and this is the one ingredient of the present section taken from outside the book.

*Where $E$ sits in cohomology.* The Hodge-theoretic half of the criterion is available here, and it locates the form $E$ inside $H^2$. The integral cohomology of the torus is the exterior algebra on $H^1(X, \mathbb{Z}) = \mathrm{Hom}(\Gamma, \mathbb{Z})$, so

$$H^2(X, \mathbb{Z}) = \bigwedge^2 \mathrm{Hom}(\Gamma, \mathbb{Z})$$

and an integral degree-2 class is the same thing as an alternating form $E$ on $\Gamma$. Write $V_{\mathbb{C}} = W \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0} \oplus V^{0,1}$ for the eigenspace decomposition of multiplication by $i$. By example 4.2.2 the summand $H^{1,0}(X)$ consists of the $\mathbb{C}$-linear functionals on $W$, and such a functional $\ell$ kills $V^{0,1}$, since $\ell(u + iJu) = \ell(u) + i\ell(iu) = \ell(u) - \ell(u) = 0$. Consequently every summand of $\bigwedge^2 H^1(X, \mathbb{C})$ other than $H^{2,0} = \bigwedge^2 H^{1,0}$ restricts to zero on $V^{1,0} \times V^{1,0}$, while $H^{2,0}$ injects into the alternating forms on $V^{1,0}$. So $E$ has vanishing $(2,0)$-component, equivalently, being real, vanishing $(0,2)$-component, if and only if $E$ vanishes on $V^{1,0} \times V^{1,0}$, which by the computation in part (3) of theorem 5.4.1 is the invariance $E(iu, iv) = E(u, v)$. By definition 4.4.3 and theorem 4.4.2 the integral $(1,1)$-classes are exactly $\mathrm{NS}(X)$, and are exactly the first Chern classes of holomorphic line bundles. Condition (3) therefore asks for an element of $\mathrm{NS}(X)$ satisfying a definiteness condition, and it is that condition, not the existence of line bundles, that separates projective tori from the rest. For $g = 2$ this recovers the description of $\mathrm{NS}(X)$ in example 4.4.5.

*The two extreme cases.* Let $g = 1$, so that $W$ is a real plane and the group of alternating integral forms on $\Gamma$ has rank 1; fix a generator $E_0$. Multiplication by $i$ acts on the one-dimensional space of alternating forms on $W$ through $\det(i)$, and $\det(i)^2 = \det(-\mathrm{id}) = 1$ forces $\det(i) = \pm 1$. The value $-1$ is impossible: a real $2 \times 2$ matrix of determinant $-1$ has real eigenvalues, whereas an eigenvalue $\lambda$ of $i$ would satisfy $\lambda^2 = -1$. So $\det(i) = 1$, the action is trivial, and every alternating form satisfies the invariance in (3). Moreover $u$ and $iu$ are linearly independent over $\mathbb{R}$ for $u \neq 0$, so $E_0(iu, u) \neq 0$; this quantity is continuous and nonvanishing on the connected set $W \setminus \{0\}$, hence of constant sign, and one of $\pm E_0$ satisfies (3).

Let $g \geq 2$ and let $X$ be very general. Such a torus has $\rho(X) = 0$: exercise 4.4.1 (3) gives only the bound $\rho \leq h^{1,1}$, and that the floor is attained on a very general torus is proved in [3], which owns the complex-torus theory this example borrows. So $\mathrm{NS}(X) = 0$, so $\mathrm{NS}(X) = 0$ and, by the previous paragraph, the only invariant integral alternating form on $\Gamma$ is 0, which fails the positivity in (3). Hence $X$ is not projective, completing the proof.

The criterion is a positivity condition on a lattice, and nothing in it refers to maps into projective space. The question "which complex tori are algebraic" has become the question "which lattices with a complex structure carry an invariant positive alternating form", answerable by counting integral solutions of a linear condition and then testing a sign. For $g = 1$ the linear condition is vacuous and the sign can always be arranged, so every one-dimensional complex torus is an elliptic curve; this agrees with exercise 4.4.1 (4), where $\mathrm{NS}(E) = H^2(E, \mathbb{Z}) \cong \mathbb{Z}$ for every elliptic curve. For $g \geq 2$ the linear condition generically has no nonzero integral solution, and the generic complex torus carries no divisors at all.

**Example 5.4.3: The Weight-One Structure of an Elliptic Curve**

Fix $\tau = s + it$ with $t = \operatorname{Im} \tau > 0$, and let $E_\tau = \mathbb{C}/\Lambda_\tau$ with $\Lambda_\tau = \mathbb{Z} \cdot 1 \oplus \mathbb{Z}\tau$. This is the $n = 1$ case of example 4.2.2, with $h^{1,0} = h^{0,1} = 1$ and $b_1 = 2$. Both sides of theorem 5.4.1 can be written down.

*The lattice side.* Take $V_{\mathbb{Z}} = \Lambda_\tau = H_1(E_\tau, \mathbb{Z})$ with $J$ multiplication by $i$. In the real basis $(1, \tau)$ of $W = \mathbb{C}$ one has $i = -\frac{s}{t} + \frac{1}{t}\tau$ and $i\tau = -\frac{s^2+t^2}{t} + \frac{s}{t}\tau$, so $J$ has matrix

$$M = \begin{pmatrix} -s/t & -(s^2+t^2)/t \\ 1/t & s/t \end{pmatrix}, \qquad \det M = \frac{-s^2 + s^2 + t^2}{t^2} = 1$$

Let $Q_0$ be the alternating form with $Q_0(1, \tau) = 1$; every alternating integral form on $\Lambda_\tau$ is an integer multiple of it. Writing $u = x + y\tau$ in the coordinates $(x, y)$, so that $Q_0(u, v)$ is the determinant of the coordinate matrix of $(u, v)$, a direct computation gives

$$Q_0(u, Ju) = \frac{1}{t}\big(x^2 + 2sxy + (s^2+t^2)y^2\big) = \frac{(x+sy)^2 + (ty)^2}{t} = \frac{|u|^2}{\operatorname{Im}\tau}$$

so $Q_0(Ju, u) < 0$, and it is $Q := -Q_0$, normalized by $Q(1, \tau) = -1$, that satisfies (5.8). Relation (5.7) is automatic because $\det M = 1$. The associated Hermitian form of theorem 5.4.1 is

$$H(u, v) = Q(Ju, v) + i\, Q(u, v) = \frac{u\bar{v}}{\operatorname{Im}\tau}$$

as one checks on the basis, and $\operatorname{Im} H = Q$ is integral on $\Lambda_\tau$. So $\Lambda_\tau$ carries a polarization for every $\tau$ in the upper half-plane, and corollary 5.4.2 returns the classical fact that every one-dimensional complex torus is an algebraic curve.

*The cohomological side.* Take instead $V_{\mathbb{Z}} = H^1(E_\tau, \mathbb{Z}) = \mathrm{Hom}(\Lambda_\tau, \mathbb{Z})$ with the weight-1 structure of example 5.1.2, and let $\ell_1, \ell_2$ be the basis dual to $1, \tau$. The space $H^{1,0}(E_\tau) = H^0(E_\tau, \Omega^1)$ is spanned by $dz$, whose class in $H^1(E_\tau, \mathbb{C}) = \mathrm{Hom}(\Lambda_\tau, \mathbb{C})$ is recorded by its periods,

$$\int_1 dz = 1, \qquad \int_\tau dz = \tau, \qquad \text{so} \qquad dz = \ell_1 + \tau\,\ell_2$$

and $H^{0,1} = \overline{H^{1,0}}$ is spanned by $\ell_1 + \bar\tau\,\ell_2$. The two lines are transverse precisely because $\bar\tau - \tau = -2i\,t \neq 0$, which is (5.1) for this structure. With the cup product $Q'(\alpha, \beta) = \int_{E_\tau} \alpha \wedge \beta$, normalized by $Q'(\ell_1, \ell_2) = 1$ for the complex orientation,

$$Q'\big(dz, \overline{dz}\big) = Q'\big(\ell_1 + \tau\ell_2,\ \ell_1 + \bar\tau\ell_2\big) = \bar\tau - \tau = -2i\,\mathrm{Im}\,\tau$$

so $i\,Q'(dz, \overline{dz}\,) = 2\,\mathrm{Im}\,\tau > 0$. The second Hodge-Riemann relation for $H^1(E_\tau, \mathbb{Z})$ is therefore literally the condition $\mathrm{Im}\,\tau > 0$, and the first is automatic because $F^1$ is a line and $Q'$ is alternating. The period line $\mathbb{C} \cdot (1, \tau) \subseteq \mathbb{C}^2$ is the point of the period domain that example 5.3.3 identifies with $\tau$ in the upper half-plane.

*Moduli.* An isomorphism of polarized weight-1 structures of rank 2 is a change of $\mathbb{Z}$-basis of the lattice preserving the alternating form, that is, an element of $\mathrm{SL}_2(\mathbb{Z})$, and replacing the basis $(1, \tau)$ of $\Lambda_\tau$ by $(c\tau + d,\ a\tau + b)$ rescales the lattice to $\Lambda_{\tau'}$ with $\tau' = (a\tau + b)/(c\tau + d)$. The isomorphism classes are therefore the orbits of $\mathrm{SL}_2(\mathbb{Z})$ on the upper half-plane, with the $j$-invariant as the coordinate on the quotient [5].

The example shows the mechanism in miniature. The Hodge structure is one line in a two-dimensional space; the first Hodge-Riemann relation is automatic because that line is isotropic for dimension reasons; the second is a single inequality; and the parameter recording the structure is the ratio of the two periods of $dz$. The higher-weight theory repeats this pattern with lines replaced by subspaces and the two relations no longer automatic.

Weight 2 is the first weight in which the polarization has both signs to distribute, and theorem 4.3.3 already computed the distribution on a surface: positive on $H^{2,0} \oplus H^{0,2}$ and along the Kähler class, negative on the primitive $(1,1)$-classes. A K3 surface is the case $h^{2,0} = 1$, so that the top step $F^2$ of the filtration is again a line and the whole structure is again recorded by a single point of a projective space.

**Example 5.4.4: The Weight-Two Structure of a K3 Surface**

Let $X$ be a K3 surface, a compact Kähler surface with trivial canonical bundle and $h^{1,0} = 0$. Example 4.3.4 recorded its Hodge diamond: $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 20$, $b_2 = 22$, and $h^{1,0} = h^{0,1} = 0$. So $H^1(X, \mathbb{Z}) = 0$ and every Hodge-theoretic invariant of $X$ lives in $H^2$.

Take $V_{\mathbb{Z}} = H^2(X, \mathbb{Z})$ with the weight-2 structure of example 5.1.2 and $Q(\alpha, \beta) = \int_X \alpha \wedge \beta$. By example 4.3.4 the pair $(V_{\mathbb{Z}}, Q)$ is an even unimodular lattice of signature $(3, 19)$, isometric to $U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$; the identification of the isometry class is cited to [4].

Since $h^{2,0} = 1$, the top step of the Hodge filtration is a line,

$$F^2 V_{\mathbb{C}} = H^{2,0}(X) = \mathbb{C}\,\sigma$$

for a nonzero holomorphic 2-form $\sigma$ (a trivialization of $K_X$), and the whole structure is determined by that line. Indeed $H^{0,2} = \overline{\mathbb{C}\sigma}$ by definition 5.1.1, and

$$H^{1,1}(X) = \{\, x \in H^2(X, \mathbb{C}) \mid Q(x, \sigma) = Q(x, \bar{\sigma}) = 0 \,\}$$

For the last identity, note that $\alpha \wedge \beta$ has type $(p + p', q + q')$ and only a $(2, 2)$-form has nonzero integral over a surface, so $Q$ pairs $H^{2,0}$ with $H^{0,2}$, pairs $H^{1,1}$ with itself, and kills every other combination of summands (Step 1 in the proof of theorem 4.3.3). Writing $x = a\sigma + y + b\bar{\sigma}$ with $y \in H^{1,1}$, this gives $Q(x, \sigma) = b\, Q(\bar{\sigma}, \sigma)$ and $Q(x, \bar{\sigma}) = a\, Q(\sigma, \bar{\sigma})$, and $Q(\sigma, \bar{\sigma}) \neq 0$ by the relation below, so both vanish exactly when $a = b = 0$.

The Hodge-Riemann relations, applied to the line $F^2$ and written out for the cup form $Q$ rather than for the polarizing form $-Q$ of proposition 5.2.8, become two conditions on $\sigma$ alone:

$$Q(\sigma, \sigma) = 0, \qquad Q(\sigma, \bar{\sigma}) > 0 \tag{5.10}$$

The first is the relation $Q(F^2, F^2) = 0$, and it is automatic here, since $\sigma \wedge \sigma$ is a form of type $(4, 0)$ on a surface and hence zero. The second is part (1) of lemma 4.3.2, proved in chapter 4 by a computation in a unitary coframe. It is stated here for the cup form, whose negative is the polarizing form in weight 2 (example 5.2.9): a $(2, 0)$-class on a surface is primitive, since $\omega \wedge \sigma$ has type $(3, 1)$ and vanishes, and condition (2) of definition 5.2.1 for $-Q$ at $\sigma$ reads $i^{2-0}\big(-Q(\sigma, \bar{\sigma})\big) > 0$, which is the displayed inequality because $i^2 = -1$. Neither $Q$ nor $-Q$ polarizes all of $H^2$ (exercise 5.2.1 (5)), so (5.10) is a statement about the line $F^2$ rather than an instance of theorem 5.2.3 applied to $(V_{\mathbb{Z}}, Q)$; the weight-2 relations on that line were already verified for surfaces in Part I.

Three consequences follow at once.

1. *The Néron-Severi group is a hyperplane section of the lattice.* An integral class $\gamma$ is real, so $\gamma^{2,0} = \overline{\gamma^{0,2}}$ and $\gamma$ is of type $(1, 1)$ as soon as $Q(\gamma, \sigma) = 0$. Hence

   $$\mathrm{NS}(X) = \sigma^{\perp} \cap H^2(X, \mathbb{Z})$$

   which by theorem 4.4.2 is the group of first Chern classes of line bundles on $X$. Its rank satisfies $\rho(X) \leq h^{1,1}(X) = 20$ (exercise 4.4.1 (3)). The bound is attained, by the K3 surfaces classically called *singular* (a misnomer: they are smooth, and the name records only that $\rho$ is as large as the Hodge numbers allow). That such surfaces exist is not proved here and is not needed below; where a $\rho = 20$ surface is used it is taken as a hypothesis.

2. *The structure is a point of a quadric.* By (5.10) the line $[\sigma]$ lies in the subset of $\mathbb{CP}\big(H^2(X, \mathbb{C})\big) \cong \mathbb{CP}^{21}$ cut out by the quadratic equation $Q(x, x) = 0$ together with the open condition $Q(x, \bar{x}) > 0$, an open subset of the 20-dimensional quadric $Q(x, x) = 0$; the point $[\sigma]$ is the period point of $X$. That subset is not itself a period domain in the sense of definition 5.3.2, which imposes positivity at $p = 1$ as well: there the condition asks $-Q$ to be positive definite on the real $(1, 1)$-classes, where by theorem 4.3.3 it has signature $(19, 1)$, so it holds at no point. Equivalently, the two positivity conditions force the polarizing form to have signature $(n, 2)$ with $n = h^{1,1}$ (example 5.3.6), and $-Q$ on $H^2$ has signature $(19, 3)$, not $(20, 2)$. Cutting the lattice down repairs this.

3. *Polarizing drops the dimension by one.* If $X$ is projective, fix an ample class $h$ with $Q(h, h) = 2d > 0$. The sublattice $h^{\perp}$, the primitive cohomology of definition 5.2.5, then

has rank 21 and signature $(2,19)$, since removing one positive direction from $(3,19)$ leaves $(2,19)$; and $\sigma$ lies in $h^\perp \otimes \mathbb{C}$ because $h$ is of type $(1,1)$. So $-Q$ has signature $(19,2)$ on $h^\perp$, and the period point of a polarized K3 varies in the period domain of example 5.3.6 with $n = 19$, of dimension 19.

The K3 case is the weight-2 analogue of the elliptic curve. In both, the Hodge structure is a single subspace of the complexified lattice: a $g$-plane in rank $2g$ for weight 1, a line in rank 22 for a K3. In both, the first Hodge-Riemann relation is an algebraic condition on that subspace and the second is an open positivity condition. And in both, the resulting parameter space is a concrete open subset of a projective variety, which is what makes the moduli theory of the two classes of surfaces tractable.

The difference is where the extra Hodge classes sit. On an elliptic curve every integral class in $H^2$ is of type $(1,1)$, so $\rho = 1$ always. On a K3 surface, $\mathrm{NS}(X) = \sigma^\perp \cap H^2(X,\mathbb{Z})$ ranges from 0, on a non-projective K3, where theorem 4.4.2 produces no line bundle with nonzero Chern class, up to a lattice of rank 20; and the rank jumps exactly where the period point becomes orthogonal to extra integral vectors. Those loci are the Hodge loci taken up in chapter 8, where the recovery of a K3 surface from its period point, the global Torelli theorem surveyed in [4], is also treated.

## Exercise 5.4.1

1. Let $V$ be an effective weight-1 Hodge structure of rank 2 with complex structure $J$ on $V_\mathbb{R}$, and let $Q_0$ generate the rank-one group of alternating integral forms on $V_\mathbb{Z}$. Show that (5.7) holds for every alternating form, that $Q_0(Ju,u) \neq 0$ for all $u \neq 0$, and that exactly one of $\pm Q_0$ satisfies (5.8). Then show that for $g \geq 2$ condition (5.7) is a nontrivial linear condition, by exhibiting a weight-1 structure of rank 4 with no polarization.

2. For the elliptic curve $E_\tau$ of example 5.4.3, show that the polarizations of the weight-1 structure on $\Lambda_\tau$ are exactly the forms $nQ$ with $n$ a positive integer and $Q(1,\tau) = -1$, and compute the Hermitian form attached to $nQ$ by theorem 5.4.1. Deduce that every elliptic curve carries a polarization inducing a surjection $\bigwedge^2 \Lambda_\tau \to \mathbb{Z}$, a *principal* polarization.

3. Let $X = W/\Gamma$ and $X' = W'/\Gamma'$ be complex tori. Use theorem 5.4.1 to show that $\mathrm{Hom}(X,X')$ is the group of $\mathbb{C}$-linear maps $F\colon W \to W'$ with $F(\Gamma) \subseteq \Gamma'$. Then compute $\mathrm{End}(E_\tau)$ for $E_\tau = \mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\tau)$, showing that it is $\mathbb{Z}$ unless $\tau$ satisfies a quadratic equation over $\mathbb{Q}$, in which case it is a subring of $\mathbb{Q}(\tau)$ free of rank 2 over $\mathbb{Z}$.

4. Let $X$ be a projective K3 surface with $\rho(X) = 20$. Using example 5.4.4 and theorem 4.3.3, show that the intersection form on $\mathrm{NS}(X)$ has signature $(1,19)$, that the *transcendental lattice* $T(X) = \mathrm{NS}(X)^\perp \subseteq H^2(X,\mathbb{Z})$ has rank 2, and that $Q$ is positive definite on $T(X)$. Where in $T(X) \otimes \mathbb{C}$ does $\sigma$ sit?

5. Let $X = W/\Gamma$ be a complex torus of dimension $g$ and let $V = H^1(X,\mathbb{Z})$ carry its weight-1 Hodge structure. Show that $H^k(X,\mathbb{Z}) = \bigwedge^k V_\mathbb{Z}$ carries a pure Hodge structure of weight $k$ with
$$\left(\bigwedge\nolimits^k V\right)^{p,q} = \bigwedge\nolimits^p V^{1,0} \otimes \bigwedge\nolimits^q V^{0,1}, \qquad p+q=k$$
and check that the resulting Hodge numbers agree with those computed in example 4.2.2.

# 6

# Hard Lefschetz and Lefschetz Pencils

The Kähler class has appeared so far as a source of hypotheses. It closed the gap between Dolbeault and de Rham cohomology, it supplied the form that polarizes primitive cohomology, and in every case what was used was that $\omega$ is closed. This chapter uses something else: that wedging with $\omega$ is an operator on cohomology, and that this operator has a great deal of structure.

The structure is a representation of $\mathfrak{sl}_2$. The Lefschetz operator $L$, its adjoint $\Lambda$, and the operator that records degree satisfy the commutation relations of that Lie algebra, and the representation theory of $\mathfrak{sl}_2$ is rigid enough to force conclusions about $H^*(X, \mathbb{C})$ that no direct argument supplies. Two follow immediately. The map $L^{n-k}\colon H^k(X, \mathbb{C}) \to H^{2n-k}(X, \mathbb{C})$ is an isomorphism, which is the Hard Lefschetz theorem; and cohomology decomposes into primitive pieces, on which the form of proposition 5.2.8 is definite with the alternating signs corollary 5.2.4 predicted. The positivity that chapter 5 deferred, and the identity of Weil (theorem 5.2.7) it rested on, are both proved here.

The second half of the chapter changes the question from what holds on one manifold to how one manifold can be seen as a family. A pencil of hyperplane sections fibres a variety, after a blow-up, over $\mathbb{CP}^1$, with finitely many singular fibres. Following a loop around a singular value carries the cohomology of a nearby fibre back to itself by a nontrivial automorphism, and the Picard-Lefschetz formula computes it exactly. That automorphism is monodromy, and it is the mechanism by which a family of varieties acts on the Hodge structures of its members, which is what chapter 7 takes up.

## 6.1 The $\mathfrak{sl}_2$-Action on Cohomology

The operators $L$ and $\Lambda$ of definition 3.3.1 entered chapter 3 as ingredients of the Kähler identities and were then set aside. This section takes them as the objects of study. Adjoining a third operator, the one that records the degree of a form, turns the pair into a Lie algebra acting on the exterior algebra at a single point.

Everything here is pointwise linear algebra, valid on any Hermitian manifold: the commutation relations, the decomposition of the exterior algebra into primitive pieces, and Weil's identity for the Hodge star are all statements about a single Hermitian vector space, and none of them uses $d\omega = 0$. The Kähler condition is spent in exactly one place, the transport of the action from forms to cohomology carried out in section 6.2, and it is spent through theorem 3.3.5.

Throughout, $x$ denotes a point of a Hermitian manifold $(X, \omega)$ of complex dimension $n$, and $\bigwedge_x^k$ denotes the space of complex-valued $k$-covectors at $x$, that is, the degree-$k$ part of $\bigwedge^\bullet T_x^* X \otimes \mathbb{C}$, with its bigrading $\bigwedge_x^{p,q}$ and the Hermitian inner product $\langle -, - \rangle$ induced by the metric. The operators $L$ and $\Lambda$ are algebraic, so each acts on $\bigwedge_x^\bullet$ one point at a time, and an identity between such operators holds for differential forms as soon as it holds at every point.

The third operator is forced by the shape of the first two. Both $L$ and $\Lambda$ change the degree, $L$ by $+2$ and $\Lambda$ by $-2$, so their commutator preserves degree and the only candidate for it, if the three are to close into a Lie algebra, is an operator that reads the degree off.

**Definition 6.1.1: The Lefschetz Triple and Primitive Forms**

Let $(X, \omega)$ be a Hermitian manifold of complex dimension $n$ and let $x \in X$. The *counting operator* at $x$ is the linear map

$$H \colon \textstyle\bigwedge_x^\bullet \to \bigwedge_x^\bullet, \qquad H\alpha = (k-n)\,\alpha \quad \text{for } \alpha \in \bigwedge_x^k$$

The *Lefschetz triple* at $x$ is the triple of operators $(L, \Lambda, H)$ on $\bigwedge_x^\bullet$, where $L$ and $\Lambda$ are the Lefschetz operators of definition 3.3.1.
A covector $\alpha \in \bigwedge_x^k$ is *primitive* if $\Lambda\alpha = 0$. The space of primitive $k$-covectors is written $P_x^k$, and $P_x^{p,q} := P_x^{p+q} \cap \bigwedge_x^{p,q}$. A differential form is primitive if it is primitive at every point.

For $n = 1$ the triple is already visible: $H$ is $-1$ on functions, $0$ on 1-covectors and $+1$ on 2-covectors, and every 1-covector is primitive, since $\Lambda$ would send it to degree $-1$. The full computation in dimensions one and two is example 6.1.3.

The counting operator carries no geometry: it is the grading of the exterior algebra, shifted so that the middle degree $k = n$ sits at 0. The shift is what makes the three relations below come out symmetric, and it is why $n$, the complex dimension, appears in a purely algebraic statement.

Primitivity is defined here by $\Lambda\alpha = 0$, while definition 5.2.5 defined primitive cohomology by $L^{n-k+1} = 0$. The two conditions agree on $\bigwedge_x^k$ for $k \le n$, and that is part (1) of theorem 6.1.4 below; the present definition is the one that makes the relations usable, and the other is the one that makes sense on cohomology, where $\Lambda$ is not visible until a metric is chosen.

The relations themselves are a computation in the flat model of example 3.3.3. Nothing beyond the Clifford relations recorded there is needed, and no differentiation occurs anywhere, which is why the Kähler condition plays no part.

**Proposition 6.1.2: The Commutation Relations of the Lefschetz Triple**

Let $(X,\omega)$ be a Hermitian manifold of complex dimension $n$ and $x \in X$. On $\bigwedge_x^\bullet$ the Lefschetz triple satisfies

$$[H,L]=2L, \qquad [H,\Lambda]=-2\Lambda, \qquad [L,\Lambda]=H$$

Consequently $e\mapsto L$, $f\mapsto\Lambda$, $h\mapsto H$ defines a representation of $\mathfrak{sl}_2(\mathbb{C})$ on $\bigwedge_x^\bullet$, where $e,f,h$ is the standard basis with $[h,e]=2e$, $[h,f]=-2f$, $[e,f]=h$. The operators $L$ and $\Lambda$ shift the bidegree by $(1,1)$ and $(-1,-1)$, and $H$ preserves it.

***Proof*:**
The first two relations are degree bookkeeping. For $\alpha\in\bigwedge_x^k$ the covector $L\alpha$ lies in $\bigwedge_x^{k+2}$, so

$$[H,L]\alpha = HL\alpha - LH\alpha = (k+2-n)L\alpha - (k-n)L\alpha = 2L\alpha$$

and the same computation with $\Lambda$, which lowers the degree by two, gives $[H,\Lambda]\alpha=-2\Lambda\alpha$. Bidegrees behave as claimed because $\omega$ is of type $(1,1)$ and $\Lambda$ is its pointwise adjoint, so $H$, being a scalar on each $\bigwedge_x^k$, preserves $\bigwedge_x^{p,q}$.

The third relation is the content. Since $\omega$ is a positive Hermitian form at $x$, it can be diagonalized: there is a unitary coframe $\varphi^1,\dots,\varphi^n$ of $(1,0)$-covectors at $x$ with

$$\omega = i\sum_{j=1}^n \varphi^j\wedge\bar\varphi^j$$

and with the monomials $\varphi^I\wedge\bar\varphi^J$ orthonormal, exactly the frame used in lemma 4.3.2. Write $e_j=\varphi^j\wedge(-)$ and $\bar e_j=\bar\varphi^j\wedge(-)$ for the wedge operators and $\iota_j=e_j^*$, $\bar\iota_j=\bar e_j^*$ for their pointwise adjoints, the contractions. These satisfy the Clifford relations of example 3.3.3,

$$\{e_j,\iota_k\}=\delta_{jk}, \qquad \{\bar e_j,\bar\iota_k\}=\delta_{jk}$$

with every other anticommutator among $e_j,\bar e_j,\iota_k,\bar\iota_k$ equal to zero, and in these terms

$$L = i\sum_j e_j\bar e_j, \qquad \Lambda = -i\sum_k \bar\iota_k\iota_k$$

Write $P_j=e_j\bar e_j$ and $R_k=\bar\iota_k\iota_k$, so that $[L,\Lambda]=\sum_{j,k}[P_j,R_k]$, the scalars $i$ and $-i$ cancelling.

*Step 1: the off-diagonal terms vanish.* Fix $j\neq k$. Each of $e_j$ and $\bar e_j$ anticommutes with each of $\bar\iota_k$ and $\iota_k$: the four anticommutators are $\{e_j,\iota_k\}=\delta_{jk}=0$, $\{\bar e_j,\bar\iota_k\}=\delta_{jk}=0$, and the two mixed-type ones, which vanish for all indices. Moving the pair $R_k$ past the pair $P_j$ therefore costs four sign changes, so $P_jR_k=R_kP_j$ and $[P_j,R_k]=0$.

*Step 2: the diagonal terms.* Fix $j$ and drop the index, writing $e,\bar e,\iota,\bar\iota$ and $P=e\bar e$, $R=\bar\iota\iota$. Using $\iota e = 1-e\iota$,

$$RP = \bar\iota\,\iota\,e\,\bar e = \bar\iota\,(1-e\iota)\,\bar e = \bar\iota\bar e - \bar\iota\,e\,\iota\,\bar e$$

The first term is $\bar\iota\bar e = 1-\bar e\bar\iota$. For the second, move $e$ out to the left past $\bar\iota$ and then $\bar e$ out to the left past $\iota$, each move costing a sign, and then use $\bar\iota\bar e=1-\bar e\bar\iota$ again:

$$\bar\iota\,e\,\iota\,\bar e = -e\,\bar\iota\,\iota\,\bar e = e\,\bar\iota\,\bar e\,\iota = e(1-\bar e\bar\iota)\iota = e\iota - e\bar e\bar\iota\iota$$

Combining, and writing $N = e\iota$ and $\bar N = \bar e\bar\iota$,

$$RP = (1 - \bar N) - (N - PR) = 1 - N - \bar N + PR$$

so that $[P, R] = PR - RP = N + \bar N - 1$.

*Step 3: assembling.* By Steps 1 and 2,

$$[L, \Lambda] = \sum_j \bigl(N_j + \bar N_j - 1\bigr) = \sum_j (N_j + \bar N_j) \ - \ n$$

The operator $N_j = e_j\iota_j$ acts on a monomial $\varphi^I \wedge \bar\varphi^J$ by removing the factor $\varphi^j$ and putting it back, which is the identity when $j \in I$ and zero otherwise; likewise $\bar N_j$ is the identity when $j \in J$ and zero otherwise. On $\varphi^I \wedge \bar\varphi^J$ the sum $\sum_j (N_j + \bar N_j)$ is therefore multiplication by $|I| + |J|$, the total degree. Hence $[L, \Lambda]$ acts on $\bigwedge^k_x$ by $k - n$, which is $H$.

The three relations are those of the standard basis of $\mathfrak{sl}_2(\mathbb{C})$, so the assignment $e \mapsto L$, $f \mapsto \Lambda$, $h \mapsto H$ extends to a Lie algebra homomorphism, completing the proof.

The representation is compatible with the metric, since $\Lambda = L^*$ and $H^* = H$; and it is compatible with the bigrading, since $L$ and $\Lambda$ move $p$ and $q$ in lockstep. The first will convert representation-theoretic statements into orthogonality statements, and the second will convert them into statements about Hodge structures. Both are used below.

**Example 6.1.3: The Lefschetz Triple in Dimensions One and Two**

Take a unitary coframe as in the proof above and write $\theta_j = i\,\varphi^j \wedge \bar\varphi^j$, so that $\omega = \theta_1 + \cdots + \theta_n$, each $\theta_j$ is real of degree 2, distinct $\theta_j$ commute, and $\theta_j \wedge \theta_j = 0$. In this notation $L$ is $\sum_j \theta_j \wedge (-)$ and $\Lambda\theta_j = 1$, $\Lambda\varphi^j = \Lambda\bar\varphi^j = 0$.

*The case $n = 1$.* Here $\bigwedge^\bullet_x$ has dimensions $1, 2, 1$ in degrees $0, 1, 2$, and the $H$-eigenvalues are $-1$ on $\mathbb{C}$, $0$ on $\bigwedge^1_x = \mathbb{C}\varphi^1 \oplus \mathbb{C}\bar\varphi^1$, and $+1$ on $\bigwedge^2_x = \mathbb{C}\theta_1$. The relation $[L, \Lambda] = H$ is visible on the constant 1: $\Lambda 1 = 0$ and $L1 = \theta_1$, so $[L, \Lambda]1 = -\Lambda\theta_1 = -1 = H1$. The primitive spaces are $P^0_x = \mathbb{C}$ and $P^1_x = \bigwedge^1_x$, the latter because $\Lambda$ lands in degree $-1$; and $\bigwedge^2_x = L\,P^0_x$. As an $\mathfrak{sl}_2(\mathbb{C})$-module, $\bigwedge^\bullet_x$ is the sum of the two-dimensional irreducible spanned by 1 and $\theta_1$ and two copies of the trivial module, spanned by $\varphi^1$ and by $\bar\varphi^1$.

*The case $n = 2$.* The dimensions are $1, 4, 6, 4, 1$, totalling 16, and the $H$-eigenvalues on $\bigwedge^k_x$ are $k - 2$, running from $-2$ to 2. The primitive spaces are $P^0_x = \mathbb{C}$, $P^1_x = \bigwedge^1_x$ of dimension 4, and

$$P^2_x = \ker\bigl(\Lambda\colon \ \textstyle\bigwedge^2_x \to \bigwedge^0_x\bigr)$$

of dimension 5, since $\Lambda$ is onto in this degree ($\Lambda\omega = 2$). A basis of $P^2_x$ is

$$\varphi^1 \wedge \varphi^2, \qquad \bar\varphi^1 \wedge \bar\varphi^2, \qquad \varphi^1 \wedge \bar\varphi^2, \qquad \varphi^2 \wedge \bar\varphi^1, \qquad \theta_1 - \theta_2$$

The first two are of type $(2, 0)$ and $(0, 2)$ and are primitive because $\Lambda$ needs one factor of each of $\varphi^j$ and $\bar\varphi^j$ to act; the next two are of type $(1, 1)$ and are primitive for the same reason; the last is of type $(1, 1)$ and is primitive because $\Lambda\theta_1 = \Lambda\theta_2 = 1$. So $P^{2,0}_x$ and $P^{0,2}_x$ are lines and $P^{1,1}_x$ has dimension 3, one less than $\bigwedge^{1,1}_x$, the missing direction being $\mathbb{C}\omega$. Since $\omega \wedge \omega = 2\theta_1\theta_2 \neq 0$, the line $\mathbb{C}\omega = L\,P^0_x$ meets $P^2_x$ only in 0, so

$$\textstyle\bigwedge^2_x = L\,P^0_x \oplus P^2_x, \qquad \bigwedge^3_x = L\,P^1_x, \qquad \bigwedge^4_x = L^2\,P^0_x$$

by comparison of dimensions ($6 = 1+5$, $4 = 4$, $1 = 1$). In the top degree there is no term $L\,P_x^2$, since $L$ annihilates $P_x^2$: each of the first four basis covectors repeats an index against $\theta_1$ and against $\theta_2$, and $(\theta_1+\theta_2)\wedge(\theta_1-\theta_2) = 0$ because $\theta_j \wedge \theta_j = 0$ and the $\theta_j$ commute. This is the vanishing recorded in part (3) of theorem 6.1.4. As a module, $\bigwedge_x^\bullet$ is one copy of the three-dimensional irreducible (generated by 1), four copies of the two-dimensional one (generated by $\bigwedge_x^1$) and five trivial ones (generated by $P_x^2$), and $3+8+5 = 16$ as it must.

The pattern in both cases is the same: each primitive covector generates an irreducible module by repeated application of $L$, and the whole exterior algebra is the sum of those modules. That is a general fact, and it is a consequence of the classification of finite-dimensional $\mathfrak{sl}_2(\mathbb{C})$-modules rather than of anything about forms. The classification is not redeveloped here. Its statement, in the form used below, is that a finite-dimensional complex representation of $\mathfrak{sl}_2(\mathbb{C})$ is a direct sum of irreducibles; that the irreducibles are classified by their highest weight $m \geq 0$, with $V_m$ of dimension $m+1$ and $h$-eigenvalues $m, m-2, \dots, -m$ each occurring once; and that in $V_m$ a vector $v_0$ of weight $-m$ is annihilated by $f$, the vectors $v_j = e^j v_0$ for $0 \leq j \leq m$ form a basis, $e^{m+1}v_0 = 0$, and a weight vector $v$ of weight $\mu$ satisfies $e^j v \neq 0$ exactly when $\mu + 2j \leq m$. Complete reducibility, the classification of the irreducibles of $\mathrm{SU}(2)$ and hence, after complexification, of $\mathfrak{sl}_2(\mathbb{C})$, the highest weight theorem and the decomposition into irreducibles are all proved in *Lie Groups* [32].

Feeding the relations of proposition 6.1.2 into that classification and reading off what it says about $\bigwedge_x^\bullet$ gives the decomposition the rest of the chapter runs on.

**Theorem 6.1.4: The Lefschetz Decomposition of the Exterior Algebra**

Let $(X, \omega)$ be a Hermitian manifold of complex dimension $n$ and let $x \in X$. Set $P_x^j = 0$ for $j < 0$. Then

1. $P_x^j = 0$ for $j > n$, and for $0 \leq k \leq n$,
$$P_x^k = \ker\Big(L^{n-k+1}\colon \textstyle\bigwedge_x^k \to \bigwedge_x^{2n-k+2}\Big)$$
2. for $0 \leq k \leq n$ the map $L^{n-k}\colon \bigwedge_x^k \to \bigwedge_x^{2n-k}$ is an isomorphism;
3. for every $k$,
$$\textstyle\bigwedge_x^k = \bigoplus_{r\geq 0} L^r\, P_x^{k-2r}$$
the summands are mutually orthogonal, $L^r$ is injective on $P_x^{k-2r}$ for each $r \geq \max(0,\,k-n)$, and $L^r\,P_x^{k-2r} = 0$ for $0 \leq r < k-n$;
4. the decomposition respects the bigrading: $P_x^k = \bigoplus_{p+q=k} P_x^{p,q}$ and $\bigwedge_x^{p,q} = \bigoplus_{r\geq 0} L^r\, P_x^{p-r,q-r}$.

***Proof*:**

By proposition 6.1.2 the space $\bigwedge_x^\bullet$ is a finite-dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$ in which $\bigwedge_x^k$ is exactly the weight space of weight $k-n$, since $H$ acts there by $k-n$. Decompose it into irreducibles, $\bigwedge_x^\bullet = \bigoplus_i W_i$ with $W_i \cong V_{m_i}$, using complete reducibility [32]. In $W_i$ the weight $\mu$ occurs with multiplicity one when $|\mu| \leq m_i$ and $\mu \equiv m_i$ modulo 2, and not at all otherwise; the vectors killed by $\Lambda$ are exactly the multiples of the weight-$(-m_i)$ vector; and $L^j$ is injective on the weight-$\mu$ line precisely when $\mu + 2j \le m_i$.

*Step 1: primitive covectors and the two descriptions.* Let $\alpha \in \bigwedge_x^k$ and write $\alpha = \sum_i \alpha_i$ with $\alpha_i \in W_i$. Each $\alpha_i$ has weight $k-n$, so $\alpha_i \ne 0$ forces $m_i \ge |k-n|$ and $m_i \equiv k-n$ modulo 2.

If $\alpha$ is primitive then each $\alpha_i$ is, since $\Lambda$ preserves each $W_i$; so each nonzero $\alpha_i$ is a lowest weight vector, whence $k-n = -m_i \le 0$. This gives $P_x^j = 0$ for $j > n$ and, for $k \le n$, $m_i = n-k$ for every nonzero component. Then $L^{n-k+1}\alpha_i = 0$, because $(k-n)+2(n-k+1) = n-k+2 > n-k = m_i$; so $\alpha \in \ker L^{n-k+1}$.

Conversely let $k \le n$ and $L^{n-k+1}\alpha = 0$. Each component satisfies $L^{n-k+1}\alpha_i = 0$, which for $\alpha_i \ne 0$ means $n-k+2 > m_i$, that is $m_i \le n-k+1$. Combined with $m_i \ge n-k$ and $m_i \equiv n-k$ modulo 2 this forces $m_i = n-k$, so $\alpha_i$ has weight $-m_i$ and is annihilated by $\Lambda$. Hence $\alpha$ is primitive, and part (1) is proved.

*Step 2: the isomorphism in complementary degrees.* Let $k \le n$. The operator $L^{n-k}$ carries the weight-$(k-n)$ space of $W_i$ to its weight-$(n-k)$ space. If $m_i < n-k$ then $W_i$ has neither of these weight spaces and there is nothing to check. If $m_i \ge n-k$, both are lines, and $L^{n-k}$ is injective on the first because $(k-n)+2(n-k) = n-k \le m_i$; an injective map between lines is an isomorphism. Summing over $i$ gives part (2).

*Step 3: the decomposition.* Let $W_i \cong V_{m_i}$ with lowest weight vector $v_i$, which has weight $-m_i$ and so lies in $\bigwedge_x^{n-m_i}$ and is primitive. The basis $v_i, Lv_i, \ldots, L^{m_i}v_i$ of $W_i$ consists of covectors of degrees $n-m_i, n-m_i+2, \ldots, n+m_i$, one in each. So $W_i \cap \bigwedge_x^k$ is $\mathbb{C}\, L^r v_i$ with $k = n-m_i+2r$ when $0 \le r \le m_i$, and $0$ otherwise. Since the $v_i$ span $P_x^\bullet$ by Step 1 (a primitive covector has all its components primitive, hence a multiple of the relevant $v_i$), summing over $i$ gives $\bigwedge_x^k = \sum_{r\ge 0} L^r P_x^{k-2r}$, and the sum is direct because the summand $L^r P_x^{k-2r}$ collects exactly the $W_i$ with $m_i = n-k+2r$. For the injectivity and the vanishing, note that $P_x^{k-2r}$ is spanned by those $v_i$ and that $L^r v_i \ne 0$ exactly when $r \le m_i$, which for $m_i = n-k+2r$ says precisely $r \ge k-n$. So for $r \ge \max(0,\, k-n)$ the operator $L^r$ is injective on each line $\mathbb{C}v_i$ making up $P_x^{k-2r}$, hence on $P_x^{k-2r}$ itself, the $W_i$ being independent and stable under $L$; while for $0 \le r < k-n$ the same criterion gives $L^r v_i = 0$ for every such $v_i$, that is $L^r\, P_x^{k-2r} = 0$.

For orthogonality, take $r > s$, $\pi \in P_x^{k-2r}$ and $\pi' \in P_x^{k-2s}$. The submodule generated by $\pi$ is spanned by the $L^j\pi$, so $\Lambda^s L^r \pi$, having degree $k-2s$, is a multiple of $L^{r-s}\pi$, say $\Lambda^s L^r \pi = c\, L^{r-s}\pi$. Using $\Lambda = L^*$ twice,

$$\langle L^r\pi, L^s\pi'\rangle = \langle \Lambda^s L^r \pi, \pi'\rangle = c\,\langle L^{r-s}\pi, \pi'\rangle = c\,\langle \pi, \Lambda^{r-s}\pi'\rangle = 0$$

since $r-s \ge 1$ and $\pi'$ is primitive. This is part (3).

*Step 4: bidegrees.* The operator $\Lambda$ shifts the bidegree by $(-1,-1)$, so it maps $\bigwedge_x^{p,q}$ into $\bigwedge_x^{p-1,q-1}$ and its kernel in $\bigwedge_x^k$ is the direct sum of its kernels in the summands $\bigwedge_x^{p,q}$ with $p+q = k$; this is $P_x^k = \bigoplus_{p+q=k} P_x^{p,q}$. Applying $L^r$, which shifts the bidegree by $(r,r)$, to the decomposition of part (3) and separating types gives $\bigwedge_x^{p,q} = \bigoplus_{r\ge 0} L^r P_x^{p-r,q-r}$, completing the proof.

Part (2) is the Hard Lefschetz theorem at the level of the exterior algebra at a point, and it is gave entirely with representation theory: in an $\mathfrak{sl}_2(\mathbb{C})$-module the weight spaces of $\mu$ and $-\mu$ have the same dimension, and the power of $L$ carrying one to the other realizes the isomorphism. What section 6.2 has to supply is not a new argument of this kind but a way of transporting this one to cohomology.

Part (3) also settles the bookkeeping, since $\bigoplus_{r\ge 1} L^r P_x^{k-2r} = L \bigwedge_x^{k-2}$ turns the decomposition into

$\bigwedge_x^k = L\bigwedge_x^{k-2} \oplus P_x^k$, from which the primitive dimensions follow by subtraction (exercise 6.1.1 (5)). For $n = 2$ this returns the numbers found by hand in example 6.1.3: $\dim P_x^2 = \binom{4}{2} - \binom{4}{0} = 5$ and $\dim P_x^{1,1} = \binom{2}{1}\binom{2}{1} - \binom{2}{0}\binom{2}{0} = 3$.

Everything so far is linear algebra at a point, hence available on any Hermitian manifold and on any of its differential forms. Cohomology is another matter: $L$ descends to cohomology on any complex manifold with a closed $\omega$, since wedging with a closed form preserves closedness and exactness, but $\Lambda$ has no reason to. The device that carries the whole triple across is the harmonic representative, and it works precisely because theorem 3.3.5 makes the Laplacian commute with $L$ and $\Lambda$; section 6.2 carries out that transport.

The transport is also what licenses the use of harmonic representatives already made in section 5.2. The geometric polarization proposition 5.2.8 was evaluated on a harmonic representative of a primitive class, and the evaluation used Weil's identity, which is a statement about forms that are primitive at each point. That the two notions of primitivity agree on cohomology is part (1) of corollary 6.2.4, and they agree because $L$ commutes with the Laplacian, which is a Kähler statement and fails without that hypothesis.

What remains is that identity. Theorem 5.2.7 computes the Hodge star of a primitive form of type $(p, q)$ as an explicit multiple of $L^{n-k}$ applied to it, and section 5.2 deferred its proof here on the grounds that the proof is a statement about the exterior algebra at a point. The statement proved below is exactly that pointwise assertion; applied at every point of a compact Kähler manifold it is theorem 5.2.7.

**Theorem 6.1.5: Weil's Identity on the Exterior Algebra**

Let $(X, \omega)$ be a Hermitian manifold of complex dimension $n$, let $x \in X$, and let $\alpha \in P_x^{p,q}$ be a primitive covector of type $(p, q)$ with $k = p + q \leq n$. Then

$$\star\,\alpha = (-1)^{k(k+1)/2}\, i^{p-q}\, \frac{1}{(n-k)!}\, L^{n-k}\alpha$$

where $\star$ is the Hodge star of the metric.

***Proof of theorem 5.2.7, deferred from section 5.2*:**
Both sides are algebraic in $\alpha$, so it is enough to argue at the fixed point $x$, and theorem 5.2.7 follows by applying the result at every point. Choose a unitary coframe $\varphi^1, \ldots, \varphi^n$ at $x$ as in the proof of proposition 6.1.2, so that the monomials $\varphi^I \wedge \bar\varphi^J$ are orthonormal, and set

$$\theta_j = i\,\varphi^j \wedge \bar\varphi^j, \qquad \theta_T = \bigwedge\nolimits_{j\in T}\theta_j \quad (T \subseteq \{1, \ldots, n\})$$

The $\theta_j$ are real of degree 2, so they commute; $\theta_j \wedge \theta_j = 0$; the $\theta_T$ are orthonormal; $\theta_T \wedge \theta_{T'} = \theta_{T\cup T'}$ when $T \cap T' = \emptyset$ and 0 otherwise; and

$$\omega = \sum_j \theta_j, \qquad \omega^m = m! \sum_{|T|=m} \theta_T, \qquad \mathrm{vol} = \frac{\omega^n}{n!} = \theta_{\{1,\ldots,n\}}$$

Write $m_0 = n - k$.

*Step 1: reduction to a pairing identity.* The Hodge star is the $\mathbb{C}$-linear operator determined by

$$\gamma \wedge \star\bar\delta = \langle \gamma, \delta\rangle\,\mathrm{vol} \qquad \text{for all } \gamma, \delta \in \textstyle\bigwedge_x^k$$

the polarized form of the identity $\alpha \wedge \star\bar{\alpha} = |\alpha|^2 \operatorname{vol}$ used in section 5.2; it determines $\star\bar{\delta}$ uniquely because the wedge pairing $\bigwedge_x^k \times \bigwedge_x^{2n-k} \to \bigwedge_x^{2n}$ is nondegenerate. Since $\star$ is the $\mathbb{C}$-linear extension of a real operator, $\star\bar{\alpha} = \overline{\star\alpha}$, so the asserted formula is equivalent to

$$\star\bar{\alpha} = (-1)^{k(k+1)/2}\, i^{q-p}\, \frac{1}{m_0!}\, L^{m_0}\bar{\alpha}$$

and hence, by the characterization above and by multiplying through by the inverse of the scalar, to the assertion

$$\gamma \wedge \bar{\alpha} \wedge \frac{\omega^{m_0}}{m_0!} = (-1)^{k(k+1)/2}\, i^{p-q}\, \langle \gamma, \alpha\rangle \operatorname{vol} \qquad \text{for all } \gamma \in \textstyle\bigwedge_x^k \tag{6.1}$$

Both sides of (6.1) are $\mathbb{C}$-linear in $\gamma$ and conjugate-linear in $\alpha$.

*Step 2: a decomposition adapted to* $\Lambda$. For disjoint subsets $I, J \subseteq \{1, \dots, n\}$ let $R = R(I, J)$ be the complement of $I \cup J$ and let $\Theta_R$ be the span of the $\theta_T$ with $T \subseteq R$. Every monomial $\varphi^{I'} \wedge \bar{\varphi}^{J'}$ equals a unit scalar times $\varphi^I \wedge \bar{\varphi}^J \wedge \theta_T$ with $T = I' \cap J'$, $I = I' \setminus J'$ and $J = J' \setminus I'$, and this correspondence between triples $(I, J, T)$ and pairs $(I', J')$ is a bijection. So there is an orthogonal decomposition

$$\textstyle\bigwedge_x^\bullet = \bigoplus_{I \cap J = \emptyset} \varphi^I \wedge \bar{\varphi}^J \wedge \Theta_{R(I,J)}$$

Each summand is stable under $L$ and $\Lambda$, which act on it through the second factor. Indeed $\theta_j \wedge (-)$ kills the summand when $j \in I \cup J$, because the index $j$ then repeats, and $\Lambda_j := -i\, \bar{\iota}_j \iota_j$ kills it as well, because for $j \in I$ there is no factor $\bar{\varphi}^j$ for $\bar{\iota}_j$ to remove and for $j \in J$ there is no factor $\varphi^j$ for $\iota_j$ to remove; while for $j \in R$ the two contractions each pass the prefix $\varphi^I \wedge \bar{\varphi}^J$, of degree $|I| + |J|$, so the two signs cancel. Writing $L_R = \sum_{j \in R} \theta_j \wedge (-)$ and $\Lambda_R$ for its adjoint on $\Theta_R$, one therefore has, for $\eta \in \Theta_R$,

$$L(\varphi^I \wedge \bar{\varphi}^J \wedge \eta) = \varphi^I \wedge \bar{\varphi}^J \wedge L_R\eta, \qquad \Lambda(\varphi^I \wedge \bar{\varphi}^J \wedge \eta) = \varphi^I \wedge \bar{\varphi}^J \wedge \Lambda_R\eta$$

and $\Lambda_R\theta_T = \sum_{j \in T} \theta_{T \setminus j}$. In particular a covector is primitive if and only if each of its components in this decomposition is annihilated by the corresponding $\Lambda_R$.

*Step 3: distinct summands contribute nothing.* Suppose $\gamma$ lies in the summand indexed by $(I, J)$ and $\alpha$ in the summand indexed by $(I_1, J_1)$, with $(I, J) \neq (I_1, J_1)$. The right-hand side of (6.1) vanishes because the decomposition is orthogonal. For the left-hand side, expand $\gamma \wedge \bar{\alpha} \wedge \omega^{m_0}$ into monomials; each term is a scalar multiple of

$$\varphi^I \wedge \bar{\varphi}^J \wedge \theta_T \wedge \bar{\varphi}^{I_1} \wedge \varphi^{J_1} \wedge \theta_{T_1} \wedge \theta_S$$

for some $T \subseteq R(I, J)$, $T_1 \subseteq R(I_1, J_1)$ and $S$ with $|S| = m_0$. This product has degree $2n$, and a nonzero $2n$-covector must use each index exactly once as a $\varphi$ and once as a $\bar{\varphi}$. The $\varphi$-indices are $I \sqcup J_1 \sqcup T \sqcup T_1 \sqcup S$ and the $\bar{\varphi}$-indices are $J \sqcup I_1 \sqcup T \sqcup T_1 \sqcup S$, so both must exhaust $\{1, \dots, n\}$ and hence $I \sqcup J_1 = J \sqcup I_1$. Now any $j \in I$ lies in $J \sqcup I_1$ and not in $J$, so $j \in I_1$; and any $j \in I_1$ lies in $I \sqcup J_1$ and not in $J_1$, so $j \in I$. Thus $I = I_1$ and then $J = J_1$, contrary to assumption. Every term vanishes, so the left-hand side does too.

By Step 3 and the bilinearity noted in Step 1 it suffices to prove (6.1) when $\gamma$ and $\alpha$ lie in the same summand. Fix $I, J$ disjoint, put $a = |I|$, $b = |J|$, $R = R(I, J)$ and $r = |R| = n - a - b$, and take

$$\gamma = \varphi^I \wedge \bar{\varphi}^J \wedge \rho, \qquad \alpha = \varphi^I \wedge \bar{\varphi}^J \wedge \pi, \qquad \rho, \pi \in \Theta_R$$

Both $\gamma$ and $\alpha$ have total degree $k$, and inside this summand the degree-$k$ part is $\varphi^I \wedge \bar{\varphi}^J$ wedged with the span of the $\theta_T$ with $|T| = s$ for the single value $s = (k-a-b)/2$; so $\rho$ and $\pi$ are both combinations of such $\theta_T$, and $p = a+s$, $q = b+s$, $k = a+b+2s$, $m_0 = r-2s$, with $\Lambda_R\pi = 0$ by Step 2.

*Step 4: the computation inside one summand.* Write $\pi = \sum_{|T|=s} c_T\theta_T$ and $\rho = \sum_{|U|=s} d_U\theta_U$, the sums over subsets of $R$. Since $\theta_U \wedge \theta_T$ vanishes unless $U \cap T = \emptyset$, and since a term $\theta_S$ of $\omega^{m_0}$ survives only for $S = R \setminus (U \sqcup T)$,

$$\rho \wedge \bar{\pi} \wedge \frac{\omega^{m_0}}{m_0!} = \Big(\sum_{U\cap T=\emptyset} d_U\,\overline{c_T}\Big)\,\theta_R$$

using that $\bar{\theta}_T = \theta_T$. The claim is that primitivity of $\pi$ turns the constraint $U \cap T = \emptyset$ into the constraint $U = T$ at the cost of a sign:

$$\sum_{U\cap T=\emptyset} d_U\,\overline{c_T} = (-1)^s \sum_U d_U\,\overline{c_U} \tag{6.2}$$

To see this, note first what $\Lambda_R\pi = 0$ says. From $\Lambda_R\theta_T = \sum_{j\in T}\theta_{T\setminus j}$,

$$\Lambda_R\pi = \sum_{|T'|=s-1}\Big(\sum_{j\in R\setminus T'} c_{T'\cup j}\Big)\theta_{T'}$$

so primitivity says that $\sum_{j\in R\setminus T'} c_{T'\cup j} = 0$ for every $T' \subseteq R$ with $|T'| = s-1$. For $V \subseteq R$ set $\sigma_V = \sum_{T\supseteq V,\,|T|=s} c_T$. The relation just displayed is $\sigma_{T'} = 0$ for $|T'| = s-1$. Counting each $T \supseteq V$ once for every element of $T \setminus V$ gives

$$\sum_{j\in R\setminus V}\sigma_{V\cup j} = (s-|V|)\,\sigma_V$$

so a downward induction starting at $|V| = s-1$ shows $\sigma_V = 0$ whenever $|V| < s$; and $\sigma_V = c_V$ when $|V| = s$. Now fix $U$ with $|U| = s$. Since $\sum_{V\subseteq W}(-1)^{|V|}$ is 1 for $W = \emptyset$ and 0 otherwise, inclusion and exclusion gives

$$\sum_{T\cap U=\emptyset} c_T = \sum_{V\subseteq U}(-1)^{|V|}\sigma_V = (-1)^s\sigma_U = (-1)^s c_U$$

only the term $V = U$ surviving. Conjugating and summing against $d_U$ yields (6.2), and the right-hand side of (6.2) is $(-1)^s\langle\rho,\pi\rangle$.

It remains to restore the prefix. Since $\rho$ has even degree it commutes past $\bar{\varphi}^I \wedge \varphi^J$, so

$$\gamma \wedge \bar{\alpha} = \varphi^I \wedge \bar{\varphi}^J \wedge \bar{\varphi}^I \wedge \varphi^J \wedge \rho \wedge \bar{\pi}$$

Moving $\bar{\varphi}^I$ left past $\bar{\varphi}^J$ costs $(-1)^{ab}$ and then $\bar{\varphi}^J \wedge \varphi^J = (-1)^{b^2}\varphi^J \wedge \bar{\varphi}^J = (-1)^b\varphi^J \wedge \bar{\varphi}^J$, so the prefix is $(-1)^{ab+b}\,\varphi^I \wedge \bar{\varphi}^I \wedge \varphi^J \wedge \bar{\varphi}^J$. For a set $I = \{i_1 < \cdots < i_a\}$, bringing each $\bar{\varphi}^{i_t}$ next to its $\varphi^{i_t}$ costs $a-1, a-2, \ldots, 0$ transpositions of odd covectors, and $\varphi^j \wedge \bar{\varphi}^j = -i\,\theta_j$, so

$$\varphi^I \wedge \bar{\varphi}^I = (-1)^{a(a-1)/2}(-i)^a\,\theta_I$$

and likewise for $J$. Now wedge with $\omega^{m_0}/m_0!$. The prefix $\theta_I \wedge \theta_J$ kills every term $\theta_S$ of $\omega^{m_0}$ with $S$ meeting $I \cup J$, so only the terms already accounted for in the $\Theta_R$ computation survive, and (6.2) applies verbatim. Using $\theta_I \wedge \theta_J \wedge \theta_R = \mathrm{vol}$ together with $\langle \gamma, \alpha \rangle = \langle \rho, \pi \rangle$, which holds because the $\varphi^I \wedge \bar{\varphi}^J \wedge \theta_U$ are orthonormal,

$$\gamma \wedge \bar{\alpha} \wedge \frac{\omega^{m_0}}{m_0!} = \varepsilon\,(-1)^s \,\langle \gamma, \alpha \rangle\,\mathrm{vol}, \qquad \varepsilon = (-1)^{ab+b+a(a-1)/2+b(b-1)/2}\,(-i)^{a+b}$$

*Step 5: the sign.* It remains to check that $\varepsilon\,(-1)^s = (-1)^{k(k+1)/2} i^{p-q}$. Since $i^{a+b} = i^{a-b}\,i^{2b} = (-1)^b i^{a-b}$,

$$(-i)^{a+b} = (-1)^{a+b} i^{a+b} = (-1)^a\,i^{a-b} = (-1)^a\,i^{p-q}$$

so $\varepsilon\,(-1)^s = (-1)^N i^{p-q}$ with $N = ab + b + \frac{a(a-1)}{2} + \frac{b(b-1)}{2} + a + s$. On the other side, with $k = a + b + 2s$,

$$\frac{k(k+1)}{2} = \frac{(a+b)(a+b+1)}{2} + 2s(a+b) + 2s^2 + s \equiv \frac{(a+b)(a+b+1)}{2} + s \pmod 2$$

and expanding the first term,

$$\frac{(a+b)(a+b+1)}{2} = \frac{a(a+1)}{2} + \frac{b(b+1)}{2} + ab = \frac{a(a-1)}{2} + a + \frac{b(b-1)}{2} + b + ab$$

so $k(k+1)/2 \equiv N$ modulo 2. Hence $\varepsilon(-1)^s = (-1)^{k(k+1)/2} i^{p-q}$, which is (6.1) inside a single summand. With Step 3 this establishes (6.1) for every $\gamma \in \bigwedge_x^k$, and by Step 1 that is the asserted formula for $\star\alpha$, completing the proof.

The identity says that on primitive covectors the Hodge star, which is defined by the metric and the orientation, is computed by the Lefschetz operator, which is defined by $\omega$ alone, with a correction that depends only on the type. That is why it converts the cup product into an inner product: inserting it into $\alpha \wedge \star\bar{\alpha} = |\alpha|^2\,\mathrm{vol}$ turns $\int_X \alpha \wedge \bar{\alpha} \wedge \omega^{n-k}$, a topological quantity when the classes are fixed, into a positive multiple of the $L^2$-norm of the harmonic representative. Section 5.2 used exactly that step, and section 6.2 uses it again to prove the positivity in general.

One of the two debts left by chapter 5 is thereby settled. Proposition 5.2.8 carried a proof labelled as key ideas for the single reason that its second Hodge-Riemann relation rested on theorem 5.2.7, and that identity now has a proof.

### Exercise 6.1.1

1. Verify $[H, L] = 2L$ and $[H, \Lambda] = -2\Lambda$ directly from the fact that $L$ raises the total degree by 2 and $\Lambda$ lowers it by 2. Then show that in the decomposition of $\bigwedge_x^\bullet$ into irreducible $\mathfrak{sl}_2(\mathbb{C})$-modules no summand $V_m$ with $m > n$ occurs, and that the multiplicity of $V_m$ is $\dim \bigwedge_x^{n-m} - \dim \bigwedge_x^{n-m-2}$.

2. Let $\alpha \in P_x^k$ be primitive with $k \le n$, and set $m = n - k$. Show first that for a covector $v$ of $H$-weight $\mu$ and every $j \ge 1$,

$$[\Lambda, L^j]\,v = -\big(j\mu + j(j-1)\big)L^{j-1}v$$

(expand $[\Lambda, L^j] = \sum_{t=0}^{j-1} L^t[\Lambda, L]L^{j-1-t}$ and use $[\Lambda, L] = -H$). Deduce that $\Lambda L^j\alpha = j(m - j + 1)L^{j-1}\alpha$, that

$$\|L^r\alpha\|^2 = r!\,\frac{m!}{(m-r)!}\,\|\alpha\|^2 \qquad (0 \le r \le m)$$

and that $L^{m+1}\alpha = 0$. Which part of theorem 6.1.4 does the last statement recover?

3. Show that every covector of type $(p, 0)$ is primitive. Using theorem 6.1.5, deduce that $\star\alpha = (-1)^{n(n+1)/2} i^n \alpha$ for every $\alpha \in \bigwedge_x^{n,0}$, and verify this directly for $n = 1$ by computing $\star\varphi^1$ from the defining property of the Hodge star in a unitary coframe.

4. Let $X$ be a compact Kähler surface, so $n = 2$, and let $\beta$ be a real primitive form of type $(1, 1)$. Use theorem 6.1.5 to compute $\star\beta$, and deduce that

$$\beta \wedge \beta = -|\beta|^2 \operatorname{vol}$$

pointwise. Compare with part (3) of lemma 4.3.2 and say which of the two arguments uses the dimension of $X$.

5. Prove that $\bigwedge_x^k = L \bigwedge_x^{k-2} \oplus P_x^k$ for every $k$, and deduce that for $k \leq n + 1$

$$\dim P_x^k = \binom{2n}{k} - \binom{2n}{k-2}, \qquad \dim P_x^{p,q} = \binom{n}{p}\binom{n}{q} - \binom{n}{p-1}\binom{n}{q-1}$$

Compute the primitive dimensions and the multiplicities of the irreducibles for $n = 3$, and check that the total dimension comes out to $2^6$.

6. Identify precisely which statements of this section require the Kähler condition and which hold on an arbitrary Hermitian manifold. Name the single input that fails without $d\omega = 0$, and explain what would go wrong if one tried to transport the triple to the cohomology of a compact Hermitian manifold such as the Hopf surface.

## 6.2 The Hard Lefschetz Theorem

Section 6.1 worked at a single point throughout: the triple $(L, \Lambda, H)$ acts on the exterior algebra of a cotangent space, and the classification of finite-dimensional representations of $\mathfrak{sl}_2$ splits that algebra into primitive pieces. Nothing there used $d\omega = 0$, and nothing there reached cohomology. This section moves the action off the point, making $H^*(X, \mathbb{C})$ itself a module over $\mathfrak{sl}_2(\mathbb{C})$, and then reads that module.

With $H^*(X, \mathbb{C})$ a representation of $\mathfrak{sl}_2$ and $H^k$ sitting in weight $k - n$, three statements become weight bookkeeping: that $L^{n-k}$ carries $H^k$ isomorphically onto $H^{2n-k}$, that cohomology is the direct sum of the images of the primitive parts under powers of $L$, and that in degree $k \leq n$ the kernel of $\Lambda$ is the primitive cohomology of definition 5.2.5. The positivity that proposition 5.2.8 deferred then follows from Weil's identity, proved as theorem 6.1.5.

The bookkeeping is cleaner done once, abstractly, than three times on cohomology, so the representation theory comes first. Three facts about a finite-dimensional $\mathfrak{sl}_2$-module are needed, and all three are read off the classification of the irreducible modules rather than proved afresh: that the raising operator, applied enough times, carries a weight space isomorphically onto its mirror image; that every weight space is generated from the kernel of the lowering operator; and that being killed by the lowering operator is detected by a power of the raising operator. Write $\mathfrak{sl}_2(\mathbb{C})$ for the Lie algebra with basis $e, f, h$ and brackets

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h$$

which is the abstract shape of the relations that proposition 6.1.2 verified for $(L, \Lambda, H)$.

**Lemma 6.2.1: Weight Spaces and Powers of the Raising Operator**

Let $V$ be a finite-dimensional complex representation of $\mathfrak{sl}_2(\mathbb{C})$. Then $h$ acts semisimply with integer eigenvalues, so that $V = \bigoplus_{j\in\mathbb{Z}} V_j$ with $V_j$ the $j$-eigenspace of $h$. Write $P_j = V_j \cap \ker f$. Then:

1. $P_j = 0$ for every $j > 0$;
2. for every integer $m \geq 0$ the map $e^m\colon V_{-m} \to V_m$ is an isomorphism;
3. for every integer $j$,
$$V_j = \bigoplus_{r\geq\max(0,\,j)} e^r\big(P_{j-2r}\big)$$
and $e^r$ is injective on $P_{j-2r}$ for each $r$ in that range, while $e^r(P_{j-2r}) = 0$ for $0 \leq r < j$;
4. for $m \geq 0$ and $v \in V_{-m}$, one has $fv = 0$ if and only if $e^{m+1}v = 0$.

***Proof*:**
Every finite-dimensional complex representation of $\mathfrak{sl}_2(\mathbb{C})$ is a direct sum of irreducible ones, and for each integer $N \geq 0$ there is exactly one irreducible module $V(N)$ of dimension $N+1$ up to isomorphism. On $V(N)$ the operator $h$ acts semisimply with eigenvalues $N, N-2, \ldots, -N$, each eigenspace a line; $e$ carries $V(N)_j$ isomorphically onto $V(N)_{j+2}$ for $-N \leq j \leq N-2$ and annihilates $V(N)_N$; and $f$ carries $V(N)_j$ isomorphically onto $V(N)_{j-2}$ for $-N+2 \leq j \leq N$ and annihilates $V(N)_{-N}$. Complete reducibility and this classification are proved in *Lie Groups* [32], where the irreducible representations of SU(2) are determined and the passage to $\mathfrak{sl}_2(\mathbb{C})$ is by complexification; nothing about them is re-derived here.

Fix a decomposition $V = \bigoplus_i V^{(i)}$ into irreducibles. Since $h$ acts semisimply on each summand, it does on $V$, and $V_j = \bigoplus_i V_j^{(i)}$ for every $j$; consequently $P_j = \bigoplus_i \big(V_j^{(i)} \cap \ker f\big)$, and each of the four assertions is a statement about subspaces and maps that respect the decomposition. It therefore suffices to prove them for $V = V(N)$.

1. In $V(N)$ the operator $f$ annihilates only the line $V(N)_{-N}$, so $P_j = 0$ unless $j = -N$, and $-N \leq 0$.
2. The space $V(N)_{-m}$ is nonzero exactly when $m \leq N$ and $N-m$ is even, and the same condition characterizes $V(N)_m \neq 0$; so either both are zero, in which case there is nothing to prove, or both are lines. In the latter case $e^m$ restricted to $V(N)_{-m}$ is the composite
$$V(N)_{-m} \to V(N)_{-m+2} \to \cdots \to V(N)_m$$
of $m$ maps, each of which starts in a weight $j \leq m-2 \leq N-2$ and is therefore an isomorphism of lines. Hence $e^m\colon V(N)_{-m} \to V(N)_m$ is an isomorphism.
3. By part (1) the only $r$ for which $P_{j-2r}$ can be nonzero is the one with $j-2r = -N$, that is $r = (j+N)/2$, and this is a nonnegative integer exactly when $j \geq -N$ and $j \equiv N$ modulo 2. Suppose first that $V(N)_j \neq 0$, so that $|j| \leq N$ and $j \equiv N$. Then $r = (j+N)/2$ satisfies $r \geq 0$ and $r \geq j$ (the latter because $r - j = (N-j)/2 \geq 0$), so it lies in the stated range,

and the composite argument of part (2) makes $e^r$ carry $V(N)_{-N}$ isomorphically onto $V(N)_{-N+2r} = V(N)_j$, all intermediate weights being at most $j \le N$. So the right-hand side is $V(N)_j$ and $e^r$ is injective on $P_{j-2r}$.

If instead $V(N)_j = 0$, there are two cases. When $j < -N$ or $j \not\equiv N$ there is no admissible $r$ at all and both sides vanish. When $j > N$ the admissible index is $r = (j+N)/2 < j$, which the range $r \ge \max(0, j)$ excludes, so the right-hand side is again zero. The last assertion, that $e^r(P_{j-2r}) = 0$ for $0 \le r < j$, holds throughout that range: for the admissible index one has $r = (j+N)/2 > N$ whenever $j > N$, and $e^{N+1}$ annihilates $V(N)$; for every other $r$ the space $P_{j-2r}$ is itself zero.

4. Both conditions are trivially satisfied when $V(N)_{-m} = 0$, so assume $m \le N$ and $N \equiv m$ modulo 2, and take $0 \ne v \in V(N)_{-m}$. Then $fv = 0$ holds exactly when $V(N)_{-m}$ is the bottom line, that is when $N = m$. On the other side, $e^{m+1}v$ lies in $V(N)_{m+2}$, which is nonzero exactly when $m + 2 \le N$; and when it is nonzero the composite argument of part (2) shows $e^{m+1}$ is injective on $V(N)_{-m}$, so $e^{m+1}v \ne 0$. Hence $e^{m+1}v = 0$ holds exactly when $N < m + 2$, which given $N \ge m$ and $N \equiv m$ means $N = m$. The two conditions coincide, completing the proof.

Part (2) is the abstract shape of the theorem this section is named for, and part (3) the abstract shape of its decomposition; part (4) is what reconciles the two descriptions of primitivity, one by a vanishing power of the raising operator and one by the lowering operator. Only part (1) has no geometric counterpart worth naming: it records that primitive classes live in degrees at most $n$, which is why definition 5.2.5 was stated only there.

The module itself is not yet in hand. The triple acts on forms at every point, but only $L$ has any reason to descend to cohomology: wedging with the closed form $\omega$ preserves closed forms and exact ones, whereas $\Lambda$ is built from the metric and bears no relation to $d$. What carries the other two across is the harmonic representative, and the Kähler condition is spent there and nowhere else in this section.

**Proposition 6.2.2: The Lefschetz Triple on Cohomology**

Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$, let $\mathcal{H}^k$ be the space of harmonic $k$-forms for the associated metric, and set $\mathcal{H}^\bullet = \bigoplus_{k=0}^{2n} \mathcal{H}^k$. Then:

1. $L$, $\Lambda$ and $H$ carry $\mathcal{H}^\bullet$ into itself, and $\mathcal{H}^\bullet$ is a finite-dimensional $\mathfrak{sl}_2(\mathbb{C})$-module under $e = L$, $f = \Lambda$, $h = H$, in which $\mathcal{H}^k$ is the weight space of weight $k - n$;

2. the Hodge isomorphism $\mathcal{H}^k \to H^k(X, \mathbb{C})$ of theorem 4.2.1 transports that structure to cohomology: $H^\bullet(X, \mathbb{C}) = \bigoplus_{k=0}^{2n} H^k(X, \mathbb{C})$ is a finite-dimensional $\mathfrak{sl}_2(\mathbb{C})$-module in which $H^k(X, \mathbb{C})$ is the weight space of weight $k - n$, the operator $e$ acts as cup product with $[\omega]$, and $f$ acts by $[\alpha] \mapsto [\Lambda\alpha]$ on harmonic representatives, with $f[\alpha] = 0$ if and only if $\Lambda\alpha = 0$ at every point;

3. for $0 \le k \le n$ and $\alpha \in \mathcal{H}^k$, the class $[\alpha]$ lies in $H^k(X, \mathbb{C})_{\mathrm{prim}}$ in the sense of definition 5.2.5 if and only if $\alpha$ is a primitive form, that is $\Lambda\alpha = 0$ at every point.

***Proof*:**
*Step 1: the triple preserves harmonic forms.* By theorem 3.3.5 the Laplacian of a compact Kähler manifold commutes with $L$ and with $\Lambda$, and it commutes with $H$ as well, since it preserves the degree while $H$ is a scalar in each degree. So $\Delta\alpha = 0$ forces $\Delta L\alpha = L\Delta\alpha = 0$, and likewise for $\Lambda\alpha$ and $H\alpha$: all three operators carry $\mathcal{H}^\bullet$ into itself. This is the one step that uses $d\omega = 0$, through the Kähler identities that theorem 3.3.5 rests on.

*Step 2: the module structure on harmonic forms.* The relations $[H, L] = 2L$, $[H, \Lambda] = -2\Lambda$ and $[L, \Lambda] = H$ of proposition 6.1.2 hold at every point, hence hold between the induced operators on differential forms, hence hold on the subspace $\mathcal{H}^\bullet$ that Step 1 shows is preserved by all three. So $e \mapsto L$, $f \mapsto \Lambda$, $h \mapsto H$ makes $\mathcal{H}^\bullet$ a representation of $\mathfrak{sl}_2(\mathbb{C})$, finite dimensional because each $\mathcal{H}^k$ is (theorem 4.2.1). By definition 6.1.1 the operator $H$ is multiplication by $k - n$ on $\mathcal{H}^k$, so $\mathcal{H}^k$ lies in the weight space of weight $k - n$; the degrees $0 \leq k \leq 2n$ give distinct weights $k - n$ and their spaces exhaust $\mathcal{H}^\bullet$, so the inclusion is an equality. This is part (1).

*Step 3: transport.* By theorem 4.2.1 every de Rham class has exactly one harmonic representative, so $\alpha \mapsto [\alpha]$ is an isomorphism $\mathcal{H}^k \to H^k(X, \mathbb{C})$; carrying the operators of Step 2 through it makes $H^\bullet(X, \mathbb{C})$ a module with $H^k(X, \mathbb{C})$ the weight space of weight $k - n$. The transported $e$ is cup product with $[\omega]$: for $\alpha$ harmonic the form $L\alpha = \omega \wedge \alpha$ is harmonic by Step 1 and represents $[\omega] \cup [\alpha]$, both factors being closed. The description of $f$ is the definition of the transported operator, and for the last clause of part (2), $\Lambda\alpha$ is harmonic by Step 1, while a harmonic form whose class vanishes is exact and therefore orthogonal to itself in the decomposition of theorem 4.2.1, hence zero.

*Step 4: primitive classes and primitive forms.* Let $0 \leq k \leq n$ and $\alpha \in \mathcal{H}^k$. By Step 1 the form $L^{n-k+1}\alpha$ is harmonic, and it represents $L^{n-k+1}[\alpha]$; by the vanishing criterion of Step 3 it is therefore zero as a form exactly when that class is zero, which by definition 5.2.5 is exactly the condition $[\alpha] \in H^k(X, \mathbb{C})_{\text{prim}}$. On the other hand part (1) of theorem 6.1.4, applied to $\alpha_x \in \bigwedge_x^k$ at a point $x$, says that $L^{n-k+1}\alpha_x = 0$ if and only if $\Lambda\alpha_x = 0$. Reading this at every point, $L^{n-k+1}\alpha$ vanishes identically if and only if $\Lambda\alpha$ does, so the two conditions of part (3) agree, completing the proof.

The purchase has two halves. Part (2) puts the action on $H^*(X, \mathbb{C})$ at all, which is what the weight bookkeeping of lemma 6.2.1 is applied to below. Part (3) says more: primitivity of a class, a condition written with $[\omega]$ alone, is realized pointwise by one distinguished representative, and a pointwise primitive form is what theorem 6.1.5 accepts. The transported operator $f = \Lambda$, by contrast, depends on the metric and not only on its class; exercise 6.2.1 (5) takes that up.

Part (2) of the lemma, applied at the weight of $H^k(X, \mathbb{C})$, is the theorem over $\mathbb{C}$; the real, rational and bigraded refinements follow it by inspection.

**Theorem 6.2.3: The Hard Lefschetz Theorem**

Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$, and let $L$ denote the induced Lefschetz operator on cohomology. For each $k$ with $0 \le k \le n$ the map

$$L^{n-k}\colon H^k(X,\mathbb{R}) \to H^{2n-k}(X,\mathbb{R}), \qquad L^{n-k}[\alpha] = [\omega^{n-k} \wedge \alpha]$$

is an isomorphism. The same holds with $\mathbb{R}$ replaced by $\mathbb{C}$, and by $\mathbb{Q}$ when $[\omega]$ lies in the image of $H^2(X,\mathbb{Q})$. Moreover $L^{n-k}$ respects the Hodge decomposition, restricting for each $p + q = k$ to an isomorphism

$$L^{n-k}\colon H^{p,q}(X) \to H^{p+n-k,\, q+n-k}(X)$$

***Proof*:**
*Step 1: the module.* By proposition 6.2.2 the graded space

$$H^\bullet(X,\mathbb{C}) = \bigoplus_{k=0}^{2n} H^k(X,\mathbb{C})$$

is a finite-dimensional $\mathfrak{sl}_2(\mathbb{C})$-module under $e = L$, $f = \Lambda$, $h = H$, in which $H^k(X,\mathbb{C})$ is the weight space of weight $k - n$ and $e$ acts as the Lefschetz operator induced on cohomology, that is, as cup product with $[\omega]$. Of the three operators only $\Lambda$ passes through a choice of metric, being transported along harmonic representatives; that transport is where the Kähler condition is spent.

*Step 2: the weight count.* Fix $k$ with $0 \le k \le n$ and set $m = n - k \ge 0$. The weight space $V_{-m}$ of the module of Step 1 is $H^k(X,\mathbb{C})$, and the weight space $V_m$ is $H^{k'}(X,\mathbb{C})$ for the degree $k'$ with $k' - n = m$, that is $k' = 2n - k$. Part (2) of lemma 6.2.1 therefore gives that

$$L^{n-k}\colon H^k(X,\mathbb{C}) \to H^{2n-k}(X,\mathbb{C})$$

is an isomorphism.

*Step 3: real and rational coefficients.* The form $\omega$ is real, so $L$ carries $H^\bullet(X,\mathbb{R})$ into itself, and $H^k(X,\mathbb{C}) = H^k(X,\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$ realizes the map of Step 2 as the complexification of the real map $L^{n-k}\colon H^k(X,\mathbb{R}) \to H^{2n-k}(X,\mathbb{R})$. A linear map between real vector spaces is bijective if and only if its complexification is, so the real statement follows. If $[\omega]$ lies in the image of $H^2(X,\mathbb{Q})$, then $L$ carries $H^\bullet(X,\mathbb{Q})$ into itself and the same argument applies with $\mathbb{Q}$ in place of $\mathbb{R}$.

*Step 4: compatibility with the bigrading.* The operator $L$ raises the bidegree by $(1,1)$, so $L^{n-k}$ carries $H^{p,q}(X)$ into $H^{p+n-k,q+n-k}(X)$ for $p+q = k$. Distinct summands of $H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$ therefore land in distinct summands of $H^{2n-k}(X,\mathbb{C}) = \bigoplus_{p'+q'=2n-k} H^{p',q'}(X)$, and every summand of the target is reached: given $p' + q' = 2n - k$ with $p', q' \le n$, the inequality $q' \le n$ forces $p' \ge n-k$, so $p = p'-(n-k)$ and $q = q'-(n-k)$ are nonnegative and satisfy $p+q = k$. The isomorphism of Step 2 is thus a direct sum of the maps $H^{p,q}(X) \to H^{p+n-k,q+n-k}(X)$ over $p + q = k$, and a direct sum of linear maps is bijective precisely when every summand map is, so each of them is an isomorphism, as we sought to show.

Poincaré duality already gives $b_k = b_{2n-k}$ on any closed oriented $2n$-manifold, proved in *Algebraic Topology* [24], so the equality of dimensions is not new. What the theorem adds is an isomorphism realizing it, and one of a restricted kind: cup product with a fixed class. That the $(n-k)$-fold cup product with $[\omega]$ should be invertible is a strong constraint, and it is not available without the Kähler hypothesis. For $2 \le k \le n$ the map $L\colon H^{k-2}(X,\mathbb{C}) \to H^k(X,\mathbb{C})$ is injective, because $L^{n-k+2} = L^{n-k+1} \circ L$ is an isomorphism on $H^{k-2}(X,\mathbb{C})$ by the theorem; hence $b_{k-2} \le b_k$ for $k \le n$, and combined with Poincaré duality the even-indexed Betti numbers of a compact Kähler manifold increase up to the middle degree and decrease after it, as do the odd-indexed ones (exercise 6.2.1 (2)). The two parity classes are unrelated, so the full sequence $b_0, b_1, \dots, b_{2n}$ need not be unimodal: a K3 surface carries $b_2 = 22$ between the vanishing $b_1$ and $b_3$. The Hopf surface of box 4.2.4 has $b_0 = 1$ and $b_2 = 0$, so it violates the inequality at $k = 2 = n$, which is a second proof that it carries no Kähler metric.

Projective space is the case where the module of proposition 6.2.2 can be named outright. By example 2.4.5 the Hodge numbers of $\mathbb{CP}^n$ are $h^{p,q} = 1$ for $p = q \le n$ and zero otherwise, so corollary 4.2.3 gives $b_{2j} = 1$ for $0 \le j \le n$ and $b_k = 0$ for $k$ odd. Each $H^{2j}(\mathbb{CP}^n,\mathbb{C})$ is therefore a line, and it is spanned by $[\omega]^j$ for the Fubini-Study form: the class $[\omega]^j$ is nonzero because $[\omega]^j \cup [\omega]^{n-j}$ integrates to $\int_{\mathbb{CP}^n} \omega^n > 0$. So $H^*(\mathbb{CP}^n,\mathbb{C})$ has one line in each of the weights $-n, -n+2, \dots, n$ and none elsewhere, which forces it to be the irreducible module $V(n)$ of dimension $n+1$ (exercise 6.2.1 (1)), generated from the class of the constant function 1 in degree zero.

Part (3) of the lemma delivers the corresponding decomposition in general, and it is the geometric statement that definition 5.2.5 was set up to receive.

**Corollary 6.2.4: The Lefschetz Decomposition**

Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$.

1. For $0 \le k \le n$,
$$H^k(X,\mathbb{C})_{\text{prim}} = \ker\bigl(\Lambda\colon H^k(X,\mathbb{C}) \to H^{k-2}(X,\mathbb{C})\bigr)$$
where $\Lambda$ acts through harmonic representatives, and $\ker\Lambda \cap H^k(X,\mathbb{C}) = 0$ for $k > n$.
2. For every $k$ with $0 \le k \le 2n$,
$$H^k(X,\mathbb{C}) = \bigoplus_{r \ge \max(0,\,k-n)} L^r\, H^{k-2r}(X,\mathbb{C})_{\text{prim}}$$
the sum running over those $r$ with $2r \le k$, and $L^r$ is injective on each summand shown.
3. The decomposition is compatible with the Hodge decomposition, in that $L^r H^{p,q}(X)_{\text{prim}} \subseteq H^{p+r,q+r}(X)$, and it holds verbatim with $\mathbb{C}$ replaced by $\mathbb{R}$, or by $\mathbb{Q}$ when $[\omega]$ is rational.
4. For $0 \le k \le n$ one has $\dim_{\mathbb{C}} H^k(X,\mathbb{C})_{\text{prim}} = b_k - b_{k-2}$, with $b_{-2} = b_{-1} = 0$.

***Proof*:**
Work throughout in the $\mathfrak{sl}_2(\mathbb{C})$-module $H^\bullet(X,\mathbb{C})$ of proposition 6.2.2, where $e = L$, $f = \Lambda$, and $H^k(X,\mathbb{C})$ is the weight space of weight $k-n$. Write $P_j$ for $V_j \cap \ker f$ as in lemma 6.2.1, so that $P_{k-n} = \ker\Lambda \cap H^k(X,\mathbb{C})$.

1. Let $0 \le k \le n$ and put $m = n-k \ge 0$. Part (4) of lemma 6.2.1 says that a class

$\alpha \in H^k(X, \mathbb{C})$ satisfies $\Lambda\alpha = 0$ if and only if $L^{m+1}\alpha = 0$; and $L^{m+1} = L^{n-k+1}$ is exactly the operator whose kernel defines $H^k(X, \mathbb{C})_{\mathrm{prim}}$ in definition 5.2.5. Hence $H^k(X, \mathbb{C})_{\mathrm{prim}} = P_{k-n}$. For $k > n$ the weight $k - n$ is positive, so $P_{k-n} = 0$ by part (1) of the lemma.

2. Part (1) of the lemma lets the notation $H^{k'}(X, \mathbb{C})_{\mathrm{prim}} = P_{k'-n}$ be extended by zero to all degrees $k'$, consistently with part (1) of the present corollary. Applying part (3) of lemma 6.2.1 at $j = k - n$ gives

$$H^k(X, \mathbb{C}) = V_{k-n} = \bigoplus_{r \geq \max(0,\, k-n)} L^r\big(P_{k-n-2r}\big) = \bigoplus_{r \geq \max(0,\, k-n)} L^r\, H^{k-2r}(X, \mathbb{C})_{\mathrm{prim}}$$

since $P_{k-n-2r} = P_{(k-2r)-n}$, with $L^r$ injective on each summand. The terms with $2r > k$ vanish because $H^{k-2r}(X, \mathbb{C}) = 0$ in negative degrees, and the lower bound $r \geq k - n$ is what the lemma supplies: below it the summand is zero, a primitive class in degree $k' \leq n$ being annihilated by $L^r$ as soon as $r > n - k'$.

3. The operator $L$ raises the bidegree by $(1, 1)$, and proposition 5.2.6 gives $H^{k'}(X, \mathbb{C})_{\mathrm{prim}} = \bigoplus_{p+q=k'} H^{p,q}(X)_{\mathrm{prim}}$, so each summand $L^r H^{k-2r}(X, \mathbb{C})_{\mathrm{prim}}$ decomposes into bihomogeneous pieces of the stated types. For the real statement, $\omega$ is a real form, so $L$ and $\Lambda$ are real operators and each subspace appearing in part (2) is stable under complex conjugation. The projections of a direct sum whose summands are all conjugation-stable commute with conjugation, since the components of $\bar{\alpha}$ are then the conjugates of the components of $\alpha$; taking real points of both sides gives the decomposition of $H^k(X, \mathbb{R})$. When $[\omega]$ is rational the same argument applies over $\mathbb{Q}$, where $H^{k'}(X, \mathbb{Q})_{\mathrm{prim}}$ is defined by definition 5.2.5.

4. Let $0 \leq k \leq n$. Separating the term $r = 0$ in part (2) and re-indexing the rest by $r' = r - 1$,

$$H^k(X, \mathbb{C}) = H^k(X, \mathbb{C})_{\mathrm{prim}} \oplus L\Big(\bigoplus_{r' \geq 0} L^{r'} H^{(k-2)-2r'}(X, \mathbb{C})_{\mathrm{prim}}\Big) = H^k(X, \mathbb{C})_{\mathrm{prim}} \oplus L\big(H^{k-2}(X, \mathbb{C})\big)$$

the inner sum being the decomposition of $H^{k-2}(X, \mathbb{C})$, whose lower bound is $\max(0, k - 2 - n) = 0$. Since $k - 2 \leq n - 2 < n$, the map $L$ is injective on $H^{k-2}(X, \mathbb{C})$, as recorded after theorem 6.2.3, so the second summand has dimension $b_{k-2}$ and $\dim H^k(X, \mathbb{C})_{\mathrm{prim}} = b_k - b_{k-2}$, completing the proof.

Part (1) reconciles the two descriptions of primitivity that the book has used. Definition 5.2.5 defined the primitive part by the vanishing of $L^{n-k+1}$, a condition visibly independent of the metric and depending on the Kähler class alone; the kernel of $\Lambda$ is the description that makes the representation theory work but refers to harmonic representatives. They agree, so the Lefschetz decomposition is determined by $[\omega]$ even though its proof runs through a metric. Part (2) is what chapter 5 promised when it introduced the primitive cohomology and then had nothing to decompose: cohomology in every degree is assembled from primitive pieces sitting in degrees of the same parity, each shifted upward by a power of $L$. Part (4) makes the count explicit, and shows that the primitive cohomology is not a small correction: on a surface with $b_2 = 22$ it accounts for 21 of the 22 dimensions.

In complex dimension two the whole apparatus fits in a few lines, and a flat torus makes every number in it explicit.

**Example 6.2.5: Hard Lefschetz on a Compact Kähler Surface**

Let $X$ be a compact Kähler surface, so $n = 2$. Theorem 6.2.3 asserts two isomorphisms,

$$L^2 \colon H^0(X, \mathbb{R}) \to H^4(X, \mathbb{R}), \qquad L \colon H^1(X, \mathbb{R}) \to H^3(X, \mathbb{R})$$

and in the middle degree $k = n = 2$ it asserts only that the identity is an isomorphism, the content there being the decomposition instead. The first map sends 1 to $[\omega^2]$, which is nonzero because $\int_X \omega \wedge \omega > 0$ by part (2) of lemma 4.3.2; since $b_0 = b_4 = 1$ this alone makes it an isomorphism.

*The middle degree.* Here $L^{n-k+1} = L \colon H^2 \to H^4$, so $H^2(X, \mathbb{R})_{\mathrm{prim}}$ is the cup-orthogonal complement of $[\omega]$, as recorded after definition 5.2.5, and corollary 6.2.4 reads

$$H^2(X, \mathbb{R}) = H^2(X, \mathbb{R})_{\mathrm{prim}} \oplus \mathbb{R}\,[\omega]$$

with $\dim H^2(X, \mathbb{R})_{\mathrm{prim}} = b_2 - b_0 = b_2 - 1$. In degree one, $L^{n-k+1} = L^2$ lands in $H^5(X, \mathbb{R}) = 0$, so $H^1(X, \mathbb{R})_{\mathrm{prim}} = H^1(X, \mathbb{R})$ and $\dim H^1_{\mathrm{prim}} = b_1 - b_{-1} = b_1$; in degrees three and four the decomposition reads $H^3 = L\, H^1_{\mathrm{prim}}$ and $H^4 = L^2 H^0_{\mathrm{prim}}$, with no primitive cohomology of its own, in agreement with part (1) of the corollary.

*The isomorphism $L \colon H^1 \to H^3$ on a torus.* Take $X = \mathbb{C}^2/\Lambda$ with the flat Kähler form $\omega = i\,(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2)$. By example 4.2.2 the harmonic forms are the constant-coefficient ones, so $H^1(X, \mathbb{C})$ has basis $dz_1, dz_2, d\bar{z}_1, d\bar{z}_2$ and $H^3(X, \mathbb{C})$ has basis $dz_1 \wedge dz_2 \wedge d\bar{z}_1$, $dz_1 \wedge dz_2 \wedge d\bar{z}_2$, $dz_1 \wedge d\bar{z}_1 \wedge d\bar{z}_2$, $dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2$. Wedging with $\omega$ kills every term containing a repeated differential, leaving

$$\omega \wedge dz_1 = i\, dz_1 \wedge dz_2 \wedge d\bar{z}_2, \qquad \omega \wedge dz_2 = -i\, dz_1 \wedge dz_2 \wedge d\bar{z}_1$$
$$\omega \wedge d\bar{z}_1 = -i\, dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2, \qquad \omega \wedge d\bar{z}_2 = i\, dz_1 \wedge d\bar{z}_1 \wedge d\bar{z}_2$$

each sign coming from moving the surviving factor into increasing order. The four images are the four basis elements of $H^3(X, \mathbb{C})$ up to the scalar $\pm i$, so $L$ is an isomorphism $H^1 \to H^3$, as the theorem requires.

*The numbers on the torus.* Here $b_0 = 1$, $b_1 = 4$, $b_2 = 6$, $b_3 = 4$, $b_4 = 1$, so the primitive dimensions are 1, 4 and $6 - 1 = 5$ in degrees 0, 1, 2. Each primitive class in degree $k$ generates a module $V(n-k)$ of dimension $n-k+1$, so the total dimension of $H^*(X, \mathbb{C})$ is $1\cdot3+4\cdot2+5\cdot1 = 16$, which is $\sum_k b_k = 1+4+6+4+1$. In the middle degree the primitive Hodge numbers are $h^{2,0}_{\mathrm{prim}} = h^{0,2}_{\mathrm{prim}} = 1$ and $h^{1,1}_{\mathrm{prim}} = 4 - 1 = 3$, the Kähler class accounting for the missing $(1,1)$-dimension, exactly as in example 5.2.9.

The surface also shows what is still owed. Example 5.2.9 checked that $Q = -\int_X \alpha \wedge \beta$ polarizes the primitive part of $H^2$, but that verification ran through lemma 4.3.2, three signs computed in a unitary coframe on a four-manifold, and every step of it was tied to the dimension. In arbitrary weight the corresponding positivity was stated in proposition 5.2.8 and left unproved. It comes out of Weil's identity, which is a statement about forms that are primitive at each point, and the form to feed it is fixed by part (3) of proposition 6.2.2: the harmonic representative of a primitive class is pointwise primitive, and no other representative need be, since an arbitrary one can be modified by any exact form. What is left is the accounting of signs.

**Corollary 6.2.6: Positivity on Primitive Cohomology**

Let $X$ be a compact Kähler manifold of complex dimension $n$ with Kähler form $\omega$, let $0 \leq k \leq n$, and let

$$Q(\alpha, \beta) = (-1)^{k(k-1)/2} \int_X \alpha \wedge \beta \wedge \omega^{n-k}$$

be the form of proposition 5.2.8. If $\alpha$ is a nonzero class in $H^{p,q}(X)_{\mathrm{prim}}$ with $p+q=k$, and $\alpha$ also denotes its harmonic representative, then

$$i^{p-q}\, Q\big(\alpha, \bar{\alpha}\big) = (n-k)! \int_X |\alpha|^2 \,\mathrm{vol} > 0$$

So $Q$ satisfies the second Hodge-Riemann bilinear relation of definition 5.2.1 on $H^k(X, \mathbb{C})_{\mathrm{prim}}$, and proposition 5.2.8 holds as stated: whenever $H^k(X, \mathbb{Q})_{\mathrm{prim}}$ is defined over $\mathbb{Q}$, the form $Q$ is a polarization of it.

***Proof*:**
*Step 1: the forms involved are primitive.* By part (3) of proposition 6.2.2 the harmonic representative $\alpha$ satisfies $\Lambda\alpha = 0$ at every point, and it has type $(p,q)$ because the Hodge decomposition assigns each class of type $(p,q)$ a harmonic representative of that type (theorem 4.2.1). Complex conjugation preserves harmonicity by Step 4 of the proof of that theorem and commutes with $\Lambda$, the form $\omega$ being real, so $\bar{\alpha}$ is a harmonic primitive form of type $(q,p)$ and again of total degree $k$.

*Step 2: applying Weil's identity.* Theorem 5.2.7, proved in theorem 6.1.5, evaluates the Hodge star of a primitive form of type $(q,p)$ with $p+q=k$. Applied to $\bar{\alpha}$ it gives

$$\star\,\bar{\alpha} = (-1)^{k(k+1)/2}\, i^{q-p}\, \frac{1}{(n-k)!}\, L^{n-k}\bar{\alpha}$$

*Step 3: pairing against $\alpha$.* The Hodge star of definition 3.3.2 is characterized by $\alpha\wedge\star\bar{\beta} = \langle \alpha, \beta\rangle \,\mathrm{vol}$ for the pointwise Hermitian inner product on forms; with $\beta = \alpha$ this reads $\alpha \wedge \star\bar{\alpha} = |\alpha|^2 \,\mathrm{vol}$. Substituting Step 2,

$$|\alpha|^2 \,\mathrm{vol} = (-1)^{k(k+1)/2}\, i^{q-p}\, \frac{1}{(n-k)!}\, \alpha \wedge \bar{\alpha} \wedge \omega^{n-k}$$

and solving for the wedge, using $(-1)^{-a} = (-1)^a$ and $i^{-(q-p)} = i^{p-q}$,

$$\alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)!\, (-1)^{k(k+1)/2}\, i^{p-q}\, |\alpha|^2 \,\mathrm{vol} \tag{6.3}$$

*Step 4: collecting the signs.* Multiply (6.3) by $i^{p-q}(-1)^{k(k-1)/2}$ and integrate. The two sign factors combine as

$$(-1)^{k(k-1)/2}\, (-1)^{k(k+1)/2} = (-1)^{k(k-1)/2 + k(k+1)/2} = (-1)^{k^2} = (-1)^k$$

while the two powers of $i$ combine as $i^{2(p-q)} = (-1)^{p-q} = (-1)^{p+q} = (-1)^k$, the integers $p-q$ and $p+q$ having the same parity. Their product is $(-1)^{2k} = 1$, so

$$i^{p-q}\, Q(\alpha, \bar{\alpha}) = i^{p-q}(-1)^{k(k-1)/2} \int_X \alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)! \int_X |\alpha|^2 \,\mathrm{vol}$$

The integrand is a nonnegative function vanishing exactly where $\alpha$ does, and $\alpha$ is not the zero form because its class is nonzero, so the integral is strictly positive.

*Step 5: the polarization.* The value of $Q$ depends only on the classes, being an integral of a product of closed forms, so the inequality just proved is a statement about $H^k(X, \mathbb{C})_{\text{prim}}$ and is condition (2) of definition 5.2.1. The symmetry $Q(\alpha, \beta) = (-1)^k Q(\beta, \alpha)$ and condition (1) were verified in proposition 5.2.8, and in theorem 5.2.3 for the underlying bidegree statement, by counts that used nothing deferred. All three conditions now hold, so $Q$ is a polarization of $H^k(X, \mathbb{Q})_{\text{prim}}$ whenever that space is defined over $\mathbb{Q}$, completing the proof.

The corollary closes the last gap in proposition 5.2.8, and the gap was of one specific kind. The symmetry type and the first Hodge-Riemann relation were bookkeeping with bidegrees, valid on any compact Kähler manifold and requiring no machinery. Positivity was different: it needed the Hodge star of a primitive form to be expressible through $L$, and that identity is a statement about the $\mathfrak{sl}_2$-decomposition of the exterior algebra, which is why the verification had to wait for section 6.1. The normalizing sign $(-1)^{k(k-1)/2}$ carried by $Q$ is what meets the $(-1)^{k(k+1)/2}$ produced by Weil's identity, and Step 4 is the computation that the two, together with the factor $i^{p-q}$ prescribed by definition 5.2.1, cancel exactly.

Two consequences follow at once. First, on a projective manifold with $\omega$ the restriction of the Fubini-Study form, the class $[\omega]$ is integral (proposition 3.2.6), so $H^k(X, \mathbb{Q})_{\text{prim}}$ is a rational Hodge structure and $Q$ polarizes it; corollary 5.2.4 then computes the signature of $Q$ from the primitive Hodge numbers alone, with no further geometry, and exercise 5.2.1 (3) makes every sub-Hodge structure of it a direct summand. Second, combining with corollary 6.2.4, the whole of $H^k(X, \mathbb{Q})$ is a direct sum of copies of polarizable structures shifted by powers of $L$ (exercise 6.2.1 (4)), so every cohomology group of a projective manifold is polarizable, not only the primitive part. That is the property chapter 7 carries into families.

## Exercise 6.2.1

1. Let $\omega$ be the Fubini-Study form on $\mathbb{C}\mathrm{P}^n$.

    1. Using example 2.4.5 and corollary 4.2.3, compute $\dim H^k(\mathbb{C}\mathrm{P}^n, \mathbb{C})_{\text{prim}}$ for every $k$ with $0 \leq k \leq n$, and write out the Lefschetz decomposition of $H^{2j}(\mathbb{C}\mathrm{P}^n, \mathbb{C})$.
    2. Verify theorem 6.2.3 directly on $\mathbb{C}\mathrm{P}^3$ by exhibiting the isomorphisms $L^3 \colon H^0 \to H^6$ and $L \colon H^2 \to H^4$ on explicit bases, having first shown that $[\omega]^j \neq 0$ for $0 \leq j \leq 3$.
    3. Deduce that $H^*(\mathbb{C}\mathrm{P}^n, \mathbb{C})$ is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$, and identify which one.

2. Let $X$ be a compact Kähler manifold of complex dimension $n$.

    1. Show that $b_0(X) \leq b_2(X) \leq \cdots$ and $b_1(X) \leq b_3(X) \leq \cdots$, the inequalities running up to degree $n$, and combine this with Poincaré duality to describe the shape of the sequence $b_0, b_1, \ldots, b_{2n}$.
    2. The Hopf surface $S$ of box 4.2.4 is diffeomorphic to $S^1 \times S^3$. Compute its Betti numbers and show that they violate the inequalities of part (1), giving a proof that $S$ is not Kähler that does not use the parity of $b_1$.

3. Let $X = \mathbb{C}^n/\Lambda$ be a complex torus with a flat Kähler metric.

1. Using example 4.2.2, show that $\dim H^k(X,\mathbb{C})_{\mathrm{prim}} = \binom{2n}{k} - \binom{2n}{k-2}$ for $0 \le k \le n$, and evaluate this for $n = 3$ and $k = 0, 1, 2, 3$.
2. Check that $\sum_{k=0}^{n}(n-k+1)\dim H^k(X,\mathbb{C})_{\mathrm{prim}} = 2^{2n}$ for $n = 3$, and explain why this identity holds for every compact Kähler manifold with $2^{2n}$ replaced by $\sum_k b_k$.

4. Let $X$ be a compact Kähler manifold of dimension $n$ whose Kähler class is rational, and let $0 \le k \le 2n$.

    1. Show that for each admissible $r$ the map $L^r$ is an injective morphism of rational Hodge structures
    $$L^r\colon H^{k-2r}(X,\mathbb{Q})_{\mathrm{prim}}(-r) \to H^k(X,\mathbb{Q})$$
    where $(-r)$ is the Tate twist of definition 5.1.10.
    2. Deduce that corollary 6.2.4 is an isomorphism of rational Hodge structures, and that $H^k(X,\mathbb{Q})$ is polarizable.

5. Let $X$ be a compact Kähler manifold of complex dimension $n$.

    1. Show that for $0 \le k \le n$ the map $\Lambda^{n-k}\colon H^{2n-k}(X,\mathbb{C}) \to H^k(X,\mathbb{C})$ is an isomorphism, where $\Lambda$ acts through harmonic representatives. (Argue as in lemma 6.2.1, with $f$ in place of $e$.)
    2. Show that the subspaces $H^k(X,\mathbb{R})_{\mathrm{prim}}$ and the summands of corollary 6.2.4 depend on the Kähler metric only through its class $[\omega] \in H^2(X,\mathbb{R})$, and explain why the same is not immediate for the operator $\Lambda$ of part (1).

## 6.3 Lefschetz Pencils and Vanishing Cycles

Section 6.2 settled what a Kähler class does to the cohomology of a single manifold. The rest of the chapter needs a second object: the hyperplane sections of a projective variety, taken not one at a time but as a family. This section builds that family and identifies the homology classes its degenerations produce.

The first result concerns a single section, and it is what makes the family worth building. Below the middle degree a smooth hyperplane section already carries the cohomology of the variety containing it, so everything peculiar to a smooth projective $X$ of dimension $n$ sits in degree $n$. The pencil construction, the blow-up that turns a pencil into a fibration over $\mathbb{CP}^1$, and the vanishing cycles attached to its singular fibres are the machinery for reaching that degree.

Throughout, $X \subseteq \mathbb{CP}^N$ is a smooth connected projective variety of complex dimension $n$, carrying the Kähler structure restricted from the Fubini-Study metric (corollary 3.2.5). Hyperplanes of $\mathbb{CP}^N$ are parametrized by the dual projective space $(\mathbb{CP}^N)^\vee$, a point of which is the line spanned by a nonzero linear form $s$ on $\mathbb{C}^{N+1}$, and $H_s = \{s = 0\}$ is the corresponding hyperplane. For $X \not\subseteq H$ the intersection $Y = X \cap H$ is a hypersurface in $X$, of dimension $n-1$.

The comparison between $X$ and such a $Y$ rests on a single geometric fact: the complement $X \setminus Y$ is affine. An affine variety of complex dimension $n$ is a very thin object topologically, having no homology at all above degree $n$, and the relative cohomology of the pair $(X,Y)$ is exactly the cohomology of that complement, read backwards.

**Theorem 6.3.1: The Weak Lefschetz Theorem**

Let $X \subseteq \mathbb{CP}^N$ be a smooth connected projective variety of dimension $n \geq 1$, let $H$ be a hyperplane with $X \not\subseteq H$, and suppose $Y = X \cap H$ is smooth. Write $i\colon Y \hookrightarrow X$ for the inclusion. Then

$$i^*\colon H^k(X,\mathbb{Z}) \to H^k(Y,\mathbb{Z})$$

is an isomorphism for $k \leq n-2$ and injective for $k = n-1$, and

$$i_*\colon H_k(Y,\mathbb{Z}) \to H_k(X,\mathbb{Z})$$

is an isomorphism for $k \leq n-2$ and surjective for $k = n-1$.

***Proof*:**
Set $U = X \setminus Y$.

*Step 1: $U$ is a closed complex submanifold of $\mathbb{C}^N$ of dimension $n$.* The complement $\mathbb{CP}^N \setminus H$ is biholomorphic to $\mathbb{C}^N$: after a linear change of homogeneous coordinates $H = \{z_0 = 0\}$, and $[z_0 : \cdots : z_N] \mapsto (z_1/z_0, \ldots, z_N/z_0)$ is a biholomorphism onto $\mathbb{C}^N$. Since $X$ is closed in $\mathbb{CP}^N$, the set $U = X \cap (\mathbb{CP}^N \setminus H)$ is closed in $\mathbb{CP}^N \setminus H = \mathbb{C}^N$, and it is a complex submanifold of dimension $n$ because $X$ is one. It is nonempty because $X \not\subseteq H$.

*Step 2: every Morse index of a squared-distance function on $U$ is at most $n$.* For $a \in \mathbb{C}^N$ put $\varphi_a(z) = |z-a|^2$ and consider its restriction to $U$. It is proper: the preimage of $[0,R]$ is the intersection of $U$ with a closed ball, which is closed and bounded in $\mathbb{C}^N$, hence compact. For all $a$ outside a set of measure zero the restriction $\varphi_a|_U$ is a Morse function [26]; fix such an $a$.

Let $p$ be a critical point of $\varphi_a|_U$, let $\mathrm{Hess}_p$ denote the Hessian of $\varphi_a|_U$ at $p$, a real quadratic form on the real tangent space $T_pU$, and let $J$ be the complex structure on $T_pU$. The claim is that

$$\mathrm{Hess}_p(v,v) + \mathrm{Hess}_p(Jv,Jv) = 4|v|^2 \tag{6.4}$$

for every $v \in T_pU$, the norm on the right being the Euclidean norm of $v$ regarded as a vector in $\mathbb{C}^N$. To see it, choose a holomorphic chart of $U$ centred at $p$ and compose it with a complex line through the origin to produce a holomorphic map $u\colon \Delta \to U \subseteq \mathbb{C}^N$ from a disc with $u(0) = p$ and $u'(0) = v$. Because $p$ is a critical point of $\varphi_a|_U$, the chain rule for second derivatives loses its second-order term and gives

$$\mathrm{Hess}_0(\varphi_a \circ u)(\xi,\eta) = \mathrm{Hess}_p\big(du_0(\xi), du_0(\eta)\big)$$

for all real tangent vectors $\xi, \eta$ of $\Delta$ at 0. Writing $\zeta = \sigma + i\tau$ for the coordinate on $\Delta$ and $\psi = \varphi_a \circ u$, the left side evaluated on the pairs $(\partial_\sigma, \partial_\sigma)$ and $(\partial_\tau, \partial_\tau)$ sums to the Laplacian $\Delta_\zeta\psi(0)$. Now $\psi(\zeta) = \sum_{m=1}^N |u_m(\zeta) - a_m|^2$ with each $u_m$ holomorphic, so $\partial_\zeta\partial_{\bar\zeta}\psi = \sum_m |u_m'|^2$ and

$$\Delta_\zeta\psi(0) = 4\sum_m |u_m'(0)|^2 = 4|v|^2$$

On the right side, $du_0(\partial_\sigma) = v$ and $du_0(\partial_\tau) = Jv$, since $u$ is holomorphic and therefore intertwines the complex structures. This is (6.4).

Let $W \subseteq T_pU$ be a real subspace on which $\mathrm{Hess}_p$ is negative definite. If some nonzero $v$ lay in $W \cap JW$, say $v = Jw$ with $w \in W$, then both $\mathrm{Hess}_p(w,w) < 0$ and $\mathrm{Hess}_p(Jw,Jw) =$

$\mathrm{Hess}_p(v,v) < 0$, contradicting (6.4) applied to $w$. So $W \cap JW = 0$, whence $2\dim_{\mathbb{R}} W = \dim_{\mathbb{R}}(W \oplus JW) \leq \dim_{\mathbb{R}} T_pU = 2n$ and $\dim_{\mathbb{R}} W \leq n$. The Morse index at $p$ is the dimension of a maximal such $W$, so it is at most $n$.

*Step 3: $H_j(U,\mathbb{Z}) = H^j(U,\mathbb{Z}) = 0$ for $j > n$.* The function $\varphi_a|_U$ is proper, bounded below and Morse with all indices at most $n$, so $U$ has the homotopy type of a CW complex whose cells all have dimension at most $n$ [26]. Cellular chains and cochains vanish above dimension $n$, so the homology and cohomology do too.

*Step 4: the relative cohomology of the pair.* Fix nested closed tubular neighbourhoods $N' \subseteq N^\circ$ of the smooth compact submanifold $Y \subseteq X$ [26]. Each retracts onto $Y$, so $H^k(X,Y;\mathbb{Z}) \cong H^k(X,N;\mathbb{Z})$. Excising the open set $N'^\circ$, whose closure lies in the interior of $N$, and then retracting the collar $N \setminus N'^\circ$ onto $\partial N'$ gives

$$H^k(X,N;\mathbb{Z}) \cong H^k\big(X \setminus N'^\circ,\ \partial N';\mathbb{Z}\big)$$

The space $M = X \setminus N'^\circ$ is a compact oriented $2n$-manifold with boundary $\partial N'$, oriented by the complex structure of $X$, so Lefschetz duality [24] gives $H^k(M,\partial M;\mathbb{Z}) \cong H_{2n-k}(M,\mathbb{Z})$. Finally $M$ is a deformation retract of $U$, since $U$ is $M$ together with the half-open collar $N' \setminus Y$. Combining,

$$H^k(X,Y;\mathbb{Z}) \cong H_{2n-k}(U,\mathbb{Z})$$

and the same chain with the roles of homology and cohomology exchanged gives $H_k(X,Y;\mathbb{Z}) \cong H^{2n-k}(U,\mathbb{Z})$. By Step 3 both vanish as soon as $2n-k > n$, that is for $k \leq n-1$.

*Step 5: the long exact sequences.* For the pair $(X,Y)$,

$$H^k(X,Y;\mathbb{Z}) \to H^k(X,\mathbb{Z}) \xrightarrow{i^*} H^k(Y,\mathbb{Z}) \to H^{k+1}(X,Y;\mathbb{Z})$$

is exact. For $k \leq n-2$ both outer terms vanish by Step 4, so $i^*$ is an isomorphism; for $k = n-1$ the left term vanishes, so $i^*$ is injective. The homology sequence

$$H_{k+1}(X,Y;\mathbb{Z}) \to H_k(Y,\mathbb{Z}) \xrightarrow{i_*} H_k(X,\mathbb{Z}) \to H_k(X,Y;\mathbb{Z})$$

gives the second statement by the same two vanishings, completing the proof.

The theorem is a statement about integral cohomology, and that is worth separating from what the Kähler package alone delivers. Comparing the two needs a map running opposite to restriction, which Poincaré duality supplies. Write $D_X$ and $D_Y$ for the duality isomorphisms of $X$ and $Y$ ([24, Poincaré duality]) and set

$$i_! := D_X^{-1} \circ i_* \circ D_Y \colon H^k(Y,\mathbb{Z}) \to H^{k+2}(X,\mathbb{Z}),$$

the *Gysin map*, the homology pushforward of theorem 6.3.1 read through duality on both sides; the degree shifts by 2 because $Y$ has complex codimension 1. Note this is a map on cohomology and is not the $i_*$ of theorem 6.3.1, which acts on homology. By construction $i_!(1)$ is the class dual to $Y$, and the class of $Y$ in $H^2(X,\mathbb{Z})$ is the restriction of the hyperplane class, which is the Fubini-Study Kähler class (proposition 3.2.6). The projection formula for cap products ([24, the projection formula for cap products]) then transposes to $i_!(i^*\alpha) = \alpha \smile i_!(1) = L\alpha$: the Lefschetz operator factors through restriction to a hyperplane section. Hard Lefschetz (theorem 6.2.3) makes $L^{n-k}$ injective on $H^k(X,\mathbb{Q})$ for $k \leq n$, and $L^{n-k} = (i_!i^*)^{n-k}$, so $i^*$ is already injective on $H^k(X,\mathbb{Q})$ for $k \leq n-1$ on those grounds alone. What that argument does not give is surjectivity, nor anything over $\mathbb{Z}$; both come from the affineness of the complement.

Applied to a variety whose ambient cohomology is known, the theorem computes everything outside the middle degree.

**Example 6.3.2: The Cohomology of a Smooth Hypersurface**

Let $X \subseteq \mathbb{CP}^{n+1}$ be a smooth hypersurface of degree $d$. It is not a hyperplane section of $\mathbb{CP}^{n+1}$ unless $d = 1$, but it becomes one after a change of embedding: the $d$-th Veronese map $v_d \colon \mathbb{CP}^{n+1} \to \mathbb{CP}^M$ is a closed embedding whose image is smooth, and hyperplanes of $\mathbb{CP}^M$ pull back to exactly the degree-$d$ hypersurfaces of $\mathbb{CP}^{n+1}$ [4]. So $v_d(X)$ is a smooth hyperplane section of the smooth projective variety $v_d(\mathbb{CP}^{n+1})$ of dimension $n+1$.

Theorem 6.3.1, applied with $n+1$ in place of $n$, gives $H^k(\mathbb{CP}^{n+1}, \mathbb{Z}) \cong H^k(X, \mathbb{Z})$ for $k \leq n-1$ and $H_k(X, \mathbb{Z}) \cong H_k(\mathbb{CP}^{n+1}, \mathbb{Z})$ for $k \leq n-1$. Hence, for $k \leq n-1$,

$$H^k(X, \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k \text{ even} \\ 0 & k \text{ odd} \end{cases}$$

For $k \geq n+1$ Poincaré duality on the compact oriented $2n$-manifold $X$ gives $H^k(X, \mathbb{Z}) \cong H_{2n-k}(X, \mathbb{Z})$, and $2n - k \leq n - 1$, so the same table applies; note $2n - k$ and $k$ have the same parity. Every Betti number of $X$ outside degree $n$ is therefore 1 in even degrees and 0 in odd degrees, independent of $d$.

The middle Betti number is then forced by the Euler characteristic. If $n$ is even, the degrees $k \neq n$ contributing to $\chi_{\text{top}}(X) = \sum_k (-1)^k b_k(X)$ are the $n$ even degrees other than $n$ itself, each contributing 1, so $b_n(X) = \chi_{\text{top}}(X) - n$. If $n$ is odd there are $n+1$ even degrees, all different from $n$, and $b_n$ enters with a minus sign, so $b_n(X) = n + 1 - \chi_{\text{top}}(X)$.

Two instances. A smooth plane curve of degree $d$ has $n = 1$ and $\chi_{\text{top}} = 2 - 2g$ with $g = (d-1)(d-2)/2$ [4], giving $b_1 = 2 - (2 - 2g) = 2g$, the classical count. A smooth quartic surface in $\mathbb{CP}^3$ is a K3 surface ([4]), so by the Hodge numbers recorded in example 4.3.4 it has $b_0 = b_4 = 1$, $b_1 = b_3 = 0$ and $b_2 = 22$, whence $\chi_{\text{top}} = 1 - 0 + 22 - 0 + 1 = 24$. Read the other way, the formula above recovers $b_2 = \chi_{\text{top}} - 2 = 22$, the rank of the K3 intersection lattice.

Everything the hyperplane section does not see is concentrated in $H^n(X)$. To reach that degree a single section is useless, since it is precisely the degree where restriction fails to be an isomorphism. The device that does reach it lets the hyperplane move.

A line $\ell \subseteq (\mathbb{CP}^N)^\vee$ is spanned by two linear forms $s_0, s_1$, and the hyperplanes it parametrizes are the zero sets of $\lambda_t = t_0 s_0 + t_1 s_1$ for $t = [t_0 : t_1] \in \mathbb{CP}^1$. All of them contain the codimension-2 linear subspace $A = H_{s_0} \cap H_{s_1}$, called the *axis*, and conversely every hyperplane containing $A$ occurs. Cutting with $X$ produces a family $X_t = X \cap H_t$ of hypersurfaces in $X$, all containing the fixed locus $B = X \cap A$, and sweeping out $X$: every point of $X \setminus B$ lies on exactly one member, and every point of $B$ lies on all of them.

Some members of the family will be singular, and the definition below restricts how badly. A point $x$ of a hypersurface $Z \subseteq X$ is an *ordinary double point* if $Z$ is cut out near $x$ by a holomorphic function $g$ with $g(x) = 0$, $dg_x = 0$, and Hessian $\text{Hess}_x(g)$ a nondegenerate quadratic form on $T_x X$. The condition does not depend on the choice of $g$: two local defining functions differ by a nowhere-zero holomorphic factor $u$, and at a point where $g$ and $dg$ both vanish one has $\text{Hess}_x(ug) = u(x)\,\text{Hess}_x(g)$.

**Definition 6.3.3: Lefschetz Pencil**

Let $X \subseteq \mathbb{C}\mathrm{P}^N$ be a smooth connected projective variety of dimension $n \geq 2$. A *pencil of hyperplane sections* of $X$ is the family $\{X_t = X \cap H_t\}_{t \in \mathbb{C}\mathrm{P}^1}$ attached to a line $\ell \subseteq (\mathbb{C}\mathrm{P}^N)^\vee$ as above, with *axis* $A$ and *base locus* $B = X \cap A$. The pencil is a *Lefschetz pencil* if

1. $A$ meets $X$ transversally, that is $T_xX + T_xA = T_x\mathbb{C}\mathrm{P}^N$ for every $x \in B$;
2. the set $\Delta \subseteq \mathbb{C}\mathrm{P}^1$ of parameters $t$ for which $X_t$ is singular is finite;
3. for each $t \in \Delta$ the hypersurface $X_t$ has exactly one singular point, and that point is an ordinary double point.

The members $X_t$ with $t \in \Delta$ are the *singular fibres* of the pencil, and $\mu = \#\Delta$ is their number.

Condition (1) is a genuine restriction, not a smoothness convention. Since $A$ has codimension 2 and $n \geq 2$, the intersection $B$ is nonempty [4], and transversality forces $B$ to be a smooth submanifold of $X$ of codimension 2; it also rules out the degenerate possibility that $X \subseteq H_t$ for some $t$, since then $T_xX$ and $T_xA$ would both lie in the hyperplane $T_xH_t$ at a point $x \in B$. Condition (3) is what makes the family tractable: an ordinary double point has a normal form, derived below, and every singular member of a Lefschetz pencil looks the same near its singular point.

Before showing that such pencils exist, here is one written out completely.

**Example 6.3.4: A Pencil of Conics on the Plane**

Take $X = \mathbb{C}\mathrm{P}^2$ and embed it by the complete system of conics, that is by the Veronese map $v_2 \colon \mathbb{C}\mathrm{P}^2 \to \mathbb{C}\mathrm{P}^5$ [4]. Hyperplane sections of $v_2(\mathbb{C}\mathrm{P}^2)$ are the conics in $\mathbb{C}\mathrm{P}^2$, and a pencil of hyperplane sections is a pencil of conics. Take the one spanned by

$$F_0 = x_0^2 - x_1^2, \qquad F_1 = x_0^2 - x_2^2$$

so that $F_t = t_0F_0 + t_1F_1$ and $X_t = \{F_t = 0\}$.

*The base locus.* A point lies on every member exactly when $x_0^2 = x_1^2$ and $x_0^2 = x_2^2$. If $x_0 = 0$ then $x_1 = x_2 = 0$, which is not a point of $\mathbb{C}\mathrm{P}^2$, so $x_0 \neq 0$ and, normalizing $x_0 = 1$, the solutions are the four points $[1 : \pm1 : \pm1]$. The four are distinct, and at each of them the differentials of $F_0$ and $F_1$ are independent: at $[1 : 1 : 1]$, for instance, in the affine chart $x_0 = 1$ they are $-2\,dx_1$ and $-2\,dx_2$. That independence is exactly transversality of the axis with $X$, so condition (1) holds and $B$ consists of four points.

*The singular members.* Writing $F_t$ as $x^\top M_t x$ with

$$M_t = \begin{pmatrix} t_0 + t_1 & 0 & 0 \\ 0 & -t_0 & 0 \\ 0 & 0 & -t_1 \end{pmatrix}$$

the conic $X_t$ is singular exactly when $\det M_t = (t_0+t_1)t_0t_1$ vanishes, that is at the three parameters $[1 : 0]$, $[0 : 1]$ and $[1 : -1]$. So $\mu = 3$ and condition (2) holds.

At $t = [1 : 0]$ the member is $x_0^2 - x_1^2 = (x_0 - x_1)(x_0 + x_1)$, a pair of distinct lines meeting at $[0 : 0 : 1]$. In the affine chart $x_2 = 1$ with coordinates $(y_0, y_1)$ the defining function is $y_0^2 - y_1^2$, whose Hessian $\mathrm{diag}(2, -2)$ is nondegenerate: an ordinary double point, and the only singular point of that member. The point $[0 : 0 : 1]$ is not one of the four base points, as it must not be. The parameters $[0:1]$ and $[1:-1]$ behave identically, the latter giving $x_1^2 - x_2^2$ with singular point $[1:0:0]$. So condition (3) holds and the pencil is a Lefschetz pencil.

*The associated fibration.* Blowing up the four base points turns the rational map $\mathbb{C}\mathrm{P}^2 \dashrightarrow \mathbb{C}\mathrm{P}^1$, $x \mapsto [F_1(x) : -F_0(x)]$, into a morphism whose fibres are the members of the pencil; this is the content of proposition 6.3.6 below. The blow-up has Euler characteristic $3+4=7$, since each of the four points is replaced by a $\mathbb{C}\mathrm{P}^1$. On the other side, three of the fibres are pairs of lines ($\chi_{\text{top}} = 2+2-1 = 3$) and the rest are smooth conics ($\chi_{\text{top}} = 2$), and a count of the fibres over $\mathbb{C}\mathrm{P}^1$ gives $(2-3)\cdot 2 + 3\cdot 3 = 7$. The two computations agree.

That pencil was arranged by hand, but it exhibits the general mechanism: the members singular at a prescribed point form a linear subsystem of large codimension, and a line in the parameter space meets the union of these subsystems in a controlled way. Turning that into a proof requires knowing that the linear system is large enough to impose independent conditions on second derivatives at a point and on first derivatives at two points at once.

**Proposition 6.3.5: Existence of Lefschetz Pencils**

Let $X \subseteq \mathbb{C}\mathrm{P}^N$ be a smooth connected projective variety of dimension $n \geq 2$, and let $V$ be the space of linear forms on $\mathbb{C}^{N+1}$. Assume

1. for every $x \in X$ the restrictions to $X$ of the forms in $V$ realize every 2-jet at $x$;
2. for every pair of distinct points $x, x' \in X$ they realize every pair of 1-jets at $x$ and $x'$.

Then the lines $\ell \subseteq (\mathbb{C}\mathrm{P}^N)^\vee$ whose pencil is a Lefschetz pencil form a dense open subset of the Grassmannian of lines. Moreover every smooth projective variety satisfies (1) and (2) after re-embedding by $|\mathcal{O}_X(d)|$ for $d$ large [4], so every smooth projective variety carries Lefschetz pencils.

Here the 2-jet of a section at $x$ records its value, its first derivatives along $X$ and its second derivatives along $X$, so the space $J_x^2$ of 2-jets has dimension $1+n+\binom{n+1}{2}$ and the space $J_x^1$ of 1-jets has dimension $n+1$; both are jets of sections of $\mathcal{O}_X(1)$, so the statements are independent of local trivializations.

***Proof*:**
Write $\mathbb{G}$ for the Grassmannian of lines in $(\mathbb{C}\mathrm{P}^N)^\vee$, an irreducible projective variety of dimension $2(N-1)$. For each of the three conditions the locus of lines where it fails is closed, so it suffices to show each failure locus is a proper subset; the intersection of three dense open subsets of an irreducible variety is dense open.

*Step 1: the variety of tangent hyperplanes.* For $x \in X$ let $\mathbb{T}_x X \subseteq \mathbb{C}\mathrm{P}^N$ be the embedded projective tangent space, a linear subspace of dimension $n$. A hyperplane $H$ makes $X \cap H$ singular at $x$ exactly when $\mathbb{T}_x X \subseteq H$: the section $X \cap H$ is cut out near $x$ by the linear form defining $H$, which is singular at $x$ precisely when the form and its differential along $X$ both vanish there. Put

$$\tilde{Y} = \left\{(x,H) \in X \times (\mathbb{C}\mathrm{P}^N)^\vee \;\middle|\; \mathbb{T}_x X \subseteq H\right\}$$

The hyperplanes containing a fixed linear subspace of dimension $n$ form a linear $\mathbb{C}\mathrm{P}^{N-n-1}$ inside $(\mathbb{C}\mathrm{P}^N)^\vee$, so the fibre of $\tilde{Y}$ over $x$ is such a subspace, and $\tilde{Y}$ is a projective bundle over $X$, hence smooth and irreducible of dimension $n+(N-n-1) = N-1$. Its image $X^\vee \subseteq (\mathbb{C}\mathrm{P}^N)^\vee$ under the second projection is closed, being the image of a projective variety, irreducible, and of dimension at most $N-1$. In particular $X^\vee \neq (\mathbb{CP}^N)^\vee$, so a hyperplane section of $X$ is generically smooth. Only smoothness of $X$ entered, so the same conclusion holds for any smooth projective variety in $\mathbb{CP}^N$.

*Step 2: condition (2) of definition 6.3.3.* The singular parameters of the pencil attached to $\ell$ lie in $\ell \cap X^\vee$, a closed subvariety of the line $\ell$, hence either finite or all of $\ell$. Lines contained in the proper closed subset $X^\vee$ form a closed subset of $\mathbb{G}$, and it is proper: any line joining two points outside $X^\vee$ is not contained in it, and such points exist by Step 1.

*Step 3: condition (3).* Fix $(x,H) \in \tilde{Y}$ and let $s$ be a linear form defining $H$. Since $s$ and its differential along $X$ vanish at $x$, its Hessian along $X$ is a well-defined quadratic form on $T_xX$, and $x$ is an ordinary double point of $X \cap H$ exactly when that form is nondegenerate. Hypothesis (1) says the map sending a form vanishing to first order at $x$ to its Hessian, from an $(N-n)$-dimensional space onto $\mathrm{Sym}^2 T_x^* X$, is surjective. Degenerate quadratic forms are the zero locus of the determinant, a nonzero polynomial, hence a proper closed subset; its preimage is therefore a proper closed subset of the fibre of $\tilde{Y}$ over $x$, of dimension at most $N-n-2$. So

$$\tilde{Y}_{\mathrm{deg}} = \{(x,H) \in \tilde{Y} \mid \mathrm{Hess}_x(s) \text{ is degenerate on } T_xX\}$$

is closed of dimension at most $n + (N-n-2) = N-2$.

For the second half of condition (3), let

$$\tilde{Y}_2 = \{(x,x',H) \mid x \neq x',\ (x,H) \in \tilde{Y},\ (x',H) \in \tilde{Y}\}$$

Hypothesis (2) makes the $2(n+1)$ linear conditions defining the fibre over a pair $(x,x')$ independent, so that fibre has dimension $N-2n-2$ when nonnegative and is empty otherwise. Hence $\tilde{Y}_2$ has dimension at most $2n + (N-2n-2) = N-2$.

Let $\Sigma$ be the union of the image of $\tilde{Y}_{\mathrm{deg}}$ in $(\mathbb{CP}^N)^\vee$ and the closure of the image of $\tilde{Y}_2$. It is closed of dimension at most $N-2$, and a pencil fails condition (3) exactly when $\ell$ meets $\Sigma$. The incidence variety $\{(\ell,H) \big| H \in \ell \cap \Sigma\}$ fibres over $\Sigma$ with fibre the lines through a fixed point, a family of dimension $N-1$, so it has dimension at most $(N-2)+(N-1) = 2N-3$. Its image in $\mathbb{G}$ is closed of dimension at most $2N-3 < 2(N-1)$, hence proper.

*Step 4: condition (1).* The set of pairs $(x,\ell)$ with $x \in X$ lying on the axis $A_\ell$ and $T_xX + T_xA_\ell \neq T_x\mathbb{CP}^N$ is closed in $X \times \mathbb{G}$, and $X$ is complete, so its image in $\mathbb{G}$ is closed. It is proper: choose $H_0 \notin X^\vee$ by Step 1, so that $X_0 = X \cap H_0$ is smooth of dimension $n-1$, then apply Step 1 to $X_0$ to choose $H_1$ with $X_0 \cap H_1$ smooth of dimension $n-2$. The line through $H_0$ and $H_1$ has axis $A = H_0 \cap H_1$ with $X \cap A = X_0 \cap H_1$ smooth of codimension 2, cut out in $X$ by two forms with independent differentials, which is transversality.

The three failure loci are proper closed subsets of the irreducible variety $\mathbb{G}$, so their complements intersect in a dense open subset, as we sought to show.

The proof also counts the singular members. When $X^\vee$ is a hypersurface, a generic line meets it transversally in $\deg X^\vee$ points [4], so $\mu = \deg X^\vee$ for a generic pencil; when $X^\vee$ has codimension at least 2, a generic line misses it and $\mu = 0$, the pencil being a family of smooth hypersurfaces. The second case is not exotic: a linearly embedded $\mathbb{CP}^n \subseteq \mathbb{CP}^N$ has only smooth hyperplane sections (exercise 6.3.1 (3)).

A pencil is not yet a family in the sense that admits topological arguments, because the members all meet along $B$ and the assignment $x \mapsto t$ is undefined there. Separating the members along $B$ is exactly what a blow-up does.

**Proposition 6.3.6: The Fibration Attached to a Pencil**

Let $\{X_t\}$ be a Lefschetz pencil on $X \subseteq \mathbb{C}\mathrm{P}^N$, with axis $A$, base locus $B$, defining forms $s_0, s_1$ and singular parameters $\Delta = \{t_1, \dots, t_\mu\}$. Put

$$\tilde{X} = \left\{(x,t) \in X \times \mathbb{C}\mathrm{P}^1 \;\middle|\; t_0 s_0(x) + t_1 s_1(x) = 0\right\}$$

and let $\sigma\colon \tilde{X} \to X$ and $f\colon \tilde{X} \to \mathbb{C}\mathrm{P}^1$ be the two projections. Then

1. $\tilde{X}$ is a smooth connected projective variety of dimension $n$, and $\sigma$ is the blow-up of $X$ along $B$, an isomorphism over $X \setminus B$ with exceptional divisor $E = \sigma^{-1}(B) = B \times \mathbb{C}\mathrm{P}^1$;
2. $f$ is a proper holomorphic map with $f^{-1}(t) = X_t \times \{t\}$, and $f|_E$ is the projection $B \times \mathbb{C}\mathrm{P}^1 \to \mathbb{C}\mathrm{P}^1$;
3. the critical points of $f$ are exactly the $\mu$ singular points of the singular fibres; each is a nondegenerate critical point, none lies on $E$, and no two lie in the same fibre;
4. over $\mathbb{C}\mathrm{P}^1 \setminus \Delta$ the map $f$ is a locally trivial fibration with fibre a smooth hyperplane section of $X$.

***Proof*:**

1. The defining equation is bihomogeneous of degree $(1,1)$, so $\tilde{X}$ is a closed subvariety of $X \times \mathbb{C}\mathrm{P}^1$, hence compact and projective [4]. Let $(x,t) \in \tilde{X}$ and write $F(x,t) = t_0 s_0(x) + t_1 s_1(x)$ for the defining section.

   If $x \notin B$ then $(s_0(x), s_1(x)) \neq (0,0)$, and the derivative of $F$ in the $\mathbb{C}\mathrm{P}^1$ directions is $(s_0(x), s_1(x))$ in a local trivialization, so $dF \neq 0$ and $\tilde{X}$ is smooth of dimension $n$ at $(x,t)$. If $x \in B$ then $s_0(x) = s_1(x) = 0$, the differentials $ds_0, ds_1$ restricted to $T_x X$ are independent by transversality of $A$ with $X$, and the derivative of $F$ in the $X$ directions is $t_0\, ds_0 + t_1\, ds_1$, nonzero because $(t_0, t_1) \neq (0,0)$. Again $\tilde{X}$ is smooth of dimension $n$ at $(x,t)$.

   Over $x \notin B$ the equation determines $t$ uniquely, namely $t = [s_1(x) : -s_0(x)]$, so $\sigma$ is bijective there; the inverse $x \mapsto (x, [s_1(x) : -s_0(x)])$ is holomorphic, so $\sigma$ is an isomorphism over $X \setminus B$. Over $x \in B$ the equation is vacuous, so $\sigma^{-1}(B) = B \times \mathbb{C}\mathrm{P}^1$, of dimension $(n-2)+1 = n-1$. Since $\tilde{X}$ is smooth of pure dimension $n$, no component of $\tilde{X}$ lies over $B$, so $\tilde{X}$ is the closure of the graph of $x \mapsto [s_1(x) : -s_0(x)]$; as $X \setminus B$ is connected, $\tilde{X}$ is connected. Composing with the automorphism $[u_0 : u_1] \mapsto [u_1 : -u_0]$ of $\mathbb{C}\mathrm{P}^1$ identifies that graph closure with the closure of the graph of $x \mapsto [s_0(x) : s_1(x)]$, which is the blow-up of $X$ along the smooth centre $B$ cut out by $s_0$ and $s_1$ [4]. For $n = 2$ the centre is a finite set of points and this is literally the construction of definition 1.4.12, whose local picture is example 1.4.13.

2. Properness is compactness of $\tilde{X}$. For fixed $t$ the equation cuts out $X \cap H_t = X_t$, so $f^{-1}(t) = X_t \times \{t\}$, and $f$ restricted to $E = B \times \mathbb{C}\mathrm{P}^1$ is the second projection by construction.

3. Since $\tilde{X}$ has dimension $n$ and $f$ maps to a curve, a point is critical for $f$ exactly when $df$

vanishes there, which happens exactly when the fibre $f^{-1}(t) = X_t$ is singular at that point.

No such point lies on $E$: for $x \in B$ the differential of the defining form $\lambda_t = t_0 s_0 + t_1 s_1$ along $T_x X$ is $t_0\, ds_0 + t_1\, ds_1 \neq 0$, so $X_t$ is smooth at every point of $B$. By condition (3) of definition 6.3.3 each singular fibre has exactly one singular point, so the critical points are the $\mu$ ordinary double points, one in each singular fibre, and no two share a fibre.

For nondegeneracy fix a critical point $p$, lying over the singular parameter $t_i$, and normalize $t_i = [a_0 : 1]$ (after exchanging $s_0, s_1$ if necessary). Since $p \notin B$ and $\lambda_{t_i}(p) = 0$, one has $s_0(p) \neq 0$. Let $\zeta$ be the affine coordinate on $\mathbb{CP}^1$ centred at $t_i$ for which $t = [a_0 + \zeta : 1]$. Near $p$ the map $\sigma$ is an isomorphism, and the equation $(a_0 + \zeta)s_0 + s_1 = 0$ of $\tilde{X}$ reads $\zeta = -g$ with

$$g = \frac{\lambda_{t_i}}{s_0}$$

a holomorphic function on a neighbourhood of $p$ in $X$. So $f$ corresponds to $-g$, and $X_{t_i}$ is cut out near $p$ by $g$. Both $g$ and $dg$ vanish at $p$, and $p$ is a nondegenerate critical point of $f$ exactly when $\mathrm{Hess}_p(g)$ is nondegenerate, which is the statement that $p$ is an ordinary double point of $X_{t_i}$.

4. Over $\mathbb{CP}^1 \setminus \Delta$ the map $f$ is a proper holomorphic submersion between smooth manifolds, hence a locally trivial fibration [26] whose fibres are the smooth members $X_t$, completing the proof.

The blow-up costs nothing that matters here. It changes $X$ only along $B$, replacing each point of $B$ by a $\mathbb{CP}^1$, and $\sigma$ is an isomorphism elsewhere; in exchange, the pencil becomes a genuine map to $\mathbb{CP}^1$, and the topology of $X$ can be studied by moving in the base. What now needs description is what happens over a point of $\Delta$, and by part (3) that question is local and identical at every one of the $\mu$ critical points.

The description is a holomorphic analogue of the Morse lemma. The real statement, that a nondegenerate critical point admits coordinates in which the function is a sum of signed squares, is proved in [26, the Morse lemma]; over $\mathbb{C}$ the signs disappear, since every nonzero complex number has a square root, and with them the index. That is why a complex nondegenerate critical point has no discrete invariant and every node looks like every other. The proof is short enough to give directly, and is not the real lemma in disguise: it is a completing-the-square induction performed with holomorphic square roots.

**Lemma 6.3.7: The Local Model at a Node**

Let $M$ be a complex manifold of dimension $n \geq 1$, let $f\colon M \to \mathbb{C}$ be holomorphic, and let $p$ be a nondegenerate critical point of $f$ with $f(p) = 0$.

1. There are local holomorphic coordinates $z = (z_1, \ldots, z_n)$ centred at $p$ in which
$$f = z_1^2 + \cdots + z_n^2$$
2. In these coordinates fix $\epsilon > 0$ with the closed ball $\bar{B}_\epsilon$ contained in the coordinate domain, and for $0 < |t| < \epsilon^2$ set $F_t = \{z \in \bar{B}_\epsilon \big| f(z) = t\}$. Then $F_t$ is diffeomorphic to a closed disc bundle of the tangent bundle $TS^{n-1}$, with the sphere
$$S_t = \left\{\sqrt{t}\, x \;\middle|\; x \in \mathbb{R}^n,\ |x| = 1\right\}$$
as the zero section, for either choice of square root. In particular $F_t$ deformation retracts onto $S_t$, a smooth $(n-1)$-sphere of Euclidean radius $|t|^{1/2}$.
3. $F_0 = \{z \in \bar{B}_\epsilon \big| f(z) = 0\}$ is contractible.

***Proof*:**

1. Choose holomorphic coordinates $w$ centred at $p$ on a polydisc, in which $f(0) = 0$ and $df_0 = 0$. Since the polydisc is convex, Taylor's formula with integral remainder gives
$$f(w) = \sum_{i,j} h_{ij}(w)\, w_i w_j, \qquad h_{ij}(w) = \int_0^1 (1-s)\, \partial_i \partial_j f(sw)\, ds$$
The functions $h_{ij} = h_{ji}$ are holomorphic, being integrals of holomorphic families, and $h_{ij}(0) = \frac{1}{2}\partial_i \partial_j f(0)$, so the matrix $h(0)$ is invertible.
The claim, proved by induction on $k$, is that for each $0 \leq k \leq n$ there are holomorphic coordinates $u$ centred at $p$ with
$$f = u_1^2 + \cdots + u_k^2 + \sum_{i,j>k} c_{ij}(u)\, u_i u_j$$
where $c_{ij} = c_{ji}$ is holomorphic and the matrix $(c_{ij}(0))_{i,j>k}$ is invertible. The case $k = 0$ is the display above. Assume the statement for $k < n$. A nonzero symmetric bilinear form over $\mathbb{C}$ has a nonisotropic vector, so after a constant linear change of the coordinates $u_{k+1}, \ldots, u_n$, which does not disturb the first $k$ squares, one may assume $c_{k+1,k+1}(0) \neq 0$. Shrinking the neighbourhood so that $c_{k+1,k+1}$ takes values in a disc avoiding the origin, choose a holomorphic branch $\rho$ of $\sqrt{c_{k+1,k+1}}$ there and set
$$v_{k+1} = \rho(u)\Big(u_{k+1} + \sum_{j>k+1} \frac{c_{k+1,j}(u)}{c_{k+1,k+1}(u)}\, u_j\Big)$$
Completing the square in $u_{k+1}$ gives
$$\sum_{i,j>k} c_{ij} u_i u_j = v_{k+1}^2 + \sum_{i,j>k+1} c'_{ij}(u)\, u_i u_j, \qquad c'_{ij} = c_{ij} - \frac{c_{k+1,i}\, c_{k+1,j}}{c_{k+1,k+1}}$$

with $c'$ symmetric holomorphic, and $\det c(0) = c_{k+1,k+1}(0) \det c'(0)$ shows $c'(0)$ is invertible. The map $u \mapsto (u_1, \ldots, u_k, v_{k+1}, u_{k+2}, \ldots, u_n)$ has Jacobian at the origin equal to the identity except in row $k+1$, whose diagonal entry is $\rho(0) \neq 0$; the determinant is $\rho(0)$, so the map is a local biholomorphism and the new functions form a coordinate system. Expressing $c'$ in the new coordinates completes the inductive step, and $k = n$ is the assertion.

2. Replacing $z$ by $cz$ with $|c| = 1$ multiplies $f$ by $c^2$ and preserves $\bar{B}_\epsilon$, so it suffices to treat $t > 0$ real. Write $z = x + iy$ with $x, y \in \mathbb{R}^n$. Then

$$\sum_j z_j^2 = |x|^2 - |y|^2 + 2i\, x \cdot y$$

so $F_t$ is the set of pairs $(x, y)$ with $|x|^2 = t + |y|^2$, $x \cdot y = 0$ and $|x|^2 + |y|^2 \leq \epsilon^2$, the last condition being $2|y|^2 \leq \epsilon^2 - t$. Writing $x = |x|\, \nu$ with $\nu \in S^{n-1}$, the map

$$(\nu, y) \longmapsto \left(\sqrt{t + |y|^2}\ \nu,\ y\right)$$

is a diffeomorphism from $\{(\nu, y) | \nu \in S^{n-1},\ y \perp \nu,\ 2|y|^2 \leq \epsilon^2 - t\}$ onto $F_t$. Identifying $T_\nu S^{n-1}$ with $\nu^\perp$, the source is the closed disc bundle of radius $\sqrt{(\epsilon^2 - t)/2}$ in $TS^{n-1}$, and $y = 0$ corresponds to $x = \sqrt{t}\, \nu$, that is to $S_t$. Scaling $y$ to zero deformation retracts the disc bundle onto its zero section.

3. The set $F_0$ is preserved by $z \mapsto \lambda z$ for real $\lambda \in [0, 1]$, since that scales $f$ by $\lambda^2$ and does not increase $|z|$. The map $(z, \lambda) \mapsto \lambda z$ is a continuous contraction of $F_0$ to the origin, as required.

Part (2) is the geometric content of the whole construction. As $t$ approaches a singular parameter the nearby fibre carries a sphere of real dimension $n - 1$, half the real dimension $2n - 2$ of the fibre, and that sphere contracts to the node. Since the fibres over the complement of $\Delta$ are identified with one another by the fibration of proposition 6.3.6, the sphere can be carried back to a fixed fibre, where it becomes a homology class.

**Definition 6.3.8: Vanishing Cycle**

Let $\{X_t\}$ be a Lefschetz pencil on $X$, with associated fibration $f \colon \tilde{X} \to \mathbb{CP}^1$, singular parameters $\Delta = \{t_1, \ldots, t_\mu\}$ and critical points $p_1, \ldots, p_\mu$. Fix a base parameter $t_* \in \mathbb{CP}^1 \setminus \Delta$. Fix $i$, and let $z$ be coordinates near $p_i$ as in lemma 6.3.7, taken with respect to a holomorphic coordinate on $\mathbb{CP}^1$ centred at $t_i$, so that $f = t_i + \sum_j z_j^2$ near $p_i$. Let $\gamma$ be a path in $\mathbb{CP}^1 \setminus \Delta$ from $t_*$ to a parameter $t'$ close enough to $t_i$ that the sphere $S_{t'-t_i}$ of lemma 6.3.7 lies in $f^{-1}(t')$, and orient that sphere. The *vanishing cycle* attached to $p_i$ and $\gamma$ is the class

$$\delta_i \in H_{n-1}\big(f^{-1}(t_*), \mathbb{Z}\big) = H_{n-1}(X_{t_*}, \mathbb{Z})$$

obtained from the fundamental class of that sphere by the isomorphism that parallel transport along $\gamma$ induces on the homology of the fibres. The *Lefschetz thimble* attached to $p_i$ and $t'$ is the $n$-disc in $\tilde{X}$ swept out by the spheres $S_s$ as $s$ runs along a segment from 0 to $t' - t_i$; its boundary is the sphere above.

The class $\delta_i$ is well defined once the orientation of the sphere is fixed, and changing that orientation changes its sign. It depends on $\gamma$ only through the homotopy class of $\gamma$ rel endpoints, since homotopic paths induce the same map on the homology of the fibres, and it is independent of the choice of normal-form coordinates up to sign: the local fibre retracts onto the sphere, so its middle homology is infinite cyclic with the sphere class as a generator, and any other choice produces a generator of the same group. The dependence on $\gamma$ is genuine and is the subject of section 6.4: different paths to the same critical value produce different classes, and comparing them is what the monodromy of the family records.

Two features of $\delta_i$ follow immediately from the local model. First, the sphere bounds in $\tilde{X}$: the thimble of definition 6.3.8 is, in the local coordinates with $t' - t_i > 0$, the real ball $\{y \in \mathbb{R}^n \,\big|\, |y|^2 \leq t' - t_i\}$, an embedded $n$-disc whose boundary is the vanishing sphere. So $\delta_i$ maps to zero in $H_{n-1}(\tilde{X}, \mathbb{Z})$: a class that is nonzero on a fibre may die in the total space, and the vanishing cycles are the visible supply of such classes. Second, a tubular neighbourhood of the sphere inside the fibre is the disc bundle of $TS^{n-1}$, so the self-intersection of $\delta_i$ is the Euler number of the tangent bundle of a sphere, up to orientation: it is zero when $n$ is even and has absolute value 2 when $n$ is odd (exercise 6.3.1 (4)).

The word “vanishing” is literal: as the parameter approaches $t_i$ the sphere shrinks to the node and disappears from the singular fibre. The degeneration of a cubic curve makes the picture concrete.

**Example 6.3.9: A Pencil of Cubics and the Rational Elliptic Surface**

Take $X = \mathbb{C}\mathrm{P}^2$ embedded by the complete system of cubics, $v_3 \colon \mathbb{C}\mathrm{P}^2 \to \mathbb{C}\mathrm{P}^9$, and let $\{X_t\}$ be a generic pencil of plane cubics, which is a Lefschetz pencil by proposition 6.3.5. Two distinct cubics meet in 9 points counted with multiplicity, and for a generic pencil these are 9 distinct points at which the two curves meet transversally [4], so $B$ consists of 9 points and $\tilde{X}$ is $\mathbb{C}\mathrm{P}^2$ blown up at them.

The smooth members are smooth plane cubics, hence elliptic curves, with $\chi_{\mathrm{top}} = 0$. The singular members have a single node; such a curve is the image of $\mathbb{C}\mathrm{P}^1$ with two points identified, so $\chi_{\mathrm{top}} = 2 - 1 = 1$, one more than the smooth value, in agreement with the general relation of exercise 6.3.1 (2) at $n = 2$.

Counting Euler characteristics both ways determines $\mu$. On one side $\chi_{\mathrm{top}}(\tilde{X}) = 3 + 9 = 12$, each blown-up point contributing a $\mathbb{C}\mathrm{P}^1$ in place of a point. On the other side, the fibration over $\mathbb{C}\mathrm{P}^1$ has $2-\mu$ parameters carrying a smooth fibre and $\mu$ carrying a nodal one, so $\chi_{\mathrm{top}}(\tilde{X}) = (2-\mu)\cdot 0 + \mu \cdot 1 = \mu$. Hence $\mu = 12$: a generic pencil of plane cubics has exactly twelve nodal members, which is the classical statement that the discriminant hypersurface of plane cubics has degree 12.

Here $n = 2$, so a vanishing cycle is a class in $H_1(X_{t_*}, \mathbb{Z}) \cong \mathbb{Z}^2$ of the elliptic curve $X_{t_*}$, represented by an embedded circle that pinches to the node as $t \to t_i$. That circle is nonseparating, and what forces this is the irreducibility of the degenerate member. A reducible plane cubic has either more than one singular point or a singular point that is not an ordinary double point, so condition (3) of definition 6.3.3 leaves $X_{t_i}$ irreducible, hence the image of $\mathbb{C}\mathrm{P}^1$ with two points identified, a single sphere glued to itself. A separating circle on a torus bounds a disc, so collapsing one would instead exhibit the degenerate member as two closed surfaces meeting at a point. Hence $\delta_i \neq 0$; its self-intersection is zero, as it must be, the intersection form on the first homology of a curve being alternating.

Example 6.3.4 sits at the opposite extreme. There the fibres were rational curves with $H_1 = 0$, so every vanishing cycle is zero even though the pencil has three singular members: a degeneration always produces a vanishing sphere in the nearby fibre, but that sphere need not carry a nonzero homology class. What survives in general is the sphere itself and the reflection it determines, which is what section 6.4 computes.

## Exercise 6.3.1

1. Let $S \subseteq \mathbb{C}\mathrm{P}^3$ be a smooth surface of degree $d$. Using theorem 6.3.1 in the form applied in example 6.3.2, show that $S$ is connected, that $H^1(S, \mathbb{Z}) = 0$ and $H^3(S, \mathbb{Z}) = 0$, and deduce that $b_2(S) = \chi_{\text{top}}(S) - 2$. Then use theorem 4.2.1 and corollary 4.3.1 to conclude $h^{1,0}(S) = 0$.

2. Let $\{X_t\}$ be a Lefschetz pencil on $X$ with $\mu$ singular fibres, smooth member $X_{t_*}$ and base locus $B$, and let $f\colon \tilde{X} \to \mathbb{C}\mathrm{P}^1$ be as in proposition 6.3.6.

    1. Using lemma 6.3.7 and the Mayer-Vietoris sequence, show that $\chi_{\text{top}}(X_{t_i}) = \chi_{\text{top}}(X_{t_*}) + (-1)^n$ for each singular parameter $t_i$. (The boundaries of the two local pieces are closed manifolds of odd real dimension $2n-3$, whose Euler characteristics vanish by Poincaré duality; the parts of the two fibres outside the local pieces are diffeomorphic.)
    2. Deduce $\chi_{\text{top}}(\tilde{X}) = 2\chi_{\text{top}}(X_{t_*}) + (-1)^n\mu$ and then $\chi_{\text{top}}(X) = 2\chi_{\text{top}}(X_{t_*}) + (-1)^n\mu - \chi_{\text{top}}(B)$. (Cover $\mathbb{C}\mathrm{P}^1$ by small closed discs $D_i$ around the singular parameters and the closure $C$ of the complement, and use the two facts that Euler characteristic is multiplicative in a fibre bundle and that $f^{-1}(D_i)$ deformation retracts onto $X_{t_i}$.)
    3. Check both formulas against example 6.3.4 and example 6.3.9.

3. Let $X = \mathbb{C}\mathrm{P}^n \subseteq \mathbb{C}\mathrm{P}^N$ be a linearly embedded projective space with $2 \leq n < N$. Identify the variety $X^\vee$ of Step 1 in the proof of proposition 6.3.5 and compute its dimension. Conclude that a generic pencil of hyperplane sections of $X$ is a Lefschetz pencil with $\mu = 0$, describe $\tilde{X}$ and $f$ explicitly, and verify the Euler characteristic formula of exercise 6.3.1 (2) in this case.

4. Let $\delta \in H_{n-1}(X_{t_*}, \mathbb{Z})$ be a vanishing cycle of a Lefschetz pencil on a variety of dimension $n \geq 2$.

    1. Using lemma 6.3.7, show that a tubular neighbourhood of the vanishing sphere inside the fibre is the disc bundle of $TS^{n-1}$, and deduce that the self-intersection number $\delta \cdot \delta$ equals $\pm\chi_{\text{top}}(S^{n-1})$.
    2. Conclude that $\delta \cdot \delta = 0$ when $n$ is even and $|\delta \cdot \delta| = 2$ when $n$ is odd.
    3. Interpret part (2) for $n = 2$ against the intersection form on the first homology of a compact Riemann surface, and for $n = 3$ against the intersection form of a K3 surface as recorded in example 4.3.4.

5. Work in the local coordinates of lemma 6.3.7 at a critical point $p_i$, with $t' - t_i$ real and positive.

    1. Show that the set of points $\sqrt{s}\,x$ with $0 \leq s \leq t' - t_i$ and $x \in S^{n-1}$ is the closed real ball of radius $(t' - t_i)^{1/2}$ in $\mathbb{R}^n$, that $f$ maps it onto the segment $[t_i, t']$, and that its boundary is the vanishing sphere.
    2. Deduce that every vanishing cycle maps to 0 in $H_{n-1}(\tilde{X}, \mathbb{Z})$.
    3. In example 6.3.4 show that all three vanishing cycles are already zero in $H_1(X_{t_*}, \mathbb{Z})$, and identify the circle in a smooth conic that pinches at $t = [1 : 0]$.

## 6.4 Monodromy and the Picard-Lefschetz Formula

A Lefschetz pencil presents a smooth projective variety $X$ of dimension $n$ as a family of hyperplane sections over $\mathbb{CP}^1$ with finitely many singular members (definition 6.3.3, proposition 6.3.6). Over the complement of the critical values the family is a fibre bundle, so a loop in that complement carries the cohomology of a fixed smooth member back to itself. This section makes the resulting action precise and computes it for a loop encircling one critical value.

The computation is local. Everything that happens as the loop is traversed is concentrated in a small ball around the double point of the singular fibre, and inside that ball the map is $\sum z_j^2$ up to a change of coordinates. The answer is that the action moves a class only in the direction of the vanishing cycle (definition 6.3.8), by an amount read off from a single intersection number.

**6.4.1 Convention: Orientations, Loops, and the Intersection Pairing** Every complex manifold carries its complex orientation. On the middle homology $H_{n-1}(Y, \mathbb{Z})$ of a compact oriented $2(n-1)$-manifold $Y$ the symbol $\langle -, - \rangle$ denotes the intersection pairing, which satisfies $\langle a, b\rangle = (-1)^{n-1}\langle b, a\rangle$ and is unimodular modulo torsion by Poincaré-Lefschetz duality [24, integral unimodularity of the intersection pairing]. Loops in the base are traversed counterclockwise unless stated otherwise, and the product $\gamma\gamma'$ in $\pi_1$ means "traverse $\gamma'$ first, then $\gamma$", so that transport along a product is the composite of the transports in the same order.

The bundle structure over the complement of the critical values is what makes transport possible, and it comes from a single fact about proper submersions.

**Definition 6.4.2: Monodromy Representation**

Let $S$ be a connected complex manifold, $Y$ a complex manifold, and $f : Y \to S$ a proper holomorphic submersion, with fibres $Y_s = f^{-1}(s)$. Fix $s_0 \in S$. For a loop $\gamma$ at $s_0$, the pullback of $Y$ along $\gamma$ is a fibre bundle over $[0, 1]$ and hence trivial, and a trivialization determines a diffeomorphism $h_\gamma : Y_{s_0} \to Y_{s_0}$, the *geometric monodromy* of $\gamma$, well defined up to isotopy. The *monodromy representation* in degree $k$ is

$$\rho_k : \pi_1(S, s_0) \to \mathrm{Aut}\big(H^k(Y_{s_0}, \mathbb{Z})\big), \qquad \rho_k(\gamma) = \big(h_\gamma^{-1}\big)^*$$

and its image $\Gamma_k$ is the *monodromy group*. If the fibres have complex dimension $d$, Poincaré duality identifies $H_k(Y_{s_0}, \mathbb{Z})$ with $H^{2d-k}(Y_{s_0}, \mathbb{Z})$ and carries the homological monodromy $\gamma \mapsto (h_\gamma)_*$ on $H_k(Y_{s_0}, \mathbb{Z})$ to $\rho_{2d-k}$, the two indices agreeing exactly in the middle degree $k = d$. The homological action is written $\rho$ as well when the group it acts on is named.
For a Lefschetz pencil with associated fibration $f : \tilde{X} \to \mathbb{CP}^1$ (proposition 6.3.6) and critical values $t_1, \ldots, t_\mu$ (definition 6.3.3), the monodromy representation of the pencil is the one attached to the restriction of $f$ over $U = \mathbb{CP}^1 \setminus \{t_1, \ldots, t_\mu\}$.

Three points make the definition legitimate. A proper submersion is a locally trivial fibration [26, Ehresmann's theorem], so the pullback along $\gamma$ is a bundle over an interval and any two trivializations differ by a path of diffeomorphisms starting at the identity; the induced map on homology is therefore independent of the choice. Homotopic loops give isotopic diffeomorphisms, so $\rho_k$ descends to $\pi_1$. And the composition convention of box 6.4.1 makes $h_{\gamma\gamma'} = h_\gamma \circ h_{\gamma'}$, so $\rho_k$ is a homomorphism rather than an anti-homomorphism.

The smallest case already shows the mechanism, and it is the one where the vanishing cycle is a pair of points rather than a sphere.

**Example 6.4.3: Monodromy of a Double Cover**

Let $f\colon \mathbb{C} \to \mathbb{C}$ be $f(z) = z^2$. It is proper, since $|f(z)| = |z|^2$, and it is a submersion away from the origin, so its restriction over $S = \mathbb{C}^*$ is a proper holomorphic submersion with fibre $F_t = \{\pm\sqrt{t}\}$, two points. Take $s_0 = 1$, so $F_1 = \{1, -1\}$.

Along the loop $\gamma(\vartheta) = e^{2\pi i\vartheta}$ the trivialization is given by continuing the square root: the point $1 \in F_1$ travels as $e^{i\pi\vartheta}$ and arrives at $e^{i\pi} = -1$. So $h_\gamma$ exchanges the two points of $F_1$, and $\rho_0(\gamma)$ acts on $H^0(F_1, \mathbb{Z}) = \mathbb{Z}^2$ by swapping the coordinates.

On the reduced group $\tilde{H}_0(F_1, \mathbb{Z}) = \mathbb{Z}\,\delta$ with $\delta = [1] - [-1]$ the action is $\rho(\gamma)\delta = -\delta$, and the monodromy group is $\{\pm 1\}$. The class $\delta$ is the vanishing cycle of the critical point at the origin: as $t \to 0$ the two points of $F_t$ collide, and $\delta$ is the difference of the two branches, the zero-sphere $S^0$ shrinking to the origin.

The example is $\sum_{j=1}^n z_j^2$ with $n = 1$, and every case of the local computation looks like it with $S^0$ replaced by $S^{n-1}$. Lemma 6.3.7 already describes the piece of a nearby fibre that surrounds the vanishing sphere, the *Milnor fibre* of the singularity: it is a disc bundle of the tangent bundle of that sphere. What the proof of the Picard-Lefschetz formula needs beyond that description is the pair of homology classes the piece supports and the two intersection numbers between them, which come from comparing the complex orientation with the orientation of the disc bundle.

**Lemma 6.4.4: Generators and Intersection Numbers in the Local Fibre**

Let $n \geq 2$, let $f\colon \mathbb{C}^n \to \mathbb{C}$ be $f(z) = \sum_{j=1}^n z_j^2$, let $B = \{z \,\big|\, |z| \leq \epsilon\}$ and, for $0 < |\tau| < \epsilon^2$, put $F_\tau = f^{-1}(\tau) \cap B$. Write $m = n - 1$ and

$$\sigma_n = (-1)^{(n-1)(n-2)/2}$$

By lemma 6.3.7 the fibre $F_\tau$ is a compact complex $m$-manifold with boundary, diffeomorphic to a disc bundle of the tangent bundle $TS^m$ with the vanishing sphere as zero section; write $(u, v)$ for the coordinates of that disc bundle, scaled so that $|v| \leq 1$, and $\delta$ for the class of the zero section, which for $\tau > 0$ is the set of real points $\{x \in \mathbb{R}^n \,\big|\, |x|^2 = \tau\}$ of $F_\tau$.

1. $H_m(F_\tau, \mathbb{Z}) = \mathbb{Z}\,\delta$ and $H_k(F_\tau, \mathbb{Z}) = 0$ for $k \neq 0, m$, while $H_m(F_\tau, \partial F_\tau; \mathbb{Z})$ is infinite cyclic, generated by the class $\Delta$ of a fibre of the disc bundle.
2. With $\delta$ and $\Delta$ suitably oriented,

$$\langle \delta, \Delta \rangle = \sigma_n, \qquad \langle \delta, \delta \rangle = \sigma_n\big(1 + (-1)^m\big)$$

***Proof*:**

1. By lemma 6.3.7 the fibre $F_\tau$ deformation retracts onto the vanishing sphere, so its homology is that of $S^m$ [24] and $\delta$ generates in degree $m$. Poincaré-Lefschetz duality [24, Poincaré-Lefschetz duality] gives $H_m(F_\tau, \partial F_\tau; \mathbb{Z}) \cong H^m(F_\tau, \mathbb{Z}) \cong \mathbb{Z}$. The fibre disc $\Delta_{u_0} = \{(u_0, v) \,\big|\, v \perp u_0,\ |v| \leq 1\}$ over a point $u_0 \in S^m$ is a relative cycle meeting $\delta$ transversally in the single point $(u_0, 0)$, so $\langle \delta, \Delta_{u_0} \rangle = \pm 1$. Since the pairing between

$H_m(F_\tau)$ and $H_m(F_\tau, \partial F_\tau)$ is unimodular and $\delta$ generates the first group, $\Delta_{u_0}$ generates the second.

2. Multiplying the coordinates by a unit scalar $\lambda$ with $\lambda^2 = \bar{\tau}/|\tau|$ carries $F_\tau$ onto $F_{|\tau|}$, preserves $B$ and preserves complex orientations, so we may assume $\tau > 0$. Both numbers require comparing the complex orientation of $F_\tau$ with the orientation of the disc bundle. Fix an orientation $\mathfrak{o}$ of $S^m$ and let $\mathcal{O}$ be the orientation of the disc bundle given by $\mathfrak{o}$ on the base followed by $\mathfrak{o}$ on the fibre. At a point $(u, 0)$ of the zero section, where $z = \sqrt{\tau}\,u$ is real, the tangent space of $F_\tau$ is $\{w \in \mathbb{C}^n | u \cdot w = 0\} = T_u S^m \oplus i\, T_u S^m$, and by the parametrization in the proof of lemma 6.3.7 the first summand is the base directions and the second the fibre directions. For a real basis $a_1, \ldots, a_m$ of $T_u S^m$ the complex orientation is $(a_1, ia_1, \ldots, a_m, ia_m)$, while $\mathcal{O}$ is $(a_1, \ldots, a_m, ia_1, \ldots, ia_m)$; the shuffle relating the two has sign $(-1)^{m(m-1)/2} = \sigma_n$. As $F_\tau$ is connected, the complex orientation equals $\sigma_n\, \mathcal{O}$ everywhere.

   With respect to $\mathcal{O}$, the orientation of $\delta$ followed by that of $\Delta_{u_0}$ is by construction $\mathcal{O}$ itself, so the transverse intersection at $(u_0, 0)$ has sign $+1$ for $\mathcal{O}$ and sign $\sigma_n$ for the complex orientation. This is the first formula.

   For the second, the normal bundle of $\delta$ in $F_\tau$ is the vertical bundle along the zero section, that is $TS^m$, and the self-intersection is the Euler number of the normal bundle oriented so that the orientation of $\delta$ followed by that of the normal bundle gives the orientation of $F_\tau$ [26, self-intersection via the normal bundle]. The $\mathfrak{o}$-orientation of the fibres achieves this only up to the factor $\sigma_n$ just computed, so

$$\langle \delta, \delta \rangle = \sigma_n \int_{S^m} e(TS^m) = \sigma_n\, \chi_{\text{top}}(S^m) = \sigma_n \big(1 + (-1)^m\big)$$

   as we sought to show.

The self-intersection already separates the two parities. For $n$ even the vanishing sphere is odd-dimensional and its self-intersection vanishes, which will make the Picard-Lefschetz transformation a transvection; for $n$ odd the self-intersection is $\pm 2$ and the transformation is a reflection of order 2. The case $n = 2$ is small enough to draw.

**Example 6.4.5: The Vanishing Annulus**

Take $n = 2$, so $f(z_1, z_2) = z_1^2 + z_2^2$ and $m = 1$. The change of coordinates $w_1 = z_1 + iz_2$, $w_2 = z_1 - iz_2$ turns $f$ into $w_1 w_2$, and the fibre over $\tau \neq 0$ is $\{w_1 w_2 = \tau\}$, the graph of $w_2 = \tau / w_1$, which inside $B$ is an annulus. Lemma 6.3.7 says the same thing: a disc bundle of $TS^1$ is $S^1 \times [-1, 1]$.

The vanishing cycle is the core circle. In the coordinates $(u, v)$ it is $v = 0$; in the original coordinates and for $\tau > 0$ it is the real circle $x_1^2 + x_2^2 = \tau$. The two intersection numbers of lemma 6.4.4 read $\sigma_2 = 1$, so $\langle \delta, \Delta \rangle = 1$ and $\langle \delta, \delta \rangle = 0$, the latter because a circle on a surface can always be pushed off itself.

The singular fibre $F_0 = \{w_1 w_2 = 0\}$ is a pair of discs meeting at the origin, and the annulus degenerates to it by collapsing the core circle: this is part (3) of lemma 6.3.7 made visible, since a pair of discs glued at a point is contractible.

The monodromy of a loop around the critical value can be chosen to do nothing away from the double point, so that the entire effect of the loop is a diffeomorphism of the local fibre, and that diffeomorphism can be written down.

**Lemma 6.4.6: Localization of the Local Monodromy**

Let $\mathcal{X}$ be a complex manifold of dimension $n \geq 2$, let $D \subseteq \mathbb{C}$ be a disc centred at 0, and let $f\colon \mathcal{X} \to D$ be a proper holomorphic map which is a submersion over $D \setminus \{0\}$ and whose only critical point $p$ is an ordinary double point of $X_0 = f^{-1}(0)$. Choose holomorphic coordinates near $p$ in which $f = \sum_{j=1}^{n} z_j^2$, let $B$ be a ball as in lemma 6.4.4, and fix a small $\tau > 0$. Write $F_t = X_t \cap B$ and $F = F_\tau$.

1. The restriction of $f$ to $f^{-1}(\bar{D}_\tau) \setminus \mathring{B}$ is a proper submersion onto the closed disc $\bar{D}_\tau$ of radius $\tau$, and is therefore trivial over it.

2. The geometric monodromy of the counterclockwise loop $|t| = \tau$ may be chosen to be a diffeomorphism $h$ of $X_\tau$ equal to the identity on $X_\tau \setminus \mathring{F}$.

3. The diffeomorphism $h$ of (2) may in addition be chosen to equal on $F$, in the coordinates $(u, v)$ of lemma 6.4.4, the map

$$h_\psi(u, v) = \big(\cos\theta\, u - \sin\theta\, \hat{v}, \quad |v| \sin\theta\, u + \cos\theta\, v\big), \qquad \theta = \pi\, \psi(|v|)$$

where $\hat{v} = v/|v|$ and $\psi\colon [0, 1] \to [0, 1]$ is smooth, identically 1 near 0, identically 0 near 1, and satisfies $\psi' < 0$ at every point where its value lies in $(0, 1)$, so that $\psi$ is strictly decreasing there. Such a $\psi$ exists: take $\psi(x) = 1 - \int_0^x \chi$ for a smooth $\chi \geq 0$ of integral 1 which is strictly positive exactly on an interval $(a, b) \subseteq (0, 1)$. Near the zero section $h_\psi$ is $(u, v) \mapsto (-u, -v)$ and near $\partial F$ it is the identity, so it is smooth on all of $F$ despite the appearance of $\hat{v}$.

***Proof***:

1. Shrinking $\epsilon$ and then $\tau$, the sphere $\partial B$ is transverse to every fibre $X_t$ with $|t| \leq \tau$, since transversality holds at $t = 0$ for the model $\sum z_j^2$ and is an open condition on a compact set. Hence $f^{-1}(\bar{D}_\tau) \setminus \mathring{B}$ is a manifold with corners on which $f$ is a proper submersion, and Ehresmann's theorem in the form valid for manifolds with boundary [26, Ehresmann's theorem for manifolds with boundary] trivializes it over the contractible base $\bar{D}_\tau$. In particular the family of links $L_t = X_t \cap \partial B$ is trivial over $\bar{D}_\tau$, including over $t = 0$.

2. Let $A$ be an annulus neighbourhood of the circle $|t| = \tau$ inside $D \setminus \{0\}$ and let $w$ be the vector field on $A$ generating the counterclockwise rotation. On $f^{-1}(A)$ the map $f$ is a proper submersion, so a partition of unity produces a vector field $\tilde{w}_1$ with $df(\tilde{w}_1) = w$. On $f^{-1}(A) \setminus \mathring{B}$ the trivialization of (1) supplies a second such lift $\tilde{w}_2$, namely the one whose flow preserves the product decomposition. Choose a slightly larger ball $B' \supset B$, of radius $\epsilon'$ and still transverse to the fibres, and a cutoff $\chi$ that equals 1 on $B$ and vanishes off $B'$, and set
$$\tilde{w} = \chi\, \tilde{w}_1 + (1 - \chi)\, \tilde{w}_2$$
A convex combination of lifts of $w$ is again a lift of $w$, so $\tilde{w}$ is $f$-related to $w$, and its time-$2\pi$ flow is a diffeomorphism $h$ of $X_\tau$ realizing the geometric monodromy. Off $B'$ one has $\tilde{w} = \tilde{w}_2$, whose flow returns each point to its starting position after a full turn because the trivialization of (1) extends over the disc. So $h$ is the identity off $B'$, and after replacing $B$ by $B'$ it is the identity on $X_\tau \setminus \mathring{F}$.

3. Inside the ball the family has an explicit trivialization: since $f$ is homogeneous of degree 2, the maps
$$\nu_\vartheta\colon F_\tau \to F_{\tau e^{2\pi i\vartheta}}, \qquad \nu_\vartheta(z) = e^{i\pi\vartheta} z$$
preserve $|z|$, hence preserve $B$, and form a trivialization of the local family over the circle. Its monodromy is $\nu_1 = -\mathrm{id}$, which in the coordinates of lemma 6.4.4 is $(u, v) \mapsto (-u, -v)$, the rotation by $-\pi$ in the plane spanned by the orthonormal pair $(u, \hat{v})$.

That is not yet a monodromy of the kind (2) asks for, because $\nu_\vartheta$ does not agree near $\partial B$ with the trivialization of (1); the two differ by a path of diffeomorphisms of the link, which the model computes. Writing $z = x + iy$ and $t = t_1 + it_2$, the link $L_t$ is the set of pairs $(x, y)$ with
$$|x| = a(t_1), \qquad |y| = b(t_1), \qquad x \cdot y = t_2/2, \qquad 2a^2 = \epsilon^2 + t_1, \quad 2b^2 = \epsilon^2 - t_1$$
so $(\xi, \eta) = (x/a,\ y/b)$ identifies $L_t$ with
$$\Sigma_\beta = \big\{(\xi, \eta) \in S^m \times S^m \;\big|\; \xi \cdot \eta = \beta\big\}, \qquad \beta = \beta(t) := \frac{t_2}{2ab}$$
and $|\beta| < 1$ for $|t| \le \tau$. The maps
$$\Theta_\beta\colon \Sigma_0 \to \Sigma_\beta, \qquad (\xi, \eta) \longmapsto \big(\xi,\ \beta\,\xi + \sqrt{1 - \beta^2}\,\eta\big)$$
are diffeomorphisms depending smoothly on $\beta$, so $t \mapsto \Theta_{\beta(t)}$ trivializes the family of links over $\bar{D}_\tau$, and $\beta(\tau) = 0$. Any other such trivialization differs from this one by a map $\bar{D}_\tau \to \mathrm{Diff}(L_\tau)$, whose restriction to the boundary circle is a null-homotopic loop at the identity, so the comparison path
$$c_\vartheta = \Theta^{-1}_{\beta(t(\vartheta))} \circ \nu_\vartheta \in \mathrm{Diff}(L_\tau), \qquad t(\vartheta) = \tau e^{2\pi i\vartheta}$$
is independent of the trivialization chosen in (1), up to homotopy with fixed endpoints.

That homotopy class is the class of a rotation. On $L_\tau$ one has $\xi \perp \eta$, and $\nu_\vartheta$ carries $(x, y)$ to $(\cos\theta\, x - \sin\theta\, y,\ \sin\theta\, x + \cos\theta\, y)$ with $\theta = \pi\vartheta$, which in the normalized coordinates is the map sending $(\xi, \eta)$ to the pair
$$p_s = \cos\theta\, a_s\, \xi - \sin\theta\, b_s\, \eta, \qquad q_s = \sin\theta\, a_s\, \xi + \cos\theta\, b_s\, \eta$$
divided by their lengths, at $s = 0$, where $a_s = (1 - s)a(\tau) + s$ and $b_s = (1 - s)b(\tau) + s$ for $s \in [0, 1]$. The matrix carrying $(\xi, \eta)$ to $(p_s, q_s)$ has determinant $a_s b_s > 0$, and the lengths of $p_s$ and $q_s$ and their inner product depend only on $s$ and $\vartheta$, so normalizing gives a diffeomorphism of $\Sigma_0$ onto $\Sigma_{\beta_{s,\vartheta}}$ for the resulting inner product $\beta_{s,\vartheta}$; composing it with $\Theta^{-1}_{\beta_{s,\vartheta}}$ produces a family $c^s_\vartheta \in \mathrm{Diff}(\Sigma_0)$, continuous in $(s, \vartheta)$, with $c^0 = c$. At $\vartheta = 0$ and at $\vartheta = 1$ one has $\sin\theta = 0$, hence $\beta_{s,\vartheta} = 0$ and $c^s_\vartheta = \mathrm{id}$ respectively $-\mathrm{id}$, for every $s$: the homotopy fixes the endpoints. At $s = 1$, where $a_1 = b_1 = 1$, the pair $(p_1, q_1)$ is already orthonormal and $c^1_\vartheta$ is the rotation of $(\xi, \eta)$ by the angle $-\pi\vartheta$ in the plane they span. So $c$ is homotopic with fixed endpoints to that path of rotations.

Correct $\nu$ by the rotation in the opposite sense, damped towards the zero section. Let $g_\vartheta$ be the rotation by $\pi\vartheta\big(1 - \psi(|v|)\big)$ in the plane spanned by $(u, \hat{v})$; it preserves $|v|$ and is the identity near the zero section, where $\psi = 1$, so it is a diffeomorphism of $F_\tau$, and $g_0 = \mathrm{id}$. The maps $\nu_\vartheta \circ g_\vartheta$ again trivialize the local family over the circle, and near $\partial B$, where

$\psi = 0$, their comparison with the trivialization of (1) is the pointwise composite $c_\vartheta \circ g_\vartheta$: a path of diffeomorphisms of the link which starts and ends at the identity and which, by the previous paragraph, is null-homotopic with fixed endpoints. Every sphere $|z| = \varrho$ with $\epsilon \leq \varrho \leq \epsilon'$ is transverse to the fibres, so $(f, |z|)$ is a submersion on the collar $B' \setminus \mathring{B}$ and Ehresmann's theorem [26, Ehresmann's theorem] allows both trivializations to be taken there to preserve $|z|$; reading the parameter of a null-homotopy of $c_\vartheta \circ g_\vartheta$ as the radius $\varrho$ then interpolates between the two across the collar, and they glue to a trivialization of the family over the circle.

Its monodromy is the identity outside $\mathring{B}$: beyond the collar it is the monodromy of the trivialization of (1), which is trivial because that trivialization extends over the disc, and on the collar the interpolating correction is the identity at $\vartheta = 1$, the null-homotopy being stationary at the endpoints. On $F_\tau$ the monodromy is $\nu_1 \circ g_1$, the rotation by $\pi(1 - \psi(|v|))$ followed by the rotation by $-\pi$ in the same plane, that is the rotation by $-\pi\psi(|v|)$. Writing that rotation out in coordinates produces the displayed formula for $h_\psi$, as we sought to show.

Real rotations of $\mathbb{R}^n$ preserve $\sum z_j^2$, so $h_\psi$ does map $F_\tau$ to itself; at the zero section it is the antipodal map of $S^{n-1}$, and near the boundary it is the identity. It is the higher-dimensional Dehn twist: on the annulus of example 6.4.5 it is the classical twist that rotates the core circle once and tapers to the identity at the two boundary circles.

The monodromy differs from the identity only through a diffeomorphism supported in the local fibre; that fibre carries exactly one class in each of the two relevant homology groups; and the diffeomorphism is explicit.

**Theorem 6.4.7: The Picard-Lefschetz Formula**

Let $\mathcal{X}$ be a complex manifold of dimension $n \geq 2$, let $D \subseteq \mathbb{C}$ be a disc centred at 0, and let $f\colon \mathcal{X} \to D$ be a proper holomorphic map which is a submersion over $D \setminus \{0\}$ and whose only critical point is an ordinary double point of $X_0$. Fix $\tau \in D \setminus \{0\}$ small enough that lemma 6.4.6 applies, let $\delta \in H_{n-1}(X_\tau, \mathbb{Z})$ be the vanishing cycle, that is the image of the class of the vanishing sphere in the local fibre (lemma 6.4.4), and let $T = h_*$ be the action on $H_*(X_\tau, \mathbb{Z})$ of the geometric monodromy $h$ of the counterclockwise loop $|t| = |\tau|$. Then $T$ is the identity on $H_k(X_\tau, \mathbb{Z})$ for every $k \neq n-1$, while on $H_{n-1}(X_\tau, \mathbb{Z})$

$$T(\alpha) = \alpha + \varepsilon_n \langle \alpha, \delta \rangle \delta, \qquad \varepsilon_n := (-1)^{n(n+1)/2} \tag{6.5}$$

Moreover $\langle \delta, \delta \rangle = -\varepsilon_n\big(1 + (-1)^{n-1}\big)$ and $T(\delta) = (-1)^n \delta$.

***Proof*:**
Replacing $t$ by $t|\tau|/\tau$ throughout, we may assume $\tau > 0$. Write $m = n-1$, let $B$ and $F = X_\tau \cap B$ be as in lemma 6.4.6, and let $h$ be the geometric monodromy that lemma supplies: the identity on $X_\tau \setminus \mathring{F}$ and equal on $F$ to the map $h_\psi$ of its part (3). Any two geometric monodromies of the loop are isotopic (definition 6.4.2), so $T = h_*$ is unaffected by the choice. Since $\psi$ vanishes near 1, the map $h$ is the identity on the collar $\{\psi = 0\}$ of $\partial F$; write $F'' \subseteq \mathring{F}$ for the complement of that collar, so that $h$ is the identity on all of $X_\tau \setminus F''$. Set $\sigma_n = (-1)^{(n-1)(n-2)/2}$ as in lemma 6.4.4.

The identity

$$\frac{n(n+1)}{2} - \frac{(n-1)(n-2)}{2} = 2n-1$$

shows that $\varepsilon_n = -\sigma_n$, and the assertion about $\langle\delta,\delta\rangle$ is then part (2) of lemma 6.4.4: the self-intersection may be computed inside $F$, because $\delta$ is supported there and a representative may be pushed off itself without leaving $F$. The proof of (6.5) occupies Steps 1 to 4, and Step 5 treats the remaining degrees.

*Step 1: The variation map.* Let $a$ be a cycle representing $\alpha \in H_k(X_\tau,\mathbb{Z})$. The sets $\mathring{F}$ and $X_\tau \setminus F''$ form an open cover of $X_\tau$, so after barycentric subdivision [24] every simplex of $a$ lies in one of them. Those of the second kind are fixed by $h$ and cancel in $h_\# a - a$; those of the first kind are carried by $h$ into $F$, so their contribution is a chain in $F$. Thus $h_\# a - a$ is a chain in $F$ with zero boundary, hence a cycle in $F$. Its class depends only on the image of $\alpha$ in $H_k(X_\tau, X_\tau \setminus \mathring{F})$, because a class vanishing there is represented by a cycle supported in $X_\tau \setminus \mathring{F}$, which $h$ fixes. This defines the *variation*

$$\mathrm{var}\colon H_k\big(X_\tau, X_\tau \setminus \mathring{F}\big) \to H_k(F,\mathbb{Z}), \qquad T - \mathrm{id} = \mathrm{var}\circ j$$

where $j$ is the natural map from $H_k(X_\tau,\mathbb{Z})$. Excision identifies the source with $H_k(F,\partial F;\mathbb{Z})$ [24]. The collar needed to apply that duality is supplied by [26, the collar neighbourhood theorem].

*Step 2: The image of $j$.* Take $k=m$. By lemma 6.4.4 the group $H_m(F,\partial F;\mathbb{Z})$ is generated by the fibre disc $\Delta$, and the intersection pairing between $H_m(F,\mathbb{Z}) = \mathbb{Z}\delta$ and $H_m(F,\partial F;\mathbb{Z}) = \mathbb{Z}\Delta$ satisfies $\langle\delta,\Delta\rangle = \sigma_n$. Since $\delta$ is supported in $F$, the intersection number $\langle\delta,\alpha\rangle$ computed in $X_\tau$ agrees with the pairing of $\delta$ against $j(\alpha)$. Writing $j(\alpha) = c\,\Delta$ and pairing with $\delta$ gives $c\,\sigma_n = \langle\delta,\alpha\rangle$, so

$$j(\alpha) = \sigma_n\,\langle\delta,\alpha\rangle\,\Delta$$

*Step 3: The variation of the fibre disc.* The claim is that $\mathrm{var}(\Delta) = (-1)^n\delta$. Work in the coordinates $(u,v)$ of lemma 6.4.4 and take $h = h_\psi$ from lemma 6.4.6, which acts on the point $(u,v)$ as the rotation by $-\theta$, $\theta = \pi\psi(|v|)$, in the plane spanned by the orthonormal pair $(u,\hat{v})$. Choose an orthonormal basis $e_0,\dots,e_m$ of $\mathbb{R}^n$, set $u_0 = e_0$, and orient $S^m$ by declaring $u\wedge \mathrm{or}(T_uS^m)$ to be the standard orientation of $\mathbb{R}^n$, so that $\mathrm{or}(T_{e_0}S^m) = e_1\wedge\cdots\wedge e_m$. Let $\Delta = \Delta_{u_0}$ be the fibre disc over $u_0$, oriented by that basis. Fix $\varphi\in(0,\pi)$ and let $R_\varphi$ be the rotation by $\varphi$ in the plane $(e_0,e_1)$, so that $u_1 := R_\varphi e_0$ and $w' := R_\varphi e_1$ are orthonormal and $R_\varphi$ fixes $e_2,\dots,e_m$.

The chain $h(\Delta)-\Delta$ is a cycle in $F$ because $h$ is the identity near $\partial F$, and its class is $\lambda\,\delta$ for some $\lambda\in\mathbb{Z}$. Pair it with the relative class $\Delta_{u_1}$. The pairing may be computed at the chain level, since neither $\Delta$ nor $h(\Delta)$ meets $\partial\Delta_{u_1}$: both meet $\partial F$ in the fibre over $u_0$, and $u_0\neq u_1$. As $\Delta$ and $\Delta_{u_1}$ are disjoint fibres, only $h(\Delta)$ contributes, so $\langle\delta,\Delta_{u_1}\rangle = \sigma_n$ gives $\lambda\sigma_n = \langle h(\Delta),\Delta_{u_1}\rangle$.

The intersection $h(\Delta)\cap\Delta_{u_1}$ consists of one point. Indeed $h(u_0,v)$ has first coordinate $\cos\theta\,e_0 - \sin\theta\,\hat{v}$, and setting this equal to $u_1 = \cos\varphi\,e_0 + \sin\varphi\,e_1$ forces $\cos\theta = \cos\varphi$, hence $\theta=\varphi$ because $\theta\in[0,\pi]$, and then $\hat{v} = -e_1$. Since $\psi$ is strictly decreasing where it takes values in $(0,1)$, the equation $\pi\psi(s) = \varphi$ has the unique solution $s\in(0,1)$, so the only preimage is $v_* = -s\,e_1$, and

$$P := h(u_0,v_*) = \big(u_1,\ -s\,w'\big)$$

which lies in $\Delta_{u_1}$ because $w'\perp u_1$.

It remains to compute the sign of that intersection. Differentiating $v\mapsto h(u_0,v)$ at $v_*$ along $e_1$ and along $e_j$ for $j\geq 2$ gives, with $\kappa := \pi\psi'(s)$, which is negative because $\psi(s) = \varphi/\pi$ lies in

$(0,1)$ and lemma 6.4.6 takes $\psi' < 0$ at every such point,

$$X_1 = \big(-\kappa\, w',\ w' - s\kappa\, u_1\big), \qquad X_j = \big(-\sin\varphi\, e_j,\ s\cos\varphi\, e_j\big)$$

the second up to the positive factor $1/s$. The tangent space of $\Delta_{u_1}$ at $P$ is spanned by $Y_1 = (0, w')$ and $Y_j = (0, e_j)$, in that order for the chosen orientation, since $w', e_2, \ldots, e_m$ is the positively oriented basis $R_\varphi(e_1, \ldots, e_m)$ of $T_{u_1}S^m$. Splitting $T_P F$ into horizontal and vertical parts for the Levi-Civita connection of the round sphere, the horizontal lift of $a \in T_{u_1}S^m$ at the point $(u_1, v_P)$ is $(a, -(v_P \cdot a)u_1)$, so with $v_P = -sw'$

$$H_1 = (w',\, s\, u_1), \qquad H_j = (e_j, 0), \qquad V_1 = (0, w'), \qquad V_j = (0, e_j)$$

is a basis adapted to the orientation $\mathcal{O}$ of the disc bundle used in the proof of lemma 6.4.4, the base orientation followed by the fibre orientation, for which the complex orientation is $\sigma_n\mathcal{O}$; explicitly $\mathcal{O} = H_1 \wedge \cdots \wedge H_m \wedge V_1 \wedge \cdots \wedge V_m$. In this basis $X_1 = -\kappa H_1 + V_1$ and $X_j = -\sin\varphi\, H_j + s\cos\varphi\, V_j$, while $Y_1 = V_1$ and $Y_j = V_j$. Reordering basis and vectors by the same shuffle to $(H_1, V_1, H_2, V_2, \ldots)$ and $(X_1, Y_1, X_2, Y_2, \ldots)$ leaves the determinant unchanged and makes it block diagonal with blocks

$$\begin{pmatrix} -\kappa & 0 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} -\sin\varphi & 0 \\ s\cos\varphi & 1 \end{pmatrix}$$

of determinants $-\kappa > 0$ and $-\sin\varphi < 0$. The determinant therefore has sign $(-1)^{m-1}$, so the intersection sign at $P$ is $(-1)^{m-1}$ for $\mathcal{O}$ and $\sigma_n(-1)^{m-1}$ for the complex orientation. Comparing with $\lambda\sigma_n$ gives $\lambda = (-1)^{m-1} = (-1)^n$.

*Step 4: Assembling.* Combining Steps 1 to 3,

$$T(\alpha) - \alpha = \operatorname{var}\big(j(\alpha)\big) = \sigma_n\, \langle\delta, \alpha\rangle \operatorname{var}(\Delta) = \sigma_n (-1)^n\, \langle\delta, \alpha\rangle\, \delta$$

The pairing on middle homology satisfies $\langle\delta, \alpha\rangle = (-1)^m\langle\alpha, \delta\rangle$, and $(-1)^{n+m} = (-1)^{2n-1} = -1$, so the coefficient is $-\sigma_n = \varepsilon_n$. This is (6.5). Substituting $\alpha = \delta$ and using $\langle\delta, \delta\rangle = \sigma_n(1 + (-1)^m)$ gives $T(\delta) = \delta - (1 + (-1)^{n-1})\delta = (-1)^n\delta$.

*Step 5: The other degrees.* For $k \neq m$ and $k \neq 2m$, Lefschetz duality and lemma 6.4.4 give $H_k(F, \partial F; \mathbb{Z}) \cong H^{2m-k}(F, \mathbb{Z}) = 0$, so $j = 0$ and $T$ is the identity on $H_k(X_\tau, \mathbb{Z})$ by Step 1. For $k = 2m$ the group $H_{2m}(X_\tau, \mathbb{Z})$ is free on the fundamental classes of the connected components of $X_\tau$, a compact oriented $2m$-manifold, which need not be connected. The local fibre $F$ is connected, so it lies in a single component $C_0$, and $h$ is the identity on the open set $X_\tau \setminus F$. That set meets every component: the components other than $C_0$ are contained in it, and $C_0$ is not contained in $F$, since $F \subseteq C_0$ would force $C_0 = F$, whereas $F$ has nonempty boundary and $C_0$ has none. So $h$ fixes a point of each component, hence carries each component to itself, and restricts on it to an orientation-preserving diffeomorphism; it therefore fixes each of the generating fundamental classes, completing the proof.

The content of (6.5) is that a full turn around a critical value changes a class only in the direction of the vanishing cycle, and by an amount that a single intersection number determines. Two consequences are immediate from the formula. The transformation $T$ fixes the hyperplane $\delta^\perp = \{\alpha \,|\, \langle\alpha, \delta\rangle = 0\}$ pointwise, and it preserves the intersection pairing, as it must, being induced by an orientation-preserving diffeomorphism. And the parity of $n$ decides its nature: for $n$ odd, $\langle\delta, \delta\rangle = \pm 2$ and $T$ is the reflection in $\delta^\perp$, of order 2; for $n$ even, $\langle\delta, \delta\rangle = 0$ and $T$ is a transvection, unipotent with

$(T - \mathrm{id})^2 = 0$ and of infinite order whenever $\delta \neq 0$ in $H_{n-1}(X_\tau, \mathbb{Q})$. A $\delta$ that is nonzero but torsion gives $T = \mathrm{id}$, since the integral intersection pairing vanishes against torsion classes.

The formula also takes a clean shape in the normalization chapter 5 fixed. Suppose the fibres are compact Kähler, as they are for a Lefschetz pencil on a projective manifold. The fibre $X_\tau$ then has dimension $n-1$, and the form proposition 5.2.8 attaches to its middle degree $k = n-1$ carries the factor $\omega^{(n-1)-k} = 1$, so it is

$$Q(\alpha, \beta) = (-1)^{(n-1)(n-2)/2} \int_{X_\tau} \alpha \wedge \beta = \sigma_n \langle \alpha, \beta \rangle$$

Writing $\delta$ also for the Poincaré dual of the vanishing cycle, (6.5) becomes

$$T(\alpha) = \alpha - Q(\alpha, \delta)\, \delta$$

with no residual sign. The normalizing factor that proposition 5.2.8 introduced for the sake of positivity is the same factor that makes the Picard-Lefschetz formula sign-free.

The case of a pencil of curves on a surface is the classical one, and there the formula can be checked against a count of singular fibres.

**Example 6.4.8: The Nodal Degeneration of a Curve**

Let $n = 2$, so the fibres are compact Riemann surfaces and $\varepsilon_2 = (-1)^3 = -1$. Theorem 6.4.7 reads

$$T(\alpha) = \alpha - \langle \alpha, \delta \rangle\, \delta$$

on $H_1(X_\tau, \mathbb{Z})$, with $\langle \delta, \delta \rangle = 0$. If $\delta$ is part of a symplectic basis, say $\langle \delta, \beta \rangle = 1$, then $T$ fixes $\delta$ and sends $\beta$ to $\beta - \langle \beta, \delta \rangle \delta = \beta + \delta$, so in the basis $(\delta, \beta)$ of the plane they span the matrix is

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

unipotent of infinite order.

*A concrete pencil.* Take the general pencil of plane cubics of example 6.3.9: the associated fibration is $f \colon \tilde{X} \to \mathbb{C}\mathrm{P}^1$ with $\tilde{X}$ the blow-up of $\mathbb{C}\mathrm{P}^2$ at the nine base points (proposition 6.3.6), the smooth fibres are elliptic curves, and the Euler characteristic count of exercise 6.3.1 (2), carried out there, gives $\mu = 12$ nodal members.

Each of the twelve local monodromies is a transvection along its own vanishing cycle, and the fundamental group of the complement $U$ of the twelve critical values is generated by loops $\gamma_1, \ldots, \gamma_{12}$ around the punctures subject to the single relation that their product in a suitable order is trivial, so the monodromy group $\Gamma \subseteq \mathrm{Sp}(2, \mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})$ is generated by twelve transvections satisfying that one relation.

*What one transvection does to the periods.* Let $\delta$ and $\beta$ be as above, let $\eta$ be a nonzero holomorphic 1-form on the fibre, and let $\lambda = \int_\beta \eta / \int_\delta \eta$ be the period ratio. Transporting once around the puncture replaces the pair $(\delta, \beta)$ by $(T\delta, T\beta) = (\delta, \beta + \delta)$, so

$$\lambda \longmapsto \frac{\int_{\beta+\delta} \eta}{\int_\delta \eta} = \lambda + 1$$

This is the classical monodromy of a degenerating family of elliptic curves, and it is the substitution $\tau \mapsto \tau + 1$ on the lattice parameter of exercise 5.2.1 (6). Its Hodge-theoretic form, in which $\lambda$ becomes the value of a period map and $\lambda \mapsto \lambda + 1$ the monodromy of that map, belongs to chapter 7; the elliptic-curve background is in [5].

The example used a point that the formula makes general: conjugating a local monodromy transports its vanishing cycle. This keeps the vanishing cycles a single stable set rather than $\mu$ unrelated classes, and it identifies the classes on which the whole monodromy group acts trivially.

**Corollary 6.4.9: Conjugate Vanishing Cycles and the Invariant Classes**

Let $f\colon \tilde{X} \to \mathbb{C}\mathrm{P}^1$ be the fibration of a Lefschetz pencil with critical values $t_1, \dots, t_\mu$, let $t_0 \in U$, and for each $i$ let $\gamma_i \in \pi_1(U, t_0)$ be a simple loop around $t_i$ along a chosen path, with $T_i = \rho(\gamma_i)$ and with $\delta_i \in H_{n-1}(X_{t_0}, \mathbb{Z})$ the vanishing cycle at $t_i$ transported to $X_{t_0}$ along that path, so that $T_i$ satisfies (6.5) with $\delta$ replaced by $\delta_i$. Let $\Gamma$ be the monodromy group.

1. For every $g \in \pi_1(U, t_0)$ the element $\rho(g\gamma_i g^{-1})$ satisfies (6.5) with $\delta_i$ replaced by $\rho(g)\delta_i$. In particular the set $\mathcal{V}$ of all classes arising as vanishing cycles, over all choices of connecting paths, is stable under $\Gamma$ and under $\delta \mapsto -\delta$, and every element of $\mathcal{V}$ has the same self-intersection $-\varepsilon_n(1 + (-1)^{n-1})$.

2. $\Gamma$ is generated by $T_1, \dots, T_\mu$.

3. The invariant classes are the classes orthogonal to every vanishing cycle:

$$H_{n-1}(X_{t_0}, \mathbb{Z})^\Gamma = \{\alpha \big| \langle \alpha, \delta_i \rangle = 0 \text{ for } i = 1, \dots, \mu\}$$

   and $T_i = \mathrm{id}$ if and only if $\delta_i = 0$ in $H_{n-1}(X_{t_0}, \mathbb{Q})$.

***Proof*:**

1. Put $A = \rho(g)$, an automorphism preserving the intersection pairing because it is induced by an orientation-preserving diffeomorphism. Then for every $\alpha$,

$$AT_iA^{-1}(\alpha) = A\big(A^{-1}\alpha + \varepsilon_n \langle A^{-1}\alpha, \delta_i \rangle \delta_i\big) = \alpha + \varepsilon_n \langle \alpha, A\delta_i \rangle A\delta_i$$

   using $\langle A^{-1}\alpha, \delta_i \rangle = \langle \alpha, A\delta_i \rangle$. Geometrically $A\delta_i$ is the vanishing cycle at $t_i$ transported along the path modified by $g$, so conjugates of the $T_i$ are again Picard-Lefschetz transformations and $\mathcal{V}$ is $\Gamma$-stable. Stability under sign change holds because the vanishing sphere carries two orientations and (6.5) is unchanged by $\delta \mapsto -\delta$. The self-intersection is the value computed in theorem 6.4.7, and $A$ preserves it.

2. The complement $U$ of $\mu$ points in $\mathbb{C}\mathrm{P}^1$ has fundamental group generated by simple loops around the punctures [24], and changing the connecting path replaces $\gamma_i$ by a conjugate. Applying $\rho$ gives the claim.

3. By (2) a class is $\Gamma$-invariant if and only if it is fixed by every $T_i$, and by (6.5) the condition $T_i\alpha = \alpha$ says $\varepsilon_n\langle \alpha, \delta_i \rangle \delta_i = 0$. If $\delta_i$ is not torsion, that is if $\delta_i \neq 0$ in $H_{n-1}(X_{t_0}, \mathbb{Q})$, an integer multiple of $\delta_i$ vanishes only when the integer does, so the condition reads $\langle \alpha, \delta_i \rangle = 0$. If $\delta_i$ is torsion, say $N\delta_i = 0$ with $N \geq 1$, then $N\langle \alpha, \delta_i \rangle = \langle \alpha, N\delta_i \rangle = 0$ in $\mathbb{Z}$, so $\langle \alpha, \delta_i \rangle = 0$ and both conditions hold automatically. That is the displayed identity, the case $\delta_i = 0$ being the case $N = 1$.

   The same dichotomy gives the last statement. If $\delta_i = 0$ in $H_{n-1}(X_{t_0}, \mathbb{Q})$, that is if $\delta_i$ is torsion, the computation just made yields $\langle \alpha, \delta_i \rangle = 0$ for every $\alpha$ and hence $T_i = \mathrm{id}$. Conversely if $\delta_i \neq 0$ in $H_{n-1}(X_{t_0}, \mathbb{Q})$ then, the intersection pairing being nondegenerate over $\mathbb{Q}$ by Poincaré duality, some $\alpha$, integral after clearing denominators, has $\langle \alpha, \delta_i \rangle \neq 0$; then $\langle \alpha, \delta_i \rangle \delta_i$ is nonzero already in $H_{n-1}(X_{t_0}, \mathbb{Q})$, so $T_i \alpha \neq \alpha$, as we sought to show.

Part (3) is the statement the Hodge theory of families will use, and chapter 7 takes it up: the invariant classes are exactly the classes no vanishing cycle pairs against, so intersection theory alone determines them. Everything in this section is topology. The representation $\rho$ records how the lattice $H^{n-1}(X_{t_0}, \mathbb{Z})$ is carried around $U$, and is insensitive to the Hodge decomposition of the fibres, which moves holomorphically over $U$ while the lattice stays flat.

## Exercise 6.4.1

1. The hypotheses of theorem 6.4.7 exclude $n = 1$, where the fibre is a finite set of points. Carry out the case $n = 1$ by hand for $f(z) = z^2$, as in example 6.4.3: with $\delta = [1] - [-1]$ in $\tilde{H}_0(F_\tau, \mathbb{Z})$ and the intersection pairing on $H_0$ of a finite set of points declared to be $\langle [p], [q] \rangle = 1$ if $p = q$ and 0 otherwise, compute $\langle \delta, \delta \rangle$ and check both that $\langle \delta, \delta \rangle = -\varepsilon_1(1 + (-1)^0)$ and that $T(\alpha) = \alpha + \varepsilon_1 \langle \alpha, \delta \rangle \delta$ holds for every $\alpha \in \tilde{H}_0(F_\tau, \mathbb{Z})$.

2. Let $T$ be as in theorem 6.4.7 and suppose $\delta \neq 0$ in $H_{n-1}(X_\tau, \mathbb{Q})$.

    1. Show that for $n$ odd, $T$ is the orthogonal reflection $\alpha \mapsto \alpha - 2\frac{\langle \alpha, \delta \rangle}{\langle \delta, \delta \rangle}\delta$ in the hyperplane $\delta^\perp$, and that $T^2 = \mathrm{id}$.
    2. Show that for $n$ even, $(T - \mathrm{id})^2 = 0$ and $T^k(\alpha) = \alpha + k\varepsilon_n \langle \alpha, \delta \rangle \delta$ for all $k \in \mathbb{Z}$, so that $T$ has infinite order.
    3. Deduce in both cases that $T$ preserves the intersection pairing, by direct computation from (6.5) rather than from the fact that $T$ is induced by a diffeomorphism.

3. Let $Q \subseteq \mathbb{C}\mathrm{P}^3$ be a smooth quadric surface and let $(Q_t)_{t \in \mathbb{C}\mathrm{P}^1}$ be a Lefschetz pencil of hyperplane sections of $Q$, so $n = 2$ and each smooth member is a conic, a rational curve. The base locus is $Q \cap H_1 \cap H_2$, which consists of $\deg Q = 2$ points. Using $\chi_{\mathrm{top}}(Q) = 4$, the fact that blowing up a point raises the Euler characteristic by 1, and the count $\chi_{\mathrm{top}}(\tilde{Q}) = 2\chi_{\mathrm{top}}(Q_{t_0}) + \mu(-1)^n$ established in exercise 6.3.1 (2), determine the number $\mu$ of singular members. Then explain, without any further computation, why the monodromy representation on $H_1$ of a smooth member is trivial.

4. Let $f \colon \tilde{X} \to \mathbb{C}\mathrm{P}^1$ be the fibration of a Lefschetz pencil, with vanishing cycles $\delta_1, \dots, \delta_\mu$ in $H_{n-1}(X_{t_0}, \mathbb{Q})$, and let $W \subseteq H_{n-1}(X_{t_0}, \mathbb{Q})$ be the subspace they span.

    1. Show that $W^\perp = H_{n-1}(X_{t_0}, \mathbb{Q})^\Gamma$ and that both are $\Gamma$-stable.
    2. Show that $W$ is $\Gamma$-stable, and that $W \cap W^\perp$ consists of the classes of $W$ orthogonal to all of $W$.
    3. Suppose $n$ is odd. Show that $\Gamma$ is generated by reflections and that $W \cap W^\perp$ contains no nonzero $\delta_i$.

5. Let $n = 2$, so the fibres of the pencil are compact Riemann surfaces of some genus $g$.

    1. Show that $\rho$ takes values in $\mathrm{Sp}(2g, \mathbb{Z})$, the automorphisms of $H_1(X_{t_0}, \mathbb{Z}) \cong \mathbb{Z}^{2g}$ preserving the intersection pairing.
    2. Show that each $T_i$ is a symplectic transvection, that is of the form $\alpha \mapsto \alpha + c\,\langle \alpha, \delta \rangle \delta$ with $c = -1$, and compute $\det(T_i)$ and the characteristic polynomial of $T_i$.

3. Assume $g = 1$ and that the pencil has $\mu$ singular fibres whose vanishing cycles are not all proportional. Show that the invariant part $H_1(X_{t_0}, \mathbb{Q})^\Gamma$ is zero.

# 7

# Variations of Hodge Structure

Every theorem so far has been about one compact Kähler manifold. The Hodge decomposition, the Lefschetz operators, the polarization: each takes a single $X$ and produces structure on its cohomology. Chapter 6 ended somewhere else. A Lefschetz pencil presents $X$ as a family, and following a loop in the base carried the cohomology of a fibre back to itself by a nontrivial automorphism. Cohomology moved.

This chapter takes that motion as the object of study. Let $f\colon \mathcal{X} \to S$ be a smooth projective family. Each fibre carries a polarized Hodge structure by chapter 5, and the fibres are all diffeomorphic, so the underlying vector spaces may be identified locally over $S$. What cannot be identified is the Hodge decomposition: the subspaces $H^{p,q}$ move as the complex structure of the fibre moves, and it is exactly this motion that a variation of Hodge structure records.

Three facts organize the chapter. The cohomology of the fibres assembles into a local system, which is to say a bundle with a flat connection, the Gauss-Manin connection; flatness is what makes "the same cohomology class" meaningful along a path, and its monodromy is the automorphism of section 6.4. The Hodge filtration, by contrast, is not flat, and its failure to be flat is not arbitrary: the connection moves $F^p$ into $F^{p-1}$ and no further. That is Griffiths transversality, and it is the central theorem here. Finally, recording the filtration as a point of the period domain of section 5.3 turns the family into a map from $S$, the period map, holomorphic and constrained by transversality to be horizontal.

The chapter closes with the case where all of this can be computed. For a smooth hypersurface in projective space the Hodge filtration on primitive cohomology is visible in the Jacobian ring of its defining polynomial, and the period map becomes a statement in commutative algebra.

## 7.1 Local Systems and the Gauss-Manin Connection

Everything in this chapter is attached to one piece of data, a family of smooth projective varieties, and this section fixes that data and builds the first two objects it carries. What comes out is a holomorphic vector bundle on the base whose fibre over $s$ is $H^k(X_s, \mathbb{C})$, together with a connection on it whose flat sections are the cohomology classes that do not move. The Hodge filtration plays no part until section 7.2.

**Definition 7.1.1: Smooth Projective Family**

A *smooth projective family* consists of complex manifolds $\mathcal{X}$ and $S$ with $S$ connected, a proper holomorphic submersion $f\colon \mathcal{X} \to S$, and a closed holomorphic embedding $\iota\colon \mathcal{X} \hookrightarrow \mathbb{CP}^N \times S$ over $S$ for some $N$. Its *fibres* are the compact complex manifolds $X_s = f^{-1}(s)$ for $s \in S$, all of the same dimension $n = \dim_{\mathbb{C}} \mathcal{X} - \dim_{\mathbb{C}} S$, and each is a smooth projective variety by way of $\iota$.

The *relative hyperplane class* of the family is $h := \iota^* c_1(\mathcal{O}_{\mathbb{CP}^N}(1)) \in H^2(\mathcal{X}, \mathbb{Z})$, and $h_s \in H^2(X_s, \mathbb{Z})$ denotes its restriction to a fibre: the class of the Kähler form that $X_s$ inherits from the Fubini-Study metric of $\mathbb{CP}^N$. Each fibre is therefore compact Kähler, and its cohomology carries the polarized Hodge structures of chapter 5.

The two families that this chapter computes with are both cut out by a single equation, and the same argument shows in both cases that the projection is a submersion.

**Example 7.1.2: Two Smooth Projective Families**

Suppose $Z \subseteq \mathbb{CP}^m \times S$ is the zero locus of a holomorphic section $G$ of a line bundle, with $\pi\colon Z \to S$ the projection, and let $p = (x, s) \in Z$ be a point at which the fibre $Z_s \subseteq \mathbb{CP}^m$ is smooth of dimension $m-1$, that is, at which the derivative of $G$ in the $\mathbb{CP}^m$-directions is nonzero. Then $dG_p \neq 0$, so $Z$ is a complex manifold near $p$ with $T_pZ = \ker dG_p$, and $\pi$ is a submersion at $p$: otherwise some nonzero $\xi \in T_s^*S$ would annihilate $d\pi(T_pZ)$, so that $\pi^*\xi$ vanishes on $\ker dG_p$ and hence equals $c\, dG_p$ for a scalar $c$; restricting to the $\mathbb{CP}^m$-directions, where $\pi^*\xi$ vanishes and $dG_p$ does not, forces $c = 0$ and then $\xi = 0$.

1. *The Legendre family.* Let $S = \mathbb{C} \setminus \{0, 1\}$ with coordinate $\lambda$, and let $\mathcal{X} \subseteq \mathbb{CP}^2 \times S$ be cut out by
$$G = Y^2 Z - X(X - Z)(X - \lambda Z)$$
In the affine chart $Z \neq 0$ the fibre is $y^2 = x(x-1)(x-\lambda)$, whose three roots $0, 1, \lambda$ are distinct for $\lambda \in S$, so each fibre is a smooth plane cubic, and the criterion above makes $f$ a proper submersion. This is a smooth projective family of elliptic curves, $n = 1$, and it is the family computed in example 7.1.9.

2. *The universal smooth hypersurface.* Fix $n \geq 1$ and $d \geq 1$, let $\mathbb{CP}^M$ parametrize the nonzero degree-$d$ forms $F$ in $x_0, \dots, x_{n+1}$ up to scale, and let $\Delta \subseteq \mathbb{CP}^M$ be the locus of forms whose zero scheme is singular. Put $S = \mathbb{CP}^M \setminus \Delta$ and let $\mathcal{X} \subseteq \mathbb{CP}^{n+1} \times S$ be cut out by $G([x], [F]) = F(x)$. Every fibre is a smooth hypersurface of degree $d$, so $f$ is again a proper submersion, and $S$ is connected because $\Delta$ is a proper closed analytic subset of $\mathbb{CP}^M$. Section 7.4 computes the Hodge filtration of this family.

The fibres of a smooth projective family are all diffeomorphic, and more is true: the family is locally a product in the $C^\infty$ category. A proper submersion is a locally trivial smooth fibre bundle [26, Ehresmann's theorem], which is the fact definition 6.4.2 was built on. So every $s_0 \in S$ has a neighbourhood $U$ carrying a diffeomorphism $f^{-1}(U) \cong U \times X_{s_0}$ commuting with the projections to $U$. Cohomology cannot distinguish the fibres over such a $U$, and the sheaf that records this is the following.

**Definition 7.1.3: The Local System of a Family**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family and $k \geq 0$. The *local system of the family in degree $k$* is the higher direct image sheaf

$$\mathbb{H}^k := R^k f_* \mathbb{Q}$$

on $S$, that is, the sheaf associated to the presheaf $U \mapsto H^k(f^{-1}(U), \mathbb{Q})$ with the restriction maps of singular cohomology [4, higher direct images].

A *local system* of $\mathbb{Q}$-vector spaces on a space $S$ is a sheaf that is locally isomorphic to a constant sheaf with finite-dimensional stalk [24, local systems as locally constant sheaves]. The name given in definition 7.1.3 is justified by the following, and the underlying mechanism is exactly the one that produces the fibre-homology local system of a fibration [24, the fibre-homology local system of a fibration].

**Proposition 7.1.4: Fibre Cohomology Is Locally Constant**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family. Then $\mathbb{H}^k$ is a local system of $\mathbb{Q}$-vector spaces on $S$, and for every $s \in S$ the restriction maps induce an isomorphism $\mathbb{H}^k_s \cong H^k(X_s, \mathbb{Q})$ on stalks. Its rank is the Betti number $b_k$ of a fibre.

***Proof*:**
Fix $s_0 \in S$, write $X = X_{s_0}$, and choose by Ehresmann's theorem an open $U \ni s_0$ together with a diffeomorphism $\Phi\colon f^{-1}(U) \to U \times X$ over $U$; shrinking $U$, assume it is contractible. Let $\mathcal{B}$ be the collection of contractible open subsets $V \subseteq U$, a basis for the topology of $U$ because $U$ is an open subset of a manifold.

Let $p\colon f^{-1}(U) \to X$ be $\Phi$ followed by the projection $U \times X \to X$, and write $\varphi_s := p|_{X_s}\colon X_s \to X$, a diffeomorphism for every $s \in U$. For $V \in \mathcal{B}$ the restriction of $p$ to $f^{-1}(V) \cong V \times X$ is a homotopy equivalence, $V$ being contractible, so

$$p_V^*\colon H^k(X, \mathbb{Q}) \xrightarrow{\sim} H^k\big(f^{-1}(V), \mathbb{Q}\big)$$

is an isomorphism, and $p_{V'} = p_V$ on $f^{-1}(V')$ for $V' \subseteq V$ gives $\mathrm{res}^V_{V'} \circ p_V^* = p_{V'}^*$. On the basis $\mathcal{B}$ the presheaf is therefore isomorphic to the constant presheaf with value $H^k(X, \mathbb{Q})$, and a sheaf is determined by its values on a basis, so $\mathbb{H}^k|_U$ is the constant sheaf with that value. Since $X$ is compact, $H^k(X, \mathbb{Q})$ is finite dimensional, of dimension $b_k$.

For the stalks, fix $s \in U$ and let $r_s^V\colon H^k(f^{-1}(V), \mathbb{Q}) \to H^k(X_s, \mathbb{Q})$ be restriction along the inclusion of the fibre, for $V \in \mathcal{B}$ containing $s$. That inclusion is a homotopy equivalence as well, so $r_s^V$ is an isomorphism, and $r_s^V \circ p_V^* = \varphi_s^*$. The stalk of $\mathbb{H}^k$ at $s$ is the colimit of the $H^k(f^{-1}(V), \mathbb{Q})$, on which the $r_s^V$ are compatible isomorphisms, so the induced map $\mathbb{H}^k_s \to H^k(X_s, \mathbb{Q})$ is an isomorphism, as we sought to show.

The rational vector space underlying the Hodge structure of a fibre (example 5.1.2) therefore does not move: over a small $U$ the groups $H^k(X_s, \mathbb{Q})$ for $s \in U$ are canonically identified with one another. The bigrading is another matter. It is not defined over $\mathbb{Q}$, nothing above constrains it, and it will move. The content of this chapter lies in the gap between the two.

The identifications are canonical only locally. Along a loop in $S$ they compose to an automorphism that is generally not the identity, and this is the phenomenon section 6.4 computed by hand for a Lefschetz pencil.

**Proposition 7.1.5: Local Systems and Monodromy**

Let $S$ be a connected complex manifold with base point $s_0$.

1. Sending a local system $\mathbb{L}$ of $\mathbb{Q}$-vector spaces on $S$ to the transport of $\mathbb{L}$ along loops is an equivalence between the category of such local systems and the category of representations of $\pi_1(S, s_0)$ on finite-dimensional $\mathbb{Q}$-vector spaces.
2. Let $f\colon \mathcal{X} \to S$ be a smooth projective family. The representation attached in this way to $\mathbb{H}^k$ is $\rho_k \otimes \mathbb{Q}$, where $\rho_k$ is the monodromy representation of definition 6.4.2.

***Proof*:**

1. This is the sheaf form of the local-coefficient dictionary, [24, local systems as locally constant sheaves and as $\pi_1$-representations]. Its hypotheses ask only that $S$ be path connected and locally simply connected, and a connected manifold is both.
2. Let $\gamma\colon [0,1] \to S$ be a loop at $s_0$ and write $X = X_{s_0}$. The pullback of $\mathcal{X}$ along $\gamma$ is a smooth fibre bundle over $[0,1]$, hence trivial, and a trivialization amounts to a continuous family of diffeomorphisms $\varphi_t\colon X \to X_{\gamma(t)}$ with $\varphi_0 = \mathrm{id}$; the geometric monodromy of definition 6.4.2 is $h_\gamma = \varphi_1$.

   Fix $\alpha \in H^k(X, \mathbb{Q})$ and consider $t \mapsto (\varphi_t^{-1})^*\alpha \in H^k(X_{\gamma(t)}, \mathbb{Q})$. This is a section of the pullback of $\mathbb{H}^k$ along $\gamma$. Indeed, for $t$ in a subinterval $J$ with $\gamma(J)$ contained in an open $U$ as in the proof of proposition 7.1.4, the identifications of the stalks over $\gamma(J)$ with one another are induced by a trivialization of $\mathcal{X}$ over $U$, and any two trivializations of a bundle over an interval differ by a path of diffeomorphisms of the fibre starting at the identity, so they induce the same isomorphisms on cohomology (the argument recorded after definition 6.4.2). Under those identifications the displayed family is constant on $J$.

   The transport of $\mathbb{H}^k$ along $\gamma$ therefore carries $\alpha$ to the value of that section at $t = 1$, namely $(\varphi_1^{-1})^*\alpha = (h_\gamma^{-1})^*\alpha$. That is the action of $\rho_k(\gamma)$, read on rational rather than integral cohomology, completing the proof.

For a Lefschetz pencil, part (2) turns theorem 6.4.7 into a statement about $\mathbb{H}^n$, the middle degree of an $n$-dimensional fibre: the local monodromy at a critical value acts on the fibre cohomology by the Picard-Lefschetz transformation along the vanishing cycle, a transvection when $n$ is odd and a reflection of order two when $n$ is even, and corollary 6.4.9 identifies the classes on which the whole group acts trivially. Those are the classes that extend to a global section of the local system, and the question of whether such a section can be recognized Hodge-theoretically is one of the questions chapter 8 takes up.

A local system is a topological object, and the base $S$ is a complex manifold whose complex structure it ignores. To bring that structure to bear, tensor the local system with the holomorphic functions.

**Definition 7.1.6: The Cohomology Bundle and the Gauss-Manin Connection**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family. The *cohomology bundle* of the family in degree $k$ is the $\mathcal{O}_S$-module

$$\mathcal{H}^k := \mathbb{H}^k \otimes_{\mathbb{Q}} \mathcal{O}_S$$

which is locally free of rank $b_k$, since an isomorphism of $\mathbb{H}^k|_U$ with the constant sheaf $\underline{\mathbb{Q}}^{b_k}$ induces $\mathcal{H}^k|_U \cong \mathcal{O}_U^{b_k}$.

A *holomorphic connection* on a locally free $\mathcal{O}_S$-module $\mathcal{E}$ is a map of sheaves of $\mathbb{C}$-vector spaces $\nabla\colon \mathcal{E} \to \mathcal{E} \otimes_{\mathcal{O}_S} \Omega^1_S$ satisfying the Leibniz rule

$$\nabla(g\sigma) = g\,\nabla\sigma + \sigma \otimes dg$$

for local holomorphic $g$ and local sections $\sigma$ of $\mathcal{E}$. This is the holomorphic version of a connection on a vector bundle [9, connections on a vector bundle], with $\mathcal{O}_S$ in place of the smooth functions and $\Omega^1_S$ in place of the smooth cotangent bundle.

The *Gauss-Manin connection* $\nabla$ on $\mathcal{H}^k$ is defined as follows. Let $U \subseteq S$ be connected with $\mathbb{H}^k|_U$ constant and let $e_1, \dots, e_{b_k}$ be a $\mathbb{Q}$-basis of $\mathbb{H}^k(U)$, so that every section of $\mathcal{H}^k$ over an open $V \subseteq U$ is uniquely $\sum_i g_i e_i$ with $g_i \in \mathcal{O}_S(V)$. Set

$$\nabla\Big(\sum_i g_i e_i\Big) := \sum_i e_i \otimes dg_i$$

Theorem 7.1.7 shows that this is independent of the basis chosen and that the local definitions glue. For a holomorphic vector field $u$ on $U$, write $\nabla_u\sigma$ for the contraction of $\nabla\sigma$ with $u$ in the $\Omega^1_S$ factor.

The same formula defines $\nabla$ on the underlying $C^\infty$ bundle. Write $A^0(\mathcal{H}^k)$ for the sheaf of smooth sections and extend by $\nabla(\sum_i g_i e_i) := \sum_i e_i \otimes dg_i$ for smooth $g_i$, with $d$ now the full exterior derivative. This is a connection on a smooth complex vector bundle in the sense of [9, connections on a vector bundle]; it agrees with the holomorphic connection on holomorphic sections, and its $(0,1)$-part $\nabla^{0,1}$ is the $\bar{\partial}$-operator of the holomorphic structure of $\mathcal{H}^k$, since a smooth section is holomorphic exactly when its coefficients $g_i$ in a flat frame are. Both readings of $\nabla$ are used: the holomorphic one states flatness and Griffiths transversality most cleanly, while the smooth one is what section 7.2 differentiates in a $C^\infty$ trivialization.

Evaluating a section at a point recovers a cohomology class of the fibre there: in the frame above, $\mathcal{H}^k \otimes_{\mathcal{O}_S} \mathbb{C}(s) \cong \mathbb{H}^k_s \otimes_{\mathbb{Q}} \mathbb{C} \cong H^k(X_s, \mathbb{C})$, using proposition 7.1.4 and the fact that $H^k(X_s, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} \cong H^k(X_s, \mathbb{C})$. So $\mathcal{H}^k$ is a holomorphic vector bundle whose fibre over $s$ is the complex cohomology of $X_s$, and the rest of the chapter works on it.

**Theorem 7.1.7: The Gauss-Manin Connection Is Well Defined and Flat**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family and $k \geq 0$.

1. The local recipe of definition 7.1.6 is independent of the chosen basis and the local definitions agree on overlaps, so they glue to a holomorphic connection $\nabla$ on $\mathcal{H}^k$. It is the unique holomorphic connection on $\mathcal{H}^k$ that annihilates every local section of $\mathbb{H}^k$.
2. $\ker \nabla = \mathbb{H}^k \otimes_{\mathbb{Q}} \mathbb{C}$, the complexification of the local system, viewed as a subsheaf of $\mathcal{H}^k$.
3. $\nabla$ is flat: extending it to $\nabla\colon \mathcal{H}^k \otimes \Omega^p_S \to \mathcal{H}^k \otimes \Omega^{p+1}_S$ by $\nabla(\sigma \otimes \alpha) = \nabla\sigma \wedge \alpha + \sigma \otimes d\alpha$, where $\nabla\sigma \wedge \alpha := \sum_i \tau_i \otimes (\beta_i \wedge \alpha)$ for $\nabla\sigma = \sum_i \tau_i \otimes \beta_i$, the curvature $\nabla \circ \nabla\colon \mathcal{H}^k \to \mathcal{H}^k \otimes \Omega^2_S$ is zero.

***Proof*:**

1. Fix $U$ connected with $\mathbb{H}^k|_U$ constant, which is possible by proposition 7.1.4, and let $(e_i)$ be a $\mathbb{Q}$-basis of $\mathbb{H}^k(U)$. The recipe satisfies the Leibniz rule, because $d(gg_i) = g\,dg_i + g_i\,dg$ gives
$$\nabla\Big(g \sum_i g_i e_i\Big) = \sum_i e_i \otimes d(gg_i) = g \sum_i e_i \otimes dg_i + \Big(\sum_i g_i e_i\Big) \otimes dg$$
and it annihilates the sections of $\mathbb{H}^k$ over any $V \subseteq U$, since such a section has locally constant coefficients $g_i$.

   Let $(e'_j)$ be a second $\mathbb{Q}$-basis of $\mathbb{H}^k(U)$, say $e'_j = \sum_i c_{ij} e_i$ with $c_{ij} \in \mathbb{Q}$. A section written $\sum_j h_j e'_j$ has coefficients $g_i = \sum_j c_{ij} h_j$ in the first basis, and since the $c_{ij}$ are constants,
$$\sum_i e_i \otimes dg_i = \sum_i e_i \otimes \sum_j c_{ij}\, dh_j = \sum_j \Big(\sum_i c_{ij} e_i\Big) \otimes dh_j = \sum_j e'_j \otimes dh_j$$
so the two recipes agree. If $U$ and $U'$ are two such opens and $W$ is a connected component of $U \cap U'$, then restriction identifies both $\mathbb{H}^k(U)$ and $\mathbb{H}^k(U')$ with $\mathbb{H}^k(W)$, because each is identified with a stalk; the two frames therefore restrict to two $\mathbb{Q}$-bases of $\mathbb{H}^k(W)$, and the computation just made shows the two connections agree on $W$, hence on $U \cap U'$. The local definitions glue.

   Uniqueness is the Leibniz rule read backwards: a connection annihilating the $e_i$ must send $\sum_i g_i e_i$ to $\sum_i e_i \otimes dg_i$.

2. Work over $U$ with a frame as above. A section $\sum_i g_i e_i$ is killed by $\nabla$ if and only if every $dg_i$ vanishes, that is, if and only if every $g_i$ is locally constant. The sections with locally constant complex coefficients in a flat frame are exactly the local sections of $\mathbb{H}^k \otimes_{\mathbb{Q}} \mathbb{C}$.

3. The curvature is computed in a flat frame. For $\sigma = \sum_i g_i e_i$,
$$\nabla(\nabla\sigma) = \nabla\Big(\sum_i e_i \otimes dg_i\Big) = \sum_i \big(\nabla e_i \wedge dg_i + e_i \otimes d(dg_i)\big) = 0$$
since $\nabla e_i = 0$ by part (1) and $d \circ d = 0$. As the flat frames cover $S$, the curvature vanishes identically, completing the proof.

Flatness is what makes the connection usable, and part (2) says how. A section of $\mathcal{H}^k$ is flat exactly when it is a locally constant family of cohomology classes, so continuing a flat section along a path in $S$ is the same operation as transporting a class in the local system; by proposition 7.1.5 the resulting automorphism of $H^k(X_{s_0}, \mathbb{C})$ attached to a loop is $\rho_k(\gamma) \otimes \mathbb{C}$. The connection and the monodromy representation are two presentations of one object.

Part (3) also says something about $\mathcal{H}^k$ that is not automatic. A holomorphic vector bundle rarely admits a holomorphic connection at all, since the obstruction to constructing one is a cohomology class that need not vanish; the bundle $\mathcal{H}^k$ admits one because it was manufactured from a local system, and having done so it is flat for the same reason. What the converse direction would require, namely that a flat holomorphic connection always comes from a local system, is the Riemann-Hilbert correspondence, and it is not needed here: the local system came first.

For computations one needs a way to test a class against something, and the test is integration over cycles. The dual local system $(\mathbb{H}^k)^\vee$ has stalk $H^k(X_s, \mathbb{Q})^\vee \cong H_k(X_s, \mathbb{Q})$ by the universal coefficient theorem over a field [24, universal coefficients], so a section of it over $U$ is a locally constant family $s \mapsto c_s \in H_k(X_s, \mathbb{Q})$ of homology classes, and the pairing with a class $\alpha \in H^k(X_s, \mathbb{C})$ is the period $\langle c_s, \alpha \rangle = \int_{c_s} \alpha$. Call such a family a *flat family of cycles.*

**Proposition 7.1.8: Periods Compute the Gauss-Manin Connection**

Let $f \colon \mathcal{X} \to S$ be a smooth projective family and let $U \subseteq S$ be connected and simply connected, so that $\mathbb{H}^k|_U$ is constant. For a flat family of cycles $c$ over $U$ write $\langle c, - \rangle$ for the induced pairing, applied to the $\mathcal{H}^k$ factor when the argument lies in $\mathcal{H}^k \otimes \Omega^1_S$.

1. An assignment $s \mapsto \alpha_s$ of a class $\alpha_s \in H^k(X_s, \mathbb{C})$ to each $s \in U$ is a holomorphic section of $\mathcal{H}^k$ if and only if $s \mapsto \langle c_s, \alpha_s \rangle$ is holomorphic on $U$ for every flat family of cycles $c$.
2. For every section $\sigma$ of $\mathcal{H}^k$ over $U$ and every flat family of cycles $c$,
$$\langle c, \nabla\sigma \rangle = d \langle c, \sigma \rangle \tag{7.1}$$
3. A section of $\mathcal{H}^k$ over $U$ is determined by the functions $\langle c, \sigma \rangle$ as $c$ runs over the flat families of cycles, and $\nabla\sigma = 0$ if and only if all of them are constant.

***Proof*:**
Since $U$ is connected and simply connected, the monodromy representation of $\mathbb{H}^k|_U$ is trivial, so $\mathbb{H}^k|_U$ is constant by proposition 7.1.5. Fix a $\mathbb{Q}$-basis $e_1, \ldots, e_r$ of $\mathbb{H}^k(U)$ and let $c^1, \ldots, c^r$ be the dual basis of flat families of cycles, so that $\langle c^i, e_j \rangle = \delta_{ij}$ is constant. Every flat family of cycles over $U$ is a $\mathbb{Q}$-linear combination of the $c^i$ with constant coefficients, and a section $\sigma$ of $\mathcal{H}^k$ over $U$ is uniquely $\sum_i g_i e_i$ with $g_i = \langle c^i, \sigma \rangle$.

1. Each $\alpha_s$ lies in the fibre of $\mathcal{H}^k$ at $s$, so the assignment has coefficient functions $g_i(s) = \langle c^i_s, \alpha_s \rangle$ in the frame $(e_i)$, and it is a holomorphic section exactly when every $g_i$ is holomorphic. If all periods are holomorphic then in particular the $g_i$ are; conversely, if the $g_i$ are holomorphic then so is $\langle c, \alpha \rangle = \sum_i a_i g_i$ for any flat $c = \sum_i a_i c^i$ with $a_i$ constant.

2. With $\sigma = \sum_i g_i e_i$ one has $\nabla\sigma = \sum_i e_i \otimes dg_i$, so for $c$ flat,

$$\langle c, \nabla\sigma\rangle = \sum_i \langle c, e_i\rangle\, dg_i = d\Big(\sum_i \langle c, e_i\rangle\, g_i\Big) = d\,\langle c, \sigma\rangle$$

the middle equality because each $\langle c, e_i\rangle$ is a constant.

3. The functions $\langle c^i, \sigma\rangle = g_i$ determine $\sigma$, and by (7.1) they are all constant if and only if every $dg_i$ vanishes, which by theorem 7.1.7 is the condition $\nabla\sigma = 0$, as we sought to show.

Part (2) says that differentiating a period differentiates the class. It converts a question about $\nabla$ into a question about ordinary derivatives of integrals depending on a parameter, and the following example carries out that conversion in full for the family of example 7.1.2.

**Example 7.1.9: Gauss-Manin for the Legendre Family**

Let $f\colon \mathcal{X} \to S = \mathbb{C} \setminus \{0, 1\}$ be the Legendre family, whose fibre $E_\lambda$ is the smooth projective cubic with affine equation $y^2 = P(x)$, where

$$P(x) = x(x-1)(x-\lambda) = q(x)\,(x-\lambda), \qquad q(x) := x(x-1)$$

Each fibre is a compact Riemann surface of genus 1, so $b_1 = 2$ and $\mathcal{H}^1$ is a rank-two holomorphic bundle on $S$.

*A holomorphic section.* The form $\omega_\lambda := dx/y$ is a holomorphic 1-form on $E_\lambda$ with no zeros. Away from $y = 0$ and from the point at infinity this is clear; at a point with $y = 0$ the coordinate $y$ is a local parameter and $2y\,dy = P'(x)\,dx$ gives $dx/y = 2\,dy/P'(x)$ with $P'(x) \neq 0$ there; and at infinity, in the chart $u = X/Y$, $v = Z/Y$, the curve reads $v = u(u-v)(u-\lambda v)$, so $v = u^3(1+O(u))$ with $u$ a local parameter, and $dx/y = (v\,du - u\,dv)/v = (-2 + O(u))\,du$. Being closed, $\omega_\lambda$ has a class in $H^1(E_\lambda, \mathbb{C})$.

*Flat cycles as contours.* Fix $\lambda_0 \in S$. Every class in $H_1(E_{\lambda_0}, \mathbb{Q})$ is a $\mathbb{Q}$-combination of classes of loops avoiding the four branch points of the double cover $E_{\lambda_0} \to \mathbb{CP}^1$, and such a loop $\sigma$ projects to a loop $\Gamma$ in $\mathbb{C} \setminus \{0, 1, \lambda_0\}$ of which it is a lift, the projection being a covering map there. Choose a disc $U \ni \lambda_0$ disjoint from $\Gamma$. Then $P(x)$ is nonvanishing for $x$ near $\Gamma$ and $\lambda \in U$, and the branch of $y = \sqrt{P}$ along $\sigma$ continues to a branch $y(x, \lambda)$ holomorphic near $\Gamma \times U$, since the winding number of $P$ along $\Gamma$ is even, as it must be for $\sigma$ to close up. The resulting lifts $c_\lambda \subseteq E_\lambda$ form a flat family of cycles over $U$ with $c_{\lambda_0} = \sigma$, and

$$\pi(\lambda) := \int_{c_\lambda} \omega_\lambda = \int_\Gamma \frac{dx}{y(x, \lambda)}$$

is holomorphic on $U$, the contour being fixed and the integrand holomorphic in $\lambda$ uniformly on the compact set $\Gamma$. Since $H_1(E_{\lambda_0}, \mathbb{Q})$ is spanned by finitely many such loop classes and flat continuation is linear, after shrinking $U$ every flat family of cycles over $U$ is a $\mathbb{Q}$-combination of families of this shape. So proposition 7.1.8(1) makes $\omega\colon \lambda \mapsto [\omega_\lambda]$ a holomorphic section of $\mathcal{H}^1$ over $U$, and the local descriptions patch because they all assign to $\lambda$ the class of $dx/y$.

*Differentiating.* Differentiation under the integral sign is legitimate for the same reason, and $\partial_\lambda P = -q$ gives $\partial_\lambda y = -q/2y$, hence

$$\partial_\lambda \frac{1}{y} = \frac{q}{2y^3}, \qquad \partial_\lambda^2 \frac{1}{y} = \partial_\lambda\Big(\frac{q}{2} y^{-3}\Big) = -\frac{3q}{2} y^{-4} \cdot \frac{-q}{2y} = \frac{3q^2}{4y^5}$$

So by (7.1), applied twice,

$$\langle c, \nabla_{\partial_\lambda}\omega\rangle = \int_\Gamma \frac{q\,dx}{2y^3}, \qquad \langle c, \nabla^2_{\partial_\lambda}\omega\rangle = \int_\Gamma \frac{3q^2\,dx}{4y^5}$$

*One exact form.* Put $A := q^2/2$ and consider the rational function $R := A/y^3$ on $E_\lambda$. Using $2y\,dy = P'\,dx$ and $y^2 = P$,

$$dR = \Big(\frac{A'}{y^3} - \frac{3AP'}{2y^5}\Big)dx = \frac{2A'P - 3AP'}{2y^5}\,dx$$

Now $A' = qq'$, $P = q(x-\lambda)$ and $P' = q'(x-\lambda) + q$, so

$$2A'P - 3AP' = 2q^2q'(x-\lambda) - \frac{3q^2}{2}\big(q'(x-\lambda)+q\big) = \frac{q^2}{2}\big(q'(x-\lambda) - 3q\big)$$

and $q'(x-\lambda) - 3q = (2x-1)(x-\lambda) - 3x^2 + 3x = -x^2 + 2(1-\lambda)x + \lambda$, which gives

$$dR = \frac{q^2}{4y^5}\big(-x^2 + 2(1-\lambda)x + \lambda\big)\,dx \tag{7.2}$$

The same expression arises from the three period integrands in play: those of $\nabla^2_{\partial_\lambda}\omega$ and $\nabla_{\partial_\lambda}\omega$ just computed, together with $1/y$ for $\omega$ itself. Combining them with the coefficients $\lambda(1-\lambda)$, $1-2\lambda$ and $-\frac14$ and using $y^2 = P = q(x-\lambda)$ to put everything over $y^5$,

$$\frac{3\lambda(1-\lambda)q^2}{4y^5} + \frac{(1-2\lambda)q}{2y^3} - \frac{1}{4y} = \frac{q^2}{4y^5}\Big(3\lambda(1-\lambda) + 2(1-2\lambda)(x-\lambda) - (x-\lambda)^2\Big)$$

and substituting $x = t+\lambda$ turns the last bracket into $-t^2 + (2-4\lambda)t + 3\lambda - 3\lambda^2$, which is the bracket of (7.2) written in $t$.

*The Picard-Fuchs equation.* The only pole of $R$ on $E_\lambda$ is the point $(\lambda, 0)$: at $(0,0)$ and $(1,0)$ the numerator $q^2$ vanishes to order 4 in the local parameter $y$ while $y^3$ vanishes to order 3, and at infinity $R$ has a zero, since $q^2$ has a pole of order 8 there and $y^3$ one of order 9. The contour $\Gamma$ avoids $x = \lambda$, so $c_\lambda$ avoids that pole and $\int_{c_\lambda} dR = 0$. Combining the computed periods with (7.2),

$$\big\langle c,\ \lambda(1-\lambda)\nabla^2_{\partial_\lambda}\omega + (1-2\lambda)\nabla_{\partial_\lambda}\omega - \tfrac14\omega\big\rangle = \int_{c_\lambda} \frac{q^2\big(-x^2+2(1-\lambda)x+\lambda\big)}{4y^5}\,dx = \int_{c_\lambda} dR = 0$$

for every flat family of cycles, so by proposition 7.1.8(3) the section itself vanishes:

$$\lambda(1-\lambda)\,\nabla^2_{\partial_\lambda}\omega + (1-2\lambda)\,\nabla_{\partial_\lambda}\omega - \tfrac14\,\omega = 0 \tag{7.3}$$

Pairing (7.3) with a flat family of cycles and using (7.1) shows that every period $\pi(\lambda) = \int_{c_\lambda}\omega_\lambda$ satisfies the scalar equation

$$\lambda(1-\lambda)\,\pi'' + (1-2\lambda)\,\pi' - \tfrac14\,\pi = 0$$

the *Picard-Fuchs equation* of the family. It is the hypergeometric equation with parameters $\frac12, \frac12; 1$, and exercise 7.1.1 (5) produces a solution as a power series.

*Where the monodromy is.* The equation is singular exactly at $\lambda = 0, 1, \infty$, the values excluded from $S$, and at each of them the fibre degenerates: at $\lambda = 0$ the cubic $y^2 = x^2(x-1)$ has a node at the origin, the branch points 0 and $\lambda$ having collided. Near that node the family is a smoothing. Writing $w := (x-\lambda/2)\sqrt{1-x}$ for the branch of the square root equal to 1 at $x=0$, which for $\lambda$ small is a holomorphic change of coordinate near $x=0$ since $\partial w/\partial x = 1+\lambda/4 \neq 0$ there, the identity $x(x-1)(x-\lambda) = (x-1)(x-\lambda/2)^2 - (x-1)\lambda^2/4$ turns the equation of the fibre into

$$y^2 + w^2 = \frac{\lambda^2}{4}\big(1 - x(w,\lambda)\big)$$

The smoothing parameter of the node is $\lambda^2/4$ rather than $\lambda$, and theorem 6.4.7 attaches one transvection $T(\alpha) = \alpha - \langle\alpha,\delta\rangle\delta$ along the vanishing cycle to each full turn of that parameter, as in example 6.4.8. One turn of $\lambda$ about 0 turns $\lambda^2/4$ twice, so the local monodromy of the Legendre family at $\lambda=0$ is $T^2$.[a] On the period side this is the assertion that continuing a solution of the Picard-Fuchs equation once around $\lambda=0$ returns a different solution, so no basis of solutions there consists of single-valued functions, and the second solution carries a logarithm.

[a]The classical form of this computation says that the image of the monodromy of the Legendre family in $\mathrm{PSL}_2(\mathbb{Z})$ is the level-two principal congruence subgroup, of which $\lambda$ is the modular function; see [5].

The example runs the whole apparatus of this section at the level of $\mathbb{H}^1$ and $\nabla$. A family of varieties produced a flat bundle; a geometrically defined family of classes produced a section of it; and flatness turned a statement about that section into a linear differential equation whose monodromy is the topological monodromy of the family. What the section does not record is that $\omega_\lambda$ is not an arbitrary class but a holomorphic form, so that it lies in the first step of the Hodge filtration of the fibre.

## Exercise 7.1.1

1. Let $f\colon \mathcal{X} \to S$ be a smooth projective family with connected fibres of dimension $n$. Show that $\mathbb{H}^0$ is the constant sheaf $\mathbb{Q}$ and that under the resulting identification $\mathcal{H}^0 = \mathcal{O}_S$ the Gauss-Manin connection is $\nabla = d$. Then show that $\mathbb{H}^{2n}$ has trivial monodromy, by exhibiting the class $\eta_s \in H^{2n}(X_s,\mathbb{Q})$ with $\langle \eta_s, [X_s]\rangle = 1$ as a global flat section, where $[X_s]$ is the fundamental class for the complex orientation. (For the flatness, compare orientations along a local trivialization normalized to be the identity at one point.)

2. Let $X$ be a smooth projective variety, $S$ a connected complex manifold, and $\mathcal{X} = X \times S$ with $f$ the projection. Show that $\mathbb{H}^k$ is the constant sheaf with value $H^k(X,\mathbb{Q})$, that the monodromy representation is trivial, and that $\nabla$ is $1 \otimes d$ under the resulting identification $\mathcal{H}^k = H^k(X,\mathbb{C}) \otimes_{\mathbb{C}} \mathcal{O}_S$. Deduce that a family whose monodromy representation in some degree is nontrivial admits no fibre-preserving diffeomorphism onto a product family, and check that the Legendre family of example 7.1.9 is such a family.

3. Let $f\colon \mathcal{X} \to S$ be a smooth projective family with $S$ connected and let $s_0 \in S$. Show that evaluation at $s_0$ is an isomorphism from the space of global flat sections of $\mathcal{H}^k$ onto the subspace of $H^k(X_{s_0},\mathbb{C})$ fixed by the monodromy representation. Deduce that a global flat section vanishing at a single point is identically zero, and that $\dim_{\mathbb{C}} \Gamma(S, \ker\nabla) \le b_k$ with equality if and only if the monodromy representation is trivial.

4. Fibrewise cup product defines a map $\mathcal{H}^k \otimes_{\mathcal{O}_S} \mathcal{H}^l \to \mathcal{H}^{k+l}$.

    1. Show that it is flat for the Gauss-Manin connections, that is, $\nabla(\sigma\cup\tau) = \nabla\sigma\cup\tau + \sigma\cup\nabla\tau$.
    2. Show that the relative hyperplane class of definition 7.1.1 gives a global flat section $h$ of $\mathbb{H}^2$. Combining this with (1) and exercise 7.1.1 (1), show that for $0 \le k \le n$, where $n$ is

the dimension of the fibres, the bilinear form

$$Q_s(\alpha, \beta) = (-1)^{k(k-1)/2} \int_{X_s} \alpha \wedge \beta \wedge h_s^{n-k}$$

of proposition 5.2.8 is flat, in the sense that $d\,Q(\sigma, \tau) = Q(\nabla\sigma, \tau) + Q(\sigma, \nabla\tau)$ for sections $\sigma, \tau$ of $\mathcal{H}^k$.

5. Write $\pi(\lambda) = \sum_{m \geq 0} a_m \lambda^m$ and show that the Picard-Fuchs equation of example 7.1.9 is equivalent to the recursion $a_{m+1}(m+1)^2 = a_m\big(m + \frac{1}{2}\big)^2$. Verify that $a_m = \binom{2m}{m}^2 16^{-m}$ satisfies it, and determine the radius of convergence of the resulting solution. Why does the existence of this single-valued solution not contradict the last paragraph of example 7.1.9?
6. Let $f\colon \mathcal{X} \to S$ be a smooth projective family, let $\psi\colon S' \to S$ be a holomorphic map from a connected complex manifold, and let $\mathcal{X}' = \mathcal{X} \times_S S'$ with $f'$ the projection. Show that $\mathcal{X}'$ is again a smooth projective family, that $\mathbb{H}'^k = \psi^{-1}\mathbb{H}^k$, and that $\nabla'$ is the pullback of $\nabla$, meaning that $\nabla'(\psi^*\sigma) = (\psi^* \otimes \psi^*)(\nabla\sigma)$ for every local section $\sigma$ of $\mathcal{H}^k$. Deduce that the periods of $\psi^*\sigma$ are the periods of $\sigma$ composed with $\psi$.

## 7.2 Griffiths Transversality

The bundle $\mathcal{H}^k$ of section 7.1 carries a flat connection, and flatness makes it a topological object: over a simply connected base the Gauss-Manin connection identifies the fibres $H^k(X_s, \mathbb{C})$ with one another, and no feature of the complex structure of an individual fibre survives that identification. Each fibre is nonetheless a compact Kähler manifold, so each $H^k(X_s, \mathbb{C})$ carries a Hodge filtration, and under the flat identification those filtrations move.

Two facts govern the motion, and this section proves both. The filtration varies holomorphically, so the subspaces $F^pH^k(X_s, \mathbb{C})$ are the fibres of holomorphic subbundles of $\mathcal{H}^k$; and the connection carries each of those subbundles into the next one down, never further. The second statement is Griffiths transversality, and it is where the geometry of a family constrains the linear algebra of chapter 5.

Throughout, $f\colon \mathcal{X} \to S$ is a smooth projective family (definition 7.1.1) with fibres $X_s = f^{-1}(s)$, and $\mathbb{H}^k$, $\mathcal{H}^k$, $\nabla$ are the local system of definition 7.1.3 together with the cohomology bundle and the Gauss-Manin connection of definition 7.1.6. One consequence of flatness (theorem 7.1.7) is used repeatedly below: a local frame of flat sections is a local holomorphic frame, so $\nabla^{0,1}$ is the $\bar\partial$-operator of the holomorphic structure on $\mathcal{H}^k$.

Projectivity enters in one way only. Each fibre is compact Kähler (definition 7.1.1) because the embedding into $\mathbb{CP}^N \times S$ restricts the Fubini-Study form of theorem 3.2.3 to a Kähler form $\kappa_s$ on it (corollary 3.2.5); what matters below is that these forms vary smoothly with $s$, being restrictions of one fixed form on $\mathbb{CP}^N$. So theorem 4.2.1 applies to every fibre and $H^k(X_s, \mathbb{Z})$ modulo torsion carries a pure Hodge structure of weight $k$ (example 5.1.2). That structure is the object whose variation is to be measured, and by theorem 5.1.5 it is equivalent to record it by its Hodge filtration rather than by its bigrading. The filtration is the presentation that behaves well in a family, and the following definition collects the subspaces it produces.

**Definition 7.2.1: The Hodge Bundles of a Family**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family and fix $k \geq 0$. For each $p \in \mathbb{Z}$ let

$$F^p\mathcal{H}^k \subseteq \mathcal{H}^k$$

be the subset whose intersection with the fibre $H^k(X_s, \mathbb{C})$ over $s \in S$ is the step $F^pH^k(X_s, \mathbb{C})$ of the Hodge filtration of definition 5.1.4, that is $\bigoplus_{r\geq p} H^{r,k-r}(X_s)$. These subsets are the *Hodge bundles* of the family. They are nested,

$$\mathcal{H}^k = F^0\mathcal{H}^k \supseteq F^1\mathcal{H}^k \supseteq \cdots \supseteq F^k\mathcal{H}^k \supseteq F^{k+1}\mathcal{H}^k = 0$$

the outer equalities holding because the Hodge structure on the cohomology of a compact Kähler manifold is effective.

Calling these bundles is an anticipation: nothing said so far rules out the fibre dimension jumping from point to point, and even with constant dimension a family of subspaces need not vary holomorphically. Both are settled below. The smallest case is worth having in hand first, since there the subspaces are lines and can be written down.

**Example 7.2.2: The Hodge Bundles of a Family of Curves**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family whose fibres are compact Riemann surfaces of genus $g$, and take $k = 1$. By example 5.2.2 each $H^1(X_s, \mathbb{C})$ splits as $H^{1,0}(X_s) \oplus H^{0,1}(X_s)$ with $H^{1,0}(X_s) \cong H^0(X_s, \Omega^1_{X_s})$ of dimension $g$. The Hodge bundles are therefore

$$\mathcal{H}^1 = F^0\mathcal{H}^1 \supseteq F^1\mathcal{H}^1 \supseteq F^2\mathcal{H}^1 = 0$$

with $\mathcal{H}^1$ of rank $2g$ and $F^1\mathcal{H}^1$ of rank $g$, its fibre at $s$ being the space of holomorphic 1-forms on $X_s$.

For the Legendre family this is completely explicit. Let $S = \mathbb{C} \setminus \{0, 1\}$ and let $\mathcal{X} \subseteq \mathbb{CP}^2 \times S$ be the family of projective plane cubics

$$y^2z = x(x - z)(x - \lambda z), \qquad \lambda \in S$$

which is smooth for every $\lambda \neq 0, 1$, so each fibre $X_\lambda$ is an elliptic curve and $g = 1$. Then $\mathcal{H}^1$ has rank 2 and $F^1\mathcal{H}^1$ has rank 1: it is the line bundle whose fibre at $\lambda$ is spanned by the class of the holomorphic 1-form

$$\omega_\lambda = \frac{dx}{y}$$

in the affine chart $z = 1$. The Gauss-Manin derivative of this section is computed in example 7.1.9, and example 7.2.9 returns to what transversality does, and does not, say about it.

Both open questions about definition 7.2.1, constancy of the fibre dimension and holomorphy, are questions about how harmonic forms behave when the complex structure of the fibre moves. On a single compact Kähler manifold the harmonic theory of chapter 4 came from applying the elliptic package to $\Delta_{\bar\partial}$; in a family one needs the corresponding statements for the whole family of operators at once, and those belong to the theory of families of elliptic operators, which neither this book nor its prerequisites develop. Chapter 4 already imports the single-manifold package from [9, Hodge theory for the Laplacian on a compact Riemannian manifold]; the family version is imported here, and it is the only analytic input this section takes from outside.

Two conventions before the statement. The harmonic spaces of chapter 4 always carry a manifold in their argument, as in $\mathcal{H}^{p,q}(X_s)$, so no confusion arises with the bundle $\mathcal{H}^k$. And a *smooth family of forms on the fibres* means a smooth section of the bundle $\bigwedge^k T^*_{\mathcal{X}/S}$ of relative $k$-forms on $\mathcal{X}$, whose restriction to $X_s$ is a $k$-form $\beta_s$ on $X_s$; such a family has *relative type* $(r,q)$ when each $\beta_s$ is of type $(r,q)$ on $X_s$.

**Proposition 7.2.3: Hodge Numbers and Harmonic Projections in a Family**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family with $S$ connected, and equip each fibre with the Kähler form $\kappa_s$ above. Write $\Pi_s$ for the projection of the $k$-forms on $X_s$ onto the harmonic ones determined by $\kappa_s$, which by theorem 3.3.5 agrees in each bidegree with the $\bar\partial$-harmonic projection of theorem 4.1.4.

1. Each Hodge number $h^{p,q}(X_s)$ is independent of $s$.
2. If $\{\beta_s\}$ is a smooth family of $k$-forms on the fibres, then $\{\Pi_s(\beta_s)\}$ is again a smooth family.

***Proof Key ideas*:**
The two ingredients below are theorems about a smooth family of formally self-adjoint elliptic operators, proved by the perturbation theory of such families. That theory is not developed in this book, and the deduction of (1) and (2) from it is what is given here; for the ingredients themselves see [33, families of elliptic operators and the smooth variation of their harmonic projectors]. This is the one place in the chapter where an input comes from outside the series, and it is taken because no EYNTKA volume develops the perturbation theory of families of elliptic operators: [9] supplies the elliptic package for a *fixed* metric, which is what chapter 4 ran on, and [7] is not upstream of this book.

1. By proposition 4.1.3 each $\Delta_{\bar\partial,s}$ is a formally self-adjoint elliptic operator of order two. In a $C^\infty$ trivialization of $f$ over a ball these operators pull back to operators on one fixed manifold, and the resulting family varies smoothly with $s$ because $\kappa_s$ does. The first ingredient is that $s \mapsto \dim\ker\Delta_{\bar\partial,s}$ is upper semicontinuous: an eigenvalue may leave 0 under a small perturbation, but the spectral gap above 0 prevents one from arriving at 0. Since $\ker\Delta_{\bar\partial,s} = \mathcal{H}^{p,q}_{\bar\partial}(X_s)$ has dimension $h^{p,q}(X_s)$ by theorem 4.1.4, each $h^{p,q}$ is upper semicontinuous.

   Their sum is constant. By corollary 4.2.3, $\sum_{p+q=k} h^{p,q}(X_s) = b_k(X_s)$, and $b_k(X_s) = \dim \mathbb{H}^k_s \otimes \mathbb{C}$ is independent of $s$ because $\mathbb{H}^k$ is a local system on the connected space $S$ (definition 7.1.3). A finite collection of upper semicontinuous integer-valued functions with constant sum is locally constant: near $s_0$ every term satisfies $h^{p,q}(s) \le h^{p,q}(s_0)$ while the totals agree, so every inequality is an equality. Local constancy on a connected space is constancy.

2. Let $\Delta_s = 2\Delta_{\bar\partial,s}$ be the Laplacian of $\kappa_s$ on $k$-forms, which preserves the bidegree by theorem 3.3.5, so that part (1) makes $\dim\ker\Delta_s$ constant as well. The harmonic projection

is then the spectral projection

$$\Pi_s = \frac{1}{2\pi i} \oint_{|\lambda|=\epsilon} (\lambda - \Delta_s)^{-1}\, d\lambda$$

for any $\epsilon$ below the first nonzero eigenvalue, which by constancy of the kernel dimension may be chosen locally uniformly in $s$. The second ingredient is that the resolvent of a smooth family of elliptic operators depends smoothly on the parameter away from the spectrum, so the contour integral does as well, and applying $\Pi_s$ to a smooth family of forms returns a smooth family, as we sought to show.

The proposition is a statement about metrics, and both theorems to come are statements about the filtration, which is metric-free. The bridge between them is a single observation about which classes are represented by which forms. Write

$$F^p\mathcal{A}^k(X) := \bigoplus_{r\geq p} \mathcal{A}^{r,k-r}(X)$$

for the $k$-forms on a complex manifold $X$ all of whose components have holomorphic degree at least $p$. This is a filtration of the forms, and the question is how it compares with the filtration of definition 5.1.4 on cohomology.

**Lemma 7.2.4: Filtration Level of a Closed Form**

Let $X$ be a compact Kähler manifold and $\alpha \in \mathcal{A}^k(X)$ a closed $k$-form. If $\alpha \in F^p\mathcal{A}^k(X)$ then $[\alpha] \in F^pH^k(X,\mathbb{C})$. Conversely, every class in $F^pH^k(X,\mathbb{C})$ is represented by a harmonic form lying in $F^p\mathcal{A}^k(X)$.

***Proof*:**
Fix a Kähler metric on $X$ and let $\Pi$ be the associated harmonic projection. By theorem 3.3.5 the Laplacian $\Delta_d = 2\Delta_{\bar\partial}$ preserves the bidegree, so its kernel is the direct sum of the spaces $\mathcal{H}^{r,k-r}(X)$ of harmonic forms of pure type, and the orthogonal decomposition of theorem 4.1.4 splits along bidegrees. Consequently $\Pi$ commutes with the projection of a form onto its $(r,k-r)$-component, and $\Pi\big(\mathcal{A}^{r,k-r}(X)\big) \subseteq \mathcal{H}^{r,k-r}(X)$.

Let $\alpha$ be closed with $\alpha = \sum_{r\geq p} \alpha^{r,k-r}$. Every de Rham class has a unique harmonic representative, and for a closed form that representative is its harmonic projection [9, the Hodge-de Rham theorem], so $[\alpha] = [\Pi(\alpha)]$. By the previous paragraph,

$$\Pi(\alpha) = \sum_{r\geq p} \Pi\big(\alpha^{r,k-r}\big), \qquad \Pi\big(\alpha^{r,k-r}\big) \in \mathcal{H}^{r,k-r}(X)$$

The class of a harmonic form of type $(r,k-r)$ lies in $H^{r,k-r}(X)$ by theorem 4.2.1, so $[\alpha] \in \bigoplus_{r\geq p} H^{r,k-r}(X)$, which is $F^pH^k(X,\mathbb{C})$ by definition 5.1.4.

For the converse, let $\xi \in F^pH^k(X,\mathbb{C})$ and let $\gamma$ be its harmonic representative. Decomposing $\gamma$ into pure types gives $\gamma = \sum_r \gamma^{r,k-r}$ with each $\gamma^{r,k-r}$ harmonic, so $\xi = \sum_r [\gamma^{r,k-r}]$ with $[\gamma^{r,k-r}] \in H^{r,k-r}(X)$. This is the decomposition of $\xi$ under theorem 4.2.1, and $\xi \in F^p$ says exactly that its components with $r < p$ vanish. A harmonic form whose class vanishes is itself zero, so $\gamma^{r,k-r} = 0$ for $r < p$ and $\gamma \in F^p\mathcal{A}^k(X)$, as we sought to show.

The lemma converts a statement about the position of a class in the filtration into a statement about the type of a differential form, and differential forms on the fibres can be assembled into forms on the total space $\mathcal{X}$, where the complex structure is fixed once and for all. That is the mechanism behind both theorems. Making it work requires knowing how to differentiate a section of $\mathcal{H}^k$ presented by forms, which is the content of the next lemma: the Gauss-Manin derivative of a class represented by a form is computed by contracting the differential of that form against a lift of the direction of differentiation.

**Lemma 7.2.5: The Gauss-Manin Derivative of a Family of Forms**

Let $U \subseteq S$ be an open ball over which $f$ admits a $C^\infty$ trivialization, let $v$ be a smooth complex vector field on $U$, and let $\tilde{v}$ be any smooth complex vector field on $\mathcal{X}_U = f^{-1}(U)$ with $df(\tilde{v}) = v$. Let $\tilde{\alpha}$ be a smooth complex $k$-form on $\mathcal{X}_U$ whose restriction to every fibre is closed, and let $\sigma$ be the section of $\mathcal{H}^k$ over $U$ given by $\sigma(s) = [\tilde{\alpha}|_{X_s}]$. Then $\sigma$ is smooth, the form $(\iota_{\tilde{v}}\, d\tilde{\alpha})|_{X_s}$ is closed for every $s \in U$, and

$$\nabla_v \sigma(s) = \left[(\iota_{\tilde{v}}\, d\tilde{\alpha})|_{X_s}\right]$$

***Proof*:**
Both sides are $\mathbb{C}$-linear in $v$, and every smooth complex vector field on $U$ is a combination of real ones with complex coefficients, so it suffices to treat $v$ real.

*Step 1: the flat trivialization in terms of forms.* Let $\Psi\colon X \times U \to \mathcal{X}_U$ be a $C^\infty$ trivialization over $U$, where $X = X_{s_0}$ for a chosen $s_0 \in U$; one exists because $f$ is a proper submersion [26, Ehresmann's theorem]. Write $j_s\colon X \to X \times U$ for $x \mapsto (x, s)$ and $\varphi_s = \Psi \circ j_s\colon X \to X_s$, a diffeomorphism onto the fibre.

The maps $\varphi_s^*\colon H^k(X_s, \mathbb{C}) \to H^k(X, \mathbb{C})$ trivialize $\mathcal{H}^k|_U$ flatly. By theorem 7.1.7 the flat sections of $\mathcal{H}^k$ over $U$ are the sections of $\mathbb{H}^k \otimes_{\mathbb{Q}} \mathbb{C}$, and since $U$ is a ball the trivialization makes the fibre inclusion $X_s \hookrightarrow \mathcal{X}_U$ a homotopy equivalence, so those sections are exactly the maps $s \mapsto c|_{X_s}$ for $c \in H^k(\mathcal{X}_U, \mathbb{C})$ (definition 7.1.3). For such a section, $\varphi_s^*(c|_{X_s}) = (\Psi \circ j_s)^* c$, and the maps $\Psi \circ j_s\colon X \to \mathcal{X}_U$ for different $s$ are homotopic, $U$ being connected. So $s \mapsto \varphi_s^*(c|_{X_s})$ is constant, and a section $\tau$ of $\mathcal{H}^k|_U$ is flat if and only if $s \mapsto \varphi_s^* \tau(s)$ is constant. Under this trivialization $\nabla$ becomes the ordinary derivative.

*Step 2: differentiating.* Put $A = \Psi^* \tilde{\alpha}$, a $k$-form on $X \times U$. Since $\Psi \circ j_s$ is the fibre inclusion composed with $\varphi_s$,

$$\varphi_s^*\big(\tilde{\alpha}|_{X_s}\big) = j_s^* A$$

which depends smoothly on $s$; hence so does its de Rham class, and $\sigma$ is smooth. Let $\psi_t$ be the local flow of $v$ on $U$ and $\Phi_t(x, s) = (x, \psi_t(s))$ the induced flow on $X \times U$, whose generator is the vector field $\hat{v} = (0, v)$. From $j_{\psi_t(s)} = \Phi_t \circ j_s$,

$$\frac{d}{dt}\bigg|_{t=0} j_{\psi_t(s)}^* A = j_s^*\big(L_{\hat{v}} A\big)$$

By Step 1 the left-hand side represents $\varphi_s^*\big(\nabla_v \sigma(s)\big)$; it is also the derivative of a smooth family of closed forms, hence closed. Set $\tilde{v}_0 = \Psi_* \hat{v}$, a lift of $v$. Pullback commutes with the Lie derivative, so $j_s^*(L_{\hat{v}}A) = \varphi_s^*\big((L_{\tilde{v}_0}\tilde{\alpha})|_{X_s}\big)$, and therefore

$$\nabla_v \sigma(s) = \big[(L_{\tilde{v}_0}\tilde{\alpha})|_{X_s}\big]$$

with $(L_{\tilde{v}_0}\tilde{\alpha})|_{X_s}$ closed.

*Step 3: Cartan's formula and independence of the lift.* By $L_{\tilde{v}_0} = \iota_{\tilde{v}_0} d + d\,\iota_{\tilde{v}_0}$ and the fact that restriction to a submanifold commutes with $d$,

$$(L_{\tilde{v}_0}\tilde{\alpha})|_{X_s} = (\iota_{\tilde{v}_0} d\tilde{\alpha})|_{X_s} + d\big((\iota_{\tilde{v}_0}\tilde{\alpha})|_{X_s}\big)$$

The second term is exact on $X_s$, so it does not change the class, and subtracting it from a closed form leaves $(\iota_{\tilde{v}_0} d\tilde{\alpha})|_{X_s}$ closed with $\nabla_v \sigma(s) = [(\iota_{\tilde{v}_0} d\tilde{\alpha})|_{X_s}]$.

Finally, let $\tilde{v}$ be any other lift of $v$ and put $w = \tilde{v} - \tilde{v}_0$, a vector field tangent to the fibres. Contraction by a vector field tangent to a submanifold commutes with restriction to it, so

$$(\iota_w\, d\tilde{\alpha})|_{X_s} = \iota_w\big((d\tilde{\alpha})|_{X_s}\big) = \iota_w\, d\big(\tilde{\alpha}|_{X_s}\big) = 0$$

the last equality because $\tilde{\alpha}|_{X_s}$ is closed. Hence $(\iota_{\tilde{v}} d\tilde{\alpha})|_{X_s} = (\iota_{\tilde{v}_0} d\tilde{\alpha})|_{X_s}$, completing the proof.

The formula makes the whole of the Gauss-Manin connection computable from forms on $\mathcal{X}$, and it is the only computation the two theorems need. Both are then a count of types: $d$ never lowers the holomorphic degree, and contraction with a lift lowers it by one when the lift is of type $(1,0)$ and not at all when it is of type $(0,1)$. The second case gives holomorphy of the Hodge bundles, the first gives transversality.

Assembling a family of fibre forms into a form on $\mathcal{X}$ is done once, and the construction is recorded here so that both proofs may quote it. Fix a Hermitian metric on $\mathcal{X}_U$ and let $H \subseteq T\mathcal{X}_U$ be the orthogonal complement of the relative tangent bundle $T_{\mathcal{X}/S} = \ker df$. Since $f$ is holomorphic, $\ker df$ is a complex subbundle, hence so is $H$, and $df$ carries $H$ isomorphically onto $f^*TS$. Let $\pi\colon T\mathcal{X}_U \to T_{\mathcal{X}/S}$ be the projection with kernel $H$; it is complex linear. For a smooth family $\{\beta_s\}$ of $k$-forms on the fibres, the form $\pi^*\beta$ on $\mathcal{X}_U$ is smooth, restricts to $\beta_s$ on each $X_s$, and has the same type as $\beta$ has relatively. In particular a family of relative type at least $p$ extends to a form in $F^p\mathcal{A}^k(\mathcal{X}_U)$. The subbundle $H$ also supplies, for each vector field $v$ on $U$, a lift $\tilde{v}$ lying in $H$, whose $(1,0)$-part lifts the $(1,0)$-part of $v$ and whose $(0,1)$-part lifts the $(0,1)$-part.

**Theorem 7.2.6: The Hodge Bundles Are Holomorphic Subbundles**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family with $S$ connected and fix $k$ and $p$. Then $F^p\mathcal{H}^k$ is a holomorphic subbundle of $\mathcal{H}^k$, of rank

$$f^p = \sum_{r \geq p} h^{r,k-r}$$

independent of the point of $S$. Moreover every smooth local section of $F^p\mathcal{H}^k$ is of the form $s \mapsto [\gamma_s]$ for a smooth family $\{\gamma_s\}$ of closed $k$-forms on the fibres of relative type at least $p$.

***Proof*:**
All assertions are local on $S$, so fix $s_0 \in S$ and work over a ball $U$ around it, small enough to carry a $C^\infty$ trivialization of $f$ and the data $H$, $\pi$ constructed above.

*Step 1: the rank is constant.* By definition 5.1.4 the dimension of $F^pH^k(X_s, \mathbb{C})$ is $\sum_{r\geq p} h^{r,k-r}(X_s)$, and each $h^{r,k-r}(X_s)$ is independent of $s$ by proposition 7.2.3. So the fibre dimension of $F^p\mathcal{H}^k$ is the constant $f^p$.

*Step 2: a smooth frame, and the representation by forms.* Fix $\xi_0 \in F^pH^k(X_{s_0}, \mathbb{C})$. By lemma 7.2.4 it has a harmonic representative $\alpha_0 \in F^p\mathcal{A}^k(X_{s_0})$. Each pure-type component $\alpha_0^{r,k-r}$ with $r \geq p$ is a smooth section over the closed submanifold $X_{s_0}$ of the smooth vector bundle $\bigwedge^{r,k-r} T^*_{\mathcal{X}/S}$, so it extends, after shrinking $U$, to a smooth section over $\mathcal{X}_U$; summing the extensions gives a smooth family $\{\beta_s\}$ of $k$-forms on the fibres, of relative type at least $p$, with $\beta_{s_0} = \alpha_0$.

Put $\gamma_s = \Pi_s(\beta_s)$. By proposition 7.2.3 this is a smooth family, by theorem 3.3.5 the harmonic projection preserves bidegree so it again has relative type at least $p$, and each $\gamma_s$ is harmonic, hence closed. By lemma 7.2.4 the class $[\gamma_s]$ lies in $F^pH^k(X_s, \mathbb{C})$, and $\gamma_{s_0} = \alpha_0$ because $\alpha_0$ is already harmonic. Lemma 7.2.5, applied to the form $\pi^*\gamma$, shows that $\sigma_{\xi_0}\colon s \mapsto [\gamma_s]$ is a smooth section of $\mathcal{H}^k$; by construction it takes values in $F^p\mathcal{H}^k$ and $\sigma_{\xi_0}(s_0) = \xi_0$.

Choosing $\xi_0$ over a basis of $F^pH^k(X_{s_0}, \mathbb{C})$ produces $f^p$ smooth sections, linearly independent at $s_0$ and hence on a neighbourhood. By Step 1 they span $F^pH^k(X_s, \mathbb{C})$ at every point of that neighbourhood, so $F^p\mathcal{H}^k$ is a $C^\infty$ subbundle of $\mathcal{H}^k$ and these sections are a smooth frame. Any smooth section of $F^p\mathcal{H}^k$ is a combination $\sum_i g_i\sigma_{\xi_i}$ with smooth coefficients $g_i$, and the family $\{\sum_i g_i(s)\gamma_s^i\}$ then represents it and is smooth, closed and of relative type at least $p$. This proves the last assertion of the theorem.

*Step 3: $\nabla^{0,1}$ preserves the subbundle.* Let $\sigma$ be a smooth section of $F^p\mathcal{H}^k$ over $U$ and $\bar{v}$ a vector field on $U$ of type $(0,1)$. By Step 2 write $\sigma(s) = [\gamma_s]$ with $\{\gamma_s\}$ smooth, closed and of relative type at least $p$, and set $\tilde{\alpha} = \pi^*\gamma$, a form in $F^p\mathcal{A}^k(\mathcal{X}_U)$ restricting to $\gamma_s$ on each fibre. Since $d = \partial + \bar{\partial}$ and neither operator lowers the holomorphic degree, $d\tilde{\alpha} \in F^p\mathcal{A}^{k+1}(\mathcal{X}_U)$. Let $\tilde{\bar{v}}$ be the lift of $\bar{v}$ lying in $H$, of type $(0,1)$; contraction with a $(0,1)$-vector field maps $\mathcal{A}^{r,q}$ into $\mathcal{A}^{r,q-1}$ and so preserves $F^p\mathcal{A}^\bullet$. Hence $\iota_{\tilde{\bar{v}}}d\tilde{\alpha} \in F^p\mathcal{A}^k(\mathcal{X}_U)$. Restriction of forms to the complex submanifold $X_s$ preserves bidegree, so $(\iota_{\tilde{\bar{v}}}d\tilde{\alpha})|_{X_s}$ lies in $F^p\mathcal{A}^k(X_s)$ and is closed by lemma 7.2.5. That lemma and lemma 7.2.4 now give

$$\nabla_{\bar{v}}\sigma(s) = \left[(\iota_{\tilde{\bar{v}}}d\tilde{\alpha})|_{X_s}\right] \in F^pH^k(X_s, \mathbb{C})$$

for every $s \in U$, which is the assertion of this step.

*Step 4: holomorphy.* Shrink $U$ so that $\mathcal{H}^k|_U$ is trivialized by a flat frame; this identifies $\mathcal{H}^k|_U$ with $\Xi \otimes_{\mathbb{C}} \mathcal{O}_U$ for the fixed vector space $\Xi = H^k(X_{s_0}, \mathbb{C})$, with $\nabla$ the ordinary derivative and $\nabla^{0,1} = \bar{\partial}$ acting on coefficient functions. Under this identification $F^p\mathcal{H}^k$ becomes a smooth family $s \mapsto W(s) \subseteq \Xi$ of subspaces of dimension $f^p$, and Step 3 says that $\bar{\partial}\sigma$ is a section of $W$ whenever $\sigma$ is.

Choose a complement $\Xi = W(s_0) \oplus C$. The projection $\rho\colon \Xi \to W(s_0)$ along $C$ restricts to an injection on $W(s)$ for $s$ near $s_0$, since $W(s) \cap C = 0$ is an open condition satisfied at $s_0$, and by equality of dimensions it is then an isomorphism $W(s) \to W(s_0)$. So $W(s)$ is the graph

$$W(s) = \{\, w + A(s)w \big| w \in W(s_0) \,\}$$

of the linear map $A(s) = (\mathrm{id} - \rho) \circ (\rho|_{W(s)})^{-1} \colon W(s_0) \to C$, and $A$ is smooth in $s$ because $W$ is a smooth subbundle. For fixed $w \in W(s_0)$ the assignment $\sigma_w(s) = w + A(s)w$ is a smooth section of $F^p\mathcal{H}^k$, so $\bar\partial\sigma_w(s) = \bar\partial\big(A(s)w\big)$ lies in $W(s)$. It also lies in $C$, the subspace $C$ being independent of $s$, and $W(s) \cap C = 0$. Hence $\bar\partial(A(s)w) = 0$, so $A(\cdot)w$ is holomorphic and $\sigma_w$ is a holomorphic section of $\mathcal{H}^k$. Letting $w$ range over a basis of $W(s_0)$ gives $f^p$ holomorphic sections spanning $W(s)$ at every point near $s_0$, so $F^p\mathcal{H}^k$ is a holomorphic subbundle, completing the proof.

The theorem is what licenses the name in definition 7.2.1, and it also fixes which of the two presentations of a Hodge structure is the holomorphic one. The individual pieces $H^{p,q}(X_s)$ of the Hodge decomposition do *not* form a holomorphic subbundle: they are recovered from the filtration only by the formula $H^{p,q} = F^p \cap \overline{F^q}$ of (5.2), and conjugation is not holomorphic. What is holomorphic is the filtration, and this is the reason chapter 5 went to the trouble of proving the two presentations equivalent (theorem 5.1.5) before any family appeared.

With the subbundles in hand, the question is how far they fail to be flat. A flat subbundle would satisfy $\nabla F^p \subseteq F^p \otimes \Omega^1_S$, which by the argument of exercise 7.2.1 (1) would force the Hodge structures of the fibres to be constant in the flat trivialization. That does not happen for a family with varying complex structure. The theorem below says the failure is as small as it can be while still occurring: one step of the filtration, never two.

**Theorem 7.2.7: Griffiths Transversality**

Let $f \colon \mathcal{X} \to S$ be a smooth projective family with $S$ connected. Then for every $k$ and every $p$,

$$\nabla\big(F^p\mathcal{H}^k\big) \subseteq F^{p-1}\mathcal{H}^k \otimes_{\mathcal{O}_S} \Omega^1_S$$

That is, if $\sigma$ is a local holomorphic section of $F^p\mathcal{H}^k$ and $v$ a local holomorphic vector field on $S$, then $\nabla_v\sigma$ is a section of $F^{p-1}\mathcal{H}^k$.

***Proof*:**
The assertion is local on $S$, so fix $s_0$ and work over a ball $U$ carrying a $C^\infty$ trivialization of $f$ together with the splitting data $H$, $\pi$ fixed before theorem 7.2.6. It suffices to prove the second formulation, since a section of $\mathcal{H}^k \otimes \Omega^1_S$ lies in $F^{p-1}\mathcal{H}^k \otimes \Omega^1_S$ exactly when its contraction with every local holomorphic vector field lies in $F^{p-1}\mathcal{H}^k$.

*Step 1: represent the section by forms of type at least $p$.* Let $\sigma$ be a holomorphic, in particular smooth, section of $F^p\mathcal{H}^k$ over $U$. By theorem 7.2.6 there is a smooth family $\{\gamma_s\}$ of closed $k$-forms on the fibres, of relative type at least $p$, with $[\gamma_s] = \sigma(s)$. Set $\tilde\alpha = \pi^*\gamma$. By the construction of $\pi$ this is a smooth $k$-form on $\mathcal{X}_U$ with $\tilde\alpha|_{X_s} = \gamma_s$ for every $s$, and it lies in $F^p\mathcal{A}^k(\mathcal{X}_U)$ because $\pi$ is complex linear and $\gamma$ has relative type at least $p$.

*Step 2: the type of $\iota_{\tilde v} d\tilde\alpha$.* Write $\tilde\alpha = \sum_{r \ge p} \tilde\alpha^{r,k-r}$. Since $d = \partial + \bar\partial$ on $\mathcal{X}_U$ (theorem 1.2.6) and $\partial$

raises the holomorphic degree by one while $\bar\partial$ leaves it unchanged,

$$d\tilde\alpha \in F^p\mathcal{A}^{k+1}(\mathcal{X}_U)$$

Let $\tilde v$ be the lift of $v$ lying in $H$. Because $v$ is of type $(1,0)$ and $H$ is a complex subbundle mapped isomorphically to $f^*TS$, the lift $\tilde v$ is of type $(1,0)$ as well. Contraction with a $(1,0)$-vector field maps $\mathcal{A}^{r,q}$ into $\mathcal{A}^{r-1,q}$, so it lowers the level of the filtration by exactly one:

$$\iota_{\tilde v}\, d\tilde\alpha \in F^{p-1}\mathcal{A}^k(\mathcal{X}_U)$$

Every other operation in the argument preserves the filtration of forms.

*Step 3: restrict to a fibre.* The fibre $X_s$ is a complex submanifold of $\mathcal{X}_U$, so restriction of forms to it preserves the bidegree and hence carries $F^{p-1}\mathcal{A}^k(\mathcal{X}_U)$ into $F^{p-1}\mathcal{A}^k(X_s)$. By lemma 7.2.5 the restricted form $(\iota_{\tilde v} d\tilde\alpha)|_{X_s}$ is closed and

$$\nabla_v\sigma(s) = \left[(\iota_{\tilde v}\, d\tilde\alpha)|_{X_s}\right]$$

A closed form in $F^{p-1}\mathcal{A}^k(X_s)$ has class in $F^{p-1}H^k(X_s,\mathbb{C})$ by lemma 7.2.4. Hence $\nabla_v\sigma(s) \in F^{p-1}H^k(X_s,\mathbb{C})$ for every $s \in U$, which is the claim, completing the proof.

The transversality condition is a first-order differential equation satisfied by the Hodge filtration of every family of smooth projective varieties, and it is a restriction on the family only because $F^{p-1}$ is smaller than $\mathcal{H}^k$. For $p \le 1$ it says nothing, since $F^{p-1} = \mathcal{H}^k$ there (exercise 7.2.1 (2)); the first case with content is $k = 2$, $p = 2$. Two readings are worth separating. Read as a constraint, it says that a family of varieties cannot deform its Hodge structure in an arbitrary direction: most motions of a flag are forbidden, and the permitted ones form a distinguished subbundle of the tangent bundle of the parameter space of flags. That is the geometric form of the statement, and section 7.3 puts it in those terms. Read as a computation, it says that the interesting derivative is the induced map on the graded pieces, since transversality guarantees that this map is defined at all. The corollary records it.

**Corollary 7.2.8: The Graded Derivative**

Write $\mathrm{gr}_F^p\mathcal{H}^k = F^p\mathcal{H}^k/F^{p+1}\mathcal{H}^k$, a holomorphic bundle whose fibre at $s$ is $H^{p,k-p}(X_s)$. The Gauss-Manin connection induces an $\mathcal{O}_S$-linear map of holomorphic bundles

$$\bar\nabla^p\colon \mathrm{gr}_F^p\mathcal{H}^k \to \mathrm{gr}_F^{p-1}\mathcal{H}^k \otimes_{\mathcal{O}_S} \Omega^1_S$$

whose value at $s$ is a linear map $H^{p,k-p}(X_s) \to H^{p-1,k-p+1}(X_s) \otimes T_s^*S$, equivalently a bilinear map

$$T_sS \otimes H^{p,k-p}(X_s) \to H^{p-1,k-p+1}(X_s)$$

***Proof*:**
The fibre of $\mathrm{gr}_F^p\mathcal{H}^k$ at $s$ is $F^pH^k(X_s,\mathbb{C})/F^{p+1}H^k(X_s,\mathbb{C})$, which is $H^{p,k-p}(X_s)$ by definition 5.1.4. Both $F^p\mathcal{H}^k$ and $F^{p+1}\mathcal{H}^k$ are holomorphic subbundles by theorem 7.2.6, so the quotient is a holomorphic bundle (definition 1.3.13).

By theorem 7.2.7 the connection gives a map of sheaves

$$F^p\mathcal{H}^k \to F^{p-1}\mathcal{H}^k \otimes \Omega^1_S \to \left(F^{p-1}\mathcal{H}^k / F^p\mathcal{H}^k\right) \otimes \Omega^1_S$$

the second arrow being the quotient. This composite kills $F^{p+1}\mathcal{H}^k$: a section $\sigma$ of $F^{p+1}\mathcal{H}^k$ has $\nabla\sigma \in F^p\mathcal{H}^k \otimes \Omega^1_S$, again by theorem 7.2.7 applied with $p+1$ in place of $p$, and that is exactly what the quotient annihilates. So the composite factors through $\mathrm{gr}^p_F\mathcal{H}^k$.

It remains to check $\mathcal{O}_S$-linearity, which is where transversality is used a second time. For a local function $g$ and a section $\sigma$ of $F^p\mathcal{H}^k$, the Leibniz rule gives

$$\nabla(g\sigma) = g\,\nabla\sigma + \sigma \otimes dg$$

The correction term $\sigma \otimes dg$ lies in $F^p\mathcal{H}^k \otimes \Omega^1_S$, which the quotient by $F^p\mathcal{H}^k$ annihilates. Hence the induced map satisfies $\bar{\nabla}^p(g\,\bar{\sigma}) = g\,\bar{\nabla}^p(\bar{\sigma})$, so it is $\mathcal{O}_S$-linear and therefore a morphism of holomorphic bundles, with the stated value on fibres, as we sought to show.

A connection is not tensorial, so it has no value at a point; the graded derivative $\bar{\nabla}^p$ is tensorial, so it does, and that value is an invariant attached to the fibre $X_s$ together with a tangent direction at $s$. It measures the first-order motion of the Hodge structure of $X_s$ as the parameter moves, and it is the object that section 7.3 identifies with the differential of the period map. Chapter 8 takes up what happens when it vanishes.

In weight one the condition is empty, and the reason for that is worth recording before the general theory is put to work.

**Example 7.2.9: Transversality in Weight One**

Take $k = 1$, so that the Hodge bundles are

$$\mathcal{H}^1 = F^0\mathcal{H}^1 \supseteq F^1\mathcal{H}^1 \supseteq F^2\mathcal{H}^1 = 0$$

as in example 7.2.2. Every instance of theorem 7.2.7 is empty, for one of two reasons. For $p \leq 1$ the right-hand side is everything, since $F^{p-1}\mathcal{H}^1 = \mathcal{H}^1$ by exercise 7.2.1 (2): at $p = 1$ the condition reads $\nabla F^1\mathcal{H}^1 \subseteq F^0\mathcal{H}^1 \otimes \Omega^1_S = \mathcal{H}^1 \otimes \Omega^1_S$, and at $p \leq 0$ it reads $\nabla\mathcal{H}^1 \subseteq \mathcal{H}^1 \otimes \Omega^1_S$; every section satisfies both. For $p \geq 2$ the left-hand side vanishes, since $F^p\mathcal{H}^1 = 0$. So transversality imposes no condition whatever on a family of Hodge structures of weight 1.

The reason is structural rather than accidental. A weight-1 structure is a filtration with a single interesting step, $F^1$, and by example 5.1.3 it amounts to no more than a complex structure on the underlying real vector space. There is nothing for the filtration to be transverse to. The only graded derivative that can be nonzero is

$$\bar{\nabla}^1 \colon \mathrm{gr}^1_F\mathcal{H}^1 \to \mathrm{gr}^0_F\mathcal{H}^1 \otimes \Omega^1_S, \qquad H^{1,0}(X_s) \to H^{0,1}(X_s) \otimes T^*_s S$$

and theorem 7.2.7 places no restriction on it.

That does not make the graded derivative arbitrary. What constrains it is the polarization, and the constraint is a symmetry. Write $g$ for the rank of $F^1\mathcal{H}^1$, so that $\mathcal{H}^1$ has rank $2g$, and let $Q$ be the polarization of proposition 5.2.8, which is alternating in odd weight, flat for $\nabla$ (exercise 7.1.1 (4)) and zero on $F^1\mathcal{H}^1 \times F^1\mathcal{H}^1$ by the first Hodge-Riemann relation (theorem 5.2.3). For a local vector field $v$ and local sections $\sigma, \tau$ of $F^1\mathcal{H}^1$, flatness gives

$$0 = v\,Q(\sigma,\tau) = Q(\nabla_v\sigma,\tau) + Q(\sigma,\nabla_v\tau)$$

so that $Q(\nabla_v\sigma,\tau) = Q(\nabla_v\tau,\sigma)$: the form $(\sigma,\tau) \mapsto Q(\nabla_v\sigma,\tau)$ on $F^1\mathcal{H}^1$ is symmetric. Being isotropic of half the rank, $F^1\mathcal{H}^1$ is its own annihilator under $Q$, so $Q$ pairs $\mathcal{H}^1/F^1\mathcal{H}^1$ nondegenerately against it and that symmetric form determines $\bar{\nabla}^1_v$ and is determined by it. Hence $\bar{\nabla}^1_v$ ranges over a space of dimension $g(g+1)/2$ inside the space $\operatorname{Hom}\big(H^{1,0}(X_s), H^{0,1}(X_s)\big)$ of all linear maps, of dimension $g^2$. For $g \geq 2$ that is a proper subspace, and its dimension is that of the period domain $\mathfrak{H}_g$ of this numerical type (example 5.3.3). None of it comes from transversality.

The graded derivative vanishes identically exactly when $F^1\mathcal{H}^1$ is a flat subbundle, that is when the family has locally constant Hodge structure (exercise 7.2.1 (1)).

This is why the classical theory of abelian varieties never encounters transversality. By theorem 5.4.1 a polarizable weight-1 Hodge structure is the same thing as a complex torus with a polarization, and by corollary 5.4.2 the polarizable ones are the abelian varieties; a family of abelian varieties is a family of weight-1 structures, on whose variation the preceding paragraphs put no transversality constraint at all. The constraint such a family does satisfy is the symmetry computed above, and that one belongs to the polarization. For the Legendre family of example 7.2.2 the bundle $F^1\mathcal{H}^1$ is a line bundle and $\bar{\nabla}^1$ is a map of line bundles twisted by $\Omega^1_S$; the Gauss-Manin computation of example 7.1.9 is a computation of that map, and transversality contributes nothing to it.

The condition first acquires content at $k = 2$ and $p = 2$, where $F^1$ is a proper subbundle of $\mathcal{H}^2$ and $\nabla F^2 \subseteq F^1 \otimes \Omega^1_S$ is a genuine restriction. That case is worked out for K3 surfaces in exercise 7.2.1 (5).

## Exercise 7.2.1

1. Show that the following are equivalent for a smooth projective family $f\colon \mathcal{X} \to S$ and a fixed $k$: (i) the graded derivatives $\bar{\nabla}^p$ of corollary 7.2.8 vanish for every $p$; (ii) each $F^p\mathcal{H}^k$ satisfies $\nabla F^p\mathcal{H}^k \subseteq F^p\mathcal{H}^k \otimes \Omega^1_S$. Then show that when these hold and $U \subseteq S$ is a ball, the flat trivialization of $\mathcal{H}^k|_U$ carries $F^pH^k(X_s,\mathbb{C})$ to a subspace of $H^k(X_{s_0},\mathbb{C})$ independent of $s$. (For the last part, imitate Step 4 of the proof of theorem 7.2.6, replacing $\bar{\partial}$ by the full derivative.)

2. Let $f\colon \mathcal{X} \to S$ be a smooth projective family and fix $k$. Show that $F^p\mathcal{H}^k = \mathcal{H}^k$ for $p \leq 0$ and $F^p\mathcal{H}^k = 0$ for $p > k$. Deduce that the transversality condition of theorem 7.2.7 at level $p$ is automatically satisfied unless $2 \leq p \leq k$, and conclude that it says nothing at all when $k \leq 1$.

3. Keep the product family $f\colon X \times S \to S$ of exercise 7.1.1 (2), which identified $\mathbb{H}^k$ with the constant local system and $\nabla$ with $1 \otimes d$. Show that each $F^p\mathcal{H}^k$ is likewise the constant subbundle with fibre $F^pH^k(X,\mathbb{C})$. Conclude that $\bar{\nabla}^p = 0$ for every $p$, consistently with exercise 7.2.1 (1).

4. Let $s_1,\dots,s_m$ be holomorphic coordinates on an open set $U \subseteq S$ and write $\nabla_i = \nabla_{\partial/\partial s_i}$. Using flatness (theorem 7.1.7) and theorem 7.2.7, show that for a holomorphic section $\sigma$ of $F^p\mathcal{H}^k$ the classes of $\nabla_i\nabla_j\sigma$ in $\operatorname{gr}_F^{p-2}\mathcal{H}^k$ are symmetric in $i$ and $j$, and that they vanish when $\sigma$ is a section of $F^{p+1}\mathcal{H}^k$. Deduce that the composite $(\bar{\nabla}^{p-1} \otimes \mathrm{id}) \circ \bar{\nabla}^p$ takes values in $\operatorname{gr}_F^{p-2}\mathcal{H}^k \otimes \operatorname{Sym}^2\Omega^1_S$.

5. Let $f\colon \mathcal{X} \to S$ be a smooth projective family of K3 surfaces, whose degree-2 Hodge numbers are $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 20$ (see section 5.4), and take $k = 2$. Compute the ranks of $F^2\mathcal{H}^2 \subseteq F^1\mathcal{H}^2 \subseteq \mathcal{H}^2$ and of the three graded pieces. State what theorem 7.2.7 asserts at $p = 2$ and at $p = 1$, and say which of the two has content. Finally, let $Q$ denote the cup-product

pairing on the fibres, which is defined by the local system and is therefore flat for $\nabla$, and let $\sigma$ be a local holomorphic section of $F^2\mathcal{H}^2$. Show that $Q(\sigma, \nabla_v \sigma) = 0$ for every local holomorphic vector field $v$.

## 7.3 The Period Map

Section 7.1 put a flat connection on the bundle of fibre cohomology and section 7.2 put a filtration of that bundle by holomorphic subbundles which the connection moves by exactly one step. Both are statements about objects living over $S$. This section converts them into a single map out of $S$. Each fibre carries a polarized Hodge structure, the polarized structures of a fixed numerical type are the points of the period domain of section 5.3, and the assignment sending $s$ to the structure on $X_s$ is the period map.

Two things have to be arranged before the assignment is a map at all. The identification of $H^k(X_s, \mathbb{Z})$ with a fixed lattice depends on a path from a base point, so the target must be a quotient of the period domain by the monodromy group; and the quotient must be a space in which holomorphy makes sense. Once both are in place, the two theorems of the section say that the map is holomorphic and that its differential is constrained: it lands in a distinguished subbundle of the tangent bundle of $D$. That constraint is Griffiths transversality read on the domain rather than on the base.

Throughout, $f\colon \mathcal{X} \to S$ is a smooth projective family (definition 7.1.1), with fibres $X_s = f^{-1}(s)$ of complex dimension $n$, with relative hyperplane class $h \in H^2(\mathcal{X}, \mathbb{Z})$, and with $h_s = h|_{X_s}$ the class of the Fubini-Study Kähler form of $X_s$ (proposition 3.2.6, corollary 3.2.5). Fix an integer $0 \le k \le n$ and let $\mathbb{H}^k$, $\mathcal{H}^k$, $\nabla$ and $F^\bullet$ be the local system, the holomorphic bundle, the Gauss-Manin connection and the Hodge filtration attached to $f$ in section 7.1 and section 7.2.

Cup product with $h$ is a map of local systems $L\colon \mathbb{H}^j \to \mathbb{H}^{j+2}$, since $h$ is a single class on $\mathcal{X}$ restricted to the fibres, and on each fibre it is the Lefschetz operator of definition 3.3.1. The part of the cohomology that this section works with is cut out by a power of $L$.

**Definition 7.3.1: The Primitive Part of the Cohomology of a Family**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family and let $0 \le k \le n$. The *primitive part* of $\mathbb{H}^k$ is the subsheaf

$$\mathbb{H}^k_{\text{prim}} := \ker\big(L^{n-k+1}\colon \mathbb{H}^k \to \mathbb{H}^{2n-k+2}\big)$$

whose stalk at $s$ is the primitive cohomology $H^k(X_s, \mathbb{Q})_{\text{prim}}$ of definition 5.2.5, and the *primitive cohomology bundle* is the $\mathcal{O}_S$-submodule

$$\mathcal{H}^k_{\text{prim}} := \mathbb{H}^k_{\text{prim}} \otimes_{\mathbb{Q}} \mathcal{O}_S$$

of $\mathcal{H}^k$. Part (3) of proposition 7.3.2 shows that the first is a sub-local system of $\mathbb{H}^k$ and the second a holomorphic subbundle of $\mathcal{H}^k$.

The period domain of section 5.3 was attached to numerical data fixed once and for all: a lattice, a weight, a form and a list of Hodge numbers. A family supplies that data only after a fibre is singled out, and it supplies it consistently only because parallel transport moves everything rigidly. The following proposition collects the four facts that make the construction legal, and it is the only place where the geometry of $f$ is used.

**Proposition 7.3.2: Polarized Period Data of a Family**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family as above, let $0 \le k \le n$, fix $s_0 \in S$, and put

$$V_{\mathbb{Z}} = H^k(X_{s_0}, \mathbb{Z})_{\mathrm{prim}}/(\text{torsion}), \qquad h^{p,q} = h^{p,q}_{\mathrm{prim}}(X_{s_0}), \qquad f^p = \sum_{r \ge p} h^{r,k-r}$$

with $Q$ the polarization of proposition 5.2.8 on $V_{\mathbb{Z}}$. The $f^p$ are the level dimensions attached to these data in section 5.3; they are formed from the primitive Hodge numbers and are in general smaller than the ranks of the Hodge bundles $F^p \subseteq \mathcal{H}^k$. Then:

1. For every $s \in S$ and every homotopy class of path $c$ from $s_0$ to $s$, pullback along the inverse of the diffeomorphism $\varphi_c\colon X_{s_0} \to X_s$ that a trivialization of $f$ along $c$ provides is an isomorphism $T_c = (\varphi_c^{-1})^*\colon H^k(X_{s_0}, \mathbb{Z}) \to H^k(X_s, \mathbb{Z})$, depending only on $[c]$ and agreeing with the parallel transport of definition 7.1.3 after tensoring with $\mathbb{Q}$. It commutes with cup product and carries $h_{s_0}$ to $h_s$, hence restricts to an isomorphism $V_{\mathbb{Z}} \to H^k(X_s, \mathbb{Z})_{\mathrm{prim}}/(\text{torsion})$ carrying $Q$ to $Q$.

2. The primitive Hodge numbers $h^{p,q}_{\mathrm{prim}}(X_s)$ do not depend on $s$.

3. The primitive cohomology bundle $\mathcal{H}^k_{\mathrm{prim}}$ of definition 7.3.1 is a holomorphic subbundle of $\mathcal{H}^k$ preserved by $\nabla$; the subsheaves $F^p_{\mathrm{prim}} := F^p \cap \mathcal{H}^k_{\mathrm{prim}}$ are holomorphic subbundles of it, of rank $f^p$; and
$$\nabla F^p_{\mathrm{prim}} \subseteq F^{p-1}_{\mathrm{prim}} \otimes \Omega^1_S$$

4. For every $s$ and every path class $c$ as in (1), the filtration $T_c^{-1}\big(F^\bullet_{\mathrm{prim},s}\big)$ of $V_{\mathbb{C}}$ is a point of the period domain $D$ attached to $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$ by definition 5.3.2.

***Proof*:**

1. As in the proof of proposition 7.1.5, Ehresmann's theorem trivializes the pullback of $f$ along $c$, and $T_c = (\varphi_c^{-1})^*$ for the diffeomorphism $\varphi_c\colon X_{s_0} \to X_s$ arising from a trivialization $\Psi\colon X_{s_0} \times [0,1] \to f^{-1}(c)$; two trivializations differ by a path of diffeomorphisms of the fibre starting at the identity, so $T_c$ depends only on $[c]$, and with rational coefficients it is the transport of $\mathbb{H}^k$. A pullback along a diffeomorphism is a ring homomorphism for cup product, so $T_c$ is.

   For the class $h$, write $\iota_s\colon X_s \hookrightarrow \mathcal{X}$ for the inclusion. The maps $u \mapsto \iota_{c(u)} \circ \Psi_u$ form a homotopy from $\iota_{s_0}$ to $\iota_s \circ \varphi_c$ as maps $X_{s_0} \to \mathcal{X}$, so $\varphi_c^* h_s = \varphi_c^* \iota_s^* h = \iota_{s_0}^* h = h_{s_0}$, that is $T_c(h_{s_0}) = h_s$. Consequently $T_c \circ L = L \circ T_c$ and $T_c$ carries $\ker L^{n-k+1}$ onto $\ker L^{n-k+1}$.

   Finally $\varphi_c$ preserves the complex orientations. The pullback $\Psi_u^*$ of the complex orientation of $X_{c(u)}$ is an orientation of the connected manifold $X_{s_0}$ depending continuously on $u$, hence is constant, hence equals its value at $u = 0$, which is the complex orientation. So $\varphi_c = \Psi_1$ is orientation-preserving, and therefore
$$\int_{X_s} T_c\alpha \wedge T_c\beta \wedge h_s^{n-k} = \int_{X_{s_0}} \alpha \wedge \beta \wedge h_{s_0}^{n-k}$$
   using $T_c\, h_{s_0} = h_s$ for the third factor. With the sign $(-1)^{k(k-1)/2}$ of proposition 5.2.8 attached to both sides this says $Q(T_c\alpha, T_c\beta) = Q(\alpha, \beta)$.

2. Part (1) of proposition 7.2.3 already gives that each $h^{p,k-p}(X_s)$ is independent of $s$, so the ranks of the $F^p$ inside $\mathcal{H}^k$ are constant. (Those ranks are not the $f^p$ of the statement, which count primitive Hodge numbers only.)

   For the primitive numbers, fix $s$ and $p+q=k\leq n$. The operator $L$ raises the bidegree by $(1,1)$ and the primitive parts are sub-Hodge structures (proposition 5.2.6), so the Lefschetz decomposition corollary 6.2.4 of $H^k(X_s,\mathbb{C})$ refines along the bigrading into

   $$H^{p,q}(X_s)=\bigoplus_{r\geq 0} L^r\, H^{p-r,q-r}(X_s)_{\mathrm{prim}}$$

   with each $L^r$ injective on the summand it is applied to. Counting dimensions gives $h^{p,q}(X_s)=\sum_{r\geq 0} h^{p-r,q-r}_{\mathrm{prim}}(X_s)$, and the same identity in bidegree $(p-1,q-1)$ subtracts off all but the first term:

   $$h^{p,q}_{\mathrm{prim}}(X_s)=h^{p,q}(X_s)-h^{p-1,q-1}(X_s)$$

   Both terms on the right are independent of $s$, by the first paragraph applied in degree $k$ and in degree $k-2$, so the left side is.

3. By (2) the kernel of $L^{n-k+1}$ has constant rank, so $\mathbb{H}^k_{\mathrm{prim}}$ is a sub-local system of $\mathbb{H}^k$ and $\mathcal{H}^k_{\mathrm{prim}}$ is the holomorphic subbundle it generates. It is spanned locally by flat sections, so $\nabla$ carries its sections into $\mathcal{H}^k_{\mathrm{prim}}\otimes\Omega^1_S$: a local section is $\sum_j g_je_j$ with $g_j$ holomorphic and $e_j$ flat, and $\nabla(\sum_j g_je_j)=\sum_j e_j\otimes dg_j$ by theorem 7.1.7.

   The fibre of $F^p\cap\mathcal{H}^k_{\mathrm{prim}}$ at $s$ is $F^pH^k(X_s,\mathbb{C})\cap H^k(X_s,\mathbb{C})_{\mathrm{prim}}$, which by proposition 5.2.6 is the $p$-th step of the Hodge filtration of the primitive structure and so has dimension $\sum_{r\geq p} h^{r,k-r}_{\mathrm{prim}}(X_s)=f^p$, independent of $s$ by (2). Now $F^p\cap\mathcal{H}^k_{\mathrm{prim}}$ is the kernel of the composite holomorphic bundle map $F^p\to\mathcal{H}^k\to\mathcal{H}^k/\mathcal{H}^k_{\mathrm{prim}}$, whose rank is the constant $\mathrm{rank}F^p-\dim(F^p\cap\mathcal{H}^k_{\mathrm{prim}})$; the kernel of a holomorphic bundle map of constant rank is a holomorphic subbundle, so $F^p_{\mathrm{prim}}$ is one.

   For transversality, let $\sigma$ be a local holomorphic section of $F^p_{\mathrm{prim}}$. Then $\nabla\sigma\in F^{p-1}\otimes\Omega^1_S$ by theorem 7.2.7, and $\nabla\sigma\in\mathcal{H}^k_{\mathrm{prim}}\otimes\Omega^1_S$ by the previous paragraph; intersecting the two gives $\nabla\sigma\in F^{p-1}_{\mathrm{prim}}\otimes\Omega^1_S$.

4. Fix $s$ and $c$. By proposition 5.2.6 the group $H^k(X_s,\mathbb{Z})_{\mathrm{prim}}$ modulo torsion carries a pure Hodge structure of weight $k$, and by proposition 5.2.8 the form $Q$ polarizes it; its Hodge numbers are the $h^{p,q}$ by (2), and its Hodge filtration is $F^\bullet_{\mathrm{prim},s}$. The map $T_c^{-1}$ is an isomorphism of lattices carrying $Q$ to $Q$ by (1), and its complexification commutes with conjugation because it is defined over $\mathbb{Z}$; so transporting the bigrading along $T_c^{-1}$ produces a pure Hodge structure of weight $k$ on $V_\mathbb{Z}$ with Hodge numbers $h^{p,q}$, polarized by $Q$, whose Hodge filtration is $T_c^{-1}(F^\bullet_{\mathrm{prim},s})$. By part (1) of proposition 5.3.4 the Hodge filtration of such a structure is a point of $D$, completing the proof.

Part (1) is the reason the target is a quotient. Transport is an isomorphism of everything discrete, the lattice and the form, but it depends on the path only through its homotopy class, and different classes differ by a loop. The ambiguity is therefore exactly the monodromy group, and dividing by it is the smallest correction that removes it.

**Definition 7.3.3: The Period Map**

Keep the data of proposition 7.3.2 and set

$$G_{\mathbb{Z}} = \{g \in \mathrm{GL}(V_{\mathbb{Z}}) \mid Q(gu, gv) = Q(u, v) \text{ for all } u, v\}$$

a subgroup of the group $G_{\mathbb{R}}$ of proposition 5.3.5. Let

$$\rho_k \colon \pi_1(S, s_0) \to G_{\mathbb{Z}}, \qquad \rho_k(\gamma) = T_\gamma$$

be the monodromy representation of definition 6.4.2 acting on $V_{\mathbb{Z}}$, which lands in $G_{\mathbb{Z}}$ by part (1) of proposition 7.3.2, and let $\Gamma = \rho_k\big(\pi_1(S, s_0)\big)$ be the *monodromy group* of the family in degree $k$.
For $s \in S$ and a homotopy class $[c]$ of path from $s_0$ to $s$ put

$$\Phi_{[c]}(s) = T_{[c]}^{-1}\big(F^{\bullet}_{\mathrm{prim},s}\big) \in D$$

which lies in $D$ by part (4) of proposition 7.3.2. The *period map* of the family in degree $k$ is

$$\Phi \colon S \to \Gamma \backslash D, \qquad \Phi(s) = \Gamma \cdot \Phi_{[c]}(s)$$

Realizing the universal cover $\pi \colon \tilde{S} \to S$ as the set of homotopy classes of paths starting at $s_0$, the *lifted period map* is

$$\tilde{\Phi} \colon \tilde{S} \to D, \qquad \tilde{\Phi}\big([c]\big) = \Phi_{[c]}\big(c(1)\big)$$

That $\Phi$ is well defined is one line. If $c$ and $c'$ are two paths from $s_0$ to $s$, then $T_{[c']}^{-1} = T_{[\bar{c}']} = T_{[\bar{c}'c]} \circ T_{[c]}^{-1}$, where $\bar{c}'$ is $c'$ reversed and $\bar{c}'c$ is the loop at $s_0$ obtained by traversing $c$ and then $\bar{c}'$ (the composition convention of box 6.4.1). Hence

$$\Phi_{[c']}(s) = \rho_k\big([\bar{c}'c]\big)\, \Phi_{[c]}(s)$$

and the two points lie in the same $\Gamma$-orbit.

The definition should be read as a statement about what a family of varieties determines. A single fibre determines a point of $D$ only after a marking, an identification of its primitive cohomology with $V_{\mathbb{Z}}$ respecting $Q$; the family supplies markings, one for each homotopy class of path, and they differ by monodromy. What is intrinsic is the orbit. The classical case is the one where $D$ is the upper half plane and the orbit is the period ratio of an elliptic curve read modulo the substitutions monodromy performs on it.

**Example 7.3.4: The Period Map of a Family of Elliptic Curves**

Let $f \colon \mathcal{X} \to S$ be a smooth projective family of curves of genus 1 over a connected base, and take $k = n = 1$. The Legendre family $y^2z = x(x-z)(x-tz)$ over $S = \mathbb{C} \setminus \{0, 1\}$ of example 7.1.9 is one such, and so is the restriction, over the complement of its twelve critical values, of the fibration attached to a Lefschetz pencil of plane cubics (example 6.4.8).

*The numerical data.* All of $H^1$ is primitive (exercise 5.2.1 (2)), so $V_{\mathbb{Z}} = H^1(X_{s_0}, \mathbb{Z}) \cong \mathbb{Z}^2$, and $h^{1,0} = h^{0,1} = 1$. The normalizing sign of proposition 5.2.8 is $(-1)^{k(k-1)/2} = 1$ and the exponent $n - k$ is 0, so $Q$ is the cup product itself, a unimodular alternating form (example 5.2.2). Taking the genus parameter of example 5.3.3 to be 1, the compact dual is $\check{D} \cong \mathbb{CP}^1$ and the period domain is the upper half plane $\mathfrak{H}_1$ (exercise 5.3.1 (2)).

*The map in coordinates.* Fix a symplectic basis $a, b$ of $H_1(X_{s_0}, \mathbb{Z})$ and the dual basis $u, v$ of $H^1(X_{s_0}, \mathbb{Z})$, normalized as in exercise 5.2.1 (6) so that $Q(u, v) = \int u \wedge v = 1$. Let $U \subseteq S$ be connected and simply connected, and transport $a, b$ and $u, v$ over $U$; the transported cycles $a_s, b_s$ and classes $u_s, v_s$ are flat frames. The line $F^1_s = H^{1,0}(X_s)$ is spanned by the class of the holomorphic 1-form $\omega_s$ of the fibre, unique up to scale, and

$$[\omega_s] = A(s)\, u_s + B(s)\, v_s, \qquad A(s) = \int_{a_s} \omega_s, \quad B(s) = \int_{b_s} \omega_s$$

are its two periods. Expanding $Q$ in the frame,

$$i\, Q\big([\omega_s], \overline{[\omega_s]}\big) = i\big(A\bar{B} - \bar{A}B\big) = -2\,\mathrm{Im}\big(A\bar{B}\big)$$

and the second Hodge-Riemann relation makes this positive, so $\mathrm{Im}(\bar{A}B) > 0$. In particular $A$ never vanishes, the line $F^1_s$ is spanned by $u_s + \tau(s)\, v_s$ with

$$\tau(s) = \frac{B(s)}{A(s)} = \frac{\int_{b_s} \omega_s}{\int_{a_s} \omega_s}$$

and $\mathrm{Im}\, \tau = \mathrm{Im}(\bar{A}B)/|A|^2 > 0$. Under the identification of exercise 5.3.1 (2) the period map on $U$ is $s \mapsto \tau(s)$: the classical period ratio.

*The ambiguity.* Here $G_{\mathbb{Z}} = \mathrm{Sp}(V_{\mathbb{Z}}, Q) = \mathrm{SL}(2, \mathbb{Z})$, so $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ and $\Phi$ takes values in $\Gamma \backslash \mathfrak{H}_1$. Concretely, if the monodromy of a loop replaces the flat frame $(a, b)$ by $(a,\ b + a)$, then $A$ is unchanged while $B$ becomes $A + B$, so

$$\tau \longmapsto \tau + 1$$

That transformation of the frame is exactly the Picard-Lefschetz transvection computed in example 6.4.8 for a nodal degeneration of a curve, and the displayed substitution is the Hodge-theoretic form promised there. Horizontality, the second theorem of this section, says nothing here: in weight 1 the transversality condition is vacuous (example 7.2.9). The reading of $\Gamma \backslash \mathfrak{H}_1$ as a moduli space of elliptic curves, and the modular curves it produces, are not developed here; see [5, the analytic theory of elliptic curves, and modular curves].

The target needs a word, because a quotient of a complex manifold by a group is not automatically a space in which holomorphy makes sense. What rescues the situation is that monodromy is integral: $\Gamma$ sits inside $G_{\mathbb{Z}}$, which is discrete, while $G_{\mathbb{R}}$ acts on $D$ with compact isotropy.

**Proposition 7.3.5: The Monodromy Group Acts Properly Discontinuously**

Let $D$ be the period domain attached to $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$ and let $\Gamma \subseteq G_{\mathbb{Z}}$ be any subgroup. Then for every compact $C \subseteq D$ the set

$$\{\gamma \in \Gamma \,\big|\, \gamma C \cap C \neq \emptyset\}$$

is finite, and the quotient $\Gamma \backslash D$ is Hausdorff.

***Proof*:**
*Step 1: $G_{\mathbb{Z}}$ meets every compact subset of $G_{\mathbb{R}}$ in a finite set.* Choose a basis of $V_{\mathbb{Z}}$. In it the elements of $G_{\mathbb{Z}}$ are integer matrices, and $\mathrm{M}_m(\mathbb{Z})$ is a closed discrete subset of $\mathrm{M}_m(\mathbb{R})$; a closed discrete subset of a compact set is finite.

*Step 2: the action of $G_{\mathbb{R}}$ on $D$ is proper.* Fix $F_o \in D$ and let $P$ be its stabilizer in $G_{\mathbb{C}}$, so that by proposition 5.3.5 the orbit map $\pi\colon G_{\mathbb{R}} \to D$, $g \mapsto g\,F_o$, is surjective, identifies $D$ with the quotient space $G_{\mathbb{R}}/K$, and has fibres the cosets of the compact group $K = G_{\mathbb{R}} \cap P$. The map $\pi$ is open: for $W \subseteq G_{\mathbb{R}}$ open the saturation $\pi^{-1}(\pi(W)) = WK$ is open, so $\pi(W)$ is open in the quotient topology, which is the topology of $D$.

Let $C \subseteq D$ be compact. For each $x \in C$ choose $g_x \in G_{\mathbb{R}}$ with $\pi(g_x) = x$ and a relatively compact open $W_x \ni g_x$; the sets $\pi(W_x)$ form an open cover of $C$. Take a finite subcover and let $\tilde{C}$ be the union of the corresponding closures $\overline{W_x}$, a compact subset of $G_{\mathbb{R}}$ with $\pi(\tilde{C}) \supseteq C$.

If now $g \in G_{\mathbb{R}}$ satisfies $gC \cap C \neq \emptyset$, pick $x \in C$ with $gx \in C$ and write $x = \pi(c_1)$, $gx = \pi(c_2)$ with $c_1, c_2 \in \tilde{C}$. Then $gc_1K = c_2K$, so $g \in \tilde{C}K\tilde{C}^{-1}$, a compact subset of $G_{\mathbb{R}}$.

*Step 3: conclusion.* Combining the two steps, $\{\gamma \in \Gamma \,\big|\, \gamma C \cap C \neq \emptyset\}$ is contained in $G_{\mathbb{Z}} \cap \tilde{C}K\tilde{C}^{-1}$, which is finite.

For the Hausdorff property, let $x, y \in D$ lie in distinct $\Gamma$-orbits and choose compact neighbourhoods $N_x \ni x$ and $N_y \ni y$. Applying the finiteness just proved to the compact set $N_x \cup N_y$, only finitely many $\gamma \in \Gamma$ satisfy $\gamma N_x \cap N_y \neq \emptyset$; call them $\gamma_1, \ldots, \gamma_r$. For each $j$ one has $\gamma_j x \neq y$, since $x$ and $y$ lie in distinct orbits, so $D$ being Hausdorff there are open $U_j \ni x$ and $V_j \ni y$ with $\gamma_j U_j \cap V_j = \emptyset$. Then $U = N_x^\circ \cap \bigcap_j U_j$ and $V = N_y^\circ \cap \bigcap_j V_j$ are open neighbourhoods of $x$ and $y$ with $\gamma U \cap V = \emptyset$ for every $\gamma \in \Gamma$, so their images separate the two orbits in $\Gamma\backslash D$, completing the proof.

**7.3.6 Note: What "Holomorphic" Means on $\Gamma\backslash D$** Proposition 7.3.5 makes $\Gamma\backslash D$ a Hausdorff topological space, but the action need not be free: the stabilizer of a point of $D$ in $\Gamma$ is finite and may be nontrivial, and it is never trivial when $-\mathrm{id}_V \in \Gamma$ (exercise 7.3.1 (5)). Where $\Gamma$ acts freely the quotient map $q\colon D \to \Gamma\backslash D$ is a holomorphic covering and $\Gamma\backslash D$ is a complex manifold; the same holds when only the quotient of $\Gamma$ by the subgroup acting trivially does, since that quotient has the same orbits (exercise 7.3.1 (5)). In general $\Gamma\backslash D$ carries only the structure of a quotient with finite stabilizers. For that reason the statement "$\Phi$ is holomorphic" is always to be read through the lift: it means that $\tilde{\Phi}\colon \tilde{S} \to D$ is a holomorphic map of complex manifolds, equivalently that every local determination of the period map on a simply connected open subset of $S$, obtained by fixing a path class as in definition 7.3.3, is holomorphic. Nothing in this section depends on giving $\Gamma\backslash D$ more structure than that.

Holomorphy of the period map is the first place where the input of section 7.2 is used, and it is used in the weakest possible form: only that the $F^p$ are holomorphic subbundles, not that the connection moves them by one step. In a flat frame a holomorphic subbundle is a holomorphically varying family of subspaces of a fixed vector space, and such a family is a holomorphic map to a flag manifold.

**Theorem 7.3.7: The Period Map Is Holomorphic**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family and let $\Phi$ be its period map in degree $k$.

1. Let $U \subseteq S$ be open, connected and simply connected, fix $s_1 \in U$ and a homotopy class $[c]$ of path from $s_0$ to $s_1$, and for $s \in U$ let $[c_s]$ be the class of $c$ followed by the unique homotopy class of path from $s_1$ to $s$ inside $U$. Then

$$\Phi_U\colon U \to D, \qquad \Phi_U(s) = \Phi_{[c_s]}(s)$$

is holomorphic.

2. The lifted period map $\tilde{\Phi}\colon \tilde{S} \to D$ is holomorphic. If $\Gamma$ acts freely on $D$, then $\Gamma\backslash D$ is a complex manifold and $\Phi\colon S \to \Gamma\backslash D$ is holomorphic.

***Proof*:**
*Step 1: a flat trivialization.* Since $U$ is simply connected, the local system $\mathbb{H}^k_{\mathrm{prim}}|_U$ is trivial (proposition 7.1.5): parallel transport from $s_1$ along the unique homotopy class of path in $U$ identifies every fibre with $H^k(X_{s_1},\mathbb{Q})_{\mathrm{prim}}$, and composing with $T_{[c]}^{-1}$ identifies it with $V_{\mathbb{Q}}$. Tensoring with $\mathcal{O}_U$ gives an isomorphism of holomorphic bundles

$$\mathcal{H}^k_{\mathrm{prim}}\big|_U \;\cong\; V_{\mathbb{C}} \otimes_{\mathbb{C}} \mathcal{O}_U$$

under which the flat sections are the constant ones and $\nabla$ becomes the exterior derivative $d$ (theorem 7.1.7). In this trivialization $\Phi_U(s)$ is the flag $\left(F^p_{\mathrm{prim},s}\right)_p$ of subspaces of $V_{\mathbb{C}}$, of dimensions $f^p$ independent of $s$ by part (3) of proposition 7.3.2.

*Step 2: an adapted holomorphic frame.* Fix $s_2 \in U$. The subbundles $F^p_{\mathrm{prim}}$ are holomorphic and nested, so near $s_2$ there is a holomorphic frame $\sigma_1,\dots,\sigma_m$ of $V_{\mathbb{C}} \otimes \mathcal{O}$ adapted to the flag: choosing first a local holomorphic frame of the smallest nonzero step and extending it successively across each inclusion $F^{p+1}_{\mathrm{prim}} \subseteq F^p_{\mathrm{prim}}$, which is possible because a holomorphic subbundle of a holomorphic bundle is locally a direct summand, one obtains holomorphic sections $\sigma_1,\dots,\sigma_m$ of $V_{\mathbb{C}} \otimes \mathcal{O}$ such that $\sigma_1(s),\dots,\sigma_{f^p}(s)$ span $F^p_{\mathrm{prim},s}$ for every $p$ and every $s$ near $s_2$. Shrinking the neighbourhood, the $\sigma_j(s)$ are a basis of $V_{\mathbb{C}}$ for every $s$.

Let $e_1,\dots,e_m$ be the basis $\sigma_j(s_2)$ of $V_{\mathbb{C}}$ and define $g(s) \in \mathrm{GL}(V_{\mathbb{C}})$ by $g(s)e_j = \sigma_j(s)$. Its matrix in the basis $(e_j)$ has holomorphic entries and is invertible, so $g$ is a holomorphic map into $\mathrm{GL}(V_{\mathbb{C}})$, and by construction $g(s)$ carries the flag $\Phi_U(s_2)$ to the flag $\Phi_U(s)$.

*Step 3: into the flag manifold and into $D$.* The group $\mathrm{GL}(V_{\mathbb{C}})$ acts holomorphically on $\mathrm{Fl}$ (section 5.3), so the composite $s \mapsto g(s)\cdot\Phi_U(s_2)$ is a holomorphic map from a neighbourhood of $s_2$ into $\mathrm{Fl}$, and it equals $\Phi_U$ there. As $s_2 \in U$ was arbitrary, $\Phi_U\colon U \to \mathrm{Fl}$ is holomorphic.

Its image lies in $D$ by part (4) of proposition 7.3.2. By proposition 5.3.5 the compact dual $\check{D}$ is a closed complex submanifold of $\mathrm{Fl}$ and $D$ is open in it (proposition 5.3.4). A holomorphic map into a complex manifold whose image lies in a closed complex submanifold is holomorphic into that submanifold, since in a chart adapted to the submanifold the equations cutting it out are satisfied identically and the remaining coordinates of the map are holomorphic. Hence $\Phi_U\colon U \to D$ is holomorphic.

*Step 4: globalizing.* The projection $\pi\colon \tilde{S} \to S$ is a local biholomorphism, and each point of $\tilde{S}$ has a neighbourhood mapped biholomorphically onto a simply connected open $U \subseteq S$ on which $\tilde{\Phi}$ agrees with a map $\Phi_U$ of the kind treated above. So $\tilde{\Phi}$ is holomorphic.

Suppose finally that $\Gamma$ acts freely. By proposition 7.3.5 it then acts freely and properly discontinuously by biholomorphisms, so $\Gamma\backslash D$ inherits a complex structure for which $q\colon D \to \Gamma\backslash D$ is a holomorphic covering. Since $\Phi \circ \pi = q \circ \tilde{\Phi}$ is holomorphic and $\pi$ is a surjective local biholomorphism, $\Phi$ is holomorphic, as we sought to show.

The theorem is what makes the period map an object of complex geometry rather than of topology, and the proof shows precisely which hypothesis carries it: the holomorphy of the subbundles $F^p$, and nothing else. In particular the bigrading itself plays no role, which is the concrete payoff of the passage from bigradings to filtrations in theorem 5.1.5. The subspaces $H^{p,q}(X_s)$ vary only smoothly with $s$, and a map recording them would not be holomorphic; the sums $F^p$ do vary holomorphically, and the map recording them is.

Holomorphy leaves the second half of section 7.2 unused. Griffiths transversality constrains the derivative, so what it constrains is the differential of $\Phi$, and reading that constraint on the domain requires a description of the tangent spaces of Fl. Each step of a flag is a point of a Grassmannian, and the tangent space of a Grassmannian is $T_W\mathrm{Gr}(d,V) \cong \mathrm{Hom}(W, V/W)$, characterized by $dg_x(\xi)(\sigma(x)) = \partial_\xi\sigma \bmod g(x)$ for any local section $\sigma$ of $g$, and $\mathrm{GL}(V)$-equivariant [26, the tangent space of a Grassmannian]. That description is used in exactly the form just stated.

Each step of a flag is a point of a Grassmannian, so a tangent vector to Fl has a component in each $\mathrm{Hom}(F^p, V_{\mathbb{C}}/F^p)$, and the derivative of a moving flag is the collection of the derivatives of its steps. Griffiths transversality bounds how far a step can move: not into all of $V_{\mathbb{C}}/F^p$, but only into the part coming from one step down.

**Definition 7.3.8: The Horizontal Subbundle and the Infinitesimal Period Relation**

For each $p$ let $\pi_p\colon \mathrm{Fl} \to \mathrm{Gr}(f^p, V_{\mathbb{C}})$ be the holomorphic map $F \mapsto F^p$, and for a tangent vector $\theta \in T_F\mathrm{Fl}$ write

$$\theta_p := d(\pi_p)_F(\theta) \in \mathrm{Hom}\big(F^p,\ V_{\mathbb{C}}/F^p\big)$$

for its $p$-th component under the Grassmannian identification. The *horizontal subspace* of $\check{D}$ at $F \in \check{D}$ is

$$T^h_F\check{D} = \big\{\, \theta \in T_F\check{D} \;\big|\; \theta_p(F^p) \subseteq F^{p-1}/F^p \quad \text{for every } p \,\big\}$$

where $F^{p-1}/F^p$ is regarded as a subspace of $V_{\mathbb{C}}/F^p$. Its restriction to $D$ is the *horizontal subbundle* $T^hD \subseteq TD$.
A holomorphic map $g\colon M \to \check{D}$ from a complex manifold is *horizontal* if $dg_x(T_xM) \subseteq T^h_{g(x)}\check{D}$ for every $x \in M$. This condition on a map is classically called the *infinitesimal period relation.*

The horizontal subspaces do form a subbundle. The defining condition is stated purely in terms of the flag $F$, and by the equivariance clause of $\mathrm{GL}(V_{\mathbb{C}})$-equivariance an element $h \in G_{\mathbb{C}}$ carries $T^h_F\check{D}$ onto $T^h_{hF}\check{D}$; since $G_{\mathbb{C}}$ acts transitively on $\check{D}$ (proposition 5.3.5), the dimension of $T^h_F\check{D}$ is independent of $F$. The local holomorphic sections of the orbit map $G_{\mathbb{C}} \to \check{D}$ supplied by section 5.3 then transport a basis of $T^h_{F_o}\check{D}$ to a holomorphic frame of $T^h\check{D}$ near any point, so $T^h\check{D}$ is a holomorphic subbundle of $T\check{D}$.

Whether the inclusion $T^h\check{D} \subseteq T\check{D}$ is proper depends on the Hodge numbers, and in the two cases the reader has already met it is not.

**Example 7.3.9: The Horizontal Bundle in the Weight-Two K3 Case**

Take $k = 2$ with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = r$, the numerical type of example 5.3.6. A point of $\check{D}$ is a line $F^2 = \mathbb{C}\sigma$ with $Q(\sigma, \sigma) = 0$, and $F^1 = \sigma^\perp$ is determined by it, so

$$\check{D} \cong \big\{[\sigma] \in \mathbb{P}(V_\mathbb{C}) \big| Q(\sigma, \sigma) = 0\big\}$$

a smooth quadric hypersurface.

*The tangent space.* By the Grassmannian identification with $d = 1$, $T_{[\sigma]}\mathbb{P}(V_\mathbb{C}) = \mathrm{Hom}(\mathbb{C}\sigma, V_\mathbb{C}/\mathbb{C}\sigma)$. Differentiating the equation $Q(\sigma, \sigma) = 0$ along a holomorphic curve $[\sigma(s)]$ with section $\sigma$ gives $2Q(\sigma, \partial\sigma) = 0$, so a tangent vector $\theta_2 \in \mathrm{Hom}(\mathbb{C}\sigma, V_\mathbb{C}/\mathbb{C}\sigma)$ is tangent to the quadric exactly when $Q(\sigma, \theta_2(\sigma)) = 0$; the quadric being smooth, this cuts out the tangent space exactly. Hence

$$T_{[\sigma]}\check{D} = \mathrm{Hom}\big(\mathbb{C}\sigma,\ \sigma^\perp/\mathbb{C}\sigma\big) = \mathrm{Hom}\big(F^2,\ F^1/F^2\big)$$

*Horizontality.* The condition at $p = 2$ reads $\theta_2(F^2) \subseteq F^1/F^2$, which the display just established for every tangent vector. The condition at $p = 1$ reads $\theta_1(F^1) \subseteq F^0/F^1 = V_\mathbb{C}/F^1$, which is no condition at all, and the levels $p \le 0$ have $F^p = V_\mathbb{C}$ and $F^{p-1} = V_\mathbb{C}$. So $T^h\check{D} = T\check{D}$ here: a period map of this numerical type is horizontal for free. The same happens in weight 1, where the flag is a single subspace $F^1$ and the only condition that could bite reads $\theta_1(F^1) \subseteq F^0/F^1 = V_\mathbb{C}/F^1$, again vacuous; this is the period-domain form of the observation in example 7.2.9. Weight 2 with $h^{2,0} \ge 2$ already behaves differently (exercise 7.3.1 (3)).

With the horizontal subbundle in place, Griffiths transversality translates verbatim, and the translation changes what the statement is about. On the base it constrains a connection acting on subbundles, so it refers to the family; on the domain it says that the period map is everywhere tangent to a fixed distribution, one attached to the numerical data alone and shared by every family with those Hodge numbers.

**Theorem 7.3.10: The Period Map Is Horizontal**

Let $f : \mathcal{X} \to S$ be a smooth projective family and let $\tilde{\Phi} : \tilde{S} \to D$ be its lifted period map in degree $k$. Then $\tilde{\Phi}$ is horizontal:

$$d\tilde{\Phi}_{\tilde{s}}\big(T_{\tilde{s}}\tilde{S}\big) \subseteq T^h_{\tilde{\Phi}(\tilde{s})}D \qquad \text{for every } \tilde{s} \in \tilde{S}$$

Equivalently, every local determination $\Phi_U$ of the period map on a simply connected open subset of $S$, as in theorem 7.3.7, satisfies the infinitesimal period relation.

***Proof*:**

The statement is local on $S$, since $\pi : \tilde{S} \to S$ is a local biholomorphism and $\tilde{\Phi}$ agrees with a local determination $\Phi_U$ near every point. So let $U \subseteq S$ be open, connected and simply connected, fix the flat trivialization $\mathcal{H}^k_{\mathrm{prim}}|_U \cong V_\mathbb{C} \otimes_\mathbb{C} \mathcal{O}_U$ of Step 1 of the proof of theorem 7.3.7, and write $\Phi_U$ for the resulting holomorphic map $U \to D$. Fix $s \in U$ and $\xi \in T_sU$.

*Step 1: the components of $d\Phi_U$.* Fix $p$. The composite $\pi_p \circ \Phi_U$ is the map $U \to \mathrm{Gr}(f^p, V_\mathbb{C})$

sending $s'$ to $F^p_{\mathrm{prim},s'}$, and the holomorphic maps $\sigma\colon U' \to V_{\mathbb{C}}$ with $\sigma(s') \in F^p_{\mathrm{prim},s'}$ for all $s'$ are exactly the local holomorphic sections of the subbundle $F^p_{\mathrm{prim}}$, read in the trivialization. By the derivative formula characterizing the Grassmannian tangent space, for such a $\sigma$,

$$\big(d(\Phi_U)_s(\xi)\big)_p\big(\sigma(s)\big) = \partial_\xi \sigma \bmod F^p_{\mathrm{prim},s}$$

Because $F^p_{\mathrm{prim}}$ is a holomorphic subbundle, every vector of $F^p_{\mathrm{prim},s}$ is $\sigma(s)$ for some such $\sigma$, so the displayed formula determines the component $\big(d(\Phi_U)_s(\xi)\big)_p$ completely.

*Step 2: the derivative is the connection.* In the flat trivialization the flat sections are the constant ones, so $\nabla$ acts as the exterior derivative: writing $\sigma = \sum_j g_j e_j$ with $e_j$ a constant frame, theorem 7.1.7 gives $\nabla_\xi \sigma = \sum_j (\partial_\xi g_j)\, e_j = \partial_\xi \sigma$. Hence

$$\big(d(\Phi_U)_s(\xi)\big)_p\big(\sigma(s)\big) = \nabla_\xi \sigma \bmod F^p_{\mathrm{prim},s}$$

for every local holomorphic section $\sigma$ of $F^p_{\mathrm{prim}}$.

*Step 3: transversality.* By part (3) of proposition 7.3.2, which is Griffiths transversality (theorem 7.2.7) restricted to the primitive part, $\nabla_\xi \sigma$ lies in $F^{p-1}_{\mathrm{prim},s}$. Therefore

$$\big(d(\Phi_U)_s(\xi)\big)_p\big(F^p_{\mathrm{prim},s}\big) \subseteq F^{p-1}_{\mathrm{prim},s}/F^p_{\mathrm{prim},s}$$

for every $p$, which is exactly the condition defining $T^h_{\Phi_U(s)}\check{D}$ in definition 7.3.8. Since $d(\Phi_U)_s(\xi)$ is tangent to $D$ and $D$ is open in $\check{D}$ (proposition 5.3.4), it lies in $T^h_{\Phi_U(s)}D$, as we sought to show.

The proof is short because all the content was already extracted. Step 1 is linear algebra, Step 2 is the definition of the Gauss-Manin connection, and Step 3 is the theorem of section 7.2; nothing about the family enters that was not already in proposition 7.3.2. Read backwards, the theorem also says that transversality is not a property of any particular family: it is a property of the numerical data, expressed by a distribution on $\check{D}$ that every period map with those Hodge numbers must be tangent to. The geometry of $D$ that this distribution carries, and the curvature estimates along it that turn horizontality into rigidity statements about families, are taken up in chapter 8.

The graded form of the differential is the primitive counterpart of the map isolated in corollary 7.2.8. By Step 3 the component $\big(d\Phi_U(\xi)\big)_p$ takes $F^p_{\mathrm{prim}}$ into $F^{p-1}_{\mathrm{prim}}/F^p_{\mathrm{prim}}$, and it kills $F^{p+1}_{\mathrm{prim}}$ modulo $F^p_{\mathrm{prim}}$, since a section of $F^{p+1}_{\mathrm{prim}}$ has $\nabla_\xi\sigma \in F^p_{\mathrm{prim}}$ by transversality one level up. It therefore factors through a map $F^p_{\mathrm{prim}}/F^{p+1}_{\mathrm{prim}} \to F^{p-1}_{\mathrm{prim}}/F^p_{\mathrm{prim}}$, and as $\xi$ varies these assemble into the $\mathcal{O}_S$-linear map

$$F^p_{\mathrm{prim}}/F^{p+1}_{\mathrm{prim}} \to \big(F^{p-1}_{\mathrm{prim}}/F^p_{\mathrm{prim}}\big) \otimes \Omega^1_S$$

This is the construction of corollary 7.2.8 run on the primitive filtration $F^\bullet_{\mathrm{prim}}$ instead of on $F^\bullet$, which is legitimate because that construction used nothing but transversality, and part (3) of proposition 7.3.2 supplies transversality for the primitive part. Equivalently it is the restriction of the corollary's map to the primitive graded pieces, a strictly smaller map whenever the primitive cohomology is proper in $H^k$: on a fibre it reads $H^{p,q}_{\mathrm{prim}} \to H^{p-1,q+1}_{\mathrm{prim}} \otimes T^*_s S$. So the derivative of the period map is a collection of maps between adjacent primitive Hodge pieces of a single fibre, and computing it never requires leaving that fibre.

When $T^hD$ is a proper subbundle, horizontality is a genuine restriction on what a family can do, and it restricts by dimension. A period map can never be submersive onto an open subset of $D$ in that case, and if $d\Phi$ is injective at some point, which is the local Torelli condition, then $\dim_{\mathbb{C}} S$ is at most the rank of $T^hD$ (exercise 7.3.1 (4)). The first numerical type where this bites is weight 2 with $h^{2,0} = 2$, where the horizontal bundle has codimension 1 (exercise 7.3.1 (3)); the effect grows quickly with the number of nonzero Hodge numbers.

## Exercise 7.3.1

1. Let $f\colon \mathcal{X} \to S$ be a smooth projective family with period map $\Phi$ in degree $k$. Show that $\Phi$ is constant if and only if $\nabla F^p_{\mathrm{prim}} \subseteq F^p_{\mathrm{prim}} \otimes \Omega^1_S$ for every $p$, that is, if and only if every step of the Hodge filtration of the primitive part is a flat subbundle. Deduce that in that case parallel transport along any path from $s_0$ to $s$ is an isomorphism of polarized Hodge structures from $H^k(X_{s_0}, \mathbb{Q})_{\mathrm{prim}}$ to $H^k(X_s, \mathbb{Q})_{\mathrm{prim}}$. (Nothing follows about the Hodge structure on the whole of $H^k(X_s)$, which the period map in degree $k$ does not see.)

2. In the setting of example 7.3.4, suppose the monodromy of a loop in $\pi_1(S, s_0)$ replaces the flat frame $(a, b)$ of $H_1$ by $(\kappa a + \lambda b,\ \mu a + \nu b)$ with $\kappa, \lambda, \mu, \nu \in \mathbb{Z}$ and $\kappa\nu - \lambda\mu = 1$.

    1. Show that the period ratio computed in the new frame is $\tau' = (\mu + \nu\tau)/(\kappa + \lambda\tau)$, and that $\kappa + \lambda\tau \neq 0$.
    2. Verify directly that $\operatorname{Im} \tau' > 0$, so that the substitution preserves $\mathfrak{H}_1$.
    3. Recover the substitution $\tau \mapsto \tau + 1$ of example 6.4.8 as a special case, and identify the matrix.

3. Take $k = 2$ with $h^{2,0} = h^{0,2} = 2$ and $h^{1,1} = r$, so that $m = r + 4$ and $Q$ is symmetric and nondegenerate.

    1. Show that $F \mapsto F^2$ identifies $\check{D}$ with the set of 2-dimensional isotropic subspaces $W \subseteq V_{\mathbb{C}}$, with $F^1 = W^\perp$ determined by $W$.
    2. Using the Grassmannian tangent-space identification [26, the tangent space of a Grassmannian], show that

    $$T_W\check{D} = \{\theta \in \operatorname{Hom}(W, V_{\mathbb{C}}/W) \mid \beta_\theta \text{ is alternating}\}, \qquad \beta_\theta(w, w') := Q\big(\theta(w), w'\big)$$

    (note that $\beta_\theta$ is well defined because $W$ is isotropic), and that $\theta \mapsto \beta_\theta$ maps $T_W\check{D}$ onto $\Lambda^2 W^*$ with kernel $\operatorname{Hom}(W, W^\perp/W)$.
    3. Deduce that $T^h_W\check{D} = \operatorname{Hom}(W, W^\perp/W)$ and that it has codimension 1 in $T_W\check{D}$. What happens when $h^{2,0} = 1$?

4. Let $\Phi$ be the period map of a smooth projective family over a connected $S$, and suppose $d\tilde{\Phi}$ is injective at some point of $\tilde{S}$.

    1. Show that $\dim_{\mathbb{C}} S \leq \operatorname{rank} T^hD$.
    2. Show that if $T^hD \neq TD$, then $\tilde{\Phi}$ is nowhere a local biholomorphism onto an open subset of $D$, whatever the dimension of $S$.
    3. Combine (1) with exercise 7.3.1 (3) to bound $\dim_{\mathbb{C}} S$ for a family of the numerical type considered there, in terms of $r$.

5. Let $D$ be a period domain attached to data $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$.

    1. Show that $-\mathrm{id}_V$ lies in $G_{\mathbb{Z}}$ and acts trivially on $\check{D}$, so the action of $G_{\mathbb{R}}$ on $D$ is never effective.
    2. Deduce that if $-\mathrm{id}_V \in \Gamma$, then no point of $D$ has trivial $\Gamma$-stabilizer, so $\Gamma$ never acts freely and part (2) of theorem 7.3.7 never applies to $\Gamma$ itself. Show nevertheless that $\Gamma\backslash D = \bar{\Gamma}\backslash D$ for $\bar{\Gamma} = \Gamma/\{\pm\mathrm{id}_V\}$, that $\bar{\Gamma}$ inherits the conclusion of proposition 7.3.5, and hence that $q\colon D \to \Gamma\backslash D$ is a covering map as soon as $\bar{\Gamma}$ acts freely, with deck group $\bar{\Gamma}$ rather than $\Gamma$.
    3. Identify the phenomenon in weight 1 with $h^{1,0} = 1$, using example 5.3.3.

## 7.4 Hypersurfaces and the Jacobian Ring

The three preceding sections attach to a smooth projective family a local system, a filtration by holomorphic subbundles and a period map. None of them computes any of it. This section computes all of it for one family, the smooth hypersurfaces of a fixed degree in a fixed projective space, where the answer is a statement about polynomials: the Hodge filtration on the primitive cohomology of $X = \{F = 0\}$ is read off the graded ring $\mathbb{C}[x_0, \ldots, x_{n+1}]/J(F)$, where $J(F)$ is the ideal generated by the partial derivatives of $F$.

Two conventions before starting. The defining polynomial is written $F$ and the Hodge filtration $F^p$; the superscript is what distinguishes them. And the cohomology in play is always the primitive part in the middle degree: by example 6.3.2 every other cohomology group of a smooth hypersurface is that of the ambient projective space, so a family of hypersurfaces can vary nothing outside $H^n_{\mathrm{prim}}$.

Throughout, $n \geq 1$ and $d \geq 2$ are fixed, $\mathbb{C}[x] = \mathbb{C}[x_0, \ldots, x_{n+1}]$ is the polynomial ring in $n+2$ variables with its usual grading, and $\mathbb{C}[x]_e$ denotes the space of homogeneous polynomials of degree $e$, taken to be 0 for $e < 0$. For $0 \neq F \in \mathbb{C}[x]_d$ write

$$X_F = \{F = 0\} \subseteq \mathbb{CP}^{n+1}, \qquad U_F = \mathbb{CP}^{n+1} \setminus X_F$$

and abbreviate these to $X$ and $U$ when $F$ is fixed. Partial derivatives are written $\partial_i = \partial/\partial x_i$.

The object that carries the whole theory is the ideal generated by those derivatives. Its first role is to detect smoothness, through the Euler relation $\sum_i x_i\, \partial_i F = d \cdot F$, which is what ties the vanishing of the derivatives to the vanishing of $F$ itself.

**Definition 7.4.1: The Jacobian Ideal and the Jacobian Ring**

Let $F \in \mathbb{C}[x]_d$. The *Jacobian ideal* of $F$ is the homogeneous ideal

$$J(F) = \big(\partial_0 F,\ \partial_1 F,\ \ldots,\ \partial_{n+1} F\big) \subseteq \mathbb{C}[x]$$

generated by the $n+2$ partial derivatives, each of degree $d-1$. The *Jacobian ring* of $F$ is the quotient

$$R(F) = \mathbb{C}[x]/J(F)$$

with its induced grading $R(F) = \bigoplus_{e \geq 0} R(F)_e$, where $R(F)_e = \mathbb{C}[x]_e/J(F)_e$. When $F$ is fixed the ring is written $R$.

Because $J(F)$ is generated in degree $d-1$, the low-degree pieces of $R$ are not quotients at all: $R_e = \mathbb{C}[x]_e$ for every $e \leq d-2$. The interesting behaviour is in the middle range, and it is there that smoothness of $X_F$ makes itself felt.

**Proposition 7.4.2: Smoothness and Finite-Dimensionality of the Jacobian Ring**

Let $F \in \mathbb{C}[x]_d$ be nonzero, with $d \geq 1$. The following are equivalent.

1. The derivatives $\partial_0 F, \ldots, \partial_{n+1} F$ have no common zero in $\mathbb{C}\mathrm{P}^{n+1}$.
2. At every point of $X_F$ the dehomogenization of $F$ in some, hence any, affine chart containing that point has nonvanishing differential.

When they hold, $F$ is called *smooth* and $X_F$ is a complex submanifold of $\mathbb{C}\mathrm{P}^{n+1}$ of dimension $n$; moreover $R(F)$ is a finite-dimensional $\mathbb{C}$-vector space, so $R(F)_e = 0$ for all sufficiently large $e$.

***Proof*:**
*Step 1: (1) implies (2).* Let $x \in X_F$, choose $j$ with $x_j \neq 0$ and work in the affine chart $x_j = 1$ with coordinates $u_k = x_k/x_j$ for $k \neq j$, in which $X_F$ is the zero set of $f(u) = F(u)|_{u_j=1}$ and $\partial f/\partial u_k = (\partial_k F)|_{u_j=1}$. If every $\partial_k F$ with $k \neq j$ vanished at $x$, then the Euler relation

$$\sum_k x_k\, \partial_k F = d \cdot F$$

evaluated at $x$ would give $x_j\, \partial_j F(x) = d \cdot F(x) = 0$ and hence $\partial_j F(x) = 0$ as well, so all $n+2$ derivatives would vanish at $x$, contrary to (1). So $df_x \neq 0$, and the implicit function theorem presents $X_F$ near $x$ as a complex submanifold of codimension one.

*Step 2: (2) implies (1).* Suppose the derivatives have a common zero $x \in \mathbb{C}\mathrm{P}^{n+1}$. The Euler relation gives $d \cdot F(x) = 0$, so $F(x) = 0$ and $x \in X_F$; and in a chart around $x$ every partial derivative of the dehomogenization is a restriction of some $\partial_k F$ and so vanishes at $x$, contradicting (2).

*Step 3: finite-dimensionality.* Dispose first of $d = 1$, where the derivatives are constants, not all zero because $F \neq 0$: there $J(F) = \mathbb{C}[x]$ and $R = 0$. So assume $d \geq 2$, in which case the derivatives are homogeneous of positive degree and vanish at the origin. By (1) they have no common zero in $\mathbb{C}\mathrm{P}^{n+1}$, so their common zero locus in $\mathbb{C}^{n+2}$ is exactly the origin. The Nullstellensatz [4] gives $\sqrt{J(F)} = (x_0, \ldots, x_{n+1})$, so there is $k \geq 1$ with $x_i^k \in J(F)$ for every $i$. Consequently $R$ is spanned by the images of the monomials $x^a$ with every exponent $a_i < k$, of which there are $k^{n+2}$, and $R_e = 0$ once $e > (n+2)(k-1)$, completing the proof.

Condition (2) is a condition on $F$ and not only on the point set $X_F$. For $F = x_0^2$ in $\mathbb{C}\mathrm{P}^2$ the zero set is a line, hence a smooth curve, while every partial derivative of $F$ vanishes along it; that $F$ is not smooth in the sense above, and its Jacobian ring $\mathbb{C}[x_0, x_1, x_2]/(x_0)$ is infinite dimensional, as the last part of the proposition requires. From here on, smoothness of a hypersurface means smoothness of its defining polynomial.

The last assertion says that the Jacobian ring of a smooth hypersurface is a finite package of numbers. Theorem 7.4.11 will say that this package is the Hodge structure of $X$, so it is worth having one case in which every graded piece can be written down.

**Example 7.4.3: The Jacobian Ring of a Fermat Hypersurface**

Let $F = x_0^d + x_1^d + \cdots + x_{n+1}^d$, the *Fermat* polynomial of degree $d \geq 2$. Then $\partial_i F = d\, x_i^{d-1}$, and these vanish simultaneously only at the origin of $\mathbb{C}^{n+2}$, so $X_F$ is smooth by proposition 7.4.2. The Jacobian ideal is the monomial ideal

$$J(F) = \left(x_0^{d-1},\ x_1^{d-1},\ \ldots,\ x_{n+1}^{d-1}\right)$$

and a monomial $x^a$ lies in $J(F)$ exactly when $a_i \geq d-1$ for some $i$. Since $J(F)$ is spanned by the monomials it contains, the images of the remaining monomials form a basis of $R$:

$$R \ \text{ has basis } \ \left\{\, x^a \ : \ 0 \leq a_i \leq d-2 \text{ for every } i \,\right\}$$

Hence $\dim_{\mathbb{C}} R_e$ is the number of ways to write $e = a_0 + \cdots + a_{n+1}$ with every $a_i$ in $\{0, 1, \ldots, d-2\}$, that is,

$$\sum_{e \geq 0} (\dim_{\mathbb{C}} R_e)\, t^e = \left(1 + t + \cdots + t^{d-2}\right)^{n+2} \tag{7.4}$$

Three consequences are immediate from (7.4). The top nonzero degree is $\sigma = (n+2)(d-2)$, attained only by the single monomial with all $a_i = d-2$, so $\dim R_\sigma = 1$. The map $a \mapsto (d-2, \ldots, d-2) - a$ is a bijection of the basis carrying degree $e$ to degree $\sigma - e$, so

$$\dim R_e = \dim R_{\sigma - e} \qquad (0 \leq e \leq \sigma)$$

And for $d = 2$ the polynomial (7.4) is 1: the Jacobian ring of a smooth quadric is $\mathbb{C}$, concentrated in degree 0.

For $n = 2$ and $d = 4$, a Fermat quartic surface in $\mathbb{CP}^3$, the series is $(1 + t + t^2)^4 = 1 + 4t + 10t^2 + 16t^3 + 19t^4 + 16t^5 + 10t^6 + 4t^7 + t^8$, with $\sigma = 8$.

Every smooth hypersurface of degree $d$ now sits in one family, and it is that family to which sections 7.1 to 7.3 apply. Write $M = \dim_{\mathbb{C}} \mathbb{C}[x]_d - 1$ and identify $\mathbb{CP}^M$ with the projective space of nonzero forms of degree $d$ up to scale, a point of which is written $[F]$.

**Proposition 7.4.4: The Family of Smooth Hypersurfaces of Fixed Degree**

Let $n \geq 1$, $d \geq 2$, and let

$$S = \left\{[F] \in \mathbb{CP}^M \ : \ F \text{ is smooth in the sense of proposition 7.4.2}\right\}$$

Then $S$ is a nonempty connected Zariski-open subset of $\mathbb{CP}^M$; the incidence variety

$$\mathcal{X} = \left\{(x, [F]) \in \mathbb{CP}^{n+1} \times S \ : \ F(x) = 0\right\}, \qquad f\colon \mathcal{X} \to S \ \text{ the second projection}$$

is a smooth projective family with fibre $f^{-1}([F]) = X_F$; and every point of $S$ has a neighbourhood $V$ over which the pair $(\mathbb{CP}^{n+1} \times V, \mathcal{X}|_V)$ is diffeomorphic to $(\mathbb{CP}^{n+1}, X_{F_0}) \times V$ over $V$.

***Proof*:**
*Step 1: $S$ is open, nonempty and connected.* Let

$$\Sigma = \{(x,[F]) \in \mathbb{C}\mathrm{P}^{n+1} \times \mathbb{C}\mathrm{P}^{M} \ : \ \partial_0 F(x) = \cdots = \partial_{n+1} F(x) = 0\}$$

which is closed, being cut out by equations bihomogeneous in $x$ and in the coefficients of $F$. Its image $\Delta \subseteq \mathbb{C}\mathrm{P}^M$ under the second projection is closed, since $\mathbb{C}\mathrm{P}^{n+1}$ is complete [4], and by proposition 7.4.2 a point $[F]$ lies in $\Delta$ exactly when $F$ fails to be smooth. So $S = \mathbb{C}\mathrm{P}^M \setminus \Delta$ is Zariski open, and it is nonempty because the Fermat polynomial of example 7.4.3 lies in it. For connectedness, let $[F_0],[F_1] \in S$ and let $\ell \cong \mathbb{C}\mathrm{P}^1$ be the line they span. Then $\ell \not\subseteq \Delta$, since $[F_0] \notin \Delta$, so $\ell \cap \Delta$ is a proper Zariski-closed subset of $\ell$ and hence finite; and $\ell$ with finitely many points removed is path connected and contained in $S$.

*Step 2: $f$ is a proper submersion with connected fibres.* This is example 7.1.2(2), which set up exactly this incidence variety and checked the submersion criterion there; the argument is repeated in outline only because the smoothness locus is now described through proposition 7.4.2. For fixed $x$ the condition $F(x) = 0$ is one nonzero linear condition on the coefficients of $F$, so the first projection exhibits the incidence variety inside $\mathbb{C}\mathrm{P}^{n+1} \times \mathbb{C}\mathrm{P}^M$ as a bundle with fibre $\mathbb{C}\mathrm{P}^{M-1}$; in particular it is a complex manifold, and $\mathcal{X}$ is the open subset lying over $S$. The map $f$ is proper, being the restriction to a closed subset of the projection $\mathbb{C}\mathrm{P}^{n+1} \times S \to S$ with compact fibre $\mathbb{C}\mathrm{P}^{n+1}$.

For submersivity, fix $(x,[F]) \in \mathcal{X}$ and a tangent vector to $\mathbb{C}\mathrm{P}^M$ at $[F]$, represented by $G \in \mathbb{C}[x]_d$. Since $X_F$ is smooth, $dF_x \neq 0$ in a chart around $x$, so the implicit function theorem solves $F_t(x_t) = 0$ for a holomorphic curve $x_t$ with $x_0 = x$, where $F_t = F + tG$. The curve $t \mapsto (x_t,[F_t])$ lies in $\mathcal{X}$ and covers the given tangent vector, so $df$ is surjective. Finally $H^0(X_F,\mathbb{Z}) = \mathbb{Z}$ by example 6.3.2, since $n \geq 1$, so the fibres are connected.

*Step 3: local triviality of the pair.* Fix $[F_0] \in S$ and a neighbourhood $V \ni [F_0]$ identified with an open cube in $\mathbb{R}^N$ by real coordinates $t_1,\dots,t_N$ vanishing at $[F_0]$. For each coordinate vector field $\partial_{t_j}$ on $V$ choose a smooth vector field $w_j$ on $\mathbb{C}\mathrm{P}^{n+1} \times V$ that projects to $\partial_{t_j}$ and is tangent to $\mathcal{X}$: on $\mathcal{X}$ such a lift exists because $f$ is a submersion, it extends to a neighbourhood of $\mathcal{X}$, and it is patched with an arbitrary lift on the complement by a partition of unity, which preserves the property of projecting to $\partial_{t_j}$ because the lifts of a fixed vector field form an affine space.

The flow of $w_j$ is not complete: the trajectory through $(x,t)$ has base component $t + s\,e_j$, with $e_j$ the $j$-th standard basis vector, and that leaves $V$ once $|s|$ is large. What compactness of $\mathbb{C}\mathrm{P}^{n+1}$ gives is completeness in the fibre direction, which is enough. Suppose the maximal interval of a trajectory has a finite endpoint $s_+$ with $t + s_+ e_j$ still in $V$, and let $L \subseteq V$ be the closed segment from $t$ to $t + s_+ e_j$. The trajectory stays in the compact set $\mathbb{C}\mathrm{P}^{n+1} \times L$, on which $w_j$ is bounded, so it has bounded speed and converges as $s \to s_+$; restarting the flow at its limit extends it past $s_+$, contradicting maximality. So the time-$s$ flow $\Theta^j_s$ of $w_j$ is defined on $\mathbb{C}\mathrm{P}^{n+1} \times \{t\}$ whenever the segment from $t$ to $t + s\,e_j$ lies in $V$, and it carries that fibre onto $\mathbb{C}\mathrm{P}^{n+1} \times \{t + s\,e_j\}$ and $\mathcal{X}$ to $\mathcal{X}$.

Now set

$$\Theta\big(x,(t_1,\dots,t_N)\big) \ = \ \Theta^N_{t_N} \circ \cdots \circ \Theta^1_{t_1}(x,0)$$

Every segment traversed here is a coordinate segment inside the cube, so $\Theta$ is defined on all of $\mathbb{C}\mathrm{P}^{n+1} \times V$; it is smooth, covers the identity of $V$, and is inverted by running the flows backwards in the opposite order, so it is a diffeomorphism of $\mathbb{C}\mathrm{P}^{n+1} \times V$ carrying $X_{F_0} \times V$ onto $\mathcal{X}|_V$. That is the asserted trivialization of the pair; this is the argument that proves Ehresmann's theorem [26], run with the extra requirement that the lifted field be tangent to $\mathcal{X}$, completing the proof.

Proposition 7.4.4 puts the whole apparatus of this chapter at the disposal of hypersurfaces. The local system $\mathbb{H}^n = R^n f_*\mathbb{Q}$ of definition 7.1.3 carries the Gauss-Manin connection $\nabla$ of definition 7.1.6; the Hodge filtration $F^\bullet$ consists of holomorphic subbundles of $\mathcal{H}^n$ by theorem 7.2.6 and satisfies Griffiths transversality by theorem 7.2.7; and the period map $\Phi$ of definition 7.3.3 is holomorphic and horizontal. The polarization is by the hyperplane class $h_X$, the restriction of the Fubini-Study class, and proposition 5.2.6 makes $H^n_{\mathrm{prim}}(X,\mathbb{Q}) = \ker\big(h_X \smile -\big)$ a sub-Hodge structure of $H^n(X,\mathbb{Q})$. Everything that varies is inside it.

The route to computing that Hodge structure runs through the complement rather than through $X$ itself. The reason is that $U = \mathbb{C}\mathrm{P}^{n+1} \setminus X$ is affine (the $d$-uple Veronese embedding of example 6.3.2 carries $X$ to a hyperplane section, so $U$ becomes a closed submanifold of an affine space), and on an affine variety cohomology classes have global rational representatives. Those representatives can be written down.

Set

$$\Omega \;=\; \sum_{i=0}^{n+1} (-1)^i\, x_i\; dx_0 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_{n+1}$$

the contraction of the Euler vector field $E = \sum_i x_i \partial_i$ into $dx_0 \wedge \cdots \wedge dx_{n+1}$, a holomorphic $(n+1)$-form on $\mathbb{C}^{n+2}$ whose coefficients are homogeneous of degree 1.

**Proposition 7.4.5: Rational Forms with Poles Along a Hypersurface**

Let $X = X_F \subseteq \mathbb{C}\mathrm{P}^{n+1}$ be a smooth hypersurface of degree $d$ and let $m \geq 1$. The assignment

$$A \;\longmapsto\; \frac{A\,\Omega}{F^m}$$

is a linear isomorphism from $\mathbb{C}[x]_{md-n-2}$ onto the space of meromorphic $(n+1)$-forms on $\mathbb{C}\mathrm{P}^{n+1}$ that are holomorphic on $U$ and have a pole of order at most $m$ along $X$, that is, forms $\eta$ such that $f^m \eta$ is holomorphic near each point of $X$, where $f$ is a local defining function for $X$.

***Proof*:**

*Step 1: $\Omega$ trivializes $\Omega^{n+1}_{\mathbb{C}\mathrm{P}^{n+1}}(n+2)$.* Work in the chart $x_0 \neq 0$ with affine coordinates $u_i = x_i/x_0$, so that $du_i = (x_0\, dx_i - x_i\, dx_0)/x_0^2$. Wedging these,

$$du_1 \wedge \cdots \wedge du_{n+1} = x_0^{-2(n+1)} \bigwedge_{i=1}^{n+1} \big(x_0\, dx_i - x_i\, dx_0\big)$$

Expanding the product, the terms containing two or more factors of $dx_0$ vanish, so what survives is $x_0^{n+1} dx_1 \wedge \cdots \wedge dx_{n+1}$ together with, for each $i$, the term $-x_0^n x_i\, dx_1 \wedge \cdots \wedge dx_0 \wedge \cdots \wedge dx_{n+1}$ with $dx_0$ in the $i$-th slot; moving that $dx_0$ to the front costs $(-1)^{i-1}$. The bracket is therefore $x_0^n\,\Omega$, and

$$\Omega = x_0^{n+2}\; du_1 \wedge \cdots \wedge du_{n+1} \tag{7.5}$$

The same computation holds in every coordinate chart, so $\Omega$ vanishes nowhere on $\mathbb{C}^{n+2} \setminus \{0\}$ and (7.5) exhibits it as a nowhere-vanishing holomorphic section of $\Omega^{n+1}_{\mathbb{C}\mathrm{P}^{n+1}}(n+2)$; this is a restatement of $K_{\mathbb{CP}^{n+1}} = \mathcal{O}(-n-2)$ (corollary 1.3.17).

*Step 2: the map is well defined and injective.* Let $A \in \mathbb{C}[x]_{md-n-2}$. In the chart $x_0 \neq 0$, (7.5) and $f = F/x_0^d$ give

$$\frac{A\,\Omega}{F^m} = \frac{A\,x_0^{n+2}}{F^m}\,du_1 \wedge \cdots \wedge du_{n+1} = \frac{A\,x_0^{n+2-md}}{f^m}\,du_1 \wedge \cdots \wedge du_{n+1} \tag{7.6}$$

whose coefficient is $f^{-m}$ times a holomorphic function of $u$, because $A\,x_0^{n+2-md}$ is homogeneous of degree 0 and holomorphic where $x_0 \neq 0$. So $A\Omega/F^m$ is a meromorphic $(n+1)$-form on $\mathbb{CP}^{n+1}$, holomorphic on $U$, with pole order at most $m$ along $X$. It is nonzero for $A \neq 0$ because $\Omega$ vanishes nowhere, so the map is injective.

*Step 3: surjectivity.* Let $\eta$ be a form of the stated kind. In the chart $x_0 \neq 0$ write $\eta = \varphi\,du_1 \wedge \cdots \wedge du_{n+1}$; the hypothesis says that $g := f^m\varphi$ is holomorphic on the whole chart. Reading (7.6) backwards, $\eta = A_0\Omega/F^m$ on that chart with

$$A_0 = g \cdot x_0^{md-n-2}$$

a holomorphic function on $\{x_0 \neq 0\}$, homogeneous of degree $md - n - 2$. The same computation in the chart $x_j \neq 0$ produces $A_j$, and $A_i = A_j$ on overlaps because $\Omega$ vanishes nowhere; so the $A_j$ glue to a holomorphic function $A$ on $\mathbb{C}^{n+2} \setminus \{0\}$, homogeneous of degree $e = md - n - 2$. Expanding $g$ as a power series $\sum_\alpha c_\alpha u^\alpha$ converging on all of $\mathbb{C}^{n+1}$ gives $A = \sum_\alpha c_\alpha\, x^\alpha x_0^{e-|\alpha|}$, and holomorphy of $A$ at points with $x_0 = 0$ and some other coordinate nonzero forces $c_\alpha = 0$ whenever $|\alpha| > e$. Hence $A$ is a homogeneous polynomial of degree $e$, and $A = 0$ when $e < 0$, as we sought to show.

Such a form is automatically closed: it is a holomorphic $(n+1)$-form on the complex manifold $U$ of dimension $n+1$, so its exterior derivative, of type $(n+2, 0)$, vanishes. Each therefore defines a class in $H^{n+1}(U, \mathbb{C})$, and the whole problem becomes one of transporting classes from $U$ back to $X$. The transport is the residue, and the definition used here is topological.

**Definition 7.4.6: The Tube Map and the Residue**

Let $X \subseteq \mathbb{CP}^{n+1}$ be a smooth hypersurface with complement $U$. Fix a closed tubular neighbourhood $\pi\colon \mathcal{N} \to X$ of $X$ in $\mathbb{CP}^{n+1}$ [26], a closed disc bundle whose boundary $\partial\mathcal{N}$ is a circle bundle contained in $U$, with the fibre circles oriented positively for the complex structure of the normal direction. The *tube map*

$$\tau\colon H_k(X,\mathbb{Z}) \to H_{k+1}(U,\mathbb{Z})$$

is the composite

$$H_k(X,\mathbb{Z}) \;\cong\; H_{k+2}(\mathcal{N},\partial\mathcal{N};\mathbb{Z}) \;\cong\; H_{k+2}(\mathbb{CP}^{n+1},U;\mathbb{Z}) \;\xrightarrow{\partial}\; H_{k+1}(U,\mathbb{Z})$$

whose first arrow is Poincaré duality on $X$ followed by Lefschetz duality on $\mathcal{N}$, whose second is excision, and whose third is the connecting homomorphism of the pair. On the class of a cycle $\gamma$ carried by a submanifold of $X$ it is the class of the restriction of the circle bundle $\partial\mathcal{N}$ over $\gamma$. The *residue*

$$\operatorname{Res}\colon H^{k+1}(U,\mathbb{C}) \to H^k(X,\mathbb{C})$$

is the map dual to $\tau$, normalized by

$$\int_\gamma \operatorname{Res}\eta \;=\; \frac{1}{2\pi i}\int_{\tau(\gamma)} \eta \qquad \text{for all } \gamma \in H_k(X,\mathbb{C})$$

The normalization is chosen so that the residue of a form with a first-order pole is computed by the same recipe as a residue in one variable, with the $2\pi i$ already divided out.

**Proposition 7.4.7: The Residue of a Form with a First-Order Pole**

Let $\eta$ be a closed $(k+1)$-form on $U$ for which there is an open cover $\{W_\alpha\}$ of a neighbourhood of $X$ in $\mathbb{CP}^{n+1}$ and expressions

$$\eta\big|_{W_\alpha\cap U} = \frac{dg_\alpha}{g_\alpha}\wedge\psi_\alpha + \theta_\alpha$$

with $g_\alpha$ a holomorphic defining function for $X$ on $W_\alpha$ and $\psi_\alpha, \theta_\alpha$ holomorphic forms on $W_\alpha$. Then the pullbacks of the $\psi_\alpha$ to $X$ agree on overlaps and glue to a closed holomorphic $k$-form $\psi$ on $X$, and

$$\operatorname{Res}\eta = [\psi] \quad \text{in } H^k(X,\mathbb{C})$$

***Proof*:**
*Step 1: the local pieces glue.* On an overlap write $g' = ug$ with $u$ a nowhere-zero holomorphic function, so that $dg'/g' = dg/g + du/u$ and, setting $\rho = \psi - \psi'$,

$$\frac{dg}{g}\wedge\rho = \theta' - \theta + \frac{du}{u}\wedge\psi'$$

The right-hand side is holomorphic on the overlap, so multiplying by $g$ gives $dg\wedge\rho = g\beta$ with $\beta$ holomorphic. At a point of $X$ the right-hand side vanishes, so $dg\wedge\rho = 0$ there; and $dg \neq 0$ at such a point, so $\rho = dg\wedge\lambda$ for some covector-valued $\lambda$. Pulling back to $X$ kills $dg$, hence kills $\rho$, so the pullbacks of $\psi$ and $\psi'$ agree.

*Step 2: the glued form is closed.* Run the same argument on $d\eta = 0$. Since $d(dg_\alpha/g_\alpha) = 0$, expanding $d\eta$ on $W_\alpha \cap U$ gives

$$0 \;=\; -\frac{dg_\alpha}{g_\alpha} \wedge d\psi_\alpha + d\theta_\alpha$$

so multiplying by $g_\alpha$ gives $dg_\alpha \wedge d\psi_\alpha = g_\alpha\, d\theta_\alpha$ on $W_\alpha \cap U$; both sides are holomorphic on $W_\alpha$ and agree off the hypersurface $X$, so the identity holds on all of $W_\alpha$. At a point of $X$ the right-hand side vanishes, and $dg_\alpha \neq 0$ there, so $d\psi_\alpha = dg_\alpha \wedge \lambda$ at that point for some $\lambda$; pulling back to $X$ kills $dg_\alpha$, hence kills $d\psi_\alpha$. Pullback commutes with $d$, so the pullback of $\psi_\alpha$ is closed on $X \cap W_\alpha$ and $\psi$ is closed on $X$; in particular $[\psi] \in H^k(X, \mathbb{C})$ is defined. In the only case used below, $k = n = \dim_{\mathbb{C}} X$, this is automatic: $d\psi$ would be a holomorphic $(n+1)$-form on an $n$-dimensional complex manifold.

*Step 3: the tube integral.* Shrink $\mathcal{N}$ so that over $W_\alpha$ the circle bundle $\partial\mathcal{N}$ is $\{|g_\alpha| = \varepsilon\}$. Let $\gamma$ be a $k$-cycle in $X$, subdivided so that each piece $\gamma_\alpha$ lies in a single $W_\alpha$. The tube over $\gamma_\alpha$ fibres over it in circles on which $g_\alpha$ traverses $|g_\alpha| = \varepsilon$ once positively, so integrating over the fibres first,

$$\int_{\tau(\gamma_\alpha)} \frac{dg_\alpha}{g_\alpha} \wedge \psi_\alpha = \int_{\gamma_\alpha} \left( \oint_{|g_\alpha|=\varepsilon} \frac{dg_\alpha}{g_\alpha} \right) \psi_\alpha = 2\pi i \int_{\gamma_\alpha} \psi_\alpha$$

the inner integral being $2\pi i$ by the Cauchy integral formula. For the remaining term, $\theta_\alpha$ is bounded near $X$ while the $(k+1)$-dimensional volume of the tube over $\gamma_\alpha$ is $O(\varepsilon)$, the tube being a circle bundle of radius $\varepsilon$ over a fixed compact chain; so those contributions tend to 0 as $\varepsilon \to 0$. Summing over $\alpha$ and using Step 1,

$$\int_{\tau(\gamma)} \eta = 2\pi i \int_\gamma \psi + O(\varepsilon)$$

The left-hand side does not depend on $\varepsilon$, because tubes of different radii are homologous in $U$, so the error term is zero. Dividing by $2\pi i$ and comparing with definition 7.4.6 gives $\mathrm{Res}\,\eta = [\psi]$, completing the proof.

For $n = 1$ the recipe already produces the classical objects.

**Example 7.4.8: Holomorphic Differentials on a Smooth Plane Curve**

Let $n = 1$, so $X \subseteq \mathbb{CP}^2$ is a smooth plane curve of degree $d$ defined by $F \in \mathbb{C}[x_0, x_1, x_2]_d$. By proposition 7.4.5 the forms with a first-order pole along $X$ are $A\Omega/F$ with $\deg A = d - 3$.

Work in the chart $x_0 = 1$ with coordinates $x = x_1$, $y = x_2$, in which (7.5) reads $\Omega = dx \wedge dy$; write $f(x, y) = F(1, x, y)$ and $a(x, y) = A(1, x, y)$. Since $df = f_x\, dx + f_y\, dy$, one has $dx \wedge dy = -\frac{df \wedge dx}{f_y}$ wherever $f_y \neq 0$, so on that open set

$$\frac{A\,\Omega}{F} = -\frac{df}{f} \wedge \frac{a\,dx}{f_y}$$

while on $\{f_x \neq 0\}$ the same computation with $x$ and $y$ exchanged gives $\frac{df}{f} \wedge \frac{a\,dy}{f_x}$. These two open sets, together with the analogous pair in each of the charts $x_1 \neq 0$ and $x_2 \neq 0$, cover a neighbourhood of $X$, because smoothness of $F$ puts a nonvanishing partial derivative of the dehomogenization at every point of $X$ (proposition 7.4.2). So proposition 7.4.7 applies and

$$\operatorname{Res}\frac{A\,\Omega}{F} = -\left.\frac{a\,dx}{f_y}\right|_X = \left.\frac{a\,dy}{f_x}\right|_X$$

the two expressions agreeing on the overlap because $f_x\,dx + f_y\,dy = 0$ along $X$. This is the classical holomorphic differential on a plane curve. For the Fermat cubic $f = 1 + x^3 + y^3$ and $A = 1$ it is $-dx/(3y^2)$ where $y \neq 0$ and $dy/(3x^2)$ where $x \neq 0$; the curve has genus 1 by example 6.3.2, and $\dim \mathbb{C}[x]_{d-3} = \dim \mathbb{C}[x]_0 = 1$ counts its holomorphic differentials correctly.

What makes the residue useful is not the local formula but the global statement that it loses nothing: it identifies the cohomology of the complement with exactly the part of the cohomology of $X$ that the family can move.

**Theorem 7.4.9: The Residue Is an Isomorphism onto Primitive Cohomology**

Let $X \subseteq \mathbb{C}\mathrm{P}^{n+1}$ be a smooth hypersurface of degree $d$ with complement $U$, and let $n \geq 1$. Then

$$\operatorname{Res}\colon H^{n+1}(U,\mathbb{C}) \xrightarrow{\ \sim\ } H^n_{\mathrm{prim}}(X,\mathbb{C})$$

is an isomorphism.

***Proof*:**
Write $Y = \mathbb{C}\mathrm{P}^{n+1}$ and $h$ for its hyperplane class, $h_X = h|_X$. Coefficients are in $\mathbb{C}$ throughout, and all homology and cohomology of the spaces in sight is finite-dimensional, so cohomology is the dual of homology.

*Step 1: the pair $(Y,U)$ and the tube map.* Excising the complement of the tubular neighbourhood and retracting $U \cap \mathcal{N}$ onto $\partial\mathcal{N}$ gives $H_{k+2}(Y,U) \cong H_{k+2}(\mathcal{N},\partial\mathcal{N})$. Lefschetz duality on the compact oriented $(2n+2)$-manifold with boundary $\mathcal{N}$ ([24, Poincaré-Lefschetz duality]) gives $H_{k+2}(\mathcal{N},\partial\mathcal{N}) \cong H^{2n-k}(\mathcal{N}) \cong H^{2n-k}(X)$, the last step because $\mathcal{N}$ retracts onto $X$; and Poincaré duality on the closed oriented $2n$-manifold $X$ gives $H^{2n-k}(X) \cong H_k(X)$. The composite is the isomorphism $H_k(X) \cong H_{k+2}(Y,U)$ used in definition 7.4.6, and it sends the class of a cycle $\gamma$ to the class of the disc bundle over $\gamma$, whose boundary is the circle bundle over $\gamma$. So $\tau = \partial \circ (\text{this isomorphism})$, and the homology sequence of the pair reads

$$H_{n+2}(Y) \xrightarrow{\iota} H_n(X) \xrightarrow{\tau} H_{n+1}(U) \to H_{n+1}(Y) \xrightarrow{\iota} H_{n-1}(X) \tag{7.7}$$

where $\iota$ denotes the composite of the map to relative homology with the isomorphism of Step 1.

*Step 2: $\iota$ intersects with $X$.* Let $Z \subseteq Y$ be a cycle carried by a closed submanifold transverse to $X$. Choosing $\mathcal{N}$ small enough, $Z \cap \mathcal{N}$ is a disc bundle over $Z \cap X$, and $Z$ differs from $Z \cap \mathcal{N}$ by a chain in $U$; so the class of $Z$ in $H_\bullet(Y,U)$ is the image of $[Z \cap X]$ under the isomorphism of Step 1. That is, $\iota[Z] = [Z \cap X]$.

*Step 3: the two ends of (7.7).* Both $H_{n+2}(Y)$ and $H_{n+1}(Y)$ are generated by linear subspaces [24]: one of them vanishes and the other is a line, according to the parity of $n$. Suppose first that $n$ is even. Then $H_{n+1}(Y) = 0$, so $\tau$ is surjective, and $H_{n+2}(Y)$ is the line spanned by $[\Lambda]$ for $\Lambda \cong \mathbb{C}\mathrm{P}^{(n+2)/2}$ a linear subspace, which by the parametric transversality theorem [26] may be moved by an element of $\mathrm{PGL}_{n+2}$ so as to meet $X$ transversally. By Step 2, $\iota[\Lambda] = [\Lambda \cap X]$, whose Poincaré dual in $H^n(X)$ is $h_X^{n/2}$, the restriction of the class $h^{n/2}$ of $\Lambda$. Since $\int_X h_X^n =$

$\deg X = d \neq 0$, the class $h_X^{n/2}$ is nonzero; and $\ker\tau = \operatorname{im}\iota$ by exactness of (7.7), so $\ker\tau$ is the line spanned by the Poincaré dual of $h_X^{n/2}$.

Now suppose $n$ is odd. Then $H_{n+2}(Y) = 0$, so $\tau$ is injective. For surjectivity, $H_{n+1}(Y)$ is the line spanned by $[\Lambda]$ with $\Lambda \cong \mathbb{C}\mathrm{P}^{(n+1)/2}$, again taken transverse to $X$, and $\iota[\Lambda] = [\Lambda \cap X]$ has Poincaré dual $h_X^{(n+1)/2}$ in $H^{n+1}(X)$. This is nonzero because $h_X^{(n+1)/2} \smile h_X^{(n-1)/2} = h_X^n$ integrates to $d$. So the last map in (7.7) is injective, and exactness makes $\tau$ surjective.

*Step 4: dualizing.* In both cases $\tau$ is surjective, so the dual map Res is injective, and the image of Res is the annihilator of $\ker\tau$ inside $H^n(X) = H_n(X)^\vee$. For $n$ odd $\ker\tau = 0$ and $H^n(X) = H^n_{\mathrm{prim}}(X)$: indeed corollary 6.2.4 writes $H^n(X) = \bigoplus_{r\geq 0} L^r H^{n-2r}_{\mathrm{prim}}(X)$, and for $r \geq 1$ the group $H^{n-2r}(X)$ vanishes by example 6.3.2, the degree being odd and below $n$. For $n$ even the same decomposition together with example 6.3.2 gives $H^{n-2r}(X) = \mathbb{C}\, h_X^{(n-2r)/2}$ for $r \geq 1$, and $h_X^j$ is not primitive for $j \geq 1$ because $L^{n-2j+1} h_X^j = h_X^{n-j+1}$ pairs to $d$ against $h_X^{j-1}$; so only the terms $r = 0$ and $r = n/2$ survive and

$$H^n(X, \mathbb{C}) = H^n_{\mathrm{prim}}(X, \mathbb{C}) \oplus \mathbb{C}\, h_X^{n/2}$$

The annihilator of $\ker\tau$ is $\{\alpha : \int_X \alpha \smile h_X^{n/2} = 0\}$. A primitive $\alpha$ satisfies $h_X \smile \alpha = 0$ and so lies in it; conversely if $\alpha = \alpha_{\mathrm{prim}} + c\, h_X^{n/2}$ lies in it then $0 = c \int_X h_X^n = cd$, so $c = 0$. Hence the annihilator is exactly $H^n_{\mathrm{prim}}(X, \mathbb{C})$, completing the proof.

The theorem converts a question about $X$ into a question about $U$, where proposition 7.4.5 supplies explicit representatives. What is not yet visible is the Hodge filtration, and the mechanism that produces it is the order of the pole: the deeper the pole, the further down the filtration the residue sits. For that to be well defined, one needs to know when a pole can be reduced, and the answer is exactly the Jacobian ideal.

**Lemma 7.4.10: Reduction of Pole Order by the Jacobian Ideal**

Let $X = X_F$ be a smooth hypersurface of degree $d$, let $m \geq 2$, and let $A_0, \ldots, A_{n+1} \in \mathbb{C}[x]_a$ with $a = (m-1)d - n - 1$. Write $v = \sum_i A_i\, \partial_i$. Then on $U$

$$\frac{\left(\sum_i A_i\, \partial_i F\right)\Omega}{F^m} = \frac{1}{m-1}\, \frac{\left(\sum_i \partial_i A_i\right)\Omega}{F^{m-1}} - \frac{1}{m-1}\, d\!\left(\frac{\iota_v \Omega}{F^{m-1}}\right)$$

where $\iota_v\Omega/F^{m-1}$ is a holomorphic $n$-form on $U$. In particular, if $A \in J(F)_{md-n-2}$ then the class of $A\Omega/F^m$ in $H^{n+1}(U, \mathbb{C})$ is represented by a form with a pole of order at most $m-1$.

***Proof*:**

All computations take place on $\mathbb{C}^{n+2} \setminus \{F = 0\}$, and the result is transported to $U$ at the end.

*Step 1: two Cartan identities.* Write $\mu = dx_0 \wedge \cdots \wedge dx_{n+1}$ and $E = \sum_i x_i \partial_i$, so that $\Omega = \iota_E \mu$. For any polynomial vector field $w$ one has $\mathcal{L}_w \mu = (\operatorname{div} w)\,\mu$ with $\operatorname{div} w = \sum_i \partial_i w_i$, and Cartan's formula $\mathcal{L}_w = d\iota_w + \iota_w d$ [26] gives $d(\iota_w \mu) = (\operatorname{div} w)\mu$ because $d\mu = 0$. Applying $\mathcal{L}_v$ to $\Omega = \iota_E \mu$ and using $\mathcal{L}_v \iota_E = \iota_{[v,E]} + \iota_E \mathcal{L}_v$,

$$d(\iota_v \Omega) = \mathcal{L}_v(\iota_E \mu) - \iota_v\, d(\iota_E \mu) = \iota_{[v,E]}\mu + (\operatorname{div} v)\,\Omega - (n+2)\,\iota_v \mu$$

since $\mathrm{div}E = n+2$. The coefficients of $v$ are homogeneous of degree $a$, so $[E,v] = (a-1)v$ and $\iota_{[v,E]}\mu = (1-a)\iota_v\mu$, whence

$$d(\iota_v\Omega) = (1-a-n-2)\,\iota_v\mu + (\mathrm{div}v)\,\Omega \tag{7.8}$$

Next, $dF \wedge \mu = 0$ for degree reasons, so contracting with $E$ gives $dF \wedge \Omega = (EF)\,\mu = d\cdot F\,\mu$ by the Euler relation; contracting that with $v$ gives

$$dF \wedge \iota_v\Omega = v(F)\,\Omega - d\cdot F\,\iota_v\mu, \qquad v(F) = \sum_i A_i\,\partial_i F \tag{7.9}$$

*Step 2: the identity.* Differentiating the quotient and inserting (7.8) and (7.9),

$$d\left(\frac{\iota_v\Omega}{F^{m-1}}\right) = \frac{d(\iota_v\Omega)}{F^{m-1}} - (m-1)\frac{dF\wedge\iota_v\Omega}{F^m} = \frac{\big(1-a-n-2+(m-1)d\big)\,\iota_v\mu}{F^{m-1}} + \frac{(\mathrm{div}v)\Omega}{F^{m-1}} - (m-1)\frac{v(F)\,\Omega}{F^m}$$

The bracketed scalar is $1-\big((m-1)d-n-1\big)-n-2+(m-1)d = 0$, so the terms involving $\iota_v\mu$ cancel and rearranging gives the asserted identity.

*Step 3: descent to $U$.* Let $\pi\colon \mathbb{C}^{n+2}\setminus\{0\} \to \mathbb{CP}^{n+1}$ be the projection. A holomorphic form $\zeta$ is a pullback along $\pi$ if and only if $\iota_E\zeta = 0$ and $\mathcal{L}_E\zeta = 0$: the fibres of $\pi$ are the orbits of the $\mathbb{C}^*$-action generated by $E$, and the two conditions say that $\zeta$ annihilates the fibre directions and is invariant along them. Now $\iota_E\iota_v\Omega = -\iota_v\iota_E\Omega = 0$ because $\iota_E\Omega = \iota_E\iota_E\mu = 0$; and $\mathcal{L}_E\Omega = (n+2)\Omega$ together with $[E,v] = (a-1)v$ gives $\mathcal{L}_E(\iota_v\Omega) = (a-1+n+2)\,\iota_v\Omega$, so

$$\mathcal{L}_E\left(\frac{\iota_v\Omega}{F^{m-1}}\right) = \big(a-1+n+2-(m-1)d\big)\frac{\iota_v\Omega}{F^{m-1}} = 0$$

by the choice of $a$. So $\iota_v\Omega/F^{m-1}$ descends to a holomorphic $n$-form on $U$, both other terms of the identity descend by proposition 7.4.5, and the identity holds on $U$.

Finally, an element $A \in J(F)_{md-n-2}$ is of the form $\sum_i A_i\partial_i F$ with $A_i$ homogeneous of degree $md-n-2-(d-1) = a$, so the identity writes $A\Omega/F^m$ as a form of pole order $m-1$ plus an exact form, completing the proof.

The lemma already delivers half of the description. For each $m$ let

$$P_m \;=\; \left\{\left[\frac{A\,\Omega}{F^m}\right] \;:\; A \in \mathbb{C}[x]_{md-n-2}\right\} \;\subseteq\; H^{n+1}(U,\mathbb{C})$$

be the set of classes represented by a form of pole order at most $m$, with the convention $P_0 = 0$; these are subspaces, and $P_1 \subseteq P_2 \subseteq \cdots$ because $A\Omega/F^m = (FA)\Omega/F^{m+1}$. Lemma 7.4.10 says that $A \mapsto [A\Omega/F^m]$ carries $J(F)_{md-n-2}$ into $P_{m-1}$ for $m \geq 2$, and the same holds vacuously at $m = 1$, where $J(F)_{d-n-2} = 0$ because $J(F)$ starts in degree $d-1$. So the assignment induces a surjection

$$R_{md-n-2} \;\twoheadrightarrow\; P_m/P_{m-1} \tag{7.10}$$

Griffiths' theorem is the assertion that (7.10) is an isomorphism and that the filtration $P_\bullet$ is the Hodge filtration read through the residue. Two inputs from algebraic geometry make it work, and both are imported rather than rebuilt: Grothendieck's theorem that a smooth affine variety's cohomology is computed by global algebraic differential forms [4, Grothendieck's comparison theorem for smooth affine varieties], applied to the complement $U = \mathbb{CP}^{n+1}\setminus X$, which is affine; and the vanishing of

$H^q(\mathbb{CP}^{n+1}, \Omega^p(k))$ in the middle degrees [4, Bott vanishing], which is what collapses the pole-order filtration onto the graded pieces of the Jacobian ring. Both were added to [4] for this purpose; the assembly below is where they are used, and the full argument in this form is in *Period Mappings and Period Domains* [1, the cohomology of hypersurfaces].

**Theorem 7.4.11: Griffiths' Description of the Hodge Filtration**

Let $X = X_F \subseteq \mathbb{CP}^{n+1}$ be a smooth hypersurface of degree $d \geq 2$, with $n \geq 1$, and let $R = R(F)$ be its Jacobian ring. Then

1. $\bigcup_{m \geq 1} P_m = H^{n+1}(U, \mathbb{C})$;
2. $\operatorname{Res}(P_m) = F^{n+1-m} H^n_{\text{prim}}(X, \mathbb{C})$ for every $m \geq 1$, and in particular $P_{n+1} = H^{n+1}(U, \mathbb{C})$;
3. for every $m \geq 1$ the map $A \mapsto \operatorname{Res}[A\Omega/F^m]$ induces an isomorphism of vector spaces

$$R_{md-n-2} \xrightarrow{\sim} \operatorname{gr}_F^{n+1-m} H^n_{\text{prim}}(X, \mathbb{C}) \;=\; H^{n+1-m,\, m-1}_{\text{prim}}(X)$$

Equivalently, writing $m = q+1$, for every $q$ with $0 \leq q \leq n$,

$$H^{n-q,\, q}_{\text{prim}}(X) \;\cong\; R_{(q+1)d-n-2} \tag{7.11}$$

***Proof Key ideas*:**
What is proved here in full is the surjection (7.10), which is lemma 7.4.10: the assignment $A \mapsto [A\Omega/F^m]$ is linear, its restriction to $J(F)_{md-n-2}$ lands in $P_{m-1}$ by that lemma, and it is surjective onto $P_m$ by the definition of $P_m$ together with proposition 7.4.5. Since Res is injective (theorem 7.4.9), statement (3) is equivalent to the assertion that (7.10) is injective and that Res carries $P_m/P_{m-1}$ onto $\operatorname{gr}_F^{n+1-m}$, which is what (2) provides.

Two inputs are taken from outside. The first is *Grothendieck's algebraic de Rham theorem*: for a smooth affine variety $V$ over $\mathbb{C}$, the complex of global algebraic differential forms on $V$ computes $H^\bullet(V, \mathbb{C})$. Applied to the affine variety $U$ it gives statement (1), since a global algebraic $(n+1)$-form on $U$ extends meromorphically across $X$ with a pole of some finite order and is therefore some $A\Omega/F^m$ by proposition 7.4.5. Granting (2), its case $m = n+1$ reads $\operatorname{Res}(P_{n+1}) = F^0 H^n_{\text{prim}} = H^n_{\text{prim}}$, and injectivity of Res then forces $P_{n+1} = H^{n+1}(U, \mathbb{C})$.

The second input is the identification of the pole filtration with the Hodge filtration, statement (2), together with the injectivity of (7.10). The proof compares the filtration of the algebraic de Rham complex of $U$ by order of pole with its filtration by degree of form; the two induce the same filtration on cohomology, and the comparison rests on the vanishing $H^q(\mathbb{CP}^{n+1}, \Omega^p_{\mathbb{CP}^{n+1}}(k)) = 0$ for $q \geq 1$ and $k > 0$. That vanishing, and the spectral-sequence bookkeeping that consumes it, are not developed in this book; the argument is given in full in [1, the cohomology of hypersurfaces].

The final equivalence is bookkeeping. The Hodge structure on $H^n_{\text{prim}}(X)$ has weight $n$, so $\operatorname{gr}_F^{n+1-m} = H^{n+1-m,\, n-(n+1-m)} = H^{n+1-m,\, m-1}$, and substituting $m = q+1$ turns this into (7.11), as we sought to show.

The formula (7.11) is a complete answer to a question that on the face of it requires solving an elliptic equation on $X$. The Hodge decomposition of chapter 4 produces $H^{p,q}$ as a space of harmonic forms, defined by a metric and computed by analysis; here the same space appears as a graded piece of a quotient of a polynomial ring, defined by the equation of $X$ and computed by counting monomials. Two features of the answer are worth naming. Hodge symmetry $h^{p,q} = h^{q,p}$ becomes the symmetry of the graded dimensions of $R$ about the middle degree, because $\sigma - \big((q+1)d-n-2\big) = (n-q+1)d-n-2$ when $\sigma = (n+2)(d-2)$. And the Hodge numbers of $X$ depend on $F$ only through $R(F)$, so two hypersurfaces with isomorphic graded Jacobian rings have the same Hodge numbers. The rational structure of $H^n(X, \mathbb{Q})$ is invisible in $R(F)$, so this says nothing about the Hodge structures themselves being isomorphic.

Combining theorem 7.4.11 with the local constancy supplied by theorem 7.2.6 turns the single Fermat computation of example 7.4.3 into a formula for every smooth hypersurface.

**Corollary 7.4.12: The Hodge Numbers of a Smooth Hypersurface**

Let $X \subseteq \mathbb{CP}^{n+1}$ be any smooth hypersurface of degree $d \geq 2$, with $n \geq 1$. Then for $0 \leq q \leq n$,

$$h^{n-q,\,q}_{\text{prim}}(X) \;=\; \#\Big\{(a_0,\ldots,a_{n+1}) \;:\; 0 \leq a_i \leq d-2,\; \sum_i a_i = (q+1)d-n-2\Big\}$$

that is, the coefficient of $t^{(q+1)d-n-2}$ in $(1+t+\cdots+t^{d-2})^{n+2}$. In particular these numbers depend only on $n$, $d$ and $q$.

***Proof*:**
Let $S$ be the parameter space of proposition 7.4.4 and $F^\bullet \subseteq \mathcal{H}^n_{\text{prim}}$ the Hodge filtration of the associated variation. Part (3) of proposition 7.3.2 makes each $F^p$ a holomorphic subbundle of $\mathcal{H}^n_{\text{prim}}$ (note that theorem 7.2.6 is stated for the full bundle $\mathcal{H}^n$ and does not by itself cover the primitive filtration), and part (2) of the same proposition already gives that $h^{p,q}_{\text{prim}}$ is independent of $s$. It therefore suffices to evaluate the numbers at one point of $S$, and the Fermat point is one. There theorem 7.4.11 gives $h^{n-q,q}_{\text{prim}} = \dim R_{(q+1)d-n-2}$, which example 7.4.3 computes as the stated monomial count, completing the proof.

**Example 7.4.13: The Quintic Threefold**

Take $n = 3$ and $d = 5$: a smooth quintic hypersurface $X \subseteq \mathbb{CP}^4$. Here $n$ is odd, so $H^3(X,\mathbb{C}) = H^3_{\text{prim}}(X,\mathbb{C})$ by the computation in Step 4 of the proof of theorem 7.4.9, and the relevant degrees are $(q+1)\cdot 5 - 5 = 5q$ for $q = 0,1,2,3$. By corollary 7.4.12 each Hodge number is a count of $(a_0,\ldots,a_4)$ with $0 \leq a_i \leq 3$ and prescribed sum.

*Degree* 0. Only $a = 0$, so $h^{3,0} = 1$: the quintic threefold carries a one-dimensional space of holomorphic 3-forms, and by proposition 7.4.5 a generator is $\operatorname{Res}(\Omega/F)$.

*Degree* 5. Without the upper bound there are $\binom{5+4}{4} = 126$ solutions of $\sum a_i = 5$ in nonnegative integers. Those violating the bound have $a_i \geq 4$ for exactly one $i$, since two such entries would already force $\sum a_i \geq 8$; subtracting 4 from that entry leaves $\sum a_i = 1$, with $\binom{1+4}{4} = 5$ solutions, and there are 5 choices of $i$. Hence

$$h^{2,1} = 126 - 25 = 101$$

*Degrees* 10 *and* 15. The socle degree is $\sigma = (n+2)(d-2) = 15$, and the symmetry $a \mapsto (3,3,3,3,3) - a$ of example 7.4.3 gives $h^{1,2} = h^{2,1} = 101$ and $h^{0,3} = h^{3,0} = 1$, as Hodge symmetry requires.

So the Hodge numbers of the middle cohomology are $(1, 101, 101, 1)$ and $b_3(X) = 204$. The period map of definition 7.3.3 for the family of quintics therefore takes values in the period domain of section 5.3 for polarized weight-3 Hodge structures with these Hodge numbers, and the number 101 reappears as the dimension of the space of quintics modulo projective changes of coordinates (exercise 7.4.1 (5)).

One piece of the chapter has not yet been made explicit for this family: the derivative. Griffiths transversality says that $\nabla$ moves $F^p$ into $F^{p-1}$, and corollary 7.2.8 extracts from it an $\mathcal{O}_S$-linear map on the graded pieces, which is the differential of the period map. Under (7.11) both source and target are graded pieces of $R$, and the map between them is multiplication in that ring.

**Theorem 7.4.14: The Differential of the Period Map as Multiplication**

Let $[F] \in S$ and let $G \in \mathbb{C}[x]_d$ represent a tangent vector to $\mathbb{CP}^M$ at $[F]$. Under the isomorphisms (7.11), the map

$$\bar{\nabla}_G \colon H^{n-q,\,q}_{\text{prim}}(X) \to H^{n-q-1,\,q+1}_{\text{prim}}(X)$$

obtained from corollary 7.2.8 by contracting with $G$ is

$$[A] \;\longmapsto\; -(q+1)\,[G \cdot A]$$

multiplication by $G$ in the graded ring $R$, up to the nonzero scalar $-(q+1)$. In particular $\bar{\nabla}_G$ depends only on the class of $G$ in $R_d$, and vanishes when $G \in J(F)_d$.

***Proof*:**
*Step 1: the residue is flat.* Let $\mathcal{U} \to S$ be the family of complements, $\mathcal{U} = (\mathbb{CP}^{n+1} \times S) \setminus \mathcal{X}$. By Step 3 of proposition 7.4.4 the pair $(\mathbb{CP}^{n+1}, X_F)$ is locally trivial over $S$, so both $\mathcal{X} \to S$ and $\mathcal{U} \to S$ are locally trivial fibre bundles and the tube map of definition 7.4.6 is defined fibrewise by a construction that a trivialization of the pair carries along. So $t \mapsto \tau_t$ is a morphism of local systems from $t \mapsto H_n(X_t, \mathbb{C})$ to $t \mapsto H_{n+1}(U_t, \mathbb{C})$, and dually Res is a morphism of local systems from $t \mapsto H^{n+1}(U_t, \mathbb{C})$ to $\mathbb{H}^n \otimes \mathbb{C}$, an isomorphism onto $\mathbb{H}^n_{\text{prim}} \otimes \mathbb{C}$ by theorem 7.4.9. In particular Res commutes with parallel transport, and it carries flat families of tubes to flat families. By theorem 7.1.7 the flat sections of $\nabla$ are the sections of the local system, so in a flat frame $\nabla$ is differentiation of coefficients, and for any flat family of cycles $\gamma_t$ and any section $\sigma$ of $\mathcal{H}^n_{\text{prim}}$,

$$\frac{d}{dt}\big\langle \sigma(t), \gamma_t \big\rangle = \big\langle \nabla_{\partial_t}\sigma, \gamma_t \big\rangle \tag{7.12}$$

*Step 2: differentiating a rational representative.* Let $F_t = F + tG$, which lies in $S$ for small $t$, and fix $A \in \mathbb{C}[x]_{(q+1)d-n-2}$. Put

$$\eta_t = \frac{A\,\Omega}{F_t^{\,q+1}}, \qquad \sigma(t) = \operatorname{Res}[\eta_t]$$

a holomorphic section of $\mathcal{H}^n_{\text{prim}}$ along the curve, whose value at $t = 0$ is the class corresponding to $[A] \in R_{(q+1)d-n-2}$. Choose a flat family of cycles $\gamma_t \in H_n(X_t, \mathbb{C})$, so that $\tau(\gamma_t)$ is a flat family in $H_{n+1}(U_t, \mathbb{C})$ by Step 1. Trivializing the family of complements over a disc by a diffeomorphism

$\varphi_t$, the tube may be taken to be a fixed cycle $c$, and

$$\frac{d}{dt}\int_{\tau(\gamma_t)}\eta_t = \frac{d}{dt}\int_c \varphi_t^*\eta_t = \int_c \varphi_t^*\Big(\frac{\partial \eta_t}{\partial t}\Big) + \int_c \varphi_t^*(\mathcal{L}_w\eta_t)$$

where $w$ is the vector field generating $\varphi_t$. Each $\eta_t$ is closed, so $\mathcal{L}_w\eta_t = d(\iota_w\eta_t)$ is exact and the second integral vanishes. Hence differentiation passes under the integral sign, and

$$\frac{\partial \eta_t}{\partial t}\bigg|_{t=0} = -(q+1)\frac{A\,G\,\Omega}{F^{\,q+2}}$$

by the chain rule applied to $F_t^{-(q+1)}$.

*Step 3: conclusion.* Combining Step 2 with (7.12) and the definition of Res,

$$\big\langle \nabla_{\partial_t}\sigma, \gamma_0\big\rangle = \frac{1}{2\pi i}\int_{\tau(\gamma_0)} -(q+1)\frac{A\,G\,\Omega}{F^{\,q+2}} = \Big\langle \mathrm{Res}\Big[-(q+1)\frac{(GA)\,\Omega}{F^{\,q+2}}\Big], \gamma_0\Big\rangle$$

for every $\gamma_0$, so $\nabla_{\partial_t}\sigma = -(q+1)\mathrm{Res}[(GA)\Omega/F^{q+2}]$. The right-hand side lies in $\mathrm{Res}(P_{q+2}) = F^{n-q-1}H^n_{\mathrm{prim}}$ by theorem 7.4.11, which is Griffiths transversality made explicit, and its class modulo $F^{n-q}$ is $-(q+1)$ times the class of $GA$ in $R_{(q+2)d-n-2}$, again by theorem 7.4.11. Since $\bar{\nabla}_G$ is exactly the induced map on these graded pieces, the formula follows. If $G \in J(F)_d$ then $GA \in J(F)$, so $[GA] = 0$ and $\bar{\nabla}_G = 0$, completing the proof.

The vanishing in the last sentence is forced. An element of $J(F)_d$ is $\sum_i \ell_i\,\partial_i F$ with $\ell_i$ linear, which is the derivative of $F$ along the linear vector field $\sum_i \ell_i\partial_i$: it is a tangent direction along the orbit of $F$ under $\mathrm{PGL}_{n+2}$, and moving $F$ in such a direction changes the coordinates rather than the hypersurface. So $d\Phi$ annihilates exactly the directions in which nothing happens, and what the period map for hypersurfaces records is the multiplication

$$R_d \otimes R_{(q+1)d-n-2} \to R_{(q+2)d-n-2}$$

of the Jacobian ring. This is also the sharpest form of transversality available here: multiplication by an element of $R_d$ raises the degree by $d$, which is one step in the numbering (7.11), and never two. Whether $d\Phi$ is injective, a question about that multiplication, is the infinitesimal Torelli problem; exercise 7.4.1 (6) settles it for the quintic threefold, and the global question is taken up for K3 surfaces in chapter 8.

## Exercise 7.4.1

1. Let $X \subseteq \mathbb{CP}^3$ be a smooth cubic surface, so $n = 2$ and $d = 3$. Compute the graded dimensions of the Jacobian ring of the Fermat cubic and use corollary 7.4.12 to determine $h^{2,0}(X)$, $h^{1,1}_{\mathrm{prim}}(X)$ and $h^{0,2}(X)$. Deduce $b_2(X)$, and say what the vanishing of $h^{2,0}$ means about holomorphic 2-forms on $X$.

2. Repeat the computation for a smooth quartic surface in $\mathbb{CP}^3$ ($n = 2$, $d = 4$) and check the result against the Hodge numbers of a K3 surface recorded in example 4.3.4. Which degree of the Jacobian ring carries the holomorphic 2-form, and which carries the primitive $(1,1)$-classes?

3. Show that $h^{n,0}(X) = \binom{d-1}{n+1}$ for every smooth hypersurface $X \subseteq \mathbb{CP}^{n+1}$ of degree $d \geq 2$, without computing any Jacobian ring beyond its lowest degrees. Deduce that $X$ carries no nonzero holomorphic $n$-form exactly when $d \leq n+1$.

4. Let $X \subseteq \mathbb{C}\mathrm{P}^2$ be a smooth plane curve of degree $d$. Using exercise 7.4.1 (3), recover $g = (d-1)(d-2)/2$, and write down a basis of $H^0(X, \Omega^1_X)$ in the affine chart of example 7.4.8. Explain why example 7.2.9 makes Griffiths transversality vacuous here, and say which statement of this section still has content in weight one.

5. For the Fermat quintic threefold $F = x_0^5 + \cdots + x_4^5$, compute $\dim_{\mathbb{C}} J(F)_5$ directly from the monomial description of $J(F)$, and deduce $\dim R_5$ a second way, independently of the count in example 7.4.13. Compare $\dim R_5$ with $\dim \mathbb{C}\mathrm{P}^M - \dim \mathrm{PGL}_5$, and explain the coincidence using theorem 7.4.14.

6. Continuing with the quintic threefold, show that the composite
$$R_5 \to \mathrm{Hom}\big(H^{3,0}_{\mathrm{prim}}(X),\ H^{2,1}_{\mathrm{prim}}(X)\big)$$
of theorem 7.4.14 is injective at the Fermat point, and conclude that the period map of the family of quintics has injective differential there, modulo the directions along the $\mathrm{PGL}_5$-orbit.

# 8

# Period Domains and Mumford-Tate Groups

Chapter 7 put a family into a period domain and showed the resulting map is holomorphic and horizontal. Neither adjective has yet been cashed. Horizontality was a constraint on a derivative, and a constraint is only useful once something is known about the space it constrains motion in. That is the first business of this chapter.

Period domains are homogeneous, so their geometry is determined at a single point, and section 5.3 already recorded that. What it did not record is the curvature, and the curvature is peculiar. A period domain is negatively curved in the horizontal directions and not, in general, anywhere else: it is almost never a Hermitian symmetric space, and the familiar picture of a bounded domain with a complete negatively curved metric is available only in weight one, where $D$ is the Siegel upper half space. What survives in general is exactly enough. Horizontal maps are distance-decreasing, and that alone forces rigidity.

The second business is a group. A single Hodge structure carries more information than its Hodge numbers, and the invariant that captures it is the Mumford-Tate group: the smallest algebraic group over $\mathbb{Q}$ whose real points contain the circle action defining the structure. Its representation theory answers a question the earlier chapters could only pose. The Hodge classes in every tensor construction on $V$ are precisely the vectors it fixes, so the search for Hodge classes becomes the search for invariants of a reductive group.

Those two threads meet in the locus where a family acquires an unexpected Hodge class. Griffiths transversality makes that locus analytic, and a theorem of Cattani, Deligne and Kaplan makes it algebraic. The chapter closes with the case where every object in it can be written down: polarized K3 surfaces, whose period domain is again Hermitian symmetric, whose lattice is a fixed unimodular form of signature $(3, 19)$, and for which the period map is injective.

## 8.1 Curvature and Rigidity of Period Domains

Proposition 5.3.5 presents $D$ as $G_{\mathbb{R}}/(G_{\mathbb{R}} \cap P)$ with compact isotropy, and definition 7.3.8 singles out the subbundle $T^hD \subseteq TD$ that every period map is tangent to. Neither statement says anything metric. This section supplies the metric: the polarization produces a $G_{\mathbb{R}}$-invariant Hermitian metric on $D$, and the curvature of that metric is negative along $T^hD$ and not, in general, anywhere else.

A map into a negatively curved target is constrained by the Ahlfors-Schwarz lemma; a horizontal map into $D$ therefore inherits those constraints even though the negativity is unavailable in the remaining directions. The consequences are rigidity statements about families, and the first of them says that a smooth projective family over $\mathbb{CP}^n$ cannot move at all.

Throughout, $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$ is the numerical data fixed in section 5.3, and $G_{\mathbb{C}}$ and $G_{\mathbb{R}}$ are the automorphism groups of $Q$ from proposition 5.3.5. For a fixed $F \in D$ the letter $P$ denotes the stabilizer of $F$ in $G_{\mathbb{C}}$ and $K = G_{\mathbb{R}} \cap P$ the (compact) isotropy group. Write

$$\mathfrak{g}_{\mathbb{C}} = \{X \in \operatorname{End}(V_{\mathbb{C}}) \big| Q(Xu, v) + Q(u, Xv) = 0 \text{ for all } u, v \in V_{\mathbb{C}}\}$$

for the Lie algebra of $G_{\mathbb{C}}$, and $\mathfrak{g}_{\mathbb{R}}$, $\mathfrak{g}_{\mathbb{Q}}$ for the analogous spaces of real and rational endomorphisms, so that $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$. Finally $C_F$ denotes the Weil operator of the Hodge structure at $F$, acting on $V_F^{p,q}$ as $i^{p-q}$ (definition 5.2.1).

Everything below is computed inside $\mathfrak{g}$. The reason is that the Hodge structure carried by $V$ propagates to a Hodge structure on $\mathfrak{g}$, and once it does, the tangent bundle of $D$, its horizontal subbundle and its invariant metrics all become statements about a single graded Lie algebra.

**Proposition 8.1.1: The Lie Algebra of a Point of the Period Domain**

Fix $F \in D$ with bigrading $V^{p,q} = F^p \cap \overline{F^q}$, and for $r \in \mathbb{Z}$ set

$$\mathfrak{g}_F^{r,-r} = \left\{X \in \mathfrak{g}_{\mathbb{C}} \;\middle|\; X\,V^{p,q} \subseteq V^{p+r,\,q-r} \text{ for all } p, q\right\}$$

Then:

1. $\mathfrak{g}_{\mathbb{C}} = \bigoplus_{r \in \mathbb{Z}} \mathfrak{g}_F^{r,-r}$, with $\overline{\mathfrak{g}_F^{r,-r}} = \mathfrak{g}_F^{-r,r}$ and $[\mathfrak{g}_F^{r,-r}, \mathfrak{g}_F^{s,-s}] \subseteq \mathfrak{g}_F^{r+s,-r-s}$. In particular this is a pure Hodge structure of weight 0 on $\mathfrak{g}_{\mathbb{Q}}$ in the sense of definition 5.1.1, and the bracket is a morphism of Hodge structures.
2. $\operatorname{Lie}(P) = \mathfrak{p}_F := \bigoplus_{r \geq 0} \mathfrak{g}_F^{r,-r}$ and $\operatorname{Lie}(K) = \mathfrak{g}_{\mathbb{R}} \cap \mathfrak{g}_F^{0,0}$.
3. The differential at the identity of the orbit map $g \mapsto gF$ induces $\mathbb{C}$-linear isomorphisms

$$T_F^{1,0}\check{D} \cong \mathfrak{g}_{\mathbb{C}}/\mathfrak{p}_F \cong \bigoplus_{r<0} \mathfrak{g}_F^{r,-r}$$

under which the horizontal subspace of definition 7.3.8 is

$$T_F^h\check{D} \cong \mathfrak{g}_F^{-1,1}$$

***Proof*:**
Write $\mathfrak{g}^{r,-r}$ for $\mathfrak{g}_F^{r,-r}$ and $\mathfrak{p}$ for $\mathfrak{p}_F$.

1. Decompose $\mathrm{End}(V_\mathbb{C}) = \bigoplus_r E^r$, where $E^r = \bigoplus_{p,q} \mathrm{Hom}(V^{p,q}, V^{p+r,q-r})$; this is the decomposition of an endomorphism into its homogeneous components for the bigrading, and it is a direct sum because the $V^{p,q}$ span $V_\mathbb{C}$. Let $X \in \mathfrak{g}_\mathbb{C}$ and write $X = \sum_r X_r$ with $X_r \in E^r$. Fix $u \in V^{p,q}$ and $w \in V^{p',q'}$. By condition (1) of definition 5.2.1 the form $Q$ pairs $V^{a,b}$ nontrivially only with $V^{b,a}$, so $Q(X_r u, w) = 0$ unless $(p', q') = (q - r, p + r)$, that is unless $r = q - p'$; and $Q(u, X_s w) = 0$ unless $(p' + s, q' - s) = (q, p)$, again unless $s = q - p'$. Setting $r_0 = q - p'$, the identity $Q(Xu, w) + Q(u, Xw) = 0$ therefore reads
$$Q(X_{r_0} u, w) + Q(u, X_{r_0} w) = 0$$
while for $r \neq r_0$ both terms $Q(X_r u, w)$ and $Q(u, X_r w)$ vanish individually. As $u$ and $w$ range over the bigraded pieces this shows $Q(X_r u, w) + Q(u, X_r w) = 0$ for every $r$ and every pair $u, w$, so each $X_r$ lies in $\mathfrak{g}_\mathbb{C}$. Hence $\mathfrak{g}_\mathbb{C} = \bigoplus_r \mathfrak{g}^{r,-r}$.

   For conjugation, let $X \in \mathfrak{g}^{r,-r}$ and $u \in V^{p,q}$. Then $\bar{u} \in V^{q,p}$ and $\bar{X}(\bar{u}) = \overline{Xu} \in \overline{V^{p+r,q-r}} = V^{q-r,p+r}$, so $\bar{X}$ carries $V^{q,p}$ into $V^{q-r,p+r}$ and thus lies in $\mathfrak{g}^{-r,r}$. That $\bar{X} \in \mathfrak{g}_\mathbb{C}$ is automatic because $Q$ is defined over $\mathbb{Q}$. The bracket statement is immediate: a composite $XY$ with $X \in \mathfrak{g}^{r,-r}$ and $Y \in \mathfrak{g}^{s,-s}$ shifts the bigrading by $r + s$, and so does $YX$.

   Finally $\mathfrak{g}_\mathbb{Q}$ is a $\mathbb{Q}$-vector space with $\mathfrak{g}_\mathbb{Q} \otimes_\mathbb{Q} \mathbb{C} = \mathfrak{g}_\mathbb{C}$, and the displayed decomposition satisfies exactly the conjugation condition of definition 5.1.1 in weight 0, the summand $\mathfrak{g}^{r,-r}$ playing the role of the $(r, -r)$-piece. The bracket is a morphism of Hodge structures (definition 5.1.6) because it is defined over $\mathbb{Q}$ and respects the bigrading by the previous sentence.

2. An element $X \in \mathfrak{g}_\mathbb{C}$ lies in $\mathrm{Lie}(P)$ precisely when $XF^p \subseteq F^p$ for every $p$, since $P$ is the subgroup of $G_\mathbb{C}$ fixing every step of the flag and a one-parameter subgroup preserves a subspace exactly when its generator does. Now $F^p = \bigoplus_{a \geq p} V^{a,k-a}$, so for $X \in \mathfrak{g}^{s,-s}$ one has $XF^p \subseteq F^{p+s}$, and $F^{p+s} \subseteq F^p$ for all $p$ if and only if $s \geq 0$. Since $\mathrm{Lie}(P)$ is a sum of homogeneous components by part (1), this gives $\mathrm{Lie}(P) = \bigoplus_{s \geq 0} \mathfrak{g}^{s,-s} = \mathfrak{p}$.

   For the isotropy, $\mathrm{Lie}(K) = \mathfrak{g}_\mathbb{R} \cap \mathfrak{p}$. If $X$ is real and lies in $\mathfrak{p}$ then $\bar{X} = X$ lies in $\bar{\mathfrak{p}} = \bigoplus_{s \leq 0} \mathfrak{g}^{s,-s}$ as well, so $X \in \mathfrak{p} \cap \bar{\mathfrak{p}} = \mathfrak{g}^{0,0}$. Conversely $\mathfrak{g}_\mathbb{R} \cap \mathfrak{g}^{0,0} \subseteq \mathfrak{g}_\mathbb{R} \cap \mathfrak{p}$.

3. By proposition 5.3.5 the orbit map $G_\mathbb{C} \to \check{D}$ induces a biholomorphism $G_\mathbb{C}/P \to \check{D}$; its differential at the identity coset is the standard isomorphism $\mathfrak{g}_\mathbb{C}/\mathfrak{p} \cong T_F^{1,0}\check{D}$ for a complex homogeneous space ([32, homogeneous spaces]), $\mathbb{C}$-linear because the orbit map is holomorphic. Composing with the splitting $\mathfrak{g}_\mathbb{C} = \mathfrak{p} \oplus \bigoplus_{r<0} \mathfrak{g}^{r,-r}$ of part (1) gives the second isomorphism.

   Concretely, the tangent vector attached to $X \in \bigoplus_{r<0} \mathfrak{g}^{r,-r}$ is $\frac{d}{dt}\big|_0 \exp(tX)F$. Its $p$-th component in the sense of definition 7.3.8 is computed on a local section: for $u \in F^p$ the curve $t \mapsto \exp(tX)u$ is a section of the moving subspace $\exp(tX)F^p$ with derivative $Xu$ at $t = 0$, so
$$\theta_p(u) = Xu \bmod F^p$$

Horizontality asks $\theta_p(F^p) \subseteq F^{p-1}/F^p$ for every $p$, that is $XF^p \subseteq F^{p-1}$ for every $p$. Write $X = \sum_{r<0} X_r$. If $X_r = 0$ for all $r \leq -2$ then $X = X_{-1}$ and $XF^p \subseteq F^{p-1}$ for every $p$, so $X$ is horizontal. Conversely suppose $X_{r_0} \neq 0$ for some $r_0 \leq -2$, and choose a bidegree $(p,q)$ and a vector $u \in V^{p,q}$ with $X_{r_0}u \neq 0$. The vectors $X_r u$ lie in the distinct summands $V^{p+r,q-r}$, and $F^{p-1} = \bigoplus_{a \geq p-1} V^{a,k-a}$ is a sum of summands, so $Xu \in F^{p-1}$ would force $X_r u \in F^{p-1}$ for each $r$ separately, hence $p + r_0 \geq p-1$ and $r_0 \geq -1$. That contradicts $r_0 \leq -2$, so $X$ is not horizontal. Hence $T^h_F\check{D}$ corresponds to $\mathfrak{g}^{-1,1}$, as we sought to show.

The proposition converts three separate constructions into one grading. The tangent bundle of $\check{D}$ is the negative part of $\mathfrak{g}$, the horizontal subbundle is its lowest graded piece, and the isotropy algebra is the piece of degree zero. The condition $T^h\check{D} = T\check{D}$, which example 7.3.9 verified by hand in weight one and in the K3 numerical type, now reads $\mathfrak{g}^{r,-r} = 0$ for all $r \leq -2$: horizontality is vacuous exactly when the graded Lie algebra has only three nonzero pieces. Each $\mathfrak{g}^{r,-r}$ is also computable from the Hodge numbers alone, since it is a subspace of $\bigoplus_p \mathrm{Hom}(V^{p,q}, V^{p+r,q-r})$ cut out by the skew-symmetry condition against $Q$.

A metric on $D$ is now a choice of Hermitian inner product on $\bigoplus_{r<0} \mathfrak{g}_F^{r,-r}$ for each $F$, compatible with the group action. The polarization supplies one, through the positive definite form $H_F$ of (5.6) and the Weil operator.

**Definition 8.1.2: The Hodge Metric**

Let $\beta(X,Y) = \mathrm{tr}(XY)$ be the trace form on $\mathfrak{g}_{\mathbb{C}}$, and for $F \in D$ let $C_F \in \mathrm{GL}(V_{\mathbb{C}})$ be the Weil operator at $F$. The *Hodge form* at $F$ is

$$\langle X, Y\rangle_F = -\beta\big(X,\ C_F\,\bar{Y}\,C_F^{-1}\big) \qquad (X, Y \in \mathfrak{g}_{\mathbb{C}})$$

The *Hodge metric* on $D$ is the Hermitian metric $h$ whose value at $F$ is the restriction of $\langle -, - \rangle_F$ to

$$T_F^{1,0}D \cong \bigoplus_{r<0} \mathfrak{g}_F^{r,-r}$$

under the identification of proposition 8.1.1(3). The associated Riemannian metric on the real tangent bundle is $\mathrm{Re}\,h$, as in definition 3.1.1.

The definition has three things to verify: that $\langle -, - \rangle_F$ is a positive definite Hermitian form, that the resulting assignment is $G_{\mathbb{R}}$-invariant, and that it varies smoothly. All three come out of one identity, which also explains the sign.

**Proposition 8.1.3: The Hodge Form Is a Positive $G_{\mathbb{R}}$-Invariant Metric**

Fix $F \in D$ and write $C = C_F$, $H = H_F$ for the form (5.6). Then:

1. $C \in G_{\mathbb{R}}$, $H(u,v) = Q(Cu,\bar{v})$, and for $Y \in \mathfrak{g}_{\mathbb{C}}$ the $H$-adjoint of $Y$ is $Y^* = -C\bar{Y}C^{-1}$. Consequently $\langle X, Y\rangle_F = \mathrm{tr}(XY^*)$, and $\langle -, -\rangle_F$ is a positive definite Hermitian form on $\mathfrak{g}_{\mathbb{C}}$.
2. Conjugation by $C$ acts on $\mathfrak{g}_F^{r,-r}$ as multiplication by $(-1)^r$: for $X \in \mathfrak{g}_F^{r,-r}$ one has $CXC^{-1} = (-1)^r X$. The summands $\mathfrak{g}_F^{r,-r}$ are mutually orthogonal for $\langle -, -\rangle_F$, and on $\mathfrak{g}_F^{r,-r}$ one has $\langle X, X\rangle_F = (-1)^{r+1}\mathrm{tr}(X\bar{X})$.
3. For $g \in G_{\mathbb{R}}$ one has $C_{gF} = gC_F g^{-1}$ and $\langle gXg^{-1}, gYg^{-1}\rangle_{gF} = \langle X, Y\rangle_F$. The Hodge metric is therefore $G_{\mathbb{R}}$-invariant, and it a smooth Hermitian metric on $D$.

***Proof*:**
Write $\mathfrak{g}^{r,-r}$ for $\mathfrak{g}_F^{r,-r}$ and $V^{p,q}$ for $V_F^{p,q}$.

1. For $u \in V^{p,q}$ and $v \in V^{p',q'}$, the product $Q(Cu, Cv) = i^{p-q} i^{p'-q'} Q(u,v)$ vanishes unless $(p',q') = (q,p)$, in which case the scalar is $i^{p-q}i^{q-p} = 1$. So $Q(Cu,Cv) = Q(u,v)$ for all $u, v$, and $C$ preserves $V_{\mathbb{R}}$ by exercise 5.2.1 (1)(1); hence $C \in G_{\mathbb{R}}$. Next, expanding $Q(Cu,\bar{v})$ over the bigrading and using condition (1) of definition 5.2.1 kills every cross term, leaving $\sum_{p+q=k} i^{p-q} Q(u^{p,q}, \overline{v^{p,q}})$, which is (5.6).

   For the adjoint, let $Y \in \mathfrak{g}_{\mathbb{C}}$ and put $Y' = CYC^{-1}$, again in $\mathfrak{g}_{\mathbb{C}}$ because $C \in G_{\mathbb{C}}$. Then $CY = Y'C$, so
   $$H(Yu, v) = Q(CYu, \bar{v}) = Q(Y'Cu, \bar{v}) = -Q(Cu, Y'\bar{v})$$
   using that $Y'$ is $Q$-skew. Since $C$ is real, $\overline{Y'} = C\bar{Y}C^{-1}$, and $Y'\bar{v} = \overline{\overline{Y'}v}$; therefore $-Q(Cu, Y'\bar{v}) = -H\big(u, (C\bar{Y}C^{-1})v\big)$, which identifies $Y^* = -C\bar{Y}C^{-1}$ and gives $\langle X, Y\rangle_F = -\mathrm{tr}(XC\bar{Y}C^{-1}) = \mathrm{tr}(XY^*)$.

   Positivity and Hermitian symmetry are now matrix statements. The form $H$ is positive definite (proposition 5.3.4), so choose an $H$-orthonormal basis of $V_{\mathbb{C}}$; in it $Y^*$ is the conjugate transpose of the matrix of $Y$. Hence $\langle X, Y\rangle_F = \sum_{a,b} x_{ab}\overline{y_{ab}}$, which is $\mathbb{C}$-linear in $X$, conjugate-linear in $Y$, satisfies $\overline{\langle X, Y\rangle_F} = \langle Y, X\rangle_F$, and has $\langle X, X\rangle_F = \sum_{a,b} |x_{ab}|^2 > 0$ for $X \neq 0$.
2. For $X \in \mathfrak{g}^{r,-r}$ and $u \in V^{p,q}$,
   $$CXC^{-1}u = i^{-(p-q)}\, C(Xu) = i^{q-p} i^{(p+r)-(q-r)} Xu = i^{2r} Xu = (-1)^r Xu$$
   so conjugation by $C$ acts on $\mathfrak{g}^{r,-r}$ as $(-1)^r$. Thus for $Y \in \mathfrak{g}^{s,-s}$ one has $\bar{Y} \in \mathfrak{g}^{-s,s}$ and $Y^* = -(-1)^{-s}\bar{Y} = (-1)^{s+1}\bar{Y}$, whence $\langle X, Y\rangle_F = (-1)^{s+1}\mathrm{tr}(X\bar{Y})$. For $X \in \mathfrak{g}^{r,-r}$ the composite $X\bar{Y}$ shifts the bigrading by $r - s$, and an endomorphism shifting the bigrading by a nonzero amount has zero trace, so the pairing vanishes unless $r = s$. Setting $Y = X$ with $r = s$ gives the stated formula.

3. The bigrading at $gF$ is $V^{p,q}_{gF} = gV^{p,q}_F$, because $g$ is real and carries $F^\bullet$ to $(gF)^\bullet$; so $C_{gF}$ acts on $gV^{p,q}_F$ as $i^{p-q}$, which is exactly $gC_Fg^{-1}$. Since $g$ is real, $\overline{gYg^{-1}} = g\bar{Y}g^{-1}$, and therefore
$$(gXg^{-1})\, C_{gF}\, \overline{gYg^{-1}}\, C_{gF}^{-1} = g\big(XC_F\bar{Y}C_F^{-1}\big)g^{-1}$$
whose trace is unchanged. This is the asserted invariance; combined with the fact that $g$ acts on tangent vectors by $X \mapsto gXg^{-1}$ under the identification of proposition 8.1.1(3), which follows from $g\exp(tX)F = \exp(t\, gXg^{-1})gF$, it says that $g$ acts on $D$ by isometries.

   For smoothness, fix $F_o \in D$ and set $\mathfrak{m} = \mathfrak{g}_{\mathbb{R}} \cap \bigoplus_{r\neq 0} \mathfrak{g}^{r,-r}_{F_o}$. Both $\mathfrak{g}^{0,0}_{F_o}$ and $\bigoplus_{r\neq 0}\mathfrak{g}^{r,-r}_{F_o}$ are stable under conjugation, so $\mathfrak{g}_{\mathbb{R}} = (\mathfrak{g}_{\mathbb{R}} \cap \mathfrak{g}^{0,0}_{F_o}) \oplus \mathfrak{m}$, and $\mathfrak{m}$ is stable under conjugation by $K$ because an element of $K$ fixes $F_o$ and hence preserves each $\mathfrak{g}^{r,-r}_{F_o}$. This exhibits $(\mathfrak{g}_{\mathbb{R}}, \mathfrak{m})$ as a reductive decomposition for $D = G_{\mathbb{R}}/K$ in the sense of [9, reductive decomposition]. The projection $\mathfrak{m} \to \bigoplus_{r<0}\mathfrak{g}^{r,-r}_{F_o}$, $X \mapsto \sum_{r<0} X_r$, is an isomorphism of real vector spaces, and for $X, Y \in \mathfrak{m}$ with images $\xi, \eta$ the orthogonality in part (2) gives $\langle X, Y\rangle_{F_o} = 2\,\mathrm{Re}\langle \xi, \eta\rangle_{F_o}$. So $\mathrm{Re}\, h$ corresponds to the inner product $\frac{1}{2}\,\mathrm{Re}\langle -, -\rangle_{F_o}$ on $\mathfrak{m}$, which is $K$-invariant by the invariance just proved. By [9, invariant metrics and $\mathrm{Ad}(H)$-invariant inner products] there is a unique $G_{\mathbb{R}}$-invariant Riemannian metric on $D$ with that value at $F_o$, and it is smooth; by transitivity of $G_{\mathbb{R}}$ and the invariance already proved it coincides with $\mathrm{Re}\, h$, completing the proof.

Part (2) records the sign pattern used below. On $\mathfrak{g}^{r,-r}$ the trace form $\beta(X, \bar{X})$ is positive for odd $r$ and negative for even $r$, so conjugation by $C_F$ splits $\mathfrak{g}_{\mathbb{R}}$ the way a Cartan involution does: its $+1$-eigenspace is the sum of the even pieces, where $\beta$ is negative definite, and its $-1$-eigenspace is the sum of the odd pieces, where $\beta$ is positive definite (exercise 8.1.1 (5)). The horizontal directions $\mathfrak{g}^{-1,1}$ sit in the odd part, and their brackets $[\mathfrak{g}^{-1,1}, \mathfrak{g}^{1,-1}] \subseteq \mathfrak{g}^{0,0}$ land in the even part. That is the mechanism behind the curvature estimate: the negative sign of $\beta$ on the isotropy directions is what makes the horizontal curvature negative.

Carrying the mechanism through to a formula requires the Levi-Civita connection of a $G_{\mathbb{R}}$-invariant metric on the reductive homogeneous space $G_{\mathbb{R}}/K$. The series develops invariant metrics on such spaces and the curvature of a bi-invariant metric on a Lie group, but not the connection of a general invariant metric, so the computation is taken from Griffiths and Schmid rather than reproduced. What is proved below is the passage from their pointwise estimate to a uniform one, and the two structural statements about when $D$ is symmetric. For a nonzero real tangent vector $v \in T_FD$ write $K^{\mathrm{hol}}(v)$ for the *holomorphic sectional curvature* of the Hermitian metric $h$ in the direction $v$: the quantity computed from the Chern connection of $h$, equivalently the Gaussian curvature that $-\partial\bar{\partial}\log h$ assigns to a holomorphic disc through $F$ tangent to $v$.

The Hermitian reading is the one meant, and the distinction matters. The Hodge metric is in general *not* Kähler: for $X, Y \in \mathfrak{g}^{-1,1}_F$ one has $[X, Y] \in \mathfrak{g}^{-2,2}_F$ by proposition 8.1.1(1), and the invariant form pairs $\mathfrak{g}^{-2,2}_F$ nontrivially with $\mathfrak{g}^{2,-2}_F$, so the associated $(1,1)$-form has $d\omega \neq 0$ as soon as $\mathfrak{g}^{-2,2}_F \neq 0$. That is exactly the case in which $D$ fails to be Hermitian symmetric, which is the case this section exists to treat. For a Kähler metric the Chern and Levi-Civita connections agree and $K^{\mathrm{hol}}(v)$ coincides with the Riemannian sectional curvature ([9, sectional curvature]) of the real 2-plane spanned by $v$ and $Jv$; away from that case the two differ, the Chern connection having torsion, and it is the Chern quantity that fact (2) below bounds. Example 8.1.5 is a Kähler case, so the two readings agree there.

**Theorem 8.1.4: Curvature of the Hodge Metric**

Let $D$ carry the Hodge metric of definition 8.1.2.

1. There is a constant $c > 0$, depending only on the numerical data $(V_{\mathbb{Z}}, Q, k, \{h^{p,q}\})$, such that
$$K^{\text{hol}}(v) \le -c \qquad \text{for every } F \in D \text{ and every nonzero } v \in T^h_F D$$

2. The vanishing $\mathfrak{g}_F^{r,-r} = 0$ for all $r \le -2$ is independent of $F$ and is equivalent to $T^h D = TD$. When it holds, the Weil operator $C_F$ acts on $D$ as an isometry fixing $F$ with differential $-\mathrm{id}$, so the geodesic symmetry at every point of $D$ extends to a global isometry and $D$ is a Hermitian symmetric space.

3. If $\mathfrak{g}_F^{r,-r} \neq 0$ for some even $r \neq 0$, then $D$ contains a compact connected complex submanifold of positive dimension. In particular $D$ is not biholomorphic to a bounded domain in $\mathbb{C}^N$, so it is not a bounded symmetric domain.

***Proof*:**

1. Griffiths and Schmid compute the curvature tensor of the Hodge metric on $D$ and show that $K^{\text{hol}}(v) < 0$ for every nonzero horizontal $v$ [16]; that computation is the input taken from outside, for the reason recorded above. Granting it, the uniformity is a compactness argument. Fix $F_o \in D$ and let $S \subseteq T^h_{F_o} D$ be the unit sphere of the Hodge metric, a compact set since $T^h_{F_o} D$ is finite dimensional. The holomorphic sectional curvature of a Hermitian metric is a continuous function of $v$, being assembled from the Chern curvature tensor and $h$ itself, and is unchanged under real scaling of $v$, numerator and denominator both being homogeneous of degree four. It is strictly negative on the compact set $S$, so it attains a maximum $-c < 0$ there. The Chern connection is natural, so $K^{\text{hol}}$ is invariant under holomorphic isometries and $K^{\text{hol}}$ is invariant under scaling of $v$, so for arbitrary $F \in D$ and nonzero $v \in T^h_F D$ choose $g \in G_{\mathbb{R}}$ with $gF_o = F$ (proposition 5.3.5); then $dg^{-1}(v)$ is horizontal, because $g$ preserves the horizontal distribution by the equivariance noted after definition 7.3.8, and normalizing it into $S$ gives $K^{\text{hol}}(v) \le -c$.

2. Independence of $F$ holds because $\mathfrak{g}_{gF}^{r,-r} = g\,\mathfrak{g}_F^{r,-r}\,g^{-1}$ for $g \in G_{\mathbb{R}}$, which follows from $V_{gF}^{p,q} = gV_F^{p,q}$, and $G_{\mathbb{R}}$ acts transitively on $D$. The equivalence with $T^h D = TD$ is proposition 8.1.1(3): the two spaces agree at $F$ exactly when $\bigoplus_{r<0} \mathfrak{g}_F^{r,-r} = \mathfrak{g}_F^{-1,1}$.

   Assume the vanishing. Conjugation preserves the grading up to sign, so $\mathfrak{g}_F^{r,-r} = 0$ for all $|r| \ge 2$ and $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_F^{-1,1} \oplus \mathfrak{g}_F^{0,0} \oplus \mathfrak{g}_F^{1,-1}$. By part (2) of proposition 8.1.3, conjugation by $C_F$ acts as $(-1)^r$ on $\mathfrak{g}_F^{r,-r}$, hence as $-\mathrm{id}$ on $\mathfrak{g}_F^{-1,1} \oplus \mathfrak{g}_F^{1,-1}$ and as $+\mathrm{id}$ on $\mathfrak{g}_F^{0,0}$. Now $C_F \in G_{\mathbb{R}}$ by proposition 8.1.3(1) and $C_F$ preserves every $V_F^{p,q}$, hence every $F^p$, so $C_F$ fixes $F$; and it acts on $D$ by an isometry, by proposition 8.1.3(3). Its differential at $F$ is $X \mapsto C_F X C_F^{-1}$ on $T_F^{1,0} D \cong \mathfrak{g}_F^{-1,1}$, which is $-\mathrm{id}$.

   A homogeneous Riemannian manifold is complete ([9, homogeneous manifolds are complete]), so normal balls around $F$ are available, and an isometry commutes with the exponential map. Hence for $v$ in a star-shaped neighbourhood of 0 in $T_F D$,

$$C_F\big(\exp_F(v)\big) = \exp_F\big(d(C_F)_F v\big) = \exp_F(-v)$$

so $C_F$ agrees on that ball with the geodesic symmetry $\exp_F \circ(-\mathrm{id}) \circ \exp_F^{-1}$ of [9, geodesic symmetry and symmetric space]. The geodesic symmetry at $F$ therefore extends to a global isometry, and since $F$ was arbitrary $D$ is a symmetric space. It is Hermitian symmetric because $h$ is a Hermitian metric and each $C_F$ acts holomorphically, being an element of $G_{\mathbb{C}}$ acting on $\check{D}$.

3. Let $Z = \{g \in G_{\mathbb{R}} \big| gC_F = C_F g\}$ be the centralizer of the Weil operator. If $g \in Z$ then, using $H = H_F$ and proposition 8.1.3(1),

$$H(gu, gv) = Q\big(Cgu, \overline{gv}\big) = Q\big(gCu, g\bar{v}\big) = Q(Cu, \bar{v}) = H(u, v)$$

so $Z$ is a closed subset of the unitary group of the positive definite form $H$, hence compact. An element $X \in \mathfrak{g}_{\mathbb{R}}$ lies in $\mathrm{Lie}(Z)$ if and only if $\exp(tX)C\exp(-tX) = C$ for all $t$, that is if and only if $XC = CX$; by proposition 8.1.3(2) this is $\mathrm{Lie}(Z) = \mathfrak{g}_{\mathbb{R}} \cap \bigoplus_{r \text{ even}} \mathfrak{g}_F^{r,-r}$. Also $K \subseteq Z$, because an element of $K$ fixes $F$ and hence preserves each $V_F^{p,q}$, on which $C$ is a scalar; so $\mathrm{Lie}(K) = \mathfrak{g}_{\mathbb{R}} \cap \mathfrak{g}_F^{0,0} \subseteq \mathrm{Lie}(Z)$.

Let $Y = Z^0 \cdot F$ be the orbit of the identity component. The orbit map descends to an injection $Z^0/(Z^0 \cap K) \to D$ of the quotient manifold ([32, homogeneous spaces]), whose differential at the base point is the inclusion $\mathrm{Lie}(Z)/\mathrm{Lie}(K) \hookrightarrow \mathfrak{g}_{\mathbb{R}}/\mathrm{Lie}(K)$ and hence injective; by $Z^0$-equivariance it is injective at every point, so the map is an injective immersion. Its source is compact, so it is a homeomorphism onto its image and $Y$ is a compact connected embedded submanifold of $D$ of dimension

$$\dim_{\mathbb{R}} Y = \dim_{\mathbb{R}}\Big(\mathfrak{g}_{\mathbb{R}} \cap \bigoplus_{r \text{ even},\, r\neq 0} \mathfrak{g}_F^{r,-r}\Big)$$

This is positive under the hypothesis, because a conjugation-stable complex subspace of $\mathfrak{g}_{\mathbb{C}}$ of complex dimension $d$ has real points of real dimension $d$.

The tangent space $T_F Y$ corresponds, under proposition 8.1.1(3), to $\bigoplus_{r<0,\ r \text{ even}} \mathfrak{g}_F^{r,-r}$, a complex subspace of $T_F^{1,0}D$; and $Z^0$ acts transitively on $Y$ by biholomorphisms of $D$, so every tangent space of $Y$ is stable under the complex structure $J$ of $D$. The Nijenhuis tensor of $J$ vanishes on $D$ (corollary 1.1.17), and brackets of vector fields tangent to $Y$ remain tangent to $Y$, so the almost complex structure induced on $Y$ has vanishing Nijenhuis tensor; theorem 1.1.18 makes $Y$ a complex manifold, and the inclusion is holomorphic because its differential is $\mathbb{C}$-linear.

Suppose finally that $\varphi\colon D \to \Omega \subseteq \mathbb{C}^N$ were a biholomorphism onto a bounded domain. Each coordinate function $\varphi_j$ restricts to a holomorphic function on the compact connected complex manifold $Y$, hence is constant there by proposition 1.1.11. Then $\varphi(Y)$ is a single point, contradicting the injectivity of $\varphi$ and $\dim Y > 0$. So no such $\varphi$ exists, completing the proof.

Parts (2) and (3) mark the boundary of the classical theory. A Hermitian symmetric space of noncompact type is biholomorphic to a bounded symmetric domain (Harish-Chandra's realization, [1, Hermitian symmetric spaces and their bounded realizations]), so part (3) says that as soon as $\mathfrak{g}$ has a nonzero even piece in a degree other than 0, the domain $D$ leaves that world, and with it leaves the setting in which negative curvature holds in every direction. What survives is exactly part (1): an estimate on $T^hD$ and nothing more. The full statement, that $D$ is Hermitian symmetric precisely when $T^hD = TD$, is in [16]. The first numerical type meeting the hypothesis of part (3) is weight 2 with $h^{2,0} = 2$, the case in which exercise 7.3.1 (3) found $T^hD$ of codimension 1; there $\mathfrak{g}^{-2,2}$ is one-dimensional (exercise 8.1.1 (3)), so the compact complex submanifold produced by the proof is a curve.

The classical cases sit on the other side of the boundary, and the weight-one case can be computed outright.

**Example 8.1.5: The Hodge Metric on the Siegel Upper Half Space**

Take $k = 1$ and $h^{1,0} = h^{0,1} = g$, the numerical type of example 5.3.3, so that $D \cong \mathfrak{H}_g$ and $G_{\mathbb{R}} = \mathrm{Sp}(V_{\mathbb{R}}, Q)$.

*The grading has three pieces.* There are only two bidegrees, $(1, 0)$ and $(0, 1)$, so an endomorphism shifting the bigrading by $r$ is zero unless $r \in \{-1, 0, 1\}$. Hence $\mathfrak{g}^{r,-r} = 0$ for $|r| \geq 2$, and by theorem 8.1.4(2) the horizontal subbundle is everything and $\mathfrak{H}_g$ is a Hermitian symmetric space. This is the period-domain form of the observation made by hand in example 7.3.9, and it says that in weight one the infinitesimal period relation imposes nothing: every holomorphic map into $\mathfrak{H}_g$ is horizontal.

*The metric for $g = 1$.* Take $g = 1$, so $V_{\mathbb{R}} = \mathbb{R}^2$ with symplectic basis $e, f$ and $Q(e, f) = 1$, and present a point of $D$ as $F^1 = \mathbb{C}(e + \tau f)$ with $\tau \in \mathfrak{H}_1$, as in example 5.3.3. Work at $\tau = i$ and write $v = e + if$, so $V^{1,0} = \mathbb{C}v$ and $V^{0,1} = \mathbb{C}\bar{v}$. An element $X \in \mathfrak{g}^{-1,1}$ satisfies $X_a v = a\bar{v}$ and $X_a\bar{v} = 0$ for some $a \in \mathbb{C}$; conversely every such $X_a$ is $Q$-skew, since $Q(X_a v, \bar{v}) + Q(v, X_a\bar{v}) = aQ(\bar{v}, \bar{v}) = 0$ and $Q(X_a v, v) + Q(v, X_a v) = a\big(Q(\bar{v}, v) + Q(v, \bar{v})\big) = 0$ by antisymmetry of $Q$, while the remaining pairs of basis vectors give $0 = 0$. So $\mathfrak{g}^{-1,1} = \{X_a \big| a \in \mathbb{C}\}$ is a line, in accordance with $\dim_{\mathbb{C}} \mathfrak{H}_1 = 1$.

The Hodge norm is computed from proposition 8.1.3(2) with $r = -1$, where the sign is $(-1)^{r+1} = +1$. Since $\bar{X}_a$ sends $\bar{v}$ to $\bar{a}v$ and kills $v$, the composite $X_a\bar{X}_a$ kills $v$ and multiplies $\bar{v}$ by $|a|^2$, so

$$\|X_a\|^2 = \mathrm{tr}(X_a\bar{X}_a) = |a|^2$$

To read this on $\mathfrak{H}_1$, follow the curve $\exp(tX_a)F$. From $X_a^2 = 0$ one gets $\exp(tX_a)v = v + ta\bar{v} = (1 + ta)e + i(1 - ta)f$, and dividing by $1 + ta$, which changes no subspace, presents the same line as $e + \tau(t)f$ with $\tau(t) = i(1 - ta)/(1 + ta)$. Differentiating at $t = 0$ gives $\dot{\tau} = -2ia$. So the tangent vector with $\tau$-component $w$ corresponds to $a = iw/2$ and has squared Hodge norm $|a|^2 = |w|^2/4$: at $\tau = i$ the Hodge metric is $\frac{1}{4}|d\tau|^2$.

Both the Hodge metric and the Poincaré metric $|d\tau|^2/(\mathrm{Im}\,\tau)^2$ are invariant under $\mathrm{SL}(2, \mathbb{R}) = \mathrm{Sp}(V_{\mathbb{R}}, Q)$, which acts transitively on $\mathfrak{H}_1$, and they agree up to the factor $\frac{1}{4}$ at $\tau = i$. Two invariant metrics agreeing at one point of a homogeneous space agree everywhere, so

$$h = \frac{|d\tau|^2}{4\,(\operatorname{Im}\tau)^2}$$

The Poincaré metric has constant Gaussian curvature $-1$, and scaling a surface metric by $\frac{1}{4}$ multiplies its curvature by 4; so the Hodge metric on $\mathfrak{H}_1$ has constant holomorphic sectional curvature $-4$, and $c = 4$ serves in theorem 8.1.4(1). Here the bound holds in every direction, because every direction is horizontal. In the general case the same constant governs only the horizontal ones.

Negative curvature bounds the derivative of a holomorphic map into $D$. Making that precise takes two classical facts of complex hyperbolic geometry, neither of which is developed anywhere in the series; each is stated with the source it is taken from and is used here exactly as stated. Let $\Delta = \{z \in \mathbb{C} \big| |z| < 1\}$ and let

$$\rho = \frac{4\,|dz|^2}{(1 - |z|^2)^2}$$

be the Poincaré metric on $\Delta$, of constant curvature $-1$ and equal to $4|dz|^2$ at the origin. The two facts are:

1. (*Ahlfors-Schwarz lemma*) If $\sigma = \lambda|dz|^2$ is a conformal pseudo-metric on $\Delta$ with $\lambda \geq 0$ smooth, whose Gaussian curvature satisfies $K_\sigma \leq -c$ wherever $\lambda > 0$, with $c > 0$, then $\sigma \leq c^{-1}\rho$. This is Ahlfors's curvature form of the classical Schwarz lemma ([25, Schwarz's lemma]), and is taken from [1, the Ahlfors-Schwarz lemma].

2. (*curvature decreases on holomorphic curves*) If $f\colon \Delta \to (M, h)$ is holomorphic into a Hermitian manifold and $df_{z_0} \neq 0$, then the Gaussian curvature of $f^*h$ at $z_0$ is at most the holomorphic sectional curvature of $h$ at $f(z_0)$ in the direction $df_{z_0}(\partial/\partial z)$. This is the standard curvature comparison for a holomorphic curve in a Hermitian manifold, taken from [1, curvature estimates for holomorphic maps]; it is what turns the estimate of theorem 8.1.4(1), itself taken from [16], into a curvature bound on $\Delta$.

**Corollary 8.1.6: Horizontal Maps Are Distance-Decreasing**

Let $c > 0$ be as in theorem 8.1.4(1), let $\Gamma \subseteq G_{\mathbb{Z}}$ act freely on $D$, and give $\Gamma\backslash D$ the Hermitian metric descended from the Hodge metric. Then every horizontal holomorphic map $f\colon \Delta \to \Gamma\backslash D$ satisfies

$$f^*h \leq c^{-1}\rho$$

In particular $f$ is distance-decreasing from $c^{-1}\rho$ to the Hodge metric, and $\|df_0(\partial/\partial z)\|_h^2 \leq 4/c$.

***Proof***:
By proposition 7.3.5 the group $\Gamma$ acts properly discontinuously, and acting freely as well it makes $q\colon D \to \Gamma\backslash D$ a holomorphic covering map. The Hodge metric is $\Gamma$-invariant by proposition 8.1.3(3), so it descends and $q$ is a local isometry; the horizontal distribution is $G_{\mathbb{C}}$-equivariant, as noted after definition 7.3.8, so it too descends, and a map into $\Gamma\backslash D$ is called horizontal when its differential lands in the descended distribution. Since $\Delta$ is simply connected, $f$ lifts to a holomorphic map $\tilde{f}\colon \Delta \to D$ with $q \circ \tilde{f} = f$; the lift is horizontal because $q$ is a local

biholomorphism matching the two distributions, and $f^*h = \tilde{f}^*h$.

Write $\tilde{f}^*h = \lambda|dz|^2$, a conformal pseudo-metric on $\Delta$, smooth because $\tilde{f}$ and $h$ are, and vanishing exactly where $d\tilde{f}$ does. At a point $z_0$ with $d\tilde{f}_{z_0} \neq 0$, the vector $d\tilde{f}_{z_0}(\partial/\partial z)$ is a nonzero horizontal tangent vector, so fact (2) followed by theorem 8.1.4(1) gives

$$K_{\tilde{f}^*h}(z_0) \leq K^{\mathrm{hol}}\big(d\tilde{f}_{z_0}(\partial/\partial z)\big) \leq -c$$

Fact (1) then yields $\tilde{f}^*h \leq c^{-1}\rho$ on $\Delta$. Evaluating at the origin, where $\rho = 4|dz|^2$, gives the stated bound on the derivative, completing the proof.

Nothing in the corollary holds for an arbitrary holomorphic map into $D$: the estimate is available only because theorem 8.1.4(1) is restricted to $T^hD$, and by theorem 7.3.10 every period map satisfies that restriction. The consequence is a Liouville theorem, and through it a rigidity statement about families.

**Theorem 8.1.7: Rigidity of Horizontal Maps**

Let $\Gamma \subseteq G_{\mathbb{Z}}$ act freely on $D$ and give $X_\Gamma = \Gamma \backslash D$ the descended Hodge metric.

1. Every horizontal holomorphic map $\mathbb{C} \to X_\Gamma$ is constant.
2. Every horizontal holomorphic map $\mathbb{CP}^n \to X_\Gamma$ is constant, and so is every horizontal holomorphic map $M \to X_\Gamma$ from a complex manifold $M$ whose universal cover is $\mathbb{C}^n$, for instance a complex torus.
3. If $f\colon \mathcal{X} \to \mathbb{CP}^n$ is a smooth projective family, then its period map in every degree $k$ is constant.

***Proof*:**

1. Since $\mathbb{C}$ is simply connected and $D \to X_\Gamma$ is a covering, a horizontal holomorphic $f\colon \mathbb{C} \to X_\Gamma$ lifts to a horizontal holomorphic $\tilde{f}\colon \mathbb{C} \to D$. Fix $z_0 \in \mathbb{C}$; after composing with a translation of $\mathbb{C}$, which is a biholomorphism, assume $z_0 = 0$. For $R > 0$ the map $g_R\colon \Delta \to D$, $g_R(z) = \tilde{f}(Rz)$, is horizontal and holomorphic, so corollary 8.1.6 gives
$$R^2 \left\|d\tilde{f}_0(\partial/\partial z)\right\|_h^2 = \left\|d(g_R)_0(\partial/\partial z)\right\|_h^2 \leq \frac{4}{c}$$
Letting $R \to \infty$ forces $d\tilde{f}_0 = 0$. As $z_0$ was arbitrary, $d\tilde{f} \equiv 0$, and $\mathbb{C}$ being connected, $\tilde{f}$ and hence $f$ is constant.
2. Both cases reduce to (1). For $\mathbb{CP}^n$, which is simply connected, $f$ lifts to $\tilde{f}\colon \mathbb{CP}^n \to D$. Let $\ell \subseteq \mathbb{CP}^n$ be a projective line and $p \in \ell$; then $\ell \setminus \{p\} \cong \mathbb{C}$ and $\tilde{f}$ restricted to it is horizontal holomorphic, hence constant by (1), and continuity extends the constancy to $\ell$. Any two points of $\mathbb{CP}^n$ lie on a common line, so $\tilde{f}$ is constant.

   For $M$ with universal cover $\pi\colon \mathbb{C}^n \to M$, the composite $f \circ \pi$ is horizontal holomorphic and lifts to $\tilde{f}\colon \mathbb{C}^n \to D$. Restricting $\tilde{f}$ to the complex lines through a given point of $\mathbb{C}^n$

gives horizontal holomorphic maps $\mathbb{C} \to D$, constant by (1); so every directional derivative of $\tilde{f}$ vanishes, $\tilde{f}$ is constant, and $f$ is constant because $\pi$ is surjective.

3. The base $\mathbb{CP}^n$ is simply connected, so the monodromy representation is trivial and $\Gamma = \{1\}$; by theorem 7.3.7 the period map is a globally defined holomorphic map $\Phi\colon \mathbb{CP}^n \to D$, and by theorem 7.3.10 it is horizontal. Part (2) applies with $\Gamma = \{1\}$ and makes $\Phi$ constant, completing the proof.

Part (3) is a statement about geometry, not about period domains. A smooth projective family over $\mathbb{CP}^n$ has, in every degree, a Hodge filtration on the primitive cohomology that is flat for the Gauss-Manin connection (exercise 7.3.1 (1)), so parallel transport between any two fibres is an isomorphism of polarized Hodge structures on the primitive parts: the fibres are indistinguishable by transcendental invariants. Set this against the Lefschetz pencils of chapter 6, which are families over $\mathbb{CP}^1$ with nontrivial monodromy around the critical values. There is no contradiction, because a pencil is smooth only over the complement of those values, and the theorem says that this is forced: the singular fibres are not an artifact of the construction. A family that varies must degenerate.

What happens over a base that is quasi-projective rather than compact is a separate theory, and it is Schmid's. The nilpotent orbit and SL(2)-orbit theorems of [30] describe the asymptotics of a period map as the base approaches a boundary divisor, and yield the monodromy theorem, that local monodromy about a boundary component is quasi-unipotent. Those results are not developed here: the limiting objects they produce are no longer pure Hodge structures, and nothing in this chapter needs them.

### Exercise 8.1.1

1. Let $F \in D$ with Weil operator $C = C_F$. Show directly from definition 5.2.1 that $Q(Cu, Cv) = Q(u, v)$ for all $u, v \in V_{\mathbb{C}}$, and that $H_F(u, v) = Q(Cu, \bar{v})$, where $H_F$ is the form (5.6). These two identities are part of proposition 8.1.3(1); the exercise is to rederive them by hand from the definitions. Deduce that $C^2 = (-1)^k \,\mathrm{id}$ lies in the centre of $G_{\mathbb{R}}$ and that conjugation by $C$ is an involution of $\mathfrak{g}_{\mathbb{R}}$.

2. Take $k = 2$ with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = n$. This is the shape of the Hodge structure on the $H^2$ of a K3 surface (example 5.4.4), though that example also records that the cup product on the whole of $H^2$ is not a polarization, so the period domain here is the one attached to the primitive part. Show that $\mathfrak{g}_F^{r,-r} = 0$ for every $|r| \geq 2$, and conclude from theorem 8.1.4(2) that this period domain is a Hermitian symmetric space on which every holomorphic map is horizontal, recovering the computation of example 7.3.9.

3. Take $k = 2$ with $h^{2,0} = h^{0,2} = 2$ and $h^{1,1} = n$. Show that $\dim_{\mathbb{C}} \mathfrak{g}_F^{-2,2} = 1$ and that $\mathfrak{g}_F^{r,-r} = 0$ for $|r| \geq 3$. Deduce that $T^h D$ has codimension 1 in $TD$, in agreement with exercise 7.3.1 (3), and that the compact complex submanifold produced by theorem 8.1.4(3) has real dimension 2. Conclude that this $D$ is not biholomorphic to a bounded domain.

4. Show that $\mathfrak{g}_{gF}^{r,-r} = g\, \mathfrak{g}_F^{r,-r}\, g^{-1}$ for every $g \in G_{\mathbb{R}}$ and $F \in D$, and deduce that neither the hypothesis of theorem 8.1.4(3) nor the numbers $\dim_{\mathbb{C}} \mathfrak{g}_F^{r,-r}$ depend on the choice of $F$. Where does the argument use that $g$ is real, and not just that it lies in $G_{\mathbb{C}}$?

5. Let $\beta(X, Y) = \mathrm{tr}(XY)$ be the trace form on $\mathfrak{g}_{\mathbb{C}}$ and fix $F \in D$. Using proposition 8.1.3(2), show that $\beta(X, \bar{X}) > 0$ for nonzero $X \in \mathfrak{g}_F^{r,-r}$ with $r$ odd and $\beta(X, \bar{X}) < 0$ for nonzero $X$ with $r$ even. Deduce that $\beta$ is negative definite on $\mathfrak{g}_{\mathbb{R}} \cap \bigoplus_{r \text{ even}} \mathfrak{g}_F^{r,-r}$ and positive definite on

$\mathfrak{g}_{\mathbb{R}} \cap \bigoplus_{r \text{ odd}} \mathfrak{g}_F^{r,-r}$, that the two are $\beta$-orthogonal, and that writing $\theta$ for conjugation by $C_F$ the form $-\beta(X, \theta Y)$ is positive definite on $\mathfrak{g}_{\mathbb{R}}$.

6. Show that $\tau \mapsto (\tau - i)/(\tau + i)$ is a biholomorphism from $\mathfrak{H}_1$ onto the unit disc $\Delta$, so that the weight-one period domain of example 8.1.5 with $g = 1$ *is* biholomorphic to a bounded domain. Using corollary 8.1.6 and the constant $c = 4$ computed there, write down the resulting bound on $|f'(0)|$ for a holomorphic map $f \colon \Delta \to \mathfrak{H}_1$, and compare it with the bound obtained by transporting the classical Schwarz lemma through the biholomorphism.

## 8.2 Mumford-Tate Groups

A period domain records a polarized Hodge structure as a point of a homogeneous space, and the group $G_{\mathbb{R}}$ of proposition 5.3.5 moves one structure to another. That group depends only on the numerical data $(V_{\mathbb{Q}}, Q, k, \{h^{p,q}\})$, so it sees nothing about the individual structure sitting at a given point. This section attaches to a single rational Hodge structure a group that does: a $\mathbb{Q}$-algebraic subgroup of $\mathrm{GL}(V_{\mathbb{Q}}) \times \mathbb{G}_m$ whose invariants in the tensor constructions built from $V$ are exactly the Hodge classes.

The construction has two steps. A Hodge structure on $V_{\mathbb{Q}}$ is first rewritten as an action of a two-dimensional real algebraic torus on $V_{\mathbb{R}}$; the group is then the smallest subgroup defined over $\mathbb{Q}$ whose real points carry that action. The ambient language of algebraic groups, their tori and their reductivity is taken from [11] and not rebuilt here; what is developed here is the Hodge-theoretic content, which is the invariant-theoretic characterization of Hodge classes and the reductivity that a polarization forces.

Throughout, a *rational Hodge structure* of weight $k$ means the datum of definition 5.1.1 on a finite-dimensional $\mathbb{Q}$-vector space $V_{\mathbb{Q}}$: a bigrading $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$. An *algebraic subgroup* of an algebraic group over $\mathbb{Q}$ means a Zariski-closed subgroup defined over $\mathbb{Q}$ ([11, affine algebraic groups and their closed subgroups]).

The bigrading and its conjugation symmetry are, together, an action of $\mathbb{C}^\times$ on the real vector space $V_{\mathbb{R}}$, in which $z$ acts on $V^{p,q}$ by the scalar $z^p \bar{z}^q$: the symmetry $\overline{V^{p,q}} = V^{q,p}$ is precisely what makes that prescription commute with conjugation and hence preserve $V_{\mathbb{R}}$. The group carrying the action is algebraic over $\mathbb{R}$ rather than over $\mathbb{C}$, since the prescription involves $\bar{z}$ alongside $z$.

**Definition 8.2.1: The Deligne Torus**

The *Deligne torus* is the real algebraic group

$$\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$$

obtained from $\mathbb{G}_m$ over $\mathbb{C}$ by restriction of scalars: for a commutative $\mathbb{R}$-algebra $A$ it is $\mathbb{S}(A) = (A \otimes_{\mathbb{R}} \mathbb{C})^\times$, and its coordinate ring is $\mathbb{R}[x, y, (x^2 + y^2)^{-1}]$ with $z = x + iy$ the tautological coordinate. In particular $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$, and the inclusion $\mathbb{S}(\mathbb{R}) \subseteq \mathbb{S}(\mathbb{C})$ is $z \mapsto (z, \bar{z})$ under the isomorphism $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m \times \mathbb{G}_m$ induced by $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$.
Three homomorphisms of real algebraic groups are attached to it: the inclusion $\mathbb{G}_{m,\mathbb{R}} \hookrightarrow \mathbb{S}$ of real scalars, the *norm* $\mathrm{Nm} \colon \mathbb{S} \to \mathbb{G}_{m,\mathbb{R}}$, given on real points by $z \mapsto z\bar{z}$, and the *circle* $\mathbb{U} = \ker(\mathrm{Nm})$, with $\mathbb{U}(\mathbb{R}) = \{z \in \mathbb{C}^\times \,|\, z\bar{z} = 1\}$.

The characters of $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m \times \mathbb{G}_m$ are the monomials $z^p \bar{z}^q$ with $(p, q) \in \mathbb{Z}^2$, written in the coordinates supplied by the isomorphism above; complex conjugation on $\mathbb{S}$, that is the nontrivial element of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, interchanges the two factors and hence sends $z^p \bar{z}^q$ to $z^q \bar{z}^p$. That interchange is the source of the conjugation symmetry in the next statement.

**Proposition 8.2.2: Hodge Structures Are Representations of $\mathbb{S}$**

Let $V_{\mathbb{Q}}$ be a finite-dimensional $\mathbb{Q}$-vector space and $k \in \mathbb{Z}$. The following data are equivalent.

1. A rational Hodge structure of weight $k$ on $V_{\mathbb{Q}}$, that is a bigrading $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$.
2. A homomorphism of real algebraic groups $h\colon \mathbb{S} \to \mathrm{GL}(V_{\mathbb{R}})$ whose restriction to the subgroup $\mathbb{G}_{m,\mathbb{R}}$ of real scalars is $r \mapsto r^k\,\mathrm{id}$.

The passage from (1) to (2) sends the bigrading to the homomorphism acting on $V^{p,q}$ as multiplication by $z^p \bar{z}^q$; the passage from (2) to (1) takes $V^{p,q}$ to be the subspace of $V_{\mathbb{C}}$ on which $\mathbb{S}_{\mathbb{C}}$ acts through the character $z^p \bar{z}^q$. The two constructions are mutually inverse.

***Proof*:**
*Step 1: a homomorphism gives a bigrading.* Let $h$ be as in (2) and extend it to $\mathbb{S}_{\mathbb{C}} \to \mathrm{GL}(V_{\mathbb{C}})$. The group $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m \times \mathbb{G}_m$ is a torus, so $V_{\mathbb{C}}$ decomposes as the direct sum of its character eigenspaces ([11, the weight-space decomposition of a representation of a torus]):

$$V_{\mathbb{C}} = \bigoplus_{(p,q)\in\mathbb{Z}^2} V^{p,q}$$

where $V^{p,q}$ is the eigenspace on which $\mathbb{S}_{\mathbb{C}}$ acts through the character $z^p \bar{z}^q$; evaluating that character at a point of $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$ gives $h(z)v = z^p \bar{z}^q v$ for $v \in V^{p,q}$.
Only the pairs with $p + q = k$ occur. For $r \in \mathbb{R}^\times$ the operator $h(r)$ acts on $V^{p,q}$ by $r^p r^q = r^{p+q}$, while the hypothesis in (2) makes it $r^k$; taking $r = 2$ gives $2^{p+q} = 2^k$ on any nonzero $V^{p,q}$, so $V^{p,q} = 0$ unless $p + q = k$. In particular the sum is finite.
For the symmetry, note that $h(z)$ is an endomorphism of the real space $V_{\mathbb{R}}$, extended $\mathbb{C}$-linearly, so it commutes with the conjugation of $V_{\mathbb{C}}$ over $V_{\mathbb{R}}$. Let $v \in V^{p,q}$. Then

$$h(z)\bar{v} = \overline{h(z)v} = \overline{z^p \bar{z}^q v} = z^q \bar{z}^p\, \bar{v}$$

so $\bar{v} \in V^{q,p}$, and applying the same to $V^{q,p}$ gives $\overline{V^{p,q}} = V^{q,p}$. This is a Hodge structure of weight $k$.
*Step 2: a bigrading gives a homomorphism.* Let the bigrading be given and define $h(z)v = \sum_{p+q=k} z^p \bar{z}^q\, v^{p,q}$ for $v \in V_{\mathbb{C}}$ with components $v^{p,q}$. It is multiplicative in $z$ because each summand is, and it preserves $V_{\mathbb{R}}$: for $v$ real one has $\overline{v^{p,q}} = v^{q,p}$ by the symmetry, so

$$\overline{h(z)v} = \sum_{p+q=k} \bar{z}^p z^q\, \overline{v^{p,q}} = \sum_{p+q=k} \bar{z}^p z^q\, v^{q,p} = h(z)v$$

the last equality by the substitution $(p, q) \mapsto (q, p)$. So $h(z)$ restricts to an $\mathbb{R}$-linear automorphism of $V_{\mathbb{R}}$.
It is algebraic. Fix a basis of $V_{\mathbb{R}}$ and write $z = x + iy$. The function $z \mapsto z^p \bar{z}^q$ lies in $\mathbb{R}[x, y, (x^2 + y^2)^{-1}] = \mathcal{O}(\mathbb{S})$ after taking real and imaginary parts, since $z\bar{z} = x^2 + y^2$ and

$z^{-1} = \bar{z}(x^2+y^2)^{-1}$; hence so does every matrix entry of $h(z)$, these being $\mathbb{R}$-linear combinations of such functions. Finally $h(r)$ acts on $V^{p,q}$ by $r^{p+q} = r^k$ for real $r$, which is the condition in (2).
*Step 3: the constructions are inverse.* Starting from a bigrading, the homomorphism built in Step 2 acts on $V^{p,q}$ by the character $z^p\bar{z}^q$, and distinct pairs $(p,q)$ give distinct characters, so the eigenspace decomposition recovered in Step 1 is the original one. Starting from $h$, Step 1 produces the bigrading by eigenspaces and Step 2 rebuilds $h$ from it by the same formula, completing the proof.

The proposition replaces a piece of linear algebra by a representation, and the gain is that representations have automorphism groups and invariants while bigradings do not. The restriction of $h$ to real scalars is the *weight homomorphism*: it detects $k$ and nothing else, so the weight is exactly the part of the structure visible to the split subtorus $\mathbb{G}_{m,\mathbb{R}} \subseteq \mathbb{S}$. And the circle $\mathbb{U}$ sees the rest: $h|_{\mathbb{U}}$ determines the bigrading up to the weight, since $\mathbb{C}^\times$ is generated by $\mathbb{R}_{>0}$ and the unit circle.

**Example 8.2.3: The Circle Action in Weight One and on the Tate Structure**
Two instances, one in each direction of proposition 8.2.2.

1. *Weight one.* Here $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, and $h(z)$ acts by $z$ on $V^{1,0}$ and by $\bar{z}$ on $V^{0,1}$. Then $h(i)$ acts by $i$ and $-i$ on the two summands, so $h(i)$ is precisely the complex structure $J$ on $V_{\mathbb{R}}$ produced from $V^{1,0}$ in exercise 5.1.1 (1). The weight-one dictionary of theorem 5.4.1 therefore reads: the complex torus attached to $V$ is $V_{\mathbb{R}}/V_{\mathbb{Z}}$ with the complex structure $h(i)$.

   For the elliptic curve $E_\tau$ of example 5.4.3, take $V_{\mathbb{Q}} = H^1(E_\tau, \mathbb{Q})$ with the basis $\ell_1, \ell_2$ dual to $1, \tau$, and recall $H^{1,0} = \mathbb{C}\, e$ with $e = \ell_1 + \tau\ell_2$, so $H^{0,1} = \mathbb{C}\,\bar{e}$ with $\bar{e} = \ell_1 + \bar{\tau}\ell_2$. Writing $\tau = s + it$ and $z = x + iy$, the relations $e - \bar{e} = 2it\,\ell_2$ and $\tau\bar{e} - \bar{\tau}e = 2it\,\ell_1$ invert the change of basis, and $h(z)e = ze$, $h(z)\bar{e} = \bar{z}\bar{e}$ give

   $$h(z)\ell_1 = \Big(x - \frac{sy}{t}\Big)\ell_1 - \frac{(s^2+t^2)y}{t}\,\ell_2, \qquad h(z)\ell_2 = \frac{y}{t}\,\ell_1 + \Big(x + \frac{sy}{t}\Big)\ell_2$$

   using $z - \bar{z} = 2iy$, $z\tau - \bar{z}\bar{\tau} = 2i(xt + ys)$ and $\tau\bar{z} - \bar{\tau}z = 2i(tx - sy)$. The matrix of $h(z)$ in the basis $(\ell_1, \ell_2)$ has determinant $x^2 - s^2y^2/t^2 + (s^2+t^2)y^2/t^2 = x^2 + y^2 = \mathrm{Nm}(z)$, as it must, the two eigenvalues being $z$ and $\bar{z}$; and at $z = i$ it squares to $-\mathrm{id}$, exhibiting $h(i)$ as a complex structure on $H^1(E_\tau, \mathbb{R})$.

2. *The Tate structure.* The structure $\mathbb{Q}(m) := \mathbb{Z}(m) \otimes_{\mathbb{Z}} \mathbb{Q}$ of definition 5.1.10 is concentrated in bidegree $(-m, -m)$ and has weight $-2m$, so $h_{\mathbb{Q}(m)}(z)$ is multiplication by $z^{-m}\bar{z}^{-m} = \mathrm{Nm}(z)^{-m}$. Thus $h_{\mathbb{Q}(m)}$ factors through the norm, and the circle $\mathbb{U}$ acts trivially on every Tate structure. This is the precise sense in which a Tate structure carries weight and nothing else.

The group to be constructed will act not only on $V$ but on every space built from $V$ by multilinear algebra, because that is where Hodge classes other than the obvious ones live. The constructions are transparent in the representation picture: apply the operation to the representation of $\mathbb{S}$ and read off the bigrading again.

The tensor constructions on Hodge structures, and the Hodge classes they isolate, were set up in section 5.1: the tensor product and dual of definition 5.1.8, the mixed tensor $T^{a,b,m}(V)$, and $\mathrm{Hdg}(W) = W_{\mathbb{Q}} \cap W^{p,p}$ of definition 5.1.9. They are used here without further comment.

Both constructions satisfy the conjugation symmetry, so they are Hodge structures: under proposition 8.2.2 they are the tensor product and the dual of the corresponding representations of $\mathbb{S}$, and $h_{V\otimes W} = h_V \otimes h_W$, $h_{V^\vee} = (h_V^\vee)^{-1}$ are again defined over $\mathbb{R}$ with the correct weight. The bidegree formulas are then the statement that the characters of a tensor product multiply and those of a dual invert. Definition 5.1.6 and definition 5.1.10 are the cases already in use: a morphism $V \to W$ of Hodge structures is a rational element of $(V^\vee \otimes W)^{0,0}$, and $V(n) = V \otimes \mathbb{Q}(n)$.

Elements of bidegree $(p,p)$ that are rational are the classes the theory exists to detect. On a projective manifold they are the candidates for classes of algebraic subvarieties. In degree 2 this is settled: theorem 4.4.2 says an integral $(1,1)$-class is the first Chern class of a holomorphic line bundle, and on a projective manifold every line bundle is $\mathcal{O}_X(D)$ for a divisor $D$, so the integral $(1,1)$-classes are exactly the divisor classes. In higher degree the corresponding question is open, and is the Hodge conjecture.

Under the Tate twist a Hodge class of $W$ becomes a rational class of bidegree $(0,0)$: the identification $W(p)_\mathbb{Q} = W_\mathbb{Q} \otimes (2\pi i)^p$ of definition 5.1.10 carries $\mathrm{Hdg}(W)$ isomorphically onto $W(p)_\mathbb{Q} \cap W(p)^{0,0}$, and it is this weight-zero form that the group-theoretic description below uses, since bidegree $(0,0)$ is where $h(\mathbb{S})$ acts trivially and bidegree $(p,p)$ with $p \neq 0$ is not. For $X$ a compact Kähler manifold, $\mathrm{Hdg}\big(H^2(X,\mathbb{Q})\big) = \mathrm{NS}(X) \otimes_\mathbb{Z} \mathbb{Q}$ by definition 4.4.3, since a rational class of type $(1,1)$ has an integral multiple in $H^2(X,\mathbb{Z})$; for a K3 surface this is the hyperplane section $\sigma^\perp \cap H^2(X,\mathbb{Q})$ computed in example 5.4.4, of dimension $\rho(X)$.

Nothing so far uses the rational structure beyond the requirement that conjugation be defined. The group is what forces $\mathbb{Q}$ into the picture: it is the smallest algebraic group defined over $\mathbb{Q}$ that carries the action of $\mathbb{S}$. The extra factor $\mathbb{G}_m$ in the ambient group is what lets Tate twists be tracked, so that Hodge classes in every weight are covered at once.

**Definition 8.2.4: The Mumford-Tate Group**

Let $V$ be a rational Hodge structure of weight $k$, with $h\colon \mathbb{S} \to \mathrm{GL}(V_\mathbb{R})$ as in proposition 8.2.2, and consider the homomorphism of real algebraic groups

$$(h, \mathrm{Nm})\colon \mathbb{S} \to \mathrm{GL}(V_\mathbb{R}) \times \mathbb{G}_{m,\mathbb{R}}$$

The *Mumford-Tate group* $\mathrm{MT}(V)$ is the smallest algebraic subgroup $M \subseteq \mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m$ defined over $\mathbb{Q}$ whose group of real points contains $(h,\mathrm{Nm})\big(\mathbb{S}(\mathbb{R})\big)$.

The group exists. The family of algebraic subgroups of $\mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m$ defined over $\mathbb{Q}$ and containing the image in their real points is nonempty, as it contains $\mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m$; an intersection of algebraic subgroups is an algebraic subgroup, and $\big(\bigcap_i M_i\big)(\mathbb{R}) = \bigcap_i M_i(\mathbb{R})$ still contains the image, so the intersection of the whole family is the smallest member. Equivalently, $\mathrm{MT}(V)$ is the Zariski closure over $\mathbb{Q}$ of $(h,\mathrm{Nm})(\mathbb{S}(\mathbb{R}))$: that closure is an algebraic subgroup, because the closure of a subgroup is a subgroup, it is defined over $\mathbb{Q}$ by construction, and it is contained in every member of the family since each is Zariski closed.

The group is nontrivial as soon as $V \neq 0$: the real scalars contribute the one-parameter subgroup $t \mapsto (t^k\,\mathrm{id},\, t^2)$ (exercise 8.2.1 (3)). The second coordinate is not redundant even when $k = 0$, because $\mathrm{Nm}$ is nonconstant while $h$ may be trivial; it is what makes the group act on Tate structures, and hence on every twist. The Tate structure is also the smallest instance of the definition: for $V = \mathbb{Q}(n)$ the homomorphism $h$ is $z \mapsto \mathrm{Nm}(z)^{-n}$ by example 8.2.3, so $(h,\mathrm{Nm})$ has image $\{(s^{-n},\, s) \,|\, s \in \mathbb{R}_{>0}\}$ and $\mathrm{MT}(\mathbb{Q}(n))$ is the one-dimensional torus $\{(t^{-n},\, t) \,|\, t \in \mathbb{G}_m\}$, computed in exercise 8.2.1 (2).

**Theorem 8.2.5: Hodge Classes Are the Mumford-Tate Invariants**

Let $V$ be a rational Hodge structure of weight $k$ and $M = \mathrm{MT}(V)$. Let $M$ act on $T = T^{a,b,m}(V)$ through the algebraic representation

$$(g, \lambda) \longmapsto g^{\otimes a} \otimes \left((g^\vee)^{-1}\right)^{\otimes b} \otimes \lambda^{-m}$$

of $\mathrm{GL}(V_{\mathbb{Q}}) \times \mathbb{G}_m$, defined over $\mathbb{Q}$. Then

$$\left(T_{\mathbb{Q}}\right)^M = T_{\mathbb{Q}} \cap T^{0,0}$$

In particular, if $W = V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ has even weight $2p$, then $\mathrm{Hdg}(W)$ corresponds under the identification $W(p)_{\mathbb{Q}} = W_{\mathbb{Q}} \otimes (2\pi i)^p$ to the space of $M$-invariant vectors of $T^{a,b,p}(V)$: the Hodge classes of the tensor constructions on $V$ are exactly the Mumford-Tate invariants.

***Proof*:**
*Step 1: the two actions of* $\mathbb{S}$ *on* $T$ *agree.* Composing the displayed representation with $(h, \mathrm{Nm})$ gives a homomorphism $\mathbb{S} \to \mathrm{GL}(T_{\mathbb{R}})$. On the factor $V^{\otimes a}$ it is $h^{\otimes a}$, on $(V^\vee)^{\otimes b}$ it is $\left((h^\vee)^{-1}\right)^{\otimes b}$, and on $\mathbb{Q}(m)$ it is $\mathrm{Nm}^{-m}$, which is the homomorphism $h_{\mathbb{Q}(m)}$ of example 8.2.3. By definition 5.1.8 these are precisely the homomorphisms attached by proposition 8.2.2 to the tensor product, the dual and the Tate structure, so the composite is $h_T$, the homomorphism of the Hodge structure $T$.
*Step 2: an invariant is of bidegree* $(0,0)$. Let $t \in T_{\mathbb{Q}}$ be fixed by $M$. Since $(h, \mathrm{Nm})(\mathbb{S}(\mathbb{R})) \subseteq M(\mathbb{R})$, Step 1 gives $h_T(z)t = t$ for every $z \in \mathbb{C}^\times$. Decomposing $t = \sum_{p,q} t^{p,q}$ into bihomogeneous components and using that $h_T(z)$ acts on $T^{p,q}$ by $z^p \bar{z}^q$,

$$\sum_{p,q} z^p \bar{z}^q \, t^{p,q} = \sum_{p,q} t^{p,q} \qquad \text{for all } z \in \mathbb{C}^\times$$

The components lie in independent summands, so $z^p \bar{z}^q \, t^{p,q} = t^{p,q}$ for every $z$ and every $(p,q)$. Suppose $t^{p,q} \neq 0$; then $z^p \bar{z}^q = 1$ for all $z \in \mathbb{C}^\times$. Taking $z = r$ real and positive gives $r^{p+q} = 1$ for all $r > 0$, hence $p+q = 0$; taking $z = e^{i\theta}$ gives $e^{i(p-q)\theta} = 1$ for all $\theta$, hence $p = q$. The two together give $p = q = 0$. So every component with $(p,q) \neq (0,0)$ vanishes and $t = t^{0,0} \in T_{\mathbb{Q}} \cap T^{0,0}$.
*Step 3: a rational class of bidegree* $(0,0)$ *is invariant.* Let $t \in T_{\mathbb{Q}} \cap T^{0,0}$ and let

$$M_t = \{(g, \lambda) \in \mathrm{GL}(V_{\mathbb{Q}}) \times \mathbb{G}_m \big| (g, \lambda) \cdot t = t\}$$

be its stabilizer. The action is an algebraic representation defined over $\mathbb{Q}$ and $t$ is a rational vector, so the conditions defining $M_t$ are polynomial equations with rational coefficients: $M_t$ is an algebraic subgroup defined over $\mathbb{Q}$ ([11, stabilizers of vectors are closed subgroups]). Its real points contain $(h, \mathrm{Nm})(\mathbb{S}(\mathbb{R}))$, because by Step 1 the corresponding operator on $T_{\mathbb{R}}$ is $h_T(z)$, which acts on $T^{0,0}$ by $z^0 \bar{z}^0 = 1$. Definition 8.2.4 is a minimality statement, so $M \subseteq M_t$, that is $t$ is fixed by $M$.
*Step 4: the twisted form of the statement.* Let $W = V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ have weight $2p$. Then $T^{a,b,p}(V) = W(p)$ has weight 0, and $W(p)^{0,0} = W^{p,p}$ by definition 5.1.10, while $W(p)_{\mathbb{Q}} = W_{\mathbb{Q}} \otimes (2\pi i)^p$. Under this identification $W(p)_{\mathbb{Q}} \cap W(p)^{0,0}$ is $\mathrm{Hdg}(W)$, and Steps 2 and 3 identify it with the $M$-invariants of $T^{a,b,p}(V)$, as we sought to show.

The theorem is what the definition was arranged to produce. A Hodge class is defined by a transcendental condition, membership in a subspace of $V_\mathbb{C}$ cut out by the complex structure, together with a rationality condition; the theorem converts the pair into a single algebraic condition, invariance under a group defined over $\mathbb{Q}$. It also reverses the reading: a Hodge structure with many Hodge classes has a small Mumford-Tate group, since each independent class imposes an equation on $M$, and a Hodge structure whose only Hodge classes are the ones present for formal reasons has $M$ as large as the polarization allows. This is the mechanism behind the loci studied in section 8.3: the Hodge classes jump exactly where the Mumford-Tate group drops.

The invariance in Step 3 is available only in bidegree $(0,0)$, and a Hodge class of a structure of weight $2p$ with $p \neq 0$ is scaled by $h(z)$, not fixed by it. The Tate factor $\mathbb{Q}(m)$ is exactly the bookkeeping that repairs this, and it is why the ambient group carries the second coordinate.

Nothing so far has used a polarization. A polarization constrains the group in two ways at once: it confines it to a similitude group, and it makes the representation on $V_\mathbb{Q}$ semisimple, which is what forces reductivity.

**Proposition 8.2.6: The Mumford-Tate Group of a Polarized Structure**

Let $V$ be a rational Hodge structure of weight $k$ and $M = \mathrm{MT}(V)$.

1. $M$ is connected.
2. Suppose $Q$ is a polarization of $V$ (definition 5.2.1). Then $M$ is contained in the group of similitudes

$$G_Q = \big\{(g,\lambda) \in \mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m \ \big|\ Q(gu,gv) = \lambda^k\, Q(u,v) \text{ for all } u,v\big\}$$

and the $M$-stable $\mathbb{Q}$-subspaces of $V_\mathbb{Q}$ are exactly the *sub-Hodge structures* of $V$, that is the $\mathbb{Q}$-subspaces $W$ for which $W_\mathbb{C} = \bigoplus_{p+q=k}(W_\mathbb{C} \cap V^{p,q})$. Each such subspace has an $M$-stable complement, so $V_\mathbb{Q}$ is a semisimple representation of $M$.

3. Under the hypothesis of (2), $M$ is reductive.

***Proof*:**

1. Let $M^\circ \subseteq M$ be the identity component. It is an algebraic subgroup defined over $\mathbb{Q}$, and it is both Zariski open and Zariski closed in $M$, hence $M^\circ(\mathbb{R})$ is open and closed in $M(\mathbb{R})$ for the real topology. The set $(h,\mathrm{Nm})(\mathbb{S}(\mathbb{R}))$ is the continuous image of $\mathbb{C}^\times$, hence connected, and it contains the identity, so it lies in the connected component of the identity of $M(\mathbb{R})$ and therefore in $M^\circ(\mathbb{R})$. By the minimality in definition 8.2.4, $M \subseteq M^\circ$, so $M = M^\circ$ is connected. This argument uses no polarization.

2. *The similitude group.* That $G_Q$ is an algebraic subgroup defined over $\mathbb{Q}$ is immediate, the defining conditions being polynomial identities with rational coefficients, and it is a subgroup because $Q(g_1g_2u, g_1g_2v) = \lambda_1^k\lambda_2^k Q(u,v)$. Its real points contain the image of $(h,\mathrm{Nm})$: for $u \in V^{p,q}$ and $v \in V^{p',q'}$ the first Hodge-Riemann relation makes $Q(u,v)$ vanish unless $(p',q') = (q,p)$, and in that case

$$Q\big(h(z)u, h(z)v\big) = z^p\bar{z}^q\, z^q\bar{z}^p\, Q(u,v) = (z\bar{z})^{p+q}Q(u,v) = \mathrm{Nm}(z)^k\, Q(u,v)$$

When the bidegrees do not pair, both sides vanish, since $h(z)$ preserves each summand and so leaves bidegrees unchanged. The identity therefore holds on a spanning set of pairs and hence everywhere. By minimality $M \subseteq G_Q$.

*Stable subspaces.* Let $W \subseteq V_{\mathbb{Q}}$ be $M$-stable. Then $W_{\mathbb{R}}$ is stable under $h(\mathbb{S}(\mathbb{R}))$, so $W_{\mathbb{C}}$ is stable under the action of $\mathbb{S}_{\mathbb{C}}$ and therefore decomposes into character eigenspaces, $W_{\mathbb{C}} = \bigoplus_{p+q=k}(W_{\mathbb{C}} \cap V^{p,q})$; and $W_{\mathbb{C}}$ is stable under conjugation, being defined over $\mathbb{Q}$, so the symmetry $\overline{W^{p,q}} = W^{q,p}$ is inherited. Thus $W$ is a sub-Hodge structure. Conversely, let $W$ be a sub-Hodge structure. The set of $(g, \lambda)$ with $gW \subseteq W$ is an algebraic subgroup defined over $\mathbb{Q}$, and its real points contain the image of $(h, \mathrm{Nm})$ because $h(z)$ preserves each $W^{p,q}$; minimality gives $M \subseteq$ that subgroup, so $W$ is $M$-stable.

*Complements.* Let $W \subseteq V_{\mathbb{Q}}$ be $M$-stable, hence a sub-Hodge structure. The restriction $Q|_W$ satisfies both conditions of definition 5.2.1 for $W$, these being conditions on the summands $W^{p,q} \subseteq V^{p,q}$ inherited from $V$, so $Q|_W$ is a polarization of $W$. It is nondegenerate: the Weil operator $C$ acts on $V^{p,q}$ by $i^{p-q}$ and so preserves $W_{\mathbb{C}}$, and being defined over $\mathbb{R}$ it preserves $W_{\mathbb{R}}$; by part (3) of theorem 5.2.3 the form $S(u,v) = Q(Cu, v)$ is positive definite on $W_{\mathbb{R}}$, and $Q(u,v) = S(C^{-1}u, v)$, so $Q(u, W_{\mathbb{R}}) = 0$ forces $C^{-1}u = 0$ and hence $u = 0$ for $u \in W_{\mathbb{R}}$. Therefore $V_{\mathbb{Q}} = W \oplus W^{\perp}$ with $W^{\perp}$ the $Q$-orthogonal complement. That complement is $M$-stable: if $Q(W, x) = 0$ and $(g, \lambda) \in M \subseteq G_Q$, then for $w \in W$

$$Q(w, gx) = \lambda^k \, Q(g^{-1}w, x) = 0$$

since $g^{-1}w \in W$. Every $M$-stable subspace therefore has an $M$-stable complement, which is semisimplicity of $V_{\mathbb{Q}}$ as an $M$-representation.

3. The representation of $M$ on $V_{\mathbb{Q}} \oplus \mathbb{Q}(1)$, with $M$ acting on the second summand through $(g, \lambda) \mapsto \lambda^{-1}$, is faithful, since $(g, \lambda)$ acting trivially on both summands forces $g = \mathrm{id}$ and $\lambda = 1$. It is semisimple: $V_{\mathbb{Q}}$ is semisimple by part (2) and $\mathbb{Q}(1)$ is one-dimensional, hence irreducible.

   Let $U$ be the unipotent radical of $M$, the largest connected normal unipotent subgroup, formed after extension of scalars to $\overline{\mathbb{Q}}$ ([11, the unipotent radical, and reductive groups]); by part (1) the group $M$ is connected, so it is reductive precisely when $U$ is trivial. Since $U$ is canonically attached to $M_{\overline{\mathbb{Q}}}$ and $M$ is defined over $\mathbb{Q}$, the subgroup $U$ is stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, hence so is the fixed space $\big((V_{\mathbb{Q}} \oplus \mathbb{Q}(1)) \otimes \overline{\mathbb{Q}}\big)^U$. A subspace of a base-changed space that is stable under the semilinear Galois action is spanned by its Galois-invariant vectors, so that fixed space is $N \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ for a unique $\mathbb{Q}$-subspace $N \subseteq V_{\mathbb{Q}} \oplus \mathbb{Q}(1)$, and $N$ is $M$-stable because $U$ is normal in $M$.

   Suppose $N \neq V_{\mathbb{Q}} \oplus \mathbb{Q}(1)$. By semisimplicity there is a nonzero $M$-stable complement $N'$, and $N \cap N' = 0$. A unipotent group has nonzero invariants in every nonzero representation ([11, the equivalent characterizations of unipotent groups], applied over the algebraically closed field $\overline{\mathbb{Q}}$), so $(N' \otimes \overline{\mathbb{Q}})^U \neq 0$; but that space is contained in $N \otimes \overline{\mathbb{Q}}$, which meets $N' \otimes \overline{\mathbb{Q}}$ in 0. This contradiction gives $N = V_{\mathbb{Q}} \oplus \mathbb{Q}(1)$, so $U$ acts trivially on a faithful representation of $M$ and is therefore trivial. Hence $M$ is reductive, completing the proof.

Reductivity is what makes the group usable. It places $\mathrm{MT}(V)$ among the groups classified by root data ([11, the classification of reductive groups by root data]), so a polarized Hodge structure can be studied through the representation theory of a group whose structure is known, and the computation in example 8.2.9 below is exactly such an argument: it identifies the group by counting dimensions in the reductive structure theory. The polarization is what supplies it. Part (3) rests on semisimplicity of $V_{\mathbb{Q}}$, and semisimplicity in turn rests on the $Q$-orthogonal complement of part (2); without a form there is no canonical complement to a sub-Hodge structure and the argument has no starting point.

The similitude factor in part (2) suggests separating the part of the group that preserves $Q$ on the nose. That part is the group the classical literature calls the Hodge group, and it is what remains after the homotheties are removed.

**Definition 8.2.7: The Hodge Group**

Let $V$ be a rational Hodge structure of weight $k$ with $h\colon \mathbb{S} \to \mathrm{GL}(V_{\mathbb{R}})$, and let $\mathbb{U} = \ker(\mathrm{Nm}) \subseteq \mathbb{S}$ be the circle of definition 8.2.1. The *Hodge group*, or *special Mumford-Tate group*, $\mathrm{Hg}(V)$ is the smallest algebraic subgroup of $\mathrm{GL}(V_{\mathbb{Q}})$ defined over $\mathbb{Q}$ whose real points contain $h\big(\mathbb{U}(\mathbb{R})\big)$.

The group exists by the argument given after definition 8.2.4, and it is connected by the argument of proposition 8.2.6 part (1), the circle $\mathbb{U}(\mathbb{R})$ being connected. If $Q$ polarizes $V$ then $\mathrm{Hg}(V)$ preserves $Q$ exactly rather than up to a factor: the computation in proposition 8.2.6 gives $Q(h(u)x, h(u)y) = \mathrm{Nm}(u)^k Q(x,y) = Q(x,y)$ for $u \in \mathbb{U}(\mathbb{R})$, so the automorphism group of $(V_{\mathbb{Q}}, Q)$ is one of the subgroups over which the minimum is taken.

**Proposition 8.2.8: The Mumford-Tate Group as Hodge Group Times Homotheties**

Let $V$ be a rational Hodge structure of weight $k$, let $M = \mathrm{MT}(V)$ and let $\pi\colon \mathrm{GL}(V_{\mathbb{Q}}) \times \mathbb{G}_m \to \mathrm{GL}(V_{\mathbb{Q}})$ be the first projection. Then $\pi(M)$ is the smallest algebraic subgroup of $\mathrm{GL}(V_{\mathbb{Q}})$ defined over $\mathbb{Q}$ whose real points contain $h(\mathbb{S}(\mathbb{R}))$, and if $k \neq 0$ then

$$\pi(M) = \mathrm{Hg}(V) \cdot \mathbb{G}_m \,\mathrm{id}$$

the product inside $\mathrm{GL}(V_{\mathbb{Q}})$ of the Hodge group with the group of homotheties.

***Proof*:**
*Step 1: $\pi(M)$ is the minimal group for $h$.* The image of an algebraic group under a homomorphism of algebraic groups is an algebraic subgroup defined over the same field, so $\pi(M)$ is one; its real points contain $\pi\big((h, \mathrm{Nm})(\mathbb{S}(\mathbb{R}))\big) = h(\mathbb{S}(\mathbb{R}))$. Conversely let $N \subseteq \mathrm{GL}(V_{\mathbb{Q}})$ be an algebraic subgroup over $\mathbb{Q}$ with $h(\mathbb{S}(\mathbb{R})) \subseteq N(\mathbb{R})$. Then $N \times \mathbb{G}_m$ is an algebraic subgroup of $\mathrm{GL}(V_{\mathbb{Q}}) \times \mathbb{G}_m$ whose real points contain the image of $(h, \mathrm{Nm})$, so $M \subseteq N \times \mathbb{G}_m$ by definition 8.2.4 and therefore $\pi(M) \subseteq N$.
*Step 2: the homotheties lie in $\pi(M)$.* For $r \in \mathbb{R}^\times$ one has $h(r) = r^k \,\mathrm{id}$ by proposition 8.2.2, so $\pi(M)(\mathbb{R})$ contains the set $\{r^k \,\mathrm{id} | r \in \mathbb{R}^\times\}$, which is infinite because $k \neq 0$. The homotheties form a subgroup of $\mathrm{GL}(V_{\mathbb{Q}})$ isomorphic to $\mathbb{G}_m$, and $\pi(M) \cap \mathbb{G}_m \,\mathrm{id}$ is an algebraic subgroup of it containing that infinite set. A proper algebraic subgroup of $\mathbb{G}_m$ is finite, being the zero set of a nonzero polynomial in one variable, so $\pi(M) \supseteq \mathbb{G}_m \,\mathrm{id}$.
*Step 3: the two inclusions.* The group $\mathrm{Hg}(V) \cdot \mathbb{G}_m \,\mathrm{id}$ is an algebraic subgroup defined over $\mathbb{Q}$, the homotheties being central. Every $z \in \mathbb{C}^\times$ factors as $z = ru$ with $r = |z| \in \mathbb{R}_{>0}$ and $u \in \mathbb{U}(\mathbb{R})$, so

$$h(z) = h(r)h(u) = r^k \, h(u) \in \big(\mathbb{G}_m \,\mathrm{id} \cdot \mathrm{Hg}(V)\big)(\mathbb{R})$$

By Step 1 this gives $\pi(M) \subseteq \mathrm{Hg}(V) \cdot \mathbb{G}_m \,\mathrm{id}$. In the other direction, $h(\mathbb{U}(\mathbb{R})) \subseteq h(\mathbb{S}(\mathbb{R})) \subseteq \pi(M)(\mathbb{R})$, so $\mathrm{Hg}(V) \subseteq \pi(M)$ by definition 8.2.7, and $\mathbb{G}_m \,\mathrm{id} \subseteq \pi(M)$ by Step 2. The two inclusions give the equality, as we sought to show.

The proposition says the two groups differ only by the homotheties, so either can be used and the choice is a matter of which bookkeeping is wanted. The Mumford-Tate group is the one adapted to Hodge classes in all weights, since it carries the Tate twist; the Hodge group is the one adapted to a fixed polarized structure, since it lands in the automorphism group of $(V_{\mathbb{Q}}, Q)$ and is therefore visibly a subgroup of a classical group. The construction is the point of departure for Shimura data, where a period domain is replaced by the $G_{\mathbb{R}}$-conjugacy class of a homomorphism $\mathbb{S} \to G_{\mathbb{R}}$ ([10]); the standard reference for the Hodge-theoretic side is Deligne's *Hodge Cycles on Abelian Varieties* [12].

The smallest interesting case is weight one and rank two, where the whole group can be computed and the answer separates two classes of elliptic curves.

**Example 8.2.9: The Mumford-Tate Group of an Elliptic Curve**

Let $E_\tau = \mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}\tau)$ be the elliptic curve of example 5.4.3 and let $V$ be the weight-one Hodge structure on $V_{\mathbb{Q}} = H^1(E_\tau, \mathbb{Q})$, polarized by the cup form $Q$ as computed there. Here $\dim_{\mathbb{Q}} V_{\mathbb{Q}} = 2$ and $k = 1$.

*The ambient group collapses.* A nonzero alternating form on a two-dimensional space satisfies $Q(gu, gv) = \det(g)\, Q(u, v)$, so the similitude group of proposition 8.2.6 is

$$G_Q = \{(g, \lambda) \big| \lambda = \det g\} \cong \mathrm{GL}(V_{\mathbb{Q}})$$

Since $M := \mathrm{MT}(V) \subseteq G_Q$, the projection $\pi$ of proposition 8.2.8 is injective on $M$, and $M$ may be identified with the subgroup $M' = \pi(M) \subseteq \mathrm{GL}(V_{\mathbb{Q}})$, which by proposition 8.2.8 contains the homotheties.

*The endomorphism algebra.* By theorem 8.2.5 applied to $\mathrm{End}(V) = V \otimes V^\vee = T^{1,1,0}(V)$, a structure of weight 0 on which $(g, \lambda)$ acts by $f \mapsto gfg^{-1}$,

$$\mathrm{Hdg}\big(\mathrm{End}(V)\big) = \mathrm{End}_M(V_{\mathbb{Q}}) = \{f \in \mathrm{End}_{\mathbb{Q}}(V_{\mathbb{Q}}) \big| fg = gf \text{ for all } g \in M'\}$$

while the left-hand side is the algebra of endomorphisms of the Hodge structure $V$, since a rational endomorphism of bidegree $(0, 0)$ is exactly one preserving every $V^{p,q}$ (definition 5.1.6). That algebra is the opposite of the endomorphism algebra of the weight-one structure $\Lambda$ on $H_1(E_\tau, \mathbb{Q})$. The passage from one to the other is duality followed by a Tate twist and a conjugation,

$$H^1(E_\tau, \mathbb{Q}) \cong \overline{\Lambda^\vee(-1)}$$

where $\overline{W}$ denotes the Hodge structure carried by $W_{\mathbb{Q}}$ with $\overline{W}^{p,q} = W^{q,p}$. The conjugation is not optional: under the period identification $H^1(E_\tau, \mathbb{C}) = \mathrm{Hom}_{\mathbb{R}}\big(H_1(E_\tau, \mathbb{R}), \mathbb{C}\big)$ the generator $dz$ of $H^{1,0}$ is the identity map of $H_1(E_\tau, \mathbb{R}) = \mathbb{C}$, which is $\mathbb{C}$-linear and so annihilates the $(-i)$-eigenspace $\Lambda^{0,1}$ of multiplication by $i$; hence $H^{1,0}$ is the summand of bidegree $(-1, 0)$ of $\Lambda^\vee$ by definition 5.1.8, and a Tate twist, which shifts $p$ and $q$ equally, can never move it to bidegree $(1, 0)$. Neither the twist nor the conjugation changes which $\mathbb{Q}$-linear maps are morphisms of Hodge structures, and it is the dual alone that reverses the order of composition; the opposite is in any case irrelevant here because both possible answers are commutative. By theorem 5.4.1 the endomorphisms of $\Lambda$ are the holomorphic endomorphisms of $E_\tau$, so the algebra is $\mathrm{End}(E_\tau) \otimes_{\mathbb{Z}} \mathbb{Q}$, computed in exercise 5.4.1 (3) to be $\mathbb{Q}$ unless $\tau$ satisfies a quadratic equation over $\mathbb{Q}$, in which case it is the imaginary quadratic field $K = \mathbb{Q}(\tau)$.

*The generic case.* Suppose $\mathrm{End}(E_\tau) \otimes \mathbb{Q} = \mathbb{Q}$, so $\mathrm{End}_M(V_{\mathbb{Q}}) = \mathbb{Q}$. If $M'$ were commutative then every element of $M'(\overline{\mathbb{Q}})$ would lie in the commutant of $M'_{\overline{\mathbb{Q}}}$, which is $\mathrm{End}_M(V_{\mathbb{Q}}) \otimes \overline{\mathbb{Q}} = \overline{\mathbb{Q}}$ because the commutant is the solution space of a linear system and so commutes with extension of scalars;

then $M'$ would consist of homotheties, contradicting that $h(i)$ has the two distinct eigenvalues $i$ and $-i$ (example 8.2.3). So $M'$ is not commutative and its derived subgroup $[M', M']$ is nontrivial. By proposition 8.2.6 the group $M'$ is connected and reductive, so the structure theory of reductive groups ([11, the structure of reductive groups]) makes $[M', M']$ semisimple and $R(M') \cap [M', M']$ finite, whence

$$\dim M' \geq \dim R(M') + \dim[M', M']$$

Now the radical of a connected reductive group is the identity component of its center, which contains the homotheties, so $\dim R(M') \geq 1$, and a nontrivial connected semisimple group has dimension at least 3, the minimum being attained by the rank-one groups $\mathrm{SL}_2$ and $\mathrm{PGL}_2$ ([11, the classification of reductive groups by root data]). Hence $\dim M' \geq 4 = \dim \mathrm{GL}(V_{\mathbb{Q}})$, and a connected algebraic subgroup of full dimension in the connected group $\mathrm{GL}(V_{\mathbb{Q}})$ is the whole group:

$$\mathrm{MT}(V) \cong \mathrm{GL}(V_{\mathbb{Q}}) \cong \mathrm{GL}_2$$

*The complex multiplication case.* Suppose instead $\mathrm{End}(E_\tau) \otimes \mathbb{Q} = K$, an imaginary quadratic field. Then $V_{\mathbb{Q}}$ is a $K$-vector space of dimension 1, so the centralizer of $K$ inside $\mathrm{End}_{\mathbb{Q}}(V_{\mathbb{Q}})$ is $K$ itself, and the centralizer of $K^\times$ inside $\mathrm{GL}(V_{\mathbb{Q}})$ is the two-dimensional $\mathbb{Q}$-torus

$$T_K = \mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m, \qquad T_K(\mathbb{Q}) = K^\times$$

Every element of $K = \mathrm{End}_{\mathrm{HS}}(V)$ commutes with $h(z)$, since a morphism of Hodge structures preserves each $V^{p,q}$ and $h(z)$ is a scalar there; so $h(\mathbb{S}(\mathbb{R})) \subseteq T_K(\mathbb{R})$ and hence $M' \subseteq T_K$ by minimality. In fact $M' = T_K$. The map $h$ is injective, its value at $z$ having eigenvalues $z$ and $\bar{z}$, so $h(\mathbb{S}(\mathbb{R}))$ is a two-dimensional subgroup of $T_K(\mathbb{R}) = (K \otimes_{\mathbb{Q}} \mathbb{R})^\times \cong \mathbb{C}^\times$, which is connected of real dimension 2; a subgroup of full dimension is open, and an open subgroup of a connected topological group is the whole group, so $h(\mathbb{S}(\mathbb{R})) = T_K(\mathbb{R})$. Any algebraic subgroup $N \subseteq T_K$ over $\mathbb{Q}$ with $N(\mathbb{R}) \supseteq T_K(\mathbb{R})$ has $\dim N \geq \dim_{\mathbb{R}} N(\mathbb{R}) \geq 2$, while $N \subseteq T_K$ gives $\dim N \leq \dim T_K = 2$; so its identity component is a closed connected subgroup of the connected group $T_K$ of the same dimension and therefore equals $T_K$, whence $N = T_K$. Therefore

$$\mathrm{MT}(V) \cong T_K = \mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m$$

a torus of dimension 2.

*Reading the two answers.* The Hodge classes count the difference. In the generic case $\mathrm{Hdg}(\mathrm{End}V) = \mathbb{Q}$ is one-dimensional and the group is all of $\mathrm{GL}_2$, of dimension 4; in the complex multiplication case $\mathrm{Hdg}(\mathrm{End}V) = K$ is two-dimensional and the group has collapsed to a torus of dimension 2. The extra Hodge class, the endomorphism supplied by complex multiplication, is precisely the equation that cuts $\mathrm{GL}_2$ down to its maximal torus $T_K$, which is theorem 8.2.5 read from right to left.

## Exercise 8.2.1

1. Let $V$ be a rational Hodge structure of weight $k$ with associated $h\colon \mathbb{S} \to \mathrm{GL}(V_{\mathbb{R}})$. Show that $h(-1) = (-1)^k\,\mathrm{id}$ and that $h(i)$ is the Weil operator $C$ of definition 5.2.1. Deduce that $C^2 = (-1)^k\,\mathrm{id}$ and that $C$ lies in $\mathrm{MT}(V)(\mathbb{R})$ up to the second coordinate, that is $(C, 1) \in \mathrm{MT}(V)(\mathbb{R})$.
2. Compute $\mathrm{MT}\big(\mathbb{Q}(n)\big)$ for every $n \in \mathbb{Z}$, as a subgroup of $\mathrm{GL}_1 \times \mathbb{G}_m = \mathbb{G}_m \times \mathbb{G}_m$. Treat $n = 0$

separately and explain why the answer is not the trivial group.

3. Let $V \neq 0$ be a rational Hodge structure of weight $k$. Show that $\mathrm{MT}(V)$ contains the image of the homomorphism $\mathbb{G}_m \to \mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m$, $t \mapsto (t^k \,\mathrm{id},\, t^2)$, and that this image is a one-dimensional torus. Conclude that $\mathrm{MT}(V)$ is never trivial.

4. Let $X$ be a projective K3 surface and $V$ the weight-two Hodge structure on $H^2(X, \mathbb{Q})$ of example 5.4.4. Show that $\mathrm{Hdg}(V) = \mathrm{NS}(X) \otimes_\mathbb{Z} \mathbb{Q}$, of dimension $\rho(X)$, and identify this space as the fixed space of $\mathrm{MT}(V)$ acting on $V(1)$. Deduce that the Picard number is determined by the Mumford-Tate group together with the action of the group on $V$.

5. For the elliptic curve $E_\tau$ of example 8.2.9, compute $\mathrm{Hg}(V)$ in both cases: show that it is $\mathrm{SL}(V_\mathbb{Q})$ when $E_\tau$ has no complex multiplication, and the norm-one torus $T_K^1 = \ker\big(\mathrm{Nm}_{K/\mathbb{Q}} \colon T_K \to \mathbb{G}_m\big)$ when it has complex multiplication by $K$. (Use proposition 8.2.8 and the fact that $\det h(u) = \mathrm{Nm}(u)$ for $u \in \mathbb{U}(\mathbb{R})$.)

6. Let $V$ be a rational Hodge structure and $n \in \mathbb{Z}$, and let $\alpha$ be the automorphism of $\mathrm{GL}(V_\mathbb{Q}) \times \mathbb{G}_m$ given by $\alpha(g, \lambda) = (\lambda^{-n} g,\ \lambda)$. Show that $\mathrm{MT}\big(V(n)\big) = \alpha\big(\mathrm{MT}(V)\big)$ under the canonical identification $\mathrm{GL}\big(V(n)_\mathbb{Q}\big) = \mathrm{GL}(V_\mathbb{Q})$; in particular the two groups are isomorphic, so the Mumford-Tate group is insensitive to Tate twists up to isomorphism.

## 8.3 Hodge Loci and Algebraicity

Section 8.2 read the Hodge classes of a single structure off a group. In a family they are not a fixed supply. A rational class of type $(p, p)$ on one fibre transports flatly to a rational class on every nearby fibre, but the type is not carried along, and the set of parameters at which it survives is the object of this section.

Throughout, $f \colon \mathcal{X} \to S$ is a smooth projective family (definition 7.1.1) with fibres $X_s$, and $\mathbb{H}^k$, $\mathcal{H}^k$, $\nabla$, $F^\bullet\mathcal{H}^k$ are the local system, the cohomology bundle, the Gauss-Manin connection and the Hodge bundles attached to $f$ in section 7.1 and section 7.2. The degree is even, $k = 2p$, and it has to be: conjugation fixes a real class and exchanges $V^{a,b}$ with $V^{b,a}$ (definition 5.1.1), so a nonzero real class of pure type $(a, b)$ has $a = b$.

A rational class is real, and reality collapses the two ways of asking that it be of type $(p, p)$ into one. If $\mu \in H^{2p}(X_s, \mathbb{Q})$ lies in $F^p H^{2p}(X_s, \mathbb{C})$ then $\mu = \bar{\mu}$ lies in $\overline{F^p}$ as well, so $\mu \in F^p \cap \overline{F^p} = H^{p,p}(X_s)$ by the recovery formula (5.2); the converse inclusion $H^{p,p} \subseteq F^p$ is immediate from definition 5.1.4. So for a rational class the two conditions agree, and either of them says that $\mu$ is a Hodge class of $H^{2p}(X_s, \mathbb{Q})$ in the sense of definition 5.1.9. The condition of interest is therefore a condition on the position of $\mu$ in the *filtration*, which is the holomorphic datum, and not on its position in the bigrading, which is not.

**Definition 8.3.1: The Hodge Locus**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family, let $U \subseteq S$ be open, connected and simply connected, and let $\lambda$ be a nonzero flat section of $\mathbb{H}^{2p}$ over $U$. The *Hodge locus* of $\lambda$ is

$$\mathrm{NL}(\lambda) = \left\{\, s \in U \;\middle|\; \lambda_s \in F^p H^{2p}(X_s, \mathbb{C}) \,\right\}$$

the set of parameters at which $\lambda_s$ is a Hodge class (definition 5.1.9). For $p = 1$ it is also called the *Noether-Lefschetz locus* of $\lambda$.

Three features of the definition are worth separating. The section $\lambda$ is flat, so as a class it does not move at all: what moves is the filtration, and the locus records where the moving filtration catches a fixed vector. Simple connectivity of $U$ is what makes $\lambda$ exist, since a flat section over a base with monodromy need not be single valued (proposition 7.1.5); the locus is therefore local by construction, and assembling a global object from the local ones is done in corollary 8.3.4. And rationality of $\lambda$ is not decoration. Without it the condition $\lambda_s \in F^p$ says nothing about the bigrading, and the whole subject would be the study of one subbundle meeting one line.

Everything below applies verbatim with $\mathbb{H}^{2p}$ replaced by its primitive part $\mathbb{H}^{2p}_{\mathrm{prim}}$ of definition 7.3.1 and $F^\bullet\mathcal{H}^{2p}$ by the primitive filtration $F^\bullet_{\mathrm{prim}}$, part (3) of proposition 7.3.2 supplying there the holomorphy and the transversality that theorem 7.2.6 and theorem 7.2.7 supply for the full bundle. That is the version example 8.3.7 uses, and in it the locus is a condition on the period map. Fixing a path class as in definition 7.3.3 trivializes $\mathbb{H}^{2p}_{\mathrm{prim}}$ over $U$, carrying $\lambda$ to a fixed rational vector $v$ of the reference cohomology $H^{2p}_{\mathrm{prim}}(X_{s_0}, \mathbb{Q})$ and $s$ to the point $\Phi_{[c]}(s)$ of the period domain $D$; then $\mathrm{NL}(\lambda)$ is the preimage under $\Phi_{[c]}$ of the subset $\{\, F^\bullet \in D \,|\, v \in F^p \,\}$.

The definition permits the degenerate answer, and it is worth seeing which families force it before seeing which do not.

**Example 8.3.2: Classes That Are Hodge at Every Parameter**

Two mechanisms make $\mathrm{NL}(\lambda)$ all of $U$.

*The polarization class.* Let $h \in H^2(\mathcal{X}, \mathbb{Z})$ be the relative hyperplane class of section 7.3, with restrictions $h_s = h|_{X_s}$, and fix $p \le n = \dim_{\mathbb{C}} X_s$. Part (1) of proposition 7.3.2 shows that parallel transport carries $h_{s_0}$ to $h_s$ and commutes with cup product, so $s \mapsto h_s^p$ is a flat section of $\mathbb{H}^{2p}$. It is a Hodge class at every $s$: the class $h_s$ is that of the Fubini-Study Kähler form of $X_s$ (proposition 3.2.6), hence real of type $(1,1)$, and a product of $p$ classes of type $(1,1)$ has type $(p,p)$. It is nonzero for $p \le n$, since $\int_{X_s} h_s^n > 0$. So $\mathrm{NL}(h^p) = U$ for every $U$.

*Absence of room.* Take $p = 1$ and let $X \subseteq \mathbb{C}\mathrm{P}^3$ be a smooth surface of degree $d$, so $n = 2$. The class $h_X$ is of type $(1,1)$, so $H^{2,0}(X) = H^{2,0}_{\mathrm{prim}}(X)$, and by theorem 7.4.11 the latter is isomorphic to $R_{d-4}$, the degree-$(d-4)$ piece of the Jacobian ring of definition 7.4.1. The Jacobian ideal is generated in degree $d-1$, so $R_e = \mathbb{C}[x]_e$ for $e \le d-2$, and $\mathbb{C}[x]_e = 0$ for $e < 0$; hence

$$h^{2,0}(X) = \dim_{\mathbb{C}} \mathbb{C}[x]_{d-4} = \binom{d-1}{3}$$

which vanishes exactly when $d \le 3$. For those degrees $F^1\mathcal{H}^2 = \mathcal{H}^2$ by definition 7.2.1, so *every* rational class on every fibre is a Hodge class and every Hodge locus is all of $U$. Concretely $b_2 = 1 + h^{1,1}_{\mathrm{prim}} = 1 + \dim R_{2d-4}$ by corollary 7.4.12, the 1 counting $h_X$, which gives $b_2 = 2$ for a quadric and $b_2 = 7$ for a cubic surface; the Néron-Severi group of definition 4.4.3 is all of $H^2(X, \mathbb{Z})$

in both cases, so the Picard number equals $b_2$. The first degree with $h^{2,0} \neq 0$ is $d = 4$, where $h^{2,0} = 1$.

The mechanism behind the second case is the one that governs the general situation. The condition $\lambda_s \in F^p$ is the vanishing of the image of $\lambda_s$ in the quotient $\mathcal{H}^{2p}/F^p\mathcal{H}^{2p}$, and that quotient is a holomorphic bundle whose rank counts exactly the room the class has to fail. A vanishing condition on a holomorphic section is an analytic condition, and its derivative is computed by the connection.

**Proposition 8.3.3: The Hodge Locus Is a Closed Analytic Subset**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family, fix $p \geq 1$, let $U \subseteq S$ be open, connected and simply connected, and let $\lambda$ be a nonzero flat section of $\mathbb{H}^{2p}$ over $U$. Write

$$\mathcal{Q} = \mathcal{H}^{2p}/F^p\mathcal{H}^{2p}, \qquad c = \operatorname{rank}\mathcal{Q} = \sum_{r<p} h^{r,\,2p-r}$$

and let $\bar{\lambda}$ be the image of $\lambda$ under the quotient map $\pi\colon \mathcal{H}^{2p} \to \mathcal{Q}$.

1. $\bar{\lambda}$ is a holomorphic section of $\mathcal{Q}$ and $\mathrm{NL}(\lambda) = \{\, s \in U \,|\, \bar{\lambda}(s) = 0 \,\}$. In particular $\mathrm{NL}(\lambda)$ is a closed analytic subset of $U$, cut out on a neighbourhood of each of its points by $c$ holomorphic functions.

2. Let $s_0 \in \mathrm{NL}(\lambda)$, so that $\lambda_{s_0}$ defines a class in $\mathrm{gr}^p_F\mathcal{H}^{2p}_{s_0} = H^{p,p}(X_{s_0})$. Then the differential of $\bar{\lambda}$ at $s_0$ is
$$d\bar{\lambda}_{s_0}\colon T_{s_0}S \to \mathcal{Q}_{s_0}, \qquad d\bar{\lambda}_{s_0}(\xi) = -\,\bar{\nabla}^p_\xi(\lambda_{s_0})$$
with $\bar{\nabla}^p$ the graded derivative of corollary 7.2.8. Its image lies in the subspace $H^{p-1,p+1}(X_{s_0}) = F^{p-1}_{s_0}/F^p_{s_0}$ of $\mathcal{Q}_{s_0}$. Consequently every holomorphic map germ $\gamma\colon (\mathbb{D},0) \to (U,s_0)$ from the unit disc with image contained in $\mathrm{NL}(\lambda)$ satisfies $\bar{\nabla}^p_{\gamma'(0)}(\lambda_{s_0}) = 0$.

3. Suppose $p = 1$ and that $\bar{\nabla}^1(\lambda_{s_0})\colon T_{s_0}S \to H^{0,2}(X_{s_0})$ is surjective. Then $\mathrm{NL}(\lambda)$ is, on a neighbourhood of $s_0$, a closed complex submanifold of $U$ of codimension $h^{0,2}(X_{s_0})$.

***Proof*:**
One preliminary, the differential of a holomorphic section at a zero. Let $E$ be a holomorphic vector bundle on $U$ and $\mu$ a holomorphic section with $\mu(s_0) = 0$. In a local holomorphic frame $f_1, \ldots, f_m$ write $\mu = \sum_j h_j f_j$ with $h_j(s_0) = 0$. For any connection $\nabla^E$ on $E$ the Leibniz rule gives, at the point $s_0$,

$$(\nabla^E_\xi \mu)(s_0) = \sum_j (\partial_\xi h_j)(s_0)\, f_j(s_0) + \sum_j h_j(s_0)\, (\nabla^E_\xi f_j)(s_0) = \sum_j (\partial_\xi h_j)(s_0)\, f_j(s_0)$$

so the value of $\nabla^E_\xi \mu$ at $s_0$ is independent of the connection and of the frame. Write $d\mu_{s_0}\colon T_{s_0}S \to E_{s_0}$ for this map. If $\rho\colon E \to E'$ is a holomorphic bundle map then $\rho \circ \mu$ again vanishes at $s_0$ and, expanding $\rho \circ \mu = \sum_j h_j\, \rho(f_j)$ in the same way, $d(\rho \circ \mu)_{s_0} = \rho_{s_0} \circ d\mu_{s_0}$.

1. *The quotient bundle and the section.* By theorem 7.2.6 the subsheaf $F^p\mathcal{H}^{2p}$ is a holomorphic subbundle of $\mathcal{H}^{2p}$, of rank $f^p = \sum_{r \geq p} h^{r,2p-r}$; so $\mathcal{Q}$ is a holomorphic vector bundle and

$\pi$ a holomorphic bundle map (definition 1.3.13), of rank $c = \dim_{\mathbb{C}} H^{2p}(X_s, \mathbb{C}) - f^p = \sum_{r<p} h^{r,2p-r}$.

The section $\lambda$ is flat, and by theorem 7.1.7 a local frame of flat sections of $\mathcal{H}^{2p}$ is a local holomorphic frame; a flat section is a constant combination of such a frame, hence holomorphic. Therefore $\bar{\lambda} = \pi \circ \lambda$ is a holomorphic section of $\mathcal{Q}$, and $\bar{\lambda}(s) = 0$ says exactly $\lambda_s \in F^p H^{2p}(X_s, \mathbb{C})$, which is the defining condition of definition 8.3.1.

For the last clause, fix $s_0$ and a local holomorphic frame $e_1, \ldots, e_c$ of $\mathcal{Q}$ near $s_0$, and write $\bar{\lambda} = \sum_i g_i e_i$ with $g_i$ holomorphic. Then $\mathrm{NL}(\lambda)$ is the common zero locus of $g_1, \ldots, g_c$ on that neighbourhood.

2. *Computing $d\bar{\lambda}_{s_0}$.* Let $s_0 \in \mathrm{NL}(\lambda)$, so $\lambda_{s_0} \in F^p_{s_0}$. Since $F^p\mathcal{H}^{2p}$ is a holomorphic subbundle, a local holomorphic frame of it near $s_0$ produces a local holomorphic section $\sigma$ of $F^p\mathcal{H}^{2p}$ with $\sigma(s_0) = \lambda_{s_0}$. Put $\mu = \lambda - \sigma$, a holomorphic section of $\mathcal{H}^{2p}$ with $\mu(s_0) = 0$, and note $\pi \circ \mu = \bar{\lambda}$ because $\pi \circ \sigma = 0$. By the preliminary,

$$d\bar{\lambda}_{s_0}(\xi) = \pi_{s_0}\big(d\mu_{s_0}(\xi)\big) = \pi_{s_0}\big(\nabla_\xi \mu\big) = \pi_{s_0}\big(\nabla_\xi \lambda - \nabla_\xi \sigma\big) = -\,\pi_{s_0}\big(\nabla_\xi \sigma\big)$$

the connection used being the Gauss-Manin connection and the last equality holding because $\lambda$ is flat.

The right-hand side is exactly the graded derivative. By theorem 7.2.7 the class $\nabla_\xi \sigma$ lies in $F^{p-1}_{s_0}$, so $\pi_{s_0}(\nabla_\xi \sigma)$ lies in the subspace $F^{p-1}_{s_0}/F^p_{s_0}$ of $\mathcal{Q}_{s_0}$, which is $H^{p-1,p+1}(X_{s_0})$ by definition 5.1.4; and corollary 7.2.8 constructs $\bar{\nabla}^p$ as the map induced by $\sigma \mapsto \nabla\sigma \bmod F^p\mathcal{H}^{2p}$, which depends on $\sigma$ only through the class of $\sigma(s_0)$ in $\mathrm{gr}^p_F \mathcal{H}^{2p}_{s_0}$. Hence $\pi_{s_0}(\nabla_\xi \sigma) = \bar{\nabla}^p_\xi(\lambda_{s_0})$, as claimed.

If now $\gamma \colon (\mathbb{D}, 0) \to (U, s_0)$ is holomorphic with image in $\mathrm{NL}(\lambda)$, then $\bar{\lambda} \circ \gamma$ vanishes identically, so differentiating at 0 gives $d\bar{\lambda}_{s_0}(\gamma'(0)) = 0$, which is the asserted vanishing.

3. *The submersive case.* Let $p = 1$. The Hodge structure on $H^2(X_{s_0}, \mathbb{C})$ is effective, so $F^0\mathcal{H}^2 = \mathcal{H}^2$ (definition 7.2.1) and therefore

$$\mathcal{Q}_{s_0} = H^2(X_{s_0}, \mathbb{C})/F^1 = F^0_{s_0}/F^1_{s_0} = H^{0,2}(X_{s_0})$$

so that $c = h^{0,2}(X_{s_0})$ and the inclusion of part (2) is an equality of spaces. Surjectivity of $\bar{\nabla}^1(\lambda_{s_0})$ is therefore surjectivity of $d\bar{\lambda}_{s_0}$. In the frame of part (1) this says that the differentials $dg_1, \ldots, dg_c$ are linearly independent at $s_0$, so the holomorphic implicit function theorem presents $\{g_1 = \cdots = g_c = 0\}$ near $s_0$ as a closed complex submanifold of codimension $c = h^{0,2}(X_{s_0})$, completing the proof.

Part (2) is what transversality contributes. Without it, all that could be said of the defining section $\bar{\lambda}$ is that it takes values in a bundle of rank $c$; with it, the derivative of $\bar{\lambda}$ at a point of the locus lands in the single graded piece $H^{p-1,p+1}$. Choosing the frame of part (1) so that $e_1, \ldots, e_{h^{p-1,p+1}}$ span that piece, the remaining $c - h^{p-1,p+1}$ defining equations have vanishing differential at $s_0$: they vanish to order at least two there, and only $h^{p-1,p+1}$ of the $c$ equations can be independent to first order. For $p \geq 2$ this makes $\bar{\lambda}$ non-submersive whenever some $h^{r,2p-r}$ with $r < p - 1$ is nonzero. For

$p = 1$ there is no gap, since $H^{0,2}$ is the whole of $\mathcal{Q}$, and that is why weight two is the case where the codimension count of part (3) is available.

Part (2) also identifies the Zariski tangent directions concretely. In the graded description of corollary 7.2.8 the map $\xi \mapsto \bar{\nabla}^p_\xi(\lambda_{s_0})$ is a linear map $T_{s_0}S \to H^{p-1,p+1}(X_{s_0})$ attached to the fibre $X_{s_0}$ and the class $\lambda_{s_0}$ alone; the Hodge locus can move only in its kernel. When the family is one for which this map has been computed, the tangent space to the Hodge locus is computed with it, and example 8.3.7 carries that out.

A single flat section lives only over a simply connected open set, so the locus attached to it is a local object. The set where *some* nonzero rational class becomes Hodge is global, and it need not be analytic in $S$; what the analytic theory gives is a countable union of sets, each analytic in some open subset of $S$, and over an arbitrary base that is the strongest statement available.

**Corollary 8.3.4: The Locus of Extra Hodge Classes Is a Countable Union**

Let $f\colon \mathcal{X} \to S$ be a smooth projective family with $S$ connected and second countable, fix $p \geq 1$, and put

$$\mathrm{HL}^{2p}(S) = \left\{ s \in S \;\middle|\; H^{p,p}(X_s) \cap H^{2p}(X_s, \mathbb{Q}) \neq 0 \right\}$$

Then $\mathrm{HL}^{2p}(S)$ is a countable union of sets, each of which is a closed analytic subset of some open subset of $S$. If moreover no nonzero flat section of $\mathbb{H}^{2p}$ over a nonempty connected simply connected open $U \subseteq S$ is a Hodge class at every point of $U$, then each of those sets is nowhere dense in $S$ and the complement $S \setminus \mathrm{HL}^{2p}(S)$ is dense.

***Proof*:**
*Step 1: a countable family of loci.* Since $S$ is a second countable manifold it has a countable basis $U_1, U_2, \ldots$ of open sets each of which is a coordinate ball, hence connected and simply connected. Over each $U_i$ the flat sections of $\mathbb{H}^{2p}$ form a $\mathbb{Q}$-vector space isomorphic to $H^{2p}(X_{s_i}, \mathbb{Q}) \cong \mathbb{Q}^m$ for any $s_i \in U_i$, by proposition 7.1.5; that is a countable set. So the collection of pairs $(U_i, \lambda)$ with $\lambda$ a nonzero flat section over $U_i$ is countable, and each $\mathrm{NL}(\lambda) \subseteq U_i$ is a closed analytic subset of $U_i$ by proposition 8.3.3.

These loci exhaust $\mathrm{HL}^{2p}(S)$. If $s \in \mathrm{HL}^{2p}(S)$, choose a nonzero Hodge class $\mu \in H^{2p}(X_s, \mathbb{Q})$ and a basis member $U_i \ni s$; transporting $\mu$ flatly over the simply connected $U_i$ gives a nonzero flat section $\lambda$ with $\lambda_s = \mu$, so $s \in \mathrm{NL}(\lambda)$. Conversely each $\mathrm{NL}(\lambda)$ consists of points carrying the nonzero Hodge class $\lambda_s$.

*Step 2: each locus is nowhere dense.* Assume the extra hypothesis and fix a pair $(U_i, \lambda)$. Let $W \subseteq U_i$ be the set of points having a neighbourhood on which $\bar{\lambda}$ vanishes identically. Since $\mathrm{NL}(\lambda)$ is exactly the zero set of $\bar{\lambda}$, the set $W$ is the interior of $\mathrm{NL}(\lambda)$, and in particular it is open. It is also closed in $U_i$: given $s$ in its closure, choose a connected neighbourhood $N \ni s$ carrying a holomorphic chart and a holomorphic frame $e_1, \ldots, e_c$ of $\mathcal{Q}$, and write $\bar{\lambda}|_N = \sum_j g_j e_j$ with $g_j$ holomorphic on $N$. Each $g_j$ vanishes on the nonempty open set $W \cap N$, so each vanishes on all of $N$ by the identity theorem (corollary 1.1.4), whence $N \subseteq W$ and $s \in W$.

Since $U_i$ is connected, $W$ is either empty or all of $U_i$; the second case gives $\mathrm{NL}(\lambda) = U_i$, which the hypothesis excludes. Therefore $\mathrm{NL}(\lambda)$ has empty interior, and being closed in $U_i$ as well it is nowhere dense in $U_i$.

Nowhere density passes to $S$. Write $A = \mathrm{NL}(\lambda)$ and suppose $B \subseteq \overline{A}$ is open and nonempty, the closure taken in $S$. If $B$ met $U_i$ then $B \cap U_i$ would be a nonempty open subset of $\overline{A} \cap U_i$, which is the closure of $A$ in $U_i$ and has empty interior there. So $B \subseteq S \setminus U_i$, while $B \subseteq \overline{A} \subseteq \overline{U_i}$, giving $B \subseteq \overline{U_i} \setminus U_i$; but that set has empty interior, since a nonempty open subset of it would be disjoint from $U_i$ and yet meet it, $U_i$ being open and the subset lying in its closure. Hence $\overline{A}$ has empty interior in $S$.

*Step 3: density of the complement.* A complex manifold is locally compact and Hausdorff, so the Baire category theorem in its locally compact Hausdorff form applies to $S$: a countable union of nowhere dense subsets has dense complement. By Steps 1 and 2 the set $\mathrm{HL}^{2p}(S)$ is such a union, completing the proof.

The kind of conclusion at issue here has a standing name.

**Definition 8.3.5: General and Very General Points**

Let $S$ be a complex manifold and let $P$ be a property of the points of $S$. Say that $P$ holds at a *general* point of $S$ if the set where it fails is contained in a proper closed analytic subset of $S$, and that it holds at a *very general* point of $S$ if that set is contained in a countable union of proper closed analytic subsets of $S$.

The two notions differ, and not pedantically: a countable union of proper analytic subsets can be dense, as the union of the lines $\{z_2 = qz_1\} \subseteq \mathbb{C}^2$ over the Gaussian rationals $q \in \mathbb{Q}(i)$ already shows, $\mathbb{Q}(i)$ being dense in $\mathbb{C}$. Corollary 8.3.4 reaches neither notion on its own. Its pieces are analytic in *some* open subset of $S$, and such a piece need not lie in a proper closed analytic subset of $S$: the set $\{\, 1 - 1/n \,|\, n \geq 1 \,\}$ is a closed analytic subset of the unit disc $\mathbb{D} \subseteq \mathbb{C}$, being the zero set there of a holomorphic function, and is nowhere dense in $\mathbb{C}$, while a proper closed analytic subset of $\mathbb{C}$ is discrete and closed in $\mathbb{C}$ and so contains no sequence converging to 1. What survives over an arbitrary base is that $\mathrm{HL}^{2p}(S)$ is meagre, so the property of carrying no nonzero rational Hodge class in degree $2p$ holds at a dense set of parameters and fails on a set with empty interior; getting from there to a very general statement takes theorem 8.3.6.

The hypothesis of the corollary is a real restriction, and for the full local system $\mathbb{H}^{2p}$ it fails whenever $p \leq n$, the powers $h^p$ of the polarization class being Hodge at every parameter (example 8.3.2). What the corollary is applied to is therefore a sub-local system on which those classes have been removed, and the primitive part is the standard choice.

The corollary is the best that the analytic theory gives, and it leaves a gap that is visible from the shape of the statement. Each $\mathrm{NL}(\lambda)$ is analytic in an open subset of $S$, and $S$ is frequently algebraic; an analytic subset of an algebraic variety need not be algebraic, since Chow's theorem needs properness and $S$ is typically not proper (exercise 8.3.1 (6)). Closing that gap is a theorem, and it is the deepest input this chapter takes from outside.

**Theorem 8.3.6: Algebraicity of the Hodge Locus (Cattani, Deligne, Kaplan)**

Let $S$ be a smooth complex quasi-projective variety and let $f\colon \mathcal{X} \to S$ be a smooth projective family of complex algebraic varieties. Fix $p \geq 1$, let $U \subseteq S$ be open in the analytic topology, connected and simply connected, and let $\lambda$ be a nonzero flat section of $\mathbb{H}^{2p}$ over $U$. Then the closure in $S$ of every irreducible component of $\mathrm{NL}(\lambda)$ is a closed algebraic subvariety of $S$. Consequently, if no nonzero flat section of $\mathbb{H}^{2p}$ over a nonempty connected simply connected open subset of $S$ is a Hodge class at every point of that subset, then the locus $\mathrm{HL}^{2p}(S)$ of corollary 8.3.4 is contained in a countable union of *proper* closed algebraic subvarieties of $S$, so that a very general point of $S$ (definition 8.3.5) lies outside it.

The second statement follows from the first. Under its hypothesis corollary 8.3.4 makes each $\mathrm{NL}(\lambda)$ nowhere dense in $S$, so the closure of each irreducible component of $\mathrm{NL}(\lambda)$ is a closed algebraic subvariety with empty interior and hence a proper one; there are countably many pairs $(U_i, \lambda)$ in the proof of that corollary and countably many components in each locus, $S$ being second countable, so countably many closures suffice; each of them is a closed analytic subset of $S$, which is the form definition 8.3.5 asks for. The implication runs only this way. The algebraicity of the individual components is the substantive statement, and without the hypothesis the containment can fail while that statement still holds, as it does for the full local system with $p \leq n$, where $\mathrm{HL}^{2p}(S)$ is all of $S$ (example 8.3.2).

The theorem is not proved here, and the reason is specific rather than a matter of length. What an algebraicity statement requires is control of the Hodge locus near the boundary of $S$ inside a compactification: an analytic subset of $S$ is algebraic precisely when it extends analytically across that boundary, and the components of $\mathrm{NL}(\lambda)$ do so because a variation of Hodge structure degenerates in a restricted way. The tools that make that precise are Schmid's nilpotent orbit theorem, which compares a degenerating variation near a puncture with the orbit of a nilpotent monodromy logarithm, and his $\mathrm{SL}_2$-orbit theorem, which supplies the asymptotics of the Hodge metric of section 8.1 in the several-variable case. Neither is developed in this book. The proof is in Cattani, Deligne and Kaplan, *On the locus of Hodge classes* [2], and it is the source of the statement above.

Both inputs belong to Schmid's analysis of what a horizontal map into a period domain may do near a boundary point [30], which is the same phenomenon that constrains such maps in theorem 8.1.7. What distinguishes the present conclusion is its category. Analyticity is a statement about $S$ as a complex manifold and is available for any base; algebraicity is a statement about $S$ as a variety, and it gives each Hodge locus a degree, a dimension and a field of definition.

The classical instance is much older than the general theory, and it is the one where every ingredient of proposition 8.3.3 can be written down as polynomial algebra.

**Example 8.3.7: The Noether-Lefschetz Locus for Surfaces in $\mathbb{CP}^3$**

Take $n = 2$ and $d \geq 4$, let

$$S = \left\{ [F] \in \mathbb{CP}^M \;\middle|\; F \in \mathbb{C}[x_0, \dots, x_3]_d \text{ smooth} \right\}, \qquad M = \binom{d+3}{3} - 1$$

be the parameter space of proposition 7.4.4 and $f\colon \mathcal{X} \to S$ the universal family of smooth surfaces of degree $d$ in $\mathbb{CP}^3$. Then $S$ is a connected smooth quasi-projective variety of dimension $M$, and $f$ is a smooth projective family. Work with $p = 1$ and with the primitive part $\mathbb{H}^2_{\mathrm{prim}}$.

*The numerical data.* By (7.11) with $n = 2$ the three primitive Hodge pieces of a fibre $X = X_F$ are graded pieces of the Jacobian ring $R = R(F)$:

$$H^{2,0}(X) \cong R_{d-4}, \qquad H^{1,1}_{\mathrm{prim}}(X) \cong R_{2d-4}, \qquad H^{0,2}(X) \cong R_{3d-4}$$

As in example 8.3.2, $R_{d-4} = \mathbb{C}[x]_{d-4}$ and $p_g := h^{2,0}(X) = \binom{d-1}{3} \geq 1$ for $d \geq 4$. So the bundle $\mathcal{Q} = \mathcal{H}^2_{\mathrm{prim}}/F^1_{\mathrm{prim}}$ has rank $p_g$ and proposition 8.3.3 cuts out every Hodge locus locally by $p_g$ holomorphic equations.

*Picard number and primitive Hodge classes.* Since $\int_X h_X \smile h_X = d \neq 0$, the assignment $\alpha \mapsto \alpha - d^{-1}\big(\int_X \alpha \smile h_X\big)h_X$ is a projection defined over $\mathbb{Q}$, and

$$H^2(X, \mathbb{Q}) = \mathbb{Q}\, h_X \;\oplus\; H^2_{\mathrm{prim}}(X, \mathbb{Q})$$

a decomposition into Hodge substructures (proposition 5.2.6), with $h_X$ of type $(1,1)$. The Néron-Severi group is $\mathrm{NS}(X) = H^{1,1}(X) \cap H^2(X, \mathbb{Z})$ (definition 4.4.3), and every rational class has an integral multiple, so its rank is $\rho(X) = \dim_{\mathbb{Q}}\big(H^{1,1}(X) \cap H^2(X, \mathbb{Q})\big)$. Splitting off $h_X$,

$$\rho(X) = 1 + \dim_{\mathbb{Q}}\big(H^{1,1}(X) \cap H^2_{\mathrm{prim}}(X, \mathbb{Q})\big)$$

Hence $\rho(X_s) > 1$ if and only if $X_s$ carries a nonzero primitive rational Hodge class, and the *Noether-Lefschetz locus* $\mathrm{NL}_d = \{\, s \in S \,|\, \rho(X_s) > 1 \,\}$ is the set $\mathrm{HL}^2(S)$ of corollary 8.3.4 formed inside $\mathbb{H}^2_{\mathrm{prim}}$.

*No class is Hodge everywhere.* This is the hypothesis corollary 8.3.4 needs, and it is where the Jacobian ring does the work. Suppose $\lambda$ is a nonzero flat section of $\mathbb{H}^2_{\mathrm{prim}}$ over a connected simply connected $U \subseteq S$ with $\mathrm{NL}(\lambda) = U$. Let $\pi\colon \tilde{S} \to S$ be the universal cover, a holomorphic covering, and pull $f$ back along it; the result is again a smooth projective family, its local system and Hodge bundles being the pullbacks of those of $f$. Since $U$ is simply connected, $\lambda$ lifts to a flat section over a component $\tilde{U}$ of $\pi^{-1}(U)$, and since $\tilde{S}$ is connected and simply connected that section extends uniquely to a flat section $\tilde{\lambda}$ over $\tilde{S}$ (proposition 7.1.5). The associated holomorphic section $\bar{\tilde{\lambda}}$ of $\mathcal{H}^2_{\mathrm{prim}}/F^1_{\mathrm{prim}}$ vanishes on the nonempty open set $\tilde{U}$, hence vanishes identically on $\tilde{S}$ by the identity theorem, exactly as in Step 2 of corollary 8.3.4. So $\tilde{\lambda}_t$ is a Hodge class for every $t \in \tilde{S}$, and part (2) of proposition 8.3.3 applied with $\mathrm{NL}(\tilde{\lambda}) = \tilde{S}$ gives

$$\bar{\nabla}^1_\xi(\tilde{\lambda}_t) = 0 \qquad \text{for every } t \in \tilde{S} \text{ and every } \xi \in T_t\tilde{S}$$

Evaluate at a point $t_0$ lying over the Fermat surface $F_0 = x_0^d + x_1^d + x_2^d + x_3^d$, which lies in $S$ by example 7.4.3. Let $A \in R_{2d-4}$ correspond to $\tilde{\lambda}_{t_0} \in H^{1,1}_{\mathrm{prim}}(X_{F_0})$ under (7.11). Tangent vectors to $S$ at $[F_0]$ are represented by arbitrary $G \in \mathbb{C}[x]_d$, and theorem 7.4.14 with $q = 1$ computes $\bar{\nabla}_G$ as $[A] \mapsto -2\,[GA]$ in $R_{3d-4}$. So the displayed vanishing says

$$A \cdot \mathbb{C}[x]_d = 0 \quad \text{in } R_{3d-4}$$

*The monomial computation.* By example 7.4.3 the ring $R(F_0)$ has basis the monomials $x^a$ with $0 \leq a_i \leq d-2$ for every $i$, and its top degree is $\sigma = 4(d-2)$, which is at least $3d-4$ because $d \geq 4$. Suppose $A \neq 0$ and write $A = \sum_a c_a\, x^a$ over the basis monomials of degree $2d-4$; choose $a^*$ with $c_{a^*} \neq 0$ and put $m_i = d - 2 - a^*_i \geq 0$, so that

$$\sum_i m_i = 4(d-2) - (2d-4) = 2d - 4 \geq d$$

Choose integers $0 \le b_i \le m_i$ with $\sum_i b_i = d$, possible by the displayed inequality. Then $b_i \le d-2$, so $x^b$ is a basis monomial of $R_d$, and in $R$ the product $x^a x^b$ is either the basis monomial $x^{a+b}$, when $a_i + b_i \le d-2$ for every $i$, or zero. Distinct $a$ give distinct $a+b$, and $a_i^* + b_i \le a_i^* + m_i = d-2$, so the coefficient of $x^{a^*+b}$ in $A\,x^b$ is $c_{a^*} \neq 0$. Hence $A\,x^b \neq 0$ in $R_{3d-4}$, contradicting the vanishing above. Therefore $A = 0$, so $\tilde{\lambda}_{t_0} = 0$, so $\tilde{\lambda} = 0$ by flatness and $\lambda = 0$, contrary to the choice of $\lambda$.

*The Noether-Lefschetz theorem.* The hypothesis of corollary 8.3.4 therefore holds, and the corollary gives: for $d \ge 4$ the locus $\mathrm{NL}_d$ is a countable union of sets, each of them closed analytic in some open subset of $S$ and nowhere dense in $S$, so that $\mathrm{NL}_d$ is meagre and its complement is dense. Since $S$ is quasi-projective, theorem 8.3.6 applies as well: the closure of each irreducible component of each locus $\mathrm{NL}(\lambda)$ is a closed algebraic subvariety of $S$, proper because that locus is nowhere dense, and these closures are the classical Noether-Lefschetz components. They are countably many proper closed subvarieties whose union contains $\mathrm{NL}_d$, so the very general (definition 8.3.5) surface of degree $d \ge 4$ in $\mathbb{CP}^3$ satisfies $\rho = 1$, that is $\mathrm{NS}(X) \otimes \mathbb{Q} = \mathbb{Q}\,h_X$; by theorem 4.4.2 the Néron-Severi group is the group of first Chern classes of line bundles, so every divisor class on such a surface is a rational multiple of a plane section.

*Codimension.* Where the map $\bar{\nabla}^1(\lambda_s)$ is surjective, part (3) of proposition 8.3.3 makes $\mathrm{NL}(\lambda)$ smooth of codimension $p_g$. For $d = 4$ one has $p_g = 1$ and $H^{0,2}(X) \cong R_8$ is a line, so surjectivity is the same as nonvanishing: the Noether-Lefschetz locus of a quartic surface is a smooth hypersurface near every parameter at which $\bar{\nabla}^1(\lambda_s) \neq 0$, and the monomial computation above produces such a parameter. That count is carried out explicitly for one class at the Fermat quartic in exercise 8.3.1 (5).

None of this asserts that a Hodge class is the class of an algebraic cycle. That assertion is the Hodge conjecture, and **??** takes it up; a proof of it would make each Hodge locus algebraic by a spreading-out argument, so theorem 8.3.6 is one of its predictions verified in advance.

## Exercise 8.3.1

1. Let $X$ be a compact Kähler manifold and $p \ge 0$.
    1. Show that a class $\mu \in H^{2p}(X, \mathbb{Q})$ lies in $F^p H^{2p}(X, \mathbb{C})$ if and only if it lies in $H^{p,p}(X)$, so that the two descriptions of a Hodge class agree, and that the Hodge classes (definition 5.1.9) of degree $2p$ form a $\mathbb{Q}$-subspace of $H^{2p}(X, \mathbb{Q})$.
    2. Show that the corresponding statement for a class in $H^{2p}(X, \mathbb{C})$ is false, by exhibiting a compact Kähler manifold and a class in $F^1 H^2(X, \mathbb{C})$ that is not of type $(1,1)$.
    3. Let $\lambda, \mu$ be flat sections of $\mathbb{H}^{2p}$ over a connected simply connected $U$. Show that $\mathrm{NL}(\lambda) \cap \mathrm{NL}(\mu) \subseteq \mathrm{NL}(a\lambda + b\mu)$ for all $a, b \in \mathbb{Q}$ with $a\lambda + b\mu \neq 0$, and that $\mathrm{NL}(a\lambda) = \mathrm{NL}(\lambda)$ for $a \in \mathbb{Q}^\times$.
2. Let $f \colon \mathcal{X} \to S$ be a smooth projective family for which the graded derivatives $\bar{\nabla}^p$ of corollary 7.2.8 vanish in degree $2p$ for every $p$. Using exercise 7.2.1 (1), show that for every connected simply connected $U \subseteq S$ and every nonzero flat section $\lambda$ of $\mathbb{H}^{2p}$ over $U$, the locus $\mathrm{NL}(\lambda)$ is either empty or all of $U$. Check the conclusion directly for the product family of exercise 7.2.1 (3).
3. Let $f \colon \mathcal{X} \to S$ be a smooth projective family and fix $p \ge 1$.
    1. Show that if $h^{r,2p-r}(X_s) = 0$ for every $r < p$, then every Hodge locus in degree $2p$ is all of $U$, and identify the rank $c$ of proposition 8.3.3 in that case.

2. For the family of smooth surfaces of degree $d$ in $\mathbb{C}\mathrm{P}^3$ with $p = 1$, determine exactly which $d$ satisfy the hypothesis of (1), and compute $b_2(X)$ and $\rho(X)$ for those $d$.

4. Take $d = 5$ in example 8.3.7, so that the fibres are smooth quintic surfaces in $\mathbb{C}\mathrm{P}^3$. Using corollary 7.4.12 and (7.4), compute $p_g$, $h^{1,1}_{\mathrm{prim}}$, $h^{1,1}$ and $b_2$ for such a surface, and compute $\dim_{\mathbb{C}} S$. How many holomorphic equations does proposition 8.3.3 use to cut out a Noether-Lefschetz locus here, and what codimension does part (3) predict where the differential is surjective?

5. Let $F_0 = x_0^4 + x_1^4 + x_2^4 + x_3^4$ be the Fermat quartic surface, with Jacobian ring $R = R(F_0)$ as in example 7.4.3. Suppose $\lambda$ is a flat section of $\mathbb{H}^2_{\mathrm{prim}}$ defined near $[F_0]$ for which $\lambda_{[F_0]}$ is a Hodge class corresponding, up to a nonzero scalar, to $[x_0x_1x_2x_3] \in R_4$ under (7.11).

   1. Show that $R_8$ is one dimensional, spanned by $[x_0^2x_1^2x_2^2x_3^2]$, and determine every basis monomial $x^b$ of $R_4$ with $[x_0x_1x_2x_3] \cdot [x^b] \neq 0$.
   2. Compute the dimension of the Zariski tangent space $\ker\big(\bar{\nabla}^1(\lambda_{[F_0]})\big) \subseteq T_{[F_0]}S$ and its codimension.
   3. Conclude that $\mathrm{NL}(\lambda)$ is a smooth hypersurface near $[F_0]$.

6. Let $Z = \{\, (z, w) \in \mathbb{C}^2 \,|\, w = e^z \,\}$. Show that $Z$ is a closed analytic subset of $\mathbb{C}^2$ and that the only polynomial $P \in \mathbb{C}[z, w]$ vanishing identically on $Z$ is $P = 0$; deduce that $Z$ is not the common zero set of any family of polynomials. Explain in one sentence why this shows that theorem 8.3.6 does not follow formally from proposition 8.3.3.

## 8.4 Moduli of Polarized K3 Surfaces

Everything this chapter has built comes together on one class of surfaces. A K3 surface carries a weight-two Hodge structure whose entire content is a single line (example 5.4.4); a family of them has a period map (definition 7.3.3) which is horizontal for free (example 7.3.9); and the domain that map lands in is one of the classical bounded domains, so the horizontal restriction which limits theorem 8.1.4 costs nothing here. What remains is to say which domain it is, and to record the theorem that turns the period map into a classification: the global Torelli theorem, which says that a K3 surface is determined by its period point.

Example 5.4.4 works inside $H^2(S, \mathbb{Z})$ for one surface $S$ at a time. Comparing two surfaces means comparing two abstract lattices, and the comparison is only useful once a single model lattice has been fixed and each surface has been identified with it. The model is forced: by example 5.4.4 the cup form makes $H^2(S, \mathbb{Z})$ an even unimodular lattice of signature $(3, 19)$, and that isometry class has a name.

**Definition 8.4.1: The K3 Lattice, Markings and Polarizations**

Write $\langle -,-\rangle$ for the bilinear form of a lattice, extended $\mathbb{C}$-bilinearly whenever the lattice is complexified.

1. Let $U$ be the free abelian group on $e, f$ with $\langle e,e\rangle = \langle f,f\rangle = 0$ and $\langle e,f\rangle = 1$, and let $E_8$ be the root lattice of type $E_8$, realized as

$$E_8 = \left\{ x \in \mathbb{Z}^8 \cup \left(\mathbb{Z} + \tfrac{1}{2}\right)^8 \;\middle|\; \textstyle\sum_{i=1}^8 x_i \in 2\mathbb{Z} \right\}$$

with the restriction of the standard positive definite form of $\mathbb{R}^8$; it is the root lattice of type $E_8$ [32, the root system of type $E_8$], and it is positive definite, even and unimodular of rank 8. Evenness is immediate: a vector of $\mathbb{Z}^8$ with even coordinate sum has even square, and a vector of $(\mathbb{Z}+\frac{1}{2})^8$ has square $\sum x_i^2 \equiv 8 \cdot \frac{1}{4} = 2$ modulo 2. Unimodularity follows by computing the index: the sublattice $\{x \in \mathbb{Z}^8 : \sum x_i \in 2\mathbb{Z}\}$ has discriminant 4 and index 2 in $\mathbb{Z}^8$, and adjoining the half-integer vector $\frac{1}{2}(1,\dots,1)$ halves the covolume, giving discriminant 1. For a lattice $L$ write $L(-1)$ for the same group with the form negated. The *K3 lattice* is

$$\Lambda_{K3} = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$$

of rank 22; it is even and unimodular of signature $(3,19)$ (exercise 8.4.1 (1)).

2. A *marking* of a K3 surface $S$ is an isometry $\alpha\colon H^2(S,\mathbb{Z}) \to \Lambda_{K3}$ for the cup form, and a *marked K3 surface* is a pair $(S,\alpha)$.

3. A *polarized K3 surface of degree* $2d$, with $d \geq 1$, is a pair $(S,h)$ consisting of a K3 surface $S$ and a primitive class $h \in H^2(S,\mathbb{Z})$ which is the first Chern class of an ample line bundle and satisfies $\langle h,h\rangle = 2d$.

4. Fix once and for all a primitive $h_0 \in \Lambda_{K3}$ with $\langle h_0,h_0\rangle = 2d$, and put

$$\Lambda_{2d} = h_0^{\perp} \subseteq \Lambda_{K3}$$

A *marking* of a polarized K3 surface $(S,h)$ of degree $2d$ is a marking $\alpha$ of $S$ with $\alpha(h) = h_0$.

5. A *Hodge isometry* between K3 surfaces $S$ and $S'$ is an isometry $\varphi\colon H^2(S,\mathbb{Z}) \to H^2(S',\mathbb{Z})$ for the cup forms whose complexification satisfies $\varphi_{\mathbb{C}}\big(H^{2,0}(S)\big) = H^{2,0}(S')$.

A marking exists for every K3 surface, because example 5.4.4 identifies $H^2(S,\mathbb{Z})$ with $\Lambda_{K3}$ up to isometry; a marking is a choice of such an identification, and there are as many as there are isometries of $\Lambda_{K3}$. The construction depends on the choice not being canonical, and the ambiguity it creates is what the quotient in definition 7.3.3 divides out.

The lattice $\Lambda_{2d}$ has rank 21 and, since $\mathbb{R}h_0$ is positive definite, signature $(2,19)$. It is not unimodular: its discriminant group is cyclic of order $2d$ (exercise 8.4.1 (2)). Writing $h_0 = e + df$ inside the first hyperbolic summand of $\Lambda_{K3}$ produces a primitive vector of square $2d$ for every $d \geq 1$, so every degree occurs, the smallest being $2d = 2$; the case $2d = 4$ is the degree of a smooth quartic surface in $\mathbb{CP}^3$ with its hyperplane class, computed in example 8.4.4. That $\Lambda_{2d}$ depends only on $d$, and not on which primitive vector of square $2d$ was chosen, is Eichler's criterion for $\Lambda_{K3}$ ([22, Eichler's criterion for the K3 lattice]): the isometry group of $\Lambda_{K3}$ acts transitively on the primitive vectors of a fixed square.

It is a theorem of the arithmetic theory of quadratic forms, external to this series and taken here on faith, and it is what allows a marking of a polarized surface to be found at all.

A Hodge isometry is automatically an isomorphism of the whole Hodge structure. Since $\varphi$ is defined over $\mathbb{Z}$, its complexification commutes with conjugation, so $\varphi_{\mathbb{C}}\big(H^{0,2}(S)\big) = H^{0,2}(S')$; and by example 5.4.4 the summand $H^{1,1}$ is the orthogonal complement of $H^{2,0} \oplus H^{0,2}$ for the cup form, so $\varphi_{\mathbb{C}}$ carries it across as well.

With the model lattice fixed, a marked polarized surface produces a point of a fixed space. Let $(S, h, \alpha)$ be a marked polarized K3 surface of degree $2d$ and let $\sigma$ span $H^{2,0}(S)$. By part (3) of example 5.4.4 the class $\sigma$ is orthogonal to $h$, so $\alpha_{\mathbb{C}}(\sigma)$ lies in $\Lambda_{2d} \otimes \mathbb{C}$, and the two relations (5.10) of that example say

$$\big\langle \alpha_{\mathbb{C}}\sigma,\, \alpha_{\mathbb{C}}\sigma \big\rangle = 0, \qquad \big\langle \alpha_{\mathbb{C}}\sigma,\, \overline{\alpha_{\mathbb{C}}\sigma} \big\rangle > 0$$

The *period point* of $(S, h, \alpha)$, in the sense attached to a single surface in example 5.4.4, is the line

$$\mathcal{P}(S, h, \alpha) = \big[\alpha_{\mathbb{C}}(\sigma)\big] \in \mathbb{P}\big(\Lambda_{2d} \otimes \mathbb{C}\big) \tag{8.1}$$

independent of the choice of $\sigma$ because $H^{2,0}(S)$ is a line. The set of points satisfying the two displayed conditions is

$$D_{2d} = \Big\{\, [\sigma] \in \mathbb{P}\big(\Lambda_{2d} \otimes \mathbb{C}\big) \;\Big|\; \langle \sigma, \sigma \rangle = 0,\; \langle \sigma, \bar{\sigma} \rangle > 0 \,\Big\}$$

This is the period domain of definition 5.3.2 for the numerical type $k = 2$, $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 19$, computed in example 5.3.6 with $n = 19$. The polarizing form of proposition 5.2.8 in weight two is the *negative* of the cup form, so it has signature $(19, 2)$ on $\Lambda_{2d}$, and the description there in terms of $Q = -\langle -, - \rangle$ becomes the one displayed here on changing signs. Dropping the polarization is not an option: on all of $\Lambda_{K3}$ the same recipe produces a 20-dimensional open subset of a quadric which is *not* a period domain, because the positivity at $p = 1$ fails there (part (2) of example 5.4.4, and example 5.3.6).

The general theory already supplies a good deal about $D_{2d}$: it is an open complex submanifold of a smooth quadric, homogeneous under a real orthogonal group, with compact isotropy (proposition 5.3.5). What is special to this numerical type, and what the rest of the section rests on, is that the homogeneous space in question is one of the classical bounded domains. Call a connected complex manifold a *Hermitian symmetric domain* when it carries a Hermitian metric for which every point is an isolated fixed point of an involutive holomorphic isometry.

**Proposition 8.4.2: The Period Domain of a Polarized K3 Surface**

Let $V_{\mathbb{R}} = \Lambda_{2d} \otimes \mathbb{R}$ with its cup form of signature $(2, 19)$, let $G_{\mathbb{R}} = \mathrm{O}(V_{\mathbb{R}}) \cong \mathrm{O}(2, 19)$, and let $D_{2d}$ be as above.

1. Sending $[\sigma] = [x + iy]$ to the plane $\mathrm{span}_{\mathbb{R}}(x, y)$ with the orientation $(x, y)$ is a homeomorphism from $D_{2d}$ onto the set of oriented positive definite real 2-planes in $V_{\mathbb{R}}$. Under it $G_{\mathbb{R}}$ acts transitively, the isotropy group of a point is $\mathrm{SO}(2) \times \mathrm{O}(19)$, and
$$D_{2d} \cong \mathrm{O}(2, 19)\big/\big(\mathrm{SO}(2) \times \mathrm{O}(19)\big)$$
2. $D_{2d}$ has exactly two connected components, interchanged by the antiholomorphic involution $[\sigma] \mapsto [\bar{\sigma}]$, and each is homeomorphic to an open ball of real dimension 38. In particular $\dim_{\mathbb{C}} D_{2d} = 19$.
3. $T^h D_{2d} = T D_{2d}$: every holomorphic map into $D_{2d}$ is horizontal.
4. Each connected component of $D_{2d}$ is a Hermitian symmetric domain.

***Proof*:**
Write $m = 21$ for the rank of $\Lambda_{2d}$. An element of $\mathrm{GL}(V_{\mathbb{R}})$ preserves the cup form if and only if it preserves its negative, so the group $G_{\mathbb{R}}$ of the statement is the group $G_{\mathbb{R}}$ of proposition 5.3.5 attached to the polarizing form, and the results of section 5.3 apply to it verbatim.

1. *Planes and the group action.* Let $[\sigma] \in D_{2d}$ and write $\sigma = x + iy$ with $x, y \in V_{\mathbb{R}}$. Expanding as in example 5.3.6, with the signs reversed because the form used there is $-\langle -, - \rangle$, the two defining conditions read
$$\langle x, x \rangle = \langle y, y \rangle > 0, \qquad \langle x, y \rangle = 0 \tag{8.2}$$
so $x$ and $y$ are independent and $P := \mathrm{span}_{\mathbb{R}}(x, y)$ is a positive definite plane. That example also establishes that $[\sigma] \mapsto (P, (x, y))$ is a bijection onto the oriented positive definite planes: rescaling $\sigma$ by $re^{i\theta}$ rotates the ordered pair $(x, y)$ without changing the plane or its orientation, and every oriented positive definite plane has an oriented basis satisfying (8.2), unique up to that rotation and scaling.

   The bijection is a homeomorphism. On a standard affine chart of $\mathbb{P}(\Lambda_{2d} \otimes \mathbb{C})$ a holomorphic section $\sigma$ of the tautological line is available, and $(\mathrm{Re}\sigma, \mathrm{Im}\sigma)$ is then a continuous oriented frame of $P$, so the map is continuous. In the other direction, an oriented plane admits locally a continuous oriented basis $(u_1, u_2)$, and Gram-Schmidt for the positive definite form on it,
$$x = \frac{u_1}{\sqrt{\langle u_1, u_1 \rangle}}, \qquad y' = u_2 - \langle u_2, x \rangle x, \qquad y = \frac{y'}{\sqrt{\langle y', y' \rangle}}$$
   produces continuously a pair satisfying (8.2) and inducing the given orientation, hence the point $[x + iy]$.

   For the group action, let $g \in G_{\mathbb{R}}$ fix $[\sigma]$, say $g\sigma = \lambda\sigma$. Applying conjugation gives $g\bar{\sigma} = \bar{\lambda}\bar{\sigma}$, and $\langle g\sigma, g\bar{\sigma} \rangle = \langle \sigma, \bar{\sigma} \rangle$ forces $|\lambda| = 1$, say $\lambda = e^{i\theta}$. Separating real and imaginary parts of

$g(x+iy) = e^{i\theta}(x+iy)$,

$$gx = \cos\theta\, x - \sin\theta\, y, \qquad gy = \sin\theta\, x + \cos\theta\, y$$

so $g$ preserves $P$ and acts on it by a rotation in the oriented orthonormal basis $(x,y)$ supplied by (8.2). Since $g$ preserves $P$ it preserves $P^\perp$, and conversely a rotation of $P$ together with any isometry of $P^\perp$ assembles to an element of $G_{\mathbb{R}}$ fixing $[\sigma]$, by the displayed formulas read backwards. So the isotropy group is $\mathrm{SO}(P) \times \mathrm{O}(P^\perp) \cong \mathrm{SO}(2) \times \mathrm{O}(19)$, the second factor being the full orthogonal group of the negative definite 19-dimensional space $P^\perp$; this agrees with the general computation of exercise 5.3.1 (3), since $\mathrm{SO}(2) \cong \mathrm{U}(1)$. Transitivity of $G_{\mathbb{R}}$ on $D_{2d}$ is proposition 5.3.5.

2. *Components and dimension.* Fix a positive definite plane $W \subseteq V_{\mathbb{R}}$ with a chosen orientation, so that $V_{\mathbb{R}} = W \oplus W^\perp$ with $W^\perp$ negative definite of dimension 19; write $\pi\colon V_{\mathbb{R}} \to W$ for the orthogonal projection, $\|w\|^2 = \langle w, w\rangle$ on $W$ and $\|u\|^2 = -\langle u, u\rangle$ on $W^\perp$, both positive definite.

   If $P$ is a positive definite plane then $\pi|_P$ is injective, since a nonzero $v \in P \cap W^\perp$ would satisfy $\langle v, v\rangle > 0$ and $\langle v, v\rangle < 0$. Being a map between 2-dimensional spaces it is an isomorphism, so $P$ is the graph

$$P = \left\{\, w + Aw \;\middle|\; w \in W \,\right\}$$

   of the linear map $A\colon W \to W^\perp$ obtained by composing $(\pi|_P)^{-1}$ with the projection to $W^\perp$. Conversely the graph of any $A$ is a 2-plane, and for $w \neq 0$,

$$\langle w + Aw,\, w + Aw\rangle = \|w\|^2 - \|Aw\|^2$$

   so the graph is positive definite exactly when $\|Aw\| < \|w\|$ for every $w \neq 0$, that is, exactly when the operator norm satisfies $\|A\| < 1$, the supremum defining it being attained on the compact unit sphere of $W$. So $P \mapsto A_P$ is a bijection from the positive definite planes onto the open unit ball

$$B = \left\{\, A \in \mathrm{Hom}(W, W^\perp) \;\middle|\; \|A\| < 1 \,\right\}$$

   an open convex subset of a real vector space of dimension $2 \cdot 19 = 38$, hence connected. It is a homeomorphism, both directions being given by linear algebra applied to a continuously varying frame, as in part (1).

   An oriented positive definite plane carries in addition a sign $\epsilon(P) \in \{\pm 1\}$, equal to $+1$ when $\pi|_P$ is orientation preserving for the chosen orientation of $W$. Combining, $[\sigma] \mapsto (A_P, \epsilon(P))$ is a homeomorphism of $D_{2d}$ onto $B \times \{\pm 1\}$, since reversing the orientation of a plane changes $\epsilon$ and nothing else. So $D_{2d}$ has exactly two components, each an open ball of real dimension 38. The involution $[\sigma] \mapsto [\bar\sigma]$ fixes $P$ and replaces the oriented basis $(x,y)$ by $(x,-y)$, so it preserves $A_P$ and reverses $\epsilon$, interchanging the two components; it is antiholomorphic because conjugation on $\Lambda_{2d} \otimes \mathbb{C}$ is. Finally $D_{2d}$ is a complex manifold (proposition 5.3.5 and proposition 5.3.4) of real dimension 38, so $\dim_{\mathbb{C}} D_{2d} = 19$, agreeing with the count in example 5.3.6.

3. *Horizontality.* This is example 7.3.9 with $r = 19$: for the numerical type $h^{2,0} = h^{0,2} = 1$ the tangent space to the compact dual at $[\sigma]$ is

$$T_{[\sigma]}\check{D}_{2d} = \mathrm{Hom}\big(\mathbb{C}\sigma,\ \sigma^\perp/\mathbb{C}\sigma\big) \tag{8.3}$$

and the only nonvacuous horizontality condition, the one at $p = 2$, is satisfied by every tangent vector. So $T^h\check{D}_{2d} = T\check{D}_{2d}$, and the same holds on the open subset $D_{2d}$.

4. *The symmetry at a point.* Part (2) of theorem 8.1.4 already gives this conclusion for any numerical type with no $\mathfrak{g}^{r,-r}$ in $|r| \geq 2$, a hypothesis checked for the present type in exercise 8.1.1 (2); the involution is written out because it gives more, namely the symmetry in its $\mathrm{O}(2,19)$ form. Fix $[\sigma] \in D_{2d}$ with oriented plane $P$ and set

$$s_P = \mathrm{id}_P \oplus (-\mathrm{id}_{P^\perp}) \in G_\mathbb{R}$$

It is an involution, it is the identity on $P$ and so preserves both $P$ and its orientation, and it fixes $\sigma$ because $\sigma \in P \otimes \mathbb{C}$. Being induced by a linear map preserving the form, it acts holomorphically on $\mathbb{P}(\Lambda_{2d} \otimes \mathbb{C})$ and preserves both conditions defining $D_{2d}$, so it restricts to a holomorphic involution of $D_{2d}$ preserving each component; and it lies in $G_\mathbb{R}$, so it is an isometry of the $G_\mathbb{R}$-invariant Hermitian Hodge metric of definition 8.1.2.

Its differential at $[\sigma]$ is $-\mathrm{id}$. Write $P_\mathbb{C} = \mathbb{C}\sigma \oplus \mathbb{C}\bar{\sigma}$, a decomposition valid because $\langle\sigma, \sigma\rangle = 0$ while $\langle\sigma, \bar{\sigma}\rangle \neq 0$, so that $\sigma$ and $\bar{\sigma}$ are independent. Then $\sigma^\perp$ contains both $\mathbb{C}\sigma$ and $P^\perp \otimes \mathbb{C}$, the sum of the two is direct because $\mathbb{C}\sigma \subseteq P_\mathbb{C}$, and its dimension is $1 + (m-2) = m - 1 = \dim \sigma^\perp$; hence

$$\sigma^\perp = \mathbb{C}\sigma \oplus \big(P^\perp \otimes \mathbb{C}\big), \qquad \sigma^\perp/\mathbb{C}\sigma \cong P^\perp \otimes \mathbb{C}$$

Now $s_P$ acts as the identity on $\mathbb{C}\sigma$ and as $-\mathrm{id}$ on $P^\perp \otimes \mathbb{C}$, so by the equivariance of the identification (8.3) [26, the tangent space of a Grassmannian] it acts on $\mathrm{Hom}(\mathbb{C}\sigma, \sigma^\perp/\mathbb{C}\sigma)$ by $\theta \mapsto (-\mathrm{id}) \circ \theta \circ \mathrm{id}^{-1} = -\theta$.

That $[\sigma]$ is isolated in the fixed locus follows from the linearity of $s_P$ directly. The fixed locus of $s_P$ in $\mathbb{P}(\Lambda_{2d} \otimes \mathbb{C})$ is the union of the projectivized eigenspaces, $\mathbb{P}(P_\mathbb{C}) \cup \mathbb{P}(P^\perp \otimes \mathbb{C})$. No point of the second lies in $D_{2d}$: writing $\tau = u + iv$ with $u, v \in P^\perp$ gives $\langle\tau, \bar{\tau}\rangle = \langle u, u\rangle + \langle v, v\rangle \leq 0$, since $P^\perp$ is negative definite. A point $[\alpha\sigma + \beta\bar{\sigma}]$ of the first has $\langle\alpha\sigma + \beta\bar{\sigma}, \alpha\sigma + \beta\bar{\sigma}\rangle = 2\alpha\beta\langle\sigma, \bar{\sigma}\rangle$, which vanishes only when $\alpha = 0$ or $\beta = 0$, so the first contributes exactly $[\sigma]$ and $[\bar{\sigma}]$, and the latter lies in the other component by part (2). So $[\sigma]$ is the unique fixed point of $s_P$ on its own component, completing the proof.

The proposition places the K3 period domain among the objects the classical theory of bounded symmetric domains already covers, and that is the situation theorem 8.1.4 was careful not to assume. In general the negativity of the holomorphic sectional curvature holds only along $T^hD$, and the period domain of a general numerical type is not Hermitian symmetric; a period map is then constrained by a distribution of positive codimension, and the curvature estimate applies only to the directions it can move in. Here the distribution is everything and the domain is symmetric, so the estimate, the distance-decreasing property of corollary 8.1.6 and the rigidity statement of theorem 8.1.7 all apply to $D_{2d}$ and its quotients without restriction. The same degeneration was visible in weight one (example 8.1.5), where the domain is the Siegel space of example 5.3.3. What the two cases share is the shape of the numerical type: the nonzero Hodge numbers occupy two consecutive bidegrees, as in weight one, or three consecutive bidegrees with a one-dimensional space at each end, as here. Either shape leaves no $\mathfrak{g}^{r,-r}$ with $|r| \geq 2$, the computations of example 8.1.5 and exercise 8.1.1 (2) using the shape of the numerical type and nothing else, so part (2) of theorem 8.1.4 applies and the domain is symmetric; shapes of both kinds occur in higher weights as well, and what singles out these two is that their moduli theory was understood classically.

Symmetry of the domain says nothing about injectivity of the period map, and injectivity is a theorem about surfaces rather than about domains. A point of $D_{2d}$ records a line in $\Lambda_{2d} \otimes \mathbb{C}$ and nothing else, so recovering a surface from it asserts that all the geometry of $S$ is encoded in one line of transcendental data.

**Theorem 8.4.3: Global Torelli for K3 Surfaces**

Let $S$ and $S'$ be projective K3 surfaces and let $\varphi\colon H^2(S,\mathbb{Z}) \to H^2(S',\mathbb{Z})$ be a Hodge isometry in the sense of definition 8.4.1.

1. If $\varphi$ carries the first Chern class of some ample line bundle on $S$ to the first Chern class of an ample line bundle on $S'$, then there is a unique isomorphism $f\colon S' \to S$ of complex manifolds with $f^* = \varphi$.
2. Consequently, if $(S,h)$ and $(S',h')$ are polarized K3 surfaces of degree $2d$ and $\varphi(h) = h'$, then $f^* = \varphi$ for a unique isomorphism $f\colon S' \to S$, and in particular $S \cong S'$.

Part (2) is the case of part (1) in which the ample class carried across is $h$ itself. Part (1) is not proved here, and it is proved nowhere in this series. The argument of [29] establishes the statement first for Kummer surfaces, where it reduces to the corresponding assertion for complex tori of dimension two and hence to weight-one Hodge theory, and then propagates it to all K3 surfaces by a density argument inside the space of marked surfaces. Both halves need machinery outside this book: the geometry of Kummer surfaces and their resolutions on one side, and the deformation theory that makes the density argument available on the other.

The hypothesis in part (1) cannot be dropped. A Hodge isometry is linear algebra together with a compatibility with one line, and there are Hodge isometries of $H^2(S,\mathbb{Z})$ induced by no automorphism of $S$, starting with $-\mathrm{id}$, which is a Hodge isometry for every $S$ but carries an ample class to its negative. The effectivity hypothesis pins down which of the many chambers into which the classes of positive square are divided the isometry must respect, and the theorem says that once it respects the right one it is geometric.

Torelli converts the period point from an invariant into a complete invariant, and that is what turns $D_{2d}$ into a moduli space.

**Example 8.4.4: The Moduli of Degree-$2d$ Polarized K3 Surfaces**

Fix $d \geq 1$ and let $\mathcal{M}_{2d}^m$ be the set of isomorphism classes of marked polarized K3 surfaces $(S,h,\alpha)$ of degree $2d$, two being isomorphic when an isomorphism of the surfaces matches both the polarizations and the markings. Sending $(S,h,\alpha)$ to (8.1) defines

$$\mathcal{P}\colon \mathcal{M}_{2d}^m \to D_{2d}$$

*Forgetting the marking.* Two markings of the same $(S,h)$ differ by an isometry of $\Lambda_{K3}$ fixing $h_0$, so put

$$\Gamma_{2d} = \left\{ \tilde{g}|_{\Lambda_{2d}} \;\middle|\; \tilde{g} \in \mathrm{O}(\Lambda_{K3}),\ \tilde{g}(h_0) = h_0 \right\} \subseteq \mathrm{O}(\Lambda_{2d})$$

Changing the marking changes the period point by the corresponding element of $\Gamma_{2d}$, so composing with the quotient map gives a well-defined $\mathcal{M}_{2d} \to \Gamma_{2d}\backslash D_{2d}$ on isomorphism classes of polarized surfaces. Since $\Gamma_{2d}$ consists of integral isometries it is a subgroup of the group $G_{\mathbb{Z}}$ of definition 7.3.3, so it acts properly discontinuously on $D_{2d}$ by proposition 7.3.5.

*Injectivity.* Suppose $\mathcal{P}(S,h,\alpha)$ and $\mathcal{P}(S',h',\alpha')$ lie in the same $\Gamma_{2d}$-orbit, say $g\cdot\mathcal{P}(S,h,\alpha) = \mathcal{P}(S',h',\alpha')$ with $g = \tilde{g}|_{\Lambda_{2d}}$. Set

$$\varphi = (\alpha')^{-1}\circ\tilde{g}\circ\alpha\colon H^2(S,\mathbb{Z})\to H^2(S',\mathbb{Z})$$

a composite of isometries, hence an isometry. It satisfies $\varphi(h) = (\alpha')^{-1}\tilde{g}(h_0) = (\alpha')^{-1}(h_0) = h'$, and it is a Hodge isometry: with $\sigma$ spanning $H^{2,0}(S)$ and $\sigma'$ spanning $H^{2,0}(S')$, the hypothesis says $[\tilde{g}_{\mathbb{C}}\alpha_{\mathbb{C}}\sigma] = [\alpha'_{\mathbb{C}}\sigma']$, so $\varphi_{\mathbb{C}}(\mathbb{C}\sigma) = \mathbb{C}\sigma'$. Part (2) of theorem 8.4.3 makes the two polarized surfaces isomorphic. So $\mathcal{M}_{2d}\to\Gamma_{2d}\backslash D_{2d}$ is injective, and taking $\tilde{g} = \mathrm{id}$ shows that $\mathcal{P}$ itself is injective on marked surfaces.

*Surjectivity, cited.* Every point of $D_{2d}$ lying on none of the hyperplane sections $\delta^\perp$ with $\delta\in\Lambda_{2d}$ and $\langle\delta,\delta\rangle = -2$ is the period point of a marked polarized K3 surface of degree $2d$. This is the surjectivity theorem for the K3 period map, due to Todorov, Looijenga and Siu and recorded in [1, the K3 period map]; it constructs the surface from its periods by transcendental means and is not proved here. The excluded classes are the ones that would prevent the polarization from being ample. A condition of exactly this shape appeared in section 8.3 as the locus on which extra Hodge classes are acquired, and here exercise 8.4.1 (4) identifies each $\delta^\perp\cap D_{2d}$ as itself a period domain, of dimension 18.

*The count.* Injectivity and surjectivity together identify $\mathcal{M}_{2d}$ with the image in $\Gamma_{2d}\backslash D_{2d}$ of the complement of those hyperplane sections, so the moduli of polarized K3 surfaces of degree $2d$ is 19-dimensional for every $d$. The independence of the degree is visible in the lattice: changing $d$ changes the discriminant of $\Lambda_{2d}$ but not its rank or its signature, and $D_{2d}$ depends by proposition 8.4.2 only on the signature.

*The count, checked on quartics.* Let $S\subseteq\mathbb{CP}^3$ be a smooth quartic surface. Adjunction gives $K_S = \mathcal{O}_S$, and $h^{1,0}(S) = 0$, so $S$ is a K3 surface [4, hypersurfaces in projective space]; the hyperplane class $h$ is ample and primitive with $\langle h,h\rangle = \deg S = 4$, making $(S,h)$ a polarized K3 surface of degree 4, so $d = 2$. These surfaces form a family of the kind studied in section 7.4. By theorem 1.3.11 the quartic forms on $\mathbb{CP}^3$ span a space of dimension $\binom{7}{4} = 35$, so the smooth quartics are parametrized by an open subset of $\mathbb{CP}^{34}$, and two of them that differ by a projective transformation are isomorphic as polarized surfaces. Since $\dim\mathrm{PGL}_4 = 15$ and the stabilizer of a smooth quartic is finite (exercise 8.4.1 (3)), those transformations account for a 15-dimensional family of such identifications, and

$$34 - 15 = 19$$

This is a parameter count rather than a second proof, but it reproduces the dimension obtained from the period domain out of data (the dimension of a linear system and of a projective linear group) with no Hodge theory in it.

The identification is more than a bijection of sets. The quotient $\Gamma_{2d}\backslash D_{2d}$ is an arithmetic quotient of a bounded symmetric domain, and such quotients carry an algebraic structure which the moduli of polarized K3 surfaces then inherits through the period map; nothing here depends on that structure. See also [10] for arithmetic quotients, and [1, period mappings and K3 surfaces] for the K3 case. What the present section supplies is the Hodge-theoretic content: the domain, its geometry, and the two theorems that make the period point a complete invariant of the surface.

Every variety in Part II has been smooth and projective, or at worst compact Kähler, and every Hodge structure attached to one has been pure. Chapter 9 drops the compactness, and the singular case follows it. Purity fails in both, and what replaces it is a second filtration.

### Exercise 8.4.1

1. Verify the assertions about $\Lambda_{K3}$ made in definition 8.4.1: that $U$ and $E_8$ are even, that $U$ has signature $(1,1)$ and determinant $-1$, and hence, taking the unimodularity of $E_8$ as known, that $\Lambda_{K3}$ is even and unimodular of rank 22 and signature $(3,19)$. Then show that for every $d \geq 1$ the vector $h_0 = e + df$ of the first hyperbolic summand is primitive with $\langle h_0, h_0 \rangle = 2d$, and that no vector of $\Lambda_{K3}$ has odd square.

2. Let $\Lambda$ be a unimodular lattice and $h_0 \in \Lambda$ a primitive vector with $\langle h_0, h_0 \rangle = 2d \neq 0$, and set $K = h_0^{\perp}$.

    1. Show that $y \mapsto \langle y, h_0 \rangle$ is a surjection $\Lambda \to \mathbb{Z}$, and deduce that $\Lambda = \mathbb{Z}x \oplus K$ for some $x$ with $\langle x, h_0 \rangle = 1$, and that $h_0 = 2d\,x + k$ for some $k \in K$.
    2. Show that restriction of functionals gives a surjection $\Lambda \to K^{\vee}$ with kernel $\mathbb{Z}h_0$, and conclude that the discriminant group $K^{\vee}/K$ is cyclic of order $2d$.

    Deduce that $\Lambda_{2d}$ has rank 21 and signature $(2,19)$ and is unimodular for no $d \geq 1$.

3. Let $(S,h)$ be a polarized K3 surface of degree $2d$ and let $\mathrm{Aut}(S,h)$ be the group of automorphisms $f$ of $S$ with $f^*h = h$.

    1. Show that $f \mapsto f^*$ is an injective anti-homomorphism from $\mathrm{Aut}(S,h)$ into $\mathrm{O}\big(H^2(S,\mathbb{Z})\big)$, so that $f \mapsto (f^*)^{-1}$ is an injective homomorphism, and that their common image consists of isometries fixing $h$ and fixing the period point.
    2. Using proposition 7.3.5, deduce that $\mathrm{Aut}(S,h)$ is finite.

4. Let $\lambda \in \Lambda_{2d}$ be nonzero and let $D_\lambda = \{[\sigma] \in D_{2d} \big| \langle \lambda, \sigma \rangle = 0\}$.

    1. Show that $D_\lambda \neq \emptyset$ if and only if $\langle \lambda, \lambda \rangle < 0$.
    2. Show that when $\langle \lambda, \lambda \rangle < 0$ the set $D_\lambda$ is the period domain attached by the recipe of this section to the lattice $\lambda^{\perp} \cap \Lambda_{2d}$, of dimension 18.

    How does this compare with the local description of a Hodge locus in proposition 8.3.3?

5. Let $(S,h,\alpha)$ be a marked polarized K3 surface of degree $2d$ with period point $[\sigma] \in D_{2d}$. Show that
$$\alpha\big(\mathrm{NS}(S)\big) \cap \Lambda_{2d} = \big\{\, \lambda \in \Lambda_{2d} \;\big|\; \langle \lambda, \sigma \rangle = 0 \,\big\}$$
and that $\rho(S)$ is 1 plus the rank of the right-hand side. Deduce that $\rho(S) = 1$ for every period point outside a countable union of the sets $D_\lambda$ of exercise 8.4.1 (4), and compare with example 8.3.7.

6. Let $f\colon \mathcal{X} \to S$ be a smooth projective family whose fibres are polarized K3 surfaces of degree $2d$, with period map $\Phi$ in degree 2.

    1. Assuming $d\tilde{\Phi}$ is injective at some point, show that $\dim_{\mathbb{C}} S \leq 19$, and explain why part (2) of exercise 7.3.1 (4) says nothing here.

2. The smooth quartic surfaces in $\mathbb{C}\mathrm{P}^3$ are parametrized by an open subset of $\mathbb{C}\mathrm{P}^{34}$. Explain why the map to $\Gamma_{2d}\backslash D_{2d}$ attached to this family cannot separate two quartics differing by a projective transformation, and reconcile $34 - 15 = 19$ with (1).

# Part III

# Mixed Hodge Structures and Cycles

Two hypotheses have been in force since the first page. Every variety has been compact, and every variety has been smooth. Both were doing work: compactness gave the Hodge theorem its elliptic operator and its finite-dimensional harmonic spaces, and smoothness gave the Hodge decomposition a bigrading to decompose into. Together they produced purity, the statement that $H^k$ carries a Hodge structure of weight exactly $k$, and purity is what every result of Part I and Part II rests on.

Neither hypothesis survives contact with the objects algebraic geometry actually produces. The complement of a hypersurface is smooth and not compact; a nodal curve is compact and not smooth; the fibres of the families of Chapter 7 degenerate to both. On such a variety $H^k$ is not pure, and the question is not whether purity fails but what replaces it.

The answer is a second filtration. Alongside the Hodge filtration $F^\bullet$ there is a weight filtration $W_\bullet$, and while $H^k$ itself is no longer pure, the graded pieces $\mathrm{gr}_j^W H^k$ are, each of its own weight. This is a mixed Hodge structure, and the theorem of this Part is that every complex algebraic variety has one, functorially. Chapter 9 does the smooth open case, where the construction runs on differential forms with logarithmic poles along a boundary divisor. **??** drops smoothness, where the construction runs instead on simplicial resolutions. **??** then asks the question the whole book has been circling: which Hodge classes come from algebraic subvarieties.

# 9

# Mixed Hodge Structures: Smooth Open Varieties

The simplest variety that is not compact is a compact one with a point removed, and it already breaks purity. Let $E$ be an elliptic curve and $U = E \setminus \{p\}$. Then $H^1(U, \mathbb{Q})$ has rank two, as does $H^1(E, \mathbb{Q})$, but the two are not isomorphic as Hodge structures and cannot be: one of the two classes on $U$ is the restriction of a class on $E$, of weight one, while the other is a residue at the puncture and behaves like a class of weight two. A single weight does not describe $H^1(U)$.

What describes it is a pair of filtrations. The Hodge filtration $F^\bullet$ survives unchanged from chapter 5. Beside it sits an increasing filtration $W_\bullet$ by weight, arranged so that the graded pieces $\mathrm{gr}_j^W$ are pure of weight $j$ even though the total space is not pure of any weight. That pair is a mixed Hodge structure, and the theorem of this chapter is that the cohomology of every smooth complex variety carries one.

The construction is due to Deligne and it is concrete. Compactify $U$ to a smooth $X$ whose boundary $D = X \setminus U$ has normal crossings, which is possible by Hironaka's resolution theorem and is the one input taken on faith here. Then replace the de Rham complex of $X$ by the complex of forms with logarithmic poles along $D$: forms that blow up along the boundary, but only as badly as $dz/z$. That complex computes the cohomology of $U$, it carries an obvious filtration by how many logarithmic poles a form is allowed, and the filtration by pole order is the weight filtration.

Two spectral sequences then have to be shown to degenerate, one for each filtration, and their degeneration is the technical heart of the chapter. Both arguments have the same shape and it is worth naming in advance: the terms of each page carry pure Hodge structures of known weight, the differentials are morphisms of Hodge structures, and a morphism between pure structures of different weights is zero. The bookkeeping is in identifying the weights.

## 9.1 Mixed Hodge Structures

The object attached to a smooth open variety is linear algebra before it is geometry: a rational vector space carrying two filtrations, one defined over $\mathbb{Q}$ and one over $\mathbb{C}$, tied together by a purity requirement on the graded pieces. This section fixes that definition, proves the theorem that makes the resulting category usable, and records the tensor constructions. The geometry that produces such an object begins in section 9.2.

Theorem 4.2.1 attaches to a compact Kähler manifold $X$ a pure Hodge structure of weight $k$ on $H^k(X, \mathbb{Q})$, and definition 5.1.1 extracted that datum. Compactness is not a technicality there. Let $U = \mathbb{C}^*$, a smooth affine curve. It retracts onto the unit circle, so $H^1(U, \mathbb{Q}) \cong \mathbb{Q}$ is one-dimensional. A pure Hodge structure of odd weight $k$ has no summand $V^{p,q}$ with $p = q$, since $p + q = k$ is odd; the symmetry $\overline{V^{p,q}} = V^{q,p}$ of definition 5.1.1 then pairs the nonzero summands off into conjugate pairs of equal dimension, so the underlying space has even dimension. Hence $H^1(\mathbb{C}^*, \mathbb{Q})$ carries no pure Hodge structure of weight 1.

Purity can fail in two ways, and both occur. The first is a shift: $H^1(\mathbb{C}^*, \mathbb{Q})$ does carry a pure structure, the Tate structure $\mathbb{Q}(-1)$ of weight 2 (section 9.3), so the weight exceeds the cohomological degree. The second is genuine mixing, where no single weight serves and the group is assembled from pieces of several weights; the punctured curves of section 9.4 are the first geometric instance. The definition below accommodates both by filtering rather than grading: the weights are separated by an increasing filtration whose graded pieces are pure, one weight at a time.

**Definition 9.1.1: Mixed Hodge Structure**

A *mixed Hodge structure* $V = (V_{\mathbb{Q}}, W_\bullet, F^\bullet)$ consists of a finite-dimensional $\mathbb{Q}$-vector space $V_{\mathbb{Q}}$ together with

1. an increasing filtration $\cdots \subseteq W_{m-1}V \subseteq W_mV \subseteq \cdots$ of $V_{\mathbb{Q}}$ by $\mathbb{Q}$-subspaces, the *weight filtration*, with $W_mV = 0$ for $m \ll 0$ and $W_mV = V_{\mathbb{Q}}$ for $m \gg 0$;
2. a decreasing filtration $\cdots \supseteq F^pV_{\mathbb{C}} \supseteq F^{p+1}V_{\mathbb{C}} \supseteq \cdots$ of $V_{\mathbb{C}} = V_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$ by complex subspaces, the *Hodge filtration*, with $F^pV_{\mathbb{C}} = V_{\mathbb{C}}$ for $p \ll 0$ and $F^pV_{\mathbb{C}} = 0$ for $p \gg 0$,

subject to one compatibility. Write $W_mV_{\mathbb{C}} = W_mV \otimes_{\mathbb{Q}} \mathbb{C}$, write $\mathrm{gr}^W_m V = W_mV/W_{m-1}V$ for the graded pieces of the weight filtration, and equip $\mathrm{gr}^W_m V_{\mathbb{C}}$ with the filtration induced by $F$,

$$F^p\mathrm{gr}^W_m V_{\mathbb{C}} = \big((F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}) + W_{m-1}V_{\mathbb{C}}\big)/W_{m-1}V_{\mathbb{C}} \tag{9.1}$$

The requirement is that for every $m$ this filtration make $\mathrm{gr}^W_m V$ a pure Hodge structure of weight $m$ in the sense of definition 5.1.1, that is, that it satisfy the transversality condition (5.1) in weight $m$.
The structure is *pure of weight* $k$ if $\mathrm{gr}^W_m V = 0$ for $m \neq k$. The *weights* of $V$ are the integers $m$ with $\mathrm{gr}^W_m V \neq 0$. An *integral* mixed Hodge structure carries in addition a finitely generated abelian group $V_{\mathbb{Z}}$ with $V_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} = V_{\mathbb{Q}}$.

Two features of the definition deserve emphasis before any example. The first is that the two filtrations live over different fields: $W$ is rational and $F$ is only complex. Nothing forces them to be compatible beyond (9.1). Over $\mathbb{C}$ the pair $(W_{\mathbb{C}}, F)$ can always be split simultaneously, by Deligne's canonical bigrading (box 9.1.9), but no such splitting need be defined over $\mathbb{Q}$, and that failure is exactly where the new invariants of the theory sit, as example 9.1.4 shows. The second is that purity is imposed one weight at a time and says nothing about how the pieces are glued. A mixed Hodge structure is therefore an iterated extension of pure structures, of increasing weight, and the whole content beyond chapter 5 lies in the extension data.

**Definition 9.1.2: Morphisms and Sub-Structures**

Let $V$ and $V'$ be mixed Hodge structures. A *morphism of mixed Hodge structures* $\varphi\colon V \to V'$ is a $\mathbb{Q}$-linear map $\varphi\colon V_{\mathbb{Q}} \to V'_{\mathbb{Q}}$ such that

$$\varphi(W_m V) \subseteq W_m V' \quad \text{for all } m, \qquad \varphi_{\mathbb{C}}(F^p V_{\mathbb{C}}) \subseteq F^p V'_{\mathbb{C}} \quad \text{for all } p$$

A *sub-mixed Hodge structure* of $V$ is a $\mathbb{Q}$-subspace $C \subseteq V_{\mathbb{Q}}$ which, with the induced filtrations $W_m C = C \cap W_m V$ and $F^p C_{\mathbb{C}} = C_{\mathbb{C}} \cap F^p V_{\mathbb{C}}$, is itself a mixed Hodge structure. A *quotient* is $V/C$ for such a $C$, with the filtrations $W_m(V/C) = (W_m V + C)/C$ and $F^p(V/C)_{\mathbb{C}} =$ the image of $F^p V_{\mathbb{C}}$.
Between integral structures a morphism is required in addition to come from a homomorphism $V_{\mathbb{Z}} \to V'_{\mathbb{Z}}$ of the underlying lattices. Unless the integral structure is named, all statements below are about the rational category.

Where definition 5.1.6 asked a map to respect the bigrading, this definition asks only that it respect two filtrations, and it imposes no condition relating the weights of $V$ and $V'$. The consequence is that morphisms are now plentiful between structures of different weights: the projection of $\mathbb{Q}(0) \oplus \mathbb{Q}(1)$ onto $\mathbb{Q}(0)$ is one, and no nonzero map between the pure structures $\mathbb{Q}(0)$ and $\mathbb{Q}(1)$ is. Lemma 9.1.5 shows that on pure structures the two notions of morphism nevertheless agree.

**Example 9.1.3: Pure and Tate Structures as Mixed Structures**

Three instances of definition 9.1.1.

1. A pure Hodge structure $V$ of weight $k$ becomes a mixed one by setting $W_m V = 0$ for $m < k$ and $W_m V = V_{\mathbb{Q}}$ for $m \geq k$, with $F$ its Hodge filtration (definition 5.1.4). The only nonzero graded piece is $\mathrm{gr}^W_k V = V$, and (9.1) returns $F$ itself, so the requirement is exactly that $V$ was a pure structure of weight $k$. Every pure structure is read as a mixed one this way from here on.

2. The Tate structure $\mathbb{Q}(n) = \mathbb{Z}(n) \otimes_{\mathbb{Z}} \mathbb{Q}$ of definition 5.1.10 is one-dimensional, concentrated in bidegree $(-n, -n)$, pure of weight $-2n$. Its weight filtration jumps once, at $m = -2n$.

3. Let $V_{\mathbb{Q}} = \mathbb{Q}e \oplus \mathbb{Q}f$ with $W_{-2}V = \mathbb{Q}f$, $W_{-1}V = W_{-2}V$, $W_0 V = V_{\mathbb{Q}}$ and $W_m V = 0$ for $m \leq -3$, and let $F^{-1}V_{\mathbb{C}} = V_{\mathbb{C}}$, $F^0 V_{\mathbb{C}} = \mathbb{C}e$, $F^1 V_{\mathbb{C}} = 0$. Then $\mathrm{gr}^W_{-2} V = \mathbb{Q}f$ receives $F^0 = 0$ and $F^{-1} =$ everything, so it is $\mathbb{Q}(1)$; and $\mathrm{gr}^W_0 V = \mathbb{Q}e$ receives $F^0 =$ everything and $F^1 = 0$, so it is $\mathbb{Q}(0)$. This is the direct sum $\mathbb{Q}(1) \oplus \mathbb{Q}(0)$, with $W$ and $F$ the sums of the two structures.

The third instance splits: it has a sub-structure, namely $\mathbb{Q}e$, mapping isomorphically onto the top graded piece. The next example is the same pair of graded pieces glued differently, and it is the reason the theory exists.

**Example 9.1.4: A Nonsplit Extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$**

Fix $z \in \mathbb{C}$. Let $V_{\mathbb{Z}} = \mathbb{Z}e \oplus \mathbb{Z}f$, where $f$ is the generator $2\pi i$ of the lattice of $\mathbb{Z}(1)$, write $V_{\mathbb{Q}} = V_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$, and put

$$W_{-3}V = 0, \qquad W_{-2}V = W_{-1}V = \mathbb{Q}f, \qquad W_0 V = V_{\mathbb{Q}}$$
$$F^1 V_{\mathbb{C}} = 0, \qquad F^0 V_{\mathbb{C}} = \mathbb{C}\,(e + zf), \qquad F^{-1} V_{\mathbb{C}} = V_{\mathbb{C}}$$

Call the resulting datum $V_z$. It is a mixed Hodge structure. Indeed $F^0 V_{\mathbb{C}} \cap W_{-2} V_{\mathbb{C}} = \mathbb{C}(e + zf) \cap \mathbb{C}f = 0$, so $\mathrm{gr}^W_{-2} V_z$ carries $F^0 = 0$ and $F^{-1} =$ everything and is $\mathbb{Z}(1)$; and $F^0 V_{\mathbb{C}}$ maps onto $\mathrm{gr}^W_0 V_{z,\mathbb{C}}$, so that piece carries $F^0 =$ everything and $F^1 = 0$ and is $\mathbb{Z}(0)$. The inclusion $\mathbb{Z}(1) \to V_z$ and the projection $V_z \to \mathbb{Z}(0)$ are morphisms, so

$$0 \to \mathbb{Z}(1) \to V_z \to \mathbb{Z}(0) \to 0$$

is an extension of mixed Hodge structures, and its graded pieces do not depend on $z$.

Every extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ arises this way, for exactly one $z$ once a lift is fixed. Given such an extension $V$, let $f$ generate the lattice $W_{-2}V \cap V_{\mathbb{Z}} = \mathbb{Z}(1)$ and let $e \in V_{\mathbb{Z}}$ be any lift of the generator of $\mathbb{Z}(0)$. Purity of $\mathrm{gr}^W_0 V$ forces $F^0$ to surject onto it and purity of $\mathrm{gr}^W_{-2} V$ forces $F^0 \cap W_{-2} V_{\mathbb{C}} = 0$, so $F^0 V_{\mathbb{C}}$ is a line meeting $\mathbb{C}f$ trivially and surjecting onto $\mathrm{gr}^W_0 V_{\mathbb{C}}$, hence $F^0 V_{\mathbb{C}} = \mathbb{C}(e + zf)$ for a unique $z \in \mathbb{C}$. The remaining steps are forced as well: $F^{-1} V_{\mathbb{C}} \supseteq W_{-2} V_{\mathbb{C}} + F^0 V_{\mathbb{C}} = V_{\mathbb{C}}$, and $F^1 V_{\mathbb{C}}$ has zero image in both graded pieces, hence is zero.

Changing the lift replaces $e$ by $e + nf$ with $n \in \mathbb{Z}$, which replaces $z$ by $z - n$. So the invariant of the extension is the class of $z$ in $\mathbb{C}/\mathbb{Z}$, equivalently the number

$$\exp(2\pi i z) \in \mathbb{C}^*$$

and $V_z \cong V_{z'}$ by an isomorphism inducing the identity on $\mathbb{Z}(1)$ and on $\mathbb{Z}(0)$ if and only if $z - z' \in \mathbb{Z}$.

The extension splits precisely when $z \in \mathbb{Z}$. A splitting is a morphism $s\colon \mathbb{Z}(0) \to V_z$ with $\pi \circ s = \mathrm{id}$, so $s(1) = e + nf$ for some $n \in \mathbb{Z}$; compatibility with $F$ forces $e + nf \in F^0 V_{\mathbb{C}} = \mathbb{C}(e + zf)$, and comparing coefficients of $e$ makes the scalar 1, so $n = z$ and $z \in \mathbb{Z}$. Conversely if $z = n \in \mathbb{Z}$ then $\mathbb{Z}(e + nf)$ is a sub-structure mapping isomorphically onto $\mathbb{Z}(0)$, and $V_z = \mathbb{Z}(1) \oplus \mathbb{Z}(0)$.

The same computation with $\mathbb{Q}$ in place of $\mathbb{Z}$ throughout classifies extensions of $\mathbb{Q}(0)$ by $\mathbb{Q}(1)$ by the class of $z$ in $\mathbb{C}/\mathbb{Q}$, and $V_z$ splits as a rational mixed Hodge structure exactly when $z \in \mathbb{Q}$.

The graded pieces of $V_z$ are $\mathbb{Z}(1)$ and $\mathbb{Z}(0)$ for every $z$, so a mixed Hodge structure is not determined by its associated graded. What distinguishes $V_z$ from $V_0$ is the position of one complex line relative to one rational lattice, and the invariant that records it, $\exp(2\pi i z) \in \mathbb{C}^*$, is transcendental in nature: no amount of rational linear algebra produces it, as exercise 9.1.1 (6) makes precise. This is the shape every computation in this chapter takes. The pure pieces are cohomology groups of compact objects, computed by the theory of Part I, and the gluing between them is the new geometric information contributed by the removed locus; the punctured curves of section 9.4 are the first case worked out in full.

Before the main theorem, three facts about the category are needed. The first compares the two notions of morphism on pure structures, the second manufactures sub-objects and quotients, and the third is the vanishing statement that drives everything else.

**Lemma 9.1.5: Morphisms Between Pure Structures**

Let $V$ and $V'$ be pure Hodge structures of weights $k$ and $l$, regarded as mixed Hodge structures as in example 9.1.3, and let $\varphi\colon V \to V'$ be a morphism of mixed Hodge structures.

1. If $k = l$ then $\varphi_{\mathbb{C}}(V^{p,q}) \subseteq V'^{p,q}$ for all $p, q$; that is, $\varphi$ is a morphism of Hodge structures in the sense of definition 5.1.6.
2. If $k \neq l$ then $\varphi = 0$.
3. The kernel and image of a morphism of pure Hodge structures of weight $k$ are sub-Hodge structures of weight $k$, and the quotient of a pure Hodge structure of weight $k$ by a sub-Hodge structure is pure of weight $k$.

***Proof*:**
Since $\varphi$ is defined over $\mathbb{Q}$, its complexification commutes with the conjugation of $V_{\mathbb{C}}$ over $V_{\mathbb{R}}$, so $\varphi_{\mathbb{C}}(\overline{F^qV_{\mathbb{C}}}) \subseteq \overline{F^qV'_{\mathbb{C}}}$. Combining this with compatibility with $F$ and using the recovery formula (5.2),

$$\varphi_{\mathbb{C}}\big(V^{p,q}\big) = \varphi_{\mathbb{C}}\big(F^pV_{\mathbb{C}} \cap \overline{F^qV_{\mathbb{C}}}\big) \subseteq F^pV'_{\mathbb{C}} \cap \overline{F^qV'_{\mathbb{C}}} \tag{9.2}$$

for every $p, q$ with $p + q = k$.

*(1).* When $k = l$, the right-hand side of (9.2) is $V'^{p,q}$ by (5.2) again, which is the assertion.

*(2).* Suppose first $l < k$ and let $p + q = k$. Then $q \geq l - p + 1$, so $\overline{F^qV'_{\mathbb{C}}} \subseteq \overline{F^{l-p+1}V'_{\mathbb{C}}}$, and (5.1) for $V'$ gives $F^pV'_{\mathbb{C}} \oplus \overline{F^{l-p+1}V'_{\mathbb{C}}} = V'_{\mathbb{C}}$. Hence the right-hand side of (9.2) is zero, so $\varphi_{\mathbb{C}}$ kills every $V^{p,q}$ and therefore kills $V_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}$. Suppose next $l > k$. As a mixed structure $V'$ has $W_kV' = 0$, while compatibility with $W$ gives $\varphi(V_{\mathbb{Q}}) = \varphi(W_kV) \subseteq W_kV' = 0$.

*(3).* Let $\varphi\colon V \to V'$ be a morphism of pure structures of weight $k$. By (1) the map $\varphi_{\mathbb{C}}$ is bigraded, so

$$\ker\varphi_{\mathbb{C}} = \bigoplus_{p+q=k} \big(\ker\varphi_{\mathbb{C}} \cap V^{p,q}\big), \qquad \mathrm{im}\varphi_{\mathbb{C}} = \bigoplus_{p+q=k} \varphi_{\mathbb{C}}\big(V^{p,q}\big)$$

the first because a vector lies in the kernel exactly when each of its bihomogeneous components does, the second because $\varphi_{\mathbb{C}}$ carries the $(p, q)$-summand into the $(p, q)$-summand. Both $\ker\varphi$ and $\mathrm{im}\varphi$ are defined over $\mathbb{Q}$, so their complexifications are stable under conjugation, and the induced bigradings inherit $\overline{S^{p,q}} = S^{q,p}$ from $V$ and $V'$; each is therefore a pure Hodge structure of weight $k$. For a subspace $S$ carrying a bigrading in this way, $F^pS_{\mathbb{C}} = \bigoplus_{r\geq p} S^{r,k-r} = S_{\mathbb{C}} \cap F^pV_{\mathbb{C}}$, so its Hodge filtration is the one induced from $V$. For the quotient, if $S \subseteq V$ is a sub-Hodge structure then $V_{\mathbb{C}}/S_{\mathbb{C}} = \bigoplus_{p+q=k} V^{p,q}/S^{p,q}$ carries the quotient bigrading, again conjugation-symmetric and defined over $\mathbb{Q}$, completing the proof.

Part (2) is the mechanism that will kill differentials in section 9.3: a map between pure structures of different weights has no room to be nonzero, and the whole degeneration argument there consists of arranging for such a map. The proof of part (1) also shows that proposition 5.1.7, whose argument uses only the bigrading and not the lattice, applies verbatim to morphisms of rational Hodge structures; it is used in that form throughout this section.

**Lemma 9.1.6: Sub-Structures and Quotients**

Let $V$ be a mixed Hodge structure.

1. For every $r$ the subspace $W_rV$ is a sub-mixed Hodge structure, with $\mathrm{gr}^W_m(W_rV) = \mathrm{gr}^W_m V$ for $m \le r$ and $0$ for $m > r$.

2. Let $C \subseteq V_{\mathbb{Q}}$ be a sub-mixed Hodge structure. Then for each $m$ the injection $\mathrm{gr}^W_m C \to \mathrm{gr}^W_m V$ identifies $\mathrm{gr}^W_m C$ with a sub-Hodge structure of $\mathrm{gr}^W_m V$, and the quotient $V/C$ of definition 9.1.2 is a mixed Hodge structure with
$$\mathrm{gr}^W_m(V/C) = \mathrm{gr}^W_m V/\mathrm{gr}^W_m C$$
as pure Hodge structures of weight $m$.

***Proof*:**
*(1).* The induced weight filtration on $W_rV$ is $W_mV \cap W_rV = W_{\min(m,r)}V$, so the graded pieces are as stated. For $m \le r$ the induced Hodge filtration on $\mathrm{gr}^W_m(W_rV)$ is computed by (9.1) from $F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}} \cap W_rV_{\mathbb{C}} = F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}$, which is the datum computing $F^p\mathrm{gr}^W_m V$. So $\mathrm{gr}^W_m(W_rV)$ and $\mathrm{gr}^W_m V$ agree as filtered spaces, and purity is inherited.

*(2).* The map $\mathrm{gr}^W_m C \to \mathrm{gr}^W_m V$ is injective because $W_mC = C \cap W_mV$ and $W_{m-1}C = C \cap W_{m-1}V$, and it is a morphism of mixed Hodge structures between pure structures of weight $m$, hence a morphism of Hodge structures by lemma 9.1.5. By proposition 5.1.7 its image meets $F^p\mathrm{gr}^W_m V_{\mathbb{C}}$ in the image of $F^p\mathrm{gr}^W_m C_{\mathbb{C}}$, so the filtration that $\mathrm{gr}^W_m C$ carries as a subquotient of $C$ agrees with the one it inherits as a subspace of $\mathrm{gr}^W_m V$; by lemma 9.1.5(3) it is a sub-Hodge structure of $\mathrm{gr}^W_m V$.

Next the underlying spaces. Since $W_{m-1}V \subseteq W_mV$, the modular law gives
$$W_mV \cap \big(W_{m-1}V + C\big) = W_{m-1}V + \big(W_mV \cap C\big) = W_{m-1}V + W_mC$$
so that
$$\mathrm{gr}^W_m(V/C) = \frac{W_mV + C}{W_{m-1}V + C} \cong \frac{W_mV}{W_mV \cap (W_{m-1}V + C)} = \frac{W_mV}{W_{m-1}V + W_mC} = \mathrm{gr}^W_m V/\mathrm{gr}^W_m C$$
the middle isomorphism being the second isomorphism theorem. Under it, the class of $y \in W_mV$ modulo $C$ corresponds to the class of $[y] \in \mathrm{gr}^W_m V$ modulo $\mathrm{gr}^W_m C$.

It remains to identify the filtration (9.1) on $\mathrm{gr}^W_m(V/C)$ with the quotient of $F^p\mathrm{gr}^W_m V_{\mathbb{C}}$. One inclusion is immediate: an element of $F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}$ maps into $F^p(V/C)_{\mathbb{C}} \cap W_m(V/C)_{\mathbb{C}}$, so the quotient filtration is contained in the induced one. For the reverse, let $\xi \in F^p(V/C)_{\mathbb{C}} \cap W_m(V/C)_{\mathbb{C}}$. Choose a representative $y \in F^pV_{\mathbb{C}}$ of $\xi$; membership in $W_m(V/C)_{\mathbb{C}}$ gives a decomposition $y = w + c$ with $w \in W_mV_{\mathbb{C}}$ and $c \in C_{\mathbb{C}}$. Let $r$ be minimal with $c \in W_rC_{\mathbb{C}}$.

If $r \le m$ then $y = w + c \in W_mV_{\mathbb{C}}$, so $y \in F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}$ and the class of $\xi$ in $\mathrm{gr}^W_m(V/C)$ is the image of $[y] \in F^p\mathrm{gr}^W_m V_{\mathbb{C}}$, as required. If $r > m$ then $w \in W_mV_{\mathbb{C}} \subseteq W_{r-1}V_{\mathbb{C}}$, so $[y] = [c]$ in $\mathrm{gr}^W_r V_{\mathbb{C}}$; moreover $y \in F^pV_{\mathbb{C}} \cap W_rV_{\mathbb{C}}$, whence $[c] = [y] \in F^p\mathrm{gr}^W_r V_{\mathbb{C}}$. Since $[c]$ lies in $\mathrm{gr}^W_r C_{\mathbb{C}}$, the identification of the first paragraph gives $[c] \in F^p\mathrm{gr}^W_r C_{\mathbb{C}}$, so there is $c' \in F^pC_{\mathbb{C}} \cap W_rC_{\mathbb{C}}$ with

$c - c' \in W_{r-1}C_{\mathbb{C}}$. Replacing $y$ by $y - c'$, which again lies in $F^p V_{\mathbb{C}}$ and again represents $\xi$, replaces $c$ by $c - c'$ and lowers $r$. Since $W_r C = 0$ for $r \ll 0$, finitely many replacements reach the case $r \leq m$.

Finally, $\mathrm{gr}^W_m(V/C) = \mathrm{gr}^W_m V/\mathrm{gr}^W_m C$ with the quotient filtration is the quotient of a pure Hodge structure of weight $m$ by a sub-Hodge structure, hence pure of weight $m$ by lemma 9.1.5(3), which completes the proof.

**Lemma 9.1.7: Vanishing Below the Weight**

Let $P$ be a pure Hodge structure of weight $n$ and let $T$ be a mixed Hodge structure with $W_{n-1}T = T_{\mathbb{Q}}$, that is, with all weights at most $n-1$. Then every morphism $\psi\colon P \to T$ of mixed Hodge structures is zero.

***Proof*:**
Induction on the number of nonzero graded pieces of $T$. If $T = 0$ there is nothing to prove. Otherwise let $b$ be the largest weight of $T$, so $b \leq n-1$ and $W_b T = T_{\mathbb{Q}}$. By lemma 9.1.6, $W_{b-1}T$ is a sub-mixed Hodge structure and $T/W_{b-1}T$ is a mixed Hodge structure whose only graded piece is $\mathrm{gr}^W_b T$, hence pure of weight $b$. The composite of $\psi$ with the projection is a morphism of mixed Hodge structures from a pure structure of weight $n$ to a pure structure of weight $b \neq n$, hence zero by lemma 9.1.5(2). Therefore $\psi(P_{\mathbb{Q}}) \subseteq W_{b-1}T$, and $\psi$ is a morphism of mixed Hodge structures into $W_{b-1}T$, since the filtrations there are induced from $T$. That target has fewer nonzero graded pieces, so the inductive hypothesis gives $\psi = 0$, completing the proof.

For pure structures, proposition 5.1.7 obtained strictness from the bigrading: a preimage could be truncated componentwise. A mixed Hodge structure has no such bigrading available, and no reason is visible from the definition why a class deep in the weight filtration of the target should have a preimage equally deep in the source. That it does is the theorem below, and everything this chapter proves about $H^*(U)$ rests on it. Without strictness the associated graded would not be exact, the category would not be abelian, and the uniqueness assertions of section 9.3 would be unavailable.

**Theorem 9.1.8: Strictness and the Abelian Category of Mixed Hodge Structures**

Let $\varphi\colon V \to V'$ be a morphism of mixed Hodge structures.

1. $\varphi$ is *strict for the weight filtrations*:
$$\varphi(V_{\mathbb{Q}}) \cap W_m V' = \varphi(W_m V) \qquad \text{for every } m$$
2. $\varphi$ is *strict for the Hodge filtrations*:
$$\varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^p V'_{\mathbb{C}} = \varphi_{\mathbb{C}}\big(F^p V_{\mathbb{C}}\big) \qquad \text{for every } p$$
3. $\ker\varphi$ and $\mathrm{im}\varphi$, with the filtrations induced from $V$ and $V'$, are mixed Hodge structures, and for every $m$
$$\mathrm{gr}^W_m(\ker\varphi) = \ker(\mathrm{gr}^W_m\varphi), \qquad \mathrm{gr}^W_m(\mathrm{im}\varphi) = \mathrm{im}(\mathrm{gr}^W_m\varphi)$$
4. Mixed Hodge structures and their morphisms form an abelian category, and each $V \mapsto \mathrm{gr}^W_m V$ is an exact functor from it to the category of pure Hodge structures of weight $m$.

***Proof*:**
Write $\ell(V)$ for the number of $m$ with $\mathrm{gr}^W_m V \neq 0$. Throughout, $\mathrm{gr}^W_m\varphi\colon \mathrm{gr}^W_m V \to \mathrm{gr}^W_m V'$ denotes the induced map; it is $\mathbb{Q}$-linear and compatible with the filtrations (9.1), so by lemma 9.1.5 it is a morphism of pure Hodge structures of weight $m$ and proposition 5.1.7 applies to it.

*Step 1: strictness for $W$, by induction on $\ell(V)$.* The inductive hypothesis is statements (1), (2) and (3) for every morphism whose source has fewer than $\ell(V)$ nonzero graded pieces; this is legitimate because Steps 2 and 3 below derive (2) and (3) for a morphism from (1) for that same morphism, so only (1) has to be propagated.

If $\ell(V) = 0$ then $V = 0$ and there is nothing to prove. Let $n$ be the largest weight of $V$, so $W_n V = V_{\mathbb{Q}}$, and set $A = W_{n-1}V$, a sub-mixed Hodge structure with $\ell(A) = \ell(V) - 1$ by lemma 9.1.6(1). By the inductive hypothesis, $\varphi|_A$ is strict for $W$ and its image $C' = \varphi(A)$ is a sub-mixed Hodge structure of $V'$.

For $m \geq n$ the claim is trivial, since then $W_m V = V_{\mathbb{Q}}$ and $\varphi(V_{\mathbb{Q}}) \subseteq W_n V' \subseteq W_m V'$. The substance is the case $m = n-1$, from which the remaining cases follow: granting
$$\varphi(V_{\mathbb{Q}}) \cap W_{n-1}V' = \varphi(A) \tag{9.3}$$
and taking $m \leq n-1$,
$$\varphi(V_{\mathbb{Q}}) \cap W_m V' = \big(\varphi(V_{\mathbb{Q}}) \cap W_{n-1}V'\big) \cap W_m V' = \varphi(A) \cap W_m V' = \varphi(W_m A) = \varphi(W_m V)$$
the third equality by the inductive hypothesis for $\varphi|_A$ and the fourth because $W_m A = W_m V$ for $m \leq n-1$.

The inclusion $\supseteq$ in (9.3) holds because $A = W_{n-1}V$ maps into $W_{n-1}V'$. For the reverse, let $x \in V_{\mathbb{Q}}$ with $\varphi(x) \in W_{n-1}V'$, and let $[x] \in \mathrm{gr}^W_n V$ be its class. Then $\mathrm{gr}^W_n\varphi([x]) = 0$, so $[x]$ lies in $K = \ker(\mathrm{gr}_n^W \varphi)$, a sub-Hodge structure of $\mathrm{gr}_n^W V$ by lemma 9.1.5(3). Let $V_K \subseteq V_{\mathbb{Q}}$ be the preimage of $K$ under $V_{\mathbb{Q}} \twoheadrightarrow \mathrm{gr}_n^W V$.

The subspace $V_K$ is a sub-mixed Hodge structure. Its weight filtration is $W_m V_K = W_m V$ for $m \le n-1$ and $V_K$ for $m \ge n$, so its graded pieces are those of $V$ below $n$ together with $K$ in degree $n$; those below $n$ carry the same induced Hodge filtration as in $V$, since $W_m V_K = W_m V$ there. In degree $n$, (9.1) computes $F^p \mathrm{gr}_n^W V_K$ as the image of $F^p V_{\mathbb{C}} \cap V_{K,\mathbb{C}}$, and this is $F^p \mathrm{gr}_n^W V_{\mathbb{C}} \cap K_{\mathbb{C}}$: a class in the latter is represented by some $v \in F^p V_{\mathbb{C}}$ whose image lies in $K_{\mathbb{C}}$, and such a $v$ lies in $V_{K,\mathbb{C}}$ by definition of $V_K$. So $\mathrm{gr}_n^W V_K$ is $K$ with its sub-Hodge structure, which is pure of weight $n$.

Every $v \in V_K$ has $\mathrm{gr}_n^W \varphi([v]) = 0$, hence $\varphi(v) \in W_{n-1}V' =: B'$. So $\varphi$ restricts to a morphism $V_K \to B'$, and $C' = \varphi(A) \subseteq B'$ is a sub-mixed Hodge structure of $B'$, the filtrations induced from $V'$ and from $B'$ agreeing on it. By lemma 9.1.6(2) the quotients $V_K/A$ and $B'/C'$ are mixed Hodge structures, the first pure of weight $n$ (its only nonzero graded piece is $K$) and the second with all weights at most $n-1$ (its graded pieces are quotients of those of $B'$). The composite $V_K \to B' \to B'/C'$ kills $A$, so it factors through a morphism $V_K/A \to B'/C'$, which is zero by lemma 9.1.7. Hence $\varphi(V_K) \subseteq C' = \varphi(A)$; since $x \in V_K$, this proves (9.3).

*Step 2: strictness for $F$, and the combined form.* All the subspaces in (1) are defined over $\mathbb{Q}$, and intersections and images of $\mathbb{Q}$-subspaces commute with the extension of scalars to $\mathbb{C}$, so (1) may be used in the form $\varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap W_m V'_{\mathbb{C}} = \varphi_{\mathbb{C}}(W_m V_{\mathbb{C}})$. The claim to be proved is the combined statement

$$\varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^p V'_{\mathbb{C}} \cap W_m V'_{\mathbb{C}} = \varphi_{\mathbb{C}}\big(F^p V_{\mathbb{C}} \cap W_m V_{\mathbb{C}}\big) \tag{9.4}$$

for all $p$ and $m$; part (2) is the case $m \gg 0$.

The inclusion $\supseteq$ is clear. For the reverse, let $w'$ lie in the left-hand side, and assume $w' \neq 0$. Let $m_0 \le m$ be minimal with $w' \in W_{m_0} V'_{\mathbb{C}}$. By (1), $w' = \varphi_{\mathbb{C}}(v)$ for some $v \in W_{m_0} V_{\mathbb{C}}$, so the class $[w'] \in \mathrm{gr}_{m_0}^W V'_{\mathbb{C}}$ lies in the image of $\mathrm{gr}_{m_0}^W \varphi$; and $w' \in F^p V'_{\mathbb{C}} \cap W_{m_0} V'_{\mathbb{C}}$ puts $[w']$ in $F^p \mathrm{gr}_{m_0}^W V'_{\mathbb{C}}$. By proposition 5.1.7 applied to $\mathrm{gr}_{m_0}^W \varphi$ there is $\beta \in F^p \mathrm{gr}_{m_0}^W V_{\mathbb{C}}$ with $\mathrm{gr}_{m_0}^W \varphi(\beta) = [w']$, and by (9.1) the class $\beta$ lifts to some $b \in F^p V_{\mathbb{C}} \cap W_{m_0} V_{\mathbb{C}}$. Then

$$w' - \varphi_{\mathbb{C}}(b) \in \varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^p V'_{\mathbb{C}} \cap W_{m_0-1} V'_{\mathbb{C}}$$

since the difference lies in $W_{m_0} V'_{\mathbb{C}}$ and has zero class in $\mathrm{gr}_{m_0}^W V'_{\mathbb{C}}$. Repeating the construction on $w' - \varphi_{\mathbb{C}}(b)$ lowers $m_0$ by at least one each time, and $W_j V' = 0$ for $j \ll 0$, so after finitely many steps the remainder is zero. Summing the elements $b$ produced gives $v_0 \in F^p V_{\mathbb{C}} \cap W_{m_0} V_{\mathbb{C}} \subseteq F^p V_{\mathbb{C}} \cap W_m V_{\mathbb{C}}$ with $\varphi_{\mathbb{C}}(v_0) = w'$, which is (9.4).

*Step 3: kernel and image.* Give $\mathrm{im}\varphi$ the induced filtrations. By (1) and (2), $W_m(\mathrm{im}\varphi) = \varphi(W_m V)$ and $F^p(\mathrm{im}\varphi)_{\mathbb{C}} = \varphi_{\mathbb{C}}(F^p V_{\mathbb{C}})$. Since the filtration on $\mathrm{im}\varphi$ is induced, the map $\mathrm{gr}_m^W(\mathrm{im}\varphi) \to \mathrm{gr}_m^W V'$ is injective. The natural map $\mathrm{gr}_m^W V \to \mathrm{gr}_m^W(\mathrm{im}\varphi)$ induced by $\varphi$ is surjective, and its kernel consists of the classes of $v \in W_m V$ with $\varphi(v) \in \varphi(W_{m-1}V)$; by (1) this is the same as $\varphi(v) \in W_{m-1}V'$, that is, the same as $[v] \in \ker(\mathrm{gr}_m^W \varphi)$. Hence $\mathrm{gr}_m^W(\mathrm{im}\varphi) = \mathrm{im}(\mathrm{gr}_m^W \varphi)$ as subspaces of $\mathrm{gr}_m^W V'$, which by lemma 9.1.5(3) is a pure Hodge structure of weight $m$. Its induced Hodge filtration is the image of $\mathrm{im}\varphi_{\mathbb{C}} \cap F^p V'_{\mathbb{C}} \cap W_m V'_{\mathbb{C}}$, which by (9.4) is the image of $\varphi_{\mathbb{C}}(F^p V_{\mathbb{C}} \cap W_m V_{\mathbb{C}})$, that is $\mathrm{gr}_m^W \varphi\big(F^p \mathrm{gr}_m^W V_{\mathbb{C}}\big)$; by proposition 5.1.7 this is $\mathrm{im}(\mathrm{gr}_m^W \varphi) \cap F^p \mathrm{gr}_m^W V'_{\mathbb{C}}$, the filtration of the sub-Hodge structure. So $\mathrm{im}\varphi$ is a mixed Hodge structure.

For the kernel, first $\mathrm{gr}_m^W(\ker\varphi) = \ker(\mathrm{gr}_m^W \varphi)$. The inclusion $\subseteq$ is clear. Conversely let $[v] \in \ker(\mathrm{gr}_m^W \varphi)$ with $v \in W_m V$; then $\varphi(v) \in \varphi(V_{\mathbb{Q}}) \cap W_{m-1}V' = \varphi(W_{m-1}V)$ by (1), so

$\varphi(v) = \varphi(u)$ with $u \in W_{m-1}V$, and $v - u \in \ker\varphi \cap W_m V$ has class $[v]$. The induced Hodge filtration on $\mathrm{gr}_m^W(\ker\varphi)$ is the image of $\ker\varphi_{\mathbb{C}} \cap F^p V_{\mathbb{C}} \cap W_m V_{\mathbb{C}}$, which is contained in $\ker(\mathrm{gr}_m^W\varphi) \cap F^p\mathrm{gr}_m^W V_{\mathbb{C}}$ and contains it: given $\xi$ in the latter, lift it to $v \in F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}$; then $\varphi_{\mathbb{C}}(v)$ lies in $\varphi_{\mathbb{C}}(V_{\mathbb{C}}) \cap F^pV'_{\mathbb{C}} \cap W_{m-1}V'_{\mathbb{C}}$, so by (9.4) $\varphi_{\mathbb{C}}(v) = \varphi_{\mathbb{C}}(u)$ for some $u \in F^pV_{\mathbb{C}} \cap W_{m-1}V_{\mathbb{C}}$, and $v - u$ is a lift of $\xi$ inside $\ker\varphi_{\mathbb{C}} \cap F^pV_{\mathbb{C}} \cap W_mV_{\mathbb{C}}$. So $\mathrm{gr}_m^W(\ker\varphi)$ is the sub-Hodge structure $\ker(\mathrm{gr}_m^W\varphi)$ of $\mathrm{gr}_m^W V$, and $\ker\varphi$ is a mixed Hodge structure.

*Step 4: the category.* Morphisms between two fixed mixed Hodge structures form a $\mathbb{Q}$-vector space, and the direct sum of two mixed Hodge structures, filtered summandwise, is a biproduct, so the category is additive. By Step 3 the subspace $\ker\varphi$ with its induced filtrations is a mixed Hodge structure, and it is a categorical kernel: a morphism $\psi\colon T \to V$ with $\varphi\psi = 0$ factors through it as a $\mathbb{Q}$-linear map, and the factorization respects both filtrations because those on $\ker\varphi$ are induced from $V$. Dually $\mathrm{coker}\varphi = V'/\mathrm{im}\varphi$ with the quotient filtrations is a mixed Hodge structure by lemma 9.1.6(2), and is a categorical cokernel.

It remains to check that the canonical map $V/\ker\varphi \to \mathrm{im}\varphi$ is an isomorphism, which is where strictness is used. It is a bijection of $\mathbb{Q}$-vector spaces and a morphism of mixed Hodge structures. Its inverse is one too: the image of $W_m(V/\ker\varphi)$ is $\varphi(W_mV) = W_m(\mathrm{im}\varphi)$ by (1), and the image of $F^p(V/\ker\varphi)_{\mathbb{C}}$ is $\varphi_{\mathbb{C}}(F^pV_{\mathbb{C}}) = F^p(\mathrm{im}\varphi)_{\mathbb{C}}$ by (2), so the inverse carries each filtration step into the corresponding one. The category is therefore abelian.

Finally $\mathrm{gr}_m^W$ is exact. Let $0 \to V' \to V \to V'' \to 0$ be exact, so that $V' \to V$ is a kernel of $V \to V''$ and $V \to V''$ is a cokernel of $V' \to V$. By Step 3 the isomorphism $V' \cong \mathrm{im}(V' \to V)$ gives $\mathrm{gr}_m^W V' \cong \mathrm{gr}_m^W\big(\mathrm{im}(V' \to V)\big) = \mathrm{im}(\mathrm{gr}_m^W V' \to \mathrm{gr}_m^W V)$, so $\mathrm{gr}_m^W V' \to \mathrm{gr}_m^W V$ is injective with image $\mathrm{gr}_m^W\big(\ker(V \to V'')\big) = \ker(\mathrm{gr}_m^W V \to \mathrm{gr}_m^W V'')$; and surjectivity of $V \to V''$ gives $\mathrm{gr}_m^W V'' = \mathrm{gr}_m^W(\mathrm{im}(V \to V'')) = \mathrm{im}(\mathrm{gr}_m^W V \to \mathrm{gr}_m^W V'')$. The graded sequence is therefore exact, completing the proof.

Strictness is what allows weight arguments to be run inside exact sequences. A short exact sequence of mixed Hodge structures induces, for each $m$, a short exact sequence of pure Hodge structures of weight $m$; and a map between two mixed structures whose weight ranges are disjoint must vanish, by lemma 9.1.7 and its counterpart for higher weights (exercise 9.1.1 (2)). Section 9.3 uses these two facts together: the differentials of a spectral sequence built from mixed Hodge structures are morphisms between pure structures of different weights, hence zero, and the sequence degenerates. Strictness is also what makes the mixed Hodge structure on the cohomology of a variety unique once it is functorial: two structures with the same underlying filtrations related by the identity map must coincide, because the identity is strict in both directions.

**9.1.9 Note: Deligne's Canonical Bigrading** The proof above never split the two filtrations simultaneously, and no splitting is needed for strictness. One exists nevertheless. Deligne showed that

$$I^{p,q} = \left(F^pV_{\mathbb{C}} \cap W_{p+q}V_{\mathbb{C}}\right) \cap \left(\overline{F^qV_{\mathbb{C}}} \cap W_{p+q}V_{\mathbb{C}} + \sum_{j\geq 1} \overline{F^{q-j}V_{\mathbb{C}}} \cap W_{p+q-j-1}V_{\mathbb{C}}\right)$$

satisfies $V_{\mathbb{C}} = \bigoplus_{p,q} I^{p,q}$, with $W_mV_{\mathbb{C}} = \bigoplus_{p+q\leq m} I^{p,q}$ and $F^pV_{\mathbb{C}} = \bigoplus_{r\geq p}\bigoplus_s I^{r,s}$, and with $\overline{I^{p,q}} \equiv I^{q,p}$ modulo $W_{p+q-2}V_{\mathbb{C}}$. The bigrading is given by a formula in $W$, $F$ and $\overline{F}$, so it is preserved by every morphism, and theorem 9.1.8(1) and (2) follow from it by truncating a preimage componentwise exactly as in proposition 5.1.7. It fails to be a bigrading of a pure Hodge structure only in that the conjugation symmetry holds modulo lower weight rather than on the nose; see [28, the canonical bigrading of a mixed Hodge structure] and [13]. For $V_z$ of example 9.1.4 the recipe returns $I^{0,0} = \mathbb{C}(e+zf)$ and $I^{-1,-1} = \mathbb{C}f$, and the failure of $\overline{I^{0,0}} = \mathbb{C}(e+\bar{z}f)$ to equal $I^{0,0}$ is the extension class again.

The constructions of definition 5.1.8 extend to the mixed setting, and the extension is forced: the weight of a tensor product should be the sum of the weights, so the filtration steps must add. These are the operations that let morphisms, twists and duals be discussed at all, and section 9.3 uses the twist to state the weight bounds.

**Proposition 9.1.10: Tensor Products and Duals of Mixed Structures**

Let $V$ and $V'$ be mixed Hodge structures.

1. The space $V_{\mathbb{Q}} \otimes_{\mathbb{Q}} V'_{\mathbb{Q}}$ with

$$W_m(V\otimes V') = \sum_{a+b=m} W_aV\otimes W_bV', \qquad F^p(V\otimes V')_{\mathbb{C}} = \sum_{a+b=p} F^aV_{\mathbb{C}}\otimes F^bV'_{\mathbb{C}}$$

is a mixed Hodge structure, and

$$\mathrm{gr}^W_m(V\otimes V') = \bigoplus_{a+b=m} \mathrm{gr}^W_a V\otimes \mathrm{gr}^W_b V'$$

as pure Hodge structures of weight $m$, the summands being the tensor products of definition 5.1.8.

2. The space $V^\vee = \mathrm{Hom}_{\mathbb{Q}}(V_{\mathbb{Q}}, \mathbb{Q})$ with

$$W_m(V^\vee) = \left(W_{-m-1}V\right)^\perp, \qquad F^p(V^\vee)_{\mathbb{C}} = \left(F^{1-p}V_{\mathbb{C}}\right)^\perp$$

where $(\,\cdot\,)^\perp$ denotes the annihilator, is a mixed Hodge structure with $\mathrm{gr}^W_m(V^\vee) = \left(\mathrm{gr}^W_{-m}V\right)^\vee$.

3. When $V$ and $V'$ are pure, these are the structures of definition 5.1.8; in particular the Tate twist $V(n) = V\otimes\mathbb{Q}(n)$ of a mixed Hodge structure satisfies $W_m\big(V(n)\big) = W_{m+2n}V$ and $F^p\big(V(n)\big)_{\mathbb{C}} = F^{p+n}V_{\mathbb{C}}$.

***Proof*:**
*Step 1: a splitting adapted to both filtrations.* Let $E$ be a finite-dimensional complex vector space carrying a finite increasing filtration $W_\bullet$ and a finite decreasing filtration $F^\bullet$. Set $A_a^p = W_a \cap F^p$ and $B_a^p = (W_{a-1} \cap F^p) + (W_a \cap F^{p+1}) \subseteq A_a^p$, and choose a complement $E_a^p$ with $A_a^p = B_a^p \oplus E_a^p$. Then

$$W_a \cap F^p = \bigoplus_{a' \leq a,\ p' \geq p} E_{a'}^{p'} \tag{9.5}$$

for all $a, p$. That the right side spans the left follows from $A_a^p = A_{a-1}^p + A_a^{p+1} + E_a^p$ by induction on $(a, -p)$, the base case being the pairs with $A_a^p = 0$, which covers $a \ll 0$ and $p \gg 0$ because both filtrations are finite. The sum is direct by a dimension count: since $A_{a-1}^p \cap A_a^{p+1} = W_{a-1} \cap F^{p+1} = A_{a-1}^{p+1}$,

$$\dim E_a^p = \dim A_a^p - \dim A_{a-1}^p - \dim A_a^{p+1} + \dim A_{a-1}^{p+1}$$

and summing this over $a' \leq a$ and $p' \geq p$ telescopes to $\dim A_a^p$. Taking $p \ll 0$ and $a \gg 0$ in (9.5) gives $E = \bigoplus_{a,p} E_a^p$, with $W_a = \bigoplus_{a' \leq a, p} E_{a'}^p$ and $F^p = \bigoplus_{a, p' \geq p} E_a^{p'}$. Consequently $\mathrm{gr}_a^W E \cong \bigoplus_p E_a^p$, with the induced filtration $F^r \mathrm{gr}_a^W E = \bigoplus_{p \geq r} E_a^p$ under that identification.

*Step 2: the tensor product.* On $V \otimes V'$ the filtration $W$ is increasing and defined over $\mathbb{Q}$, the filtration $F$ is decreasing, and both are finite. Choose splittings $E_a^p$ of $(W_\mathbb{C}, F)$ on $V_\mathbb{C}$ and $E_a'^p$ on $V'_\mathbb{C}$ as in Step 1. Then

$$W_m(V \otimes V')_\mathbb{C} = \bigoplus_{a+a' \leq m} \bigoplus_{p,p'} E_a^p \otimes E_{a'}'^{p'}, \qquad F^r(V \otimes V')_\mathbb{C} = \bigoplus_{p+p' \geq r} \bigoplus_{a,a'} E_a^p \otimes E_{a'}'^{p'}$$

directly from the definitions and Step 1. Hence

$$\mathrm{gr}_m^W(V \otimes V')_\mathbb{C} = \bigoplus_{a+a'=m} \bigoplus_{p,p'} E_a^p \otimes E_{a'}'^{p'} = \bigoplus_{a+a'=m} \mathrm{gr}_a^W V_\mathbb{C} \otimes \mathrm{gr}_{a'}^W V'_\mathbb{C}$$

an identification defined over $\mathbb{Q}$, being the canonical map induced by $W_a V \otimes W_{a'} V' \to \mathrm{gr}_a^W V \otimes \mathrm{gr}_{a'}^W V'$. Under it, the filtration (9.1) is $\bigoplus_{a+a'=m} \bigoplus_{p+p' \geq r} E_a^p \otimes E_{a'}'^{p'}$, which by the last sentence of Step 1 is $\sum_{r_1+r_2=r} F^{r_1} \mathrm{gr}_a^W V_\mathbb{C} \otimes F^{r_2} \mathrm{gr}_{a'}^W V'_\mathbb{C}$ summed over $a + a' = m$. That is the Hodge filtration of the tensor product of the pure structures $\mathrm{gr}_a^W V$ and $\mathrm{gr}_{a'}^W V'$ of weights $a$ and $a'$, which by definition 5.1.8 is pure of weight $a + a' = m$. A direct sum of pure structures of weight $m$ is pure of weight $m$, so $V \otimes V'$ is a mixed Hodge structure with the stated graded pieces.

*Step 3: the dual.* With the same splitting of $V_\mathbb{C}$, let $(E_a^p)^\vee \subseteq V_\mathbb{C}^\vee$ be the annihilator of all summands other than $E_a^p$, so $V_\mathbb{C}^\vee = \bigoplus_{a,p} (E_a^p)^\vee$. Then

$$W_m(V^\vee)_\mathbb{C} = \left(W_{-m-1} V_\mathbb{C}\right)^\perp = \bigoplus_{a \geq -m} \bigoplus_p (E_a^p)^\vee, \qquad F^r(V^\vee)_\mathbb{C} = \left(F^{1-r} V_\mathbb{C}\right)^\perp = \bigoplus_{p \leq -r} \bigoplus_a (E_a^p)^\vee$$

since $W_{-m-1} V_\mathbb{C} = \bigoplus_{a \leq -m-1, p} E_a^p$ and $F^{1-r} V_\mathbb{C} = \bigoplus_{p \geq 1-r, a} E_a^p$. So $(E_a^p)^\vee$ sits in weight $-a$ and Hodge level $-p$, giving $\mathrm{gr}_m^W(V^\vee)_\mathbb{C} = \bigoplus_p (E_{-m}^p)^\vee$ with $F^r = \bigoplus_{p \leq -r} (E_{-m}^p)^\vee$. This is the dual of the pure structure $\mathrm{gr}_{-m}^W V$ in the sense of definition 5.1.8, whose Hodge filtration is the annihilator of $F^{1-r} \mathrm{gr}_{-m}^W V_\mathbb{C}$; the dual of a pure structure of weight $-m$ is pure of weight $m$, so

$V^\vee$ is a mixed Hodge structure with the stated graded pieces.

*Step 4: the pure case and the twist.* If $V$ is pure of weight $k$ and $V'$ pure of weight $l$, the formulas give $W_m(V \otimes V') = 0$ for $m < k + l$ and $V_\mathbb{Q} \otimes V'_\mathbb{Q}$ for $m \geq k + l$, so $V \otimes V'$ is pure of weight $k + l$ and Step 2 identifies its bigrading with that of definition 5.1.8; the dual is handled the same way. For the twist, $\mathbb{Q}(n)$ is one-dimensional with $W_m = 0$ for $m < -2n$ and $F^p = \mathbb{Q}(n)_\mathbb{C}$ for $p \leq -n$, so the two displayed sums in (1) each have a single contributing term, giving $W_m(V(n)) = W_{m+2n}V$ and $F^p(V(n))_\mathbb{C} = F^{p+n}V_\mathbb{C}$, which completes the proof.

The twist is the device that normalizes weights, and it is used in section 9.3 in exactly that role: the weight-2 piece appearing in the cohomology of a punctured curve is a Tate twist of a weight-0 piece, and naming it that way is what makes the residue map a morphism of mixed Hodge structures rather than a map that shifts the bigrading by hand.

## Exercise 9.1.1

1. Let $V_\mathbb{Q} = \mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $W_{-1}V = 0$, $W_0V = W_1V = \mathbb{Q}e_1$ and $W_2V = V_\mathbb{Q}$, and let $V'_\mathbb{Q} = \mathbb{Q}f$ with $W_{-1}V' = 0$ and $W_0V' = V'_\mathbb{Q}$. Let $\varphi\colon V_\mathbb{Q} \to V'_\mathbb{Q}$ send $e_1 \mapsto 0$ and $e_2 \mapsto f$. Show that $\varphi$ respects the weight filtrations but is not strict for them. Check that each of these weight filtrations on its own does underlie a mixed Hodge structure, namely $\mathbb{Q}(0) \oplus \mathbb{Q}(-1)$ and $\mathbb{Q}(0)$, and deduce that there is nevertheless no choice of Hodge filtrations making $(V_\mathbb{Q}, W, F)$ and $(V'_\mathbb{Q}, W, F')$ mixed Hodge structures with these weight filtrations and $\varphi$ a morphism.
2. Let $V$ and $V'$ be mixed Hodge structures with no common weight, that is, for every $m$ at least one of $\mathrm{gr}^W_m V$ and $\mathrm{gr}^W_m V'$ vanishes. Show that every morphism $V \to V'$ of mixed Hodge structures is zero. (First treat $V$ pure of weight $n$: split $V'$ using $W_{n-1}V'$, handling the weights above $n$ by compatibility with $W$ alone and those below $n$ by lemma 9.1.7. Then induct on the number of nonzero graded pieces of $V$.)
3. Let $V_z$ be as in example 9.1.4, regarded as a rational mixed Hodge structure, and suppose $z \notin \mathbb{Q}$. Show that its only sub-mixed Hodge structures are 0, $W_{-2}V_z$ and the whole structure. Conclude that $V_z$ is not isomorphic to $\mathbb{Q}(1) \oplus \mathbb{Q}(0)$, and compute $\mathrm{Hom}(\mathbb{Q}(0), V_z)$ and $\mathrm{Hom}(V_z, \mathbb{Q}(0))$.
4. Let $0 \to V' \to V \to V'' \to 0$ be a short exact sequence of mixed Hodge structures. Show that
$$0 \to F^pV'_\mathbb{C} \to F^pV_\mathbb{C} \to F^pV''_\mathbb{C} \to 0$$
is exact for every $p$, and deduce that $\dim_\mathbb{C} F^pV_\mathbb{C} = \dim_\mathbb{C} F^pV'_\mathbb{C} + \dim_\mathbb{C} F^pV''_\mathbb{C}$. Deduce further that the numbers $h^{p,q}(V) := \dim_\mathbb{C} \mathrm{gr}^p_F \mathrm{gr}^W_{p+q} V_\mathbb{C}$, where $\mathrm{gr}^p_F = F^p/F^{p+1}$, are additive in short exact sequences.
5. Let $V$ be a mixed Hodge structure with weights in the interval $[a, b]$, meaning $\mathrm{gr}^W_m V = 0$ for $m < a$ and for $m > b$. Using proposition 9.1.10, determine the intervals containing the weights of $V(n)$, of $V^\vee$ and of $V \otimes V'$ when $V'$ has weights in $[a', b']$. Then show that $\mathrm{Hom}(V, V') = 0$ whenever $b < a'$.
6. Let $V_z$ and $V_0$ be as in example 9.1.4. Exhibit an isomorphism of complex vector spaces $V_{z,\mathbb{C}} \to V_{0,\mathbb{C}}$ carrying $W_\bullet$ to $W_\bullet$ and $F^\bullet$ to $F^\bullet$, and check that it is not defined over $\mathbb{Q}$ unless $z \in \mathbb{Q}$. Explain in one sentence which ingredient of definition 9.1.1 the invariant $\exp(2\pi i z)$ therefore depends on.

## 9.2 The Logarithmic de Rham Complex

A mixed Hodge structure is two filtrations on one rational vector space, and section 9.1 supplied the definition and the formal properties but no example coming from geometry. To put one on $H^k(U,\mathbb{Q})$ for a smooth open $U$ there has to be a complex that computes $H^*(U,\mathbb{C})$ and that carries the two filtrations visibly. Smooth forms on $U$ compute the cohomology but carry no holomorphic grading, and $U$ is not compact, so nothing from Part I applies to it directly. The resolution is to move the whole computation onto a compactification $X \supseteq U$: this section constructs a complex of holomorphic sheaves on $X$, built from forms whose poles along the boundary are as mild as a pole can be, and proves that it computes $H^*(U,\mathbb{C})$. The two filtrations are read off it in section 9.3.

Everything depends on the boundary $D = X \setminus U$ being simple, and the first task is to say what simple means.

**Definition 9.2.1: Normal Crossings Divisor**

Let $X$ be a complex manifold of dimension $n$ and $D \subseteq X$ a closed analytic subset. Then $D$ is a *normal crossings divisor* if every point $x \in D$ admits a holomorphic chart $\varphi\colon V \to \Delta^n$ centred at $x$ and an integer $1 \le k \le n$ with

$$\varphi(D \cap V) \;=\; \{w \in \Delta^n \,|\, w_1 \cdots w_k = 0\}$$

Such a chart is *adapted to $D$ at $x$*, and $k = k(x)$ is the number of *branches* of $D$ at $x$. The divisor is a *simple* normal crossings divisor if in addition every irreducible component $D_1, \ldots, D_N$ of $D$ is a closed complex submanifold of $X$.

Simplicity is a global condition and it is what the residue calculus below needs: if the components are smooth then each of the $k$ branches in an adapted chart at $x$ lies in a different component, since a component containing two of them would be singular at $x$. Every normal crossings divisor in this chapter is simple, and from here on the adjective is dropped.

**Example 9.2.2: Coordinate Hyperplanes, the Node and the Cusp**

1. In $\mathbb{C}^n$ the divisor $D = \{z_1 \cdots z_k = 0\}$ is the local model itself: it is a normal crossings divisor with $N = k$ components $D_i = \{z_i = 0\}$, all smooth, and $k(x)$ is the number of coordinates vanishing at $x$. Its complement is $(\mathbb{C}^*)^k \times \mathbb{C}^{n-k}$.

2. The nodal cubic $C = \{y^2 = x^2(x+1)\} \subseteq \mathbb{C}^2$ is a normal crossings divisor. Near the origin $\sqrt{x+1}$ is a well-defined holomorphic function, so
$$y^2 - x^2(x+1) \;=\; \big(y - x\sqrt{x+1}\big)\big(y + x\sqrt{x+1}\big)$$
exhibits two smooth branches with distinct tangent lines $y = \pm x$; taking those two functions as new coordinates puts $C$ in the model with $k = 2$. It is *not* simple: $C$ is irreducible and its unique component is singular at the origin.

3. The cuspidal cubic $C = \{y^2 = x^3\} \subseteq \mathbb{C}^2$ is not a normal crossings divisor at the origin. The map $t \mapsto (t^2, t^3)$ is a homeomorphism of a disc onto a neighbourhood of $0$ in $C$, so $C \setminus \{0\}$ is connected near $0$, which rules out $k = 2$: in the model $\{w_1 w_2 = 0\}$ the punctured germ has two connected components. And $k = 1$ would make $C$ smooth at $0$, whereas the gradient $(-3x^2, 2y)$ vanishes there.

The construction needs a compactification whose boundary is of this shape, and producing one is not something this book can do. It is Hironaka's resolution of singularities, and it enters as a hypothesis.

**Theorem 9.2.3: Existence of a Good Compactification**

Let $U$ be a smooth complex algebraic variety. There exist a smooth complex algebraic variety $X$, proper over $\mathbb{C}$, and an open immersion $U \hookrightarrow X$ whose complement $D = X \setminus U$ is a simple normal crossings divisor. If $U$ is quasi-projective, then $X$ may be taken projective. Such an $X$ is called a *good compactification* of $U$.

No proof is given here, and none is given anywhere in this series. The statement is Hironaka's theorem on resolution of singularities in characteristic zero [18]: one first embeds $U$ in some proper variety, then resolves the singularities of that variety and of the boundary until the boundary is a normal crossings divisor. The argument is a book-length induction on invariants of singularities and shares no machinery with anything developed earlier in this book.[1] The compactification is very far from unique, and the independence of everything built below from the choice is the substance of theorem 9.3.11.

**Example 9.2.4: A Good Compactification of a Cubic Complement**

Let $C \subseteq \mathbb{C}\mathrm{P}^2$ be an irreducible cubic curve with a single node at $p$, for instance $\{y^2z = x^2(x+z)\}$, and let $U = \mathbb{C}\mathrm{P}^2 \setminus C$. The obvious compactification $X_0 = \mathbb{C}\mathrm{P}^2$ fails: its boundary $C$ is a normal crossings divisor, by the local computation of example 9.2.2, but not a simple one, since its only component is singular at $p$.

Blow up the node. Let $\pi\colon X = \mathrm{Bl}_p\mathbb{C}\mathrm{P}^2 \to \mathbb{C}\mathrm{P}^2$ be the blow-up of chapter 1, let $E = \pi^{-1}(p)$ be the exceptional curve and let $\widetilde{C}$ be the strict transform of $C$. The two branches of $C$ at $p$ have distinct tangent directions, so their strict transforms meet $E$ at two distinct points, each transversally, and $\widetilde{C}$ is smooth. The reduced divisor $D = \widetilde{C} + E$ therefore has two smooth components meeting transversally in two points and no triple points, so it is a simple normal crossings divisor. Since $\pi$ is an isomorphism away from $p$ and $p \in C$, one has $X \setminus D \cong \mathbb{C}\mathrm{P}^2 \setminus C = U$, and $X$ is a good compactification of $U$.

Fix a good compactification from now on: $X$ a complex manifold of dimension $n$, $D \subseteq X$ a normal crossings divisor, $U = X \setminus D$ and $j\colon U \hookrightarrow X$ the inclusion. Throughout, $H^k(U, \mathbb{C})$ means the sheaf cohomology $H^k(U, \underline{\mathbb{C}})$, which the fine resolution of $\underline{\mathbb{C}}$ by smooth forms identifies with de Rham cohomology (section 2.1, theorem 2.1.5).

The complex to be built out of $X$ has to see $U$, so it must be allowed poles along $D$; and it has to remain a complex of coherent sheaves on the compact $X$, so the poles must be bounded. Bounding the pole order of $\alpha$ alone is the wrong condition, because the exterior derivative then increases it. Bounding it for $\alpha$ and $d\alpha$ at once is the right one, and it is the smallest condition stable under $d$.

[1] Hironaka's *theorem* on resolution should not be confused with Hironaka's *example* of a smooth proper threefold that is not projective, which is a separate construction; see [4, Hironaka's example of a proper non-projective threefold].

**Definition 9.2.5: Forms with Logarithmic Poles**

Let $V \subseteq X$ be open and small enough that $D \cap V$ is the zero set of a single $f \in \mathcal{O}_X(V)$ generating its ideal. A holomorphic $p$-form $\alpha$ on $V \cap U$ has *logarithmic poles along* $D$ if both $f\alpha$ and $f\,d\alpha$ extend to holomorphic forms on $V$. The condition is unchanged when $f$ is replaced by a unit multiple, and it is local, so these forms are the sections of a subsheaf

$$\Omega^p_X(\log D) \;\subseteq\; j_*\Omega^p_U$$

of $\mathcal{O}_X$-modules. If $\alpha$ has logarithmic poles then so does $d\alpha$, since $f\,d(d\alpha) = 0$, so $d$ makes

$$\Omega^\bullet_X(\log D)\colon \quad \Omega^0_X(\log D) \xrightarrow{d} \Omega^1_X(\log D) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n_X(\log D)$$

into a complex, the *logarithmic de Rham complex* of the pair $(X, D)$. Its differentials are not $\mathcal{O}_X$-linear, so it is a complex of sheaves of $\mathbb{C}$-vector spaces whose terms happen to be $\mathcal{O}_X$-modules.

The definition is a growth condition, which is not a usable description. The next proposition converts it into a generating set: in an adapted chart the logarithmic forms are exactly the $\mathcal{O}$-combinations of wedge products of the $n$ one-forms $dz_1/z_1, \dots, dz_k/z_k, dz_{k+1}, \dots, dz_n$. Every later computation in this section is made in that frame.

**Proposition 9.2.6: The Local Frame of the Logarithmic Forms**

Let $(V; z_1, \dots, z_n)$ be a chart adapted to $D$, with $D \cap V = \{z_1 \cdots z_k = 0\}$. Put

$$\omega_i \;=\; \frac{dz_i}{z_i} \quad (1 \le i \le k), \qquad \omega_i \;=\; dz_i \quad (k < i \le n)$$

and $\omega_S = \omega_{i_1} \wedge \cdots \wedge \omega_{i_p}$ for $S = \{i_1 < \cdots < i_p\} \subseteq \{1, \dots, n\}$. Then $\Omega^p_X(\log D)|_V$ is a free $\mathcal{O}_V$-module with basis $\{\omega_S : |S| = p\}$. In particular $\Omega^p_X(\log D)$ is locally free of rank $\binom{n}{p}$, it equals $\Lambda^p\Omega^1_X(\log D)$, and $\Omega^0_X(\log D) = \mathcal{O}_X$.

***Proof*:**
Write $f = z_1 \cdots z_k$, a defining equation for $D \cap V$. For $S \subseteq \{1, \dots, n\}$ set $z_S = \prod_{i \in S,\, i \le k} z_i$, so that $dz_S = z_S\,\omega_S$, and write $S^c = \{1, \dots, k\} \setminus S$.

*Step 1: the listed forms are logarithmic.* Each $\omega_i$ is closed: $d(z_i^{-1} dz_i) = -z_i^{-2}\,dz_i \wedge dz_i = 0$. Hence $d\omega_S = 0$, and for $\alpha = \sum_{|S|=p} a_S\,\omega_S$ with $a_S \in \mathcal{O}(V)$ one has $d\alpha = \sum_S da_S \wedge \omega_S$. Both $f\omega_S$ and $f\,da_S \wedge \omega_S$ are holomorphic, because $f\omega_S = (f/z_S)\,dz_S$ and $f/z_S$ is a monomial. So $\alpha$ is a section of $\Omega^p_X(\log D)$.

*Step 2: a logarithmic form is a combination of them.* Let $\alpha$ be a section of $\Omega^p_X(\log D)$ over $V$ and set $\eta = f\alpha$, a holomorphic $p$-form. Expand $\eta = \sum_{|T|=p} b_T\,dz_T$ with $b_T \in \mathcal{O}(V)$. Since

$dz_T = z_T\,\omega_T$,

$$\alpha \;=\; \frac{\eta}{f} \;=\; \sum_{|T|=p} \frac{b_T}{z_{T^c}}\,\omega_T \tag{9.6}$$

so the claim is that $z_{T^c}$ divides $b_T$ for every $T$.

The second half of definition 9.2.5 has not been used yet, and it is what supplies the divisibility. From $\alpha = \eta/f$,

$$f\,d\alpha \;=\; d\eta - \frac{df}{f}\wedge\eta$$

and $d\eta$ is holomorphic, so $f\,d\alpha$ is holomorphic if and only if $\frac{df}{f}\wedge\eta = \sum_{i=1}^{k}\frac{1}{z_i}\,dz_i\wedge\eta$ is. Fix $S$ with $|S| = p+1$ and read off the coefficient of $dz_S$: for each $i\in S$ with $i\le k$ the summand contributes $\varepsilon_i b_{S\setminus i}/z_i$ with $\varepsilon_i = \pm 1$, and the other summands contribute nothing to that slot. So

$$g_S \;:=\; \sum_{i\in S,\ i\le k} \varepsilon_i\,\frac{b_{S\setminus i}}{z_i}$$

is holomorphic on $V$. Multiply by $f$ and restrict to $\{z_i = 0\}$ for a fixed $i\in S$ with $i\le k$. Every summand with index $j\neq i$ acquires the factor $z_i$ from $f/z_j$ and dies, so

$$0 \;=\; \varepsilon_i\, b_{S\setminus i}\prod_{l\le k,\ l\neq i} z_l \qquad \text{on } \{z_i = 0\}$$

The product $\prod_{l\neq i} z_l$ is nonzero on a dense open subset of the polydisc $\{z_i = 0\}$, so $b_{S\setminus i}$ vanishes on that dense subset and hence, by continuity, on all of $\{z_i = 0\}$; expanding $b_{S\setminus i}$ in powers of $z_i$ then shows $z_i$ divides it.

Writing $T = S\setminus i$, this says: $z_i$ divides $b_T$ for every $i\le k$ with $i\notin T$. The functions $z_i$ for distinct $i$ are coprime, and dividing out one at a time gives $z_{T^c}\big| b_T$. By (9.6) the coefficients of $\alpha$ in the frame are holomorphic.

*Step 3: independence.* If $\sum_S a_S\omega_S = 0$ with $a_S$ holomorphic, multiplying by $f$ gives $\sum_S a_S(f/z_S)\,dz_S = 0$, and the $dz_S$ are a frame for $\Omega^p_V$, so $a_S f/z_S = 0$ and hence $a_S = 0$ for every $S$. The forms $\omega_S$ are therefore a basis, of cardinality $\binom{n}{p}$; taking $p=0$ the basis is the empty product 1, so $\Omega^0_X(\log D) = \mathcal{O}_X$, and the description of the basis as the $p$-fold wedges of the $\omega_i$ is the statement $\Omega^p_X(\log D) = \Lambda^p\Omega^1_X(\log D)$, completing the proof.

The rank is the same as that of $\Omega^p_X$, and indeed $\Omega^p_X \subseteq \Omega^p_X(\log D)$ is an inclusion of locally free sheaves of the same rank; the two differ only along $D$, where the second is larger. The gain is already visible in the smallest case.

**Example 9.2.7: Logarithmic Forms on the Projective Line**

Let $X = \mathbb{CP}^1$, let $D = \{0\} + \{\infty\}$ and let $U = \mathbb{C}^*$. In the affine coordinate $z$ the divisor is $\{z=0\}$ near the origin, so $\Omega^1_X(\log D)$ is generated there by $dz/z$ by proposition 9.2.6. In the other chart $w = 1/z$ the divisor is $\{w = 0\}$ and

$$\frac{dz}{z} \;=\; \frac{d(1/w)}{1/w} \;=\; -\frac{dw}{w}$$

so the same form generates there too. Hence $dz/z$ is a global nowhere-vanishing section and

$$\Omega^1_{\mathbb{CP}^1}(\log D) \;\cong\; \mathcal{O}_{\mathbb{CP}^1}$$

In particular $H^0(\mathbb{CP}^1, \Omega^1_{\mathbb{CP}^1}(\log D)) = \mathbb{C} \cdot \frac{dz}{z}$ is one-dimensional, whereas $H^0(\mathbb{CP}^1, \Omega^1_{\mathbb{CP}^1}) = 0$ by example 2.3.5: allowing logarithmic poles at two points has manufactured a global holomorphic 1-form where there was none. That form is the source of the weight-2 class in section 9.4.

Where a logarithmic form differs from a holomorphic one is along $D$, and the measurement of the difference is its residue. In one variable the residue of $g(z)\,dz/z$ at the origin is $g(0)$, and the definition below is that formula made to work along a whole component of $D$, in every degree. The first step is to check that the recipe is well posed.

**Lemma 9.2.8: Splitting Off One Component**

Let $D_1$ be an irreducible component of $D$, let $D'$ be the union of the remaining components, and let $D_1' = D' \cap D_1$, a normal crossings divisor in the manifold $D_1$ with inclusion $\iota_1 \colon D_1 \hookrightarrow X$. Let $x \in D_1$ and let $h \in \mathcal{O}_{X,x}$ be a local equation for $D_1$ at $x$.

1. Every germ $\alpha \in \Omega^p_X(\log D)_x$ can be written

$$\alpha \;=\; \frac{dh}{h} \wedge \beta \;+\; \gamma, \qquad \beta \in \Omega^{p-1}_X(\log D')_x, \quad \gamma \in \Omega^p_X(\log D')_x$$

2. The pullback $\iota_1^*\beta \in \Omega^{p-1}_{D_1}(\log D_1')_x$ depends only on $\alpha$, and not on $h$ or on the choice of decomposition.

***Proof*:**
Choose a chart adapted to $D$ at $x$ with $D \cap V = \{z_1 \cdots z_k = 0\}$ and $D_1 \cap V = \{z_1 = 0\}$, which is possible because the components of $D$ are smooth. In it $D' \cap V = \{z_2 \cdots z_k = 0\}$, and $\iota_1$ identifies $D_1 \cap V$ with the polydisc in $z_2, \ldots, z_n$ carrying the normal crossings divisor $\{z_2 \cdots z_k = 0\}$. Pullback along $\iota_1$ kills $dz_1$ and restricts coefficients, so by proposition 9.2.6 it carries $\Omega^q_X(\log D')_x$ into $\Omega^q_{D_1}(\log D_1')_x$.

*Step 1: existence.* By proposition 9.2.6 the germ $\alpha$ is an $\mathcal{O}$-combination of the $\omega_S$. Splitting the sum according to whether $1 \in S$ and factoring $\omega_1$ out of the first part gives $\alpha = \omega_1 \wedge \beta + \gamma$ with $\beta, \gamma$ combinations of the $\omega_T$ with $1 \notin T$, hence sections of $\Omega^\bullet_X(\log D')$. This is the required decomposition for $h = z_1$. A general local equation for $D_1$ is $h = uz_1$ with $u$ a unit, and then $dh/h = \omega_1 + du/u$ with $du/u$ holomorphic, so

$$\alpha \;=\; \frac{dh}{h} \wedge \beta \;+\; \Big(\gamma - \frac{du}{u} \wedge \beta\Big)$$

is a decomposition for $h$, with the same $\beta$.

*Step 2: uniqueness for fixed $h$.* Suppose $\frac{dh}{h} \wedge \beta_1 + \gamma_1 = \frac{dh}{h} \wedge \beta_2 + \gamma_2$. Put $\rho = \beta_1 - \beta_2$ and $\sigma = \gamma_2 - \gamma_1$, both sections of $\Omega^\bullet_X(\log D')$, so that $dh \wedge \rho = h\,\sigma$. With $h = uz_1$ this reads $u\,dz_1 \wedge \rho + z_1\,du \wedge \rho = uz_1\sigma$, that is,

$$dz_1 \wedge \rho \;=\; z_1\,\tau, \qquad \tau \;=\; \sigma - u^{-1}\,du \wedge \rho$$

and $\tau$ is again a section of $\Omega^p_X(\log D')$. Write $\rho = dz_1 \wedge \rho' + \rho''$ with $\rho', \rho''$ free of $dz_1$ in the frame of proposition 9.2.6; then $dz_1 \wedge \rho'' = z_1\tau$. The coefficients of $\rho''$ and of $\tau$ in that frame are holomorphic, and the coefficients of $dz_1 \wedge \rho''$ are those of $\rho''$ up to sign, so $z_1$ divides every coefficient of $\rho''$. Hence $\iota_1^*\rho'' = 0$, while $\iota_1^*(dz_1 \wedge \rho') = 0$ because $\iota_1^* dz_1 = 0$. Therefore $\iota_1^*\rho = 0$, that is, $\iota_1^*\beta_1 = \iota_1^*\beta_2$.

*Step 3: independence of $h$.* Let $h' = vh$ with $v$ a unit be another local equation. Step 1 produced from any $h$-decomposition an $h'$-decomposition with the same $\beta$, and Step 2 applied to $h'$ says that every $h'$-decomposition has the same $\iota_1^*\beta$. So the two local equations give the same answer, completing the proof.

**Definition 9.2.9: The Residue Along a Component**

In the situation of lemma 9.2.8, the *residue along* $D_1$ is the $\mathcal{O}_X$-linear map of sheaves

$$\mathrm{Res}_{D_1} : \Omega^p_X(\log D) \;\to\; \iota_{1*}\Omega^{p-1}_{D_1}(\log D_1'), \qquad \mathrm{Res}_{D_1}(\alpha) \;=\; \iota_1^*\beta$$

where $\alpha = \frac{dh}{h} \wedge \beta + \gamma$ is any decomposition as in that lemma. In an adapted chart with $D_1 = \{z_1 = 0\}$ it is given on the frame of proposition 9.2.6 by

$$\mathrm{Res}_{D_1}(\omega_S) \;=\; \begin{cases} \iota_1^*\omega_{S\setminus 1}, & 1 \in S \\ 0, & 1 \notin S \end{cases}$$

No sign has been inserted into $\mathrm{Res}_{D_1}$. On the frame it is literally $\omega_S \mapsto \iota_1^*\omega_{S\setminus 1}$, so that in one variable $\mathrm{Res}_{\{0\}}(dz/z) = 1$, and the anticommutation with $d$ recorded in theorem 9.2.11 below is absorbed into the differential of the shifted target rather than into the map. That is the normalization used throughout this section, and it is the one that matches the tube residue wherever the two can be compared (box 9.2.10). A residue renormalized by a sign depending on the degree of the form has the same kernel and the same image, hence yields the same exact sequence, but it is a different map, and any comparison with the residue defined here has to carry that sign explicitly; the graded residue $\mathrm{Res}^m$ of proposition 9.3.6 carries it at $m = 1$.

The name is not an analogy. Away from the points where two components of $D$ meet, the residue that definition 7.4.6 attached to a class on the complement of a hypersurface, by integrating over a tube, computes exactly this.

**9.2.10 Note: The Algebraic and the Topological Residue** Proposition 7.4.7 takes a closed form $\eta$ on the complement of a smooth hypersurface $X \subseteq \mathbb{C}\mathrm{P}^{n+1}$ that can be written locally as $\frac{dg_\alpha}{g_\alpha} \wedge \psi_\alpha + \theta_\alpha$ with $g_\alpha$ a defining function and $\psi_\alpha, \theta_\alpha$ holomorphic, and concludes that the $\psi_\alpha$ glue to a form $\psi$ on $X$ with $\mathrm{Res}\,\eta = [\psi]$, where Res is the map dual to the tube map (definition 7.4.6). That local shape is precisely membership in $\Omega^{k+1}(\log X)$, by definition 9.2.5 in one direction and lemma 9.2.8 in the other, and $\psi$ is precisely $\mathrm{Res}_X \eta$ in the sense of definition 9.2.9: the gluing statement there is the well-definedness proved above. So the topological residue of the class of a logarithmic form is the class of its logarithmic residue. The normalization agrees as well, the factor $1/2\pi i$ having been built into definition 7.4.6 for exactly this reason. Nothing in the argument of proposition 7.4.7 uses more about $X$ than that it is a smooth divisor carrying a tubular neighbourhood, so the same computation identifies $\mathrm{Res}_{D_1}$ with the tube residue along a component $D_1$ meeting no other component of $D$, and in particular along every component when $D$ is smooth, which is the case of theorem 9.2.12. It goes no further than that. Where $D_1$ meets another component the hypothesis of proposition 7.4.7 already fails: by lemma 9.2.8 the $\beta$ in $\alpha = \frac{dh}{h} \wedge \beta + \gamma$ is only logarithmic along $D_1' = D' \cap D_1$, not holomorphic. Both sides of the proposed identity break down there as well. The value $\mathrm{Res}_{D_1}\alpha$ is a section of $\Omega^{\bullet-1}_{D_1}(\log D_1')$, not a holomorphic form on $D_1$, so it represents no class in $H^\bullet(D_1, \mathbb{C})$; and the tube over a cycle in $D_1$ passing through a point of $D_1'$ is not contained in $U$, so the tube map of definition 7.4.6 is not defined on that cycle. Exercise 9.2.1 (6) is the smallest instance. There $D$ is the union of the four coordinate curves in a product of two projective lines, and $\mathrm{Res}_{D_1}\left(\frac{dz}{z} \wedge \frac{dw}{w}\right) = \iota_1^* \frac{dw}{w}$ along $D_1 = \{0\} \times \mathbb{C}\mathrm{P}^1$ is a nonzero logarithmic form, whereas $H^1(D_1, \mathbb{C}) = H^1(\mathbb{C}\mathrm{P}^1, \mathbb{C}) = 0$ leaves no class for a tube residue to be. Recovering a topological reading of the residue at a crossing is what the weight filtration of section 9.3 is for.

The residue is more than a measurement: it is a surjection with a computable kernel, and that is what turns the logarithmic complex into an object one can induct on. Peeling off one component at a time is the mechanism behind every computation in the rest of the section.

**Theorem 9.2.11: The Residue Sequence Along One Component**

Let $D_1$ be an irreducible component of $D$, let $D'$ be the union of the others and $D_1' = D' \cap D_1$. Then for every $p$ the sequence of sheaves

$$0 \to \Omega^p_X(\log D') \to \Omega^p_X(\log D) \xrightarrow{\mathrm{Res}_{D_1}} \iota_{1*}\Omega^{p-1}_{D_1}(\log D_1') \to 0$$

is exact, the first map being the inclusion. Moreover $\mathrm{Res}_{D_1} \circ d = -\,d \circ \mathrm{Res}_{D_1}$, so that with the sign convention that the shifted complex $\iota_{1*}\Omega^{\bullet-1}_{D_1}(\log D_1')$ carries the differential $-d$, the display is a short exact sequence of complexes of sheaves of $\mathbb{C}$-vector spaces.

***Proof*:**
Exactness is local, so fix $x \in X$. If $x \notin D_1$ the third sheaf has zero stalk and the first map is the identity, since $D$ and $D'$ agree near $x$. So assume $x \in D_1$ and work in a chart adapted to $D$ with $D_1 = \{z_1 = 0\}$ and $D' = \{z_2 \cdots z_k = 0\}$, keeping the notation of proposition 9.2.6.

*Step 1: the composite vanishes and the kernel is no larger.* A section of $\Omega^p_X(\log D')$ admits the decomposition of lemma 9.2.8 with $\beta = 0$, so its residue vanishes. Conversely take the decomposition $\alpha = \omega_1 \wedge \beta + \gamma$ produced by the frame, in which $\beta$ and $\gamma$ are $\mathcal{O}$-combinations of the $\omega_T$ with $T \subseteq \{2, \dots, n\}$, and suppose $\iota_1^* \beta = 0$. Since no $\omega_T$ occurring in $\beta$ involves $dz_1$, that vanishing says exactly that every coefficient of $\beta$ vanishes on $\{z_1 = 0\}$, hence is divisible by $z_1$; write $\beta = z_1 \widetilde{\beta}$ with $\widetilde{\beta}$ again a section of $\Omega_X^{p-1}(\log D')$. Then

$$\omega_1 \wedge \beta \;=\; \frac{dz_1}{z_1} \wedge z_1 \widetilde{\beta} \;=\; dz_1 \wedge \widetilde{\beta}$$

which is a section of $\Omega_X^p(\log D')$, and hence so is $\alpha$.

*Step 2: surjectivity.* Let $\beta_0$ be a germ of $\Omega_{D_1}^{p-1}(\log D_1')$ at $x$. In the chart, $D_1$ is the polydisc in $z_2, \dots, z_n$ and $\Omega_{D_1}^{p-1}(\log D_1')$ is free on the $\iota_1^* \omega_T$ with $T \subseteq \{2, \dots, n\}$, again by proposition 9.2.6. Write $\beta_0 = \sum_T a_T \, \iota_1^* \omega_T$ and let $\widetilde{a}_T$ be the function on the chart obtained by regarding $a_T$ as independent of $z_1$. Then $\alpha = \omega_1 \wedge \sum_T \widetilde{a}_T \omega_T$ is a section of $\Omega_X^p(\log D)$ with $\mathrm{Res}_{D_1} \alpha = \beta_0$.

*Step 3: the sign.* Decompose $\alpha = \frac{dh}{h} \wedge \beta + \gamma$ as in lemma 9.2.8. Since $d(dh/h) = 0$,

$$d\alpha \;=\; -\frac{dh}{h} \wedge d\beta \;+\; d\gamma$$

and $d\beta$, $d\gamma$ are again sections of $\Omega_X^\bullet(\log D')$. So $\mathrm{Res}_{D_1}(d\alpha) = -\iota_1^*(d\beta) = -\,d(\iota_1^* \beta) = -\,d\,\mathrm{Res}_{D_1}(\alpha)$, pullback commuting with $d$. With the differential of the shifted target taken to be $-d$, the residue commutes with the differentials, completing the proof.

**Theorem 9.2.12: The Residue Sequence of a Smooth Boundary**

Suppose $D$ is smooth, that is, its components are pairwise disjoint, and write $\iota \colon D \hookrightarrow X$. Then

$$0 \;\to\; \Omega_X^\bullet \;\to\; \Omega_X^\bullet(\log D) \;\xrightarrow{\mathrm{Res}}\; \iota_* \Omega_D^{\bullet - 1} \;\to\; 0$$

is exact, where Res is the sum of the residues along the components.

***Proof*:**
Exactness is local and each point of $D$ lies on exactly one component, so it is enough to treat a single component $D_1$. For it, $D' = \varnothing$ near every point of $D_1$, so $\Omega_X^\bullet(\log D') = \Omega_X^\bullet$ and $D_1' = \varnothing$; theorem 9.2.11 is then exactly the displayed sequence, as we sought to show.

Theorem 9.2.12 is the shape in which the residue sequence is usually quoted. It is the special case of theorem 9.2.11 in which the components of $D$ are disjoint, not the general statement.

**9.2.13 Warning: The Total Residue Is Not Surjective at a Crossing** When components of $D$ meet, the map $\bigoplus_i \operatorname{Res}_{D_i}$ from $\Omega^p_X(\log D)$ to $\bigoplus_i \iota_{i*}\Omega^{p-1}_{D_i}(\log D_i')$ is not surjective for $p \geq 2$. Take $X = \mathbb{C}^2$, $D = \{z_1 z_2 = 0\}$ and $p = 2$. Then $\Omega^2_X(\log D)$ is free on $\omega_1 \wedge \omega_2$, so a section is $c\,\omega_1 \wedge \omega_2$ with $c$ holomorphic, and

$$\operatorname{Res}_{D_1}(c\,\omega_1 \wedge \omega_2) = c|_{D_1}\,\iota_1^*\omega_2, \qquad \operatorname{Res}_{D_2}(c\,\omega_1 \wedge \omega_2) = -\,c|_{D_2}\,\iota_2^*\omega_1$$

the second because $\omega_1 \wedge \omega_2 = -\,\omega_2 \wedge \omega_1$. The pair $(\iota_1^*\omega_2, 0)$ is therefore not in the image: it would need $c|_{D_1} = 1$ and $c|_{D_2} = 0$, giving $c(0) = 1$ and $c(0) = 0$ at once. Recovering the general statement means filtering $\Omega^\bullet_X(\log D)$ by the number of branches along which a form has a pole, which is the weight filtration of section 9.3.

What remains is the comparison theorem: the complex just built, which lives on the compact $X$, computes the cohomology of the open $U$. Both sides of that statement are the cohomology of a complex of sheaves rather than of a single sheaf, so the first thing to fix is what such cohomology means.

**9.2.14 Convention: Cohomology of a Complex of Sheaves** For a bounded-below complex of sheaves of $\mathbb{C}$-vector spaces $\mathcal{K}^\bullet$ on a space $X$, write $H^k(X, \mathcal{K}^\bullet)$ for the $k$-th hypercohomology $\mathbb{R}^k\Gamma(X, \mathcal{K}^\bullet)$, constructed and shown independent of choices in *Homological Algebra* [6, hypercohomology via Cartan-Eilenberg resolutions]. For a single sheaf placed in degree 0 this is ordinary sheaf cohomology, so the notation extends the usual one. Two properties are used below, both proved there.

1. A *quasi-isomorphism* $\mathcal{K}^\bullet \to \mathcal{L}^\bullet$, meaning a morphism of complexes inducing an isomorphism on the cohomology of the stalk complex at every point, induces isomorphisms $H^k(X, \mathcal{K}^\bullet) \cong H^k(X, \mathcal{L}^\bullet)$ for all $k$.

2. There is a spectral sequence

$$E_1^{p,q} \;=\; H^q(X, \mathcal{K}^p) \;\Longrightarrow\; H^{p+q}(X, \mathcal{K}^\bullet)$$

   whose differential $d_1$ is induced by the differential of $\mathcal{K}^\bullet$.

If every $\mathcal{K}^p$ is acyclic then (2) is concentrated in the row $q = 0$, where it reads $E_1^{p,0} = \Gamma(X, \mathcal{K}^p)$, and the sequence degenerates at $E_2$ for want of room; so

$$H^k(X, \mathcal{K}^\bullet) \;\cong\; H^k\big(\Gamma(X, \mathcal{K}^\bullet)\big)$$

This is the abstract de Rham theorem (theorem 2.1.5) for complexes.

The comparison is proved by computing both sides locally, and on the holomorphic side the local computation begins with the case of no boundary at all. The ordinary Poincaré lemma has a holomorphic refinement, for the reason that the homotopy operator producing the primitive is built from the radial contraction $z \mapsto tz$, which is holomorphic in $z$.

The holomorphic de Rham complex resolves the constant sheaf: on a complex manifold a closed holomorphic form is locally exact, so $0 \to \underline{\mathbb{C}}_X \to \mathcal{O}_X \to \Omega^1_X \to \cdots$ is a resolution and $\mathbb{H}^i(X, \Omega^\bullet_X) \cong H^i(X, \mathbb{C})$. This is [4, the holomorphic Poincaré lemma and the holomorphic de Rham resolution]; it is general complex geometry rather than Hodge theory, and is used here only through its conclusion.

This is the case $D = \varnothing$ of the theorem being aimed at. The general case needs the local cohomology of the logarithmic complex, and the residue sequence computes it by induction on the number of branches: each branch contributes one new closed form $\omega_i$, and nothing else.

**Lemma 9.2.15: Local Cohomology of the Logarithmic Complex**

Let $V = \Delta^n$ and $D_V = \{z_1 \cdots z_k = 0\}$ with $0 \le k \le n$. Then the cohomology of the complex $\Omega^\bullet_V(\log D_V)(V)$ of global sections is

$$H^q\big(\Omega^\bullet_V(\log D_V)(V)\big) \;=\; \bigoplus_{S \subseteq \{1,\dots,k\},\ |S|=q} \mathbb{C}\,[\omega_S]$$

In particular it is finite-dimensional of dimension $\binom{k}{q}$, and the exterior algebra on the classes of $\omega_1, \dots, \omega_k$ maps isomorphically onto it.

***Proof*:**
Induct on $k$. For $k = 0$ the complex is $\Omega^\bullet(V)$ and the assertion is the holomorphic Poincaré lemma ([4, the holomorphic Poincaré lemma]).

*Step 1: the residue sequence on global sections.* Let $k \ge 1$, put $D_1 = \{z_1 = 0\}$ and $D' = \{z_2 \cdots z_k = 0\}$, so $D'_1 = D' \cap D_1$. By theorem 9.2.11 the sequence of sheaves is exact; on global sections over the polydisc the only point in question is surjectivity of $\mathrm{Res}_{D_1}$, and Step 2 of that proof produces the lift globally: writing $\beta_0 = \sum_T a_T\, \iota_1^* \omega_T$ with $a_T$ holomorphic on $D_1 = \Delta^{n-1}$ and letting $\widetilde{a}_T$ be the same function viewed as independent of $z_1$, the global section $\omega_1 \wedge \sum_T \widetilde{a}_T \omega_T$ has residue $\beta_0$. So

$$0 \to \Omega^\bullet_V(\log D')(V) \to \Omega^\bullet_V(\log D_V)(V) \xrightarrow{\mathrm{Res}_{D_1}} \Omega^{\bullet-1}_{D_1}(\log D'_1)(D_1) \to 0$$

is a short exact sequence of complexes of $\mathbb{C}$-vector spaces, the third carrying the differential $-d$.

*Step 2: the induction hypothesis on the two ends.* The pair $(V, D')$ is the model with $k-1$ branches in $n$ variables, and the pair $(D_1, D'_1)$ is the model with $k-1$ branches in $n-1$ variables, since $D_1 = \Delta^{n-1}$ with coordinates $z_2, \dots, z_n$ and $D'_1 = \{z_2 \cdots z_k = 0\}$. So by induction the first complex has cohomology with basis the $[\omega_S]$ for $S \subseteq \{2, \dots, k\}$, $|S| = q$, and the third, whose $q$-th cohomology is the $(q-1)$-st cohomology of $\Omega^\bullet_{D_1}(\log D'_1)(D_1)$ because changing the sign of a differential changes neither kernels nor images, has basis the $[\iota_1^*\omega_S]$ for $S \subseteq \{2, \dots, k\}$, $|S| = q-1$.

*Step 3: the connecting maps vanish.* Let $S \subseteq \{2, \dots, k\}$ with $|S| = q-1$. The form $\omega_1 \wedge \omega_S$ is a global section of $\Omega^q_V(\log D_V)$, it is closed because each $\omega_i$ is, and $\mathrm{Res}_{D_1}(\omega_1 \wedge \omega_S) = \iota_1^*\omega_S$ by definition 9.2.9. So the map induced by $\mathrm{Res}_{D_1}$ on cohomology sends a basis to a basis and is surjective in every degree. In the long exact sequence of the short exact sequence of Step 1, a surjective residue map forces every connecting homomorphism to vanish.

*Step 4: assembly.* The long exact sequence therefore breaks into short exact sequences

$$0 \to H^q\big(\Omega^\bullet_V(\log D')(V)\big) \to H^q\big(\Omega^\bullet_V(\log D_V)(V)\big) \to H^{q-1}\big(\Omega^\bullet_{D_1}(\log D'_1)(D_1)\big) \to 0$$

The classes $[\omega_S]$ with $1 \notin S$, $|S| = q$, form a basis of the subspace by Step 2, and the classes $[\omega_1 \wedge \omega_{S'}]$ with $S' \subseteq \{2, \dots, k\}$, $|S'| = q - 1$, map to a basis of the quotient by Step 3. Together they are a basis of the middle term, and they are exactly the $[\omega_S]$ with $S \subseteq \{1, \dots, k\}$ and $|S| = q$, completing the induction and the proof.

The answer is the cohomology of a $k$-torus, which is what it must be: the complement of the local model is $(\Delta^*)^k \times \Delta^{n-k}$, homotopy equivalent to $(S^1)^k$. The next proposition upgrades that coincidence into a map.

**Proposition 9.2.16: The Logarithmic Complex Is Quasi-Isomorphic to the Push-forward**

Let $\mathcal{A}^p_U$ denote the sheaf of smooth complex-valued $p$-forms on $U$. Restriction of forms defines a morphism of complexes of sheaves on $X$,

$$\Omega^\bullet_X(\log D) \;\to\; j_*\mathcal{A}^\bullet_U$$

and it is a quasi-isomorphism.

***Proof*:**
A section of $\Omega^p_X(\log D)$ over $V$ is by definition 9.2.5 a holomorphic $p$-form on $V \cap U$, hence in particular a smooth one, so the map is defined; it commutes with $d$ because it is a restriction. What must be shown is that it induces an isomorphism on the cohomology of every stalk complex.

Fix $x \in X$ and let $k = k(x)$ be the number of branches of $D$ at $x$, with $k = 0$ when $x \notin D$. Choose an adapted chart at $x$ and let $V_\varepsilon \cong \Delta^n_\varepsilon$ be the concentric adapted polydiscs, which form a fundamental system of neighbourhoods of $x$. Cohomology of a complex of vector spaces commutes with filtered colimits, so the cohomology of either stalk complex is the colimit over $\varepsilon$ of the cohomology of the complex of sections over $V_\varepsilon$.

*Step 1: the holomorphic side.* By lemma 9.2.15 the cohomology of $\Omega^\bullet_X(\log D)(V_\varepsilon)$ has basis the classes $[\omega_S]$ with $S \subseteq \{1, \dots, k\}$ and $|S| = q$. The restriction maps between two such polydiscs carry $\omega_S$ to $\omega_S$, hence carry basis to basis and are isomorphisms, so the colimit is again $\bigoplus_{|S|=q} \mathbb{C}\,[\omega_S]$.

*Step 2: the smooth side.* By definition $\Gamma(V_\varepsilon, j_*\mathcal{A}^\bullet_U) = \mathcal{A}^\bullet(V_\varepsilon \cap U)$, whose cohomology is $H^q(V_\varepsilon \cap U, \mathbb{C})$ by theorem 2.1.5 applied to the fine resolution of $\underline{\mathbb{C}}$ by smooth forms (proposition 2.1.4). In the adapted chart $V_\varepsilon \cap U = (\Delta^*_\varepsilon)^k \times \Delta^{n-k}_\varepsilon$, which deformation retracts onto the torus $(S^1)^k$. Homotopy invariance and the Künneth formula [24, the Künneth formula] give

$$H^\bullet(V_\varepsilon \cap U, \mathbb{C}) \;\cong\; \bigotimes_{i=1}^{k} H^\bullet(\Delta^*_\varepsilon, \mathbb{C})$$

one tensor factor for each punctured disc. On the $i$-th factor, $H^1(\Delta^*_\varepsilon, \mathbb{C})$ is one-dimensional and generated by the class of $\frac{1}{2\pi i}\frac{dz_i}{z_i}$, since that form integrates to 1 over a positively oriented circle. Hence $H^q(V_\varepsilon \cap U, \mathbb{C})$ has basis the classes of $(2\pi i)^{-|S|}\omega_S$ for $S \subseteq \{1, \dots, k\}$, $|S| = q$, and again the restriction maps are isomorphisms.

*Step 3: comparison.* The map sends the class of $\omega_S$ on the holomorphic side to the de Rham class of the same form, which is $(2\pi i)^{|S|}$ times the basis element of Step 2. It therefore carries a basis to a basis and is an isomorphism in every degree, for every $x$, completing the proof.

That quasi-isomorphism is the last ingredient. It moves the computation onto the smooth side, where the sheaves are acyclic and the hypercohomology collapses to the de Rham complex of $U$, which is the comparison theorem this section was aimed at.

**Theorem 9.2.17: The Logarithmic Complex Computes the Cohomology of $U$**

Let $X$ be a complex manifold, $D \subseteq X$ a normal crossings divisor and $U = X \setminus D$. Restriction of forms induces isomorphisms

$$\mathbb{H}^k\big(X, \Omega_X^\bullet(\log D)\big) \;\cong\; H^k(U, \mathbb{C}) \qquad \text{for all } k$$

***Proof*:**
*Step 1: the sheaves $j_*\mathcal{A}_U^p$ are acyclic.* A smooth function on an open $V \subseteq X$ restricts to a smooth function on $V \cap U$, so $j_*\mathcal{A}_U^p$ is a sheaf of $\mathcal{A}_X^0$-modules. The argument of proposition 2.1.4 uses nothing about the module beyond this: a smooth partition of unity on $X$ subordinate to a locally finite cover acts on the sheaf by multiplication, giving endomorphisms with the required supports that sum to the identity, so $j_*\mathcal{A}_U^p$ is fine in the sense of definition 2.1.1. Since $X$ is paracompact, proposition 2.1.3 makes it acyclic.

*Step 2: the smooth complex computes $H^\bullet(U, \mathbb{C})$.* By Step 1 and the last clause of box 9.2.14,

$$\mathbb{H}^k\big(X, j_*\mathcal{A}_U^\bullet\big) \;\cong\; H^k\big(\Gamma(X, j_*\mathcal{A}_U^\bullet)\big) \;=\; H^k\big(\mathcal{A}^\bullet(U)\big)$$

the last equality being the definition of the pushforward. The de Rham complex of $U$ is the complex of global sections of the fine resolution of $\underline{\mathbb{C}}_U$ by smooth forms, so its cohomology is $H^k(U, \mathbb{C})$ by theorem 2.1.5.

*Step 3: transport along the quasi-isomorphism.* By proposition 9.2.16 the restriction map $\Omega_X^\bullet(\log D) \to j_*\mathcal{A}_U^\bullet$ is a quasi-isomorphism, so property (1) of box 9.2.14 gives $\mathbb{H}^k(X, \Omega_X^\bullet(\log D)) \cong \mathbb{H}^k(X, j_*\mathcal{A}_U^\bullet)$. Composing with Step 2 gives the theorem, and the composite is induced by restriction of forms, completing the proof.

Every input to that proof is internal to this book except one: the existence of hypercohomology and its first spectral sequence, imported from *Homological Algebra* [6] in box 9.2.14. The local computation on which everything rests, lemma 9.2.15, is proved here in full.

What the theorem gives is a change of ambient space. The cohomology of the noncompact $U$, which carries no Hodge theory of its own, has been rewritten as the cohomology of a complex of locally free sheaves on the compact $X$, where Part I applies. Two filtrations of that complex, one by the order of the pole and one by the holomorphic degree, will produce the weight and Hodge filtrations of a mixed Hodge structure; the spectral sequence of box 9.2.14 is already the second of them. The smallest case displays the whole mechanism.

**Example 9.2.18: The Punctured Line by the Logarithmic Complex**

Let $X = \mathbb{C}\mathrm{P}^1$, $D = \{0\} + \{\infty\}$ and $U = \mathbb{C}^*$, as in example 9.2.7. The logarithmic complex is

$$\mathcal{O}_{\mathbb{C}\mathrm{P}^1} \xrightarrow{d} \Omega^1_{\mathbb{C}\mathrm{P}^1}(\log D) \cong \mathcal{O}_{\mathbb{C}\mathrm{P}^1}$$

concentrated in degrees 0 and 1. Its hypercohomology is computed by the spectral sequence of box 9.2.14, whose first page is

$$E_1^{p,q} \;=\; H^q\big(\mathbb{C}\mathrm{P}^1, \Omega^p_{\mathbb{C}\mathrm{P}^1}(\log D)\big)$$

Both terms are $\mathcal{O}_{\mathbb{C}\mathrm{P}^1}$, and $H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}) = \mathbb{C}$ while $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}) = 0$ by example 2.3.5. So the first page is concentrated in the row $q = 0$, where it reads

$$\mathbb{C} \xrightarrow{d_1} \mathbb{C}, \qquad d_1(c) \;=\; dc \;=\; 0$$

since the global sections of $\mathcal{O}_{\mathbb{C}\mathrm{P}^1}$ are the constants. Hence $E_2 = E_\infty = E_1$ and, by theorem 9.2.17,

$$H^0(\mathbb{C}^*, \mathbb{C}) = \mathbb{C}, \qquad H^1(\mathbb{C}^*, \mathbb{C}) = \mathbb{C} \cdot \left[\frac{dz}{z}\right]$$

which is the cohomology of a circle, as it must be. The generator of $H^1$ is the logarithmic form of example 9.2.7, and its residue at 0 is $\mathrm{Res}_{\{0\}}(dz/z) = 1$ by definition 9.2.9. A single boundary point has contributed a single class, detected by its residue; the same bookkeeping on a punctured elliptic curve is section 9.4.

## Exercise 9.2.1

1. On $X = \Delta^2$ with $D = \{z_1 = 0\}$, consider $\alpha = dz_2/z_1$. Show that $z_1\alpha$ is holomorphic but that $\alpha$ is not a section of $\Omega^1_X(\log D)$, identifying which half of definition 9.2.5 fails. Confirm the answer against proposition 9.2.6.
2. Let $D = \{xy(x + y) = 0\} \subseteq \mathbb{C}^2$. Show that $D$ is not a normal crossings divisor at the origin. Then let $\pi\colon \widetilde{X} \to \mathbb{C}^2$ be the blow-up at the origin and show that the reduced preimage $\pi^{-1}(D)$, that is, the union of the exceptional curve with the three strict transforms, is a simple normal crossings divisor.
3. Let $\Delta \subseteq \mathbb{C}$ be the unit disc and $D = \{0\}$. Compute the cohomology of the two-term complex $\Omega^\bullet_\Delta(\log D)(\Delta)$ directly from power series, and check the answer against lemma 9.2.15 and against $H^\bullet(\Delta^*, \mathbb{C})$.
4. Let $D$ be a normal crossings divisor on a complex manifold $X$ of dimension $n$. Show that $\Omega^n_X(\log D) \cong K_X \otimes \mathcal{O}_X(D)$, where $K_X$ is the canonical bundle of definition 1.3.12 and $\mathcal{O}_X(D)$ is the line bundle of the divisor $D$ from chapter 1. Deduce the isomorphism of example 9.2.7 a second way.
5. Let $X = \mathbb{C}^2$, $D = \{z_1z_2 = 0\}$, $D_1 = \{z_1 = 0\}$ and $D_2 = \{z_2 = 0\}$. Compute $\mathrm{Res}_{D_1}$ on the frames of $\Omega^1_X(\log D)$ and $\Omega^2_X(\log D)$, identify the kernel of $\mathrm{Res}_{D_1}$ in degree 1 with $\Omega^1_X(\log D_2)$, and verify the sign of theorem 9.2.11 on the form $\alpha = g\,\omega_1$ with $g$ holomorphic.
6. Let $X = \mathbb{C}\mathrm{P}^1 \times \mathbb{C}\mathrm{P}^1$ and let $D$ be the union of the four curves $\{0\} \times \mathbb{C}\mathrm{P}^1$, $\{\infty\} \times \mathbb{C}\mathrm{P}^1$,

$\mathbb{CP}^1 \times \{0\}$ and $\mathbb{CP}^1 \times \{\infty\}$, so that $U = (\mathbb{C}^*)^2$. Show that $D$ is a simple normal crossings divisor and that $\Omega^p_X(\log D) \cong \mathcal{O}_X^{\oplus\binom{2}{p}}$. Then use theorem 9.2.17 and the spectral sequence of box 9.2.14 to compute $H^k((\mathbb{C}^*)^2, \mathbb{C})$, taking as given that $H^q(X, \mathcal{O}_X) = 0$ for $q \geq 1$.

## 9.3 Deligne's Mixed Hodge Structure

The logarithmic complex of section 9.2 computes $H^*(U, \mathbb{C})$, and it carries more structure than the comparison isomorphism uses. Its terms are graded by how many logarithmic poles a form is allowed, and, like any complex of sheaves of differential forms, it is filtered by form degree. This section writes down the two filtrations, transports them to $H^k(U, \mathbb{Q})$, and proves that what they produce there is a mixed Hodge structure in the sense of definition 9.1.1.

Throughout, $X$ is a smooth projective variety over $\mathbb{C}$ of dimension $n$, and $D = D_1 \cup \cdots \cup D_N \subseteq X$ is a normal crossings divisor whose irreducible components $D_i$ are smooth (definition 9.2.1); by theorem 9.2.3 every quasi-projective $U$ admits a compactification of this form. Write $U = X \setminus D$ and $j\colon U \hookrightarrow X$ for the inclusion. A point $x \in X$ lying on exactly $r$ of the components has a coordinate neighbourhood with coordinates $z_1, \ldots, z_n$ in which $D = \{z_1 \cdots z_r = 0\}$; coordinates of this kind are called *adapted* to $D$ at $x$, and every local computation below is carried out in them.

The Hodge decomposition fails on $U$ because $U$ is not compact, and the logarithmic poles are the exact measure of that failure: a form on $X$ with no poles at all restricts to a form on $U$ that extends across the boundary, while a form with $m$ logarithmic poles remembers $m$ directions in which $U$ runs off to $D$. Counting poles therefore produces a filtration whose bottom piece is the cohomology $X$ already had, and whose higher pieces are new. This is the filtration that will carry the weights.

**Definition 9.3.1: The Weight Filtration on the Logarithmic Complex**

The *weight filtration* on $\Omega^\bullet_X(\log D)$ is the increasing filtration by subsheaves

$$W_m\Omega^p_X(\log D) = \begin{cases} 0 & m < 0 \\ \Omega^{p-m}_X \wedge \Omega^m_X(\log D) & 0 \leq m \leq p \\ \Omega^p_X(\log D) & m > p \end{cases}$$

where the middle line denotes the image of $\Omega^{p-m}_X \otimes \Omega^m_X(\log D) \to \Omega^p_X(\log D)$ under the wedge product.

In adapted coordinates $\Omega^p_X(\log D)$ is free on the forms

$$\frac{dz_I}{z_I} \wedge dz_J := \frac{dz_{i_1}}{z_{i_1}} \wedge \cdots \wedge \frac{dz_{i_a}}{z_{i_a}} \wedge dz_{j_1} \wedge \cdots \wedge dz_{j_b}$$

with $I = \{i_1 < \cdots < i_a\} \subseteq \{1, \ldots, r\}$, $J = \{j_1 < \cdots < j_b\} \subseteq \{1, \ldots, n\} \setminus I$ and $a + b = p$ (definition 9.2.5). Comparing the two descriptions, $W_m\Omega^p_X(\log D)$ is the free submodule spanned by those basis elements with $|I| \leq m$: a generator $\frac{dz_I}{z_I} \wedge dz_J$ with $|I| \leq m$ is obtained by taking $J' \subseteq J$ with $|J'| = m - |I|$ and factoring the element as $\left(dz_{J \setminus J'}\right) \wedge \left(\frac{dz_I}{z_I} \wedge dz_{J'}\right)$, the second factor lying in $\Omega^m_X(\log D)$, and conversely every product of an element of $\Omega^{p-m}_X$ with an element of $\Omega^m_X(\log D)$ has at most $m$ logarithmic factors. So $W_m$ is locally free, $W_0\Omega^\bullet_X(\log D) = \Omega^\bullet_X$, and $W_p\Omega^p_X(\log D) = \Omega^p_X(\log D)$.

Each $W_m$ is a subcomplex. Indeed a section of $W_m\Omega^p_X(\log D)$ is locally a sum $\sum f_{IJ}\frac{dz_I}{z_I}\wedge dz_J$ with $|I|\le m$, the basis forms are closed, and $d$ of the sum is $\sum df_{IJ}\wedge\frac{dz_I}{z_I}\wedge dz_J$, which again has at most $m$ logarithmic factors in each term.

The second filtration needs no new idea. On the de Rham complex of a compact Kähler manifold the filtration by form degree induces the Hodge filtration (chapter 4), and the same recipe applies verbatim to the logarithmic complex.

**Definition 9.3.2: The Hodge Filtration on the Logarithmic Complex**

The *Hodge filtration* on $\Omega^\bullet_X(\log D)$ is the decreasing filtration by the subcomplexes

$$F^p\Omega^\bullet_X(\log D)=\Omega^{\ge p}_X(\log D)=\Big(0\to\cdots\to 0\to\Omega^p_X(\log D)\to\Omega^{p+1}_X(\log D)\to\cdots\Big)$$

the truncation that discards the terms in degrees below $p$.

$F^p$ is a subcomplex because $d$ raises degree, and $F^p\supseteq F^{p+1}$ with $F^0$ the whole complex and $F^{n+1}=0$. The two filtrations are of different natures and this is the point: $W$ is defined on $\mathbb{Q}$-cohomology and grades it into pure pieces, while $F$ lives only on $\mathbb{C}$-cohomology and supplies the Hodge filtration of each piece. Neither alone is a Hodge structure.

**Example 9.3.3: The Two Filtrations on $\mathbb{C}^*$**

Take $X=\mathbb{C}\mathrm{P}^1$ with affine coordinate $z$, $D=\{0\}\cup\{\infty\}$ and $U=\mathbb{C}^*$. In the coordinate $w=1/z$ near $\infty$ one has $\frac{dw}{w}=-\frac{dz}{z}$, so $\frac{dz}{z}$ is a global frame and $\Omega^1_{\mathbb{C}\mathrm{P}^1}(\log D)\cong\mathcal{O}_{\mathbb{C}\mathrm{P}^1}$, while $\Omega^1_{\mathbb{C}\mathrm{P}^1}\cong\mathcal{O}_{\mathbb{C}\mathrm{P}^1}(-2)$.

The weight filtration is

$$W_{-1}=0,\qquad W_0\Omega^\bullet_{\mathbb{C}\mathrm{P}^1}(\log D)=\Omega^\bullet_{\mathbb{C}\mathrm{P}^1},\qquad W_1\Omega^\bullet_{\mathbb{C}\mathrm{P}^1}(\log D)=\Omega^\bullet_{\mathbb{C}\mathrm{P}^1}(\log D)$$

since $W_m\Omega^0=\mathcal{O}_{\mathbb{C}\mathrm{P}^1}$ for every $m\ge 0$ and $W_1\Omega^1=\Omega^1(\log D)$. The only graded piece above $\mathrm{gr}^W_0$ is $\mathrm{gr}^W_1$, the quotient $\Omega^1_{\mathbb{C}\mathrm{P}^1}(\log D)/\Omega^1_{\mathbb{C}\mathrm{P}^1}$ placed in degree 1, which is a skyscraper of length 1 at each of 0 and $\infty$.

The Hodge filtration is $F^0=\Omega^\bullet_{\mathbb{C}\mathrm{P}^1}(\log D)$, $F^1=\Omega^1_{\mathbb{C}\mathrm{P}^1}(\log D)$ in degree 1, and $F^2=0$. The group $H^1(\mathbb{C}^*,\mathbb{C})$ is one dimensional, spanned by the class of $\frac{dz}{z}$. That class lies in $F^1$, and it is not in the image of $H^1(\mathbb{C}\mathrm{P}^1,W_0\Omega^\bullet_{\mathbb{C}\mathrm{P}^1}(\log D))=H^1(\mathbb{C}\mathrm{P}^1,\mathbb{C})=0$, so it is not reached by the bottom step of the weight filtration either. A class simultaneously as deep in $F$ and as high in $W$ as this one has Hodge type $(1,1)$ and weight 2, whereas the first cohomology of a compact Kähler manifold is pure of weight 1. Non-compactness has moved the weight.

The graded pieces of $W$ are computed by residues, and the residue of a form with $m$ logarithmic poles lives on the locus where $m$ branches of $D$ meet. Those loci are the objects the whole construction is written in terms of.

**Definition 9.3.4: The Strata of a Normal Crossings Divisor**

For $m \geq 1$ and a strictly increasing multi-index $I = \{i_1 < \cdots < i_m\} \subseteq \{1, \ldots, N\}$ put $D_I = D_{i_1} \cap \cdots \cap D_{i_m}$, and set

$$D^{(m)} = \coprod_{|I|=m} D_I, \qquad D^{(0)} = X$$

the disjoint union of the $m$-fold intersections. Write $a_m \colon D^{(m)} \to X$ for the map that is the inclusion on each summand.

Because the components meet transversally, $D_I$ is smooth of pure dimension $n - m$ or empty, so each $D^{(m)}$ is a smooth projective variety and $a_m$ is a finite morphism. In particular $D^{(m)} = \emptyset$ for $m > \min(n, N)$, and each $D^{(m)}$ carries the pure Hodge structures of chapter 4.

**Example 9.3.5: Three Lines in the Plane**

Let $X = \mathbb{C}\mathrm{P}^2$ and let $D = L_1 \cup L_2 \cup L_3$ be three lines in general position, so no two are equal and the three do not share a point. Then $D^{(1)} = L_1 \sqcup L_2 \sqcup L_3$ is a disjoint union of three copies of $\mathbb{C}\mathrm{P}^1$, $D^{(2)} = \{L_1 \cap L_2\} \sqcup \{L_1 \cap L_3\} \sqcup \{L_2 \cap L_3\}$ is three reduced points, and $D^{(3)} = \emptyset$ because no point lies on all three lines. The complement $U = \mathbb{C}\mathrm{P}^2 \setminus D$ is $(\mathbb{C}^*)^2$.

The next statement is the computational core of the section: every graded piece of the weight filtration is the de Rham complex of a stratum, shifted.

**Proposition 9.3.6: The Graded Pieces of the Weight Filtration**

For every $m \geq 0$ there is an isomorphism of complexes of sheaves on $X$

$$\mathrm{Res}^m \colon \mathrm{gr}^W_m \Omega^\bullet_X(\log D) \xrightarrow{\sim} a_{m*}\Omega^{\bullet - m}_{D^{(m)}}$$

given in adapted coordinates, on a form of degree $p$, by

$$\mathrm{Res}^m\left(\tfrac{dz_I}{z_I} \wedge \eta\right) = (-1)^{mp}\, \eta\big|_{D_I}, \qquad |I| = m$$

and by 0 on the basis elements with $|I| < m$. For $m = 0$ it is the identity of $\Omega^\bullet_X$. For $m = 1$ it is, on the summand of a component $D_i$ of $D$, the Poincaré residue $\mathrm{Res}_{D_i}$ of definition 9.2.9 multiplied by $(-1)^p$ in degree $p$; the sign is what converts the anticommutation $\mathrm{Res}_{D_i} \circ d = -\,d \circ \mathrm{Res}_{D_i}$ of theorem 9.2.11 into commutation with the unsigned differential carried by $a_{1*}\Omega^{\bullet-1}_{D^{(1)}}$ here.

***Proof*:**
*Step 1: the map is well defined.* Fix $x \in X$ and adapted coordinates $z_1, \ldots, z_n$ at $x$, labelled so that the branch of $D_i$ through $x$ is $\{z_i = 0\}$ for $1 \leq i \leq r$, and let $I$ be an $m$-element subset of $\{1, \ldots, r\}$. If $i \leq r$ and $u$ is a unit then

$$\frac{d(uz_i)}{uz_i} = \frac{dz_i}{z_i} + \frac{du}{u}$$

and $du/u$ is holomorphic. Two coordinate systems adapted to $D$ at $x$ differ, on each branch, by such a unit together with a change of the remaining coordinates, so the basis form $\frac{dz_I}{z_I} \wedge \eta$ changes by a term with strictly fewer logarithmic factors, that is by an element of $W_{m-1}$, plus a form $\frac{dz_I}{z_I} \wedge \eta'$ with $\eta' \equiv \eta$ modulo the ideal of $D_I$ and modulo $dz_i$ for $i \in I$. Both ambiguities die on $\mathrm{gr}^W_m$ and after restriction to $D_I$, so the formula defines a morphism of sheaves independent of the coordinates. The labelling of the components of $D$ fixes the ordering of $I$ and hence the sign.

*Step 2: it is an isomorphism.* In adapted coordinates $\mathrm{gr}^W_m \Omega^p_X(\log D)$ is free over $\mathcal{O}_X$ on the classes of the basis elements $\frac{dz_I}{z_I} \wedge dz_J$ with $|I| = m$ exactly. For $i \in I$,

$$z_i \cdot \frac{dz_I}{z_I} \wedge dz_J = \pm \frac{dz_{I\setminus\{i\}}}{z_{I\setminus\{i\}}} \wedge dz_i \wedge dz_J$$

lies in $W_{m-1}$, so multiplication by the ideal of $D_I$ annihilates the corresponding summand of $\mathrm{gr}^W_m$, which is therefore a module over $\mathcal{O}_{D_I}$, free on the classes of $\frac{dz_I}{z_I} \wedge dz_J$ with $J \subseteq \{1, \dots, n\} \setminus I$ and $|J| = p - m$. The submanifold $D_I$ is cut out by the equations $z_i = 0$ for $i \in I$, so the $z_j$ with $j \notin I$ restrict to coordinates on it and restriction sends the $dz_J$ to a basis of $\Omega^{p-m}_{D_I}$. Hence $\mathrm{Res}^m$ is an isomorphism onto $a_{m*}\Omega^{p-m}_{D^{(m)}}$ in every degree.

*Step 3: it commutes with the differentials.* For $\alpha = \frac{dz_I}{z_I} \wedge \eta$ of degree $p$ with $|I| = m$, the basis forms $\frac{dz_i}{z_i}$ are closed, so $d\alpha = (-1)^m \frac{dz_I}{z_I} \wedge d\eta$, of degree $p + 1$. Therefore

$$\mathrm{Res}^m(d\alpha) = (-1)^{m(p+1)}(-1)^m \, (d\eta)\big|_{D_I} = (-1)^{mp}\, d\big(\eta|_{D_I}\big) = d\big(\mathrm{Res}^m \alpha\big)$$

using $(-1)^{2m} = 1$ and the fact that restriction to a submanifold commutes with $d$. The sign $(-1)^{mp}$ in the definition exists precisely to absorb the $(-1)^m$ produced by moving $d$ past the $m$ logarithmic factors, and plays no further role.

Finally, for $m = 0$ the filtration step is $W_0 = \Omega^\bullet_X$ and the formula is the identity. For $m = 1$ the formula reads $\mathrm{Res}^1 = (-1)^p \mathrm{Res}_{D_i}$ in degree $p$ on the summand of a component $D_i$: by definition 9.2.9 the map $\mathrm{Res}_{D_i}$ sends $\frac{dz_i}{z_i} \wedge \eta$ to $\eta|_{D_i}$ and kills the basis elements without $\frac{dz_i}{z_i}$, so it carries $W_1$ into $\iota_{i*}\Omega^{\bullet-1}_{D_i}$ and kills $W_0$. The two maps differ by a sign depending only on the degree, so they have the same kernel and the same image, and that sign is exactly what replaces the differential $-d$ that theorem 9.2.11 puts on the shifted target by the differential $d$ used here. When $D$ is smooth, $W_1$ is the whole logarithmic complex, so the exact sequence $0 \to W_0 \to W_1 \to \mathrm{gr}^W_1 \to 0$ is the residue sequence $0 \to \Omega^\bullet_X \to \Omega^\bullet_X(\log D) \to a_{1*}\Omega^{\bullet-1}_{D^{(1)}} \to 0$ of theorem 9.2.12, re-signed in this way, completing the proof.

The proposition says that the logarithmic complex is assembled, in $N + 1$ layers, out of the de Rham complexes of the strata $D^{(0)} = X$, $D^{(1)}$, $D^{(2)}$, and so on, each shifted into higher degree by its codimension. Every stratum is smooth and projective, so every layer contributes cohomology that is already understood, and the only question left is how the layers are glued. A spectral sequence answers exactly that question.

Both filtrations are filtrations of a complex of sheaves by subcomplexes, so each has a spectral sequence converging to the hypercohomology $H^k(X, \Omega^\bullet_X(\log D))$, which theorem 9.2.17 identifies with $H^k(U, \mathbb{C})$. The construction, the convergence criterion and the description of the first page are those of a filtered cochain complex, taken from *Homological Algebra*: see [6, The Exact Couple of a Filtered Complex] for the construction and [6, Convergence for a Decreasing Filtration of a Cochain Complex] for the convergence, exactly as in the proof of theorem 2.4.2. Both filtrations here are finite in each degree, so both spectral sequences converge.

The machinery is stated for decreasing filtrations, and $W$ is increasing; feed it instead the decreasing filtration $p \mapsto W_{-p}$, which is the same data read backwards. With that convention the two spectral sequences are

$$ {}_F E_1^{p,q} = H^q\big(X, \Omega_X^p(\log D)\big) \Longrightarrow H^{p+q}(U, \mathbb{C}) \tag{9.7} $$

$$ {}_W E_1^{-m,\,k+m} = H^k\big(X, \mathrm{gr}_m^W \Omega_X^\bullet(\log D)\big) \Longrightarrow H^k(U, \mathbb{C}) \tag{9.8} $$

with differentials $d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$ in both cases. The abutment filtrations are ${}_F E_\infty^{p,q} = \mathrm{gr}_F^p H^{p+q}(U,\mathbb{C})$ and ${}_W E_\infty^{-m,k+m} = \mathrm{gr}_m^W H^k(U,\mathbb{C})$, where $F$ and $W$ on $H^k(U,\mathbb{C})$ denote the images of $H^k(X, F^p\Omega_X^\bullet(\log D))$ and of $H^k(X, W_m\Omega_X^\bullet(\log D))$ respectively. The indexing in (9.8) is chosen so that $p+q=k$ throughout, with $p=-m$ and $q=k+m$; the reason for grouping the terms by $q$ rather than by $k$ appears in proposition 9.3.7, where $q$ turns out to be a weight.

**Proposition 9.3.7: The First Page of the Weight Spectral Sequence**

With the notation above,

$$ {}_W E_1^{-m,\,k+m} \cong H^{k-m}\big(D^{(m)}, \mathbb{C}\big) $$

for $0 \le m$, and the term vanishes for $m < 0$ and for $m > k$. Under this isomorphism:

1. the filtration induced by $F$ is $F^{p-m}H^{k-m}(D^{(m)}, \mathbb{C})$, the Hodge filtration of the smooth projective variety $D^{(m)}$ shifted by $m$;
2. the rational structure inherited from $H^k(U,\mathbb{Q})$ is $(2\pi i)^{-m}H^{k-m}(D^{(m)}, \mathbb{Q})$, so that ${}_W E_1^{-m,k+m} \cong H^{k-m}(D^{(m)}, \mathbb{Q})(-m)$ is a pure Hodge structure of weight $k+m$;
3. $d_1$ is, on the summand indexed by $I = \{i_1 < \cdots < i_m\}$, the alternating sum $\sum_{l=1}^m (-1)^{l-1}\gamma_l$ of the Gysin maps $\gamma_l$ of the codimension one inclusions $D_I \hookrightarrow D_{I\setminus\{i_l\}}$, up to a single overall sign that does not depend on $l$ and so changes neither the kernel nor the image; and it is a morphism of pure Hodge structures of weight $k+m$.

***Proof*:**

*Step 1: the terms.* By proposition 9.3.6, $\mathrm{gr}_m^W \Omega_X^\bullet(\log D) \cong a_{m*}\Omega_{D^{(m)}}^{\bullet-m}$. The morphism $a_m$ is a closed embedding on each connected summand of $D^{(m)}$, so $a_{m*}$ is exact and preserves cohomology, and hypercohomology may be computed downstairs:

$$ H^k\big(X, a_{m*}\Omega_{D^{(m)}}^{\bullet-m}\big) = H^k\big(D^{(m)}, \Omega_{D^{(m)}}^{\bullet-m}\big) = H^{k-m}\big(D^{(m)}, \Omega_{D^{(m)}}^{\bullet}\big) = H^{k-m}\big(D^{(m)}, \mathbb{C}\big) $$

the last equality because the holomorphic de Rham complex resolves the constant sheaf $\mathbb{C}_{D^{(m)}}$, which is the input already used in theorem 9.2.17. The term vanishes for $m<0$ because $W_{-1} = 0$, and for $m > k$ because $H^{k-m}$ of anything vanishes in negative degree.

*Step 2: the induced Hodge filtration.* The filtration $F$ induces on $\mathrm{gr}_m^W\Omega_X^\bullet(\log D)$ the truncation by degree, and under $\mathrm{Res}^m$ the degree $p$ part of the source corresponds to the degree $p-m$ part of $\Omega_{D^{(m)}}^\bullet$. So $F^p$ on the graded piece corresponds to $F^{p-m}\Omega_{D^{(m)}}^\bullet$, shifted. The filtration induced on $H^{k-m}(D^{(m)}, \Omega^\bullet)$ by the truncation of the holomorphic de Rham complex is the Hodge filtration precisely because the Hodge to de Rham spectral sequence of the smooth projective variety $D^{(m)}$ degenerates at $E_1$ (corollary 4.2.3). This gives (1).

*Step 3: the rational structure and the weight.* The map $\mathrm{Res}^m$ was normalised algebraically, by stripping $\frac{dz_I}{z_I}$, and for a single stripped factor it is, up to the degree-dependent sign carried by proposition 9.3.6, the topological residue of definition 7.4.6, whose factor $\frac{1}{2\pi i}$ was built in for this very purpose (box 9.2.10). That sign is invisible to the rational structure, which is what is at issue here. It is not that residue that is rational, however, but the transpose of the tube map: definition 7.4.6 normalises by $\int_\gamma \operatorname{Res}\eta = \frac{1}{2\pi i}\int_{\tau(\gamma)}\eta$, and what takes rational values on rational classes is $\gamma \mapsto \int_{\tau(\gamma)}\eta$, that is $2\pi i \operatorname{Res}$. Stripping one factor therefore multiplies the rational structure by $(2\pi i)^{-1}$: on $U = \mathbb{C}^*$ the generator of $H^1(U,\mathbb{Q})$ dual to the loop $S^1$ is $\frac{1}{2\pi i}\frac{dz}{z}$, because $\int_{S^1}\frac{dz}{z} = 2\pi i$, and its algebraic residue is $(2\pi i)^{-1}$. Iterating over the $m$ factors, $(2\pi i)^m \operatorname{Res}^m$ carries the rational structure inherited from $H^k(U,\mathbb{Q})$ into $H^{k-m}(D^{(m)},\mathbb{Q})$, so that $\operatorname{Res}^m$ carries that structure to $(2\pi i)^{-m}H^{k-m}(D^{(m)},\mathbb{Q})$. Writing $\mathbb{Q}(-m)$ for the Tate structure $(2\pi i)^{-m}\mathbb{Q}$ of weight $2m$ and type $(m,m)$ (definition 5.1.10), and combining with (1), the term is $H^{k-m}(D^{(m)},\mathbb{Q})(-m)$: a Hodge structure pure of weight $(k-m)+2m = k+m$, which is the index $q$. That the rational structure inherited from $H^k(U,\mathbb{Q})$ is preserved by the filtration, so that the statement makes sense at all, is the rationality of $W$ discussed in theorem 9.3.11 below; the computation just made is what identifies it.

*Step 4: the differential.* The differential $d_1$ on ${}_WE_1^{-m,k+m}$ is the connecting homomorphism of the short exact sequence of complexes

$$0 \to \operatorname{gr}^W_{m-1} \to W_m/W_{m-2} \to \operatorname{gr}^W_m \to 0$$

in hypercohomology, a map $H^{k-m}(D^{(m)},\mathbb{C}) \to H^{k-m+2}(D^{(m-1)},\mathbb{C})$. Fix $I = \{i_1 < \cdots < i_m\}$ and represent a class by $\alpha = \frac{dz_I}{z_I}\wedge\eta$ with $\eta|_{D_I}$ closed. Then $d\eta$ vanishes on $D_I$, so locally $d\eta = \sum_l z_{i_l}\beta_l + \sum_l dz_{i_l}\wedge\delta_l$; the second sum is killed by $\frac{dz_I}{z_I}$, and

$$d\alpha = (-1)^m\frac{dz_I}{z_I}\wedge d\eta = (-1)^m\sum_{l=1}^m(-1)^{m-l}\frac{dz_{I\setminus\{i_l\}}}{z_{I\setminus\{i_l\}}}\wedge dz_{i_l}\wedge\beta_l$$

which lies in $W_{m-1}$ and whose class in $\operatorname{gr}^W_{m-1}$ is supported on the $m$ strata $D_{I\setminus\{i_l\}}$. The sign is that of moving $dz_{i_l}$ out of slot $l$ and to the end of the product, past the $m-l$ logarithmic factors that follow it, namely $(-1)^{m-l} = (-1)^{m+1}(-1)^{l-1}$: it alternates in $l$, and the leftover $(-1)^{m+1}$ is one of the several $l$-independent signs, along with the $(-1)^m$ above and the normalising signs of $\operatorname{Res}^m$ and $\operatorname{Res}^{m-1}$, that make up the overall sign left unspecified in (3). So $d_1$ decomposes, up to that overall sign, as an alternating sum of $m$ maps, the $l$-th of which is the connecting homomorphism attached to the single smooth divisor $D_I \subseteq D_{I\setminus\{i_l\}}$. For a single smooth divisor $Z \subseteq Y$ of smooth projective varieties, the sequence $0 \to \Omega^\bullet_Y \to \Omega^\bullet_Y(\log Z) \to \Omega^{\bullet-1}_Z \to 0$ of theorem 9.2.12 has connecting homomorphism $H^j(Z,\mathbb{C}) \to H^{j+2}(Y,\mathbb{C})$, and the resulting long exact sequence is the Gysin sequence of the pair $(Y,Z)$, because theorem 9.2.17 identifies its middle term with $H^*(Y\setminus Z,\mathbb{C})$ and definition 7.4.6 identifies the residue with the topological one; the connecting map of the Gysin sequence is the Gysin pushforward $\gamma$ ([24, the Gysin sequence of a closed submanifold]). This proves the first half of (3).

*Step 5: $d_1$ is a morphism of Hodge structures.* Let $\iota\colon Z \hookrightarrow Y$ be a smooth divisor in a smooth projective variety $Y$ of dimension $d$, and fix $s \geq 0$. The Gysin map $\gamma_\iota\colon H^s(Z,\mathbb{Q}) \to H^{s+2}(Y,\mathbb{Q})$ is by definition the transpose of $\iota^*\colon H^{2d-s-2}(Y,\mathbb{Q}) \to H^{2d-s-2}(Z,\mathbb{Q})$ under Poincaré duality ([24, Poincaré duality]); in particular it is defined over $\mathbb{Q}$. Restriction $\iota^*$ is a morphism of pure Hodge structures of weight $2d-s-2$, while Poincaré duality identifies $H^s(Z,\mathbb{Q})$ with $H^{2d-2-s}(Z,\mathbb{Q})^\vee(-(d-1))$ and $H^{s+2}(Y,\mathbb{Q})$ with $H^{2d-s-2}(Y,\mathbb{Q})^\vee(-d)$, the twists being those of the fundamental classes of $Z$ and of $Y$. Transposing $\iota^*$ and comparing the two twists,

$$\gamma_\iota\colon H^s(Z,\mathbb{Q})(-1) \to H^{s+2}(Y,\mathbb{Q})$$

is a morphism of pure Hodge structures. Apply this with $Z = D_I$ and $Y = D_{I\setminus\{i_l\}}$, so $s = k - m$ and the two groups are summands of ${}_W E_1^{-m,k+m}$ and of ${}_W E_1^{-m+1,k+m}$; twisting both sides by $(-(m-1))$ turns it into a morphism $H^{k-m}(D^{(m)},\mathbb{Q})(-m) \to H^{k-m+2}(D^{(m-1)},\mathbb{Q})(-(m-1))$ of pure Hodge structures of weight $k+m$. An alternating sum of such is again one, as claimed.

Everything on the first page is therefore cohomology of smooth projective varieties, and the first differential is built from maps of the kind Hodge theory controls. The weight $q = k+m$ attached to the term ${}_W E_1^{-m,k+m}$ is constant along $d_1$, which is why the page is grouped by $q$; along $d_r$ with $r \geq 2$ it drops by $r-1$.

Two inputs to what follows are taken from elsewhere, and it is worth stating both precisely before they are used, so that the reader knows exactly where this section stops proving things. The first is upstream in the series; the second is external. Both are due to Deligne, and [13, Théorie de Hodge II] is the original for each, with [28, the construction of a mixed Hodge complex] a modern account.

**Theorem 9.3.8: Strictness of the Weight Differentials**

Let $K^\bullet$ carry an increasing filtration $W$ and a decreasing filtration $F$, both biregular filtrations by subcomplexes. Suppose the differentials $d_s$ of the spectral sequence of $W$ are strictly compatible with $F$ for every $s$. Then, in the notation of the proof below, $\sigma_s = \mathrm{rank}\delta_s$ for every $s$, the three filtrations $F$ induces on each page agree, and the spectral sequence of $F$ degenerates at $E_1$.
This is the lemma on two filtrations, [6, the lemma on two filtrations], where it is proved and the filtered derived category it lives in is constructed.

The hypothesis is the point, and it must be read carefully. Strictness of the $d_s$ is *assumed*, not deduced. It is tempting to hope that strictness on the graded pieces $\mathrm{gr}_n^W$ alone would suffice, and it does not: strictness of $d_0$ is a statement about the graded pieces separately and cannot see how they are glued, while strictness of $d_1$ is a statement about the gluing. [6] gives a two-dimensional counterexample. What supplies the hypothesis here is Hodge theory rather than homological algebra: the terms of the weight spectral sequence carry pure Hodge structures, and a morphism of pure Hodge structures is strict for the Hodge filtration (proposition 5.1.7). That is the same mechanism as part (1) of theorem 9.3.10, applied one page at a time, and it is why the two parts of that theorem are not independent.

**Theorem 9.3.9: Imported: Rationality of the Weight Filtration**

Under the comparison isomorphism of theorem 9.2.17, the subcomplex $W_m\Omega_X^\bullet(\log D)$ corresponds to the canonical truncation $\tau_{\leq m}Rj_*\mathbb{C}_U$, and that truncation is already defined on $Rj_*\mathbb{Q}_U$. Consequently $W_{k+m}H^k(U,\mathbb{C})$ is the complexification of a $\mathbb{Q}$-subspace of $H^k(U,\mathbb{Q})$.

Neither is a detail, and they are of different kinds. Theorem 9.3.8 is *proved* upstream in [6], whose filtered derived category exists for exactly this purpose; it is cited, not taken on faith. What it does is convert strictness of the weight differentials into the statements part (2) of the degeneration needs. That strictness is a separate matter and is imported, from Deligne's mixed Hodge complex formalism, as the proof below records at the point of use. Theorem 9.3.9 is imported outright: it makes the weight filtration exist over $\mathbb{Q}$ at all, without which there is no mixed Hodge structure on $H^k(U,\mathbb{Q})$, and it needs the derived pushforward and its canonical filtration, which no book in the series develops, since section 9.2 reaches theorem 9.2.17 through the fine complex $j_*\mathcal{A}_U^\bullet$ instead.

Together with the strictness input these are the only things taken on faith beyond theorem 9.2.3.

**Theorem 9.3.10: Degeneration of the Weight and Hodge Spectral Sequences**

Let $X$ be smooth projective, $D \subseteq X$ a normal crossings divisor with smooth components, and $U = X \setminus D$. Then

1. the weight spectral sequence (9.8) degenerates at $E_2$: the differentials $d_r$ vanish for all $r \geq 2$;
2. the Hodge spectral sequence (9.7) degenerates at $E_1$, so that $\dim_{\mathbb{C}} H^k(U, \mathbb{C}) = \sum_{p+q=k} \dim_{\mathbb{C}} H^q\big(X, \Omega_X^p(\log D)\big)$ for every $k$.

***Proof*:**
*Part (1).* By proposition 9.3.7 the term ${}_W E_1^{p,q}$ carries a pure $\mathbb{Q}$-Hodge structure of weight $q$, and $d_1 \colon E_1^{p,q} \to E_1^{p+1,q}$ is a morphism of such. Kernels and images of morphisms of pure Hodge structures of a fixed weight are again pure Hodge structures of that weight, since such morphisms are strict for $F$ (proposition 5.1.7); hence

$$ {}_W E_2^{p,q} = \ker\big(d_1 \colon E_1^{p,q} \to E_1^{p+1,q}\big)/\operatorname{im}\big(d_1 \colon E_1^{p-1,q} \to E_1^{p,q}\big) $$

is a pure $\mathbb{Q}$-Hodge structure of weight $q$, with rational structure and Hodge filtration induced from those of $E_1$. Inductively, if $E_r^{p,q}$ is pure of weight $q$ for all $p$ and $d_r$ vanishes, the same holds on $E_{r+1}$.

Now fix $r \geq 2$ and consider $d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$. The source is pure of weight $q$, the target is pure of weight $q-r+1$, and $q-(q-r+1) = r-1 \geq 1$, so the target has *strictly smaller* weight than the source. The map $d_r$ is defined over $\mathbb{Q}$, since the whole page is, and it is compatible with $F$, since $F$ is a filtration of the logarithmic complex by subcomplexes and therefore induces filtrations on every page compatible with every differential. Write $A$ for the source and $B$ for the target, of weights $a$ and $b = a-(r-1) < a$. Then $d_r$ is a morphism of pure Hodge structures between structures of different weights, so $d_r = 0$ by lemma 9.1.5(2), which is proved in section 9.1 for exactly this purpose.

The hypothesis that the weight *drops* is what makes the argument work: a rational $F$-compatible map into a structure of strictly larger weight need not vanish, as the identity map $\mathbb{Q}(0) \to \mathbb{Q}(-1)$ shows. This is the content of exercise 9.3.1 (5). Since $d_r = 0$ for every $r \geq 2$, the sequence degenerates at $E_2$, proving (1).

*Part (2).* Argue by induction on the number $N$ of components of $D$. If $N = 0$ then $U = X$ and (9.7) is the Hodge to de Rham spectral sequence of a smooth projective variety, which degenerates at $E_1$ by corollary 4.2.3.

Let $N \geq 1$, write $D' = D_1 \cup \cdots \cup D_{N-1}$, $Z = D_N$, $D'_Z = D' \cap Z$, and set $U' = X \setminus D'$ and $Z^\circ = Z \setminus D'_Z$. Both $D'$ on $X$ and $D'_Z$ on $Z$ are normal crossings divisors with at most $N-1$ components, so the inductive hypothesis applies to them. In adapted coordinates in which $Z = \{z_r = 0\}$, the basis elements $\frac{dz_I}{z_I} \wedge dz_J$ of $\Omega_X^p(\log D)$ with $r \notin I$ span $\Omega_X^p(\log D')$ and those with $r \in I$ map under the residue along $Z$ to a basis of $\Omega_Z^{p-1}(\log D'_Z)$, so for every $p$

$$ 0 \to \Omega_X^p(\log D') \to \Omega_X^p(\log D) \xrightarrow{\operatorname{Res}_Z} \Omega_Z^{p-1}(\log D'_Z) \to 0 \tag{9.9} $$

is exact. Assembling (9.9) over $p$ gives a short exact sequence of complexes $0 \to K' \to K \to L \to 0$

with $K' = \Omega^\bullet_X(\log D')$, $K = \Omega^\bullet_X(\log D)$ and $L = \Omega^\bullet_Z(\log D'_Z)[-1]$, and it is a sequence of complexes filtered by $F$ whose $F$-graded pieces $0 \to \mathrm{gr}^p_F K' \to \mathrm{gr}^p_F K \to \mathrm{gr}^p_F L \to 0$ are again exact, by (9.9).

Write $a_k = \dim H^k(U', \mathbb{C})$, $c_k = \dim H^k(Z^\circ, \mathbb{C})$, $b_k = \dim H^k(U, \mathbb{C})$ and $A_k, C_k, B_k$ for the corresponding sums of $E_1$-dimensions, so that $a_k = A_k$ and $c_k = C_k$ by induction and $b_k \le B_k$ always. Taking hypercohomology of $0 \to K' \to K \to L \to 0$ gives the long exact sequence

$$\cdots \to H^k(U', \mathbb{C}) \to H^k(U, \mathbb{C}) \xrightarrow{\mathrm{Res}_Z} H^{k-1}(Z^\circ, \mathbb{C}) \xrightarrow{\delta_{k-1}} H^{k+1}(U', \mathbb{C}) \to \cdots$$

whence $b_k = a_k + c_{k-1} - \mathrm{rank}\delta_{k-1} - \mathrm{rank}\delta_{k-2}$. Taking hypercohomology of the $F$-graded sequences gives, for each $p$, the long exact sequence of coherent cohomology groups with connecting maps $\partial^{p,q} \colon H^q(Z, \Omega^{p-1}_Z(\log D'_Z)) \to H^{q+1}(X, \Omega^p_X(\log D'))$, whence $B_k = A_k + C_{k-1} - \sigma_{k-1} - \sigma_{k-2}$, where $\sigma_s$ is the sum of $\mathrm{rank}\partial^{p,q}$ over the pairs with $(p-1) + q = s$. Because the $E_1$-pages of the three spectral sequences fit into the long exact sequence just written, and because those of $K'$ and $L$ degenerate at $E_1$ by induction, $\partial^{p,q}$ is the map induced by $\delta_s$ on $\mathrm{gr}^p_F$, so

$$\sigma_s = \sum_p \mathrm{rank}\big(\mathrm{gr}^p_F \delta_s\big) \le \mathrm{rank}\delta_s$$

with equality for all $s$ if and only if every $\delta_s$ is strictly compatible with $F$. Comparing the two displayed counts,

$$B_k - b_k = \big(\mathrm{rank}\delta_{k-1} - \sigma_{k-1}\big) + \big(\mathrm{rank}\delta_{k-2} - \sigma_{k-2}\big)$$

so the induction closes exactly when the Gysin maps $\delta_s \colon H^s(Z^\circ, \mathbb{C}) \to H^{s+2}(U', \mathbb{C})$ are strict for $F$.

That strictness is what has to be supplied, and it does not follow from the material developed here. The obstacle is precise: $Z^\circ$ and $U'$ are open, their cohomology is not pure, and proposition 5.1.7 therefore does not apply to $\delta_s$. Nor does theorem 9.3.8 supply it, since that lemma *consumes* strictness of the $d_s$ rather than producing it; what the lemma then gives, once strictness is in hand, is that the three induced filtrations agree and that the $F$-sequence degenerates.

The strictness itself is Deligne's, and it comes from the mixed Hodge complex formalism: the $E_1$ terms are cohomology of smooth projective strata and so are pure, which makes $d_1$ strict by proposition 5.1.7, and the formalism bootstraps from that page to all later ones simultaneously with the construction. It is proved in [13, Théorie de Hodge II, the construction of the mixed Hodge structure], with a modern account in [28, the construction of a mixed Hodge complex], and it is not proved here. Granting it, $B_k = b_k$ for all $k$ and the Hodge spectral sequence degenerates at $E_1$, completing the proof.

The mechanism behind Part (1) is worth separating from its bookkeeping. The objects on the first page are cohomology groups of smooth projective varieties, and such groups admit no nonzero map that lowers weight. Nothing about the geometry of $U$ enters the argument. What enters is that the strata $D^{(m)}$ are projective, which is the payoff of insisting on a good compactification.

Degeneration at $E_2$ is what allows the graded pieces of the weight filtration to be read off a page already computed, and reading them off is all that remains. The next theorem puts the two filtrations on $H^k(U, \mathbb{Q})$ and checks the three things a mixed Hodge structure asks of them: that $W$ be defined over $\mathbb{Q}$, that its graded pieces be pure with the filtration induced by $F$, and that neither filtration depend on the compactification used to produce it.

**Theorem 9.3.11: Deligne's Mixed Hodge Structure on a Smooth Variety**

Let $U$ be a smooth complex algebraic variety and let $X \supseteq U$ be a smooth projective compactification with $D = X \setminus U$ a normal crossings divisor with smooth components. Define, on $H^k(U, \mathbb{C}) = H^k(X, \Omega^\bullet_X(\log D))$,

$$W_{k+m}H^k(U,\mathbb{C}) = \operatorname{im}\Big(H^k\big(X, W_m\Omega^\bullet_X(\log D)\big) \to H^k(U,\mathbb{C})\Big)$$
$$F^pH^k(U,\mathbb{C}) = \operatorname{im}\Big(H^k\big(X, F^p\Omega^\bullet_X(\log D)\big) \to H^k(U,\mathbb{C})\Big)$$

Then $W$ is defined over $\mathbb{Q}$, and $\big(H^k(U,\mathbb{Q}), W_\bullet, F^\bullet\big)$ is a mixed Hodge structure. Its graded pieces are

$$\mathrm{gr}^W_{k+m}H^k(U,\mathbb{Q}) \cong {}_WE_2^{-m,\,k+m}$$

a subquotient of $H^{k-m}(D^{(m)}, \mathbb{Q})(-m)$; in particular $\mathrm{gr}^W_w H^k(U,\mathbb{Q}) = 0$ unless $k \le w \le 2k$. The structure does not depend on the choice of $(X, D)$.

***Proof*:**
*Step 1: rationality of $W$.* The filtration $F$ is defined only on $\mathbb{C}$-cohomology and needs no rational structure, but $W$ does. This is theorem 9.3.9, imported: $W_{k+m}H^k(U,\mathbb{C})$ is the complexification of a $\mathbb{Q}$-subspace of $H^k(U,\mathbb{Q})$, and (9.8) becomes the Leray spectral sequence of $j$ with $\mathbb{Q}$-coefficients, reindexed. The local shape of the sheaves involved is visible from proposition 9.3.7: near a point of $X$ lying on exactly $r$ branches, $U$ looks like $(\mathbb{C}^*)^r \times \mathbb{C}^{n-r}$, whose $m$-th cohomology is $\Lambda^m$ of $\mathbb{Q}(-1)^r$, so $R^mj_*\mathbb{Q}_U \cong a_{m*}\mathbb{Q}_{D^{(m)}}(-m)$, and the $E_1$-terms computed in proposition 9.3.7 are exactly $H^{k-m}(X, a_{m*}\mathbb{Q}_{D^{(m)}}(-m)) = H^{k-m}(D^{(m)}, \mathbb{Q})(-m)$.

The identification of $W_m\Omega^\bullet_X(\log D)$ with that truncation is the single input this proof takes from outside the book, and it is what supplies the rationality of $W$: the truncation is intrinsic to $Rj_*\mathbb{Q}_U$, but comparing it with a filtration of the logarithmic complex is done in the filtered derived category, and section 9.2 reaches theorem 9.2.17 through the fine complex $j_*\mathcal{A}^\bullet_U$ rather than through a derived pushforward. So this one step is taken from [13, Théorie de Hodge II, the weight filtration of the logarithmic complex]. Nothing else is: the weight bound of Step 3 and the independence of the compactification of Step 4, for which such a citation is usually made, are proved here.

*Step 2: the graded pieces are pure.* By theorem 9.3.10(1) the weight spectral sequence degenerates at $E_2$, so

$$\mathrm{gr}^W_{k+m}H^k(U,\mathbb{C}) = {}_WE_\infty^{-m,k+m} = {}_WE_2^{-m,k+m}$$

and by the proof of theorem 9.3.10(1) this is a pure $\mathbb{Q}$-Hodge structure of weight $k+m$, a subquotient of the pure structure $H^{k-m}(D^{(m)}, \mathbb{Q})(-m)$ of proposition 9.3.7. Its Hodge filtration is the one induced by $F$: the filtration $F$ induces the Hodge filtration on each ${}_WE_1$ term by proposition 9.3.7(1), and it induces on the abutment the filtration $F^\bullet H^k(U,\mathbb{C})$ defined in the statement, these two being compatible because theorem 9.3.10(2) makes $F$ strict on the weight spectral sequence. Hence $\mathrm{gr}^W_{k+m}H^k(U,\mathbb{Q})$ with the filtration induced by $F$ is a pure Hodge structure of weight $k+m$, which is the defining condition of definition 9.1.1.

*Step 3: the weight bounds.* By proposition 9.3.7 the term ${}_WE_1^{-m,k+m}$ vanishes unless $0 \le m \le k$: for $m < 0$ because $W_{-1} = 0$, and for $m > k$ because $H^{k-m}(D^{(m)}, \mathbb{Q})$ sits in negative degree. The same therefore holds for the subquotient ${}_WE_2^{-m,k+m} = \mathrm{gr}^W_{k+m}H^k(U,\mathbb{Q})$, so the weights

$w = k + m$ occurring in $H^k(U, \mathbb{Q})$ satisfy $k \le w \le 2k$. The sharper bound $w \le k + \min(k, n, N)$ also follows, since $D^{(m)}$ is empty for $m > \min(n, N)$.

*Step 4: independence of the compactification.* Given two good compactifications $X_1, X_2$ of $U$, resolve the closure of the diagonal $U \hookrightarrow X_1 \times X_2$ to obtain a third good compactification $X_3$ dominating both, by [18]. It therefore suffices to compare $X$ with a good compactification $\tilde{X}$ admitting a morphism $\pi\colon \tilde{X} \to X$ restricting to the identity on $U$. Pullback of forms gives $\pi^*\Omega^\bullet_X(\log D) \to \Omega^\bullet_{\tilde{X}}(\log \tilde{D})$; it preserves $F$ because it preserves form degree, and it preserves $W$ because if $z \circ \pi = u \prod_l w_l^{e_l}$ with $u$ a unit then $\pi^* \frac{dz}{z} = \frac{du}{u} + \sum_l e_l \frac{dw_l}{w_l}$ contributes at most one logarithmic pole to each term. So the induced map on $H^k(U, \mathbb{C})$, which is the identity, is a morphism of bifiltered vector spaces, and the filtrations computed on $\tilde{X}$ are at least as large as those computed on $X$. Both are mixed Hodge structures on the same $H^k(U, \mathbb{Q})$ and the identity is a morphism between them; by theorem 9.1.8 a morphism of mixed Hodge structures is strict for both filtrations, so an identity morphism forces the filtrations to agree, completing the proof.

The theorem says that the cohomology of a smooth variety is built from the cohomology of the strata of any boundary one chooses to add, and that the assembly is canonical even though the boundary is not. The weight filtration records how far $U$ is from being compact: the bottom step is the part visible already on a compactification, the top step is the part that only the punctures see, and everything between is a subquotient of the cohomology of a stratum, twisted. On a smooth projective $U$ the divisor $D$ is empty, only $m = 0$ survives, and the theorem returns the pure Hodge structure of chapter 4.

**Corollary 9.3.12: The Lowest Weight Part is the Image from a Compactification**

With $U$, $X$, $D$ as above,

$$W_k H^k(U, \mathbb{Q}) = \operatorname{im}\Big(j^*\colon H^k(X, \mathbb{Q}) \to H^k(U, \mathbb{Q})\Big)$$

and this subspace does not depend on the choice of $(X, D)$.

***Proof*:**

By definition $W_k H^k(U, \mathbb{C})$ is the image of $H^k(X, W_0\Omega^\bullet_X(\log D)) \to H^k(U, \mathbb{C})$, and $W_0\Omega^\bullet_X(\log D) = \Omega^\bullet_X$, whose hypercohomology is $H^k(X, \mathbb{C})$; the map induced by the inclusion of complexes is the restriction $j^*$. Both sides are defined over $\mathbb{Q}$, the left by theorem 9.3.11 and the right because $j^*$ is, so the equality holds rationally. Independence of $(X, D)$ is that of $W$, as we sought to show.

**Example 9.3.13: The Punctured Projective Line**

Let $X = \mathbb{CP}^1$, let $D = \{p_1, \ldots, p_r\}$ with $r \ge 1$ distinct points, and let $U = \mathbb{CP}^1 \setminus D$. The strata are $D^{(0)} = \mathbb{CP}^1$ and $D^{(1)} = D$, a disjoint union of $r$ points, and $D^{(m)} = \emptyset$ for $m \ge 2$ since the points are distinct.

The first page of the weight spectral sequence for $k = 1$ has two entries,

$$_W E_1^{0,1} = H^1(\mathbb{CP}^1, \mathbb{Q}) = 0, \qquad _W E_1^{-1,2} = H^0(D^{(1)}, \mathbb{Q})(-1) = \mathbb{Q}(-1)^r$$

and the differential $d_1\colon {}_W E_1^{-1,2} \to {}_W E_1^{0,2} = H^2(\mathbb{CP}^1, \mathbb{Q})$ is the Gysin map, which sends the class of a point to the generator of $H^2(\mathbb{CP}^1, \mathbb{Q}) \cong \mathbb{Q}(-1)$; on the $r$ summands it is the sum map

$(c_1, \dots, c_r) \mapsto \sum_i c_i$. It is surjective, with kernel of rank $r-1$. Hence

$$\mathrm{gr}_1^W H^1(U, \mathbb{Q}) = 0, \qquad \mathrm{gr}_2^W H^1(U, \mathbb{Q}) = \ker(d_1) \cong \mathbb{Q}(-1)^{r-1}$$

so $H^1(U, \mathbb{Q})$ is pure of weight 2 and of type $(1,1)$, of rank $r-1$. This is what the topology predicts: $\mathbb{C}\mathrm{P}^1$ minus $r$ points is homotopy equivalent to a wedge of $r-1$ circles. It is also what the residues say: a class in $H^1(U, \mathbb{C})$ is represented by a rational 1-form with at worst simple poles at the $p_i$, its residues are the $r$ numbers $c_i$, and the residue theorem is precisely the statement $\sum_i c_i = 0$ that cuts out $\ker(d_1)$.

The Hodge spectral sequence confirms the count. Since $\Omega^1_{\mathbb{C}\mathrm{P}^1}(\log D) \cong \mathcal{O}_{\mathbb{C}\mathrm{P}^1}(r-2)$, the nonzero terms in total degree 1 are $H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}(r-2))$, of dimension $r-1$ for $r \geq 1$, and $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) = 0$; the total is $r-1$, matching $\dim H^1(U, \mathbb{C})$ as theorem 9.3.10(2) requires. For $k = 2$ one finds ${}_W E_2^{0,2} = \mathrm{coker}(d_1) = 0$ and ${}_W E_2^{-1,3} = 0$, so $H^2(U, \mathbb{Q}) = 0$.

The weight filtration is trivial here in the sense that $H^1(U, \mathbb{Q})$ has a single weight. That is an accident of genus 0: the piece $\mathrm{gr}_1^W$, which would carry the cohomology of the compactification, vanishes because $H^1(\mathbb{C}\mathrm{P}^1) = 0$. The first genuinely mixed example, where $\mathrm{gr}_1^W$ and $\mathrm{gr}_2^W$ are both nonzero and the extension between them carries information, is the punctured elliptic curve of section 9.4.

A construction that depended on the compactification would be of limited use; one that does not is a candidate for a functor. The last result of the section confirms that it is one on the smooth quasi-projective varieties, which are the ones for which theorem 9.2.3 produces the projective compactification that theorem 9.3.11 needs.

**Proposition 9.3.14: Functoriality**

Let $f \colon U' \to U$ be a morphism of smooth quasi-projective complex algebraic varieties. Then

$$f^* \colon H^k(U, \mathbb{Q}) \to H^k(U', \mathbb{Q})$$

is a morphism of mixed Hodge structures for the structures of theorem 9.3.11. Consequently $U \mapsto H^k(U, \mathbb{Q})$ is a contravariant functor from smooth quasi-projective complex algebraic varieties to mixed Hodge structures.

***Proof*:**
Both $U$ and $U'$ are quasi-projective, so theorem 9.2.3 supplies good compactifications with projective total space: choose one $X \supseteq U$ with boundary $D$ and $X$ projective, and one $Y' \supseteq U'$ with $Y'$ projective. Let $\Gamma \subseteq Y' \times X$ be the closure of the graph of $f$, a projective variety because $Y' \times X$ is one, and apply [18] to obtain a smooth $X'$ with a projective morphism $X' \to \Gamma$ that is an isomorphism over $U'$ and for which $D' = X' \setminus U'$ is a normal crossings divisor; being projective over the projective $\Gamma$, the variety $X'$ is itself projective. The two projections give a good compactification $X' \supseteq U'$ together with a morphism $\bar{f} \colon X' \to X$ restricting to $f$ on $U'$. Since $f(U') \subseteq U$, we have $\bar{f}^{-1}(D) \subseteq D'$.

Pullback of forms along $\bar{f}$ therefore carries forms with logarithmic poles along $D$ to forms with logarithmic poles along $D'$: if $z$ is a local equation of a branch of $D$ then $z \circ \bar{f} = u \prod_l w_l^{e_l}$ for a unit $u$ and local equations $w_l$ of branches of $D'$, so

$$\bar{f}^* \frac{dz}{z} = \frac{du}{u} + \sum_l e_l \frac{dw_l}{w_l}$$

a sum of a holomorphic form and forms with a single logarithmic pole. Hence $\bar{f}^* \colon \Omega^\bullet_X(\log D) \to \Omega^\bullet_{X'}(\log D')$ is a morphism of complexes taking $W_m$ into $W_m$, by the displayed identity applied to each of the at most $m$ logarithmic factors of a local generator, and taking $F^p$ into $F^p$, because pullback preserves form degree.

Passing to hypercohomology, the induced map $H^k(U, \mathbb{C}) \to H^k(U', \mathbb{C})$ is $f^*$, and it carries $W_{k+m}$ into $W_{k+m}$ and $F^p$ into $F^p$ by the previous paragraph. It is defined over $\mathbb{Q}$ because it is pullback on singular cohomology. By theorem 9.3.11 the filtrations so computed are the intrinsic ones, so $f^*$ is a morphism of mixed Hodge structures as required. Functoriality of the assignment is the functoriality of $f \mapsto f^*$ on singular cohomology, completing the proof.

Strictness now comes for free: by theorem 9.1.8 every $f^*$ is strict for $W$ and for $F$, so kernels, images and cokernels of maps induced by morphisms of varieties are again mixed Hodge structures. This is what makes the formalism usable in practice, and it is the property that fails for a bare pair of filtrations chosen by hand.

## Exercise 9.3.1

1. Let $X = \mathbb{C}\mathrm{P}^1$, $D = \{0\}$ and $U = \mathbb{C}$. Write down $W$ and $F$ on $\Omega^\bullet_X(\log D)$, compute the first page of the weight spectral sequence in all bidegrees, and identify $d_1$. Deduce $H^k(\mathbb{C}, \mathbb{Q})$ for all $k$, and observe that the weight spectral sequence does *not* degenerate at $E_1$ here. Why does this not contradict theorem 9.3.10?

2. Let $X = \mathbb{C}\mathrm{P}^1 \times \mathbb{C}\mathrm{P}^1$, let $D_1 = \{0\} \times \mathbb{C}\mathrm{P}^1$ and $D_2 = \mathbb{C}\mathrm{P}^1 \times \{0\}$, and let $U = X \setminus (D_1 \cup D_2)$, so $U \cong \mathbb{C}^2$. List the strata $D^{(0)}, D^{(1)}, D^{(2)}$, write out ${}_WE_1^{-m,k+m}$ for $k = 0, 1, 2$ and every $m$, and compute enough of $d_1$ to conclude that $H^k(U, \mathbb{Q}) = 0$ for $k \geq 1$. This is the smallest example in which the stratum $D^{(2)}$ contributes.

3. Take $D = \emptyset$, so that $U = X$ is smooth projective. Compute both spectral sequences and show that theorem 9.3.11 returns the pure Hodge structure of weight $k$ on $H^k(X, \mathbb{Q})$. Which step of the proof of theorem 9.3.10 becomes vacuous?

4. Let $X$ be a smooth projective curve of genus $g$, let $D = \{p_1, \ldots, p_r\}$ with $r \geq 1$, and let $U = X \setminus D$. Compute $\mathrm{gr}_1^W H^1(U, \mathbb{Q})$ and $\mathrm{gr}_2^W H^1(U, \mathbb{Q})$, and check the resulting dimension of $H^1(U, \mathbb{Q})$ against $\dim H^0(X, \Omega^1_X(\log D)) + \dim H^1(X, \mathcal{O}_X)$.

5. Let $A = \mathbb{Q}(0)$ and $B = \mathbb{Q}(-1)$, and let $f \colon A \to B$ be the identity map of the underlying $\mathbb{Q}$-vector space $\mathbb{Q}$. Show that $f$ is rational and satisfies $f(F^p A) \subseteq F^p B$ for every $p$, but is not a morphism of Hodge structures. Explain why this does not disturb the vanishing argument in the proof of theorem 9.3.10(1).

6. Let $U = \mathbb{C}^*$ and $U' = \mathbb{C}^* \setminus \{1\}$. Using example 9.3.13, identify the mixed Hodge structures on $H^1(U, \mathbb{Q})$ and $H^1(U', \mathbb{Q})$, and show that the restriction map $H^1(U, \mathbb{Q}) \to H^1(U', \mathbb{Q})$ is an injective morphism of mixed Hodge structures. Is it strict for $W$?

## 9.4 The Punctured Elliptic Curve

Everything section 9.1 through section 9.3 constructs can be written down in closed form on a curve, where the logarithmic complex has two terms and the weight filtration has two steps. This section carries that out for $E$ an elliptic curve and $U = E \setminus D$ the complement of a finite set of points, and then extracts the invariant that the graded pieces do not see.

The computation settles a question the general theorem leaves open, namely how much weight the punctures create. A single puncture creates none: $H^1(E \setminus \{p\}, \mathbb{Q})$ is the pure structure of example 5.4.3, for every $p$. Two punctures create exactly one class of weight two, and the resulting extension remembers the difference $p - q$ as a point of $E$.

Fix $\tau$ in the upper half-plane and write $E = E_\tau = \mathbb{C}/\Lambda_\tau$ with $\Lambda_\tau = \mathbb{Z} \cdot 1 \oplus \mathbb{Z}\tau$, as in example 5.4.3. Let $D \subseteq E$ be a set of $n \geq 1$ points[2] and $U = E \setminus D$, with inclusion $j \colon U \to E$. No compactification has to be produced: $E$ is smooth and projective, and a finite set of points on a curve is a normal-crossings divisor in the sense of definition 9.2.1, each point being cut out by a single coordinate. So theorem 9.2.3, the one imported hypothesis of section 9.2, is not used anywhere below; the compactification is the one the geometry hands over.

Two features of dimension one simplify the logarithmic complex of definition 9.2.5 to the point where it can be handled by hand. It has only two terms,

$$\Omega^\bullet_E(\log D) \;=\; \left[\; \mathcal{O}_E \xrightarrow{\;d\;} \Omega^1_E(\log D) \;\right] \tag{9.10}$$

and the coordinate $z$ on $\mathbb{C}$ descends to a nowhere-vanishing holomorphic 1-form $dz$ on $E$, so $\Omega^1_E \cong \mathcal{O}_E$ and a section of $\Omega^1_E(\log D)$ is $h(z)\,dz$ with $h$ a $\Lambda_\tau$-periodic meromorphic function whose poles lie in $D$ and are at most simple. Every question about the complex therefore becomes a question about elliptic functions, and the tool for those is contour integration over a fundamental parallelogram.

That contour supplies two identities at once. Integrating a form over the boundary of the parallelogram gives the residue theorem ([25, the residue theorem]), and integrating the form against the primitive $z$ of $dz$ gives a relation between the periods of the form and the positions of its poles. The second identity is what will eventually locate the extension class, so it is worth recording both together.

**Lemma 9.4.1: Residues and Periods on a Complex Torus**

Let $E = \mathbb{C}/\Lambda_\tau$ and let $\eta$ be a meromorphic 1-form on $E$, not identically zero, with pole set $S$. Choose $z_0 \in \mathbb{C}$ so that the closed parallelogram $P$ with vertices $z_0,\ z_0 + 1,\ z_0 + 1 + \tau,\ z_0 + \tau$ contains exactly one lift of each point of $S$, and each in its interior; let $\gamma_1$ and $\gamma_2$ be the images in $E$ of the segments $[z_0, z_0 + 1]$ and $[z_0, z_0 + \tau]$.

1. $\displaystyle\sum_{x \in S} \operatorname{Res}_x \eta = 0.$
2. If every pole of $\eta$ is simple, and $a_x$ denotes the lift of $x \in S$ interior to $P$, then

$$\int_{\gamma_2} \eta \;-\; \tau \int_{\gamma_1} \eta \;=\; 2\pi i \sum_{x \in S} a_x \operatorname{Res}_x \eta \tag{9.11}$$

[2] Here $n$ counts the punctures. In section 9.2 and section 9.3 the letter $n$ was the dimension of $X$ and $N$ the number of components of the boundary; on a curve the dimension is 1 and $N = n$.

***Proof*:**
Pull $\eta$ back to $\mathbb{C}$ and write it there as $h(z)\,dz$ with $h$ meromorphic and $\Lambda_\tau$-periodic. The boundary $\partial P$, traversed counterclockwise, consists of the segment $A$ from $z_0$ to $z_0+1$, the segment $B$ from $z_0+1$ to $z_0+1+\tau$, the reverse of the segment $A+\tau$, and the reverse of the segment $B-1$. Note that $B-1$ is the segment from $z_0$ to $z_0+\tau$, whose image in $E$ is $\gamma_2$, while the image of $A$ is $\gamma_1$.

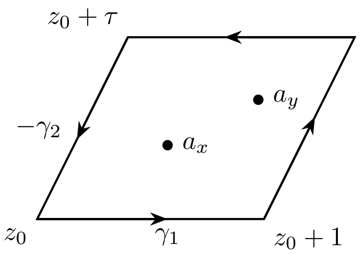


Figure 9.1: The parallelogram $P$, traversed counterclockwise, with the lifted poles in its interior; the cycle $\gamma_2$ is the left side traversed upward, so it occurs in $\partial P$ with a minus sign

*Step 1: the residue theorem.* Let $g$ be holomorphic on a neighbourhood of $P$. Pairing the two horizontal sides and then the two slanted ones,

$$\oint_{\partial P} g\,h\,dz \;=\; \int_A \big(g(z)-g(z+\tau)\big)h(z)\,dz \;+\; \int_{B-1} \big(g(z+1)-g(z)\big)h(z)\,dz \tag{9.12}$$

where periodicity of $h$ has been used to replace $h(z+\tau)$ and $h(z+1)$ by $h(z)$. Taking $g\equiv 1$ makes both integrands vanish, so $\oint_{\partial P}\eta=0$; and the residue theorem ([25, the residue theorem]) in the plane evaluates the same integral as $2\pi i$ times the sum of the residues of $h$ at the interior poles, which are the lifts $a_x$. Since $\mathrm{Res}_{a_x}h=\mathrm{Res}_x\eta$, assertion (1) follows.
*Step 2: the period relation.* Take $g(z)=z$ in (9.12). The first integrand becomes $-\tau\,h(z)$ and the second becomes $h(z)$, so

$$\oint_{\partial P} z\,h(z)\,dz \;=\; -\tau\int_A h\,dz \;+\; \int_{B-1} h\,dz \;=\; -\tau\int_{\gamma_1}\eta \;+\; \int_{\gamma_2}\eta$$

On the other hand $z\,h(z)\,dz$ has a simple pole at each $a_x$ with residue $a_x\mathrm{Res}_{a_x}h=a_x\mathrm{Res}_x\eta$, because $\eta$ has simple poles and $z$ is holomorphic; the residue theorem in the plane evaluates the contour integral as $2\pi i\sum_x a_x\mathrm{Res}_x\eta$. Comparing the two evaluations gives (9.11), as we sought to show.

Part (1) is the obstruction that governs the whole section. A residue is a local quantity, but the residues of a global form cannot be prescribed independently: their sum is forced to vanish. Part (2) refines this into an equality rather than a vanishing, and its right-hand side is the only place in the section where the positions of the punctures, and not merely their number, enter.

The first consequence is a count of the logarithmic differentials, which by (9.10) is the only piece of the logarithmic complex that is not already understood.

**Proposition 9.4.2: Logarithmic Differentials on an Elliptic Curve**

Let $E = \mathbb{C}/\Lambda_\tau$ and let $D \subseteq E$ consist of $n \geq 1$ points. The residue map of theorem 9.2.12 in degree one,

$$\mathrm{Res}\colon H^0\big(E, \Omega^1_E(\log D)\big) \to \mathbb{C}^D$$

has kernel $H^0(E, \Omega^1_E) = \mathbb{C}\,dz$ and image the hyperplane $\big\{ r \in \mathbb{C}^D \big| \sum_{x \in D} r_x = 0 \big\}$. In particular $\dim_{\mathbb{C}} H^0\big(E, \Omega^1_E(\log D)\big) = n$, and for distinct $p, q \in E$ there is a meromorphic 1-form on $E$ with simple poles exactly at $p$ and $q$, of residues $+1$ and $-1$, unique up to adding a multiple of $dz$.

***Proof*:**
The set $D$ is finite, so its points are pairwise disjoint components and theorem 9.2.12 applies. In degree one its sequence is the short exact sequence of sheaves

$$0 \to \Omega^1_E \to \Omega^1_E(\log D) \xrightarrow{\mathrm{Res}} \mathcal{O}_D \to 0 \tag{9.13}$$

the quotient being the skyscraper $\mathcal{O}_D = \bigoplus_{x \in D} \mathbb{C}_x$ because $D$ is a finite set of reduced points. Its long exact cohomology sequence begins

$$0 \to H^0(E, \Omega^1_E) \to H^0\big(E, \Omega^1_E(\log D)\big) \xrightarrow{\mathrm{Res}} \mathbb{C}^D \xrightarrow{\delta} H^1(E, \Omega^1_E)$$

Here $H^0(E, \Omega^1_E) = \mathbb{C}\,dz$: a global holomorphic 1-form is $h(z)\,dz$ with $h$ holomorphic and $\Lambda_\tau$-periodic, so $h$ descends to a holomorphic function on the compact connected $E$ and is constant by proposition 1.1.11. And $H^1(E, \Omega^1_E) \cong H^{1,1}(E)$ by theorem 2.3.3, which is one-dimensional by example 4.2.2. So the proposition amounts to identifying $\delta$ with a nonzero multiple of the sum map.

*Step 1: $\delta$ is a multiple of the sum map.* For $x \in D$ the inclusion $\Omega^1_E(\log x) \subseteq \Omega^1_E(\log D)$ and the inclusion of $\mathbb{C}_x$ as a summand of $\mathcal{O}_D$ form a morphism from (9.13) for the divisor $\{x\}$ to (9.13) for $D$, compatible with the identity of $\Omega^1_E$. Naturality of the connecting homomorphism gives $\delta(1_x) = \delta_x$, where $\delta_x \in H^1(E, \Omega^1_E)$ is the connecting map for the one-point divisor $\{x\}$ applied to 1. Hence $\delta(r) = \sum_{x \in D} r_x\, \delta_x$, and it suffices to see that $\delta_x$ does not depend on $x$.

Let $x, x' \in E$ and let $t\colon E \to E$ be the translation by $x' - x$, a biholomorphism carrying $x$ to $x'$. Pullback along $t$ carries forms with a logarithmic pole at $x'$ to forms with a logarithmic pole at $x$ and preserves residues, so $t^*$ is a morphism from (9.13) for $\{x'\}$ to (9.13) for $\{x\}$, acting as the identity on the quotient under the identifications $\mathbb{C}_{x'} \cong \mathbb{C} \cong \mathbb{C}_x$. Naturality gives $t^*\delta_{x'} = \delta_x$. Now $t$ is homotopic to the identity of $E$, since $E$ is connected and translations by a path from 0 to $x' - x$ interpolate between them, so $t^*$ is the identity on $H^2(E, \mathbb{C})$. A curve has $h^{2,0} = h^{0,2} = 0$ by example 4.2.2, so $H^2(E, \mathbb{C})$ is the single summand $H^{1,1}(E) = H^1(E, \Omega^1_E)$ and no compatibility of $t^*$ with a decomposition is needed. Hence $\delta_x = \delta_{x'}$.

*Step 2: the multiple is nonzero.* Write $\delta(r) = (\sum_x r_x)\kappa$ with $\kappa \in H^1(E, \Omega^1_E)$ the common value of the $\delta_x$. If $\kappa$ were zero then Res would be surjective for every $D$, and taking $D = \{x\}$ a single point would produce a meromorphic 1-form on $E$ with one simple pole, of residue 1. Its residues then sum to 1, contradicting lemma 9.4.1(1). So $\kappa \neq 0$, the kernel of $\delta$ is the hyperplane of sum-zero vectors, and exactness identifies it with the image of Res.

The dimension count follows: $\dim H^0(E, \Omega^1_E(\log D)) = 1 + (n-1) = n$. For the last assertion, take $D = \{p, q\}$; the vector $(1, -1)$ has sum zero, so some $\omega$ has $\mathrm{Res}_p\omega = 1$ and $\mathrm{Res}_q\omega = -1$, and two such forms differ by an element of $\ker \mathrm{Res} = \mathbb{C}\,dz$. Such an $\omega$ has poles exactly at $p$ and $q$, both residues being nonzero, completing the proof.

The last form is classically a *differential of the third kind*: a meromorphic 1-form whose only poles are simple, with residues summing to zero. On $E = \mathbb{C}/\Lambda_\tau$ it can also be written out in closed form using the Weierstrass zeta function [5, the Weierstrass zeta function], whose quasi-periods cancel in the difference $\zeta(z-a) - \zeta(z-b)$; the cohomological argument above is used here because it needs nothing beyond (9.13) and gives the dimension count at the same time.

Proposition 9.4.2 is already the whole answer, because theorem 9.3.10 converts the dimension of $H^0(E, \Omega^1_E(\log D))$ directly into the dimension of $H^1(U, \mathbb{C})$ and into the size of the Hodge filtration. What remains is to locate the single step of the weight filtration, and for that no further geometry is needed: the two filtrations of a mixed Hodge structure constrain each other, and the dimension of $F^1$ already determines where $W$ jumps.

**Theorem 9.4.3: The Mixed Hodge Structure of a Punctured Elliptic Curve**

Let $E = \mathbb{C}/\Lambda_\tau$, let $D \subseteq E$ consist of $n \geq 1$ points, let $U = E \setminus D$ and let $j\colon U \to E$ be the inclusion. Give $H^1(U, \mathbb{Q})$ the mixed Hodge structure of theorem 9.3.11, computed from the compactification $E$ and the divisor $D$. Then

1. $\dim_{\mathbb{Q}} H^1(U, \mathbb{Q}) = n + 1$;
2. writing $H^0(D, \mathbb{Q})_0$ for the kernel of the sum map $\mathbb{Q}^D \to \mathbb{Q}$, the sequence
$$0 \to H^1(E, \mathbb{Q}) \xrightarrow{j^*} H^1(U, \mathbb{Q}) \xrightarrow{\mathrm{Res}} H^0(D, \mathbb{Q})_0(-1) \to 0 \tag{9.14}$$
is exact, and both maps are morphisms of mixed Hodge structures;
3. $W_0 H^1(U, \mathbb{Q}) = 0$, $W_1 H^1(U, \mathbb{Q}) = j^* H^1(E, \mathbb{Q})$, and $W_2 H^1(U, \mathbb{Q}) = H^1(U, \mathbb{Q})$; hence $\mathrm{gr}^W_1 \cong H^1(E, \mathbb{Q})$, pure of weight one with $h^{1,0} = h^{0,1} = 1$, and $\mathrm{gr}^W_2 \cong \mathbb{Q}(-1)^{n-1}$, pure of weight two and of type $(1,1)$;
4. $F^0 H^1(U, \mathbb{C}) = H^1(U, \mathbb{C})$, $F^2 H^1(U, \mathbb{C}) = 0$, and $F^1 H^1(U, \mathbb{C})$ is the image of $H^0\big(E, \Omega^1_E(\log D)\big)$ under $\eta \mapsto [\eta]$, an injection with $n$-dimensional image.

***Proof*:**
Throughout, for $x \in D$ let $\sigma_x \subseteq U$ be a small circle around $x$, positively oriented for the complex structure, and let $\mathrm{per}_x(\alpha) = \int_{\sigma_x} \alpha$ be the corresponding period. Since $[\sigma_x] \in H_1(U, \mathbb{Z})$, the functional $\mathrm{per}_x$ is $\mathbb{Q}$-valued on $H^1(U, \mathbb{Q})$. By definition 7.4.6, whose recipe applies verbatim to a smooth divisor in a compact complex manifold and which for a point on a curve sends $x$ to the class of $\sigma_x$, the residue map on cohomology is $\mathrm{Res}_x = \frac{1}{2\pi i}\mathrm{per}_x$; and for a form $\eta$ with at most a simple pole at $x$ this agrees with the classical residue, by the Cauchy integral formula or by proposition 7.4.7. It carries $H^1(U, \mathbb{Q})$ into $\mathbb{Q}(-1)^D$, the rational structure of the Tate twist being $(2\pi i)^{-1}\mathbb{Q}$ by definition 5.1.10.
*Step 1: dimensions and the Hodge filtration.* By theorem 9.2.17 the complex (9.10) computes $H^*(U, \mathbb{C})$, and by theorem 9.3.10 the spectral sequence of its stupid filtration (definition 9.3.2) degenerates at $E_1$. In degree one this gives
$$\dim_{\mathbb{C}} H^1(U, \mathbb{C}) = \dim_{\mathbb{C}} H^1(E, \mathcal{O}_E) + \dim_{\mathbb{C}} H^0\big(E, \Omega^1_E(\log D)\big)$$
together with $F^1 H^1(U, \mathbb{C}) = \mathrm{gr}^1_F H^1(U, \mathbb{C}) \cong H^0(E, \Omega^1_E(\log D))$, the isomorphism being induced by $\eta \mapsto [\eta]$, and $F^2 = 0$ because the complex has no term in degree two. Now $H^1(E, \mathcal{O}_E) \cong$

$H^{0,1}(E)$ is one-dimensional by theorem 2.3.3 and example 4.2.2, and $H^0(E, \Omega^1_E(\log D))$ has dimension $n$ by proposition 9.4.2. This proves (1) and (4); in particular the map $\eta \mapsto [\eta]$ is injective, its source and its image both having dimension $n$.

*Step 2: the sequence* (9.14) *is exact.* Injectivity of $j^*$: a loop in $E$ can be moved off the finite set $D$, so $H_1(U, \mathbb{Z}) \to H_1(E, \mathbb{Z})$ is surjective, and dualizing gives injectivity of $j^*$ on $H^1$ with any field coefficients.

The composite $\mathrm{Res} \circ j^*$ vanishes, because $\sigma_x$ bounds a disc in $E$ and so is null-homologous there, making $\mathrm{per}_x(j^*\beta) = \int_{\sigma_x} \beta = 0$ for $\beta \in H^1(E, \mathbb{Q})$. The image of Res lies in the sum-zero subspace, because $\sum_{x \in D} [\sigma_x] = 0$ in $H_1(U, \mathbb{Z})$: removing an open disc around each point of $D$ from $E$ leaves a compact oriented surface with boundary $\coprod_x \sigma_x$, contained in $U$.

For surjectivity onto the sum-zero subspace, complexify and factor through Step 1. If $\eta \in H^0(E, \Omega^1_E(\log D))$ then $\mathrm{per}_x[\eta] = \int_{\sigma_x} \eta = 2\pi i \, \mathrm{Res}_x \eta$, so the composite $H^0(E, \Omega^1_E(\log D)) \to H^1(U, \mathbb{C}) \xrightarrow{\mathrm{Res}} \mathbb{C}^D$ is the sheaf residue map of proposition 9.4.2, whose image is the full sum-zero hyperplane. Hence Res is surjective onto that hyperplane after complexification, and therefore onto $H^0(D, \mathbb{Q})_0(-1)$, a $\mathbb{Q}$-linear map with surjective complexification onto the complexification of a rational subspace being surjective onto that subspace. Counting dimensions,

$$\dim \ker \mathrm{Res} = (n+1) - (n-1) = 2 = \dim j^* H^1(E, \mathbb{Q})$$

and $j^* H^1(E, \mathbb{Q}) \subseteq \ker \mathrm{Res}$, so the two coincide and (9.14) is exact.

*Step 3: the weight filtration.* By theorem 9.3.11 the weights of $H^1(U, \mathbb{Q})$ lie in $[1, 2]$, so $W_0 = 0$ and $W_2 = H^1(U, \mathbb{Q})$; only the single step $W_1$ is at issue. By proposition 9.3.14 the map $j^*$ is a morphism of mixed Hodge structures out of a structure pure of weight one, so it carries $W_1 H^1(E, \mathbb{Q}) = H^1(E, \mathbb{Q})$ into $W_1 H^1(U, \mathbb{Q})$; hence $j^* H^1(E, \mathbb{Q}) \subseteq W_1$ and $\dim W_1 \geq 2$.

The reverse inequality comes from counting the Hodge filtration on the graded pieces. Write $w = \dim_{\mathbb{Q}} \mathrm{gr}^W_1 = \dim_{\mathbb{Q}} W_1$, so that $\dim_{\mathbb{Q}} \mathrm{gr}^W_2 = (n+1) - w$ by (1). Since $F$ induces the quotient filtration on each graded piece,

$$\dim_{\mathbb{C}} F^1 H^1(U, \mathbb{C}) = \dim_{\mathbb{C}} F^1 \mathrm{gr}^W_1 + \dim_{\mathbb{C}} F^1 \mathrm{gr}^W_2$$

Now $\mathrm{gr}^W_1$ is pure of weight one by definition 9.1.1, so $F^1 \mathrm{gr}^W_1$ is its $(1,0)$-summand, of dimension $w/2$ because conjugation exchanges the two summands. And $\mathrm{gr}^W_2$ is pure of weight two with $F^2 \mathrm{gr}^W_2 = 0$, the whole of $F^2 H^1(U, \mathbb{C})$ being zero by (4); its $(2,0)$-summand is therefore zero, hence so is its $(0,2)$-summand, and $\mathrm{gr}^W_2$ is entirely of type $(1,1)$ with $F^1 \mathrm{gr}^W_2 = \mathrm{gr}^W_2$. Substituting and using $\dim F^1 H^1(U, \mathbb{C}) = n$ from (4),

$$n = \frac{w}{2} + (n+1) - w$$

which gives $w = 2$. So $\dim W_1 = 2 = \dim j^* H^1(E, \mathbb{Q})$ and the inclusion above is an equality, $W_1 = j^* H^1(E, \mathbb{Q})$.

*Step 4: the graded pieces and the two morphisms.* From Step 3, $\mathrm{gr}^W_1 = W_1 \cong H^1(E, \mathbb{Q})$, whose Hodge numbers are $h^{1,0} = h^{0,1} = 1$ by example 5.4.3; and $\mathrm{gr}^W_2 = H^1(U, \mathbb{Q})/W_1$, which Step 3 showed to be pure of weight two and entirely of type $(1,1)$, of dimension $n-1$. A weight-two rational Hodge structure of type $(1,1)$ is a direct sum of copies of $\mathbb{Q}(-1)$ (definition 5.1.10), one for each element of a rational basis, so $\mathrm{gr}^W_2 \cong \mathbb{Q}(-1)^{n-1}$. This proves (3).

Finally, $j^*$ is a morphism by proposition 9.3.14, and Res is one as well: its target is pure of weight two with $F^1$ everything and $F^2$ zero, so compatibility with $W$ asks only $\mathrm{Res}(W_1) = 0$, which holds because $W_1 = j^*H^1(E,\mathbb{Q}) \subseteq \ker \operatorname{Res}$ by Step 2 and Step 3, and compatibility with $F$ asks only $\operatorname{Res}(F^2H^1(U,\mathbb{C})) = 0$, which holds because $F^2 = 0$. Together with the exactness of Step 2 this completes the proof.

The statement is a complete description: the mixed structure on $H^1$ of a punctured elliptic curve is an extension of $\mathbb{Q}(-1)^{n-1}$ by the pure structure of $E$, with the Hodge filtration read off the logarithmic differentials. Both graded pieces are forced, and neither depends on where the punctures sit. What the punctures contribute to $\mathrm{gr}_2^W$ is only their number, and only through the sum condition of lemma 9.4.1(1), which removes exactly one dimension from $\mathbb{Q}^D$.

That accounting has an immediate consequence at $n = 1$, where removing one dimension from a one-dimensional space leaves nothing at all.

**Corollary 9.4.4: A Single Puncture Creates No Weight**

Let $E$ be an elliptic curve and $p \in E$. Then

$$j^* \colon H^1(E,\mathbb{Q}) \xrightarrow{\sim} H^1\big(E \setminus \{p\}, \mathbb{Q}\big)$$

is an isomorphism of mixed Hodge structures, and $H^1(E \setminus \{p\}, \mathbb{Q})$ is pure of weight one. Its Hodge filtration has $F^1 = \mathbb{C}\,dz$.

***Proof*:**
Apply theorem 9.4.3 with $n = 1$. Then $H^0(D,\mathbb{Q})_0 = 0$, so (9.14) makes $j^*$ an isomorphism of mixed Hodge structures; $\mathrm{gr}_2^W = 0$ and $\mathrm{gr}_1^W = H^1(E,\mathbb{Q})$, which is purity of weight one; and $F^1$ is the image of $H^0(E, \Omega_E^1(\log p))$, which is $\mathbb{C}\,dz$ because proposition 9.4.2 gives it dimension 1 and $dz$ already lies in it, as we sought to show.

The mechanism deserves stating in terms of forms rather than of dimensions. A class of weight two on $E \setminus \{p\}$ would have to be represented by a 1-form with a genuine logarithmic pole at $p$, that is, with nonzero residue there. Lemma 9.4.1(1) forbids it: the residue at $p$ has no other residue to cancel against. Non-compactness by itself therefore does not create weight; what creates weight is boundary along which a form can have residues that cancel. A single point is not enough boundary.

The same reading explains an earlier computation. The Legendre family of example 7.1.9 was handled through its affine model $y^2 = P(x)$, which is the fibre with one point removed, and the Gauss-Manin computation there never encountered anything of weight two. By corollary 9.4.4 it could not have, and the periods of $dx/y$ computed there are periods of a class on the compact curve.

Two punctures change the answer, and this is the smallest case in which the extension appearing in (9.14) is not determined by its ends. Example 9.1.4 produced an abstract non-split extension by writing down a period; the punctured curve produces one geometrically, and lemma 9.4.1(2) evaluates the period in terms of the two points.

**Example 9.4.5: The Extension Class of a Twice-Punctured Elliptic Curve**
Let $E = \mathbb{C}/\Lambda_\tau$, let $p \neq q$ in $E$, let $D = \{p, q\}$ and $U = E \setminus D$. By theorem 9.4.3 the group $H^1(U,\mathbb{Q})$ has dimension 3 and sits in an exact sequence of mixed Hodge structures

$$0 \to H^1(E,\mathbb{Q}) \to H^1(U,\mathbb{Q}) \xrightarrow{\mathrm{Res}} \mathbb{Q}(-1) \to 0 \tag{9.15}$$

where the copy of $\mathbb{Q}(-1)$ is spanned by the residue at $p$, the residue at $q$ being its negative.

*A basis adapted to the sequence.* Fix a parallelogram $P$ as in lemma 9.4.1, with interior lifts $\tilde{p}$ of $p$ and $\tilde{q}$ of $q$, and let $\gamma_1, \gamma_2$ be the two boundary cycles and $\sigma = \sigma_p$ a small positive loop about $p$. By proposition 9.4.2 there is a form $\omega$ on $E$ with simple poles at $p, q$ of residues $+1, -1$, unique up to adding a multiple of $dz$; normalize it by $\int_{\gamma_1} \omega = 0$, which fixes the multiple since $\int_{\gamma_1} dz = 1$. Put

$$\theta \;:=\; \frac{1}{2\pi i}\,\omega, \qquad u \;:=\; \tilde{p} - \tilde{q} \;\in \mathbb{C}$$

Lemma 9.4.1(2) evaluates the remaining period of $\omega$ as $2\pi i\,(\tilde{p}\cdot 1 + \tilde{q}\cdot(-1)) = 2\pi i\, u$, so the period matrix of the three closed forms $dz$, $d\bar{z}$ and $\theta$ against the three cycles is

$$\begin{pmatrix} \int_{\gamma_1} dz & \int_{\gamma_2} dz & \int_\sigma dz \\ \int_{\gamma_1} d\bar{z} & \int_{\gamma_2} d\bar{z} & \int_\sigma d\bar{z} \\ \int_{\gamma_1} \theta & \int_{\gamma_2} \theta & \int_\sigma \theta \end{pmatrix} \;=\; \begin{pmatrix} 1 & \tau & 0 \\ 1 & \bar{\tau} & 0 \\ 0 & u & 1 \end{pmatrix} \tag{9.16}$$

the entry $\int_\sigma \theta = 1$ because $\int_\sigma \omega = 2\pi i \mathrm{Res}_p \omega = 2\pi i$. Its determinant is $\bar{\tau} - \tau = -2i\,\mathrm{Im}\,\tau \neq 0$. Since $\dim H^1(U, \mathbb{C}) = 3 = \dim H_1(U, \mathbb{Q})$, nondegeneracy of (9.16) shows simultaneously that $\{[dz], [d\bar{z}], [\theta]\}$ is a basis of $H^1(U, \mathbb{C})$ and that $\{[\gamma_1], [\gamma_2], [\sigma]\}$ is a basis of $H_1(U, \mathbb{Q})$. A class is rational exactly when its three periods against the latter basis are rational.

*When the sequence splits.* A splitting of (9.15) in the category of mixed Hodge structures is a morphism $s\colon \mathbb{Q}(-1) \to H^1(U, \mathbb{Q})$ with $\mathrm{Res} \circ s = \mathrm{id}$. Since $\mathbb{Q}(-1)$ is pure of weight 2 with $F^1 = \mathbb{C}$ and $F^2 = 0$, such an $s$ is the same thing as a class

$$\alpha \in H^1(U, \mathbb{Q}) \cap F^1 H^1(U, \mathbb{C}) \quad \text{with} \quad \mathrm{Res}\alpha \neq 0$$

the conditions on $W$ being automatic because $W_2$ is everything and $\mathbb{Q}(-1)$ has $W_1 = 0$. By theorem 9.4.3(4) the space $F^1 H^1(U, \mathbb{C})$ is spanned by $[dz]$ and $[\theta]$, so write $\alpha = c\,[\theta] + b\,[dz]$ with $c \neq 0$. Reading off (9.16), the periods of $\alpha$ are

$$\int_{\gamma_1} \alpha = b, \qquad \int_{\gamma_2} \alpha = cu + b\tau, \qquad \int_\sigma \alpha = c$$

so $\alpha$ is rational precisely when $b \in \mathbb{Q}$, $c \in \mathbb{Q}^\times$ and $cu + b\tau \in \mathbb{Q}$. Such $b, c$ exist if and only if $u \in \mathbb{Q} + \mathbb{Q}\tau$: given $u = r + s\tau$ with $r, s \in \mathbb{Q}$, take $c = 1$ and $b = -s$, which gives $cu + b\tau = r$; conversely $cu + b\tau \in \mathbb{Q}$ with $c \in \mathbb{Q}^\times$ and $b \in \mathbb{Q}$ forces $u \in \mathbb{Q} + \mathbb{Q}\tau$.

*The class.* The condition $u \in \mathbb{Q} + \mathbb{Q}\tau$ says exactly that the point $u + \Lambda_\tau$ of $E$ is a torsion point, and $u + \Lambda_\tau = p - q$ in the group law of $E$, since $\tilde{p}$ and $\tilde{q}$ lift $p$ and $q$. So (9.15) splits over $\mathbb{Q}$ if and only if $p - q$ is a torsion point of $E$, and the invariant controlling it is

$$e(p, q) \;=\; p - q \;\in\; E \tag{9.17}$$

the image of the degree-zero divisor $p - q$ under the map that sends a difference of points to the class of $\int_q^p dz$ modulo $\Lambda_\tau$. Over $\mathbb{Z}$ the criterion is sharper. An integral splitting is a class with integral periods and $\int_\sigma \alpha = 1$, so $c = 1$, and rationality of the remaining periods is replaced by $b \in \mathbb{Z}$ and $u + b\tau \in \mathbb{Z}$, that is by $u \in \Lambda_\tau$ and $p = q$. The integral extension of $\mathbb{Z}(-1)$ by $H^1(E, \mathbb{Z})$ therefore never splits for $p \neq q$, and it is only after tensoring with $\mathbb{Q}$ that torsion points become invisible.

*Two concrete cases.* Take $\tau = i$ and $p = 0$. If $q = \frac{1}{2}$ then $u = -\frac{1}{2} \in \mathbb{Q}$, so the sequence splits over $\mathbb{Q}$; the splitting class is $\alpha = [\theta]$, with periods $(0, -\frac{1}{2}, 1)$, and $2\alpha$ has integral periods. If instead $q = \frac{1}{\pi}$, then $u = -\frac{1}{\pi}$ is real and irrational, hence not of the form $r + si$ with $r, s \in \mathbb{Q}$, so $q$ is not a torsion point of $\mathbb{C}/(\mathbb{Z} \oplus \mathbb{Z}i)$, no splitting exists over $\mathbb{Q}$, and $H^1(U, \mathbb{Q})$ is a non-split extension. The two curves $E \setminus \{0, \frac{1}{2}\}$ and $E \setminus \{0, \frac{1}{\pi}\}$ have the same Betti numbers, the same Hodge numbers, and isomorphic $\mathrm{gr}^W$, and are distinguished by the extension alone.

The example is the geometric form of the phenomenon that example 9.1.4 exhibited by hand. A mixed Hodge structure is not determined by its graded pieces, and here the residual information is a point of $E$, varying continuously with the punctures while $\mathrm{gr}^W$ stays constant. The general statement behind (9.17) is Carlson's description of $\mathrm{Ext}^1$ in the category of mixed Hodge structures: for $A$ pure of weight $-1$, extensions of $\mathbb{Q}(0)$ by $A$ are classified by $A_{\mathbb{C}}/(F^0 A_{\mathbb{C}} + A_{\mathbb{Q}})$, and the computation above is that quotient written out for $A = H^1(E, \mathbb{Q})(1)$, where it is the Jacobian of $E$, that is $E$ itself. The general theorem is not needed for the case at hand and is proved in *Period Mappings and Period Domains* [1, extensions of mixed Hodge structures]; the original construction of the mixed structure on a smooth open variety is Deligne's [13].

One curve has now been examined from four positions. Example 5.4.3 gave the pure weight-one structure on $H^1(E, \mathbb{Q})$ and its polarization, with the period ratio $\tau$ as the moduli coordinate. Example 7.1.9 put $E$ in the Legendre family and differentiated its periods, producing the Picard-Fuchs equation and the monodromy. Example 8.2.9 computed the Mumford-Tate group of that structure and found it to be all of $\mathrm{GL}_2$ except when $E$ has complex multiplication. And example 9.4.5 leaves the compact curve, where the answer is an extension class living in $E$ rather than a subspace of $H^1$.

The hypothesis still in force is smoothness of $U$. Dropping it means allowing $X$ itself to be singular, where there is no logarithmic complex to filter because there is no smooth ambient space. **??** replaces $X$ by a simplicial object built out of smooth pieces and runs the same argument on the resulting bicomplex.

## Exercise 9.4.1

1. Example 9.3.13 computed the mixed Hodge structure on $H^1(\mathbb{CP}^1 \setminus D, \mathbb{Q})$ for $D$ a set of $n \geq 1$ points, finding it pure of weight two and isomorphic to $\mathbb{Q}(-1)^{n-1}$. Take that as given. For $n = 2$ identify a generator of $H^1(\mathbb{C}^*, \mathbb{Q})$ explicitly as the class of a logarithmic form, and explain why the answer is pure while $H^1(E \setminus \{p\}, \mathbb{Q})$ of theorem 9.4.3 is not, in terms of the weights available on $\mathbb{CP}^1$ and on $E$.

2. Lemma 9.1.5(2) shows that a morphism between pure structures of different weights vanishes. Use it to deduce that $H^1(E \setminus \{p\}, \mathbb{Q})$ contains no sub-mixed-Hodge-structure isomorphic to $\mathbb{Q}(-1)$, and that there is no nonzero morphism $H^1(E, \mathbb{Q}) \to \mathbb{Q}(-1)$. Explain why this does not contradict the existence of the surjection $H^1(E \setminus \{p\}, \mathbb{Q}) \twoheadrightarrow \mathbb{Q}(-1)$ onto the top graded piece.

3. In the notation of example 9.4.5, suppose $p - q$ is a torsion point of $E$ of exact order $N$, so that $u = r + s\tau$ with $r, s \in \mathbb{Q}$. Write down a rational class $\alpha \in H^1(U, \mathbb{Q}) \cap F^1 H^1(U, \mathbb{C})$ with $\int_\sigma \alpha = 1$, compute its residues at $p$ and at $q$, and determine the smallest positive integer $m$ for which $m\alpha$ has integral periods against $[\gamma_1], [\gamma_2], [\sigma]$.

4. The affine curve $C_\lambda = \{y^2 = x(x-1)(x-\lambda)\} \subseteq \mathbb{C}^2$ is $E_\lambda \setminus \{\infty\}$ for the Legendre fibre $E_\lambda$ of example 7.1.9, the point at infinity being the origin of the group law. Use corollary 9.4.4

to show that $H^1(C_\lambda, \mathbb{Q})$ is pure of weight one. Then let $C'_\lambda = C_\lambda \setminus \{(0,0)\}$ and describe the mixed Hodge structure on $H^1(C'_\lambda, \mathbb{Q})$: give its dimension, its graded pieces, and decide whether the resulting extension splits over $\mathbb{Q}$.

5. Keep the notation of example 9.4.5 and let $c \in E$. Using proposition 9.3.14 for the translation $t_c\colon E \to E$, show that the splitting criterion for $D = \{p+c,\ q+c\}$ is the same as for $D = \{p, q\}$; then show that interchanging the roles of $p$ and $q$ replaces $e(p,q)$ by $-e(p,q)$. Conclude that (9.17) depends only on the difference $p - q$ and changes sign under the interchange.

6. Corollary 9.3.12 identifies $W_k H^k(U, \mathbb{Q})$ with the image of $j^*\colon H^k(X, \mathbb{Q}) \to H^k(U, \mathbb{Q})$ for a good compactification $X$ of $U$. Use it, together with the injectivity of $j^*$ from Step 2 of the proof of theorem 9.4.3, to obtain $W_1 H^1(U, \mathbb{Q}) = j^* H^1(E, \mathbb{Q})$ without the count of Hodge numbers in Step 3, and deduce $\dim_{\mathbb{Q}} \mathrm{gr}^W_2 H^1(U, \mathbb{Q}) = n - 1$. Say which part of Step 3 this does not replace, and check that the same corollary gives $W_1 H^1(V, \mathbb{Q}) = 0$ for the complement $V$ of exercise 9.4.1 (1).

# A

# The Newlander-Nirenberg Theorem

The Newlander-Nirenberg theorem (theorem 1.1.18) splits into a forward and a converse direction. The forward direction, that a complex manifold has vanishing Nijenhuis tensor, is essentially a calculation in holomorphic coordinates and was given as corollary 1.1.17. The converse direction, that an almost complex structure with $N_J = 0$ must arise from a complex manifold structure, is the substantive content. This appendix sets that problem up in full, isolates the two obstacles that make it hard, and cites Hörmander for the iteration that resolves them. What is proved here and what is cited is stated precisely at each step; the one deferred ingredient is the convergence of the iteration, and theorem A.0.3 is labelled accordingly.

The argument proceeds by constructing local holomorphic coordinates around each point. The hypothesis $N_J = 0$ is reformulated, via proposition 1.1.16, as the involutivity of the $(0,1)$-distribution $T^{0,1}M \subseteq TM_{\mathbb{C}}$. The task is to produce $n$ functionally independent functions $w_1, \ldots, w_n$, where $n$ is the complex dimension, satisfying $\bar{\partial} w_j = 0$, that is, killed by every section of $T^{0,1}M$. Such functions then serve as holomorphic coordinates: their differentials annihilate $T^{0,1}M$ and are linearly independent over $\mathbb{C}$. The construction reduces to solving a system of overdetermined first-order PDEs whose integrability is exactly the hypothesis $N_J = 0$, but the regularity required to pass from a formal power-series solution (which is automatic) to a smooth solution requires elliptic-PDE machinery. The proof below follows Hörmander [20, Chapter 5].

The remainder of the appendix is the proof of the converse direction.

**Local reduction.** The statement is local: holomorphic coordinates around each point assemble into a holomorphic atlas because the transition between any two such charts $w, w'$ has $\bar{\partial}^J w' = 0$ in the $w$-trivialization, hence is holomorphic (corollary 1.1.5 applied chart-locally). It therefore suffices, given $p \in M$ and the hypothesis $N_J = 0$, to produce holomorphic coordinates on a neighborhood of $p$.

Fix $p \in M$ and choose smooth real coordinates $x^1, \ldots, x^{2n}$ on a neighborhood $U \subseteq M$ of $p$, centered so that $x(p) = 0$ and the chart identifies $U$ with an open neighborhood of the origin in $\mathbb{R}^{2n}$. After an $\mathbb{R}$-linear change of coordinates on $\mathbb{R}^{2n}$, the operator $J(p)$ on $T_pM$ takes the standard form

$$J_0\big(\partial/\partial x^j\big)\big|_0 = \partial/\partial x^{j+n}\big|_0, \qquad J_0\big(\partial/\partial x^{j+n}\big)\big|_0 = -\partial/\partial x^j\big|_0$$

for $j = 1, \dots, n$. Set $z^j := x^j + i\, x^{j+n}$ and identify $U \subseteq \mathbb{R}^{2n}$ with an open neighborhood of $0 \in \mathbb{C}^n$. The Wirtinger operators of example 1.1.14 satisfy $\partial/\partial \bar{z}^j|_0 \in T^{0,1}_p M$ at the basepoint (though not generally elsewhere on $U$, since $J \neq J_0$ off the basepoint).

**The $J$-twisted Dolbeault operator.** The operator $\bar{\partial}^J$ on smooth $\mathbb{C}$-valued functions is the projection $\pi^{0,1}_J \circ d$, exactly as in definition 1.2.5 of section 1.2 but defined via the projection rather than in holomorphic coordinates (which do not yet exist). On the local chart $U$:

**Lemma A.0.1: Local Form of $\bar{\partial}^J$**

On the chart $U$, there exist smooth $\mathbb{C}$-valued functions $a^j_k, b^j_k \colon U \to \mathbb{C}$ with $a^j_k(0) = b^j_k(0) = 0$ such that for every smooth $f \colon U \to \mathbb{C}$,

$$\bar{\partial}^J f = \sum_j \left( \frac{\partial f}{\partial \bar{z}^j} + \sum_k a^j_k(z, \bar{z}) \frac{\partial f}{\partial z^k} + \sum_k b^j_k(z, \bar{z}) \frac{\partial f}{\partial \bar{z}^k} \right) d\bar{z}^j_J$$

where $\{d\bar{z}^j_J\}_{j=1}^n$ is a smooth frame for the $(0,1)$-cotangent bundle $T^{*0,1}_J U$, agreeing with $\{d\bar{z}^j\}$ at the basepoint.

***Proof*:**
The decomposition $T^*M_{\mathbb{C}} = T^{*1,0}_J \oplus T^{*0,1}_J$ has smooth projections by proposition 1.2.1. At $p$, the $(0,1)$-piece is spanned by $d\bar{z}^j|_0$. By smoothness of $J$ there exists a neighborhood of $p$ on which the projections give a smooth frame $\{d\bar{z}^j_J\}$ of $T^{*0,1}_J U$ with $d\bar{z}^j_J|_0 = d\bar{z}^j|_0$. Expressing $\pi^{0,1}_J(df)$ in this frame and the standard $\partial/\partial z^k$, $\partial/\partial \bar{z}^k$ basis for $df$ gives the formula. The coefficients $a^j_k, b^j_k$ are smooth (smooth in the frame, smooth in $J$) and vanish at the basepoint because $\pi^{0,1}_J\big|_0 = \pi^{0,1}_{J_0}$ and $\pi^{0,1}_{J_0}$ is precisely the standard projection $\sum_j (\partial f/\partial \bar{z}^j) d\bar{z}^j$ (no mixed terms), completing the proof.

The lemma rephrases $\bar{\partial}^J$ as a small perturbation of the standard $\bar{\partial}_0 = \sum_j (\partial/\partial \bar{z}^j) d\bar{z}^j$:

$$\bar{\partial}^J f = \bar{\partial}_0 f + (R \cdot df)$$

where $R$ is a smooth tensor of order zero in $df$ vanishing at the basepoint and $\bar{\partial}_0$ is identified with its image under the frame $\{d\bar{z}^j_J\}$ (which agrees with $\{d\bar{z}^j\}$ at the basepoint). Both sides are $\mathbb{C}$-linear first-order differential operators on functions.

**The integrability identity.** The hypothesis $N_J = 0$ (equivalently, the involutivity of $T^{0,1}_J M$ from proposition 1.1.16) implies that the $\bar{\partial}^J$ operator on functions extends to a $\bar{\partial}^J$ operator on $(p, q)$ forms with $(\bar{\partial}^J)^2 = 0$, exactly as in proposition 1.2.8 of section 1.2. The proof of those identities used $d = \partial^J + \bar{\partial}^J$, which held there because the manifold was complex. In the present setting $M$ is only almost complex, so $d$ on a $(p, q)$-form has four bidegree components (box 1.2.7); the two extra components vanish exactly when $N_J = 0$, restoring $d = \partial^J + \bar{\partial}^J$ and hence $(\bar{\partial}^J)^2 = 0$.

The integrability identity is the source of compatibility for the PDE system that produces holomorphic coordinates: the equation $\bar\partial^J u = f$ for smooth $f$ requires $\bar\partial^J f = 0$ as a necessary condition, and integrability ensures this is also sufficient locally.

**Hörmander's $L^2$-existence theorem.** The technical input is the following $\bar\partial$-existence theorem of Hörmander on a ball, which provides solvability with derivative control.

**Theorem A.0.2: Existence for $\bar\partial$ on a Ball, with Hölder Gain**

Let $B \subset \mathbb{C}^n$ be a ball and $f$ a $\bar\partial_0$-closed $(0,1)$-form on $\overline{B}$ whose coefficients lie in the Hölder class $\Lambda_\alpha(\overline{B})$ for some $\alpha \in (0, 1/2)$. Then the equation $\bar\partial_0 u = f$ has a solution on $B$ satisfying

$$\|u\|_{\Lambda_{\alpha+1/2}(\overline{B})} \leq C\,\|f\|_{\Lambda_\alpha(\overline{B})}$$

with $C$ depending only on $\alpha$, $n$, and the radius of $B$. If $f$ is smooth on $\overline{B}$ then $u$ may be taken smooth on $\overline{B}$.

The estimate is the standard one for the Henkin solution operator on a strictly pseudoconvex domain; the weighted $L^2$ method of [20, the weighted $L^2$ existence theorem for $\bar\partial$] gives existence, and the Hölder gain comes instead from the explicit integral kernel of the Henkin solution operator, which is a different mechanism from the $L^2$ theorem. Two features of the statement are used below and are worth marking now. The operator gains $\frac{1}{2}$ of a derivative, not a whole one; and the constant $C$ degrades in a definite way as the radius shrinks, which is what the iteration of Step 4 has to track.

**Construction of holomorphic coordinates.** The remainder of the proof produces, for every $p \in M$ with $N_J = 0$, holomorphic functions $w_1, \ldots, w_n$ on a neighborhood of $p$ with $dw_1, \ldots, dw_n$ a $\mathbb{C}$-basis of $T_p^{*1,0}M$ at the basepoint. The construction is by Picard iteration applied to a small ball around $p$.

**Theorem A.0.3: Local Holomorphic Coordinates**

Let $(M, J)$ be a smooth almost complex manifold of complex dimension $n$ with $N_J = 0$. For every $p \in M$ there exist smooth functions $w_1, \ldots, w_n \colon V \to \mathbb{C}$ on a neighborhood $V \subseteq M$ of $p$ such that:

1. $\bar\partial^J w_k = 0$ on $V$ for $k = 1, \ldots, n$;
2. $w_k(p) = 0$;
3. $dw_1|_p, \ldots, dw_n|_p$ form a $\mathbb{C}$-basis of $T_p^{*1,0}M$.

***Proof Key ideas:***
*Step 1: Reduction and ansatz.* By the local reduction at the start of this appendix, work on a chart $U$ around the origin in $\mathbb{C}^n$ with $J(0) = J_0$. By lemma A.0.1, $\bar\partial^J f = \bar\partial_0 f + R(z, \bar z) \cdot df$ where $R$ is a smooth tensor with $R(0) = 0$.

Seek $w_k = z^k + u_k$ for $u \colon U \to \mathbb{C}^n$ a smooth correction with $u(0) = 0$ and $du(0) = 0$. The equation $\bar{\partial}^J w_k = 0$ becomes

$$\bar{\partial}_0 u_k + R \cdot du_k = -\bar{\partial}^J z^k = -\bar{\partial}_0 z^k - R \cdot dz^k = -R \cdot dz^k$$

using $\bar{\partial}_0 z^k = 0$. Rearranging,

$$\bar{\partial}_0 u_k = -R \cdot dz^k - R \cdot du_k =: F_k(u)$$

a fixed-point equation for $u$ in which the right-hand side $F$ depends on $u$ through $du$.

*Step 2: Restriction to a small ball.* Fix $r > 0$ small enough that $\overline{B_r(0)} \subseteq U$ and rescale to the unit ball $B = B_1(0)$ via the dilation $z \mapsto z/r$. Under rescaling, $R(rz, \overline{rz}) = R(0,0) + O(r) = O(r)$ uniformly on $\overline{B}$, since $R(0) = 0$ and $R$ is smooth. Each derivative of $R$ in $z, \bar{z}$ is also $O(r)$ after the dilation. Specifically there exist constants $c_k$ depending only on $J$ and $k$ such that

$$\|R\|_{C^k(\overline{B})} \leq c_k\, r$$

for all $k$.

The ansatz becomes: find $u\colon \overline{B} \to \mathbb{C}^n$ smooth with $u(0) = 0$, $du(0) = 0$, and $\bar{\partial}_0 u_k = F_k(u)$ on $B$, where $F_k(u) = -R \cdot dz^k - R \cdot du_k$ in the rescaled coordinate. The dilation gives smallness of the coefficient $R$ and nothing else: the term $R \cdot du_k$ still carries one derivative of the unknown, which Step 4 shows is the obstacle smallness cannot remove.

*Step 3: Compatibility obstruction.* For the equation $\bar{\partial}_0 u_k = F_k(u)$ to be solvable via theorem A.0.2, the right-hand side must be $\bar{\partial}_0$-closed, and in general it is not. Locating the obstruction requires first fixing the operator to be squared. The perturbation $R \cdot d$ of lemma A.0.1 was defined only on functions, so the expression $(R\,d)^2$ is not yet meaningful. Both $\bar{\partial}^J$ and $\bar{\partial}_0$ are, however, defined on $(0,q)$-forms as well as on functions, so their difference

$$\mathcal{R} := \bar{\partial}^J - \bar{\partial}_0$$

is defined on forms and restricts on functions to $R \cdot d$. It is $\mathcal{R}$ that the computation below squares.

The integrability hypothesis gives $(\bar{\partial}^J)^2 = 0$. Expanding $(\bar{\partial}_0 + \mathcal{R})^2 = 0$ and using $\bar{\partial}_0^2 = 0$,

$$\bar{\partial}_0 \mathcal{R} + \mathcal{R}\, \bar{\partial}_0 + \mathcal{R}^2 = 0$$

as operators on functions. Apply this to $z^k + u_k$, and use $F_k(u) = -\mathcal{R}(z^k + u_k)$ together with $\bar{\partial}_0 z^k = 0$:

$$\bar{\partial}_0 F_k(u) = -\bar{\partial}_0 \mathcal{R}(z^k + u_k) = \mathcal{R}\, \bar{\partial}_0 u_k + \mathcal{R}^2(z^k + u_k)$$

The right-hand side is the obstruction to $\bar{\partial}_0$-closedness of $F_k(u)$. Two observations follow.

1. The obstruction vanishes at any solution. If $\bar{\partial}_0 u_k = F_k(u) = -\mathcal{R}(z^k + u_k)$, then substituting into the display gives $\mathcal{R}\big(-\mathcal{R}(z^k + u_k)\big) + \mathcal{R}^2(z^k + u_k) = 0$, the two terms cancelling exactly. The system is compatible precisely where it is solved, and nowhere else.

2. Away from a solution the obstruction is genuinely nonzero, though it is $O(r)$-small relative to $F_k(u)$ by the bound $\|R\|_{C^k} \leq c_k r$ of Step 2.

Observation (1) is the reason no naive iteration can be run directly: theorem A.0.2 needs closed data, and the data is closed only once the problem is already solved.

*Step 4: Why the iteration is delicate, and what carries it out.* Write

$$\Phi(u)_k := \bar{\partial}_0 u_k - F_k(u) = \bar{\partial}_0 u_k + \mathcal{R}(z^k + u_k) = \bar{\partial}^J(z^k + u_k)$$

so the goal is $\Phi(u) = 0$. Since $\mathcal{R}$ acts on functions as $R \cdot d$, the map $\Phi$ is *affine* in $u$, with linear part $\bar{\partial}^J = \bar{\partial}_0 + \mathcal{R}$. Three facts then determine the shape of any argument, and naming them precisely is what this appendix contributes.

*(a) The scheme loses half a derivative per pass.* Theorem A.0.2 returns $\frac{1}{2}$ a derivative, $\|v\|_{\Lambda_{\alpha+1/2}} \leq C\|f\|_{\Lambda_\alpha}$, while the perturbation costs a whole one, $\|\mathcal{R}w\|_{\Lambda_\alpha} \leq C\|R\|_{C^1}\|w\|_{\Lambda_{\alpha+1}}$, because $\mathcal{R}w = R \cdot dw$. So the naive iteration $u \mapsto T(-\mathcal{R}(z+u))$ carries $\Lambda_{\alpha+1}$ only to $\Lambda_{\alpha+1/2}$. No Hölder ball is preserved and no contraction mapping is available. Shrinking $r$ does not repair this: smallness of $R$ multiplies the estimate but does not change which norm sits on the right.

*(b) The data is closed only to within $O(r)$.* Theorem A.0.2 requires $\bar{\partial}_0$-closed data, which Step 3 showed holds only at a solution. The defect is at least controlled: from $(\bar{\partial}^J)^2 = 0$ one gets $\bar{\partial}^J\Phi(u) = 0$, hence

$$\bar{\partial}_0\Phi(u) = -\mathcal{R}\,\Phi(u)$$

so the failure of closedness is $O(r)$ in size but costs one derivative, the same currency as (a).

*(c) Affineness removes the usual engine.* The standard response to a loss of derivatives is a Nash-Moser scheme: interleave the solution operator with smoothing operators $S_\theta$, accept that smoothing reintroduces the lost derivative at a cost $\theta^1$, and rely on the *quadratic* convergence of a Newton iteration to outrun that cost. Here that engine is unavailable in its usual form. Because $\Phi$ is affine,

$$\Phi(u^{(m+1)}) = \Phi(u^{(m)}) + \bar{\partial}^J\dot{u}^{(m)}$$

holds exactly, with no quadratic remainder to exploit; a smoothed Newton step leaves an error term of size $O(r\,\theta_m)\|\Phi(u^{(m)})\|$, and since $\theta_m \to \infty$ no fixed $r > 0$ absorbs it beyond the first few stages. Any correct treatment must therefore either recover a quadratic gain by reformulating the problem nonlinearly, or replace the Newton scheme by a different mechanism.

Both routes exist in the literature. Hörmander's is the elliptic one, run through the $L^2$ theory rather than a Newton iteration [20, Chapter 5]. Webster's recasts the problem so that a Nash-Moser argument does apply, with the estimates organized economically enough to be checked by hand [35, a new proof of the Newlander-Nirenberg theorem]. Either produces, for every sufficiently small $r > 0$ depending only on $J$ and $n$, a smooth $u\colon \overline{B_r(0)} \to \mathbb{C}^n$ with $\bar{\partial}^J(z+u) = 0$, $u(0) = 0$ and $du(0) = 0$. That construction is not reproduced here; (a), (b) and (c) are its conceptual content, and (c) in particular is the reason the obvious scheme cannot simply be written down.

*Step 5: Verification of the three conclusions.* Take the solution $u$ supplied by Step 4, smooth on $\overline{B_r(0)}$ with $u(0) = 0$ and $du(0) = 0$, and set $w := z + u\colon \overline{B_r(0)} \to \mathbb{C}^n$. Then $\bar{\partial}^J w_k = 0$ on the ball, which is (1); $w_k(0) = 0$, which is (2); and $dw_k|_0 = dz^k|_0$ because $du(0) = 0$, so that $dw_1|_0, \ldots, dw_n|_0$ are the standard basis $dz^1|_0, \ldots, dz^n|_0$ of $T_0^{*1,0}M$, which is (3). Pulling back from the ball to a chart on $M$ along the original coordinate map transports all three to a neighborhood $V$ of $p$, establishing the theorem.

**Conclusion of Newlander-Nirenberg.** The functions $w_1, \ldots, w_n$ produced by theorem A.0.3 form a holomorphic coordinate chart on a neighborhood $V$ of each $p \in M$: the differentials are $\mathbb{C}$-linearly independent at $p$ by part (3), hence in a neighborhood by continuity, so $w$ is a local diffeomorphism onto an open subset of $\mathbb{C}^n$; the $\bar{\partial}^J w_k = 0$ condition of part (1) makes the coordinate chart compatible with the $J$-induced $(0,1)$-structure. Two such charts $w$, $w'$ on overlapping neighborhoods have $\bar{\partial}^J(w'_k \circ w^{-1}) = 0$ in the $w$-trivialization, hence $w' \circ w^{-1}$ is holomorphic by corollary 1.1.5. The collection $\{w\}$ defines a holomorphic atlas on $M$ inducing the original $J$, completing the converse direction of theorem 1.1.18 and finishing the proof of the theorem.

# Index

# *Bibliography*


[1] James Carlson, Stefan Müller-Stach, and Chris Peters. *Period Mappings and Period Domains*. 2nd ed. Cambridge Studies in Advanced Mathematics 168. Cambridge: Cambridge University Press, 2017. ISBN: 978-1-316-63956-6. DOI: 10.1017/9781316995846.

[2] Eduardo Cattani, Pierre Deligne, and Aroldo Kaplan. "On the Locus of Hodge Classes". In: *Journal of the American Mathematical Society* 8.2 (1995), pp. 483–506. DOI: 10.1090/s0894-0347-1995-1273413-2. URL: https://doi.org/10.1090/s0894-0347-1995-1273413-2.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Abelian Varieties*. 1st ed. 2026. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_abelian_varieties.pdf.

[4] Nathanael Chwojko-Srawley. *Everything You Need to Know about Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elliptic Curves*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_elliptic_curves.pdf.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Homological Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_homological_algebra.pdf.

[7] Nathanael Chwojko-Srawley. *Everything You Need To Know About Linear Elliptic and Parabolic PDE*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_linear_pde.pdf.

[8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Microlocal Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA-microlocal-analysis.pdf.

[9] Nathanael Chwojko-Srawley. *Everything You Need To Know About Riemannian Geometry*.

[10] Nathanael Chwojko-Srawley. *Everything You Need To Know About Shimura Varieties*.

[11] Nathanael Chwojko-Srawley. *Everyting You Need To Know About Geometric Representation-Theory*.

[12] P. Deligne. "Hodge Cycles on Abelian Varieties". In: *Lecture Notes in Mathematics*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1982, pp. 9–100. ISBN: 978-3-540-11174-0. DOI: 10.1007/978-3-540-38955-2_3. URL: https://doi.org/10.1007/978-3-540-38955-2_3.

[13] Pierre Deligne. "Théorie de Hodge : II". In: *Publications Mathématiques de l'IHÉS* 40 (Dec. 7, 1971), pp. 5–57. DOI: 10.1007/bf02684692. URL: https://doi.org/10.1007/bf02684692.

[14] Pierre Deligne. "Théorie de Hodge : III". In: *Publications Mathématiques de l'IHÉS* 44 (Jan. 7, 1974), pp. 5–77. DOI: 10.1007/bf02685881. URL: https://doi.org/10.1007/bf02685881.

[15] Phillip Griffiths and Joseph Harris. *Principles of Algebraic Geometry by Griffith and Harris*. 1st ed. John Wiley & Sons, Ltd, 1978. 824 pp. ISBN: 978-0-471-05059-9. DOI: 10.1002/9781118032527.fmatter. URL: https://onlinelibrary-wiley-com.myaccess.library.utoronto.ca/doi/book/10.1002/9781118032527.

[16] Phillip Griffiths and Wilfried Schmid. "Locally Homogeneous Complex Manifolds". In: *Acta Mathematica* 123.0 (1969), pp. 253–302. DOI: 10.1007/bf02392390. URL: https://doi.org/10.1007/bf02392390.

[17] Phillip A. Griffiths. "Periods of Integrals on Algebraic Manifolds, I. (Construction and Properties of the Modular Varieties)". In: *American Journal of Mathematics* 90.2 (Apr. 1968), p. 568. DOI: 10.2307/2373545. URL: https://doi.org/10.2307/2373545.

[18] Heisuke Hironaka. "Resolution of Singularities of an Algebraic Variety Over a Field of Characteristic Zero: I". In: *The Annals of Mathematics* 79.1 (Jan. 1964), p. 109. DOI: 10.2307/1970486. URL: https://doi.org/10.2307/1970486.

[19] W. V. D. Hodge. *The Theory and Applications of Harmonic Integrals*. Cambridge, New York: Cambridge University Press, 1989. ISBN: 978-0-521-35881-1. URL: http://openlibrary.org/books/OL1791471M/The_theory_and_applications_of_harmonic_integrals.

[20] Lars Hörmander. *An Introduction to Complex Analysis in Several Variables*. 3rd ed. North-Holland Mathematical Library 7. Amsterdam: North-Holland, 1990. ISBN: 978-0-444-88446-6.

[21] Daniel Huybrechts. *Complex Geometry: An Introduction*. Universitext. Berlin, Heidelberg: Springer, 2005. ISBN: 978-3-540-21290-4. DOI: 10.1007/b137952.

[22] Daniel Huybrechts. *Lectures on K3 Surfaces*. Cambridge University Press, 2016. ISBN: 978-1-107-15304-2. URL: http://openlibrary.org/books/OL28609296M/Lectures_on_K3_Surfaces.

[23] Erich Kähler. "Über Eine Bemerkenswerte Hermitesche Metrik". In: *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 9.1 (Dec. 1933), pp. 173–186. DOI: 10.1007/bf02940642. URL: https://doi.org/10.1007/bf02940642.

[24] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. 126 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.

[25] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. 233 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.

[26] Nathanael Chwojko-Srawley. *Everything You Need To Know About Differential Topology*. 1st ed. 233 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_differential_geometry.pdf.

[27] Nathanael Chwojko-Srawley. *Everything You Need to Know about K-theory*. Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_K-Theory.pdf.

[28] Chris A. M. Peters and Joseph H. M. Steenbrink. *Mixed Hodge Structures.* Ergebnisse Der Mathematik Und Ihrer Grenzgebiete. 3. Folge / a Series of Modern Surveys in Mathematics 52. Berlin, Heidelberg: Springer, 2008. ISBN: 978-3-540-77015-2. DOI: 10.1007/978-3-540-77017-6.

[29] I. I Pjateckiĭ-Šapiro and I R Šafarevič. "A TORELLI THEOREM FOR ALGEBRAIC SURFACES OF TYPE$K3$". In: *Mathematics of the USSR-Izvestiya* 5.3 (June 30, 1971), pp. 547–588. DOI: 10.1070/im1971v005n03abeh001075. URL: https://doi.org/10.1070/im1971v005n03abeh001075.

[30] Wilfried Schmid. "Variation of Hodge Structure: The Singularities of the Period Mapping". In: *Inventiones Mathematicae* 22.3–4 (Sept. 1973), pp. 211–319. DOI: 10.1007/bf01389674. URL: https://doi.org/10.1007/bf01389674.

[31] Jean-Pierre Serre. "Géométrie Algébrique et Géométrie Analytique". In: *Annales de l'Institut Fourier* 6 (1956), pp. 1–42. DOI: 10.5802/aif.59. URL: https://doi.org/10.5802/aif.59.

[32] Nathanael Chwojko-Srawley. *Everything You Need To Know About Lie Groups.*

[33] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry I.* Trans. by Leila Schneps. Cambridge Studies in Advanced Mathematics 76. Cambridge: Cambridge University Press, 2002. 1 p. ISBN: 978-0-511-61534-4. DOI: 10.1017/CBO9780511615344.

[34] Claire Voisin. *Hodge Theory and Complex Algebraic Geometry II.* Trans. by Leila Schneps. Cambridge Studies in Advanced Mathematics 77. Cambridge: Cambridge University Press, 2003. ISBN: 978-0-521-71802-8. DOI: 10.1017/CBO9780511615177.

[35] S. M. Webster. "A New Proof of the Newlander-Nirenberg Theorem". In: *Mathematische Zeitschrift* 201.3 (Sept. 1989), pp. 303–316. DOI: 10.1007/bf01214897. URL: https://doi.org/10.1007/bf01214897.

[36] Hermann Weyl. "On Hodge's Theory of Harmonic Integrals". In: *The Annals of Mathematics* 44.1 (Jan. 1943), p. 1. DOI: 10.2307/1969060. URL: https://doi.org/10.2307/1969060.